Luca Longo
Sebastian Lapuschkin
Christin Seifert (Eds.)

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Explainable Artificial Intelligence

Second World Conference, xAI 2024
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Proceedings, Part II

Part 2

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Luca Longo · Sebastian Lapuschkin ·
Christin Seifert
Editors

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Editors
Luca Longo
Technological University Dublin
Dublin, Ireland

Christin Seifert
University of Marburg
Marburg, Germany

Sebastian Lapuschkin
Fraunhofer Institute for Telecommunications,
Heinrich-Hertz-Institut, HHI
Berlin, Germany

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Preface

The field of eXplainable AI (XAI) has seen significant growth recently. As part of the larger field of Artificial Intelligence, it has evolved into a highly multi-disciplinary and inter-disciplinary active field of research. Artificial Intelligence and its methods have been used in many disciplines besides computer science, including medicine and neuroscience, chemistry, biology, education, psychology and philosophy. Thanks to its evolving subfield, machine learning, it has been applied in many real-world applications. This is due to its ability to learn and extract patterns automatically from sparse, complex, non-linear data. These include prediction and forecasting, classification and recommendation, to mention a few. However, in some critical applications, such as finance and health care, understanding machine-learned models and their underlying inferential mechanisms is paramount for creating trustworthy and responsible applications. In addition, the requirements imposed by the GDPR, the new AI act, and those regulations that will follow worldwide are leading scholars to develop methods for explaining AI systems and their outputs. This is evident in the hundreds of manuscripts submitted to the 2nd World Conference of Artificial Intelligence, their diverse methodology, techniques and approaches, and their application in various real-world contexts. Similarly to the first edition, the second edition has attracted considerable interest from scholars in academia and industry. The hundreds of authors, attendees and programme committee members from more than 40 countries make the conference a truly world event. With an acceptance rate of roughly ~40%, with 95 manuscripts being accepted from 204 submissions, it is our great privilege to present the proceedings of the second World Conference on eXplainable Artificial Intelligence (xAI 2024), held in Valletta, Malta, from the 17th to the 19th of July at the historic Mediterranean Conference Centre, a fascinating venue.

Split over four volumes, this book aggregates a collection of the best contributions received and presented at xAI 2024, describing recent approaches, methods and techniques for explainability. The accepted articles were selected through a rigorous, single-blind peer-review process. Each article received at least three reviews, with an average of four reviews per paper, from more than 250 scholars in academia and industry, with 99% of them holding a PhD in an area relevant to the topics of the conference. The programme committee chairs of the conference carefully selected the top contributions by ranking articles across several objective criteria and evaluating and triangulating the qualitative feedback left by the international reviewers. The peer-review process was exhaustive and intensive, ensuring that xAI-2024 adhered to the highest quality standards. All the accepted research contributions are included in these proceedings and were invited to give oral presentations.

Several special tracks and thematic sessions were organised, each proposed and chaired by various scholars, to aggregate highly innovative areas within the larger field of explainable Artificial Intelligence. A parallel track was organised for work in progress, specifically preliminary novel research studies relevant to xAI, which were presented as posters during the event. A demo track was held where scholars presented their software

prototypes on explainability or real-world applications of explainable AI-based systems. A doctoral consortium was organised, with lecturers to PhD scholars who submitted their doctoral proposals on future research related to eXplainable Artificial Intelligence. These scholars pitched their preliminary doctoral work and were matched with renowned scientists who provided guidance and constructive feedback. A separate programme committee was set up for the late-breaking work and demo and doctoral consortium tracks. Finally, a panel discussion was organised with renowned scholars in xAI, offering a multidisciplinary view while inspiring the attendees with tangible recommendations to tackle challenges toward designing responsible, trustworthy AI-based technologies through explainable AI.

A thank you to all the volunteers who helped in the organising committee for the 2nd World Conference on eXplainable Artificial Intelligence (xAI 2024). In particular, we would like to thank the local chair, Charlie Abela; the doctoral committee chair, Grégoire Montavon; the inclusion & accessibility chairs, Mario Brcic and Verena Klös; and the late-breaking work chair, Weiru Liu. Also, special thanks to the keynote speaker, Fosca Giannotti. A word of appreciation goes to the organisers of the special tracks and those who chaired them during the conference. Special thanks go to the researchers and practitioners who submitted their work, the various programme committee members who provided precious feedback during the peer-review process, and all who attended the event and turned it into a fantastic networking opportunity to share findings and learn from each other as a community.

The Mediterranean Conference Centre, a 16th-century marvel in historic Valletta, was initially built as a hospital by the Order of St. John. Known as the Sacra Infermeria, it was meant to receive Maltese and foreign patients and pilgrims travelling to the Holy Land. Analogously, explainable artificial intelligence (XAI) is intended to be a transit between current opaque AI-based technologies and the development of robust, transparent, fair and trustworthy AI for the benefit of humankind.

May 2024

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XAI for Graphs and Computer Vision

Model-Agnostic Knowledge Graph Embedding Explanations for Recommender Systems

André Levi Zanon[1(✉)], Leonardo Chaves Dutra da Rocha[2], and Marcelo Garcia Manzato[1]

[1] Universidade de São Paulo, São Paulo, Brazil
andrezanon@usp.br, mmanzato@icmc.usp.br
[2] Universidade Federal de São João del-Rei, Minas Gerais, Brazil
lcrocha@ufsj.edu.br

Abstract. Explanations in recommender systems play an essential role in enhancing transparency, trust, and persuasiveness. In that regard, Knowledge Graphs (KGs) model-agnostic explanations do not rely on user-inputted data such as reviews or require any changes in a recommendation algorithm to provide explanations. The state-of-the-art of model-agnostic KG explainable algorithms are based on syntactic approaches that consider the trade-off of attributes among the user-interacted items and the catalog to explain recommendations. In this study, we propose a novel model-agnostic KG algorithm for explanations. Our approach utilizes KG embeddings to rank explanations based on the path's similarity to the user. Specifically, we train an embedding algorithm on a KG and compare path embeddings, composed of node and edge embeddings, to the user embedding derived from previously interacted item embeddings. Our proposed method is evaluated by comparing it against three baselines representing the state-of-the-art of KG explanation algorithms. We assess explanation quality using three metrics: diversity and popularity of attributes displayed in explanations and recency of interacted items. Results indicate that the embedding approach achieves a superior balance between attribute popularity and explanation diversity. Furthermore, our analysis emphasizes the importance of tailored metrics for evaluating explanations in recommender systems.

Keywords: Recommender Systems · Model-agnostic explanation · Knowledge Graphs · Explanation Evaluation · Explainable AI

1 Introduction

Recommender Systems (RS) provide suggestions to users based on the underlying relation between interacted items, content descriptions, other users' interactions, and domain knowledge [33]. To capture such dependencies, RS architectures have become increasingly complex and consequently are considered black

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boxes concerning transparency, meaning that users can not understand why a recommendation was provided [39].

Explaining recommendations has received attention due to its ability to personalize the user experience by offering reasons for interacting with suggested items [3,19]. Recently, the demand for explainability has grown, driven by a heightened awareness of the social impact of recommender systems on user behavior in e-commerce, as well as ethical considerations affecting users' perceptions of advertising and news recommendations [10,18].

Explanations can improve recommender systems by providing transparency, scrutability, trust, effectiveness, persuasiveness, efficiency, and satisfaction [39]. Table 1 displays each of these goals and their respective definition.

Table 1. Definition of each explanation goal by [39]

Goal	Definition
Transparency	Understanding how the systems work
Scrutability	Allow the user to correct the system
Trust	Increase the confidence in the system
Effectiveness	Help the user make good decisions
Persuasiveness	Convince the user to interact with a recommendation
Efficiency	Help users make faster decisions
Satisfaction	Increase the user joy in using the system

To achieve these goals, recommender system (RS) architectures are categorized into model-agnostic and model-intrinsic approaches [34,47]. Model-agnostic methods use separate algorithms to explain black-box recommendations, while model-intrinsic approaches aim to generate explanations alongside suggestions, potentially in a white or gray-box manner. Model-agnostic approaches do not rely on the recommendation algorithm, making them compatible with any algorithm, but they may lack fidelity in explaining the recommendation process [30]. Conversely, model-intrinsic methods may impact system latency and expose it to adversarial attacks [41].

1.1 Problem Setting and Objective

Explanations in recommender systems are crucial for enhancing user understanding and satisfaction. While collaborative information from user interactions is valuable, users tend to be more receptive to explanations that utilize item metadata [21,47]. These explanations can be enriched by external sources such as Knowledge Graphs (KGs) or text reviews [41].

As most recommender systems remain opaque to users, written reviews can be cumbersome and reliant on user input. Post-hoc knowledge-graph explainable algorithms offer a solution by connecting user interactions with recommendations

based on item metadata without altering the system's architecture. However, existing post-hoc KG algorithms predominantly use syntactic approaches, which may not fully capture the semantic relationships within the data. In contrast, graph-embedding approaches offer semantic representations of nodes by projecting them into a latent space, showing promise for various applications, including recommender systems [28].

This study aims to address two research questions. **RQ1: What differences exist in generating explanations for recommendations using KG syntactic and embedding post-hoc approaches?** We conducted an offline experiment comparing three syntactic approaches with a newly proposed graph embedding algorithm. Evaluation metrics from [2] were employed to assess explanation quality, focusing on the recency of interacted items and the popularity and diversity of attributes. However, there is no evidence in the literature that these metrics fully capture enhancements in explanation goals, such as transparency and trust. As a result, the second research question (**RQ2**) is: **Do improvements in offline metrics of explanation quality directly correspond to improvements in online explanation goal metrics?** We address this question by comparing offline metric results with online user studies reported in the literature evaluating explanation goals for the same three baseline algorithms.

1.2 Overview of Main Findings and Contributions

The main contributions of this paper are as follows:

- Proposal of a KG embedding-based post-hoc explanation algorithm adaptable to any recommendation algorithm.
- Comparative analysis between embedding and syntactic approaches in KGs for generating model-agnostic explanations in recommender systems, considering three quality explanation metrics: recency of items, popularity of attributes, and diversity of attributes.
- Discussion on the relationship between offline experiment results based on explanation quality metrics and online experiments reported in the literature regarding explanation goal metrics.

Our findings demonstrate that the proposed embedding method better balances attribute popularity and diversity than syntactic approaches. In addition, while improvements in offline metrics reflect on explanation quality, they may not fully align with enhancements in online explanation goal metrics. This highlights the need for further research into metrics that accurately reflect explanation goals in explainable recommender systems.

The paper is structured as follows: Sect. 2 reviews related work on knowledge-graph explainable algorithms and offline evaluation methods. Section 3 details our experimental setup, including metrics, datasets and baselines. The proposed algorithm is described in Sect. 4. Section 5 discusses our findings, and Sect. 6 explores limitations, future directions, and open research areas in explainable recommender systems. Finally, Sect. 7 summarizes our findings.

2 Related Work

Because KG provides structured metadata about items, it has been used to generate accurate and explainable recommendations in various recommendation architectures. In [14], the authors created a KG language path modeling approach to predict new paths between interacted items and recommendations. Each node serves as a token to the language model. Recommendations are generated based on the highest-scoring paths and used as an explanation. On [23], on the other hand, a Graph Convolutional Neural Network was used to leverage path and item embeddings that predict an interaction probability. Explanations are generated based on a path-level self-attention mechanism. A KG with Reinforcement Learning (RL) was used in [4] to create a recommender system that initially finds an entity based on a user and a relation from the KG, then an actor-critic RL algorithm is employed as reasoner to generate explanations. In [22] it was also adopted a RL approach to generate sequential-aware explainable recommendations by exploring paths based on the user's previous interactions with the system using the Reinforce RL algorithm. Finally, in [43] a self-supervised approach is proposed by masking triplets from a KG and recommending items with high predictive scores from an auto-encoder architecture; collaborative information is also encoded based on a knowledge-aware contrasting mechanism.

However, while these studies often prioritize accuracy and beyond-accuracy metrics, they frequently do not comprehensively evaluate the generated explanations. As a result, explanations are often showcased in simplified examples [27]. Furthermore, their qualitative assessment may not be as robust, potentially leaving them susceptible to biases inherent in the recommendation engine [6,38].

In this context, model-agnostic or post-hoc reordering explainable algorithms re-rank recommendations based on the association between interacted and recommended items with shared attributes. For instance, the authors in [2] utilized three optimization metrics - recency of interacted items, popularity, and diversity of attributes extracted from KG explanation paths - to reorder recommendations. Similarly, in [44] recommendations were reordered by evaluating the relevance of attributes extracted from explanation paths. The relevance of attributes was determined by comparing the frequency of attribute associations of interacted items with the catalog of items. Additionally, on [15] explanations were generated through aspect extraction and sentiment analysis, enhancing recommendation accuracy by incorporating textual reviews as a recommendation algorithm's regularizer. However, similar to explainable recommenders, these approaches evaluated explanations based on ranking metrics.

Considering post-hoc KG architectures to generate explanations, the authors of [25] created an algorithm called ExpLOD, which leverages explanations from a KG based on a bipartite graph that connects interacted with recommended items by the attributes they share. The explanations are ranked based on an adaptation of the Term-Frequency Inverse Document Frequency (TF-IDF) metric, where item nodes are documents and attribute nodes are terms. This work was evaluated by comparing the proposed explanation with information extracted

from the KG in an online experiment where the method achieved improved user perception considering explanation goals. The work on [26] extended ExpLOD [25], by adding broader attributes from the KG hierarchy. The same online experiment was conducted, but explanations from ExpLOD were compared with the proposed new version. Users preferred explanations with broader attributes when analyzing under the perspective of explanation goals. More recently, the authors in [12] proposed Property-based Explanation Model (PEM), a scoring function that ranks attributes based on their connections to interacted items and the entire item catalog. PEM outperformed the second version of ExpLOD in online experiments, establishing itself as the state-of-the-art for model-agnostic KG explainable algorithms. Nevertheless, these approaches are syntactic and do not intrinsically consider the KG structure and path to generate explanations.

Evaluating explanations has also been gaining attention since explanations are mainly evaluated based on online user trials, which can be time-consuming to validate algorithms. In [7], offline metrics were implemented to evaluate explainable recommendations; nevertheless, they assess the robustness of explainable algorithms by measuring the number of items that can be explainable for users and the number of user interactions related to explanations. In contrast, some papers also consider the diversity and relevance of attributes displayed in explanations [23]. However, the relation between such metrics and online tests is not defined. In addition, offline metrics are not standardized, and papers that evaluate with user studies lack quantitative evaluation [24].

In conclusion, explainable recommender systems often do not evaluate explanations quantitatively and qualitatively since they rely on other contributions, such as improving accuracy and beyond-accuracy metrics. On the other hand, model-agnostic explanations are assessed with online user trials, which are costly and limited to the evaluation of the number of participants and, as a result, are not extensively evaluated. In terms of evaluation metrics for explanations, current approaches either employ metrics whose effectiveness in reflecting improved user perception remains unstudied or validate the robustness of the explanation algorithm by verifying if explanations are grounded in user interactions and if the algorithm can generate explanations for all recommended items.

3 Methodology

This paper introduces a model-agnostic KG embedding-based algorithm for generating explanations in recommendation systems. The path forming the explanation is determined by the similarity of its embedding, which incorporates embeddings of items, relations, and attributes, with the user embedding - a composite of embeddings from interacted items. We compare our approach with three state-of-the-art algorithms: ExpLOD [25], ExpLOD version 2 (ExpLOD v2) [26], and the Property-based Explanation Model (PEM) [12], all of which have been validated through online user studies. Specifically, [26] compared ExpLOD v2 with ExpLOD in an online experiment, while [12] conducted a similar study comparing PEM with ExpLOD v2. Although statistical significance was not observed for

all metrics in both cases, one approach consistently outperformed the baseline in transparency, persuasion, trust, and effectiveness of explanation goals. Additionally, the papers introduced the goal of engagement, defined as presenting users with new information, in line with previous work [12,25,26].

In both online studies, users were presented with explanations only for the top-ranked recommendation. However, this approach limits the user's ability to discern biases in the explanation algorithm, particularly if it favors popular attributes [6]. Users can more effectively evaluate algorithmic bias by showcasing explanations for multiple recommendations. To assess explanation quality, [2] utilized three primary metrics: attribute diversity, attribute popularity, and item recency within the explanations. In summary, effective explanations establish connections between recently interacted items and recommendations while balancing the popularity and diversity of their attributes.

To address RQ1, we assessed the quality metrics of explanations generated by the proposed graph embedding approach and compared them with three state-of-the-art explainable algorithms using the MovieLens 100k dataset [16], focusing on explaining the Top-1 and Top-5 recommendations for all users of six recommender systems algorithms. This dataset was selected due to its common usage across all baselines and its inclusion in the online experiments conducted by [12,26]. Additionally, we utilized the LastFM dataset [5].

Influenced by the emphasis on reproducibility and robustness explanation evaluation highlighted by [13,38], respectively, and to ensure rigorous evaluation, we applied six recommender system algorithms across different families, utilizing 90% of the dataset for training and 10% for testing. Specifically, we employed the Most Popular algorithm [8] for non-personalized recommendations, the Personalized Page Rank algorithm [25] augmented with Wikidata graph for graph-based recommendations, User-KNN [32] for neighborhood algorithms, Embarrassingly Shallow AutoEncoder (EASE) [35] and Bayesian Personalized Ranking Matrix Factorization (BPR-MF) [31] for non-neural algorithms, and Neural Collaborative Filtering (NCF) [17] for neural-based architectures. We used the CaseRecommender [9] library to implement Most Popular, User-KNN, and BPR-MF algorithms. The implementations of the other recommenders, along with the explanation outputs, metrics generated by the explanation algorithms, and queries created to extract the triples for the KG construction, are available in an open-source repository[1].

To answer RQ2, we compared the results of explanation quality metrics of the offline experiments with the reported online results by [12,26]. Since the evolution of algorithms is straightforward, meaning that ExpLOD v2 outperformed ExpLOD in [26] and PEM outperformed ExpLOD v2 in [12] for explanation goals, we compared this evolution of the state-of-the-art regarding explanation goals with the ordering of the algorithms' explanation quality metrics by applying the Multi-Attribute Utility Theory (MAUT) from Game Theory. MAUT is used in decision-making for options with numerous criteria. We used MAUT to compare if the explanation algorithm chosen by the method with the criteria as

[1] https://github.com/andlzanon/lod-personalized-recommender.

the explanation quantity metrics match the evolution of the explanation goals perceived by users in online experiments.

A KG from Wikidata[2] Linked Open Data (LOD) was obtained for the movie and musical artist dataset domains to implement all the baselines and the proposed algorithms. This LOD was chosen because it has more up-to-date and data completeness of entities compared to the DBPedia[3] LOD [29]. Then, the original datasets were processed to keep only interactions with items obtained from Wikidata KG. The processed dataset remained with 99% of the original interactions for the MovieLens 100k dataset and 89% for the LastFM dataset.

A summary of the MovieLens 100k and LastFM statistics before and after the pre-processing and the KG information is available in Table 2. In the Wikidata KG section of the Table, entities regard items and items' attributes nodes from the dataset found in Wikidata KG; edge types are the possible relations between an item entity and an attribute. For example, when it is extracted the path that the movie Titanic has the actress Kate Winslet, the movie Titanic is an item entity node, and Kate Winslet is an entity attribute node. The edge "has the actress" that connects the item with the attribute, is the edge type. The two entities and the relation form a triple.

Table 2. Statistics of Original and Processed Datasets, and Knowledge Graph.

		MovieLens 100k	LastFM
Original Dataset	users	610	1,892
	items	9,724	17,632
	ratings	100,836	92,834
Processed Dataset	users	610	1,875
	items	9,517	11,641
	ratings	100,521	83,017
Wikidata KG	entities	78,703	34,297
	triples	29,5787	134,197
	edges types	23	33

In the following sections, the baseline syntactic algorithms for model-agnostic recommendation explanation will be explained, along with the calculation of each quality explanation metric for item recency, diversity of attributes, and popularity of attributes in explanations. At the end of this section, the MAUT method is also presented. The proposed KG embedding for model-agnostic explanation is described in Sect. 4.

[2] https://www.wikidata.org.
[3] https://www.dbpedia.org/.

3.1 Baselines

The ExpLOD method [25] ranks properties in the Knowledge Graph (KG) using an adapted TF-IDF approach. Here, the relevance of an attribute is determined by the frequency of references to the attribute from both interacted and recommended items relative to the total references to the attribute across all items. Equation 1 illustrates the computation for the relevance score of an attribute p, where n_{p,I_u} represents the number of links from the set of items interacted by the user I_u to the attribute p. Similarly, n_{p,I_r} denotes the number of links connecting attribute p to recommended items I_r, and $IDF(p)$ signifies the Inverse Document Frequency of p, calculated as $\log(\frac{|C|}{n_{p,I_C}})$, where $|C|$ represents the total number of items in the catalog and n_{p,I_C} is the total number of items referencing attribute p.

$$score_explod(p, I_u, I_r) = (\alpha \frac{n_{p,I_u}}{|I_u|}) + (\beta \frac{n_{p,I_r}}{|I_r|}) * IDF(p) \tag{1}$$

Equation 2 displays the calculation for ranking attributes of ExpLOD v2 [26], which is very similar to its previous version; but also encompasses broader attributes. For instance, consider the movie 'La La Land' from 2016, classified with the attribute 'romance' in the Wikidata KG. This classification implies that the movie is also associated with broader attributes such as 'interpersonal relationship' and 'love,' as 'romance' is a subclass of these attributes. As a result, for broader KG attributes b that have subclasses, the score is the sum of Eq. 1 for all the attributes p_{bi} in the set of $P_c(b)$ that are children to b, multiplied by the IDF of the broader class ($IDF(b)$). Therefore, the ExpLOD algorithms rank attributes that are popular among the set of interacted items but rare to the catalog set of items.

$$score_explod(b, I_u, I_r) = \sum_{i=1}^{|P_c(b)|} score_explod(p_{bi}, I_u, I_r) * IDF(b) \tag{2}$$

The scoring mechanism used in the Property-based Explanation Model (PEM) is represented by Eq. 3 and, differently from the ExpLOD algorithms, accounts for the number of interacted items that reference the attribute rather than the number of links. To score an attribute p, firstly, it considers the number of interacted items that reference the attribute $|\overline{I(p, I_u)}|$, where I_u represents the set of items the user has interacted with. This value is then normalized by the total number of items the user has interacted with, denoted by $|I_u|$.

Additionally, the equation considers the number of items in the catalog C connected to the attribute $|\overline{I(p, C)}|$. Similar to the previous term, this value is normalized by the total number of items in the catalog, denoted by $|C|$. Finally, the logarithm of the total number of items in the catalog connected to the attribute $\log(|I(p, C)|)$ is calculated to amplify the importance of relatively rare attributes in the catalog. Similar to ExpLOD v2, PEM also considers broader properties, thereby extending its evaluation beyond the syntactic aspect of the

presence of attributes in the catalog of items and user-previously interacted items to include semantics within the hierarchy and ontology of the KG. The difference in PEM is that only broader attributes directly linked to a catalog item are considered to exclude irrelevant hierarchical attributes [12].

$$score_pem(p, I_u, I_r, C) = \frac{|\overline{I(p, I_u)}|/|I_u|}{|\overline{I(p, C)}|/|C|} * \log(|I(p, C)|) \tag{3}$$

All paths were extracted from interacted items to the recommended items for the baseline algorithms. The path with the highest mean attribute score was then chosen as an explanation. Although shorter paths represent a closer relation between interacted and recommended items [37], a maximum path length was set to five, and the maximum number of interacted items to a recommended item was limited to three to capture complex and underlying relations.

3.2 Metrics

Explanations based on KGs represent paths that connect one or more interacted items to a recommended item based on shared attributes. The Linking Interaction Recency (LIR) metric measures the recency of the user-interacted items that form an explanation; the Shared Entity Popularity (SEP) measures the popularity of attributes displayed in explanations for a single user and Explanation Type Diversity (ETD) the number of different attributes in the explanations. These metrics were proposed by [2] as explanation quality metrics and used in their work as reordering optimization metrics.

Equations 4 and 5 represent the Linking Interaction Recency (LIR) and Shared Entity Popularity (SEP) metrics, respectively. These metrics are computed based on the mean of normalized exponentially weighted moving average equations for each interacted item and attribute within an explanation.

For LIR, the values of interacted items (p^i) are calculated using their corresponding timestamps (t^i), which are min-max normalized to range between 0 and 1. The recursive nature of the function ensures that the value of a property i depends on $i-1$, with values sorted in ascending order of timestamps. Thus, $LIR(p^1, t^1)$ equals t_1. The parameter β is typically set to 0.3, as suggested in [2]. Consequently, LIR assigns higher values to explanations that connect recommendations with more recent interacted items, prioritizing their relevance.

$$LIR(p^i, t^i) = (1-\beta) * LIR(p^{i-1}, t^{i-1}) + \beta * t^i \tag{4}$$

In Eq. 5, the Shared Entity Popularity (SEP) metric quantifies the popularity of attributes by considering the number v^i of item nodes linked to the attribute e^i. Similar to the computation of LIR, the sorting and min-max normalization are applied to the number of references an attribute has to other nodes. Consequently, $SEP(e^1, v^1)$ corresponds to v_1, which represents the attribute with the lowest number of references in the KG. High values of SEP indicate that attributes in explanations are popular and, consequently, more likely to be recognized by the user.

$$SEP(e^i, v^i) = (1 - \beta) * SEP(e^{i-1}, v^{i-1}) + \beta * v^i \tag{5}$$

Finally, Eq. 6 defines the Explanation Type Diversity (ETD) metric, which quantifies the diversity of attributes in explanations associated with recommendations. It computes the ratio of the number of properties in the recommended list $|\omega_{L_u}|$ to the minimum between the length of the recommendation list k and the total number of possible attributes ω_L that could form an explanation. ETD provides insight into the variety of attributes presented in explanations and helps assess whether the explanation algorithm tends to favor repetitive attributes. Higher ETD values indicate a greater diversity of attributes, suggesting a more balanced and informative set of explanations.

$$ETD(S) = \frac{|\omega_{L_u}|}{min(k, |\omega_L|)} \tag{6}$$

3.3 Multi-Attribute Utility Theory (MAUT)

MAUT is used in Game Theory for decision-making in scenarios where multiple attributes are accounted for when choosing an option. Therefore, based on a series of criteria that may have different weights, MAUT generates a single score representing each option's utility and sorts it based on the best option according to the requirements.

Equation 7 represents the score ($maut_score$) for each option o in the set of possibilities O. MAUT comprises three main components: a set of criteria common to all options (C_o), weights assigned to each criterion to denote its importance ($weight_c$), and a utility function ($utility(c)$) applied to each criterion, which yields its utility. Consequently, the utility function is applied to all attributes of the options, and the resulting utilities are multiplied by their respective weights, generating a utility score for each criterion c. The MAUT score for each option is obtained by summing the utility scores of its attributes.

$$\forall o \in O \; maut_score(o) = \sum_{c \in C_o} utility(c) * weight_c \tag{7}$$

Considering explanation algorithms, the algorithms are the options to be ordered and eventually chosen as the algorithm for production or an online test and metrics are the criteria [40]. In the context of this work, the three quality metrics are the criteria. The utility function used was the min-max, and the weights for the criteria were equally divided ($\frac{1}{m}$, where m is the number of metrics). Therefore, for each recommendation algorithm, the three metrics are min-max normalized based on the values for different explanation algorithms and multiplied by the set of weights to obtain a utility score for each metric of an explanation algorithm. Finally, the algorithms are ordered based on the sum of the utility of each metric.

4 Embedding Proposal Algorithm

Similarly to the baselines, the explanation for a recommendation in the proposed embedding approach is constructed based on the most relevant KG path that connects an interacted item node to the recommended item node. Figure 1 illustrates the proposed KG embedding, model agnostic approach. User-interacted nodes in the KG are blue, attribute nodes are represented in yellow, and the recommended item node is red. The same colors apply to the arrays that represent the embeddings of such nodes. The black array represents the embedding of the relations between nodes displayed by the KG, represented by an arrow.

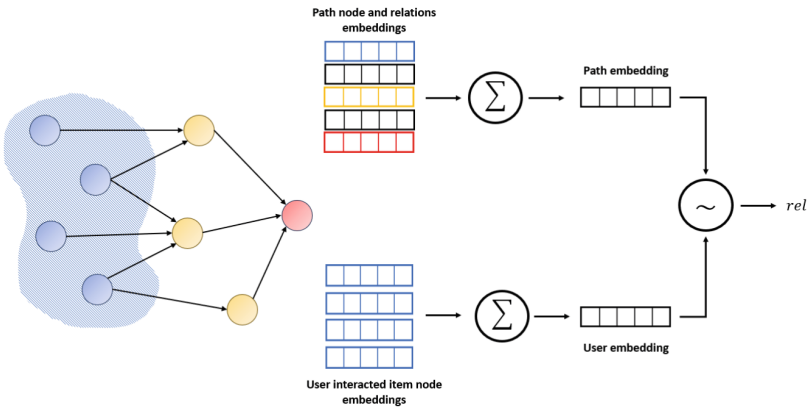

Fig. 1. Framework of the proposed model. Blue nodes in the KG represent items interacted by the user, yellow nodes represent attribute nodes, and the red node is the recommended item node. The same color schema applies to the arrays representing the embedding generated from the RotatE algorithm. Black array embeddings represent relation embedding between nodes, represented by an arrow in the graph. The $\sum$ represents a sum pooling operation, and $\sim$ is the cosine similarity between two embeddings, *rel* is the output of the cosine similarity function. (Color figure online)

Two embeddings are necessary to compute the relevance *rel* for an explanation path: the user and path. The user embedding is calculated by the sum pooling of the interacted items embeddings; the path embedding, on the other hand, is the sum pooling of all the items, attributes, and relations embeddings of the path that connects a user interacted item node with a recommendation item node in the KG. Equations 8 and 9 display the calculation for each of the embeddings where I is the set of user-interacted items nodes, and P is the set of items, relations, and attribute nodes in a path. The method *rotate_embedding* returns the embedding of the node passed as a parameter for the RotatE KG embedding algorithm [36]. All paths were extracted using the Dijkstra algorithm [11] and the one with maximum similarity is chosen as the explanation path to be displayed to the user.

$$user_e = \sum_{i \in I} rotate_embedding(i) \tag{8}$$

$$path_e = \sum_{n \in P} rotate_embedding(n) \tag{9}$$

The embedding proposal was made by computing the embedding of the KGs extracted from Wikidata LOD for the movie and artistic domains from the MovieLens and LastFM datasets, respectively. To compute the embeddings, the RotatE [36] was used due to its competitiveness in state-of-the-art performance [45] in generating KG embeddings and the low computational resources necessary to train the model. To generate latent space representations of the KG, RotatE uses the distance equation $d_r(h,t) = ||h \circ r - t||$, in which h, r, and t refer to the (h,r,t) triple, where h and t are nodes in the KG and r is the relation connecting the two nodes. The symbol $\circ$ denotes the element-wise product. During training, the model infers the tail based on the head and the relation $(t = h \circ r)$ to update the embeddings of relations (edges) and attributes (nodes). At the end of the training, every node and edge have an array that can be compared in a latent space. The method performs well in predicting symmetric, inverse, and composed relations.

The model was trained using the following parameters: Adam [20] as the optimizer, an embedding dimension of 300, and a learning rate of 0.001. A batch size of 256 was set for 50 epochs. These parameters were extracted from the original paper [36]. We also employed the self-adversarial negative sampling loss function proposed by the authors [36]. The KG triples were divided into training, validation, and test sets in a ratio of 0.8, 0.1, and 0.1, respectively. Pykeen [1] library was utilized to implement the graph embedding model. The training and test metrics for RotatE are presented in Table 10 of Appendix 7. The trained model for both KG datasets is available in the paper's code repository.

The explanation path chosen was the one with the most affinity to the user among all paths connecting at least one interacted item to the recommendation item. In that regard, the cosine similarity of the user and the path embeddings compute the path scoring. The maximum path length of 5 and the number of interacted items of 3 per explanation were set in the same way as the baselines. The relevance similarity score can be seen in Eq. 10, where $user_e$ is the user embedding and $path_e$ is the path embedding.

$$cosine_sim(user_e, path_e) = \frac{user_e.path_e}{||user_e||.||path_e||} \tag{10}$$

The selection of the explanation path respects Eq. 11, wherein, for all explanation paths within the set $Paths$ that link an interacted item node to a recommended item node by attributes shared among them, the path embedding with the highest cosine similarity with the user embedding is chosen.

$$argmax(\forall path \in Paths\, cosine_sim(user_e, path_e)) \tag{11}$$

5 Results

To answer research questions RQ1 and RQ2 and for robust results, we ran the ExpLOD, ExpLOD v2, PEM, and the proposed embedding approach (Embedding) for all users of the MovieLens 100k and LastFM datasets for the top one and five recommended items of the Most Popular, BPR-MF, Page Rank, UserKNN, EASE and NCF recommendation algorithms. Since the metrics are list-wise, they correspond to the mean of the explanation to all users. The standard deviation is also displayed in the table. Values in bold are the highest among algorithms, and values in italic are the lowest. It is worth highlighting the highest and lowest values because explanation goals and quality attributes display a trade-off [2,3,39]. Such behavior can happen because ETD measures the diversity of properties in explanation and SEP the popularity of the same attributes.

Table 3. Table of the LIR, ETD, and SEP metrics for explanations for the top five recommended items for the MovieLens100k dataset. Values in **bold** are the highest among the recommendation algorithm and *in italic* values are the lowest.

		ExpLOD	ExpLOD v2	PEM	Embedding
Most Popular	LIR	**0.0945 ± 0.15**	0.0834 ± 0.14	0.0320 ± 0.07	*0.0300 ± 0.11*
	ETD	*0.5809 ± 0.19*	0.5947 ± 0.19	**0.939 ± 0.12**	0.8045 ± 0.32
	SEP	**0.6322 ± 0.14**	0.6107 ± 0.12	*0.143 ± 0.13*	0.5392 ± 0.15
Page Rank	LIR	**0.0939 ± 0.15**	0.0872 ± 0.15	0.0323 ± 0.08	*0.0271 ± 0.10*
	ETD	*0.5563 ± 0.20*	0.6043 ± 0.19	**0.9389 ± 0.12**	0.8496 ± 0.29
	SEP	**0.6250** ± 0.16	0.5606 ± 0.15	*0.1095 ± 0.11*	0.5174 ± 0.15
UserKNN	LIR	**0.1010 ± 0.14**	0.0914 ± 0.13	0.0322 ± 0.08	*0.0290 ± 0.10*
	ETD	0.6593 ± 0.19	*0.6409 ± 0.19*	0.9452 ± 0.11	**0.9721 ± 0.26**
	SEP	**0.5803 ± 0.16**	0.5275 ± 0.14	*0.131 ± 0.12*	0.4149 ± 0.14
BPR-MF	LIR	**0.1048 ± 0.14**	0.0945 ± 0.13	*0.0306 ± 0.07*	0.0311 ± 0.11
	ETD	*0.6891 ± 0.19*	0.6937 ± 0.19	0.9583 ± 0.10	**0.9852 ± 0.28**
	SEP	**0.6113 ± 0.14**	0.5428 ± 0.14	*0.1400 ± 0.12*	0.4851 ± 0.15
EASE	LIR	**0.1000 ± 0.14**	0.0912 ± 0.14	0.03193 ± 0.08	*0.0268 ± 0.10*
	ETD	*0.6204 ± 0.20*	0.6328 ± 0.19	0.9459 ± 0.11	**0.9627 ± 0.24**
	SEP	**0.5743 ± 0.17**	0.5274 ± 0.15	*0.1350 ± 0.13*	0.3936 ± 0.14
NCF	LIR	**0.1181 ± 0.13**	0.1035 ± 0.12	0.0380 ± 0.08	*0.0285 ± 0.09*
	ETD	0.8432 ± 0.16	*0.8161 ± 0.16*	0.9885 ± 0.05	**1.1563 ± 0.27**
	SEP	**0.5868 ± 0.14**	0.5350 ± 0.13	*0.1613 ± 0.11*	0.3655 ± 0.13

Tables 3 and 4 display explanation quality metrics for the MovieLens 100k and LastFM datasets, respectively, regarding the top five recommendations of each algorithm. The recommender systems accuracy and beyond accuracy metrics for the datasets are in Appendix 7. Overall, the performances of ExpLOD in the MovieLens dataset and ExpLOD v2 in the LastFM dataset were the best

of all the other algorithms for the LIR and SEP. On the other hand, for the MovieLens 100k dataset, the Embedding approach outperformed the baselines in the ETD metric, except for the Most Popular and Page Rank algorithms, in which PEM achieved the best results for ETD. ExpLOD v2, as an evolution of ExpLOD, achieves very similar results as ExpLOD. In the MovieLens dataset, ExpLOD achieved higher LIR and SEP, while ExpLOD v2 outperformed its previous version in the LastFM dataset in the same metrics for most of the recommenders.

A trade-off behavior between diversity and popularity of attributes is also highlighted in PEM, which achieves a very high diversity but, instead, has the lowest SEP for all the metrics. This happens because, unlike ExpLOD algorithms, which are based on TF-IDF, PEM normalizes the number of items referenced by an attribute to the interacted items by the catalog. Since the catalog of items is extensive, PEM is more susceptible to displaying more diverse items.

Table 4. Table of the LIR, ETD, and SEP metrics for explanations for the top five recommended items for the LastFM dataset. Values in **bold** are the highest among the recommendation algorithm and *in italic* values are the lowest.

		ExpLOD	ExpLOD v2	PEM	Embedding
Most Popular	LIR	0.0182 ± 0.09	**0.0189 ± 0.11**	0.0143 ± 0.08	*0.0092 ± 0.06*
	ETD	0.7023 ± 0.17	*0.4927 ± 0.22*	0.9212 ± 0.12	**0.962 ± 0.36**
	SEP	0.7097 ± 0.17	**0.7537 ± 0.23**	*0.1214 ± 0.08*	0.6377 ± 0.15
Page Rank	LIR	0.0191 ± 0.10	**0.0212 ± 0.12**	0.0134 ± 0.07	*0.0093 ± 0.06*
	ETD	0.6100 ± 0.20	*0.5447 ± 0.19*	**0.9440 ± 0.10**	0.8769 ± 0.40
	SEP	0.6501 ± 0.21	**0.7164 ± 0.21**	*0.1209 ± 0.09*	0.6522 ± 0.19
UserKNN	LIR	0.0183 ± 0.10	**0.0191 ± 0.11**	0.0158 ± 0.08	*0.0076 ± 0.06*
	ETD	*0.5335 ± 0.20*	0.5355 ± 0.18	**0.9106 ± 0.14**	0.8453 ± 0.36
	SEP	0.5288 ± 0.26	0.2810 ± 0.22	*0.1417 ± 0.10*	**0.6464 ± 0.20**
BPR-MF	LIR	0.0191 ± 0.10	**0.0204 ± 0.11**	0.0164 ± 0.08	*0.0102 ± 0.06*
	ETD	*0.6145 ± 0.21*	0.6196 ± 0.19	**0.9450 ± 0.11**	0.9148 ± 0.38
	SEP	0.5605 ± 0.23	0.6302 ± 0.19	*0.1759 ± 0.12*	**0.6332 ± 0.18**
EASE	LIR	0.0188 ± 0.11	**0.0194 ± 0.11**	0.0154 ± 0.08	*0.0073 ± 0.05*
	ETD	*0.5474 ± 0.20*	0.5585 ± 0.18	**0.9246 ± 0.13**	0.8581 ± 0.37
	SEP	0.5307 ± 0.25	0.2861 ± 0.21	*0.1466 ± 0.10*	**0.6182 ± 0.21**
NCF	LIR	**0.0182 ± 0.09**	0.0162 ± 0.08	0.0162 ± 0.08	*0.0093 ± 0.06*
	ETD	*0.7775 ± 0.18*	0.7867 ± 0.18	0.9589 ± 0.09	**1.0776 ± 0.35**
	SEP	0.5904 ± 0.19	0.5514 ± 0.20	*0.2748 ± 0.15*	**0.6185 ± 0.17**

Unlike PEM, the Embedding approach achieved the highest ETD for most recommender systems in the MovieLens 100k dataset, except for the Most Popular and Page Rank algorithms, and ranked second best in the LastFM dataset for most recommenders while maintaining competitive SEP values (being even the best for four algorithms on the LastFM dataset). This indicates its ability to

provide diverse explanation paths for various recommendations and keep a certain level of attribute popularity. However, it also demonstrated the lowest level of LIR, which can be attributed to its training methodology. Unlike the three baseline methods that prioritize items to rank attributes, the embedding model was trained with a focus on path completion. Consequently, although it identifies diverse paths for explanations, it only incorporates the interacted item into the summation pooling process to generate the explanation path embedding.

Figure 2 displays explanations of each algorithm for the top five recommended items outputted by the EASE recommender for the user with ID 1 of the MovieLens 100k dataset. The text in blue represents the interacted item nodes, and the text in red represents the recommended item node. The text in yellow is the attribute node that connects the interacted and recommended node from the top-ranked explanation path obtained from the Wikidata KG. The explanations illustrate the low ETD from the ExpLOD family of algorithms since they repeat the same attribute in multiple recommendations. In addition, the explanations also highlight the diversity of PEM and the Embedding approaches that did not repeat a single attribute over recommendations.

Fig. 2. Top five explanations paths generated by the explainable recommendation algorithms for recommendations produced by the EASE recommender to user ID 1 from the MovieLens 100k dataset. Blue text denotes interacted item nodes, yellow text represents attribute nodes, and red text means the recommended item node. (Color figure online)

There are a few significant changes comparing Table 3 and Table 4. The proposed Embedding method achieved the highest SEP for the LastFM dataset for four over six recommendation algorithms, and ExpLOD v2 outperformed ExPLOD in LIR and SEP for the LastFM dataset. Such changes highlight the need to test explanation algorithms with multiple datasets and algorithms.

Table 5. Table of the LIR, ETD, and SEP metrics for explanations produced to the top recommended item in the MovieLens 100k dataset. Values in **bold** are the highest among the recommendation algorithms, and *in italic* values are the lowest.

		ExpLOD	ExpLOD v2	PEM	Embedding
Most Popular	LIR	0.0871 ± 0.17	**0.0967 ± 0.18**	0.0303 ± 0.13	*0.0245 ± 0.15*
	ETD	1.0000 ± 0.00	1.0000 ± 0.00	1.0000 ± 0.00	**1.1475 ± 0.35**
	SEP	**0.6813 ± 0.22**	0.6483 ± 0.19	*0.1465 ± 0.20*	0.5222 ± 0.33
Page Rank	LIR	0.0875 ± 0.18	**0.0945 ± 0.19**	*0.0262 ± 0.12*	0.0276 ± 0.16
	ETD	1.0000 ± 0.00	1.0000 ± 0.04	1.0000 ± 0.00	**1.1644 ± 0.37**
	SEP	**0.6562 ± 0.24**	0.6287 ± 0.22	*0.1253 ± 0.19*	0.4944 ± 0.35
UserKNN	LIR	**0.0980 ± 0.18**	0.0945 ± 0.18	*0.0270 ± 0.13*	0.0286 ± 0.16
	ETD	1.0000 ± 0.00	1.0016 ± 0.04	1.0000 ± 0.00	**1.1819 ± 0.38**
	SEP	**0.5588 ± 0.25**	0.5414 ± 0.24	*0.1339 ± 0.19*	0.43089 ± 0.31
BPR-MF	LIR	**0.1122 ± 0.19**	0.1032 ± 0.19	0.0377 ± 0.16	*0.0245 ± 0.15*
	ETD	1.0000 ± 0.00	1.0001 ± 0.04	1.0000 ± 0.00	**1.1737 ± 0.37**
	SEP	**0.5928 ± 0.24**	0.5574 ± 0.25	*0.1427 ± 0.20*	0.4861 ± 0.33
EASE	LIR	**0.0886 ± 0.17**	0.0878 ± 0.18	*0.0248 ± 0.11*	0.0260 ± 0.15
	ETD	1.0000 ± 0.00	1.0001 ± 0.04	1.0000 ± 0.00	**1.1644 ± 0.37**
	SEP	**0.6019 ± 0.23**	0.5501 ± 0.24	*0.1329 ± 0.20*	0.4253 ± 0.33
NCF	LIR	**0.1076 ± 0.20**	0.0961 ± 0.20	0.0399 ± 0.16	*0.02852 ± 0.09*
	ETD	1.0000 ± 0.00	1.0001 ± 0.04	1.0000 ± 0.00	**1.1563 ± 0.27**
	SEP	**0.5878 ± 0.27**	0.5382 ± 0.27	*0.1273 ± 0.16*	0.3655 ± 0.13

Tables 5 and 6 display the explanation metrics in the MovieLens 100k and LastFM, respectively, to the top recommended item. Most of the ETD for all recommender systems are one or close to this value because only one recommendation is explained. Consequently, one attribute was shown connecting the recommended to an interacted item for all the explanations generated. More than one attribute could be displayed in a single explanation to generate explanations as paths with a maximum length of five were considered. However, only the proposed embedding method could find longer explanations for the top recommended item. This underscores the significance of providing explanations for a set of recommendations rather than solely focusing on the top item in user online experiments, as participants may not discern diversity differences among explanation algorithms when only one recommendation is explained.

Considering the LIR and SEP metrics, the results are similar to the ones described for the top five recommendations in Tables 3 and 4. ExpLOD and ExpLOD v2 achieve improved LIR for most recommendation algorithms and higher SEP. Therefore, the algorithms were robust to changes in the number of recommended items for these metrics. Like the top five explanations, the Embedding method achieved the highest SEP for most recommendation algorithms for the LastFM dataset.

Table 6. Table of the LIR, ETD, and SEP metrics for explanations of the top recommended item in the LastFM dataset. Values in **bold** are the highest among the recommendation algorithm and *in italic* values are the lowest.

		ExpLOD	ExpLOD v2	PEM	Embedding
Most Popular	LIR	**0.0202 ± 0.14**	0.0200 ± 0.14	0.0154 ± 0.12	*0.0085 ± 0.09*
	ETD	1.0005 ± 0.02	1.0011 ± 0.03	1.0000 ± 0.00	**1.6091 ± 0.49**
	SEP	0.7882 ± 0.34	**0.8433 ± 0.30**	*0.1002 ± 0.13*	0.6024 ± 0.32
Page Rank	LIR	**0.0211 ± 0.14**	**0.0211 ± 0.14**	0.0154 ± 0.12	*0.0068 ± 0.08*
	ETD	1.0005 ± 0.02	1.0016 ± 0.04	1.0000 ± 0.00	**1.5111 ± 0.50**
	SEP	0.6351 ± 0.35	**0.7285 ± 0.32**	*0.0930 ± 0.13*	0.6452 ± 0.31
UserKNN	LIR	**0.0235 ± 0.15**	0.0219 ± 0.14	0.0148 ± 0.12	*0.0091 ± 0.09*
	ETD	0.9984 ± 0.05	0.9979 ± 0.05	1.0000 ± 0.00	**1.5248 ± 0.50**
	SEP	0.5849 ± 0.33	0.6222 ± 0.31	*0.1320 ± 0.19*	**0.6688 ± 0.30**
BPR-MF	LIR	**0.0219 ± 0.14**	0.0173 ± 0.13	0.0147 ± 0.12	*0.0093 ± 0.10*
	ETD	*0.9984 ± 0.04*	*0.9984 ± 0.04*	0.9995 ± 0.02	**1.5685 ± 0.50**
	SEP	0.6220 ± 0.32	0.6288 ± 0.30	*0.1569 ± 0.22*	**0.6464 ± 0.32**
EASE	LIR	**0.0257 ± 0.16**	0.0214 ± 0.14	0.0167 ± 0.13	*0.0084 ± 0.09*
	ETD	*0.9967 ± 0.06*	*0.9967 ± 0.06*	0.9995 ± 0.02	**1.5257 ± 0.50**
	SEP	0.5881 ± 0.33	0.6439 ± 0.30	*0.1279 ± 0.18*	**0.6591 ± 0.30**
NCF	LIR	0.0168 ± 0.13	**0.0195 ± 0.14**	0.0133 ± 0.11	*0.0114 ± 0.11*
	ETD	*0.9886 ± 0.11*	*0.9886 ± 0.11*	0.9957 ± 0.07	**1.6759 ± 0.49**
	SEP	0.6260 ± 0.33	0.5604 ± 0.33	*0.1899 ± 0.26*	**0.6798 ± 0.30**

In the online experiments described in [12,26], a clear state-of-the-art evolution for explanation goals was set where ExpLOD v2 outperformed ExpLOD and PEM outperformed ExpLOD v2, respectively. However, this evolution was not directly reflected in explanation quality metrics since one method did not outperform another in all metrics. For this reason, to answer RQ2 and verify if the explanation quality metrics are reflected in user perception of explanation goals, we applied the MAUT method in the results for each algorithm to the top five recommended items of each recommender. We compared the ordering of the MAUT regarding the explanation quality metrics to the online evolution of algorithms on explanation goals. In that regard, we did not experiment with the top-1 recommended item because the ETD metrics for all explanation algorithms were mostly equal. Therefore, the MAUT method would only consider SEP and LIR explanation metrics since the min-max normalization utility score of ETD for all explanation algorithms would be the same.

Table 7 presents the MAUT results for the LIR, ETD, and SEP metrics across the MovieLens 100k and LastFM datasets for the top 5 recommendations. Bold values indicate the highest score based on the MAUT method, representing the preferred choice for users as it considers a combination of all explanation quality metrics.

In the MovieLens 100k dataset, ExpLOD outperformed all other explanation algorithms for all recommendation algorithms, contrary to the findings in [12,26]. This disparity is primarily attributed to the significant difference in LIR between ExpLOD and other methods. ExpLOD v2 ranked second, followed by the proposed Embedding approach and PEM. The Embedding approach secured the third position despite its lower LIR than other algorithms due to its competitive ETD and SEP values. Conversely, PEM's performance can be attributed to its low SEP and LIR, which compensated for its higher ETD score. In contrast, the best-performing explanation algorithm varied for different recommendation algorithms in the LastFM dataset, as the results showed less disparity compared to the MovieLens 100k dataset. Notably, the proposed Embedding approach outperformed three out of six baselines.

However, users participating in the online experiments described in [12,26] were unable to perceive the LIR metric. This limitation is due to the experimental setup, where interacted items were added only once during the session to evaluate explanations. Therefore, users added the movies they watched, received recommendations, and received explanations within the same session. Table 8 presents the MAUT values for all recommendation and explanation algorithms for the top five recommendations, utilizing only the ETD and SEP metrics. These metrics were chosen due to their trade-off effect [2] and the potential for users to perceive differences.

In this scenario, the proposed embedding method outperformed all the other approaches since it is the most balanced regarding both metrics. Such results align with Table 3 for the top-5 recommendations, where the Embedding method achieved the highest ETD for most of the recommendations algorithms while having a close value in SEP compared to ExpLOD and ExpLOD v2, which outperformed the proposed approach in popularity. Similarly, in Table 4, the proposed approach outperformed the baselines in SEP for the LastFM dataset, maintaining a competitive ETD metric.

In addition, the results of MAUT value considering only the SEP and ETD quality explanation metrics are more closely aligned to the results of the online

Table 7. Table of the MAUT metric using the min-max normalization as a utility function and equal weights (of 0.33) to LIR, ETD, and SEP metrics for the MovieLens 100k and LastFM datasets. **Bold** values are the highest for a recommendation algorithm.

	MovieLens 100k				LastFM			
	ExpLOD	ExpLOD v2	PEM	Embedding	ExpLOD	ExpLOD v2	PEM	Embedding
Most Popular	**0.6666**	0.6075	0.3437	0.4781	**0.7665**	0.6666	0.4795	0.6054
Page Rank	**0.6666**	0.6335	0.3593	0.5193	0.6248	**0.6666**	0.4476	0.5746
UserKNN	**0.6851**	0.5828	0.3214	0.5436	0.5658	0.4270	0.5713	**0.6089**
BPR-MF	**0.6666**	0.5772	0.3030	0.5796	0.5724	**0.6696**	0.5377	0.6362
EASE	**0.6666**	0.6029	0.3401	0.5294	0.5889	0.4417	0.5573	**0.6079**
NCF	**0.6932**	0.5716	0.2045	0.4932	0.6393	0.5362	0.4620	**0.6666**

experiments in [12,26], despite still some discrepancy, since PEM achieved closer but did not outperform both ExpLOD versions in explanation quality metrics as in the online experiments evaluated under explanation goals.

Table 8. Table of the MAUT method using the min-max normalization as a utility function and equal weights (0.5) for ETD and SEP metrics in the MovieLens 100k and LastFM datasets. **Bold** values are the highest for a recommendation algorithm.

	MovieLens 100k				LastFM			
	ExpLOD	ExpLOD v2	PEM	Embedding	ExpLOD	ExpLOD v2	PEM	Embedding
Most Popular	0.5000	0.5000	0.4972	**0.7790**	0.6884	0.4999	0.4564	**0.9082**
Page Rank	0.5000	0.5002	0.5000	**0.7790**	0.5260	0.5000	0.5000	**0.8620**
UserKNN	0.5277	0.4410	0.4594	**0.8155**	0.3835	0.1405	0.5000	**0.9134**
BPR-MF	0.500	0.4351	0.4545	**0.8661**	0.4205	0.5044	0.4999	**0.9543**
EASE	0.4999	0.4647	0.4754	**0.7942**	0.4072	0.1625	0.5000	**0.9118**
NCF	0.5399	0.4391	0.2534	**0.7399**	0.4590	0.4176	0.3022	**1.000**

In conclusion, addressing RQ1, the embedding method demonstrates an improvement in the trade-off between diversity and popularity explanation quality metrics compared to the baselines, as it is trained based on path completion, which is then utilized to generate explanations. However, despite achieving the best balance in ETD and SEP, it does not prioritize interacted items, resulting in a lower LIR.

Regarding RQ2, the quality explanation metrics of LIR, SEP, and ETD did not entirely align with the online explanation goals perceived by users in the experiments of [12,26]. However, the MAUT values were closer to those reported in the online experiments when considering only SEP and ETD. This suggests that the popularity and diversity of attributes shown may significantly impact user perception more than interacted items when explaining a recommended item.

In this context, the metrics are constrained to evaluating a single user's list SEP, LIR, and ETD. Therefore, metrics do not penalize attributes and items shared among users, potentially indicating a bias toward generating identical explanation paths for different users. Consequently, introducing new metrics considering the diversity of explanations across users could enhance alignment with online results. Moreover, despite the discrepancy between explanation quality metrics and explanation goals, we highlight the importance of these metrics in assessing bias in explanation algorithms and comparing different algorithms.

6 Future Work, Limitations, and Open Directions

There are many open directions and future works regarding model-agnostic KG explanations and metrics to evaluate explanations offline. While our model utilized the RotatE algorithm [36], exploring and comparing the effects of other

embedding algorithms on explanation quality metrics is left as future work. Additionally, conducting a user study to compare the embedding approach with PEM for explanation goals would address a limitation of our work. Although our approach has shown an improved trade-off between diversity and popularity in attribute explanation quality metrics, evaluating whether the embedding approach enhances user perception of the system based on online explanation goals metrics requires user assessment. Furthermore, exploring contextual KG embedding algorithms, which have demonstrated promising results and can incorporate structural information [42,46], is also a future work.

Future research could also explore user perception of explanations for single and multiple recommendations. For example, conducting an evaluation comparing user perception between PEM and ExpLOD v2 for multiple recommendations could shed light on whether the difference in perception leans more towards PEM due to its diversity or more towards ExpLOD v2 because of its use of popular attributes that users can easily recognize.

There are many more open directions regarding offline evaluation of explanations and how they reflect in online user perception. While the ETD and SEP metrics yielded rankings closer to user perception when evaluated using the MAUT method, certain limitations of these metrics may influence their ordering. Specifically, they do not account for algorithmic attribute bias in the dataset. This means that if an algorithm consistently explains recommendations using a limited set of attributes across different users, but the attributes appear diverse and popular within each user's explanation list, the SEP and ETD metrics will not penalize this behavior. Consequently, there is a need to develop new metrics for offline explanation evaluation that are more closely aligned with online explanation goal metrics. This represents an open research direction in the field of explanations in recommender systems [24].

Finally, [38] proposed a methodological framework for explainable AI, but a specific framework for explanations in recommender systems would be beneficial for consistency in offline evaluation and improvement in the reproducibility of results. In this research, we took inspiration from the methodological aspect of [38] and adapted the guidelines for offline experiments in recommender systems to evaluate accuracy by [13] to run explanations to all users and for different recommender systems algorithms. Finally, although the metrics can be adapted to review-based explanations in which an aspect has its popularity and diversity measured, multi-domain metrics for explanations are also an open research topic.

7 Conclusion

This paper introduced a novel KG model-agnostic embedding approach for recommender systems that ranks paths according to the similarity of the explanation path embedding to the user embedding. The path embedding is the sum pooling of the embeddings from the elements that compose the path, while the user embedding is the sum pooling of all user-interacted items' embeddings. Three explanation quality metrics were used to validate explanations generated

by the proposed approach. Such metrics evaluate the sentence considering the recency of interacted items and the popularity (SEP) and diversity (ETD) of attributes displayed in explanations. The metrics were computed for all the users in the MovieLens 100k and LastFM datasets. The three baselines used represent an evolution in the state-of-the-art KG model-agnostic approaches in recommender systems and rank paths based on a syntactic score of attributes. Results reported that, although the baselines achieve improved LIR, the embedding approach achieves the best balance between SEP and ETD. Furthermore, baseline explanation goals results reported in the literature on online user studies were compared to the offline experiments on explanation quality metrics, however, despite measuring important aspects of explanation quality, studying offline metrics that reflect explanation goals remains an open research topic.

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Appendix 1 - Recommender Systems Algorithms Metrics

(See Table 9)

Table 9. Mean accuracy and beyond accuracy metrics for the MovieLens100k and LastFM datasets on a 10-fold cross-validation. **Bold** values are the highest for a dataset. The NCF algorithms do not have beyond accuracy metrics because the leave-one-out evaluation was used as proposed in the original paper.

		MovieLens						LastFM					
Metric	K	MostPop	BPR-MF	PageRank	UserKNN	EASE	NCF	MostPop	BPRMF	PageRank	UserKNN	EASE	NCF
NDCG	1	0,1690	0,2059	0,2021	0,2901	**0,3582**	0,2306	0,0706	0,1404	0,1550	0,2252	**0,2407**	0,1672
	3	0,2677	0,3507	0,3511	0,4443	**0,5118**	0,3996	0,0990	0,1885	0,1989	0,2847	**0,2988**	0,2350
	5	0,2928	0,3789	0,3776	0,4634	**0,5229**	0,4463	0,1062	0,1986	0,2078	0,2942	**0,3090**	0,2544
	10	0,3084	0,3919	0,3919	0,4659	**0,5209**	0,4957	0,1100	0,2014	0,2098	0,2903	**0,3033**	0,2695
MAP	1	0,1690	0,2059	0,2021	0,2901	**0,3582**	0,2306	0,0706	0,1404	0,1550	0,2252	**0,2407**	0,1672
	3	0,2201	0,2814	0,2800	0,3703	**0,4374**	0,3165	0,1261	0,2343	0,2411	0,3412	**0,3546**	0,3006
	5	0,2309	0,2941	0,2929	0,3770	**0,4384**	0,3392	0,1417	0,2571	0,2615	0,3654	**0,3807**	0,3404
	10	0,2278	0,2867	0,2879	0,3604	**0,4153**	0,3584	0,1600	0,2783	0,2807	0,3821	**0,3965**	0,3793
AGG-DIV	1	16,6	**185,6**	44,8	110,1	123,9	–	9,2	**184,6**	60,5	155,5	132,8	–
	3	32,8	**343,3**	89,8	187	236,2	–	15,1	**314,6**	125,1	294,6	266,6	–
	5	47,2	**446,1**	126,6	248	314,8	-	20,9	**402,6**	179,7	400	375,4	–
Entropy	1	0,6987	**1,9851**	0,8659	1,7723	1,7783	-	0,4095	**1,7667**	0,9659	1,7390	1,6393	–
	3	1,0248	**2,1690**	1,2282	1,9355	1,9838	–	0,8013	**1,9158**	1,2481	1,9392	1,8695	–
	5	1,1881	**2,2571**	1,4076	2,0266	2,0943	–	0,9799	**1,9971**	1,3946	2,0531	1,9898	–
Gini	1	0,9995	**0,9908**	0,9991	0,9948	0,9946	–	0,9997	**0,9954**	0,9991	0,9960	0,9968	–
	3	0,9990	**0,9861**	0,9983	0,9926	0,9915	–	0,9994	**0,9938**	0,9986	0,9937	0,9968	–
	5	0,9986	**0,9833**	0,9976	0,9910	0,9892	-	0,9992	**0,9926**	0,9981	0,9919	0,9930	–
Coverage	1	0,0018	**0,0202**	0,0049	0,0120	0,0135	–	0,0008	**0,0167**	0,0055	0,0141	0,0120	–
	3	0,0036	**0,0374**	0,0098	0,0204	0,0257	–	0,0014	**0,0167**	0,0113	0,0267	0,0241	–
	5	0,0051	**0,0486**	0,0138	0,0270	0,0343	–	0,0019	**0,0364**	0,0163	0,0362	0,0340	–

Appendix 2 - RotatE KG Embedding Training Statistics

Table 10. Metrics for the training and testing of the RotatE KG embeddings.

	Training Set Triples	Validation Set Triples	Test Set Triples	First Epoch Loss	Final Epoch Loss	Hit@1	Hit@3	Hit@5	Hit@10
MovieLens 100k	235,466	29,434	29,433	5.4359e-5	3.4565e-5	0.0858	0.1386	0.1676	0.2132
Last FM	101,516	12,690	12,689	5.5456e-5	3.6342e-5	0.1475	0.2260	0.2654	0.3234

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Graph-Based Interface for Explanations by Examples in Recommender Systems: A User Study

Marta Caro-Martínez[1(✉)], José L. Jorro-Aragoneses[2(✉)], Belén Díaz-Agudo[1], and Juan A. Recio-García[1]

[1] Departamento de Ingeniería del Software e Inteligencia Artificial, Universidad Complutense de Madrid, Madrid, Spain
{martcaro,belend,jareciog}@ucm.es

[2] Departamento de Ingeniería Informática, Universidad Autónoma de Madrid, Madrid, Spain
jose.jorro@uam.es

Abstract. In recent years, recommender systems have used advanced machine-learning techniques to improve recommendation precision. However, many of these approaches, like deep-learning, are considered black-box models; in other words, users struggle to comprehend why the system recommends a particular item, affecting the user's confidence in the provided recommendation. Some recommender systems include methods to explain a recommendation. A common technique is applying explanations-by-examples, but sometimes users do not understand the link between the recommendation and the examples. This paper describes a user study where we have evaluated a novel method to create graph-based explanations. It uses Formal Concept Analysis to extract the most relevant attributes to relate the recommendation with each example. Next, this method shows an interactive graph to users that explains the recommendation based on the links extracted before. Results show a high level of satisfaction regarding the explanations and their visualisation compared with other approaches found in the literature.

Keywords: Explanations-by-examples · Instance-based Explanations · User Study · Graph-based Visualisation · Formal Concept Analysis

1 Introduction

Recommender systems appeared to help users identify the best products to purchase based on their preferences when they do not know which one to choose since they have a wide offer of products. In recent years there has been an important improvement in machine learning algorithms, which have been also used in new recommendation approaches [20]. Despite the increasing precision of these recommendation algorithms, the systems that use them are considered black-box approaches: these approaches are not interpretable by end users [18].

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Recently, an active research area to resolve the interpretability and explainability problem is eXplainable Artificial Intelligence (XAI). In the context of recommender systems, explanations improve the understanding of the user who receives a recommendation. XAI methods can increase the utility of recommender systems because they can improve users' satisfaction or users' confidence, which also enhances the user experience [23]. There are already different approaches for generating explanations for a recommender system. One of the most commonly used is the explanation-by-example methodology. An example of this type of explanation would be *"the system recommends you X because it is similar to Y, and you liked Y previously"*. Explanations using examples are easy to understand for a user because they are more intuitive than other common explanations like trace-based explanations [24]. However, in some cases, the examples are not enough for users to understand a recommendation and they could need an extra component like a connection between the recommendation and the examples. These connections can adapt the explanation to the user's mental model [14].

In this paper, we propose a novel interface to enrich explanations-by-examples for black-box recommender systems. Commonly, explanations-by-examples only offer the examples as a way to explain a recommendation. Our interface enriches those examples using complementary knowledge to connect them with the recommendation and generate a final explanation. We use Formal Concept Analysis (FCA) [5] to get the most useful links between the recommended item and the most useful explanation examples for the user. Moreover, we use an interactive graph-based visualisation to show the explanation to the final users.

To prove the utility of enriched explanations-by-examples interfaces, and in particular our proposed interface, we have carried out a user study. Thirty-five real users interacted with our approach and answered a questionnaire about the explanations generated with our method. With this study, we have obtained promising results with which we can conclude that the users were satisfied with the enriched explanations shown. This is a first step to use these types of interfaces for approaches that offer explanation-by-examples in recommendation systems.

The rest of the paper is structured as follows: first, Sect. 2 includes a summary about relevant works on XAI applied in recommender systems and related to this work. Next, Sect. 3 describes the complete process of obtaining link between the recommended item and the explanation examples. In addition, it describes the graph-based visualisation approach. In Sect. 4, we describe the details of the user study and the results achieved. Finally, Sect. 5 contains a summary of conclusions extracted from this work.

2 Related Work

In recent years, the number of proposals for explainability applied to recommender systems has increased. The main goal of applying explanations in recommender systems is improving the transparency, effectiveness, trust and satisfaction of those recommendation users, increasing their utility and acceptance [23].

Indeed, there are a lot of publications about surveys and taxonomies to classify explainability methods in recommender systems [7]. Among all the explanation approaches that we can find in the literature, explanations based on examples are more natural and intuitive for users because they are adapted to human reasoning mechanisms [19].

Explanations-by-examples, also called example-based explanations, can be implemented using different methodologies. They explain to a user a prediction p obtained with an AI model using instances whose predictions obtained with that same AI model were the same or different to p. First, some methodologies use similar examples to the situation to be explained whose predictions are the same to the prediction p. This type of explanation is known as factual explanations [15]. Another approach is to provide users with counterexamples, telling the users which features from the query need to be changed to also change the prediction p [2,13]. These methods are called counterfactuals. Semifactual explanations return "even-if" examples of feature modifications that not change the prediction p. These explanations employ Near Unlike Neighbours: the least similar examples to the situation to be explained but obtaining the same prediction p [2,13]. We can find different contributions of explanations-by-examples. An example in this field is the PBI-CBR system [16], which predicts the growth of grass and provides personalised explanations-by-example to justify the reasons behind its prediction. Another approach generates explanations-by-example for a recommender system based on past user interactions [8]. Despite the large number of papers in the literature, most of them are focused on obtaining the examples to include in an explanation, leaving unstudied how to show the end user the relationship of these examples to their query.

To resolve this problem, in this work we analysed the characteristics that a correct explanation must have. A relevant work about it is the Miller survey [17]. Miller describes explanations in artificial intelligence from the point of view of social sciences. There, Miller describes a methodology based on the reasoning process carried out by humans. This methodology starts in a *cognitive process* that defines an explanation for a given event (called the *explanandum*), where some causes of the event (or *explanans*) are identified; and next *social process* transfers the knowledge between the explainer and the final user. The interface proposed in this paper is oriented to the social process. In this process, Miller remarks that it is important to include interactions between the user and explainable processes to improve user satisfaction. We note these features to apply in our visualisation.

There is also a large variety of works related to visualisation strategies for explanations in recommender systems. We found graph and data visualisation surveys where the authors describe different layouts for representing data as graphs [3,10]. These works highlight that there are many ways to display data as graphs. We can find some examples nowadays. Known websites such as LinkedIn use simple graphs to show recommendations of new contacts to a target user where links represent the users characteristics in common [6]. Another graph-based visualisation approach is Moodplay [1]. It uses a graph to explain recom-

mended music including information and a set of colors to represent where the information comes from.

3 Generating Graph-Based Explanations-by-Examples

The main goal of our proposal is to explain a recommendation using the most relevant data that connects it with examples extracted and show the complete explanation using an interactive graph. The interface receives the recommended item, a set of examples related to the recommendation. Then, the first step extracts the explanation links based on item descriptions, i.e. relevant attributes to relate the recommendation and relevant examples. The second step shows a graph-based explanation using the information obtained in the step before. Next, we explain the methodology of each step.

3.1 Obtaining the Explanation Links

In some explanations based on example approaches, the examples extracted may not be useful to understand a recommendation by the target user [11]. This problem could arise due to two reasons: the target user does not understand the relation between recommendations and examples, or not all examples are significant for the user in the explanation. To resolve both problems, we applied the Formal Concept Analysis (FCA) method to get the examples and shared attributes with the recommendation to use in the final explanation. FCA is a mathematical approach to data analysis based on lattice theory [5]. It calculates a hierarchical structure to relate objects based on their properties and the objectives are clustered according to their shared properties. The final result is a lattice, where the properties shared by more objects, i.e. the more general ones, are represented at the highest levels and the properties shared by fewer objects, i.e. the more specific ones, are at the lowest levels.

In our approach, we build a conceptual lattice with the recommended item and explanation examples as objects, and their attributes as properties. There are different alternatives and applications of FCA to support explanations, which we have already looked into [9]. For our current work, we decided to use the lattice to extract the most relevant shared attributes to relate the examples with the recommendation. However, searching the level of the lattice to extract the most relevant relationships has no trivial solution. In a previous study [12], we detected that users preferred explanations that contain as many examples as possible, especially in small lattices. In addition, users preferred explanations where shared attributes were the most specific ones, i.e., the ones in the deepest lattice levels.

Following these findings, we propose the Algorithm 1 to extract the most relevant examples and shared attributes between them and the recommended item. The algorithm receives a lattice and the item recommended. Then, the algorithm gets the immediate parents from the node that contains the recommended item. Next, it traverses through the set of those immediate parent nodes to extract (1)

Algorithm 1 Algorithm to extract the most relevant examples and attributes to include in the final explanation

Require: *lattice*: lattice obtained from FCA.
Require: $item_{rec}$: recommended item to explain.
Ensure: A dictionary where keys are shared attributes and values a list of examples.

```
dict ← {}
parents ← lattice.get_parents(item_rec)
for n in parents do
    for attribute in n.get_attributes() do
        dict[attr] ← n.get_examples()
    end for
end for
return dict
```

the list of examples related to each of those nodes (i.e. the immediate parents's children) and (2) the shared attributes (i.e. the values saved in the immediate parents nodes). The algorithm returns a dictionary that saves this information, where the keys are the attributes and the values are the list of examples that share this attribute with the recommended item.

Figure 1 shows an example of a lattice applied to a movie recommendation. In this case, the movie recommended is *E.T.* The other movies are a set of examples to explain the recommendation. Following the algorithm explained before, the first step is locating the nodes that contain the recommender movie and getting the immediate parent nodes, in this case, there are three parents. Next, the algorithm visit all immediate parents until it extracts the shared attributes and the examples related to this node. From the first parent (node in level 3),

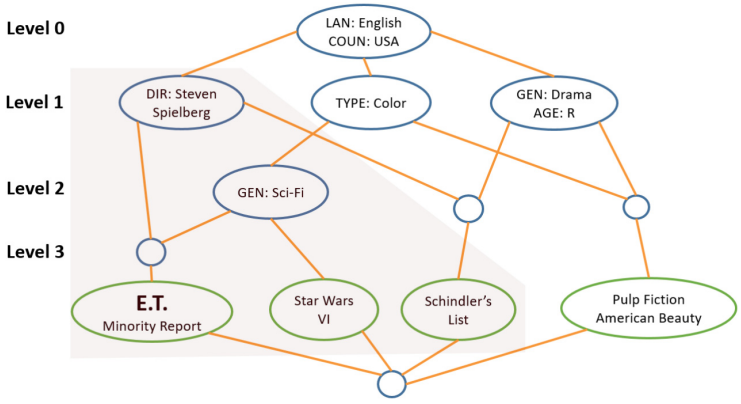


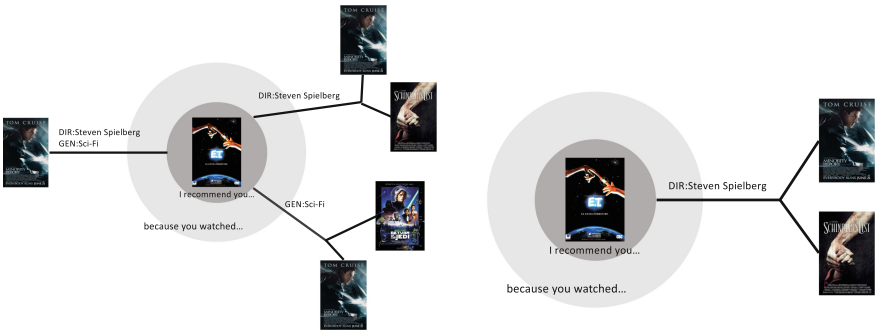
Fig. 1. Example of a lattice built with $item_{rec} = \{E.T.\}$ and *examples* = {*Minority Report, Star Wars VI, Schindler's List, Pulp Fiction, American Beauty*} to obtain their shared attributes. The shadowed area contains the examples and attributes extracted to use in the final explanation.

it extracts the first relation between *E.T.* and *Minority Report*, where shared attributes are *DIR:Steven Spielberg* and *GEN:Sci-Fi*. Next, the second parent (level 2) relates *E.T.* with *Minority Report* and *Start Wars VI* together. In this case, the share attributed is *GEN:Sic-Fi*. Finally, from the last parent (level 1), it extracts a relation between *E.T.* with *Minority Report* and *Schindler's List*, where the related attribute is *DIR:Steven Spielberg*. In this step, the algorithms finish because there are no more parents.

3.2 Visualising Explanations Through Interactive Graphs

Once we select the most relevant examples and attributes related to the recommendation, the next step is to design a visualisation that shows all the information extracted from FCA without overwhelming the user. As it is remarked in the literature [3,10], we decided to use a graph-based visualisation because it allows us to show the links between the items and the shared attributes in and appealing fashion. Moreover, the graph-based interface allows us to adapt the knowledge extracted from FCA creating a simplified and more understandable visualisation. Continuing with the example explained in the previous section, Fig. 2a shows the visualisation of the information extracted from FCA.

We use a graph with three concentric circular areas to represent an explanation. The recommended item is in the middle of the graph, in the example, the recommended movie is *E.T.* The recommended movie is connected to other nodes, which represent the shared attributes and group the explanation examples. The explanation examples will be connected to the attribute nodes that they share with the recommended item. In our example, we can observe three groups, one for each relation extracted from FCA. For instance, attribute *DIR:Steven Spielberg* appears in the link that joins *E.T.* with the group of *Minority Report* and *Schindler's List*.



(a) Complete Explanation (b) After removing attribute.

Fig. 2. Design of the visualisation proposal. On the left, the complete explanation. On the right, the explanation after removing the attribute *GEN:Sci-Fi*

Our visualisation approach includes different features that allow users to interact with explanations. Users receive k recommendations, and the system generates an explanation for each recommendation. At the top of the screen, the system marks in green colour the most explainable recommendation (Fig. 3). It is the recommendation whose explanation contains the highest number of attributes. Another key feature of our visualisation approach is that users can adapt all explanations to their preferences. This feature allows them to eliminate attributes that they do not consider relevant to understand the recommendation. For example, in Fig. 2b, we observe the new graph generated when attribute *GEN:Sci-Fi* is removed. When users change an explanation, the system recalculates the most explainable recommendation and updates the result at the top of the window, if necessary. Finally, users can obtain a complete description of the recommended item and examples when they click on this node in the graph.

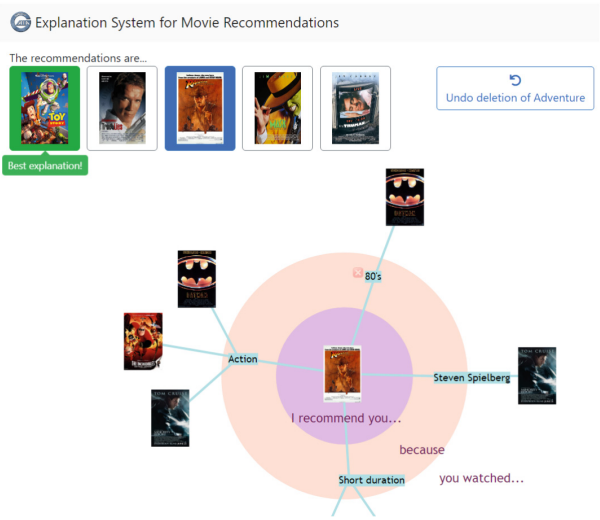


Fig. 3. Interactive interface for switching between recommendations

4 User Study

Our user study aimed to check whether our graph-based interface for enriched explanations-by-examples was useful and provided with satisfaction to users. Regarding this, our research questions were:

- **RQ-1**: Explanations generated with the proposed interface provide satisfaction to users. The explanations are scrutable and efficient.
- **RQ-2**: The graph-based visualisation is interesting and useful for users. Examples and links of explanations are useful for users.

In the next sections, we determine the methodology utilised to carry out the user study (Sect. 4.1) and the results obtained (Sect. 4.2).

4.1 Methodology

We have implemented our enriched XAI interface (Fig. 3) for a movie recommendation system based on matrix factorisation [4], which is a well-known algorithm in the literature considered as a black box. To get our explanation examples, we have applied the CBR-based approach we proposed before [11]. The dataset used to generate the recommendations was the MovieLens 100K dataset[1], while we use the IMDB[2] dataset to obtain the shared attributes between the recommendation and the examples.

The evaluation methodology proposed for the experiments with real users is twofold and combines (1) a quantitative analysis of logs interactions with the application, and (2) a qualitative analysis of the responses to subsequent questionnaires. Users perform the following tasks during the evaluation experiment. First, users follow a short tutorial. We decided to include a tutorial for users to familiarise with our interface before they could evaluate it. Otherwise they will evaluate all the functionalities and features without understanding everything properly. Then, the explanation system starts an explanation session, displaying five different recommendations and their corresponding explanations. For each explanation session, users decide if an explanation is useful or not for them by accepting or rejecting it. To make this decision, participants can interact with the alternative explanations through the process described in Sect. 3.2.

In any case, the system logs the interactions carried out by the users, such as switching between the explanations shown, removing attributes, or undoing actions. After concluding an explanation session, the participant fills out a *feedback form*. In this questionnaire, users select which features of the system they consider useful, choosing among the attributes, the explanation examples, and their visual organisation. The explanation sessions are repeated five times by each participant using different sets of recommendation-explanations. For each session, we got the following log information to use in the analysis: (1) collecting the number of acceptances in every explanation session; (2) measuring the average time spent in every explanation session; (3) counting the average number of interactions carried out in every explanation session; (4) counting the number of attributes removed taking into account the number of examples linked to them; (5) determining the type of actions (switch explanation, remove attributes, or undo actions) carried out in every explanation session; and when removing attributes (6) defining which attributes where deleted.

[1] https://grouplens.org/datasets/movielens/.
[2] https://www.imdb.com/interfaces/.

Table 1. Whole set of questions that users answered in the evaluation, considering the PSSUQ test and the custom questionnaire we created about users' satisfaction.

Id.	Sentence
System Usefulness	
I1	Overall, I am satisfied with how easy it is to use this system
I2	It was simple to use this system
I3	I was able to complete the tasks and scenarios quickly using this system
I4	I felt comfortable using this system
I5	It was easy to learn to use this system
I6	I believe I could become productive quickly using this system.
Information Quality	
I7	The system gave error messages that clearly told me how to fix problems
I8	Whenever I made a mistake using the system, I could recover easily and quickly
I9	The information (such as online help, on-screen messages, and other documentation) provided with this system was clear
I10	It was easy to find the information I needed
I11	The information was effective in helping me complete the tasks and scenarios
I12	The organisation of information on the system screens was clear.
Interface Quality	
I13	The interface of this system was pleasant
I14	I liked using the interface of this system
I15	The information (such as online help, on-screen messages, and other documentation) provided with this system was clear
I16	Overall, I am satisfied with this system.
Users' satisfaction with the system	
I17	I felt confident with the explanations the system provided for each recommendation
I18	I am satisfied with the visualization and interactivity of the system
I19	I am satisfied with the different options offered by the system (modification of explanations and switching between recommendations)
I20	The system allows to tell why an explanation is not interesting to me
I21	The attributes shown in the explanation were useful to understand the recommendation
I22	Movies shown as examples in the explanation helped me to understand the recommendation
I23	The groupings of movies with the shared attributes were useful to understand the recommendation

The second part of the evaluation is a *global questionnaire*, which is structured in two sections (see Table 1). In the first section, participants complete the Post-Study System Usability Questionnaire (PSSUQ, version 3) [21], a standardised generic test to measure the user-perceived utility of a system. This test defines 16 items to evaluate System Usefulness (SYSUSE, items 1–6), Information Quality (INFOQUAL, items 7–12), and Interface Quality (INTERQUAL, items 13–15). The work [21] analyses different ways of evaluating user experience. Here, the authors include PSSUQ as one of the main and most popular standarised usability questionnaires. Since PSSUQ can be used for any types of interfaces, we decided to use it in our work due to the demonstrated utility of PSSUQ measuring satisfaction and user experience. The second section consists of 10 custom questions regarding the users' satisfaction with the main system features. We use a 7-point Likert scale (from 1-strongly agree to 7-strongly disagree) following the structure of the PSSUQ questionnaire. The global questionnaire ends with a free-text field to include any other comments or suggestions about the explanation system.

It is important to note that what we propose in this work is a graph-based interface that allows to show instance-based explanations. Therefore, we have evaluated the interface with other interfaces found in the literature using a PSSUQ questionnaire, which is also a tool to evaluate interfaces. We have not considered to compare our proposal with explainers such as LIME, SHAP, etc.

since what we are posoposing is not an explainer itself. It is a technique to support instance-based explainers, in order to add personalised and supplementary knowledge and a way to visualise the explanation in a more appealing and satisfactory fashion.

4.2 User Study Results

Thirty-five participants collaborated in the evaluation online, approximately 34% of the participants were women, while the rest were men. Most of the people who took part in the experiment are between 20 and 40 years old, and many of them have studied or worked in the Faculty of Computer Science at the Complutense University of Madrid. Most of them are not experts in recommender systems or explanation systems. We also had people who do not work in the computer science field and who are linked to other fields such as psychology, social work, or geology, for example.

In this section, we describe the evaluation results from participants' interactions and questionnaire responses.

Regarding the interactions carried out by the participants, we analyse two observations to estimate the efficiency of the system (understood as the speed with which the system helps users decide their actions) [7]. The first one is the time spent by the users interacting with the system. The average time of interactions decreases as the user interacts with each session. This result may indicate that our system is efficient because users could understand the explanations quickly, once they have learned how the application works. The second observation we take into consideration to assess the efficiency of the explanations is the number of interactions carried out by the users for each explanation. The number of interactions is similar during the whole experiment although there is a small decrease as the experiment progresses. However, this decrease is much less pronounced than the decrease in interaction times. This could indicate that the system is efficient again, so we could rule out user fatigue as a key factor in the decrease in time employed with the system.

Next, we analysed the type of participant interactions. We divided the interactions into three groups: switch actions, remove actions, and undo actions. Switch actions represent interactions where users change the explanation in a particular session. We observed that the total number of switching interactions decreases in each session. It could indicate, again, that the system is efficient because the participants had previously found an explanation that satisfied them. Analysing the amount of remove actions, i.e. participants remove an attribute from an explanation, we did not observe a tendency. However, the low number of undo actions indicates that these deletions are conscious participant actions. Allowing this type of action makes the proposed interface scrutable. This means that the participants could interact with the system to find an explanation interesting to them, and they provided feedback to the system, so we could use this information to improve the explanations for users, showing them only the relevant information. Finally, through a small questionnaire, participants indicated what system features shown made them accept or reject the explanation. The

three predominant features behind participant acceptance are attributes, explanation examples, and example groupings. These are the main features of our proposed interface.

The second part of our experiment was a user satisfaction questionnaire (Table 1).

Table 2. Evaluation of the PSSUQ questionnaire. Lower values indicate higher usability. The number of systems evaluated in [21] is 21, thanks to the participation of 210 people in total. The interfaces evaluated are of all kinds, and not specifically about XAI.

Dimension	Items	Our Score	Std. deviation	Mean values in [21]
PSSUQ	I1-16	2.12	1.19	2.82
SYSUSE	I1-6	2.20	1.22	2.80
INFOQUAL	I7-12	2.19	1.24	3.02
INTERQUAL	I13-15	1.92	1.10	2.49

Table 2 shows results from PSSUQ questions. These results are divided into four groups: total PSSUQ score, System Usefulness (SYSUSE), Information Quality (INFOQUAL), and Interface quality (INTERQUAL). Table 2 shows the scores obtained in each section and the mean scores obtained in applications evaluated with the same methodology [21]. Our explanation approach obtained a global score of 2.12 in PSSUQ, whose standard deviation is 1.19. Table 2 shows a better total score for our explanatory interface than the average scores in PSSUQ calculated for explanation systems found in the literature. In addition, the explanation interface scores better in each of the PSSUQ dimensions. The difference between the results obtained with the PSSUQ questionnaire for our interface and the interfaces studied in [21] is not huge. However, because those interfaces were published as interfaces that provide high user experience, according to the PSSUQ questionnaire, then we can use these data as the baseline for our work. Having similar scores than them, or even better, means our interface is also providing high user experience for our participants. Then, it could indicate that the explanations generated are clear and understandable by the participants.

Table 3. Questions in the custom questionnaire (see Table 1) with their corresponding mean values (1-strongly agree to 7-strongly disagree) for the 35 answers/participants.

Item	Mean	Std. dev.
I17	2.20	1.05
I18	1.80	0.90
I19	2.00	0.98
I20	1.70	1.03
I21	2.05	1.21
I22	1.94	1.06
I23	1.83	1.12

Next, we analysed the results from the second part of the questionnaire. Table 3 shows the average scores for each question. Scores are calculated as the average of the 7-point Likert scale values provided by the users. In this scale, the lowest values are the best scores in the evaluation. The questions here were oriented to measure confidence in the explanation, and satisfaction about visualisation and interactivity. The results show that the interface proposed provided with confidence and satisfaction to the participants.

5 Conclusions

In this research work, we propose a new interface to enrich explanations-by-examples in black-box recommender systems. Explanations-by-examples are very well-known in the literature and widely used in many applications and platforms. However, those examples may not be as useful as they seem, since users may not remember those items or do not see the similarities between them and the recommendations. Then, these examples could not be enough for understanding the recommendations completely. To complement the explanation examples, we used Formal Concept Analysis to select the best examples and the remarkable attributes that the recommendations and the examples share. To show to the users all this explanation, we employed a graph-based visualisation that allows users to interact with the explanation. Although the use of concept lattices has been widely used to represent knowledge in AI [22], we have observed that it is not a common technique within XAI to optimise the search for common attributes between the AI model decision and the examples that explain it. As we demonstrated in this paper, the concept lattices can recover the most relevant attributes and they are helpful to users to understand AI decisions.

To validate the utility of interfaces that enrich explanations-by-examples for recommender systems using knowledge that link the recommendation with its examples, we have carried out a user study to evaluate our proposed interface. Results show that our interface provides satisfaction to users when they want to understand a recommendation, being the examples, the links between them and the recommendation, and the graph-visualisation, appreciated features of the interface. Our evaluation process has also some limitations that we plan to overcome in future work, for example evaluating the users' confidence in the system, which is a key aspect to value in XAI, or using real recommendations for the users who participate in the evaluation study.

This work is the first step to provide enrichment and additional knowledge in example-based explanations for recommendation systems. Our interface can be extrapolated to other approaches: using different types of links to connect the items, or generating different types of examples, like counterfactuals. We also might prove the utility of these types of interfaces in other types of black-box systems whose explanations could be carried out through explanations-by-examples.

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Explainable AI for Mixed Data Clustering

Jonas Amling[2], Stephan Scheele[1(✉)], Emanuel Slany[3], Moritz Lang[2], and Ute Schmid[1]

[1] University of Bamberg, Bamberg, Germany
{stephan.scheele,ute.schmid}@uni-bamberg.de
[2] dab: Daten-Analysen und Beratung GmbH, Deggendorf, Germany
{jonas.amling,moritz.lang}@dab-gmbh.de
[3] Fraunhofer Institute for Integrated Circuits IIS, Erlangen, Germany
emanuel.slany@iis.fraunhofer.de

Abstract. Clustering, an unsupervised machine learning approach, aims to find groups of similar instances. Mixed data clustering is of particular interest since real-life data often consists of diverse data types. The unsupervised nature of clustering emphasizes the need to understand the criteria for defining and distinguishing clusters. Current explainable AI (XAI) methods for clustering focus on intrinsically explainable clustering techniques, surrogate model-based explanations utilizing established XAI frameworks, and explanations generated from inter-instance distances. However, there exists a research gap in developing post-hoc methods that directly explain clusterings without resorting to surrogate models or requiring prior knowledge about the clustering algorithm. Addressing this gap, our work introduces a model-agnostic, entropy-based Feature Importance Score for continuous and discrete data, offering direct and comprehensible explanations by highlighting key features, deriving rules, and identifying cluster prototypes. The comparison with existing XAI frameworks like SHAP and ClAMP shows that we achieve similar fidelity and simplicity, proving that mixed data clusterings can be effectively explained solely from the distributions of the features and assigned clusters, making complex clusterings comprehensible to humans.

Keywords: XAI · Mixed Data Clustering · Model-Agnostic

1 Introduction

Clustering can be seen as one of the most important unsupervised Machine Learning (ML) problems, that is frequently encountered in the context of data mining [26]. The objective is to group instances into meaningful structures that share similar characteristics [21,26]. Given its unsupervised nature, identifying the specific shared characteristics within a cluster becomes a central question. The discipline of Explainable Artificial Intelligence (XAI) aims to render ML models comprehensible to humans [1]. Hence, it is only safe to deduce that XAI principles should apply to the problem of interpreting clusterings.

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Explaining clusterings is challenging for several reasons, primarily due to the fundamental differences between unsupervised cluster algorithms and supervised ML models that XAI methods traditionally target. Prevalent clustering algorithms like AGNES (Agglomerative Nesting), PAM (Partitioning Around Medoids), and DBSCAN (Density-Based Spatial Clustering of Applications with Noise) lack ground truth labels and a predictive function (ML model) that maps inputs to outputs [21,37], typical in supervised learning models. Unsupervised clustering focuses on categorizing data based on different principles, for instance, AGNES through hierarchical clustering of objects based on their similarity, PAM by partitioning around representative objects (medoids), and DBSCAN by identifying dense regions in the data space. Hence, traditional XAI methods that rely on a ML model are not directly applicable here.

We identify five key challenges within the current research landscape of explainable mixed data clustering: 1) Clustering benchmarks focus primarily on clustering complex shapes in numeric data, neglecting the intricacies of mixed data scenarios. Consequently, existing XAI approaches for clustering often overlook the significance of mixed data scenarios. 2) Different cluster algorithms on the same dataset may yield diverse outcomes. Model-agnostic alternatives are favoured over intrinsically explainable methods, as the latter support only a single clustering method. 3) Model-agnostic explanations based on surrogate models carry inherent risks, as improper surrogate model training could lead to potentially misleading explanations. 4) Intra-point distance-based model-agnostic explanations can be used to explain clusters but are only applicable if the intra-point distances are known. 5) XAI for clustering research often concentrates on a single specific explanation type, yet diverse users may favour multimodal explanations [2]. Thus, it is essential to investigate various explanation types for explaining clusterings.

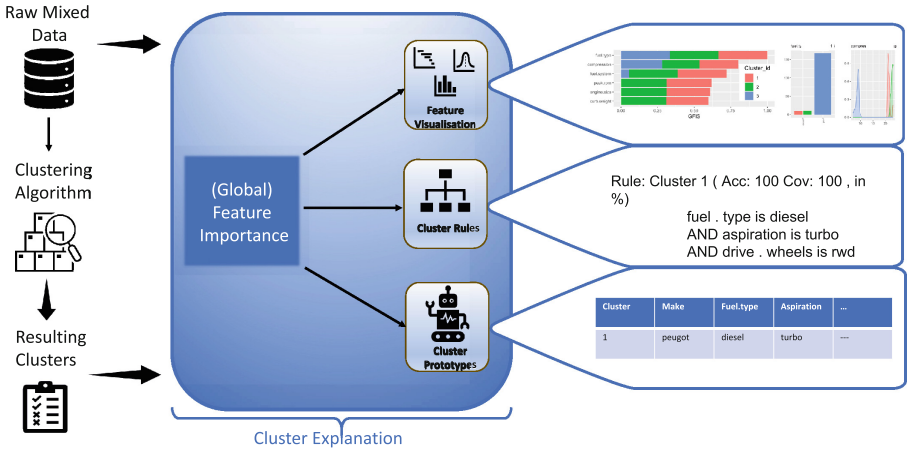
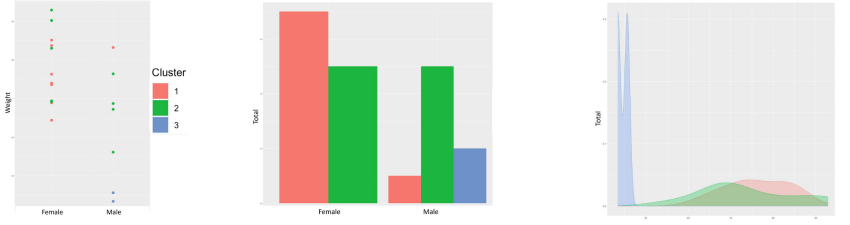


Fig. 1. Concept of our explanation methodology for clustering.

Our contribution is a novel model-agnostic post-hoc explanation method for mixed data clustering, summarized in Fig. 1. Our approach demonstrates that explanations for mixed data clustering can be directly derived from the distributions of input features and their corresponding clusters, without the need for surrogate models or prior knowledge of the clustering process. Central to our contribution is the development of an entropy-based feature importance score, a mathematical framework aimed at providing comprehensive explanations of clustering processes, supporting both, local and global explanations. Hereby, we utilize the mathematical concept of discrete and continuous entropy as the basis for assessing the importance of features. Our methodology further includes the derivation of rules from feature importance scores and the identification of prototypes for each cluster, enhancing the comprehensibility of the clustering outcomes. We have implemented our approach in the publicly available R package "clusterExplainR"[1], including our examples and a tutorial. We evaluate our method's effectiveness and the faithfulness of its explanations using synthetic datasets from clustering benchmarks, alongside two public mixed datasets, comparing our approach with the established XAI frameworks SHAP and ClAMP.

Our approach provides the following types of explanations, as illustrated on the right-hand side of Fig. 1. Firstly, we provide feature summary statistics in terms of Feature Importance Scores (FIS) that detail the importance of a feature in a global or local scope. A bar chart visually illustrates the impact and direction of each feature's effect on cluster assignments, thereby highlighting the key features that differentiate between clusters. These feature importance scores are combined with feature summary visualisations that depict the distributions of feature values within and between clusters. Secondly, our approach includes local cluster explanations by deriving a single logical rule for each cluster. Such rules are formed by a conjunction of value sets for discrete variables and specified ranges for continuous variables, respectively. Thirdly, we identify prototypes as a single characteristic representative of a cluster.

In the coming sections, we demonstrate our entropy-based explainable clustering method using a simplified mixed-type synthetic dataset. This dataset comprises two features and is categorized into three clusters, as introduced in Ex. 1. We will showcase how our method explains mixed data clusterings, by identifying key features, deriving rules, and pinpointing prototypes for each cluster.



(a) Clustered dataset. (b) Distribution of `gender`. (c) Distribution of `weight`.

Fig. 2. Synthetic example dataset for explaining mixed data clustering.

[1] R-Package clusterExplainR: https://github.com/imperonas/clusterExplainR.

Example 1. Synthetic dataset comprising a discrete random variable `gender`, a continuous variable `weight`, and three designated clusters $\mathcal{B} = \{B^{(1)}, B^{(2)}, B^{(3)}\}$. The distribution of the variables is shown in Fig. 2.

2 Related Work

Conventional XAI techniques, designed primarily for supervised learning, are not directly applicable to clustering due to the absence of ground truth labels and trained ML models. Cluster algorithms operate as a direct input-output process that returns cluster labels as output, thus requiring distinct approaches for explainability. Within the existing research on cluster explainability, the key strategies can be differentiated into three branches, as summarized in Fig. 3.

One focus is the development of intrinsically explainable clustering algorithms, with an emphasis on decision tree-based techniques [5,11,15,22,33], alongside research on interpretable neural networks [31], linear models [32], exemplars [9] and rule-based methods [13]. The remaining two branches are subsumed under post-hoc, model-agnostic XAI strategies for clustering, which typically focus on the development and application of surrogate models, either integrated with well-known XAI frameworks or being interpretable models based on decision trees or rule-based methods such as ClAMP [6]. These approaches diverge from conventional XAI methods, as they explain the decision-making process of a surrogate model rather than the clustering itself. Less explored are model-agnostic explanatory methods that operate directly on inter-point distances used for clustering, such as identifying a set of prototypes post-clustering [8] or perturbation-based identification of relevant features [10,38]. This overview, while not exhaustive, highlights several notable approaches.

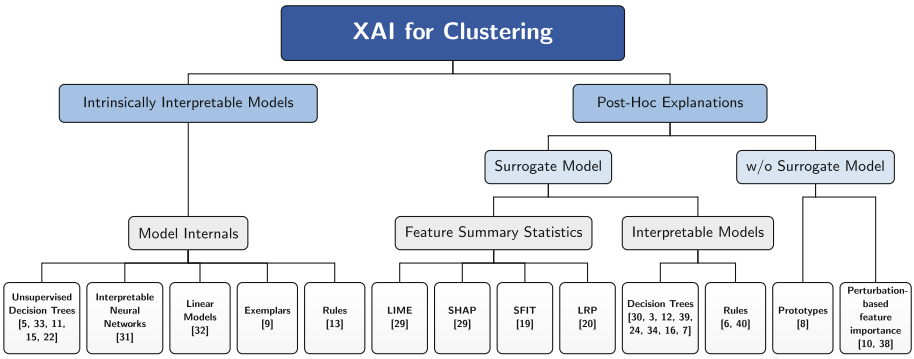


Fig. 3. Overview of existing research on cluster explanations.

Intrinsic Explainable Clustering. models are designed in such a way that they are inherently understandable to humans, incorporating an interpretability component at their core [28]. In the specific case of clustering, these comprise new types

of clustering algorithms specifically developed with an interpretability component. The types of explanations generated by these approaches are based on model internals. Most prominent among those are decision tree-based clustering algorithms that are particularly notable for their intrinsic clarity, given by the explicit structure of decision trees [5,11,15,22,33]. In addition to decision tree-based approaches, other methodologies aim to achieve intrinsic explainability in clustering. Peng et al. [31] devise a neural network architecture that mimics a clustering methodology, allowing for unsupervised clustering and subsequent cluster explanations. Another approach, proposed by Davidson et al. [9], combines clustering with exemplars (akin to prototypes), explaining compact clusters by disjoint sets of exemplars. A knowledge graph embedding is utilized for clustering in [13], offering rule sets as explanations for each cluster. Moreover, Plant et al. [32] employ the Minimum Description Length principle to identify meaningful clusters that can be efficiently compressed and explained in terms of relevant attributes.

Post-Hoc Cluster Explanations frequently employ surrogate models for generating model-agnostic explanations, utilizing the combined dataset of input features and clustering results for surrogate model training, essentially framing it as a classification task. Such models serve as proxy for the clustering algorithm to be explained. These can be differentiated into two approaches: one that emphasizes on methods deriving feature summary statistics and another that focuses on utilizing interpretable models. The former has been studied in combination with existing XAI frameworks such as SHAP [23,29], LIME [29,35], SFIT [19], and LRP [20] for explanatory clustering, based on support vector machines [29] or neural networks [19,20]. This integration with existing XAI frameworks ensures state-of-the-art explanation standards, including both model-agnostic and model-dependent XAI techniques. Nonetheless, this approach is constrained by the necessity for precise surrogate model training to ensure both accuracy and relevance. Interpretable models, such as decision trees have emerged as a well-studied choice as surrogate models for cluster explanations [3,7,12,16,24,30,34,39]. As surrogate models tend to increase in size to achieve high accuracy, this can compromise their interpretability. Research thus focuses on balancing accuracy with interpretability, through abstraction strategies like reducing the complexity of a model [30,39], removing outliers from clusterings prior surrogate model training [3], or minimizing the number of features [24]. Recent approaches such as LUX [7], LORE [16,17] and EXPLAN [34] train a classifier decision tree surrogat model using a subset of the data, derived from the neighborhood of the instance to be explained, as base for rule extraction. Additionally, LUX utilizes feature important scores based on SHAP values for optimization of the decision tree. An alternative approach [6] involves the generation of rule sets after clustering, extracting rules from instances that are located at the boundaries between clusters. Significantly less research has been devoted to post-hoc cluster explanations that do not depend on surrogate models. The method in [8] utilizes data points as prototypes, aiming to identify a set of prototypes that accurately represent each cluster post clustering, based on the

distance measures employed during clustering. Perturbation-based algorithmic-agnostic approaches focus on identifying crucial features on a local and global scope by leveraging feature perturbations [10,38]. These methods manipulate feature values for a subset of instances and then reassign these modified instances to the existing clusters. By utilizing custom metrics (such as G2PC and L2PC [10] or SMART and IDEA [38]), feature importance scores are derived based on the instance-wise cluster assignments of the manipulated sample.

Entopy-Based Mixed Data Clustering is at the core of our novel approach, drawing inspiration from a precursory concept presented in [32], demonstrating the application of the Minimum Description Length principle for clustering mixed data. At the heart of this concept lies the idea that a good cluster should demonstrate efficient data compression. In their study, the authors use entropy as a measure of the compactness within a cluster to generate good and interpretable clusters. Drawing on this principle, we introduce a novel feature importance score that is grounded in the mathematical concept of entropy.

3 Entropy-Based Cluster Explanations for Mixed Data

Feature importance scores play a crucial role in XAI for predictive models, typically obtained by permuting each feature individually and assessing its effect on the model error [28]. However, applying these traditional feature importance scores directly to unsupervised clustering poses challenges. This is primarily because clustering algorithms are not designed to predict instances post-clustering, and the ground truth clusters of instances are unknown. In previous research on perturbation-based feature importance cluster explanations [10,38], addressing this challenge required cluster processes capable of reassigning clusters to instances after the initial clustering. We employ the concept of entropy to identify key features per cluster, guide users in exploring relevant features, introduce an innovative approach to detect representative cluster instances, and produce a single rule describing each cluster without any knowledge about the underlying clustering process.

Notation. Let $x = (x_1, ..., x_i, ..., x_n) \in \mathcal{X}^n$ be a feature with identifier set $\{1, ..., i, ..., n\}$. Further, let $c \subseteq \{1, ..., n\}$ be the set of categorical and $m = \{1, ..., n\} \setminus c$ the set of metric identifiers. Let (hard) clustering be a decomposition of an input dataset $\mathcal{X}$ into k partitions (clusters) $\{B^{(1)}, ..., B^{(k)}\} = \mathcal{B}$. Finally, let X be a random process corresponding to a feature dimension in $\mathcal{X}$.

The fact that tabular datasets frequently contain categorical and metric features imposes the additional challenge to formulate an approach that likewise accounts for both data types. We use the Shannon entropy (Def. 1) for categorical and continuous entropy (Def. 2) for metric features. Entropy is a measure that quantifies the average information contained by a random variable [25]. Intuitively, heterogenous partitions – clusters with equally distributed observations – receive maximum entropy values. The entropy reaches its minimum, when the observations converge to a single-point distribution.

Definition 1 (Shannon Entropy [25]). Let X be a discrete random variable of a feature from $\{x_i \mid i \in c,\ x \in \mathcal{X}\}$ with a finite set of possible outcomes $Z = \{z_1, ..., z_m\}$. Let $p_z = Pr(X = z)$. The *Shannon entropy* is given as:

$$H(X) = -\sum_{z \in Z} p_z \log_2(p_z).$$

Definition 2 (Continuous Entropy [27]). Let X be a continuous random variable with possible outcomes $z \in]\infty, \infty[$ of a feature from $\{x_i \mid i \in m,\ x \in \mathcal{X}\}$ with probability density function p. The *continuous entropy* is given as:

$$H(X) = -\int p(z) \log_2 p(z) \mathrm{d}z.$$

Remark 1. We standardize each feature dimension in $\{x_i \mid i \in m,\ x \in \mathcal{X}\}$ to counteract the dimensionality dependence of the continuous entropy.

We utilize the entropy measure to calculate local cluster feature importance scores ($lFIS$) within the interval $[0, 1]$, see Def. 3. The basic idea is to compare the entropy of a feature within a cluster with its global entropy. The closer it aligns, the less important it is. Low entropy values of a cluster describe a high concentration of feature values in a cluster, which is why we argue that our formulation is a valid quantification of the importance of a feature for a cluster. A short example of the calculation of $lFIS$ for a discrete random variable is shown in Ex. 2.

Definition 3 (Local Cluster Feature Importance Score). Let

$$lFIS(X, B) = 1 - \min\left(H(X_B) \times H(X)^{-1}, 1\right)$$

be the *local cluster feature importance score*, where X_B is a random variable wrt. a partition $B \in \mathcal{B}$. We utilize Def. 1 if $i \in c$ and Def. 2 otherwise.

Example 2. Suppose the following distribution of outcomes of the random variable `gender` in the population $\mathcal{X}$ and the clusters $B^{(1)}$ to $B^{(3)}$ (Ex. 1):

`Gender`	$\mathcal{X}$	$B^{(1)}$	$B^{(2)}$	$B^{(3)}$
Female	60%	87.5%	50%	0%
Male	40%	12.5%	50%	100%

We calculate the local cluster feature importance score $lFIS(\texttt{gender}, B^{(1)})$, with its components, the entropy for the population $H(\texttt{gender})$ as well as for the cluster $H(\texttt{gender}_{B^{(1)}})$, as follows:

$$H(\texttt{gender}) = \left(0.6 \times \log_2 \frac{1}{0.6}\right) + \left(0.4 \times \log_2 \frac{1}{0.4}\right) \approx 0.44 + 0.53 = 0.97$$

$$H(\texttt{gender}_{B^{(1)}}) = \left(0.875 \times \log_2 \frac{1}{0.875}\right) + \left(0.125 \times \log_2 \frac{1}{0.125}\right) \approx 0.54$$

$$lFIS(\texttt{gender}, B^{(1)}) = 1 - \min\left(\frac{H(\texttt{gender}_{B^{(1)}})}{H(\texttt{gender})}, 1\right) \approx 0.44.$$

We extend the concept of feature importance to the global level by calculating the average of aggregated local importance of a respective feature over all clusters. The global feature importance score (Def. 4) is essentially the mean value of local importance given a feature.

Definition 4 (Global Feature Importance Score). Let

$$gFIS(X, \mathcal{B}) = \left(\sum_{B \in \mathcal{B}} lFIS(X, B) \right) \times k^{-1}$$

be the *global feature importance score*, calculated wrt. k partitions $B \in \mathcal{B}$.

Example 3. Calculating $lFIS(...)$ for the remaining clusters of Ex. 2 yields $lFIS(\texttt{gender}, B^{(2)}) = 1.0$ and $lFIS(\texttt{gender}, B^{(3)}) = 0.0$, from which directly follows that $gFIS(\texttt{gender}, \mathcal{B}) = (0.44 + 1 + 0) \times \frac{1}{3} = 0.48$.

The global feature importance score provides a general overview of the features that are important for the entire set of generated clusters. It allows users to obtain a concise understanding of the overall clustering results.

Representative instances such as prototypes (Def. 5) are an established explanation strategy for clusters [8], intuitively, a prototype captures the key characteristics or decisive features for a specific cluster. We select prototypes given a specific instance wrt. its feature values and the feature importance by utilizing the entity cluster matching score (Def. 6). By incorporating feature importance weighting, we guarantee that a cluster prototype is not only a representative of the overall cluster but particularly of the cluster's most important features.

Definition 5 (Prototype). A *prototype* for partition B can be retrieved by $\bar{x}_B = \arg\max_{x \in B} ECMS(x, \mathbf{X}, B)$, where $\mathbf{X}$ is the sequence of random processes $(X_1, ...X_i, ..., X_n)$ corresponding to the feature dimensions in $\mathcal{X}$.

Definition 6 (Entity Cluster Matching Score). Let X_B be a random process for cluster B. Let Z_B the possible outcomes of X_B in the categorical case. Suppose that $x_i \in Z_B$ and $x_i \in]-\infty, \infty[$ for the categorical and the continuous case, respectively. Then, the *entity cluster matching score* is defined as follows:

$$ECMS(x, \mathbf{X}, B) = \frac{\sum_{x_i \in x, X_B \in \mathbf{X}} m(x_i, X_B) \times lFIS(X, B)}{\sum_{X \in \mathbf{X}} lFIS(X, B)}, \text{ where}$$

$$m(x_i, X_B) = \begin{cases} Pr(X_B = x_i) \times \max\left(\{Pr(X_B = z) | z \in Z_B\}\right)^{-1} & \text{if } i \in c, \\ p_{X_B}(x_i) \times \max\left(p_{X_B}(z)\right)^{-1} & \text{otherwise,} \end{cases}$$

where we assume that we can approximate the outcomes of the probability density function p_{X_B} for x_i and $z \in]-\infty, \infty[$ in the continuous case.

Example 4. Suppose the realization of an additional random variable `weight`, being 75 kg. Assume that $lFIS(\texttt{weight}, B^{(1)}) = 0.33$ with $p_{\texttt{weight}_{B^{(1)}}}(75) = 0.05$

and mode($p_{\texttt{weight}_{B^{(1)}}}$) = 0.05. For now, let there be a single woman (w) and a single man (m) from cluster $B^{(1)}$, both with weight = 75 kg. The $ECMS(...)$ of both instances can be calculated as follows:

$$ECMS(\texttt{w}, (\texttt{gender}, \texttt{weight}), B^{(1)}) = \frac{1.0 \times 0.44 + 1.0 \times 0.33}{0.44 + 0.33} = 1.0,$$
$$ECMS(\texttt{m}, (\texttt{gender}, \texttt{weight}), B^{(1)}) \approx \frac{0.1429 \times 0.44 + 1.0 \times 0.33}{0.44 + 0.33} \approx 0.51,$$

which lets us select the female instance (w) as prototype for cluster $B^{(1)}$.

Rules derived from the local cluster feature importance are the final type of examined explanations. We aim to produce concise and interpretable rules for each cluster, containing only the relevant features. The rules are conjunctive statements with numerical ranges for continuous features and sets for discrete features. We aim to find only one rule per cluster that is designed to strike a balance between coverage and accuracy. The refinement involves incrementally generating a rule for the feature with the highest local importance and eventually combining the individual rules into a final rule. We use the $F1$-score as a rule selection heuristic (Def. 7) to simultaneously account for accuracy and coverage.

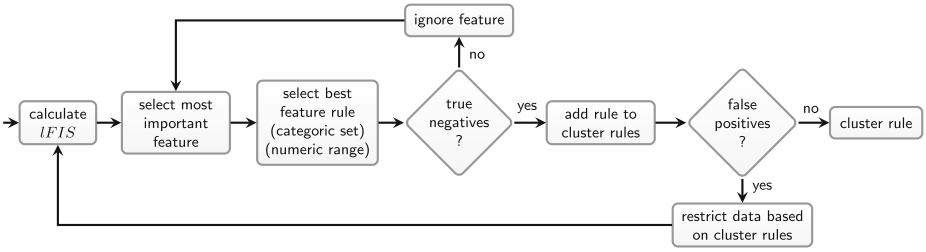


Fig. 4. Rule selection procedure involves choosing the optimal rule based on local feature importance and $F1$-score, disregarding features that don't yield true negatives and incrementally adding features to eliminate false positives.

Definition 7 (Rule Selection Heuristic). Let $r(\mathcal{X})$ and $R(\mathcal{X})$ denote the application of a single rule r and a rule set R to a dataset $\mathcal{X}$, returning a partition $B \in \mathcal{B}$. Let the *rule selection heuristic* be defined as $\arg\max_r\{F1(r(\mathcal{X}))|r \in R_{B,X}\}$, where $R_{B,X}$ is a set of proposal rules describing partition B by random variable X. Suppose that $F1(r(\mathcal{X}))$ returns the $F1$-score for r on $\mathcal{X}$.

The entire rule search procedure is depicted in Fig. 4: We select the most important feature by the local cluster feature importance score and the best proposal rule using the rule search heuristic. We apply the rule to the dataset. A feature is ignored if a resulting rule would sort all instances into the cluster. The rule is refined if its application yields false positives. Algorithm 1 contains the technical details of our RuleSearch procedure.

Algorithm 1: RuleSearch(B, $\mathcal{X}$, $\mathbf{X}$)

Input: Partition B, dataset $\mathcal{X}$, sequence of random variables $\mathbf{X}$
Output: Rule set R describing partition B

$R \leftarrow \emptyset$
while $\exists_{x \in R(\mathcal{X})} x \notin B$ **do** ▷ *Refine rule if there exist false positives.*
  $r \leftarrow \varnothing; \mathbf{X}' \leftarrow \mathbf{X}$
  while $r = \varnothing \wedge |\mathbf{X}'| > 0$ **do**
    $X \leftarrow \arg\max_X \{lFIS(X, B) | X \in \mathbf{X}'\}$ ▷ *Def. 3.*
    $r' \leftarrow \arg\max_r \{r(\mathcal{X}) | r \in R_{B,X}\}$ ▷ *Def. 7.*
    $r \leftarrow r'$ **if** $r'(\mathcal{X}) \neq \mathcal{X}$ **else** $\mathbf{X}' \leftarrow \mathbf{X}' \setminus \{X\}$ ▷ *Add rule if it produces at least one true negative, otherwise continue with the next random variable.*
  $R \leftarrow R \cup r$
return R

Example 5. Rules for the individual clusters for our running example.

```
Cluster 1 (Accuracy:80%, Coverage: 87,5%):
    Gender is  Female
    AND Weight is between  64.4 and 85.1

Cluster 2 (Accuracy:60%, Coverage: 100%):
    Weight is between  56.1 and 92.9

Cluster 3 (Accuracy:100%, Coverage: 100%):
    Gender is  Male
    AND Weight is between  43.4 and 45.6
```

4 Evaluation

We evaluate our entropy-based cluster explainer on cluster-typical, yet purely numerical benchmark datasets (wingnut, mk1, mk2, and isolation) [14]. We include evaluations on the real-world mixed datasets Customer[2] and Automobile[3], using the well-known cluster algorithms kproto[4] and hdbscan[5] in combination with Harikumars distance for mixed data[6].

Two reasons let us advocate in favour of our evaluation setup: First, to the best of our knowledge, there exists no mixed data cluster benchmark dataset. Second, the current literature on clustering algorithms does not suggest a publicly available, implemented method in the absence of a surrogate model dedicated for mixed data clustering (Sect. 2). Therefore, we benchmark our explanation with other well-established methods – SHAP[7] [23] applied to a random forest[8] surrogate model to derive feature importance scores and ClAMP[9] [6] to

[2] https://www.kaggle.com/datasets/datascientistanna/customers-dataset.
[3] https://archive.ics.uci.edu/dataset/10/automobile.
[4] https://cran.r-project.org/web/packages/clustMixType/.
[5] https://cran.r-project.org/web/packages/dbscan/.
[6] https://cran.r-project.org/web/packages/kmed/.
[7] https://github.com/shap/shap.
[8] https://scikit-learn.org/stable/modules/generated/sklearn.ensemble.RandomForestClassifier.html.
[9] https://github.com/sbobek/clamp.

generate rules – while we emphasize the necessity of a structured evaluation of XAI methods for mixed data clustering.

SHAP [23] simplifies the feature space and transforms the original into a linear model. SHAP starts with a baseline attribution ϕ_0 that is if all features are zero (missing), e.g., $\phi_0 = f(h_x(\mathbf{0}))$, where h_x is a transformation function wrt. x. It adds the remaining features, where the linear coefficients represent the attribution values. The SHAP approach approximates the original model in the sense that $f(x) = g(x') = \psi_0 + \phi^T x'$, where f is the original and g is the simplified (linear) model wrt. x, ϕ the attribution vector, and x' the simplified feature vector obtained by $x = h_x(x')$.

CLAMP [6] is a model-agnostic cluster explanation approach that generates rules by leveraging the Anchor explainer. [36]. To achieve this, a surrogate model classifier is trained using the clustered dataset. Subsequently, a set of instances is selected to form (potentially multidimensional) bounding boxes around identified clusters. Finally, CLAMP utilizes the Anchor explainer to create local "if-then" rules, derived from the bounding-box instances and the trained classifier. The rules are of the form $p \rightarrow y$, where the premise p is a conjunction of boolean conditions on feature values and the cluster y is the consequence of a rule.

Regarding the evaluation criteria, we restrict ourselves to fidelity and simplicity [4]. The former assesses the degree of which the explanations represent the underlying cluster, whereas the latter quantifies rule complexity and comprehensibility. For the importance scores, we measure the fidelity by comparing our attribution to SHAP values. For the rules, we use the accuracy and the coverage metrics, where we aim to reproduce the clustering results using our rule model. We quantify the simplicity only in context of the generated rules in both dimensions: the number of rules and the average number of feature per rule.

Part of our contribution is the R-package "clusterExplainR" (https://github.com/imperonas/clusterExplainR) that implements our eplanation approach: local and global feature importance scores, prototypes, and rules. With the use of our R implementation, all experiments can be exactly reproduced.

Benchmark Datasets. Figure 6 visualizes in which cluster the global feature importance score originates. Comparing our entropy-based global explanation score with SHAP, we can generally observe that the feature ranking and the associated magnitude is similar for both metrics. Yet, for the isolation dataset, their results differ strongly: SHAP receives almost uniformly distributed importance values from each cluster in both features, whereas the global feature importance score attributes nearly the entire importance magnitude of cluster one, basically neglecting the decision-making capabilities of the remaining clusters.

Both, graphical (Fig. 5) and quantitative (Table 1) assessments of our generated rules reveal a similar behaviour to the rules generated by ClAMP. While both approaches cover only less sophisticated cluster shapes, our RuleSearch generates only one rule per cluster with fewer features per rule on average, indicating simpler rules compared to the ones produced by ClAMP (Table 2).

Table 1. Rule fidelity evaluation on benchmark data. We compare our RuleSearch procedure with ClAMP regarding the **Accuracy** and **Coverage** metrics wrt. each **Cluster** on benchmark **Data** sets.

Data	Cluster	RuleSearch		ClAMP	
		Accuracy	Coverage	Accuracy	Coverage
wingnut	Cluster 1	100%	100%	100%	100%
	Cluster 2	100%	100%	100%	100%
mk1	Cluster 1	100%	100%	100%	100%
	Cluster 2	97%	91%	99%	98%
	Cluster 3	98%	94%	99%	100%
mk2	Cluster 1	51.3%	99.8%	49%	99%
	Cluster 2	58.2%	96.2%	49%	98%
isolation	Cluster 1	98.7%	100%	66%	90%
	Cluster 2	53%	99.9%	69%	30%
	Cluster 3	–	–	65%	33%

Table 2. Rule simplicity evaluation on benchmark data. We compare our RuleSearch procedure with ClAMP regarding the number of rules (# **Rules**) and the average number of feature per rule (⌀ **Features**) on benchmark **Data** sets.

	RuleSearch		ClAMP	
Data	# Rules	⌀ Features	# Rules	⌀ Features
wingnut	1.0	1.0	4.0	3.25
mk1	1.0	1.67	3.33	3.5
mk2	1.0	2.0	5.0	3.5
isolation	1.0	2.0	5.0	3.93

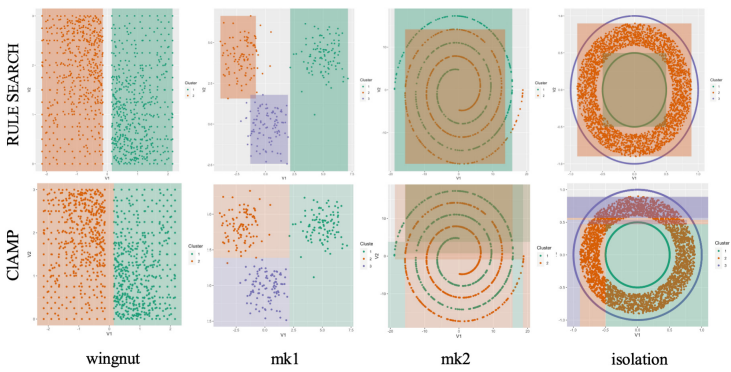


Fig. 5. Assessment of rule-based cluster explanations from RuleSearch and ClAMP on the synthetic benchmark datasets wingnut, mk1, mk2, and isolation.

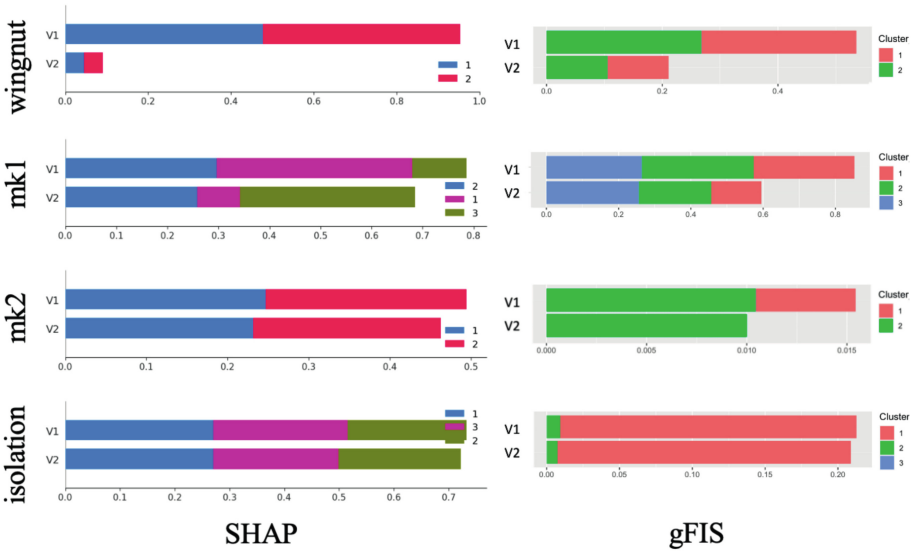


Fig. 6. Comparison of global feature importance score ($gFIS$) with SHAP values on benchmark data. We compare our global feature importance score with SHAP values, indicating which cluster is emphasized by a feature.

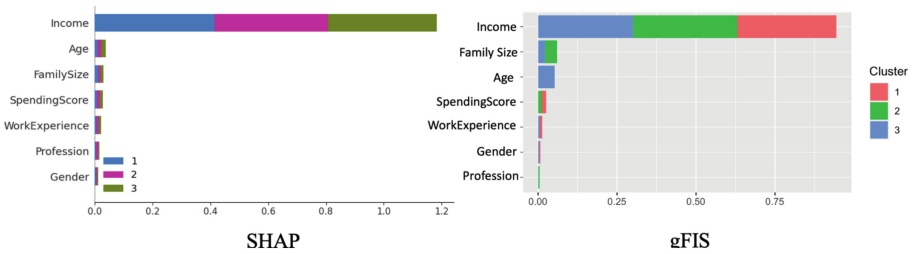


Fig. 7. Comparison of $gFIS$ with SHAP values on Customer and kproto. We compare our $gFIS$ with SHAP values, indicating which cluster is emphasized by a feature.

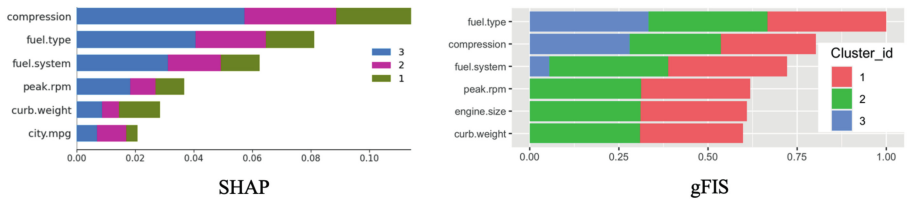


Fig. 8. Comparison of $gFIS$ with SHAP values on Automobile and hdbscan. We compare our $gFIS$ with SHAP values, indicating which cluster is emphasized by a feature.

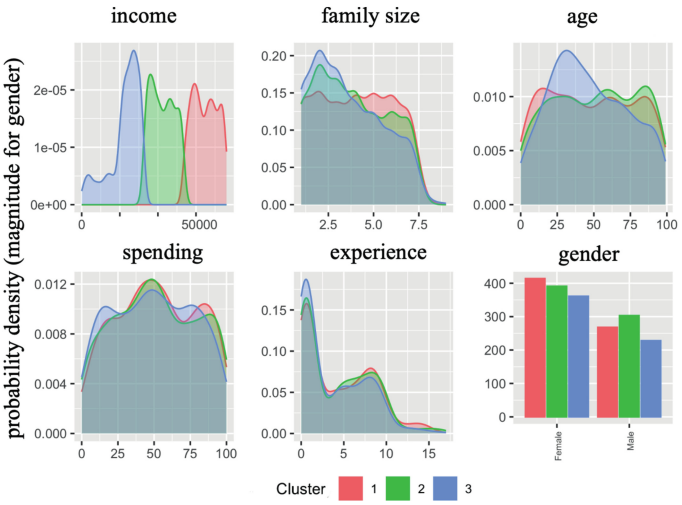


Fig. 9. Global feature importance score visualization for Customer and kproto. We select the six most important features according to the global feature importance score and visualize the distribution wrt. the clusters.

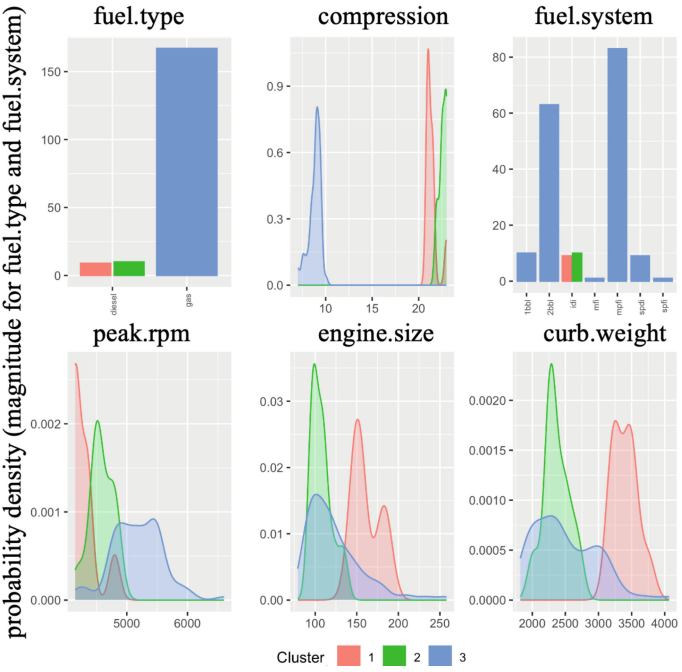


Fig. 10. Global feature importance score visualization for Automobile and hdbscan. We select the six most important features according to the global feature importance score and visualize the distribution wrt. the clusters.

Customer Dataset. We cluster the Customer dataset with kproto. Figure 9 visualizes the distributions of the six most important features according to our global feature importance score. It depicts a clear visual decomposition by the feature income into three clusters. Furthermore, we compare the origins of the global feature importance score with the ones of SHAP values (Fig. 7). Similarly to SHAP, our score identifies the income as most important feature. Both procedures differ from each other in the ranking of remaining features, which all have smaller score magnitudes. Figure 11 compares the rules obtained by RuleSearch with the ones produced by ClAMP. Note, that RuleSearch contains information on accuracy (Acc.) and coverage (Cov.). For ClAMP, we print only the first and final rules per cluster if more than two rules exist for a cluster. Our approach supersedes ClAMP in both complexity dimensions, the number of rules per cluster and the average number of features per rule. Moreover, we display accuracy and coverage values wrt. a certain cluster for an easy assessment of its distinction capabilities.

Automobile Dataset. We cluster the Automobile dataset with hdbscan and apply the identical evaluation strategy as used for the Customer dataset: First, we inspect the six most important features (Fig. 10) and visualize the global feature importance scores conditioned on the clusters in comparison to SHAP (Fig. 8). Second, we compare the rules derived by RuleSearch and ClAMP. The global feature importance score selects the identical six most important features compared to SHAP (Fig. 8). However, it prioritizes fuel as the most important feature, whereas SHAP prefers compression. Figure 10 indicates that both features offer distinction capabilities in the sense that the clusters are built along specific outcomes. Interesting are the cluster attributions to both importance scores: For instance, SHAP attributes the fuel system to cluster three, whereas $gFIS$ attributes it to clusters one and two, while clusters one and two are the only partitions containing only one distinct value for the feature fuel system. The results offered by the RuleSearch and ClAMP comparison for the hdbscan clustered Automobile dataset are in line with the previous experiments (Fig. 12). Our rule search procedure produces simpler rules regarding the number of rules per cluster as well as the average number of features per rule.

Summary. Our evaluation lets us retrieve four main findings: First, the global explanation score has a high fidelity in the sense that it selects the same important features as we do if we train a surrogate model and apply SHAP, which is a state-of-the-art cluster explanation strategy. Second, the origins of the global feature importance score occasionally differ compared to SHAP. Third, RuleSearch explanations have a low fidelity in the sense that they fail to explain complex cluster shapes. However, its performance is in line with the state-of-the-art ClAMP algorithm to generate rules. For shapes, which can be clustered by RuleSearch or ClAMP, our approach offers comparable accuracy and coverage. And fourth, RuleSearch produces simpler rules with less complexity compared to ClAMP regarding both, the number of rules needed to explain a cluster as well as the average number of features per rule.

RuleSearch

```
Rule Cluster 1 ( Acc: 100 Cov: 100 , in %)
 Income is between  134213 and 189974

Rule Cluster 2 ( Acc: 100 Cov: 100 , in %)
 Income is between  81545 and 134008

Rule Cluster 3 ( Acc: 100 Cov: 100 , in %)
 Income is between  0 and 81199
```

ClAMP

```
# Cluster 1: 4 rules
FamilySize < 7.0 AND FamilySize < 6.0 AND FamilySize < 5.0
        AND Income >= 134213.0 -> Cluster 1 ...
FamilySize >= 7.0 AND Income >= 134213.0 -> Cluster 1

# Cluster 2: 3 rules
FamilySize < 7.0 AND FamilySize < 6.0 AND Income < 134213.0
        AND Income >= 81545.0 -> Cluster 2 ...
FamilySize >= 7.0 AND Income < 134213.0
        AND Income >= 81545.0 -> Cluster 2

# Cluster 3: 3 rules
Age < 96.0 AND Income < 134213.0 AND Income < 81545.0
        AND Profession >= 6.0 -> Cluster 3 ...
Age < 96.0 AND Income < 134213.0 AND Income < 81545.0
        AND Profession < 6.0 -> Cluster 3
```

Fig. 11. Comparison of RuleSearch and ClAMP for Customer dataset.

RuleSearch

```
Rule Cluster 1 ( Acc: 100 Cov: 100 , in %)
 fuel.type is  diesel
 AND aspiration is  turbo
 AND drive.wheels is  rwd

Rule Cluster 2 ( Acc: 100 Cov: 100 , in %)
 fuel.type is  diesel
 AND engine.type is  ohc
 AND numofcylinders is  four

Rule Cluster 3 ( Acc: 100 Cov: 100 , in %)
 fueltype is  gas
```

ClAMP

```
# Cluster 1: 4 rules
compression >= 21.0 AND city.mpg < 30.0 AND highway.mpg >= 33.0
        AND peak.rpm < 4350.0 -> Cluster 1 ...
compression >= 21.0 AND city.mpg < 30.0 AND peak.rpm >= 4350.0
        AND length >= 190.9 -> Cluster 1

# Cluster 2: 3 rules
compression >= 21.0 AND compression < 23.0 AND horsepower 73.0
        AND city.mpg >= 30.0 -> Cluster 2 ...
compression >= 21.0 AND horsepower >= 73.0
        AND city.mpg >= 30.0 -> Cluster 2

# Cluster 3: 5 rules
compression < 21.0 AND horsepower < 182.0 AND stroke < 4.17 AND engine .
    size < 234.0 -> Cluster 3 ...
price >= 35550.0 AND compression < 21.0 AND stroke >= 4.17 -> Cluster 3
```

Fig. 12. Comparison of RuleSearch and ClAMP for Automobile dataset.

5 Discussion

Our experimental results demonstrate the reliability of our explanations on purely numeric benchmark datasets as well as on real-life mixed-type datasets. Our FIS-based rankings closely align with those derived by SHAP [23], a state-of-the-art XAI framework applied to a surrogate model. Additionally, our generated rules showcase simplicity, utilizing fewer features compared to the rule-based explainer ClAMP [6]. In contrast to ClAMP, our rules are less complex because we restrict our explanations to a single rule per cluster that considers only important features during the rule generation process. However, prioritizing simplicity in the rules leads to reduced performance in terms of fidelity. Based on our findings, we conclude that it is possible to derive global and local mixed data cluster explanations without the need for an additional surrogate model or prior knowledge about the underlying clustering process.

Our core contributions encompass several key advancements in explainable mixed data clustering. Firstly, we introduce a model-agnostic post-hoc explanation method that derives insights solely from the raw data feature distributions and the assigned clusters. Secondly, we address the challenge of mixed data handling by utilizing the concept of entropy applicable to both discrete and continuous data types. What sets our approach apart is the integration of multiple types of explanations, including feature importance rankings, feature visualizations, cluster rules, and prototypes, supporting both visual and textual representations, providing users with multimodal local and global cluster explanations. Finally, our methodology ensures that rules and prototypes consider the importance of features.

Although, our proposed approach shows promising results in explaining clusters based on important features, there are still several open challenges. Our method currently faces limitations in capturing feature interactions (e.g. in the mk2 dataset) and complex cluster shapes (such as the mk1 and isolation datasets). Additionally, the comparison between continuous and discrete entropy may be affected by the tendency of discrete entropy to approach zero, potentially favoring discrete features over continuous ones. Furthermore, when dealing with very small clusters, there is a risk that features will occur within a small range of values, potentially limiting the interpretability of explanations.

6 Conclusion

In the domain of unsupervised clustering, particularly with mixed data, comprehending the subtleties of clustering is crucial for human comprehensibility. Our approach tackles the complex challenge of explaining unsupervised clustering with mixed data by focusing on making the outcomes easily interpretable and comprehendsible. At its core, our method clarifies through local cluster explanations the reasons why instances are assigned to specific clusters, and pinpoints the critical features that influence the partitioning of data into distinct clusters at the global level. Moreover, our approach provides insights into the logic of clustering processes, revealing specific rules that describe the criteria and patterns

of how instances are classified into clusters. We provide multi-modal explanations, including visual representations that highlight the importance of features and illustrate the distribution of features within and across clusters, alongside rule- and prototype-based explanations. A key advantage of our approach is its model-agnostic nature, which eliminates the need for predictive surrogate models, prior knowledge about the inner workings of a cluster algorithm, or any additional information beyond the raw data and its corresponding clusters.

We envision diverse paths for future work to advance our method. Firstly, establishing benchmarks for mixed data clustering would enhance comparability and reproducibility, advancing the field and its explanations. Secondly, a broader comparative evaluation of our approach with other explanation methods and various clustering algorithms will provide deeper insights on the robustness and generalizability of our methodology. This evaluation is crucial to understand the underlying reasons for the observed divergence between $gFIS$ and mean SHAP values. Furthermore, improving our rule search algorithm by relaxing the restriction of a single ruler per cluster could strike a better balance between simplicity and fidelity. Moreover, the investigation of counterfactual explanations in the spirit of [16], but in our case based on generated rules and the Entity Cluster Matching Score promises to improve the interpretability of cluster explanations. In this respect, we plan to provide an explanation for why a particular instance was assigned to a specific cluster contrary to the user's expectations. The integration of our novel explanation methodology into the process of data exploration via clustering, can enable users to interactively refine the cluster parameters and remove unnecessary features in order to generate good clusters that meet specific user needs, including ethical and fairness considerations. Thus, we propose an architecture to combine our appproach with interactive learning, in the spirit of [18], enabling the inclusion of human feedback to enhance user engagement in the clustering workflow.

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Explaining Graph Classifiers by Unsupervised Node Relevance Attribution

Michele Fontanesi, Alessio Micheli, and Marco Podda(✉)

Department of Computer Science, University of Pisa, 56127 Pisa, Italy
michele.fontanesi@phd.unipi.it, {alessio.micheli,marco.podda}@unipi.it

Abstract. Assessing and comparing the performances of explainable Artificial Intelligence (XAI) techniques for Deep Graph Networks (DGN) is an actively researched open problem. In particular, XAI techniques that assign an importance score to the graph nodes are evaluated with standard metrics such as accuracy, which in turn entails to label a certain number of nodes as relevant according to a hard threshold. This protocol is unsuited to real-world applications where the true number of relevant nodes cannot be known in the absence of a ground truth. We attempt to fill this gap by developing unsupervised methods able to assign relevance even when the ground truth is not accessible. Specifically, we empirically show that a simple clustering-based strategy is able to recover the correct number of relevant nodes in most of the analyzed cases. Furthermore, we show that the silhouette score (which measures the clustering fit) can be used to proxy the quality of the relevance attribution. This work is a preliminary step towards reliably assessing the explanation quality in realistic settings.

Keywords: Explainable Artificial Intelligence · Deep Graph Networks · Node relevance attribution

1 Introduction

Since the pioneering works that introduced them [7,11], neural networks able to process graph structures (a.k.a. Deep Graph Networks [3]) have seen widespread adoption in multiple fields due to their capacity to flexibly handle graph data for modeling complex relational phenomena with high performances. However, despite their success, the decision process behind the mere numerical prediction remains opaque to the end user. This lack of interpretability hinders their trustworthiness [8], especially in safety-critical domains. In response to this shortcoming, researchers developed over the last few years a plethora of eXplainable Artificial Intelligence (XAI) [4] techniques able to craft explanations for DGN predictions known as graph explainers [5,16]. Specifically, we focus on graph explainers that assign a real-valued importance score to each node of the graph under study, representing their contribution to the model prediction. Typically,

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the evaluation of a graph explainer involves thresholding the importance scores into relevance (1) or irrelevance (0) indicators, a task which we term *relevance assignment*. The result of a relevance assignment is a binary vector that can be compared to a relevance ground truth vector with standard metrics such as accuracy, F_1, or more sophisticated metrics [1,2,6]. Assigning relevance is straightforward when the ground truth is known; it amounts to simply counting the number k of relevant nodes in the ground truth, and then deeming as relevant the top-k importance scores given by the explainer. However, this approach is not applicable in real-world settings where k cannot be known since the ground truth is not accessible to the end user. Thus, there is currently no systematic method for performing relevance assignment without a ground truth, which complicates the performance assessment of graph explainers outside of a benchmarking context. Here, we attempt to fill this gap by presenting and evaluating two unsupervised strategies to assign relevance when the ground truth is unavailable. Our experiments reveal that assigning relevance with a clustering-based strategy, i.e. without considering k, achieves a F_1 score close to that obtained by explicitly looking up k in the ground truth. Moreover, we observe that the silhouette score of the unsupervised relevance assignment (which measures how well-separated the relevant nodes are from the irrelevant nodes) positively correlates with its associated F_1 score. This suggests that, when the ground truth is inaccessible, one could look at the silhouette score (which essentially measures the confidence of the attribution) to reason about the relevance assignment, and in turn to the expected performance of the explainer. Specifically, if the silhouette score is low, we observe a mismatch between the k found by the unsupervised assignment and the k one would obtain by looking at the ground truth. This in turn means that the F_1 will likely be lower, which might depend on problems of the underlying model and/or explainer. In contrast, when the silhouette score is high (implying a confident, although not necessarily correct, relevance assignment), we observe that the k found by unsupervised relevance assignment is closer to the ground truth k, which in our experiments is associated to a high F_1 score. To the best of our knowledge, this is the first work to address the development of explanation metrics that can assess the quality of graph explainers in real-world applications. All the code and data used in this article is publicly available[1].

2 Preliminaries

In this section, we introduce the notation and discuss some background concepts needed to better understand our proposal.

Notation. A *graph* is a pair $G = (V, E)$ where $V = \{v_1, ..., v_{N_G}\}$ is a set of vertices or nodes of cardinality $N_G = |V|$ and $E \subseteq (V \times V) = \{(u, v) \mid u, v \in V\}$ is a set of edges. The topological structure of a graph is represented by an *adjacency matrix* $\mathbf{A} \in \mathbb{R}^{N_G \times N_G}$, with entries $a_{uv} = 1 \iff (u, v) \in E$ and 0

[1] https://github.com/Fonta1/unsupervised-node-relevance-attribution.

otherwise. Features for a node v are represented by vectors $\mathbf{x}_v \in \mathbb{R}^d$, $d \in \mathbb{N}$; the feature matrix $\mathbf{X} \in \mathbb{R}^{N_G \times d}$ is composed by all node feature vectors stacked row-wise. The set $\mathcal{N}_v = \{u \in V \mid (u, v) \in E\}$ containing the neighbors of a node v is called *neighborhood*. Throughout this work, we assume *undirected* graphs, meaning that their edges are unordered pairs, i.e. $(u, v) \in E \iff (v, u) \in E$.

2.1 Deep Graph Networks

We focus on graph classification tasks which aim to approximate a certain function that maps input graphs $G \in \mathcal{G}$ to a set of target labels $\mathcal{C}$, given point-wise estimates $\mathcal{D} = \{(G, y) \mid G \in \mathcal{G}, y \in \mathcal{C}\}$ of said target function. Deep Graph Networks [3] perform graph classification by learning a parametric function $f_{\boldsymbol{\Theta}}$ that maps a graph G to a vector of class probabilities $\hat{\mathbf{y}} = f_{\boldsymbol{\Theta}}(G)$ with parameters $\boldsymbol{\Theta}$. The parameters are learned by minimizing the cross-entropy loss function between model predictions $\hat{\mathbf{y}}$ and the targets $\mathbf{y}$, which are one-hot encoded versions of the labels $y \in \mathcal{C}$. More specifically, a DGN is a composite function:

$$f_{\boldsymbol{\Theta}}(G) = \left(f^{\text{trans}}_{\theta_1} \circ f^{\text{pool}}_{\theta_2} \circ f^{\text{out}}_{\theta_3}\right)(G), \tag{1}$$

where $f^{\text{trans}}_{\theta_1}$ performs an isomorphic transduction of the graph, transforming node features $\mathbf{X}$ into node embeddings $\mathbf{H} \in \mathbb{R}^{N_G \times d'}$; $f^{\text{pool}}_{\theta_2}$ aggregates the node embeddings $\mathbf{H}$ into a single graph embedding, and $f^{\text{out}}_{\theta_3}$ is a readout function that maps the graph embedding to the prediction vector $\hat{\mathbf{y}}$. The isomorphic transduction is performed with *message passing*, an iterative computation which updates the node embeddings $\mathbf{h}_v$ at iteration l according to the graph topology:

$$\mathbf{h}_v^{l+1} = \text{Upd}\left(\mathbf{h}_v^l,\ \text{Agg}(\{\text{Msg}(\mathbf{h}_v^l,\ \mathbf{h}_u^l) \mid u \in \mathcal{N}_v\})\right). \tag{2}$$

Basically, the Msg function computes a pairwise message between target node and each of its neighbors; Agg combines all the messages of the neighborhood in a permutation-invariant fashion; and Upd produces the updated node embedding by combining the current node embedding and the aggregated messages. Each sub-step can be parameterized, and their weights adjusted with gradient descent to approximate the downstream target function. In convolutional architectures [3], iterations $1 \leq l \leq L$ of the message-passing paradigm correspond to the number of layers of the model architecture, determining the size of the receptive field of each node embedding [7]. This work uses the Graph Isomorphism Network (GIN) [14] convolutional DGN as backbone, which is widely used for graph classification. GIN performs message passing as follows:

$$\mathbf{H}^{l+1} = \text{MLP}\left((1 + \epsilon^{l+1})\mathbf{H}^l + \mathbf{A}\mathbf{H}^l\right) \tag{3}$$

where by convention $\mathbf{H}^1 = \mathbf{X}$, MLP is a multilayer perceptron, and ϵ is a learnable or fixed parameter that scales the importance of the previous node embedding. In terms of message-passing operations, $\mathbf{H}$ represents the messages, the aggregation is a sum, and the update function adds the aggregated information to an ϵ-modified version of the initial embedding before applying the MLP.

2.2 XAI Methods for Graphs

For the purposes of this work, a *graph explainer* is a local post-hoc XAI technique that yields an *importance score* to each node in a graph, quantifying their contribution to the model outcome. More formally, it is a function $e(G, f_{\boldsymbol{\Theta}}) = \hat{\mathbf{t}}$ taking as input a graph G and a trained DGN $f_{\boldsymbol{\Theta}}$, and returning as output a real-valued explanation $\hat{\mathbf{t}} \in \mathbb{R}^{N_G}$ with one importance score for each node. Among the many methods that exist in the literature [5,16], one of the first and most used is GNNExplainer (GNNE) [15], which works by optimizing a soft mask corresponding to the subgraph that maximizes the mutual information between the output of the model given that subgraph and the original input graph. Integrated Gradients (IG) [12] computes the contributions of the input features to the output prediction relatively to a baseline point that encodes the complete absence of information. It is based on the principle that for every input and baseline that differ in one feature but have different predictions, the differing feature should be given a non-zero attribution. CAM [9] was initially developed for images and then adapted for graphs. It is based on the observation that the output logits of a DGN are the weighted sum of the contribution of each node. These contributions are the importance scores returned by the explainer.

2.3 Benchmarking XAI Methods for Graphs

Benchmarking an XAI attribution method requires a dataset $\mathcal{S} = \{(G, y_G, \mathbf{t}_G) \mid G \in \mathcal{G}, y \in \mathcal{C}\}$ where graphs G are associated to classification targets y as well as to *ground truth explanations* $\mathbf{t} \in \{0,1\}^{N_G}$. Basically, a ground truth explanation is a binary vector encoding the relevance (1) or irrelevance (0) of each node[2]. Then, the following steps are taken:

1. First, $\mathcal{S}$ is partitioned into a training set $\mathcal{S}_{\text{train}}$ and a test set $\mathcal{S}_{\text{test}}$, such that $\mathcal{S}_{\text{train}} \cup \mathcal{S}_{\text{test}} = \mathcal{S}$ and $\mathcal{S}_{\text{train}} \cap \mathcal{S}_{\text{test}} = \emptyset$. $\mathcal{S}_{\text{train}}$ is used to train a DGN $f_{\boldsymbol{\Theta}}$;
2. Graphs in the test set are processed by the explainer resulting in a set of explanations $\{\hat{\mathbf{t}}_G = e(G, f_{\boldsymbol{\Theta}}) \mid G \in \mathcal{S}_{\text{test}}\}$;
3. The quality of the explanations is assessed by comparing them sample-wise to the ground truth through a metric $\mathcal{M}$ and averaging the result:

$$\frac{1}{|\mathcal{S}_{\text{test}}|} \sum_{G \in \mathcal{S}_{\text{test}}} \mathcal{M}(\psi^*(\hat{\mathbf{t}}_G), \mathbf{t}_G), \tag{4}$$

where $\psi^* : \mathbb{R}^{N_G} \to \{0,1\}^{N_G}$ is an *attribution strategy* that performs relevance assignment, i.e., it transforms the importance scores into relevance (1) or non-relevance (0) indicators (for example by hard thresholding), and $\mathcal{M}$ is any standard evaluation metric such as accuracy. When the ground truth is available, ψ^* considers the top-k importance scores as relevant, with k being the *explanation size*, i.e., the number of relevant nodes in the ground truth $\mathbf{t}$. Notice that simply

[2] Although explanations can be defined for edges and node features as well, in this work we restrict our attention only to nodes.

knowing the explanation size does not necessarily imply a good performance, since the explainer might still assign the node importances incorrectly. For completeness, we mention that benchmarking is possible with soft attribution by employing the plausibility metric [6,13]. However, it is not considered in this study since we are interested in giving unique and "hard" explanations, which can be evaluated with standard metrics and are easier to interpret on graphs.

2.4 Assessing XAI Methods for Graphs in Real-World Contexts

Once the explainability focus moves from a benchmarking context to a real-world application, the framework above becomes unusable since it relies on the availability of a ground truth. As such, one has to resort to unsupervised metrics to assess the quality of the explanations. To our knowledge, the only unsupervised metrics available are Fidelity$^+$ and Fidelity$^-$ [2,6,10,16], which are defined as:

$$\text{Fidelity}^+ = \frac{1}{|\mathcal{S}_{\text{test}}|} \sum_{G \in \mathcal{S}_{\text{test}}} \left(f_{\boldsymbol{\Theta}}(G) - f_{\boldsymbol{\Theta}}(G^{1-\psi^*(\hat{\mathbf{t}})}) \right) \tag{5}$$

$$\text{Fidelity}^- = \frac{1}{|\mathcal{S}_{\text{test}}|} \sum_{G \in \mathcal{S}_{\text{test}}} \left(f_{\boldsymbol{\Theta}}(G) - f_{\boldsymbol{\Theta}}(G^{\psi^*(\hat{\mathbf{t}})}) \right). \tag{6}$$

In practice, Fidelity$^+$ averages, across the whole dataset, the differences between the prediction of the original graph to the prediction of the graph where the relevant subgraph has been masked out (in the formula, $1 - \psi(\hat{\mathbf{t}})$ is the complementary mask of $\psi(\hat{\mathbf{t}})$). Fidelity$^-$ is computed analogously, but measures the difference with the prediction on the graph where the irrelevant subgraph has been masked out. Nonetheless, a major drawback of Fidelity is that the masked graphs may fall outside the distribution of the input data, making the model prediction (and as a consequence, the metric itself) unreliable.

3 Methods

The problem we try to overcome in this work is that the standard benchmarking setup described in Sect. 2.3 cannot be applied in real-world contexts. This is due to the fact that the strategy ψ^* requires the ground truth $\mathbf{t}$ to assign the relevance, which is usually unavailable. Therefore, in this section, we propose two *unsupervised* attribution strategies for graph explainers, which can be used to assign hard relevance when the ground truth is not accessible to the user. We call this scenario the "realistic setting", as opposed to the "benchmark setting" where the ground truth is known or recoverable and ψ^* can be used.

3.1 Unsupervised Attribution Strategies for the Realistic Setting

Our objective is to replace ψ^* with unsupervised attribution strategies having the same signature but able to assign relevance without depending on $\mathbf{t}$. Ideally, we would like them to possess the following qualities:

– they should always be applicable in the realistic scenario;
– they should leverage the importance scores to infer the separation.

Our first proposal builds and expands existing work for our specific use case. Specifically, it exploits the idea that a good explanation should be both necessary (Fidelity$^+$) and sufficient (Fidelity$^-$) for the model to generate its outcome. Based on these criteria, we implement a **fidelity-based** attribution strategy as:

$$\psi_F(\hat{\mathbf{t}}) = \underset{\psi(\hat{\mathbf{t}})\in\mathcal{T}}{\arg\max}\, f_{\boldsymbol{\Theta}}(G^{\psi(\hat{\mathbf{t}})}) - f_{\boldsymbol{\Theta}}(G^{1-\psi(\hat{\mathbf{t}})}), \tag{7}$$

where $\mathcal{T}$ is the set of all possible relevance assignments of $\hat{\mathbf{t}}$. Basically, we search, among a large number of possible relevance assignments, the one that maximizes the difference between the two fidelity metrics applied to the importances.

Our second proposal embodies the concept that a good explanation should set apart relevant from irrelevant nodes by a non-trivial margin. Inspired by this, we define a **clustering-based** attribution strategy as follows:

$$\psi_C(\hat{\mathbf{t}}) = \begin{cases} 1 & \text{if} \quad \|\hat{t} - \mu_{\mathcal{R}}\| < \|\hat{t} - \mu_{\bar{\mathcal{R}}}\| \\ 0 & \text{otherwise} \end{cases} \quad \forall \hat{t} \in \hat{\mathbf{t}}, \tag{8}$$

where $\hat{t}$ is the generic component of vector $\hat{\mathbf{t}}$, $\mu_{\mathcal{R}}$ (resp. $\mu_{\bar{\mathcal{R}}}$) are the relevant (resp. irrelevant) cluster centroids found with the K-means algorithm (where $K = 2$), and $\|.\|$ is the L-2 norm. Basically, we first partition the importance scores with K-means; then, we mark as relevant all the nodes assigned to the cluster that contains the highest importance score.

3.2 A Measure of Quality for the Relevance Assignment

In a realistic setting, given the absence of the explanation ground truth, it is impossible to assess an attribution method performance by computing the F1 score. However, we can evaluate the quality of the relevance assignment by applying clustering metrics, i.e., we can measure how well the unsupervised attribution strategy has separated the nodes deemed to be relevant from those deemed to be irrelevant. A natural choice for this task is the *Silhouette score*, which quantifies, for each sample assigned to a cluster, how similar it is to its cluster (intra-cluster distance) compared to other clusters (nearest cluster distance). The silhouette score of a sample is defined in the range $[-1, 1]$, where 1 indicates correct clustering, 0 indicates that the point belongs to overlapping clusters, and -1 indicates that the point has been assigned to the wrong cluster. We propose to evaluate the overall quality of the explanations returned by a graph explainer by globally averaging the node-wise Silhouette score as follows:

$$sil = \frac{1}{N_G\,|\mathcal{S}_{\text{test}}|} \sum_{G\in\mathcal{S}_{\text{test}}} \sum_{r\in\psi(\hat{\mathbf{t}}_G)} \frac{\text{NCD}(r) - \text{ICD}(r)}{\max\{\text{ICD}(r), \text{NCD}(r)\}}, \tag{9}$$

where ICD is the mean intra-cluster distance and NCD is the mean nearest-cluster distance, both computed on the node relevance r obtained by applying the unsupervised attribution ψ to the explanation $\hat{\mathbf{t}}$.

4 Experiments

This section describes the experimental design, starting from the research questions that motivated our analysis.

4.1 Objective

The experiments that we conducted have been designed to answer empirically to the following research questions:

Q1 *Can we get close to the benchmark setting performance by using unsupervised attribution strategies to assign relevance*?
To answer **Q1**, we measure the quality of the explanations obtained by the two proposed unsupervised attribution strategies (ψ_F and ψ_C) in the realistic setting and check whether they are close to the quality obtained by the optimal attribution strategy ψ^* in the benchmark setting. Specifically, we evaluate the proposed strategies by computing and relating the following metrics:

$$\mathrm{F}_1^b = \frac{1}{|\mathcal{S}_{\text{test}}|} \sum_{G \in \mathcal{S}_{\text{test}}} \mathrm{F}_1(\psi^*(\hat{\mathbf{t}})_G, \mathbf{t}_G) \tag{10}$$

$$\mathrm{F}_1^r = \frac{1}{|\mathcal{S}_{\text{test}}|} \sum_{G \in \mathcal{S}_{\text{test}}} \mathrm{F}_1(\psi(\hat{\mathbf{t}})_G, \mathbf{t}_G), \quad \psi \in \{\psi_C, \psi_F\}. \tag{11}$$

In other words, F_1^b is the benchmark F_1 score that the explainer would obtain by having access to the ground truth, while F_1^r is the realistic F_1 score obtained by applying the unsupervised attribution strategy; the smaller their difference is, the better. Notice that, with respect to other works, we opted for the F_1 metric since it is not affected by class imbalance and is maximized when the explainer finds all (high recall) and only (high precision) the correct explanations. If this question is answered positively, it entails that the unsupervised strategies are capable of finding a set of relevant nodes that is close to the best possible (i.e., the relevant nodes according to the ground truth).

Q2 *Can we proxy the realistic setting performance by measuring the quality of the unsupervised relevance assignment*?
To answer **Q2**, we evaluate the Pearson correlation between the silhouette scores of the relevance assignments and the corresponding F_1 produced by the two unsupervised attribution strategies. If this finding is confirmed empirically, it might indicate that the silhouette score provides a viable unsupervised estimation of the performance that the explainer would obtain, had it been exposed to the true explanation size.

All the experiments to answer the two research questions have been run independently for each dataset and for each graph explainer considered in this study.

Table 1. Characteristics of the datasets used in this study. Top 4 rows are synthetic datasets with Barabasi-Albert structure, bottom 3 are real-world molecular datasets. The column "Samples" contains the total number of samples in each dataset; "Features" contains a description of the node feature vectors (None stands for constant features for all nodes); "Explanation" is a description of the explanatory pattern. For synthetic datasets, the explanation is a graph motif, while for molecular datasets, the explanation is the presence of functional groups.

Dataset	Samples	Features	Explanation	Structure
BA2grid	2000	None	3×3 grid	Barabasi-Albert
BA2Motif	960	None	5-ring	Barabasi-Albert
HouseColors	1000	Node color	Green house	Barabasi-Albert
GridHouse	2000	None	3×3 grid + house	Barabasi-Albert
AlkaneCarbonyl	1125	Atom type	Alkane and C=O	Molecules
Mutagenicity	1768	Atom type	NH2, NO2	Molecules
Benzene	12000	Atom type	Benzene Ring	Molecules

4.2 Experimental Details

Datasets. We tested our methods on a comprehensive set of 7 datasets, 4 synthetic (BA2Grid, BA2Motif, HouseColors, GridHouse) and 3 real-world datasets of molecules (AlkaneCarbonyl, Mutagenicity, Benzene). Table 1 lists the characteristics of each of them. Note that for the real-world molecular datasets, the classification task is to decide the presence or absence of functional groups.

Model Selection and Assessment. We rigorously trained multiple DGNs on the datasets described above. Specifically, we partitioned each dataset into a training (80%) and test (20%) set stratifying the splits to respect the target distributions. Then, we performed a 5-fold CV model selection on the training set to tune the DGN hyperparameters. The model with the best validation accuracy was selected for assessment. Our backbone model is a GIN architecture with 32 units per layer, layer-wise global sum pooling, and a linear readout mapping the layer-wise embeddings to the output class. The number of layers was chosen from appropriate ranges (2 to 5 for synthetic datasets, 2 to 7 for molecular datasets). Training was performed for 500 epochs at most, with a learning rate of 0.001, weight decay of 0.0001, and early stopping patience of 30. All models used in this study have achieved at least 0.95 AUROC on the test set.

Explainers Assessment. We experimented on 3 explainers from the literature (CAM, IG, and GNNE) plus a random explainer (i.e., one that assigns importance scores at random) as a control. We manually implemented the CAM algorithm; for IG, we use the Captum implementation available from the PyG library[3]; similarly, we used the GNNE implementation in PyG. We tested CAM,

[3] Captum: https://captum.ai/, Pyg: https://www.pyg.org/.

GNNE, and IG with default configurations, as improving their performance through hyperparameter tuning would have led to an overestimation of their effectiveness, given the absence of ground truth in realistic settings. Explanations have been computed over all correctly classified positive test samples. The evaluation followed points 2 and 3 as described in Sect. 2.3.

5 Results

To answer to **Q1**, we first evaluate which unsupervised attribution strategy (ψ_F or ψ_C) better approximates the benchmark performance. For each graph $G \in \mathcal{S}_{\text{test}}$ and a given attribution strategy ψ, we compute the squared difference between $\text{F}_1(\psi^*(\hat{\mathbf{t}})_G, \mathbf{t}_G)$, i.e. the performance obtainable with the optimal relevance attribution, and $\text{F}_1(\psi(\hat{\mathbf{t}})_G, \mathbf{t}_G)$, i.e. the performance obtained with the unsupervised relevance attribution strategy. We then average the results on the whole test set, obtaining a Mean Squared Error (MSE) which quantifies how much the realistic performance is close to the benchmark performance. The results are presented in Table 2 for the different combinations of graph explainers and datasets, clearly showing that the clustering-based attribution strategy ψ_C provides the best approximation when coupled with the CAM explainer. More precisely, the lowest MSE is obtained by ψ_C + CAM in 5 out of 7 datasets, and the second-lowest one in the remaining two. Moreover, the small magnitude of the MSE achieved by ψ_C + CAM indicates good approximation (with an average of 0.015 ± 0.014 across datasets), ultimately answering to **Q1** positively.

Therefore, based on the empirical evidence, *we conclude that it is possible (and more so when adopting ψ_C + CAM) to approximate the benchmark performance with a completely unsupervised strategy for relevance attribution.* Further visual support to the finding is provided in Fig. 1, which shows the F_1^b (x-axis) and the F_1^r (y-axis) achieved by the proposed strategies across datasets and explainers. In the figure, the blue line shows the behavior of the best possible

Table 2. Mean Squared Error obtained by the graph explainer across datasets according to the proposed attribution strategies. Best results are in **boldface**, second-best results are underlined.

Dataset	Fildelity-based (ψ_F)				Clustering-based (ψ_C)			
	RExpl	CAM	IG	GNNE	RExpl	CAM	IG	GNNE
BA2grid	0.041	0.005	0.004	0.030	0.011	**0.003**	0.013	0.026
BA2Motif	0.021	0.072	0.077	0.045	0.027	**0.011**	**0.011**	0.052
HouseColors	0.027	0.197	0.155	0.159	0.020	**0.013**	0.029	0.067
GridHouse	0.064	**0.001**	0.009	0.055	0.011	0.002	0.040	0.012
AlkaneCarbonyl	0.050	0.032	0.027	0.083	0.038	**0.025**	0.045	0.039
Mutagenicity	0.046	0.092	0.078	0.083	**0.032**	0.045	0.170	0.074
Benzene	0.043	0.101	0.066	0.063	0.018	**0.011**	0.035	0.038

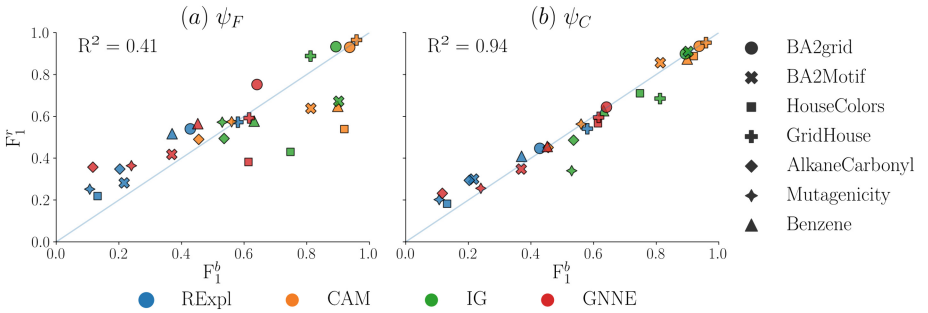

Fig. 1. Plot of the F_1^b scores against the F_1^r scores obtained with (a) the fidelity-based strategy ψ_F and (b) the clustering-based strategy ψ_C. Markers identify dataset, colors identify graph explainers. The blue diagonal shows the ideal behavior: $F_1^r = F_1^b$. The R^2 coefficients quantify the fit to the ideal case.(Color figure online)

Table 3. (a) Fidelity-based and clustering-based explanation sizes (obtained with CAM) compared to the optimal strategy ψ^*. (b) *sil* metric obtained across datasets by CAM coupled with the proposed attribution strategies, and correlation coefficient (CC) between the individual silhouette scores paired with the F_1 scores obtained in the realistic setting. Best results are in **boldface**.

(a)Explanation sizes

Dataset	ψ^*	ψ_F	ψ_C
BA2grid	$9_{\pm 0}$	$\mathbf{10}_{\pm 1}$	$\mathbf{10}_{\pm 1}$
BA2Motif	$5_{\pm 0}$	$10_{\pm 5}$	$\mathbf{7}_{\pm 2}$
HouseColors	$5_{\pm 0}$	$17_{\pm 11}$	$\mathbf{6}_{\pm 2}$
GridHouse	$14_{\pm 0}$	$\mathbf{15}_{\pm 1}$	$\mathbf{15}_{\pm 1}$
AlkaneCarbonyl	$4_{\pm 1}$	$\mathbf{9}_{\pm 4}$	$10_{\pm 5}$
Mutagenicity	$3_{\pm 2}$	$8_{\pm 5}$	$\mathbf{6}_{\pm 3}$
Benzene	$8_{\pm 3}$	$16_{\pm 4}$	$\mathbf{8}_{\pm 3}$

(b)*sil* and CC

Dataset	ψ_F		ψ_C	
	sil	CC	*sil*	CC
BA2grid	0.67	0.15	**0.69**	0.19
BA2Motif	0.53	0.76	**0.71**	0.70
HouseColors	0.53	0.88	**0.82**	0.54
GridHouse	0.71	0.32	**0.74**	0.41
AlkaneCarbonyl	0.51	-0.13	**0.63**	0.19
Mutagenicity	0.60	0.54	**0.76**	0.60
Benzene	0.26	0.75	**0.73**	0.45

strategy, which achieves $F_1^r = F_1^b$ for all datasets. The plot clearly highlights the superior ability of CAM to generate better-quality explanations when paired with ψ_C. As a further quantitative support, we measured the R^2 coefficients achieved by the two strategies as a measure of fitting F_1^b with F_1^r, showing a nearly optimal fit for the clustering-based strategy, and a significantly lower fit for the fidelity-based strategy. These results also highlight that the ψ_F strategy, even if based on the widely used Fidelity metric, leads to the generation of explanations whose F_1^r does not mimic F_1^b. This raises concerns about the limitations of Fidelity (i.e. the fact that it is based on potentially out-of-distribution predictions) and its reliability as a metric for assessing explanation quality. We conclude the analysis devoted to answer **Q1** by showing, in Table 3a, the optimal explanation size (obtainable by the ψ^* strategy, i.e. by "looking" at the

ground truth) compared to the explanation size retrieved by CAM + the proposed attribution strategies. Again, we observe that assigning relevance with ψ_C approximates the true explanation size better than ψ_F (in 6 out of 7 datasets).

To answer to **Q2**, we computed the *sil* metric presented in Sect. 3.2 for the two relevance assignment strategies using the CAM explainer. Then, we calculated the correlation between the individual silhouette scores paired with the individual F_1 scores obtained in the realistic setting. Here, the rationale is that if the average silhouette score can be used to proxy the performance in the realistic setting, the two should correlate positively. Table 3b shows the results: interestingly, we notice that ψ_C + CAM correlate positively in all the datasets used in this study. Moreover, by comparing Table 3a and 3b, we notice that the silhouette score is higher when the k found by the unsupervised relevance assignment closely matches the ground truth k. Therefore, based on this empirical evidence, we answer to **Q2** by stating that *one may use the silhouette score to get a sense of what the relevance attribution performance would be, had the ground truth been known.* This suggests that a practitioner could use the silhouette score to identify possible issues with the relevance attribution or the graph explainer; for example, a *sil* score close to 0 (indicating overlapping clusters) might be caused by an explainer that has assigned monotonically increasing and uniformly distributed importance scores, resulting in a close-to-random relevance assignment. This might also suggest appropriate counter-measures like choosing a different separation strategy or switching to a different explainer altogether.

6 Conclusion

This work considers the under-researched case of applying graph explainers to realistic scenarios, where standard assessment frameworks relying on the availability of a ground truth are not applicable. In particular, we have developed two unsupervised strategies to perform the task of relevance assignment, i.e. to transform the importance scores provided by an explainer to intuitive "yes-or-no" indicators of the relevance of each node in a graph. Through an assessment involving 3 different graph explainers and 7 datasets (3 of which consisting of real molecules), we have gathered evidence suggesting that assigning relevance with unsupervised methods is feasible, and leads to relevance assignments that resemble those obtainable when the ground truth is known. Furthermore, we proposed to look at the quality of the separation induced by the attribution strategy (through the silhouette score) as a tool to analyze the quality of the explanations given by the explainer. While the former contribution provides practitioners with a means to assign relevance in real-world use-cases, the latter provides a proxy to assess the quality of said assignment. In future works, we plan to extend this study to other explanation methods and relevance assignment strategies, as well as to test other unsupervised metrics that better correlate with the F_1 score. Moreover, we plan to identify what dataset characteristics make more appropriate the use of the silhouette score to proxy the relevance assignment quality.

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Explaining Clustering of Ecological Momentary Assessment Data Through Temporal and Feature Attention

Mandani Ntekouli[1(✉)], Gerasimos Spanakis[1], Lourens Waldorp[2], and Anne Roefs[3]

[1] Department of Advanced Computing Sciences, Maastricht University, Maastricht, The Netherlands
{m.ntekouli,jerry.spanakis}@maastrichtuniversity.nl
[2] Department of Psychological Methods, University of Amsterdam, Amsterdam, The Netherlands
L.J.Waldorp@uva.nl
[3] Faculty of Psychology and Neuroscience, Maastricht University, Maastricht, The Netherlands
a.roefs@maastrichtuniversity.nl

Abstract. In the field of psychopathology, Ecological Momentary Assessment (EMA) studies offer rich individual data on psychopathology-relevant variables (e.g., affect, behavior, etc.) in real-time . EMA data is collected dynamically, represented as complex multivariate time series (MTS). Such information is crucial for a better understanding of mental disorders at the individual- and group-level. More specifically, clustering individuals in EMA data facilitates uncovering and studying the commonalities as well as variations of groups in the population. Nevertheless, since clustering is an unsupervised task and true EMA grouping is not commonly available, the evaluation of clustering is quite challenging. An important aspect of evaluation is clustering explainability. Thus, this paper proposes an attention-based interpretable framework to identify the important time-points and variables that play primary roles in distinguishing between clusters. A key part of this study is to examine ways to analyze, summarize, and interpret the attention weights as well as evaluate the patterns underlying the important segments of the data that differentiate across clusters. To evaluate the proposed approach, an EMA dataset of 187 individuals grouped in 3 clusters is used for analyzing the derived attention-based importance attributes. More specifically, this analysis provides the distinct characteristics at the cluster-, feature- and individual level. Such clustering explanations could be beneficial for generalizing existing concepts of mental disorders, discovering new insights, and even enhancing our knowledge at an individual level.

Keywords: Clustering · Multivariate Timeseries · EMA · Explainability · Temporal Attention · Feature Attention

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1 Introduction

For quite some years now, Ecological Momentary Assessment (EMA) [2,24] has been a popular method in psychology research to collect data in daily life. During an EMA study, individual real-time data is collected over time, capturing various psychopathology-related behaviors, experiences, and symptoms in their natural environments [7]. For each individual, EMA data is then organized as a multivariate time series (MTS). Such a rich structure sets new pathways to employing advanced analytical techniques, enhancing our understanding of the complex patterns in psychopathology.

According to psychopathology theory, personalized models have been traditionally applied. A popular model for EMA data is the vector Autoregressive (VAR) model, where one or more time lags are used to predict variables at subsequent time-points. Because of heterogeneity between individuals, these models focusing on each individual separately allow for interventions and treatments that are tailored to their unique patterns of symptoms [20]. Although, personalized models can reveal quite distinct patterns, they may also reveal commonalities. Such commonalities can be quite valuable, giving insights into more general mechanisms that are valid for particular subgroups. It is, however, quite difficult to ascertain if there are such subgroups for which commonalities hold and also how to discover them.

In an EMA study, collecting data for multiple individuals at the same time could facilitate the understanding of mental disorders by studying the commonalities as well as variations present in the sample. An interesting approach for broadening EMA analysis involves enhancing models with data from more than one individual [15]. More specifically, group-based models, relying on individuals exhibiting similar temporal patterns of symptomatology or behavior, can offer potentially valuable insights for the whole sample [17]. Thus, uncovering homogeneous groups of people is of great importance. Such groups of individuals can be uncovered by clustering applied to EMA data [16]. Nevertheless, since clustering is an unsupervised task and true EMA grouping is not commonly available, the evaluation of the derived clustering labels is quite challenging. Because of the inherent variability of EMA data as well as the complexity of the real-world MTS, it is expected that clustering algorithms could produce quite different results leading to different groups. Thus, a critical point to be addressed is evaluating the clustering results.

Beyond the well-applied distance-based criteria capturing the quality of clusters, clustering interpretability is another important aspect of evaluation. Interpretability ensures that the patterns and common characteristics of the groups identified through clustering can be understood, explained and validated in the context of mental disorders. Thus, uncovering the meaningful patterns of each cluster within EMA data could provide insight into intra-individual psychopathological processes, their temporal patterns, and their inter-relationships among theoretically similar subtypes of disorders.

In this paper, to address the need for explanations on clustering results, the proposed methodology focuses on investigating interpretable deep-learning

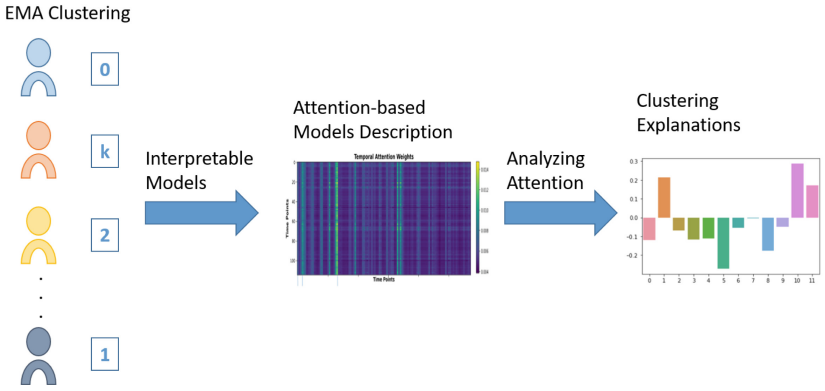


Fig. 1. An overview of our methodological approach focusing on applying attention-based interpretable models to provide clustering explanations.

mechanisms capable of handling MTS data, particularly within the domain of psychopathology. Our approach utilizes advanced interpretative deep-learning models, specifically attention-based mechanisms [25], to understand the complexities inherent in MTS data without relying on prior data transformations. Therefore, this approach ensures that our interpretations rely on the actual data dynamics rather than other transformations, providing a clearer and more accurate comprehension of MTS in psychopathology. As depicted in Fig. 1, our methodology employs a multi-aspect framework that integrates both temporal and feature-level attention. Therefore, it is designed to provide explanations by identifying the important time-points and variables that play primary roles in the domain of psychopathology [8,22]. While the proposed framework for extracting temporal and feature-level attention is fully described, a significant part of this paper is to examine ways to analyze, summarize, and interpret the attention weights as well as validate the patterns underlying the important segments of the data that differentiate between clusters. More specifically, this includes the significant and distinct characteristics in cluster-, feature- and individual-level and their evaluation. Thus, such clustering explanations could be proven beneficial for generalizing the existing concepts, uncovering new insights into psychopathology and network theory, and even enhancing our knowledge at an individual level. Furthermore, central to our framework is its independence from any specific clustering algorithms. This independence introduces a new theoretical perspective on evaluating the robustness of an examined clustering result, allowing for a more objective comparison and assessment of clustering algorithms. Overall, attention-derived interpretability, beyond contributing to a better understanding of the underlying data structures in psychopathology, could be theoretically used to benchmark clustering effectiveness.

2 Related Work

The field of time-series (TS) clustering research has attracted significant attention, with a focus on clustering explanations being introduced the recent years. First, related work on time-series clustering explanations is presented. This involves extracting meaningful representation for clusters' descriptions as well as the classical explanation methods of the field of Explainable Artificial intelligence (XAI).

2.1 Clusters' Descriptive Representation

In the context of understanding clustering results, the representation of clusters is crucial in unveiling meaningful insights into underlying constructs (connections in variables) and further facilitating decision-making. This is usually achieved by examining and visualizing all elements of a cluster. For instance, in temporal data, distinct trend lines can be observed by overlying the time series data belonging to each cluster. Summary statistics, such as mean values and variance, can be also calculated to describe each cluster. However, the difficulty of distinguishing between them increases significantly when using many clusters or high-volume datasets with multiple variables.

2.2 Explanations on TS Clustering

Beyond statistically analyzing and exploring the structures of each cluster, the majority of the work focuses on extracting the factors (e.g. important variables) that influence the cluster assignments by applying interpretable models. The methodologies employed for explaining time-series clustering vary considerably, despite the low number of publication outputs.

Following the approach of transforming the data to different time-series representations, the work of [4] applies clustering on various interpretable extracted features, including intra-signal (or within feature) and inter-signal (across different features) characteristics, such as variance and correlation, respectively. After data pre-processing and dimensionality reduction techniques, a selection of features is retained for clustering, showing the high importance of inter-signal features as always preserved in clustering. The number of features is then used as a measure of interpretability, meaning that fewer features facilitate clusters' comprehension.

Moreover, other work, like [10,18] was inspired by the general trend of training and explaining classification models to predict the derived cluster labels. On the one hand, according to [18], local interpretability methods, such as Local Interpretable Model-Agnostic Explanations (LIME), SHapley Additive exPlanations (SHAP) as well as Gradient-weighted Class Activation Mapping, are used with different classification models (e.g. XGBoost) and trained again on statistically-extracted temporal features. Examples of such features were autocorrelation and median difference. On the other hand, the work of [10] proposed a different approach to holistically perform clustering and provide explanations

through training a decision tree. In practice, to achieve this, two different data sources were used, time-series and static or baseline data (such as demographic data), for clustering and interpretation, respectively. After applying clustering on TS, the cluster labels accompanied by the associated static data were used to train an interpretable decision tree model. Therefore, it aims to optimize both objectives, clustering and interpretation, at the same time.

According to the studies above, it becomes evident that existing methodologies for providing explanations rely on the applied transformations of the original data. This means that explanations also refer to the transformed feature space rather than the raw features raising some questions regarding the actual interpretability and transparency of the explanations. Therefore, beyond the limited work on the topic, there is a need for further exploration of the level of the original data.

3 Review on Challenges of Explaining MTS Data

Due to the inherent challenges of the topic and limited available published work, the current paper was also inspired by the literature focusing on using interpretable classification models for explaining an output vector, where the classes are the distinct cluster labels. In our setting, that output can be the clustering-derived result. Thus, an additional review of the challenges of applying interpretable classification models on MTS is necessary.

3.1 Clustering Explanations

Although it is acknowledged that the interpretability of clustering is of great importance in uncovering meaningful insights about data structures, limited work has been conducted in parallel with developing clustering algorithms. All the well-known clustering methods were designed to group data, mainly considering various objective functions and dealing with different data types, but without taking into account any interpretability aspects of clustering. Therefore, when clustering results need to be explained, post-processing steps are commonly used, involving an additional classification model trying to predict and explain the clustering labels. For instance, in [1,14], interpretable threshold decision trees having k leaf nodes are applied to explain the labels of k-means or k-medoids. Therefore, in most cases, the problem of clustering interpretability can be formulated as classification interpretability. This is a quite more studied issue, with a goal to explain the cluster labels as classification targets.

Regarding classification explainability, beyond using inherently interpretable models, such as linear or decision tree models, post-hoc explainability methods are also commonly explored. These fall into two main categories, model-specific and model-agnostic methods. Model-specific techniques correspond to particular groups of models, such as extracting feature importance from tree-based models and tree ensembles or utilizing layer-specific integrated gradients for deep-learning models [23]. On the other hand, model-agnostic explanation methods

are applicable on top of various models regardless of their architecture. More specifically, these include the widely-applied LIME focusing on generating local and instance-specific explanations, as developed by [19], and SHAP for feature-based explanations developed by [12].

However, when dealing with MTS clustering, challenges arise in two aspects of such formulation. First, the aforementioned XAI techniques of supervised learning models typically focus on images, text, and tabular data, limiting their application to time-series data. Second, it is not straightforward how to input time-series data into the classical classification models, without discarding their temporal nature. Therefore, adjustments should be made in both parts to better explain the dynamics of MTS data.

Adapting XAI Methods to MTS. As already discussed, the application of the aforementioned post-hoc XAI techniques is limited when time-series data is involved. Especially, in MTS, finding meaningful constructs in high-dimensional information is not trivial. To deal with such data, segmentation techniques were mostly needed, splitting and analyzing data into valuable sub-sequences.

More specifically, LIME and SHAP extensions were necessary to be further introduced, by leveraging the strong foundations of these methods adjusted to address such directions. For instance, TS-MULE is a LIME-based method not limited to uniform segmentation using static windows, but incorporating multiple advanced segmentation algorithms, such as matrix profile and SAX [21]. This way, it is more likely that meaningful subsequences are uncovered, leading to motifs and local trends that are easily interpretable. Such methods can be also adapted to interpret forecasting output, instead of only dealing with the classical classification targets.

In the case of SHAP, multiple extensions have been proposed and adjusted to a time-series setting. An example is an extension of KernelSHAP, adapted to explain time-series models, such as AR, ARIMA, VAR, VARMAX [26]. This method focuses on computing feature importance values for time series data. Another is the TimeSHAP method, aiming to explain more complex RNN-based models [3]. TimeSHAP provides explanations on multiple levels, by computing feature-, timestep-, and cell-level attributions.

Adapting MTS as an Input to Classification Models. Another challenge of an MTS classification task arises from the complexity of time-series, which is not straightforwardly input to a classical machine learning model. The problem is the complex nature of time-series that makes it deviate from the conventional feature-vector representation. In the context of MTS, data is defined in a multi-dimensional feature space and characterized by special connections between the instances (time-points) as well as features. Thus, potential approaches focus on how to input the time-series data into the classical classification models.

The most straightforward way is to neglect any temporal dependencies by assuming instance and feature independence. This discards any time-oriented

association and perceives the data directly as a vector input. Likewise, any classification model can be used on this dataset, but using the same output for all instances of the same time-series. Moreover, a well-applied approach addressing such data is to transform the complex-structured time-series data to a simple feature-vector representation. Such transformations can be achieved by using statistical-based or shapelet- and subsequence-based characteristics of the time-series [4]. After such transformation, a feature-vector representation is retained, which can be easily used in all existing classification models.

Alternatively, the necessary transformations can come internally through models incorporating data representations and prediction. Recently, neural networks capable of handling multivariate time-series data have been used increasingly. More specifically, recurrent neural networks- (RNN) -based models, such as long-short term memory (LSTM) and gated recurrent unit (GRU) represent the state-of-the-art group of models in tasks involving sequential decision-making. Besides these, attention mechanisms have been also introduced to sequential modeling with the ability to uncover and highlight the most important parts or periods of sequences [9,25]. In other words, attention-based models offer interpretability for the results by using the learned attention weights. Utilizing attention weights is considered a form of inherent interpretability that is not commonly observed in NNs.

According to the attributes of attention-based models, our two challenges, handling MTS data and MTS interpretability, seem to be overcome. Thus, our methodology is specifically designed to enhance the explainability of MTS clustering by employing an attention-based mechanism, strategically integrating both temporal and feature-level attention.

4 Framework for Clustering Explanations

In this section, first, the structure of the examined EMA data is described along with the problem statement of our work. Specifically, the goal of providing explanations on clustering results applied to EMA MTS data is presented in detail. Then, the focus will be on the proposed architecture that utilizes an interpretable attention-based framework and how this can eventually lead to clustering explanations.

4.1 EMA Data

In the current study, EMA was used as a dynamic approach to collect data on individuals' psychopathology-related behaviors, experiences, and symptoms in real-time and in their natural environments. More specifically, through digital questionnaires on their smartphones, participants were prompted to answer sets of questions. Most questions were answered on 7-point Likert scales; after normalization at an individual level, these data can be considered continuous. As all this data was collected multiple times throughout the day, EMA is typically

structured as MTS data. In this study, the dataset examined consists of real-world EMA pilot data, collected from students of Dutch Universities [13,20]. More specifically, 187 participants were asked to complete 8 questionnaires per day at fixed time intervals for 28 days. According to the domain experts, some variables (e.g., positive affect and negative affect) were averaged across relevant variables (e.g., happy, calm, etc. and sad, angry, etc., respectively), and some variables were not included in the analyses, either due to limited within-person variance or due to relatively less relevance.

The EMA dataset, consisting of the data of $N = 187$ individuals, can be denoted as $X = \{X_1, X_2, .., X_N\}$. For each individual i, X_i is comprised of $V = 12$ variables, each captured for a number of time-points, T_i. The total number of time-points T_i ranged from 112 to 224 and was different across individuals due to differences in compliance. The minimum compliance was set to 50%. Each individual EMA dataset X_i is represented by an MTS dataset, structured as $\{V_1, \ldots, V, T_1, \ldots, T_i\}$.

4.2 EMA Clustering

As already stated, an important task in EMA analysis is clustering individuals based on their MTS data. Clustering entails assigning individuals sharing similar patterns to the same groups, while assigning individuals with different patterns to different groups. This approach enables uncovering distinct profiles among people that might not be immediately apparent. Such information could also provide insights into mechanisms maintaining or causing psychopathology.

A variety of clustering methods can be applied to EMA MTS data, each highlighting different underlying assumptions about the structure of the data [16]. These include traditional clustering techniques, such as k-means and hierarchical clustering, but are adapted to handle the temporal and multivariate nature of EMA data. For instance, clustering methods by incorporating an appropriate distance metric, like dynamic time warping (DTW) and Global Alignment Kernel (gak), are considered capable of advancing the inherent temporal dimensions of the EMA data [5,6]. Alternatively, clustering on EMA data could be approached from a multivariate functional data perspective, by representing the data with smoothed curves. The use of smoothing techniques could possibly facilitate identifying the underlying trends, which might be obscured by noise in the original time series [11]. However, its applicability to EMA data should be explored more carefully.

4.3 Proposed Framework for Clustering Explanations

An overview of the proposed framework of providing explanations on EMA clustering is given in Fig. 2. All components of the framework are described as follows.

Input: EMA Data. The framework begins with the EMA data matrix X as input. X represents the personalized data of $N = 187$ individuals, consisting of $V = 12$ variables and T_i time-points. Due to the variability observed in missing observations of individuals, we implement a padding strategy to complete the examined EMA dataset so the dimension of X is $\{187, 12, T\}$. Thus, each individual's data is transformed and filled when necessary, so that all have the same T time-points for further analysis.

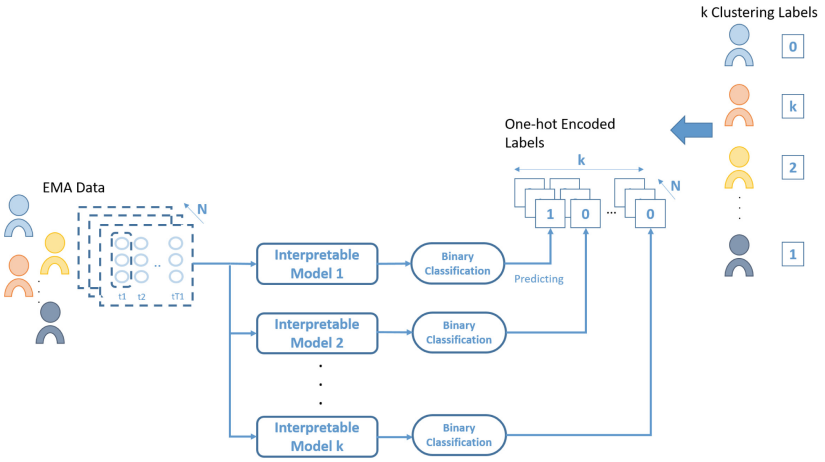


Fig. 2. An overview of the proposed framework for providing explanations on EMA clustering.

Output: Clustering Labels. In this study, we adopt an approach focusing on the explanation of clustering outcomes in EMA data without depending on a particular clustering method. That is, the approach relies on the fact that clustering is performed as a prior step, utilizing only the derived clustering labels in our framework to further provide explanations. An obvious advantage of this is the flexibility in the choice of clustering technique. Thus, as clustering is not an integrated component of the framework, it basically utilizes the clustering labels as the output to understand the reasoning behind the formation of each cluster.

***k* Interpretable Models.** Moving to the actual components of the proposed framework, in the case of k-clustering, it consists of k interpretable classification models. Each model focuses on predicting one cluster and distinguishing this from the rest of the clusters. For example, regarding the classification Model0, where the goal is to predict Cluster0, the output labels are 1 for individuals of Cluster0 and 0 for Cluster1 and Cluster2. By predicting a single cluster through an interpretable model, we could have access to the description of the model highlighting the special characteristics, dynamics, and features that differentiate that cluster from the rest. Then, fitting k models could facilitate explaining all k

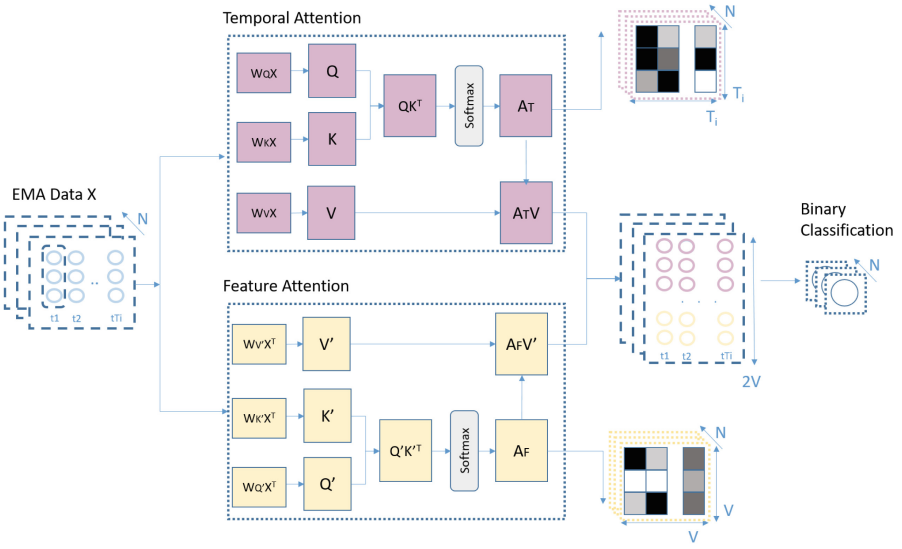


Fig. 3. The main components of each interpretable model.

derived clusters, which means exploring the important variables and time-points for each cluster.

Practically, although the input of each model is the whole EMA dataset, the difference of the k models comes into the output. This works by one-hot encoding the clustering labels of the pre-applied clustering and using a different output vector for each model. This encoding method transforms the categorical cluster labels into a binary matrix, ultimately forming k binary classification models. It should be noted that in the case of a 2-clustering, the framework consists then of only one interpretable model. As the clustering output is already binary, there is no point in fitting two models.

Cluster-Specific Binary Classification Model. As already discussed, each binary classification model aims at predicting the individuals of a single cluster over the rest of the clusters. The main components of each model used are shown in Fig. 3. Each binary classification model is interpretable mainly relying on self-attention mechanisms [25]. By nature, the attention mechanism focuses on the temporal domain of the data, identifying the most important parts of the data with respect to the prediction task. By keeping only the relevant parts of the data and minimizing or filtering out the effect of irrelevant ones, the input space is effectively sparsified. This is implemented by setting different attention weights on each time-points depending on the contribution to the output. Therefore, such weights could be beneficial in recognizing the important parts of the data for classifying an individual in one cluster [9].

In the current framework, to address the complexity of EMA data, 2-level attention is used in parallel, each focusing on different aspects of the data [8,22]. The first level of attention is dedicated to uncover the important parts in the temporal dimension, while the second one is to the important features. The value of analyzing data at the feature level is particularly evident when it comes to interpretation, as it is inherently more insightful to offer explanations based on specific features.

The description of each attention-based mechanism is as follows:

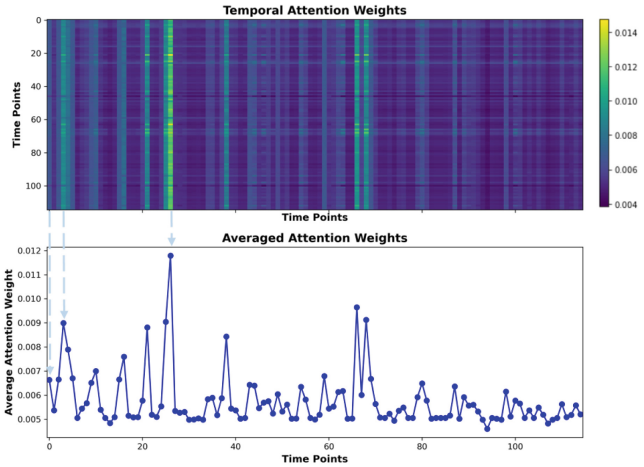


Fig. 4. The averaging process of the full Temporal Attention matrix A_T to $A_{T_{av}}$

– **Temporal Attention**
As shown in Fig. 3, the calculations of the Temporal Attention follow the procedure of a Scaled Dot-Product Attention self-attention block [25]. First, the 3 main components of the attention block, matrices query Q, key K, and value V, are calculated by a linear transformation of the input X data. Then, the product of Q and K is the attention matrix, which after a Softmax normalization produces the actual matrix of temporal attention A_T. For each individual i, the dimensions of A_T is $T_i \times T_i$. Subsequently, the attention matrix is multiplied by the matrix V (linearly transformed input X) to yield the context vector. In other words, the context vector is the weighted input based on the learned attention weights.
By learning the weights in A_T, the temporal attention component is designed to identify the most significant time points within the EMA data. These weights are different for each individual and each time-point resulting to a $T_i \times T_i$ matrix. This matrix captures the relative importance of each time point with respect to every other time point, offering rich insights into temporal dynamics. However, its size and complexity pose challenges for direct interpretation and visualization. Despite the detailed and informative structure

of the derived A_T, we employed a strategy of averaging over one dimension, while ensuring the other dimension is normalized. This approach averages the contributions of all time-points on each one, yielding only the average effect over time. This significantly simplifies the attention matrix, retaining only one dimension of size T_i or a time-series of T_i time-points.

– **Feature-Level Attention**
In parallel to temporal attention, the feature-level attention mechanism assesses the importance of each feature within the EMA data. As noticed in Fig. 3, the block of feature-level attention is the same as of the temporal attention. The only difference is that the initial Q', K' and V' matrices come after a linear transformation of the transpose of the input EMA data, X^T. As before, this component results in the attention weight matrix A_F and the context vector. While the latter has the same size as the input X, the feature-level attention is in the dimension of $V \times V$. This is designed to map the inter-feature relationships and their contributions to the model's predictions. Through such weighting, the framework can distinguish which variables play an important role in influencing the outcome, providing an additional layer of explanation that complements the temporal insights.
Regarding interpretability, it is evident that being in the feature domain makes it more straightforward to provide a deeper understanding of the role of influence for each variable. Additionally, the low feature dimensionality allows us to directly represent, visualize, and analyze the feature-level attention weights. This offers clear insights into which features are the most relevant and how these features influence each other within the context of the prediction task.

Clustering Explanations Through Attention Weights' Analysis. To provide clustering explanations, the produced averaged temporal attention weights and feature-level attention weights are further analyzed. The weights' analysis first facilitates the description of each model, aiming to differentiate each cluster, and then explaining the observed differences. In the following analysis, we adopt a multi-aspect approach to present our findings, covering a thorough exploration of the impact that attention mechanisms have at various but interconnected levels. These involve the cluster- and individual-level. From high- to low-level, each level zooms into different parts of the data, offering unique insights into the underlying relationships.

– **Cluster-level Analysis**
At the cluster level, the focus is on separately describing the individuals belonging to each cluster to get a clearer picture of the group-specific behaviors and attributes that distinguish one cluster from another. By examining the averages of attention weights across all individuals of each cluster, derived from the model describing each cluster, the aggregated behaviors and patterns of a cluster can be identified. At a high level, the average weights could uncover the special characteristics or strong effects that all these people have in common and possibly drove them to belong to the same cluster.

Although both temporal and feature-level attention weights are explored, getting access to the average effects across individuals based on the full temporal attention A_T, or even the averaged temporal attention $A_{T_{av}}$, is a bit challenging. As already discussed and shown in Fig. 4, each individual is described by a time-series, $A_{T_{av}}$, showing their important time-points. However, it is not meaningful to average over different time-series, because the important time-points differ across individuals. To address this, the correlation between the attention weights $A_{T_{av}}$ and each feature's time-series is calculated, potentially identifying which features consistently align with the attention temporal trend. Eventually, at this level of analysis, through both the temporal and feature-level attention, the importance of specific variables in leading to a particular cluster-output could be uncovered.
Subsequently, at a cluster-level, the patterns of inter-variable relationships or interactions dominant for each cluster are investigated. Based on the model's description through the attention weights, the relationship between data across individuals of one cluster can be also checked with respect to the acquired weights.

– **Individual-level Analysis**
At the individual level, the attention weights are examined separately for each individual, allowing for a personalized interpretation of the data. Without transforming or averaging the attention weights, the actual learned weights are analyzed along with the original feature space of each individual. Therefore, the focus is on understanding why or what was important to drive each individual to belong to a particular cluster. More specifically, regarding the time domain, the temporal attention weights facilitate uncovering what is happening underlying the time-points that are important for the prediction output. For instance, it is interesting to show which combinations of feature values get higher attention and which get lower.

5 Analysis and Results

As already mentioned, in our analysis, a real-world EMA dataset is used consisting of 187 individuals, 153 padded training time-points (70% of the total 224 time-points), and 12 distinct variables. After comparing different clustering methods based on different intrinsic evaluation measures, a 3-clustering result derived from a gak kernel k-means was chosen as a good clustering [16]. Then, the clustering labels derived from that particular method are further investigated to provide explanations regarding the formation of the clusters. Thus, the goal is to uncover what is different among clusters, meaning the important characteristics that drive each individual to belong to that specific cluster.

Taking as inputs the dataset X and the cluster labels, the proposed framework for describing and explaining clustering can be employed. According to the structure of the framework, for a 3-cluster grouping, a set of 3 interpretable models is trained on all individual EMA data, each aiming to predict one cluster over the rest. Therefore, the labels are one-hot encoded and each one-hot vector is the output of one model.

5.1 Performance Evaluation

Since the following analysis relies on the parameters derived from the 3 models, it is important to evaluate their effectiveness. The performance is then compared against the individual component of temporal attention and the baseline LSTM model. The assessment focuses on the ability of each approach to predict the cluster labels (provided by the pre-defined clustering) based on the EMA MTS data. After splitting the data into training (first 70% of individual time-points) and test (last 30%) sets, the number of correctly classified individuals (out of 187) is calculated. Each model was trained on the same training and test datasets, ensuring consistency in evaluation. The averaged results over the 3 models of each approach are presented in Table 1.

Table 1. Comparison of Models Performance

Model	Training Accuracy	Test Accuracy
Baseline LSTM	144/187	100/187
Temporal-Attention	159/187	100/187
Proposed Framework	187/187	105/187

This comparison shows the superior performance of the proposed framework over both the baseline LSTM model and the individual attention-based component. By effectively integrating the temporal and feature-level attention mechanisms, all individuals are identified in the correct cluster in the training set. Therefore, it is expected that the weights of a better-performing model could more accurately reflect the description of the underlying prediction task, which is the clustering. The rest of the analysis is conducted on the training set of all individual data.

5.2 Cluster-Level Explanations Through Temporal-Attention

To describe the characteristics of a cluster, the focus is on the model predicting that cluster label over the rest of the clusters. For example, for the first cluster, Cluster0, the analysis is conducted on the parameters of the first model, Model0. Regarding that model, the individuals of the two classes, which means belonging to one cluster over the rest of the clusters, are separately analyzed. Thus, by showing the average effects of the parameters of the individuals belonging to Cluster0, we could provide some description of Cluster0. A similar procedure holds when describing all different clusters. In other words, the parameters of Model1 are used for describing Cluster1 and the parameters of Model2 for Cluster2. Then, across all 3 models, the individuals belonging to the associated cluster, are separately analyzed.

As already discussed, to give an overview of the temporal effects on cluster-level, a correlation analysis is employed between the temporal attention weights

$A_{T_{av}}$ and the time-series of each feature. The average correlation scores across all individuals of each cluster are presented in Fig. 5.

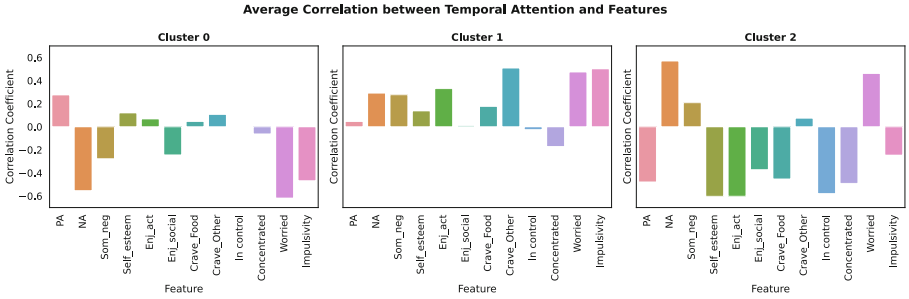


Fig. 5. Cluster-level average correlation effects between the temporal attention weights and the EMA features.

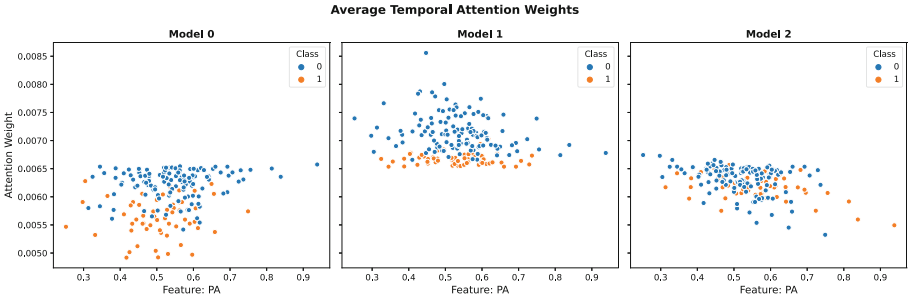
According to Fig. 5, the important effects of features in distinguishing each cluster from the rest are identified. It can easily be seen that the patterns of the correlations of EMA features and temporal attention weights are almost orthogonal for each of the clusters. Furthermore, particular high (low) correlations stand out. In Cluster0, "Negative Effect" (given as NA), "Worried" and "Impulsivity" have strong negative correlations with, meaning that the high values of these features get a lower attention weight. In Cluster1, almost all derived correlations are positive, with the highest being for "Craving_Other", "Worried" and "Impulsivity", whereas in Cluster2, almost all correlations are stronger apart from the feature Craving_Other. Although a cluster description has been uncovered by the analysis above, it still relies on the average (across all individuals of a cluster) effects. In other words, it is possible that some individuals' effects deviate from the averaged ones. Thus, explanations at the individual-level should be also further explored.

Despite the first findings on clusters' composition, the actual role of attention has still been unclear. For instance, we need to understand what it actually means to get a higher or lower attention score. Therefore, we could analyze the average attention weights of individuals belonging to each cluster (represented by Class1 in each model) or those not belonging to that cluster (represented by Class0). As before, across all models, the average attention weight of all 187 individuals is shown in Fig. 6 with respect to their average values of one feature. In each sub-figure, all 187 individuals are shown, represented by a point and colored according to their output class in each model. Regarding the coloring, these are different on the first and second row of the figure: while on the first row, the real class label (0 or 1) is depicted, on the second row the actual cluster label.

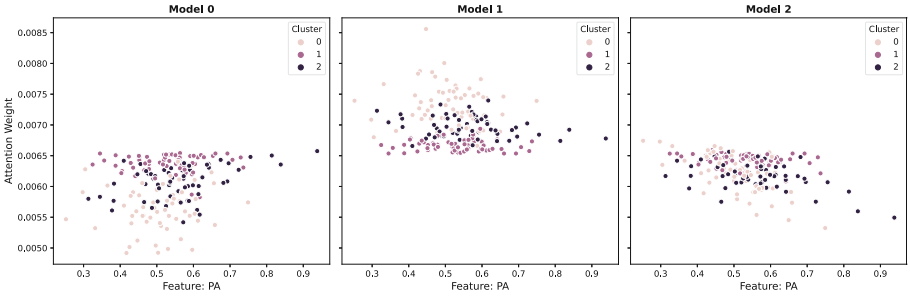
We notice that for Model0 and Model1, individuals belonging to Cluster0 and Cluster1, respectively, get lower (on average) attention scores than the rest. This could possibly reflect the fact that Class1 is always the minority class compared to Class0. Thus, each model gives more attention to the majority class.

Nevertheless, the findings of Model3 are not similarly clear. It is noticeable that the attention weights of the Cluster0 and Cluster2 are slightly mixed, but getting lower values than Cluster1. These unclear results should be further investigated as it may show that the third cluster may not be needed.

Additionally, no significant effect is observed on the feature-level. We can see that individuals with average values ranging from 0.3 to 0.9 can belong to all possible clusters. This is also apparent when plotting across any other feature, where the same patterns are found.



(a) Differentiation of attention weights across Class0 and Class1 of all 3 models.



(b) Differentiation of attention weights across the actual 3 clusters.

Fig. 6. Feature (Positive Affect) Relationship with Temporal Attention weights

In response to the overlapping attention weights of Cluster0 and Cluster2, the similarities across clusters should be analyzed. Since our clustering relies on the gak similarities, these are plotted for all individuals. In Fig. 7, the similarities of all individuals (colored by their true cluster) are depicted in Cluster1 and Cluster2. Practically, for each individual, the average similarity to all the individuals of each cluster is given. Because of a 2d plot, the similarities to Cluster0 are not shown.

As expected, it is interesting to see that there is a similarity between Cluster0 and Cluster2. Although individuals of Cluster0 have very low similarity to Cluster2, lower than 0.05, the majority of them have a similarity between 0.10 and 0.30 to Cluster2. This range is not far from the within-cluster similarity which only reaches the level of 0.40. In addition, we can observe individuals of

Cluster1 and Cluster2 exhibiting comparable levels of similarity to more than one cluster, suggesting a degree of ambiguity in their cluster membership. Such findings raise questions regarding the quality and robustness of the chosen clustering. Therefore, it is indicated that the current clustering method may not fully capture the underlying heterogeneity of the dataset.

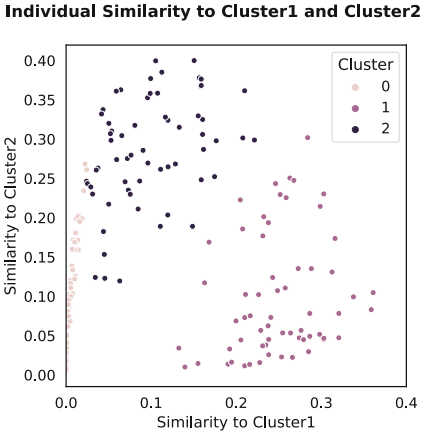


Fig. 7. Similarities of all individuals to Cluster1 and Cluster2.

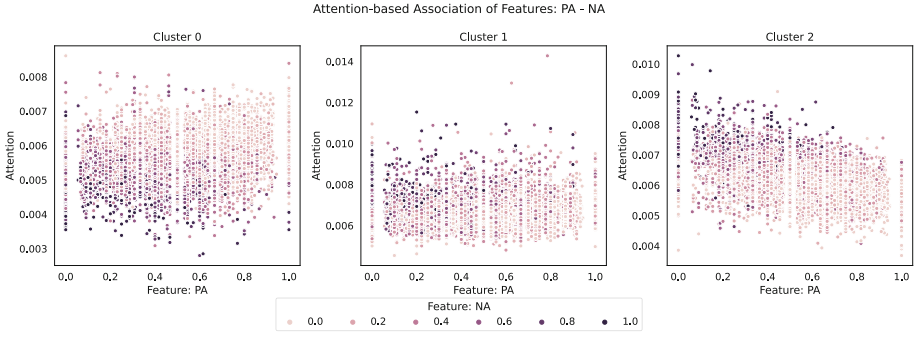


Fig. 8. Features Interaction between "Positive Affect" (PA) and "Negative Affect" (NA) with respect to the Temporal Attention.

In the context of cluster-level analysis, beyond the initial feature-related effects, we could further elaborate on the analysis using feature interactions. For example, the interaction between two features, "Positive Affect" (PA) and "Negative Affect" (NA), along with the acquired attention weight is shown in Fig. 8. For each cluster, each point represents a time-point of the individuals

belonging to that cluster. According to this figure, the different interaction patterns underlying each cluster can be identified. More specifically, for Cluster0, high NA values lead to low attention weights, whereas the opposite effect is seen for Cluster1 and Cluster2. Although for Cluster0 and Cluster1, no visible interactions were observed, a quite clear pattern is seen for Cluster2. The combination of low PA and high NA leads to high attention weights, whereas high PA and low NA lead to lower attention weights. By exploring all possible feature interactions, we have the opportunity to get more insights into the underlying structure of each cluster. Thus, the potential distinctions could point to differences in which variables are most relevant characterizing each cluster.

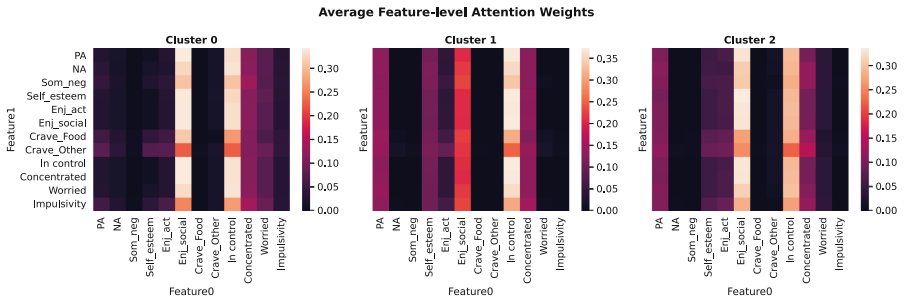


Fig. 9. Cluster-level average feature-level attention weights.

5.3 Cluster-Level Explanations Through Feature-Attention

Regarding the learned feature-level attention weights, the raw A_F weights are directly analyzed. Similarly to the previous section, to get some insights at a cluster-level, the average effects of all individuals within each cluster are aggregated. The average feature-level attention weights for each cluster are given in Fig. 9. The 12×12 heatmap presents the asymmetric inter-relations among features, specifically showing the relative contribution of Feature0 (x-axis) to Feature1 (y-axis).

The averaged attention weights can be similarly interpreted as the relative importance of various features in distinguishing one cluster from the rest. At first glance, it's noticeable that the same features emerge across all clusters, "Enjoying Social Activities" and "In Control", but to a different importance degree. For instance, this means that high values of "Enjoying Social Activities" are assigned to high attention weights in relation to any other variables. To illustrate this, an example of these interconnections is given in Fig. 10. More specifically, the association of the average "Enjoying Social Activities" value with the average "Positive Affect" and "Crave Food" across all individuals within each cluster is depicted in the first and second row, respectively. Each point represents an individual belonging to a particular cluster. As expected from

Fig. 9, we can identify the high attention contribution of high values of "Enjoying Social Activities" to the other features along with other interesting patterns. For instance, in Cluster2, high attention weights are assigned for low values of "Crave Food" and high values of "Enjoying Social Activities". By analyzing such feature interplay across all combinations, from the perspective of feature-level attention, deeper insights into the underlying dynamics can be uncovered. Thus, a more detailed examination of feature interactions could reveal some special characteristics and patterns that are important for distinguishing each cluster.

5.4 Individual-Level Explanations

While cluster-level analysis offers valuable insights into the commonalities within clusters, undoubtedly it smooths over individual differences. To more thoroughly understand the important patterns of cluster formation, it is essential to examine the data at the individual level as well.

Beyond unfolding the individual effects of the cluster-level analysis conducted before, it is important to investigate the underlying interactions of features in response to the attention weights learned by the model dedicated to the cluster each individual belongs to. The example of the first individual, Individual0, belonging to Cluster2, is used for the rest of the analysis. A detailed summary

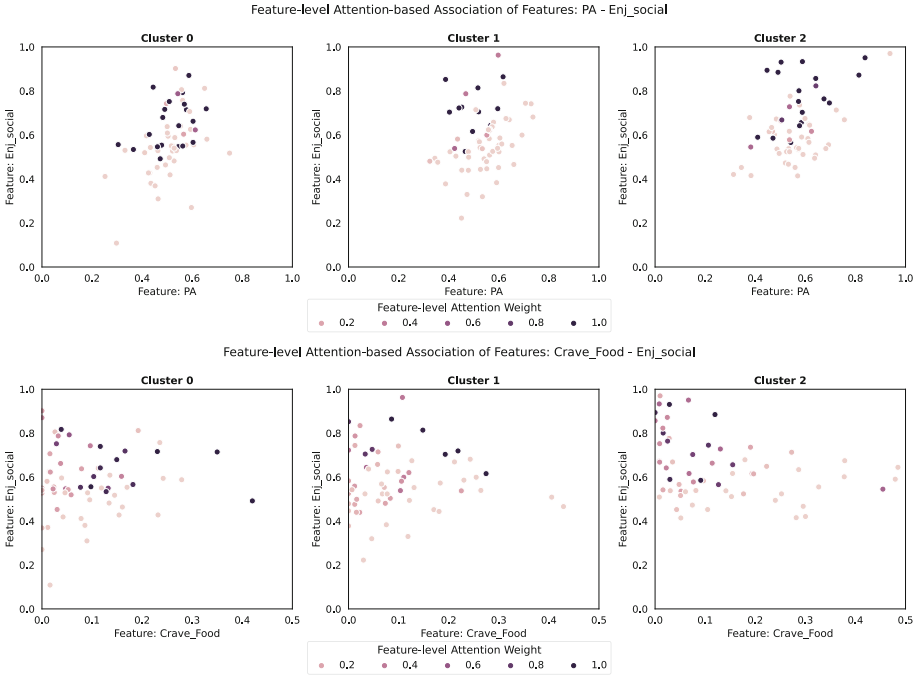


Fig. 10. Feature-level Attention: Unfolding the inter-connection of "Enjoying Social Activities" to two other features: "Positive Affect" and "Crave Food".

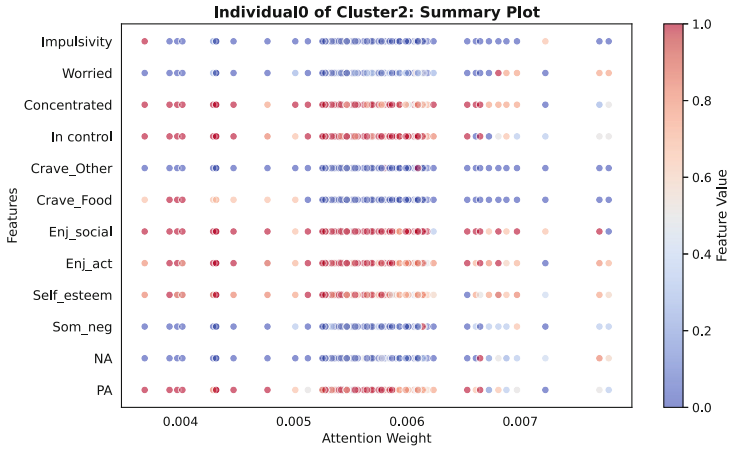


Fig. 11. The summary plot of feature interactions for an individual of Cluster2.

plot including all feature interactions and their attention weights is presented in Fig. 11. Particularly, for each feature on the y-axis, all the time-points of Individual0 are plotted, while ordered according to their attention weights, and colored by the value of each feature. Through this, we can have a thorough exploration of the combinations of feature values that lead to higher attention. According to the model, higher attention weights show the important characteristics for distinguishing individuals between clusters. Although it was indicated before that on average low attention was assigned to individuals classified as Class1, which corresponds to Cluster0, this pattern was not that dominant for Cluster2. Thus, further investigation of all individuals is necessary for a deeper understanding of the distinct feature dynamics, especially in the case of Cluster2.

To more clearly understand the role of attention on each individual, the differences in the weights learned by all models should be additionally studied. Specifically, for each individual, feature interactions could be compared against the attention weights derived from each model. This could facilitate uncovering how much each feature interplay influences each model's decisions for each individual. An example of an individual belonging to Cluster2 is shown in Fig. 12. By examining how the same feature interactions are colored based on the 3 models, some distinctions are apparent. For example, it is observed that the combination of high values in both "Positive Affect" and "Enjoying Social" leads to higher weights for Model0, whereas to lower ones for Model2. Thus, these patterns possibly reflect the impact that the specific time-points have on predicting Cluster0 and Cluster2, respectively. Also, after the previous indication that lower attention weights are linked to a particular Cluster, the lower attention of Model2 could highlight the most relevant information for predicting that individual as Cluster2. Therefore, such a comparison offers valuable insights into how different models prioritize and interpret the same set of features of an individual.

6 Discussion

In this paper, the under-studied problem of providing explanations on clustering results in the context of MTS data is explored. A novel interpretable framework is proposed and examined using a real-world EMA dataset. To address the complexity of EMA data, our framework offers interpretability by integrating 2 levels of attention mechanisms, both in the temporal and feature-level dimension. Through its multi-aspect design and analysis, this framework eventually facilitates a deeper understanding of clustering, providing interpretations of the important underlying patterns. Next, we focus on the role of the multi-aspect framework's design and analysis and the impact of prior applied clustering on interpretation and its validation.

Fig. 12. The attention weights of the interaction between "Positive Affect" and "Enjoying Social" derived from all 3 models.

6.1 The Role of the Multi-aspect Attention

Although attention mechanisms have typically been employed on the time dimension, the current approach of integrating multi-aspect attention mechanisms, focusing on both temporal and feature levels, showed that enhances the performance of the overall model in terms of accuracy. Based on Table 1, this integrated approach outperformed all baseline methodologies, by predicting more individuals in the correct clusters in both training and test sets. Its high performance is indicative of how informative the weights learned by each model are and how these could facilitate distinguishing individuals across clusters.

Moving to interpretability, the multi-aspect attention mechanism offers detailed information on the data patterns that each model considers as important. While temporal attention highlights the significance of particular time-points, such information remains meaningful on the individual level since each individual exhibits behaviors at different time-points. To enrich the learned information and our understanding on cluster-level, feature-level attention can

additionally identify the important interconnection among features. Thus, incorporating both temporal and feature-level attention enhances the interpretability of complex data like EMA, where the understanding of dynamic patterns is crucial.

6.2 The Role of the Multi-level Analysis

The analysis of all learned attention weights is conducted at 2 levels, cluster- and individual-level. The cluster-specific insights provide a deep investigation of the commonalities as well as distinctions across clusters. After aggregating the effects of all individuals in each cluster, we derived the most influential time-points as well as the underlying features and feature interactions based on a model's decision-making. Although some first cluster descriptions were derived, it should be noted that the average effects smooth over the real individual differences. Therefore, to better understand each cluster formation, it is essential to examine the data at the individual level. The individual-level analysis facilitates uncovering important feature interactions and patterns unique to each subject as well as how each person is reflected on the weights learned by models focusing on different clusters.

6.3 The Impact of Utilizing a Pre-defined Clustering Result

A key advantage of the framework is its algorithm-agnostic nature regarding the generation of clustering labels. By design, as clustering is not integrated into the framework, it shows great flexibility allowing every possible clustering technique to be potentially prior utilized. Thus, it could be used as an effective tool for evaluating the result given by any clustering method. Such a feature not only enhances the framework's utility but also broadens its applicability, allowing for the comparison and evaluation of clustering algorithms based on the quality and relevance of the insights they produce.

However, it is reasonable that the quality of the clustering results utilized in the framework plays an important role in the accuracy of the explanations. In other words, in the case that clustering labels are not robust, the proposed framework would again provide some explanations on the cluster- and individual-level importance effects, but without these being meaningful or accurate. This may be the case also for the chosen clustering of our EMA dataset. Since the real clusters are not available and the between-cluster similarities are found close to each other, the examined clustering may not be the optimal one. This can be caused by the clustering algorithm utilized or the complexity of the data. Nevertheless, even when using that clustering, the framework was capable of uncovering similarities in the influential data patterns of different clusters, suggesting that a 2-clustering result may be more possible. Therefore, regarding the evaluation of this framework, it is important for the clustering results of other clustering algorithms to be investigated. This necessity also comes since in human-centric fields, like EMA studies, developing simulation datasets or objective criteria to evaluate the quality of explanations is quite challenging.

Thus, explanations should further be assessed by domain experts on a case-by-case basis. Alternatively, explanations could be investigated using some different types of ground-truth information. This could include baseline data (such as demographic details or health-related information) collected before or during the study.

7 Conclusion

To summarize, our interpretable framework represents a significant advancement in providing explanations and evaluating an MTS clustering result. By thoroughly analyzing the attention-derived important time-points and feature interactions at both cluster- and individual-level, valuable insights into the patterns characterizing each cluster are identified. This is especially critical in fields like psychopathology, where a better understanding on clustered individuals could be beneficial for personalized interventions and mechanistic understanding.

Regarding future work, the goal is to augment personalized models by using data from similar individuals. Having evaluated various clustering results and uncovered the important data segments of an individual through the proposed framework, incorporating data from similar profiles could significantly improve the predictive performance of personalized models. Such methodology could produce more accurate predictions, enhancing the precision of personalized approaches.

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Graph Edits for Counterfactual Explanations: A Comparative Study

Angeliki Dimitriou(✉), Nikolaos Chaidos(✉), Maria Lymperaiou, and Giorgos Stamou

National Technical University of Athens, Athens, Greece
{angelikidim,nchaidos,marialymp}@ails.ece.ntua.gr, gstam@cs.ntua.gr

Abstract. Counterfactuals have been established as a popular explainability technique which leverages a set of minimal edits to alter the prediction of a classifier . When considering conceptual counterfactuals on images, the edits requested should correspond to salient concepts present in the input data. At the same time, conceptual distances are defined by knowledge graphs, ensuring the optimality of conceptual edits. In this work, we extend previous endeavors on *graph edits as counterfactual explanations* by conducting a comparative study which encompasses both supervised and unsupervised Graph Neural Network (GNN) approaches. To this end, we pose the following significant research question: should we represent input data as graphs, which is the optimal GNN approach in terms of performance and time efficiency to generate minimal and meaningful counterfactual explanations for black-box image classifiers?

Keywords: Counterfactual Explanations · Black-box Explanations · Scene Graphs · Graph Neural Networks · Graph Autoencoders

1 Introduction

In the era of large and complex neural models, ensuring trust between them and humans becomes a critical issue. Explainability literature suggests a variety of methods to probe neural models behaviors, either requiring access to their inner workings [11,31] or not [6,7,10]. There is also ample discussion regarding the nature of the features involved in a human-understandable explanation; for example, low-level features e.g. pixel-related characteristics (brightness, contrast) may be unable to provide a meaningful explanation to the end user, despite being informative for a neural model [29]. This observation is applicable in the counterfactual explanation scenario, suggesting that semantics are fundamentally essential to meaningful counterfactual explanations [4,23]. In the meanwhile, both high-level semantics as well as low-level features can be expressed in the same mathematical format (e.g. numerical vectors), allowing the transition to semantically rich explanation systems [6]. In the interest of explaining visual classification systems, and with respect to recent related work [2,6,7,10,11,31], we base our current study on *conceptual counterfactual explanations*.

A. Dimitriou and N. Chaidos—Contributed equally.

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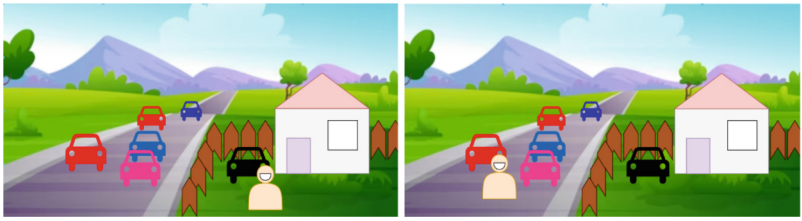

Fig. 1. Left - person in front of parked car (safe). Right - person in front of moving cars on the highway (unsafe). The relationship between the person, the car and the highway is critical for the transition to the counterfactual class.

With regard to the model accessibility an explainability system can have, most related counterfactual explanation methods belong to the white-box category [2,11,31], offering insights tailored to the model under investigation. However, in the emergence of powerful, yet proprietary models, such as ChatGPT [25] and GPT-4 [26] which are currently accessible only through APIs, black-box explanations resurface as more viable and universal solutions, able to approach a model by scrutinizing their outputs. Under these constraints, we stay within the *black-box* setting of conceptual counterfactuals [6,7,10].

We also acknowledge the importance of intra-concept roles, as associations between the same concepts via roles may result in different interpretations of situations (Fig. 1). Thus, we advocate image representation via scene graphs. To this end, we extend the framework of [7] to accommodate a larger spectrum of Graph Machine Learning algorithms to serve conceptual counterfactual explanations on scene images. We retain the brute-force calculation of Graph Edit Distance (GED) [30] as the ground truth counterfactual measure, under which the closest scene graph in terms of GED belonging to a different class is considered as the "gold" counterfactual graph. In order to overcome the computational complexity of GED calculation, we employ Graph Machine Learning algorithms, such as Graph Kernels, Graph Autoencoders (GAEs) and supervised Graph Neural Networks (GNNs), which are lightweight and significantly faster than the NP-hard deterministic GED calculation [35], even when optimizations are employed [14]. Graph edits serving as counterfactual explanations are instructed via concept distances in existing hierarchies, in our case being WordNet [22]. Merits of such explanations include but are not limited to actionability, a crucial quality of counterfactuals in several domains [21]. The outline of our approach is presented in Fig. 2. Overall, we explore the capabilities of Graph Machine Learning algorithms, contributing to the following:

- We prove quantitatively and qualitatively that both unsupervised (GAEs) and supervised (GNNs) methods can provide meaningful and accurate conceptual counterfactuals, viewing models under explanation as black-boxes.
- We demonstrate the trade-offs between unsupervised vs supervised approaches towards counterfactual explanations, and provide insights on the advantages of each method.

2 Related Work

Counterfactual Explanations have been established as an effective explanation method [33] should certain requirements, such as actionability and feasibility be met [28]. These principles are transferred in the field of counterfactual visual explanations, which aim to explain the behaviors of image classifiers. Most of these approaches leverage pixel-level edits, denoting regions that should be minimally changed [11,13,31] and often proceed to actually generate the targeted region by employing generative models [3,5,8,36].

Since semantics are inherently tied to human-interpretable counterfactuals [4], a diverging line of work suggests moving towards conceptual rather than pixel-level counterfactuals. The notion of explainable concepts was presented in [2], where fault-lines are utilized to drive concept edits. [1] produce conceptual counterfactuals to meaningfully explain model misclassifications. However, these approaches require some degree of access to the internals of the classifier. Minimal concept edits in a black-box fashion were proposed in [10], paving the way for knowledge-based model-agnostic visual counterfactuals. Some limitations were resolved in subsequent works, suggesting an indirect [6] or direct [7] incorporation of roles into concepts, while underlining the importance of correctly selecting data and representations to obtain proper explanations. The same idea is modified to explain hallucinations of image generation models rather than classifiers [20], highlighting the significance of universal black-box explainers.

3 Method

The focus of our current study centers on determining the minimal semantic alteration necessary for an image i to transition from being classified as A to B. Each image i in our dataset I is represented by a scene graph G_i, encapsulating semantic information, therefore forming an explanation dataset $\{i, G_i\}$, as previously outlined in [10]. Utilizing a reference image i_r and candidate scene graphs G_{i_c}, the GED is computed using an optimization approach to identify the most efficient path for semantic transformation between G_{i_r} and G_{i_c}, guided by predefined graph edit operations (Replacement, Deletion, Insertion). Semantic distances, drawn from the well-crafted WordNet [22] hierarchy, guide the choice of edits, ensuring meaningful transitions while penalizing irrelevant replacements. To accelerate GED calculations, we employ optimization techniques similar to [7], utilizing the Volgenant-Jonker (VJ) algorithm [14]. Finally, to obtain the counterfactual candidate $G_{i_{cc}}$ of the original scene graph G_{i_r}, we repeat the optimized GED calculation for each candidate G_{i_c}, where $c \in C = I \setminus \{i_r\}$, ensuring that the selected closest $G_{i_{cc}}$ belongs to class $B \neq A$, while G_{i_r} belongs to class A, formulated as:

$$i_{cc} = \underset{i_c \in C}{\arg\min} \left(GED\left(G_{i_r}, G_{i_c}\right)\right), \quad G_{i_r} \in A,\ G_{i_c} \in B \neq A \tag{1}$$

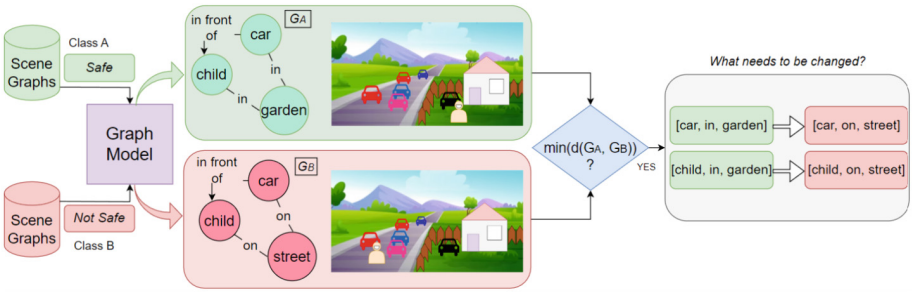

Fig. 2. Outline of our evaluation framework: Given scene graphs of classes $A \neq B$, a graph model embeds them in a low-dimensional space, allowing the retrieval of the closest graphs G_A, G_B from which we extract counterfactual edits.

3.1 The Importance of Graph Machine Learning

Considering a dataset with N graphs, calculating the counterfactual $G_{i_{cc}}$ for every G_{i_r} requires $O(N^2)$ GED operations, resulting in a prohibitive calculation time, despite using optimizations. In order to accelerate this process, we leverage Graph Machine Learning algorithms to circumvent the exhaustive calculation of N^2 GEDs, accepting an unavoidable performance sacrifice.

Graph Kernels is our first step towards testing graph matching [17]. Kernels in Machine Learning are responsible for mapping the given data in high-dimensional spaces, revealing complex patterns that cannot be observed in the original dimensionality of the input data, therefore allowing the application of linear algorithms to solve non-linear problems. Graph kernels quantify the degree of similarity between subgraphs in polynomial time by employing kernel functions, thus contributing to the overall structural similarity between two given graphs. In our experiments, we will compare different graph kernel approaches.

Graph Neural Networks (GNNs) in the supervised setting comprise architectures such as Graph Convolution Network (GCN) [15], Graph Attention Network (GAT) [32], and Graph Isomorphism Network (GIN) [34]. They are incorporated in a Siamese Network to learn similarity on input instances [19], demonstrating state-of-the-art results even when trained on a smaller subset of N^2/j labeled graph pairs (N is the number of graphs in the complete dataset, j a positive integer). During training, these GNN variants learn the relationships between counterfactual graph pairs utilizing ground truth samples calculated from Eq. 1. Embedding representations on new graphs are extracted using the trained GNNs, translating graph similarity cues to distances in the embedding space, therefore allowing retrieval of counterfactual graphs $G_{i_{cc}}$ for each G_{i_r}.

Graph Autoencoders (GAEs) [16] arise as faster and more versatile solutions by eliminating the need for labeled samples, even though their metric performance often ranks lower than GNNs. GAEs implement an encoder-decoder structure, with the encoder being tasked to produce embedding representations; these representations are useful for retrieval tasks, since similarity in the embed-

ding space corresponds to actual graph similarity. Thus, the need of GED calculations within the training process is completely avoided, dropping the computational complexity to $O(N)$ in comparison to the $O(N^2)$ operations required in the supervised case (N is the number of graphs). The encoder and decoder modules are constructed using any of the aforementioned GNN variants (same GNN modules for the encoder and the decoder when applicable).

In all cases, we retain the NP-hard GED scores for ground truth comparison to evaluate the performance of different models on the ground-truth rank.

4 Experiments

In this experimental section, we aim to provide a thorough comparison between different kernel/GAE/GNN architectures as components of the counterfactual retrieval process. We examine them both in terms of their adherence to ground truth rankings, as well as their ability to provide minimal edits regarding both quantity and cost. Throughout this analysis, we inherently contrast supervised and unsupervised GNNs, weighing the trade-offs between employing more efficient versus accurate models and quantifying the resultant incurred damages.

4.1 Experimental Setup

Data. In all our experiments, we select Visual Genome (VG) [18] as the experimental dataset, due to its abundance in real-life scenes accompanied by manually annotated scene graphs. In addition, VG offers an effortless connection with WordNet, as synset annotations of objects and roles are also provided by annotators. Since graph edges are important in our analysis, we regard two density-based dataset splits for training [7]: the first one, called VG-DENSE, includes graphs with many interconnections and fewer isolated nodes. The second one called VG-RANDOM does not impose any restrictions on scene graph density. Both splits contain $N = 500$ graphs. As VG does not contain ground truth scene labels we assign them using a pre-trained PLACES-365 classifier [37].

Graph Models. In our study, we employ various graph kernels [24] including Shortest Path (SP), Weisfeiler-Lehman (WL), Neighborhood Hash (NH), Random Walk (RW), and Graphlet Sampling (GS), implemented using the GraKel library. For supervised Graph Neural Network (GNN) models, we utilize single-layer siamese GCN, GAT, and GIN as described in [7], trained over 50 epochs with a batch size of 32 and Adam optimizer. Regarding Graph Autoencoders (GAEs), we implement vanilla GAE, Variational GAE (VGAE) [16], and Adversarially Regularized Variational Graph Autoencoder (ARVGA) [27], alongside a variation of Graph Feature Autoencoder (GFA) [12]. GAE and VGAE are both based on the generic auto-encoder structure, but differ in terms of loss function. ARVGA employs basic concepts from Generative Adversarial Networks to further boost the regularization of the latent embeddings produced by VGAE. Finally, GFA utilizes the Feature Decoder (whose goal is to predict the feature

Table 1. Counterfactual retrieval results with Supervised and Unsupervised GNN architectures. Underlined cells indicate best results per graph model (kernel, supervised, unsupervised). **Bold** denotes best results overall per dataset split.

Model	NDCG (binary)			Precision (P) (binary)			Precision (P)		
	@4	@2	@1	@4	@2	@1	@4	@2	@1
VG-DENSE									
WL kernel	0.287	0.198	0.138	0.186	0.128	0.076	0.164	0.125	0.076
SP kernel	0.334	0.251	0.194	0.232	0.174	0.122	0.154	0.145	0.122
RW kernel	0.0	0.0	0.0	0.0	0.0	0.0	0.001	0.0	0.0
NH kernel	0.282	0.192	0.131	0.124	0.086	0.05	0.088	0.073	0.05
GS kernel	0.259	0.166	0.103	0.018	0.008	0.008	0.022	0.007	0.008
Superv. GAT	0.342	0.26	0.204	0.422	0.288	0.164	0.312	0.25	0.164
Superv. GIN	0.339	0.256	0.2	0.344	0.242	0.158	0.252	0.192	0.158
Superv. GCN	**0.407**	**0.333**	**0.283**	**0.492**	**0.37**	**0.248**	**0.358**	**0.292**	**0.248**
GAE GAT	0.281	0.191	0.13	0.108	0.078	0.048	0.075	0.075	0.048
GAE GIN	0.281	0.191	0.13	0.132	0.098	0.052	0.082	0.078	0.052
GAE GCN	0.287	0.197	0.137	0.102	0.078	0.052	0.071	0.06	0.052
VGAE GAT	0.298	0.21	0.15	0.196	0.138	0.082	0.139	0.12	0.082
VGAE GIN	0.304	0.216	0.157	0.202	0.146	0.09	0.141	0.119	0.09
VGAE GCN	0.287	0.198	0.137	0.174	0.118	0.066	0.122	0.101	0.066
GFA GAT	0.302	0.214	0.155	0.182	0.14	0.086	0.122	0.109	0.086
GFA GIN	0.309	0.223	0.164	0.21	0.148	0.096	0.14	0.122	0.096
GFA GCN	0.163	0.165	0.158	0.289	0.2	0.139	0.12	0.104	0.068
ARVGA GAT	0.3	0.213	0.153	0.188	0.14	0.084	0.132	0.114	0.084
ARVGA GIN	0.317	0.232	0.174	0.218	0.164	0.106	0.135	0.124	0.106
ARVGA GCN	0.295	0.207	0.146	0.184	0.138	0.076	0.122	0.106	0.076
VG-RANDOM									
WL kernel	0.306	0.219	0.16	0.166	0.124	0.096	0.13	0.108	0.096
SP kernel	0.292	0.204	0.144	0.11	0.09	0.064	0.079	0.068	0.064
RW kernel	0.0	0.0	0.0	0.0	0.0	0.0	0.007	0.0	0.0
NH kernel	0.303	0.216	0.157	0.168	0.116	0.092	0.135	0.116	0.092
GS kernel	0.0	0.0	0.0	0.002	0.0	0.0	0.05	0.0	0.0
Superv. GAT	0.35	0.268	0.213	0.382	0.294	0.176	0.294	0.245	0.176
Superv. GIN	0.327	0.243	0.186	0.308	0.228	0.144	0.245	0.205	0.144
Superv. GCN	**0.369**	**0.29**	**0.236**	**0.424**	**0.3**	**0.2**	**0.295**	**0.246**	**0.2**
GAE GAT	0.294	0.205	0.146	0.132	0.104	0.07	0.092	0.085	0.07
GAE GIN	0.297	0.209	0.15	0.114	0.096	0.07	0.08	0.076	0.07
GAE GCN	0.291	0.203	0.143	0.114	0.1	0.066	0.079	0.078	0.066
VGAE GAT	0.305	0.218	0.159	0.16	0.122	0.09	0.103	0.1	0.09
VGAE GIN	0.304	0.217	0.158	0.152	0.118	0.088	0.11	0.101	0.088
VGAE GCN	0.303	0.216	0.157	0.156	0.122	0.088	0.101	0.089	0.088
GFA GAT	0.292	0.203	0.143	0.142	0.108	0.076	0.107	0.094	0.076
GFA GIN	0.298	0.211	0.151	0.142	0.11	0.082	0.11	0.093	0.082
GFA GCN	0.297	0.209	0.149	0.14	0.112	0.082	0.106	0.095	0.082
ARVGA GAT	0.305	0.218	0.159	0.156	0.13	0.092	0.111	0.103	0.092
ARVGA GIN	0.298	0.21	0.15	0.152	0.118	0.084	0.116	0.099	0.084
ARVGA GCN	0.306	0.22	0.161	0.154	0.13	0.094	0.113	0.097	0.094

Table 2. Average Number of node, edge and total edits, as well as Average Top-1 GED for counterfactual instances retrieved for VG-DENSE and VG-RANDOM.

Model	VG-DENSE				VG-RANDOM			
	Node ↓	Edge ↓	Total ↓	GED ↓	Node ↓	Edge ↓	Total ↓	GED ↓
WL kernel	5.24	11.668	16.908	130.146	13.44	11.946	25.386	192.824
SP kernel	4.964	12.078	17.042	129.334	14.144	12.386	26.53	208.096
RW kernel	9.162	20.006	29.168	217.684	17.814	22.36	40.174	385.726
NH kernel	5.348	11.264	16.612	141.18	12.568	11.714	24.282	190.376
GS kernel	6.444	16.748	23.192	178.434	17.216	15.344	32.56	272.744
Superv. GAT	5.278	10.892	16.17	108.636	12.81	11.978	24.788	**159.324**
Superv. GIN	5.106	10.766	15.872	116.174	12.792	11.394	24.186	173.72
Superv. GCN	4.942	**10.312**	**15.254**	**105.194**	12.392	11.84	24.232	159.356
GAE GAT	5.504	11.848	17.352	143.976	**11.952**	11.426	**23.378**	197.626
GAE GIN	5.4	12.002	17.402	143.182	12.554	11.738	24.292	201.638
GAE GCN	5.178	11.27	16.448	143.506	12.134	**11.384**	23.518	202.638
VGAE GAT	5.03	11.4	16.43	130.124	12.884	12.376	25.26	197.226
VGAE GIN	4.918	10.97	15.888	131.514	12.692	12.042	24.734	199.34
VGAE GCN	5.114	11.702	16.816	132.056	12.91	12.392	25.302	195.938
GFA GAT	5.07	11.564	16.634	131.582	12.988	12.412	25.4	198.78
GFA GIN	**4.856**	10.786	15.642	129.856	12.916	12.218	25.134	200.974
GFA GCN	5.03	11.432	16.462	132.046	12.696	12.338	25.034	201.038
ARVGA GAT	5.062	11.482	16.544	133.476	12.508	12.192	24.7	196.946
ARVGA GIN	4.894	10.912	15.806	128.944	12.234	11.794	24.028	198.654
ARVGA GCN	5.048	11.316	16.364	132.294	12.464	12.088	24.552	198.46

matrix X), alongside the original VGAE Inner-Product Decoder (whose goal is to predict the adjacency matrix A). All these models incorporate single-layer GCN, GAT, and GIN architectures in the encoder (and the decoder for GFA), employing AdamW optimizer with default parameters and a learning rate of 0.001, trained for 20 epochs with a batch size of 32.

GNNs and GAEs were implemented using PyTorch Geometric [9]. In all cases, cosine similarity was used as a measure for embedding similarity. For each scene graph in the dataset acting as the *query*, we retrieve the top k closest graphs with different PLACES-365 labels using a pretrained PLACES classifier with a ResNet50 backbone, resulting in slightly improved labels compared to [7]. We then evaluate whether the counterfactual graph, determined by the ground truth GED, is found within this ranked list of k items.

Evaluation. Information retrieval metrics are used for evaluation, specifically Precision (P@k) and Normalized Discounted Cumulative Gain (NDCG@k). P@k returns the percentage of the relevant items, when considering the first k items in the rank. We also employ the "binary" variant from [7], where only the top-1 GED item is relevant, in line with the counterfactual logic. Binary NDCG@k compares the rank of the top-1 counterfactual returned by an algorithm to the

ideal ground truth rank obtained by GED. While NDCG@k lacks direct interpretability, it's widely used for comparing retrieval system performance. Our evaluation focuses on $k = 1, 2, 4$ ranks, once again emphasizing proximity to the top-1 result. Additional metrics assess the proposed edits' minimality by GNNs, including average node, edge, and overall edits, alongside average GED of counterfactuals. Lower values indicate better model performance, with these metrics considered relative to each other.

4.2 Quantitative Results

In Table 1, we present ranking results regarding counterfactual retrieval using all proposed graph models: graph kernels, supervised GNNs and unsupervised GAEs. For both dataset subsets, all supervised models were trained using 70K scene graph pairs extracted from $N = 500$ graphs, while all unsupervised models were trained on the same 500 graphs (without pairing them).

Kernels reveal distinct performance patterns, with some closely rivaling unsupervised autoencoders. Notably, in VG-DENSE, SP and WL kernels excel, while in VG-RANDOM, WL and NH show superior performance. However, SP and NH metrics falter in VG-RANDOM and VG-DENSE, respectively, underscoring the significance of edge density. Conversely, kernels like RW consistently fail to retrieve counterfactual graphs, indicating their limitations in capturing graph structure. Supervised GNN models, particularly GCN, outperform kernels in ranking metrics, benefiting from optimized learning on ground truth GED. Unsupervised autoencoders exhibit more variability, with ARVGA GIN excelling in VG-DENSE and VG-RANDOM splits, underlining the importance of adversarial regularization for competitive performance.

In Table 2, we present results on the average number of edits, a common metric for evaluating counterfactual explanation systems, alongside the average top-1 counterfactual GED. In VG-DENSE experiments, the superior performance of supervised GCN is evident in edit numbers. While total edits appear similar across models, GFA GIN emerges as the second-best performer, indicating the limitations of relying solely on edit counts without considering their cost (GED). Comparing supervised GCN and GFA GIN, a significant gap in GED underscores their semantic divergence. Results for VG-RANDOM exhibit more variability, with GAE GAT leading in total edits. Considering both edit counts and GED provides insights into explanation minimality, highlighting the semantic relevance advantage of supervised models despite similar edit counts.

By comparing ranking metrics across the two dataset splits, we can easily observe that best results overall were achieved in the DENSE split. This is an expected behavior, since GNNs require interconnections for message passing procedures, i.e. to exchange and aggregate information from their neighbors.

Performance vs Dataset Size. Although unsupervised GAEs lag behind supervised models, there is a notable discrepancy in the required ground truth data. Supervised GNNs necessitate training on $\sim N^2/2 = 70$K pairs, maintaining a quadratic relationship with input data, entailing computationally expensive ground truth GED calculations. Conversely, unsupervised GAEs offer a

more scalable solution, requiring N training samples without optimizing on ground truth GED. This enables faster calculation of counterfactual explanations. While increasing training samples can enhance GAE performance, it may not be feasible in low-resource scenarios or where manual graph construction is required. Both approaches demonstrate computational efficiency, with GAE training taking less than 2 min for $N = 500$ graphs, whereas supervised GNN training requires $\sim$3 h for $\sim N^2/2 = 70$K graphs, with slight variations based on graph density. Despite differing training times, the retrieval, inference, and edit path computation times remain consistent between the two approaches. Training on a single GPU available in online platforms e.g. Kaggle or Google Colab is feasible for both supervised and unsupervised GNNs, while retrieval and inference operations are CPU-based, ensuring reproducibility of reported results.

4.3 Qualitative Results

Qualitative analysis is indispensable, particularly in the visual domain, where numerical metrics may not fully capture model performance. For graph retrieval, structural and semantic cues play vital roles. In Fig. 3, we present results from top-performing models: supervised GCN, unsupervised ARVGA-GIN, and WL-kernel. Note that counterfactual classes do not need to align between models.

In the case of VG-DENSE, multiple instances exhibit concordance among models, owing to its dense interconnections and the inherent ability of all models to recognize matching structures. The 1st row demonstrates unanimous agreement, as the obtained scene graph essentially forms a subgraph of the query. A human could also agree on the semantic closeness between counterfactual instances: Semantics such as "table", "cup", "phone" are preserved, while the kitchen-related "cake" and "newspaper" should be edited to allow workstation-related semantics, such as "laptop". Moving to the 2nd row, GNN methods align; successfully integrating structural and semantic information. Despite the kernel-retrieved result having a more similar structure, GNN methods successfully discern and retrieve an image featuring a cat on furniture, surpassing a mere depiction of a bed. In terms of human perception, counterfactuals depicting a "bedroom" (query) and a "living room" instance (GNN-based retrievals) respect the change of class label ("bedroom"$\rightarrow$"living room") with minimally altering semantics ("pillow" and "cat" semantics are preserved), allowing effortless high-level interpretability. Conversely, in the 3rd row with a widespread disagreement, the merits of the supervised GNN network become evident. The unsupervised GAE captures an image resembling the query with a teddy bear wearing a bow but overlooks the presence of a dog. In contrast, the supervised GNN, having more accurately approximated GED, retrieves an image centered on a dog and a different type of toy. Interestingly, the kernel fails and retrieves an image of a train, emphasizing that a sole focus on structure (star subgraph) is insufficient.

Assessing retrieved VG-RANDOM results proves challenging due to the dissimilar nature of the sparser underlying graphs. In the 4th row, the supervised GCN retrieves a structurally closer scene graph, despite visual similarities in GNN-based counterfactuals. It becomes harder even for humans to evaluate

Fig. 3. Counterfactuals from the best supervised GNN, unsupervised GNN and kernel for VG-DENSE (top 3) and VG-RANDOM (bottom 2).

their proximity, since all instances contain "chair" and "curtain" concepts, but no other discriminative structure or detail. In the final row the models reach consensus, successfully retrieving an image with palm trees, a characteristic the kernel fails to achieve despite the retrieval of very closely aligned graphs. The effectiveness of the framework relies on rich interconnections observed in VG-DENSE, underscoring the significance of expressive scene graphs, human intuition, and thoughtful data preprocessing for enhancing explanation informativeness.

Notably, none of the retrieved instances are mislead by purely visual characteristics, such as color, contrast or brightness of images. This is because graph-based algorithms naturally disregard any such characteristics, as they are not integrated into scene graphs. Using the kitchen scenario as an example, the preservation of its identity through alterations like converting to grayscale or adjusting brightness suggests that graph machine learning models, particularly GNNs, can closely align with human perception.

5 Conclusion

In this work, we unify graph-based conceptual counterfactual explanation algorithms under a common framework, upon which we conduct a comparative study. Various graph-based algorithms, such as kernels, supervised and unsupervised GNNs, highlight the value of graph representations for counterfactuals, offering insights on what needs to be changed conceptually to transit to another class efficiently. Quantitative and qualitative analysis showcases the different criteria that influence the retrieval of counterfactual instances from varying graph algorithms, and the importance of well-defined and informative semantics towards human-interpretable explanations. There is ample room for further real-world applications of this framework within various domains where we have access to graph-structured data, such as recommendation systems, transportation networks, as well as agricultural and health inspection sectors.

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Model Guidance via Explanations Turns Image Classifiers into Segmentation Models

Xiaoyan Yu[1,3(✉)], Jannik Franzen[1,2], Wojciech Samek[4,5,6], Marina M.-C. Höhne[7,8], and Dagmar Kainmueller[1,2,8(✉)]

[1] Max-Delbrueck-Center for Molecular Medicine in the Helmholtz Association (MDC), Berlin, Germany
{xiaoyan.yu,dagmar.kainmueller}@mdc-berlin.de
[2] Helmholtz Imaging, Berlin, Germany
[3] Humboldt-Universität zu Berlin, Berlin, Germany
[4] Department of Artificial Intelligence, Fraunhofer Heinrich Hertz Institute, Berlin, Germany
[5] Department of Electrical Engineering and Computer Science, Technical University of Berlin, Berlin, Germany
[6] BIFOLD - Berlin Institute for the Foundations of Learning and Data, Berlin, Germany
[7] UMI Lab, ATB Potsdam, Potsdam, Germany
[8] University of Potsdam, Potsdam, Germany

Abstract. Heatmaps generated on inputs of image classification networks via explainable AI methods like Grad-CAM and LRP have been observed to resemble segmentations of input images in many cases. Consequently, heatmaps have also been leveraged for achieving weakly supervised segmentation with image-level supervision. On the other hand, losses can be imposed on differentiable heatmaps, which has been shown to serve for (1) improving heatmaps to be more human-interpretable, (2) regularization of networks towards better generalization, (3) training diverse ensembles of networks, and (4) for explicitly ignoring confounding input features. Due to the latter use case, the paradigm of imposing losses on heatmaps is often referred to as "Right for the right reasons". We unify these two lines of research by investigating semi-supervised segmentation as a novel use case for the Right for the Right Reasons paradigm. First, we show formal parallels between differentiable heatmap architectures and standard encoder-decoder architectures for image segmentation. Second, we show that such differentiable heatmap architectures yield competitive results when trained with standard segmentation losses. Third, we show that such architectures allow for training with weak supervision in the form of image-level labels and small numbers of pixel-level labels, outperforming comparable encoder-decoder models. Code is available at https://github.com/Kainmueller-Lab/TW-autoencoder.

Keywords: Few-shot Learning · Semantic Segmentation · Layer-Wise Relevance Propagation

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1 Introduction

Ongoing advancements in explainable AI methodology aim to interpret model behavior and offer human-readable explanations. In the field of computer vision, methods that explain image classification network predictions in the form of *heatmaps*, i.e., assign relevance to each pixel of an input image, have gained wide popularity: Heatmaps are extensively leveraged to gauge the trustworthiness of classifiers, as they have been shown to detect biases and confounders in input images [4,17].

The capacity to detect biases and confounders in input images goes naturally hand in hand with the capacity to locate target objects of the semantic classes for which an image classifier has been trained. This capacity has been shown for many well-known XAI methods [1,5,11,24,32], and has motivated approaches for leveraging heatmaps towards weakly supervised segmentation, i.e., segmentation networks that train on image-level labels as opposed to expensive dense, pixel-level labels. In particular, Grad-CAM based heatmaps [32] are a popular ingredient in weakly supervised semantic segmentation models [13,15,18,19].

While the above line of research exploits heatmaps generated by trained models, an orthogonal line of research has tapped into the potential of heatmaps as a lever to influence model training in the first place [38], with the aim to yield models that are "Right for the Right Reasons" [30]. To this end, imposing losses on differentiable heatmaps has been shown to allow for aligning explanations with human interpretation [20,28,30,33,34], enhancing model generalizability [27], and ignoring undesired image features towards diverse ensembles [30,36] and confounder mitigation [31].

In our paper, we aim to integrate these two lines of research by exploring the potential of the Right for the Right Reasons paradigm for semi-supervised segmentation. To this end:

- We establish formal parallels between differentiable heatmap architectures and conventional encoder-decoder architectures commonly used for image segmentation. In particular, we show that "unrolling" LRP [23] on standard image classification architectures like a ResNet50 yields convolutional encoder-decoder architectures that resemble standard U-Nets [29], albeit with weights tied between encoder and decoder and modified, "skipped" activation functions in the decoder.
- We perform a comparative evaluation of unrolled LRP and comparable U-Nets in terms of segmentation accuracy on the *val* set of the PASCAL VOC 2012 segmentation benchmark [10]. Our experimental results reveal for multiple standard classification backbones that differentiable heatmap architectures, trained with combined classification- and segmentation loss, can achieve competitive segmentation performance.
- We evaluate semi-supervised training, with pixel-level labels ranging down to one labelled image per class, revealing that in scenarios with few pixel-wise labels, unrolled heatmap architectures outperform comparable standard UNets for segmentation by up to 10% mIoU on PASCAL.

1.1 Relation to Previous Works

The work of [19] is closely related to ours in that, to our knowledge, it is the sole previous work that also optimizes heatmaps towards improved segmentation performance. To this end, they optimize low-resolution Grad-CAM based heatmaps by means of specifically designed losses, and use resulting label maps as pseudo-labels for further training and sophisticated post-processing to increase segmentation accuracy. Instead, we optimize full-resolution LRP heatmaps by means of a vanilla segmentation loss, and establish formal parallels of this approach to conventional segmentation architectures rather than aiming at state of the art segmentation accuracy via subsequent heuristic processing. Our results outperform [19] by 9% in a comparable scenario without their CRF post-processing.

The recent work of [27] is closely related to ours as they optimize heatmaps towards improved localization performance under various loss functions and heatmap-generating methods, including IxG, which is equivalent to LRP-0 [3] as employed in our study. However, they do not aim at a formal analysis of the respective unrolled architectures, whereas we reveal strong formal parallels between unrolled LRP-0 and standard encoder-decoder architectures; Furthermore, their work focuses on localization instead of segmentation, and their losses are specifically designed for optimizing heatmaps whereas we employ standard segmentation loss. To this end, we successfully unroll heatmaps for all class scores in each batch, whereas they restrict their losses to one (randomly selected) class per batch. In a striking difference to [27], we empirically find that classifier performance does not degrade during training while segmentation performance increases significantly. This holds for all pixel-level supervision regimes we study. To the contrary, their empirical results suggest that increased localization performance has a strong tendency to go hand in hand with decreased classification performance. We discuss the hypothesis that this different behavior may be due to the concordant nature of the classification- and heatmap losses we employ.

Our derivation of unrolled heatmap architectures manifests a special case of double backpropagation on ReLU networks, which has been formally studied before by [9] and, in the context of heatmap methods, by [2]. However, to our knowledge, we are first to report formal parallels of unrolled heatmap architectures to standard segmentation networks, and we showcase their respective potential by empirical results that outperform standard segmentation baselines.

1.2 Limitations

Our work does not aim at state of the art semi-supervised segmentation: We do not employ any post-processing on optimized heatmaps [18,19,22,39], nor do we leverage any orthogonal avenues towards semi-supervised segmentation, like e.g. entropy minimization [6], consistency regularization [8,16,25,26,35], or contrastive learning [21]. However, thanks to our standard training objective, our unrolled architecture can be directly plugged into these orthogonal approaches.

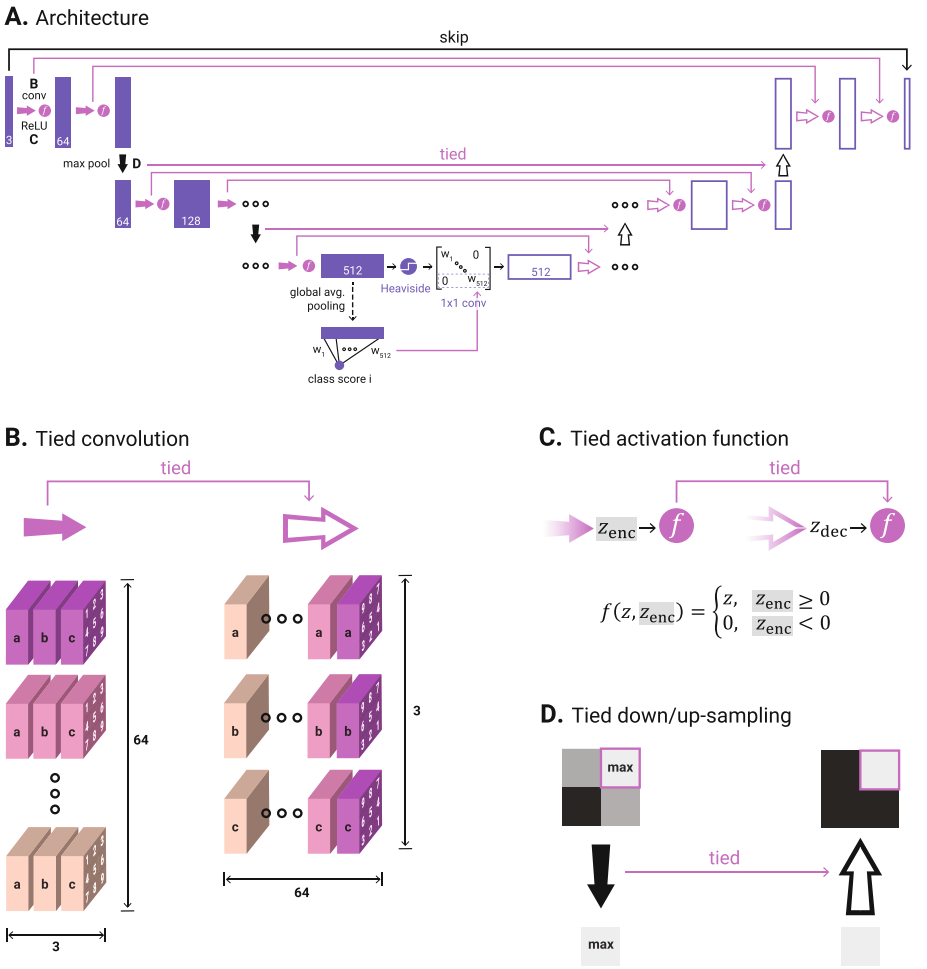

Fig. 1. Unrolled LRP-0 for standard convolutional classifiers is an encoder-decoder CNN. (A) Sketch of architectural building blocks. (B) Decoder weights are tied with corresponding encoder layers. (C) Activation functions are skipped from encoder to decoder. (D) Up-convolutions are tied to the respective pooling operation in the encoder. In (A) we sketch the decoder architecture induced by an individual semantic class i; this decoder is replicated for all classes with shared weights, except for the bottleneck 1×1 convolution, which is class-specific.

2 Unrolled Heatmap Architectures

In this Section, we present formal parallels between differentiable heatmap architectures and classical encoder-decoder architectures for image segmentation. After a brief recap of LRP-ϵ [5], we derive the building blocks of architectures entailed by "unrolling" LRP-0 on standard convolutional image classification

models, comprising ReLU nonlinearities, max pooling, and global average pooling. The resulting architectural blueprint is sketched out in Fig. 1, and forms the basis of our subsequent quantitative analysis of segmentation performance. Last, we assess formal properties of said unrolled architectures.

2.1 LRP Basics

Linear Layer. In the following, we denote the input vector of a linear layer l by $a^{(l-1)}$, its weights matrix by $w^{(l)}$, and its output vector (pre-activation) by $z^{(l)}$. Considering a single linear filter in isolation, the relevance LRP in its most basic form (LRP-0) assigns to an input variable a_i equals the variable's contribution to the (scalar) output, namely $R_i = a_i w_i$. Considering a single linear filter in an intermediate layer $l+1$ of a neural network, assuming the relevance $R_j^{(l+1)}$ of the filter's output $z_j^{(l+1)}$ has already been determined, LRP-0 distributes $R_j^{(l+1)}$ to the filter's inputs $a_i^{(l)}$ proportional to their respective contribution to $z_j^{(l+1)}$, namely as $R_j^{(l+1)} a_i^{(l)} w_{ij}^{(l+1)} / z_j^{(l+1)}$. This ensures preservation of total relevance across layers. Considering a general linear layer $l+1$ as a whole, input variables accumulate relevances from all linear filters they contribute to by summation. This entails the following recursive definition by which LRP-0 assigns relevances to inputs $a_i^{(l)}$ given relevances $R_j^{(l+1)}$ of outputs $z_j^{(l+1)}$:

$$R_i^{(l)} = a_i^{(l)} \sum_j \frac{w_{ij}^{(l+1)}}{z_j^{(l+1)} + \epsilon} \cdot R_j^{(l+1)}. \tag{1}$$

The ϵ in Eq. 1 serves for numerical stability, but may also serve to mitigate noise in the relevances in a variant of LRP termed LRP-ϵ. We focus our analysis on LRP-0, where ϵ is set to a very small constant that serves solely for numerical stability and can be reasonably ignored for simplified formal analyses. Equation 1 can be phrased as a three-step algorithm: Applying LRP to any linear layer $l+1$ amounts to the following sequence of operations applied to $R^{(l+1)}$:

Algorithm 1. LRP through a linear layer $l+1$

1: Element-wise division by $z^{(l+1)}$ (yielding $q := R^{(l+1)}/z^{(l+1)}$)
2: Weighted summation via $w^{(l+1)}$ (yielding $p_i := \sum_j w_{ij}^{(l+1)} q_j$ for all i)
3: Element-wise multiplication by $a^{(l)}$ (yielding $R_i^{(l)} = a_i^{(l)} p_i$).

ReLU and Pooling. LRP sets the relevance of the input of a ReLU to be equal to the relevance of its output, i.e., relevances are propagated through ReLUs unchanged. Pooling operations can be treated as their corresponding equivalent convolutions (where for max pooling, individual inputs entail respective individual convolution kernels).

2.2 Unrolled LRP Architectures for Convolutional Classifiers

In the following, we derive the building blocks of "unrolled LRP". To this end, we phrase LRP through standard CNN classifier/encoder building blocks as corresponding decoder building blocks.

Single Convolution. First, we consider a single convolutional filter applied to a stack of activation maps of depth c_l at some layer l. A pixel in activation map k contributes to the respective output map in layer $l+1$ as follows: (1) To a spatial region around its own location that is of the same size as the spatial extent of the convolutional kernel; (2) With weights solely from the k-th slice of the convolutional kernel; (3) To position (dx,dy) relative to its own location with weight at position (-dx,-dy) in the conv kernel (when defining the center point location of the conv kernel to be (0,0)). Thus, the weighted combination entailed by LRP (cf. Step 2 of Algorithm 1) through a single convolution of depth c_l (layer $l \rightarrow l+1$) amounts to the application of c_l convolutions of depth 1 to the relevance map obtained for layer $l+1$. These "unrolled LRP" convolutions are formed by slices of the original convolutional kernels, with weights flipped in all spatial dimensions.

Convolutional Layer. Next we consider a full convolutional layer, with c_{l+1} convolutional filters yielding c_{l+1} output maps. Here, a pixel in activation map k in layer l contributes to identical spatial regions in all resulting output maps in layer $l+1$, where weights stem from the k-th slices of each convolutional kernel in the same mirrored manner as described for a single convolutional filter above. This yields c_{l+1} "unrolled LRP" convolutional kernels, where the k-th unrolled kernel is assembled by stacking the k-th slices of all original kernels, mirrored in all spatial directions.

Altogether, the weighted combination entailed by LRP through a convolutional layer with input depth c_l and output depth c_{l+1} amounts to an "unrolled LRP" convolutional layer with input depth c_{l+1} and output depth c_l with tied weights as sketched in Fig. 1 B.

Consecutive Convolutional Layers with ReLU Activations. Consider a convolution with ReLU activation followed by another convolution. As LRP passes relevances through ReLUs unaffected, in this case, Algorithm 1 is simply applied twice in a row. In the resulting sequence of six operations, step 3 and 4 can be condensed into Element-wise multiplication by

$$a^{(l)}/z^{(l)} = \begin{cases} 1 & z^{(l)} > 0 \\ 0 & \text{otherwise} \end{cases} = H(z^{(l)}),$$

where H denotes the Heaviside function. This entails unrolled LRP "activation functions" between convolutions that do not follow the classical paradigm of a

nonlinear function applied to convolution outputs; Instead, the unrolled activation function zeroes out unrolled convolution outputs based on the output value of the respective original convolution. This can be seen as the same function being applied after original and unrolled convolution, as sketched in Fig. 1 C.

Note that pooling layers can be seen as convolutions (with appropriate kernels) and thus merged with a subsequent convolution. Consequently the above unrolled LRP "activation functions" also appear after the last unrolled convolution per downsampling level.

Final Classifier Layers. Consider the final layers of a standard classifier: The final stack of spatial activation maps, which we term a here, are fed through global average pooling, yielding an average pooling vector we term apv. apv is then fed through a linear output layer, yielding a vector of class scores. LRP is performed individually per class j. To this end, LRP is initialized by setting the relevance of the respective class score equal to its value. Thus the elements of the average pooling vector, apv_i, receive relevances $R_i = avp_i w_{ij}$. Considering avp as input, the initial steps of LRP on class score j thus play out as follows:

Algorithm 2. LRP on the class score of an individual class j through final standard classifier layers

1: Element-wise multiplication of apv with the class weights of class j from the last linear layer
2: Element-wise division by apv (i.e., step 1 of Algorithm 1 for the average poling layer)
3: Division by the (spatial) number of pixels in a, followed by nearest-neighbor upsampling to the spatial extent of a (i.e., step 2 of Algorithm 1 for the average poling layer)
4: Element-wise multiplication with a (i.e., step 3 of Algorithm 1 for the average poling layer)
5: Element-wise division by z (i.e., step 1 of Algorithm 1 for the last conv layer).

Step 3 yields a stack of activation maps of spatial extent and depth equal to a. Each activation map has a constant value, namely a weight from the output layer class weight vector divided by the (spatial) number of pixels in a. Steps 4 and 5 zero out this constant value whereever $z \leq 0$. Thus Algorithm 2 can be condensed into a Heaviside function applied to z, followed by a 1×1 convolution, where each of the conv kernels has one non-zero entry, namely one of the class weights. This is sketched out in the bottleneck of the unrolled architecture in Fig. 1 A. Note, classifiers without average pooling and/or with multiple fully connected layers before the output layer can be unrolled analogously: Average pooling is equivalent to a fully connected layer with specific weights, and fully connected layers are equivalent to convolutions with suitable kernel size.

To consider not just one class score but the class score vector as a whole, the complete decoder architecture is replicated for each class. The resulting decoder branches are class-specific solely in the first 1×1 convolution with the class-specific weights of the classifier output layer. All other weights are shared between decoder branches.

Classifier Input Layer. LRP through the first conv layer of a classifier is special as multiplication with a, which is the input image in this case, does not cancel out by supsequent propagation through another convolution. Thus the last building block of the unrolled architecture is an element-wise multiplication with the input image, depicted as skip connection in Fig. 1 A.

2.3 Losses and Training

We turn three-channel heatmaps into single-channel heatmaps by summation as is standard. We then stack together heatmaps from all decoder branches, thus forming class score vectors per pixel, on which we employ standard pixel-wise softmax cross-entropy loss. We complement this standard segmentation loss by standard classification loss on the class score vector in the bottleneck, namely sigmoid cross-entropy per class score, same as during classifier training.

This combination of losses entails interesting training behavior: LRP is designed to satisfy the conservation property, i.e., it aims at conserving total relevance across layers [23]. Assuming the conservation property holds (and thus ignoring the bias terms for the sake of a simplified argument), the total relevance of a heatmap, i.e., the sum of all pixel relevances, equals the respective class score. For negative classes, i.e., classes not present in an image, this entails the following: Softmax cross-entropy pushes down heatmaps of negative classes at all pixels. Due to conservation, this necessarily also pushes down the respective class score in the bottleneck, which is concordant with the respective classifier loss. An analogous argument holds for a single positive class that covers the whole image: Softmax cross-entropy pushes up the respective heatmap at all pixels and thus, due to conservation, also the class score in the bottleneck, which is again concordant with the respective classifier loss. For the case of multiple positive classes, which is the most common case due to the abundant background class and our multi-label scenario, the segmentation loss pushes up class-specific foreground pixels in the respective heatmap while all other pixels in this heatmap are pushed down. Consequently, some trade-off needs to be met to not cause the respective class score to decrease. We empirically find that a suitable trade-off is met, as we observe that classification performance does not degrade during training (cf. Sect. 3).

2.4 Relation to Previous Formal Analyses and Standard Architectures

Previous Formal Analyses. Backpropagtion through unrolled LRP-0 architectures as blueprinted in Fig. 1A does not reach the encoder (no matter which heatmap loss is employed): Due to the Heaviside function and its zero local derivative, global derivatives are zeroed out along all paths to the encoder. Consequently, the derivatives of the (intermediate) output directly before the final skip-multiplication with the input w.r.t. the network weights takes a simple form, namely a sum of other weights' products, where the input image determines which products are non-zero. This has been shown before by [3] as part of their proof that LRP-0 equals inputs times gradients (IxG) in ReLU networks.

It trivially follows from LRP-0 = IxG that the output of our unrolled architecture, pre skip-multiplication with the input, equals the gradient of the class score w.r.t. the input. Thus our unrolled architecture implements double backpropagation [7]. Double backpropagation has been formally analyzed in-depth by [9], who have also derived the respective simplified gradients for ReLU networks, and note that the backward pass is thus shortened.

Our architecture serves as easily accessible special case of what the above more general theory entails: When trained with classification- and heatmap loss, the gradient of the classification loss backpropagates solely through the encoder, while the gradient of the heatmap loss backpropagates solely through the decoder. This can be leveraged for efficient training; It remains unclear whether the property has further consequences, e.g. whether it has a systematic effect on training dynamics; Furthermore, in practice, note that any $\epsilon > 0$ in Eq. 1 smoothes out all Heaviside functions in the positive regime, thus yielding non-zero heatmap loss gradient flow to the encoder during backprop.

Standard Architectures. On the one hand, unrolled heatmap architectures as described above are related to U-Net architectures [29], which constitute the state of the art in semantic segmentation in a broad range of applications to date [12]. A U-Net is related in that it is an encoder-decoder convolutional architecture. It is distinct in that (1) convolution- and pooling operations are not tied between encoder and decoder whereas ours are; (2) it employs standard nonlinearities in the decoder whereas we employ "tied activations"; (3) it employs skip-concatenations of encoder activation maps to the respective decoder layer whereas we only skip-connect activations; and (4) it employs a single decoder for all classes, whereas we employ weight-shared branches per class.

On the other hand, unrolled heatmap architectures are closely related to weight-tied auto-encoders, as first described by [37]. In particular, [14] have explored weight-tied auto-encoders based on convolutional encoders. The weight-tied architecture they establish equals ours in their tied decoder convolutions and up-sampling operators, yet is distinct in that they employ standard nonlinearities in the decoder, and in that they do append a final, non-tied convolutional layer. Furthermore, like U-Net, they also employ only a single decoder for all classes.

In summary, our unrolled LRP architectures are most constrained to their encoder backbone among all previously described encoder-decoder-style convolutional segmentation architectures.

3 Unrolled Heatmap Architectures for Segmentation: Results

Data and Baselines. We evaluate the performance of our unrolled architecture on the PASCAL VOC 2012 segmentation benchmark val set. We employ a comparable U-Net, i.e., a U-Net with identical convolutional encoder, as standard segmentation baseline. We evaluate two different classifier backbones (resnet50 and vgg16 with batch norm) in four scenarios with varying levels of pixel-wise supervision, namely with 1, 5, 25, and all available labelled images per class. This amounts to respective totals of 20, 100, 500, and 1464 labelled images. We employ standard augmentations, as well as standard cropping to fit the classifiers' required input size. We initialize the classifier backbones of all architectures to their respective ImageNet pre-trained and PASCAL fine-tuned weights.

To give our UNet baseline the chance to leverage the 15,676 available image-level labels not just via pre-training but also during segmentation training, we include a modified version of the resnet50-encoder variant as additional baseline, where we extend a comparable classifier branch, i.e., simple average pooling followed by a linear output layer, from the UNet bottleneck. We train this "multi-task" UNet with same combined classification- and segmentation loss as our unrolled architecture.

We train all models with batch size 10, AdamW with learning rate $1e-5$. We give equal weight (=1) to classification- and segmentation loss in our unrolled architectures and multi-task UNet baseline.

To cover different semi-supervised scenarios we sample 20, 100, 500 and 1464 labeled images from the training set, respectively. For the scenarios with 20 and 100 labelled images we sample randomly, with fixed seed for all experiments maximally uniform distributed classes. For the 20 labeled images case, this ensures that each class is sampled at least once. However, due to the strong class imbalance within PASCAL, when dealing with larger image sets, sampling with a uniform distribution is no longer possible; Thus we relax the strategy for ≥ 500 labeled images. Here, we sample with the average of a uniform distribution and the class distribution in the training set.

Training and Testing. During training, all models' encoders are initialized with identical pretrain weights, derived from ImageNet and fine-tuned on PASCAL. Both the multi-task UNet and unrolled LRP have classification branches, which we initialize accordingly. For all models that do not have tied weights, the decoders are initialized "from scratch". Within each batch, we sample 10 pixel-labeled images for UNet and employ cross-entropy loss for the final output. For multi-task UNet and unrolled LRP, we select 10 images comprising a mixture

of pixel-labeled and image-level labeled data. We apply per-class sigmoid cross-entropy loss, with a weight of 0 for the background class, to the classification branches of these 10 images. For images with pixel-level labels (typically 4–6 per batch), we additionally apply softmax cross-entropy loss. The two losses are trained simultaneously.

During testing, the classification output from the bottleneck is disregarded for both the multi-task UNet and unrolled LRP models. We directly apply argmax on the outputs of all models to produce semantic segmentation maps.

Results. Table 1 reports results in terms of segmentation mIoU averaged over three independent training runs per model.

Table 1. Segmentation mIOU on PASCAL VOC 2012 *val* for vgg16 and ResNet50 backbones, models, and supervision scenarios, ranging from 20 to 1464 labelled images. We report average mIoU over three independent training runs for all models.

Method	1.4 % (20)	6.8 % (100)	34.2 % (500)	Full(1464)
Architecture: vgg16 backbone for encoder				
UNet	$20.38_{\pm 0.64}$	$40.50_{\pm 0.77}$	$54.60_{\pm 0.12}$	$60.53_{\pm 0.48}$
unrolled LRP (ours)	$\mathbf{33.92_{\pm 1.21}}$	$\mathbf{49.68_{\pm 0.64}}$	$\mathbf{59.96_{\pm 0.78}}$	$\mathbf{63.85_{\pm 0.18}}$
Architecture: ResNet50 backbone for encoder				
UNet	$25.89_{\pm 1.01}$	$43.40_{\pm 0.68}$	$55.10_{\pm 0.27}$	$60.07_{\pm 0.20}$
multi-task UNet	$25.06_{\pm 0.82}$	$39.77_{\pm 0.67}$	$55.13_{\pm 0.33}$	$60.37_{\pm 0.47}$
unrolled LRP (ours)	$\mathbf{39.80_{\pm 0.37}}$	$\mathbf{50.29_{\pm 0.61}}$	$\mathbf{58.30_{\pm 1.60}}$	$\mathbf{61.50_{\pm 0.27}}$

Our models outperform comparable UNets in all supervision scenarios, where the margin increases drastically with decreasing pixel-level supervision.

Table 2. Classification F1 score of our unrolled LRP models on PASCAL VOC 2012 *val* after pre-training (column "0") and after segmentation training across supervision scenarios.

Backbone	0	1.4 % (20)	6.8 % (100)	34.2 % (500)	Full (1464)
vgg16bn	81.33	$80.73_{\pm 1.10}$	$81.99_{\pm 0.80}$	$81.86_{\pm 0.59}$	$\mathbf{82.52_{\pm 0.33}}$
resnet50	81.62	$80.35_{\pm 0.82}$	$80.80_{\pm 1.49}$	$\mathbf{82.21_{\pm 0.97}}$	$81.49_{\pm 1.72}$

At the same time, interestingly and in contrast to respective empirical findings by [27], classifier performance remains largely unaffected; Table 2 lists classifier F1 scores after pre-training (with only the classification loss) as well as after training with combined classification and segmentation loss.

Our unrolled ResNet50 architecture has 23.55M trainable parameters, whereas the respective UNet baseline has 38.84M. To make sure the inferior performance of the UNet is not just due to respective higher susceptibility to overfitting, we trained another UNet baseline based on a ResNet18 backbone, which has 18.81M parameters. However, the ResNet18 UNet is outperformed by the ResNet50 UNet, i.e., the fact that our UNet baseline has more parameters than our unrolled architecture does not explain its inferior performance.

Figures 2 and 3 show exemplary segmentation results, as well as the evolution of segmentations and heatmaps over training of unrolled LRP. We observe quick convergence to crisp segmentations, also in cases with multiple labels.

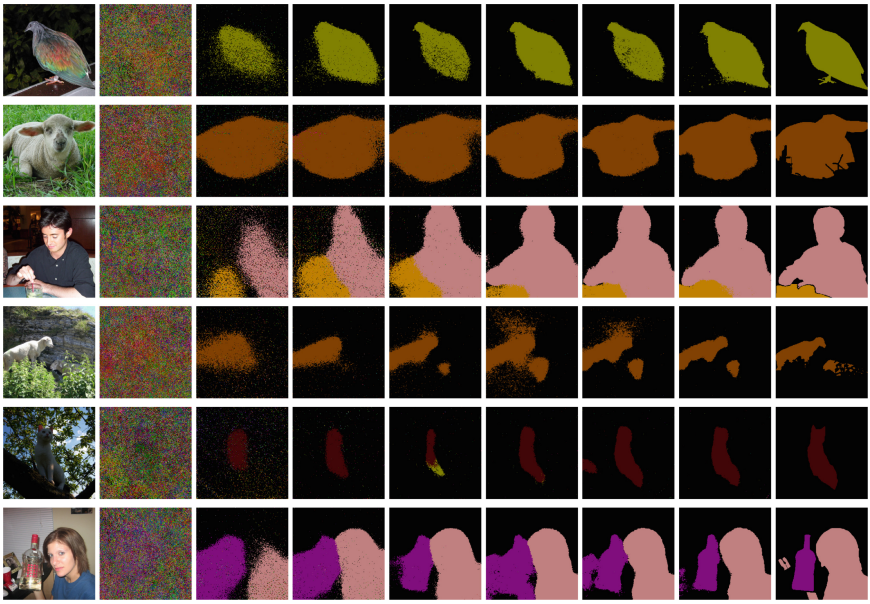

Fig. 2. Exemplary unrolled LRP segmentations on the VOC validation set, 500 labelled images scenario. Left: input image; Right: ground truth segmentation; In-between: Evolution of segmentation over training of unrolled LRP, ranging from the first iteration to peak validation mIoU.

Ablation Study. Architectural elements that turn unrolled LRP into a UNet are: (1) Tied vs standard activation functions, (2) tied vs standard decoder weights and up-sampling, and (3) concat-skip connections from encoder to decoder. Table 3 lists results of a respective ablation study. The ablation includes a weight-sharing convolutional autoencoder (WS-AE), in which tied activations as well as the bottleneck heaviside function in unrolled LRP are replaced by standard ReLUs. WS-AE is trained with segmentation- and classification loss, like unrolled LRP. The ablation further includes a standard Fully Convolutional Network (FCN), in which weights in the decoder are free, and nearest neighbor

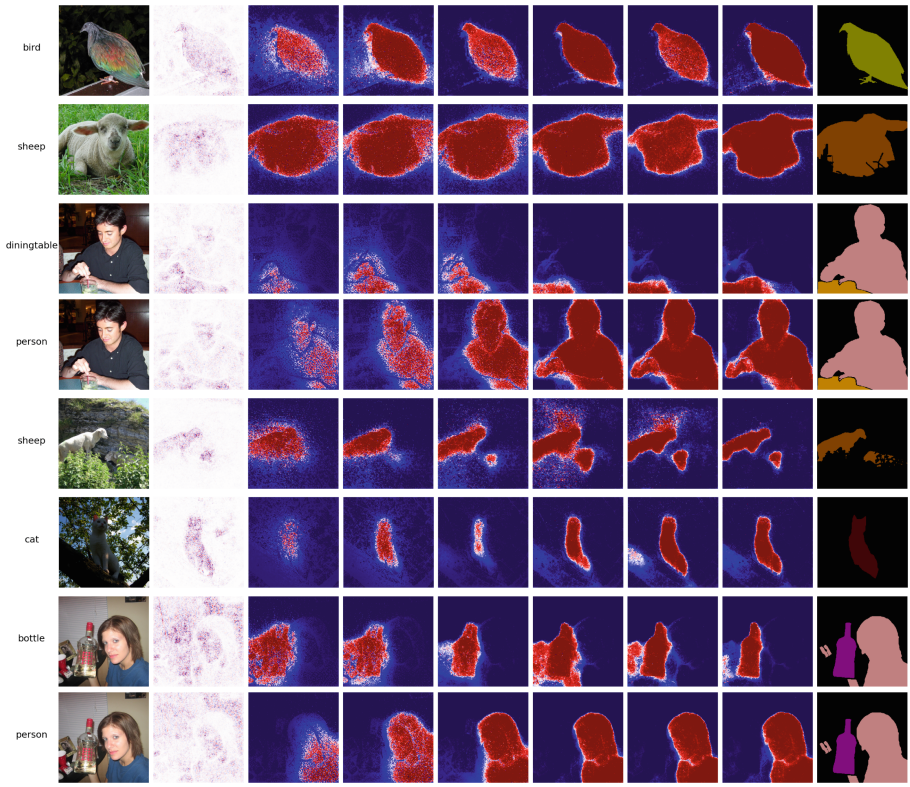

Fig. 3. Exemplary unrolled LRP heatmaps on the VOC validation set, 500 labelled images scenario (cf. Fig. 2 for corresponding segmentations). Left: input image; Right: ground truth segmentation; In-between: Evolution of heatmaps of the positive classes over training of unrolled LRP, ranging from the first iteration to peak validation mIoU. Note that all heatmaps except for the initial (left-most) one are shown after softmax to render standard blue-white-red visualization informative.

upsampling is employed. To avoid the introduction of free weights per decoder branch, FCN is reduced to a single decoder branch. This entails that FCN does not contain the class weights of the classifier output layer. FCN is trained only with segmentaiton loss, like UNet.

Discussion. We hypothesize that superior performance of unrolled LRP over a comparable standard segmentation baseline in scenarios with limited supervision is due to the strong constraints imposed by tied weights, tied up-sampling, and tied activations. Furthermore, we hypothesize that the concordant losses as described in Sect. 2.3 facilitate the full exploitation of image-level labels for segmentation, as not just the segmentation loss, but also the classification loss pushes the total relevance in the heatmaps in a meaningful direction. As a quantitative indicator of this loss concordance, Table 2 reveals, to our knowledge for

Table 3. Ablation of architectural elements between UNet and unrolled LRP. All results based on Resnet50 backbone; average mIoU on VOC val set over three independent training runs.

Method	tied ReLU	tied Conv	tied up-sample	concat skip	1.4 % (20)	6.8 % (100)	34.2 % (500)	Full (1464)
Unr. LRP (ours)	✓	✓	✓	-	**39.8**$_{\pm 0.4}$	**50.3**$_{\pm 0.6}$	**58.3**$_{\pm 1.6}$	**61.5**$_{\pm 0.3}$
WS-AE	–	✓	✓	–	29.5$_{\pm 1.7}$	46.6$_{\pm 0.5}$	54.7$_{\pm 0.5}$	57.9$_{\pm 0.5}$
FCN	–	–	–	–	26.2$_{\pm 1.0}$	40.0$_{\pm 0.5}$	51.3$_{\pm 0.5}$	56.6$_{\pm 0.5}$
UNet	–	–	–	✓	25.9$_{\pm 1.0}$	43.4$_{\pm 0.7}$	55.1$_{\pm 0.3}$	60.1$_{\pm 0.2}$

the first time, that vanilla classifiers can be trained in such a way that they condense all relevance to respective class foregrounds while classification performance remains unaffected. However, precisely how the sketched training dynamics play out for images with multiple class labels (including background) requires further study.

Our ablation study reveals that tied activation functions, an element our work newly introduces to segmentation architectures, are essential for the superior performance of LRP-0 over a standard UNet. Our study replicates previous findings that WS-AE outperforms FCN in few-labelled-samples regimes; yet it reveals that including tied activation functions yields another significant performance boost over WS-AE, which is crucial to outperforming the U-Net.

4 Conclusion

Our work establishes unrolled heatmap architectures as encoder-decoder-style convolutional architectures that can be trained for image segmentation. We observe superior performance to standard segmentation baselines in scenarios with limited pixel-level supervision, which entails the potential for practical use in semi-supervised segmentation. To this end, while not yielding state of the art semi-supervised segmentation results per se, our architecture directly lends itself to incorporation into orthogonal semi-supervised learning paradigms like augmentation consistency training. In contrast to previous work, we observe that training with a heatmap loss does not affect classification performance. This is striking in that our approach yields highly performant standard classifiers that nevertheless are trained to successfully focus on class foregrounds by means of our unrolled training objective.

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Understanding the Dependence of Perception Model Competency on Regions in an Image

Sara Pohland(✉) and Claire Tomlin

University of California, Berkeley, CA 94720, USA
{spohland,tomlin}@berkeley.edu

Abstract. While deep neural network (DNN)-based perception models are useful for many applications, these models are black boxes and their outputs are not yet well understood. To confidently enable a real-world, decision-making system to utilize such a perception model without human intervention, we must enable the system to reason about the perception model's level of competency and respond appropriately when the model is incompetent. In order for the system to make an intelligent decision about the appropriate action when the model is incompetent, it would be useful for the system to understand *why* the model is incompetent. We explore five novel methods for identifying regions in the input image contributing to low model competency, which we refer to as image cropping, segment masking, pixel perturbation, competency gradients, and reconstruction loss. We assess the ability of these five methods to identify unfamiliar objects, recognize regions associated with unseen classes, and identify unexplored areas in an environment. We find that the competency gradients and reconstruction loss methods show great promise in identifying regions associated with low model competency, particularly when aspects of the image that are unfamiliar to the perception model are causing this reduction in competency. Both of these methods boast low computation times and high levels of accuracy in detecting image regions that are unfamiliar to the model, allowing them to provide potential utility in decision-making pipelines. The code for reproducing our methods and results is available on GitHub: https://github.com/sarapohland/explainable-competency.

Keywords: Model Competency · Saliency Maps · Explainability

1 Importance of Understanding Model Competency

Deep neural networks (DNNs) have been very successful in image classification tasks. However, DNN models are considered black boxes, and there currently exists only a shallow understanding of how these models obtain their outputs [1]. This greatly limits our ability to trust these models in real-world systems, where their failure could be detrimental without human intervention. To confidently employ DNN-based perception models in the real world, we need to develop more robust and generalizable models, but we also need to gain a deeper understanding

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of how these models work and when they might fail. There has been much work on quantifying uncertainty in perception systems [2], as well as identifying when an input is out-of-distribution (OOD) [3]. However, with only knowledge that the model is uncertain or that an input is OOD, the systems that utilize these perception models are limited in their ability to appropriately respond when the perception model lacks confidence. It would be useful to understand *why* a model is believed to be uncertain and *why* an input is deemed to be OOD before the output of a perception model is used for decision-making.

We explore these ideas in this paper, developing methods to identify particular regions in an image that lead to a lack of model confidence. Towards this end, we explore five approaches for identifying regions in an image contributing to low model competency. We begin by considering a naive approach, in which images are partitioned into regions and the competency associated with each cropped region is assessed. We then explore segment masking, pixel perturbation, and competency gradient approaches that are related to methods present in artificial intelligence (AI) explainability work. Finally, we present a novel method for understanding regions contributing to model incompetency that is based on autoencoder reconstruction loss. We develop methods for evaluating each of these approaches and compare their performance across three diverse datasets. We present an analysis of the results and conclude with avenues for future work.

2 Background and Related Work

In this work, we explore explanations for *model competency*–a generalized form of predictive uncertainty. Predictive uncertainty generally arises from three factors: data (or aleatoric) uncertainty, model (or epistemic) uncertainty, and distributional shift [4]. Data uncertainty refers to uncertainty arising from complexities of the data (i.e., noise, class overlap, etc.), while model uncertainty reflects the ability of the perception model to capture the true underlying model of the data [5]. These types of uncertainty are explored in work referred to as uncertainty quantification (Sect. 2.1). Distributional shift refers to mismatched training and test distributions [6]. Extensive work has explored methods to detect inputs that are out-of-distribution (OOD) (Sect. 2.2). This paper aims to expand upon work on uncertainty quantification and OOD detection with the goal of offering explanations for why a perception model's prediction is uncertain. This idea of explaining perception model competency is related to previous work within the area of explainable image classification (Sect. 2.3), which seeks to offer explanations for the classification decisions of perception models. Our work differs from existing work in this area in that we seek to offer explanations for low model competency, rather than explanations for a model's prediction.

2.1 Uncertainty Quantification

Extensive research has been done on methods to understand and quantify uncertainty in a neural network's prediction. The modeling of these uncertainties

can generally be divided into methods based on (1) deterministic neural networks, (2) Bayesian neural networks (BNNs), and (3) ensembles of neural networks [2]. Approaches that use a single deterministic neural network estimate model uncertainty based on a single forward pass within a deterministic network. These approaches are often either distance-based [7,8] or gradient-based [9,10], although many other approaches exist [2]. A BNN is a stochastic model whose output is a probability distribution over its predictions [11,12]. Approaches that employ BNNs extract uncertainty as a statistical measure, often using sampling methods [13], variational inference [14–16], or Laplace approximation [17]. Ensemble methods combine the predictions of multiple deterministic networks to form a probability density function [18]. While all of these methods provide an estimate of the uncertainty associated with a model's prediction, none of these offer rationale for *why* the model is uncertain.

2.2 Out-of-Distribution (OOD) Detection

Many recent approaches have focused on quantifying distributional uncertainty–uncertainty that is caused by a change in the input data distribution [6]. Approaches that are specifically focused on determining if an input falls outside of the input-data distribution are referred to as OOD detection methods [3]. These approaches are generally either (1) classification-based, (2) density-based, (3) distance-based, or (4) reconstruction-based. Classification-based methods is a broad category that captures methods that largely rely on classifier predictions. These methods generally seek to revise the overly confident softmax scores at the output of neural networks to detect OOD samples more robustly [19–21]. Density-based methods model the training distribution with some probabilistic model and flag test data in low-density regions as OOD. These approaches tend to rely on Gaussian mixture models (GMMs) [22], kernel density estimation (KDE) [23], normalizing flows [24,25], or likelihood ratios [26]. Distance-based methods use distance metrics in the feature space with the assumption that OOD samples should lie relatively far from the centroids or prototypes of the training classes. Many popular approaches utilize the Mahalanobis distance [27,28], cosine similarity [29], or a nearest-neighbor distance [30]. Finally, reconstruction-based methods rely on the reconstruction loss of autoencoders (AEs) [31,32], variational autoencoders (VAEs) [33], convolutional autoencoders (CAEs) [34], or generative adversarial networks (GANs) [35–37]. While all these methods have been used to successfully identify OOD samples, they do not offer a clear explanation as to *why* the samples are deemed OOD.

2.3 Explainable Image Classification

The field of explainable artificial intelligence (XAI) seeks to offer explanations for why a model makes the decisions that it does with the goal of developing more interpretable and trustworthy models [1,38]. Within the area of explainable image classification, there exist (1) posthoc techniques, which explain the

behavior of an AI system after it has been trained without modifying the underlying model architecture, and (2) antehoc methods, which incorporate the aspect of explainability into the model design process to generate classification models that are inherently more interpretable [39]. Our work relates more closely to posthoc techniques, which employ various methods to understand the workings of black-box models and explain their decision-making process.

The most common posthoc approach for explaining an image classifier is to identify key image regions contributing to the model's predictions and display them using a saliency map. These methods display the regions salient toward the prediction of a class through gradient-based approaches [40–42], perturbation-based approaches [43–45], layerwise-relevance propagation [46–48], masking-based approaches [49], or deconvolution methods [50]. Other approaches aim to construct an inherently interpretable pseudo-classifier that approximates the working mechanism of the black box classifier, offering either local interpretations around a particular instance [51,52] or global interpretations [53,54]. There are also approaches that provide explanations based on abstract vector representations, which can be linked to human-understandable concepts. These methods tend to use concept bottleneck models [55] or concept activation vectors [56] if the categories of interest are provided, or they focus on discovering task-relevant concepts automatically [57,58]. Finally, there are methods that offer counterfactual explanations by generating alternative scenarios to explain the behavior of the classifier [59]. While all of these methods enhance the understanding of image classifiers, none of them explicitly deals with model *competency*. These methods do not offer an estimate of or explanation for how confident a model is in its prediction (as we demonstrate in Appendix B).

2.4 Explainable Competency Estimation

While there has been work on quantifying model uncertainty, identifying OOD inputs, and explaining model predictions, to the best of our knowledge, there has not been an effort to offer explanations for the lack of competency associated with a model's prediction. We seek to bridge this gap by bringing together work on uncertainty quantification, OOD detection, and explainability. We see this as an important step in developing decision-making systems that can reason about model competency and respond safely in the face of uncertainty.

3 Approach for Understanding Model Competency

Recall that *model competency* is a generalized form of predictive uncertainty that captures data (or aleatoric) uncertainty, model (or epistemic) uncertainty, and distributional shift [4]. To gain a deeper understanding of perception model competency, we explore five methods for visualizing regions in the input image that contribute most to a lack of competency (similar to saliency methods discussed in Sect. 2.3). Towards this end, we first compute a competency score, estimating the confidence in a model's prediction for a given input image (Sect. 3.1).

If the model is deemed incompetent (i.e., the competency score is below some threshold), we seek to understand which regions in the image most lead to this lack of competency by estimating the dependence of the competency score on segmented regions in the image (Sect. 3.2).

3.1 Estimating Model Competency

Let f be the true underlying model from which our images are drawn and $\hat{f}$ be the predicted model (or the perception model). For an input image, $\boldsymbol{X}$, the perception model aims to estimate the true class of the image, $f(\boldsymbol{X})$, from the set of all classes. The competency of the model for this image is given by

$$C(\boldsymbol{X}) := P\big(\{\hat{f}(\boldsymbol{X}) = f(\boldsymbol{X})\}|\boldsymbol{X}\big). \tag{1}$$

To simplify our notation, let $\hat{c}$ be the class predicted by the deterministic perception model (i.e., $\hat{f}(\boldsymbol{X}) = \hat{c}$), so we can express the model competency as

$$C(\boldsymbol{X}) = P\big(\{f(\boldsymbol{X}) = \hat{c}\}|\boldsymbol{X}\big). \tag{2}$$

It is difficult to estimate this probability directly because we are limited by the data contained in the training sample. Let D be the event that the input image is in-distribution (i.e., drawn from the same distribution as the training samples) and write the following lower bound on competency:

$$\begin{aligned} C(\boldsymbol{X}) &\geq P\big(D \cap \{f(\boldsymbol{X}) = \hat{c})\}|\boldsymbol{X}\big) &(3)\\ &= P\big(\{f(\boldsymbol{X}) = \hat{c}\}|\boldsymbol{X}, D\big)P(D|\boldsymbol{X}). &(4)\end{aligned}$$

These probabilities are not readily available but can be approximated [4]. To estimate the first probability, we can fit a calibrated transfer classifier, such as the logistic regression classifier, and use its class probability output to obtain an estimate, $\hat{p}_{\hat{c}|D}(\boldsymbol{X})$. To estimate the probability that an image is from a class distribution, we can model the training distribution as a Gaussian mixture model (GMM) and compute the Mahalanobis distance of the feature vector to each of the class distributions. We then use these distances as input to another logistic regression classifier to estimate the probability that the input image came from the training distribution, $\hat{p}_D(\boldsymbol{X})$. Using these approximations, the lower bound on the probability of model competency can be approximated as

$$C(\boldsymbol{X}) \geq \hat{p}_{\hat{c}|D}(\boldsymbol{X})\hat{p}_D(\boldsymbol{X}). \tag{5}$$

We refer to this expression as the *competency score*. This metric ties together work on uncertainty estimation (Sect. 2.1) and OOD detection (Sect. 2.2), using a deterministic classification-based approach to estimate model uncertainty assuming a sample is in-distribution and a distance-based approach to estimate the probability that a sample is in-distribution. Note that the competency score estimates a probability and is thus always between zero and one. If the model is 100% competent on a given image, the probability that the model's prediction

is correct is one. If the model is entirely incompetent, the probability that the prediction is correct is zero. The competency score is thus between zero and one, with higher scores corresponding to higher levels of competency. Another important aspect of this metric, that distinguishes is from many existing uncertainty and OOD estimates, is that it is differentiable with respect to the input image.

3.2 Identifying Regions Contributing to Low Competency

We explore five approaches for estimating the dependence of the competency score on regions in the input image to understand the extent to which different regions in the image contribute to low model competency.

Approach 1: Image Cropping. The first naive approach we consider is referred to as *Cropping* and is described in Fig. 1. For this approach, we simply compute the competency score for each region in the image within a grid. We tune the height and width of the grid cells to achieve the best performance.

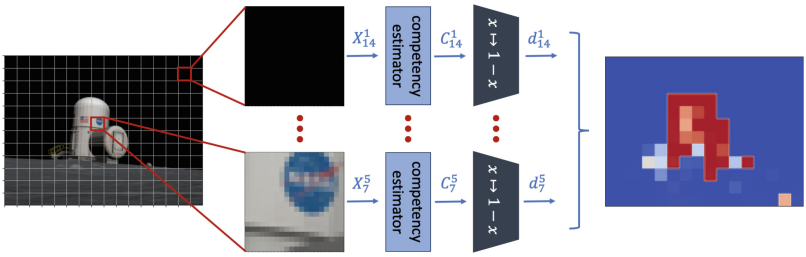

Fig. 1. Image cropping approach for identifying low competency regions. We partition the image into grid cells, crop the image around each grid cell to obtain a new image, X_j^i, and compute the competency score, C_j^i, for each cropped image. Regions with lower scores are associated with lower levels of competency and are said to contribute more to the overall low model competency, resulting in a higher dependency score, d_j^i.

Approach 2: Segment Masking. We refer to the second approach considered as *Masking*. In this approach, we compute the competency score for each segment in the image, while masking out the rest of the image, as shown in Fig. 2. We explore various methods for masking in an attempt to capture "feature missingness" [60]–blurring regions that are not of interest, adding noise to those regions, or replacing those pixels with zeros, ones, average values, uniform random values, or Gaussian random values–and select the highest-performing method.

Approach 3: Pixel Perturbation. The third approach, which is closely related to the second, is referred to as *Perturbation* and is described in Fig. 3. In this approach, we measure the increase in competency achieved by perturbing segments of the image. This approach is related to perturbation approaches in

XAI that predict the importance of pixels on a model's prediction by perturbing those pixels and observing the resulting change in output [43]. We consider various methods for perturbing regions of interest, as discussed for *Masking*.

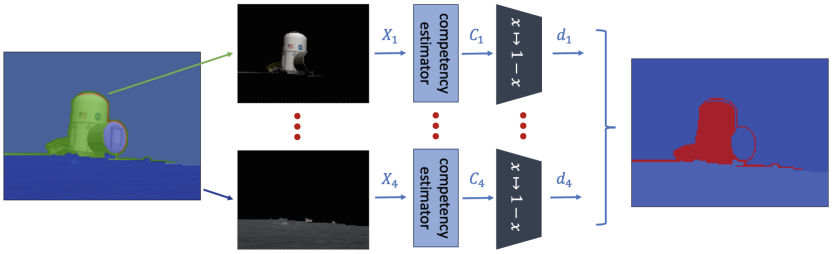

Fig. 2. Segment masking approach for identifying low competency regions. We begin by segmenting the image using the Felzenszwalb segmentation algorithm [61]. For each segment determined by this algorithm, we mask out the rest of the image to obtain a new image, X_i, and compute the competency score, C_i, of that masked image. Segments with lower corresponding competency scores are said to contribute more to the overall low model competency for that image, resulting in a higher dependency score, d_i.

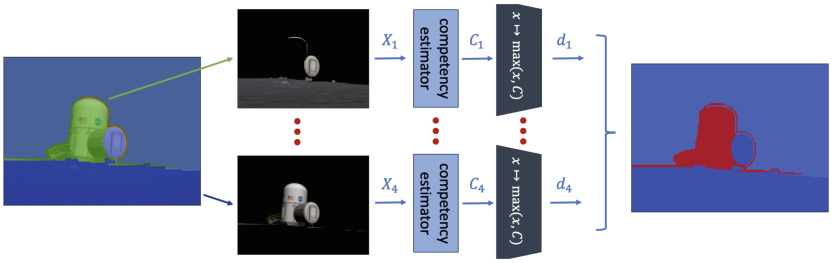

Fig. 3. Pixel perturbation approach for identifying low competency regions. We begin by segmenting the image using the Felzenszwalb algorithm [61]. We then perturb each segment determined by this algorithm to obtain a new image, X_i, and compute the competency score, C_i, for each of these perturbed images. If the competency score increases (above C) after perturbing a given region, that region is believed to contribute more to the overall low model competency, resulting in a higher dependency score, d_i.

Approach 4: Competency Gradients. The fourth approach, which we refer to as *Gradients*, relies on gradient information relating the input image to the estimated competency score and is described in Fig. 4. The competency score (defined as C in Sect. 3.1) is a differentiable function of the input image, $\boldsymbol{X}$. Therefore, we can compute the partial derivative of the competency score with respect to each of the pixel values in the input image. If the derivative is large, then small changes in this pixel value will significantly affect the competency score, and we say that model competency is more dependent on this pixel. This approach is similar to gradient-based approaches in XAI [41] but considers the dependence of the competency score on segmented regions, rather than the dependence of the model output on individual pixels.

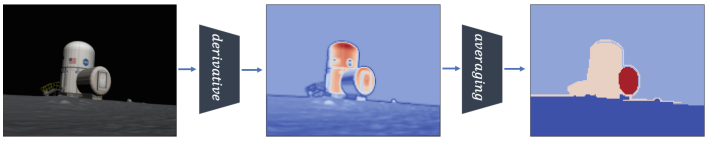

Fig. 4. Competency gradients approach for identifying low competency regions. We begin by computing the partial derivative of the overall competency score with respect to each of the pixel values in the input image. We then calculate the average derivative over segmented regions in the image, using the Felzenszwalb algorithm [61].

Approach 5: Reconstruction Loss. The fifth and final approach we considered is referred to as *Reconstruction* because it depends on the reconstruction loss of an autoencoder trained to fill in missing regions of an input image, as shown in Fig. 5. The autoencoder is designed to reconstruct the original image from the feature vector used for prediction by the perception model. We assume that the ability of the autoencoder to accurately reconstruct a region of an image reflects the familiarity of the model with that image region and thus the dependence of low model competency on that region. Reconstruction-based approaches have been explored for anomaly detection [32] and localization [62] but have not yet been explored as an explainability tool. For determining the reconstruction loss across segmented regions in an image, we considered different methods and loss functions and selected those that achieved the best performance.

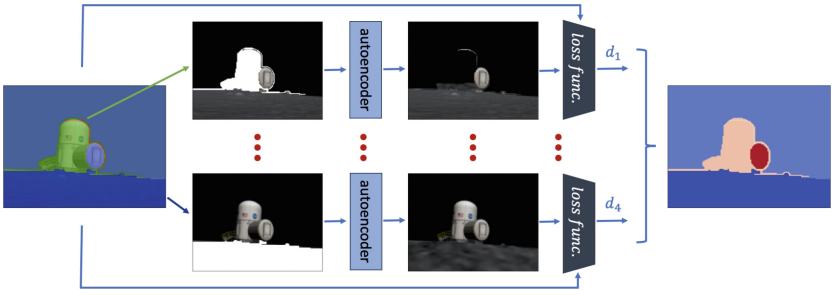

Fig. 5. Reconstruction loss approach for identifying low competency regions. We begin by segmenting the image using the Felzenszwalb algorithm [61], then generate masked images for each segment. Given an image with a segment masked out using ones, an autoencoder aims to predict the pixels of the original image. The difference between the original image and the predicted image is the reconstruction loss. If the reconstruction loss is high for a given image segment, this segment is believed to be unfamiliar to the perception model, and the overall low model competency is said to be more dependent on this region in the image, resulting in a higher dependency score, d_i.

4 Method Evaluation and Analysis

We provide a thorough analysis of the performance and usefulness of each of these five approaches for identifying regions contributing to low model competency. We conduct analysis across three unique datasets that assess the ability to identify unfamiliar objects (Sect. 4.2), detect regions associated with unseen classes (Sect. 4.3), and recognize unexplored areas in an environment (Sect. 4.4).

4.1 Metrics for Comparison

Following much of the work in explainable image classification (Sect. 2.3), we provide a *visual* comparison of the outputs of the five approaches. This visual analysis allows us to better understand the method outputs and determine the extent to which we trust each of the methods for explaining model competency. It also helps to determine whether the outputs of these methods offer explanations that are helpful to and interpretable by a human user. While we cannot display every example image in this paper, we include a few representative images.

While visual analysis is useful, it may be misleading to rely solely on a qualitative comparison [60]. In an effort to provide a more rigorous quantitative comparison, we develop a metric for *accuracy* that evaluates the ability of each of the five approaches to identify regions that are unfamilar to the model. Each image in the test set is manually labeled to indicate which regions contain features that were not present in the training set. A pixel that corresponds to a feature that was not present in the training set is considered a positive sample, and a pixel that corresponds to a feature that was present is considered a negative sample. We then assess the ability of the five approaches to label the dataset in the same way, measuring the overall accuracy, true positive rate (TPR), true negative rate (TNR), positive predictive value (PPV), and negative predictive value (NPV) averaged across all of the images in the test set.

Finally, in order for any of these approaches to truly be useful for decision-making, we must consider the *computation time* required to identify regions contributing to low competency. While it is valuable to provide users a better understanding of how their model works, when it is incompetent, and what leads to incompetency, it is desirable for systems to recognize their level of competency and respond appropriately without the need for human intervention. In order for this autonomy to be realistic, the system needs to reason about its competency fast enough that doing so does not interrupt the decision-making process.

4.2 Dataset 1: Lunar Environment

The first dataset we consider was obtained from a lunar environment developed by the Johns Hopkins University Applied Physics Laboratory in the Gazebo simulator, in which the training data contains images from an uninhabited moon and the test data contains images from a habited moon. While the training images only contain the lunar surface, the sky, and shadows, the test images additionally contain astronauts and human-made structures. This dataset assesses the

ability of the regional competency approaches discussed in Sect. 3.2 to identify unfamiliar objects. A visual comparison of these approaches is shown in Fig. 6 and a quantitative comparison is given in Table 1. We also consider the performance of existing class activation mapping techniques for identifying unfamiliar objects in Fig. 11 and Table 4 of Appendix B.1.

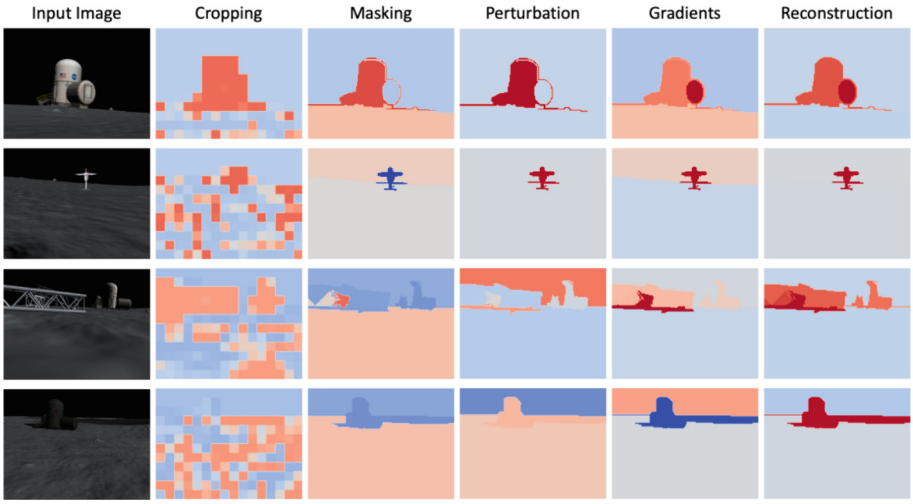

Fig. 6. Visual comparison of regional competency explanation methods for a lunar environment. Rows correspond to different input images and columns to different techniques. More red regions are more associated with low levels of model competency. (Color figure online)

Upon visual inspection, it is apparent that the *Cropping* method can roughly identify regions in the lunar images with unfamiliar objects, but this method is quite noisy, limiting its utility. This method has the worst overall accuracy, as it is quite conservative and often classifies regions that should be familiar to the model as ones leading to low model competency. In addition, this method is the slowest by far and may be too slow for many decision-making problems.

The four approaches that use segmentation appear more useful for identifying unfamiliar objects. The *Masking* method does well on some examples in the lunar dataset, but it also often inverts the expected order of dependency, giving low dependency scores to many regions that contain unfamiliar objects. The outputs of the *Perturbation* method more often align with what is expected, but this method also fails to identify several unfamiliar objects. The *Gradients* method tends to perform quite well when the unfamiliar structures look significantly different from elements of the uninhabited moon, but its performance deteriorates when the image contains dark structures that blend into the background.

These three methods perform comparably in terms of overall accuracy, but the *Gradients* method more often identifies regions as containing unfamiliar

features, resulting in a higher TPR and lower TNR compared to the *Masking* and *Perturbation* methods. Visually, the *Reconstruction* method appears to be the most useful and reliable; it consistently assigns high scores to unfamiliar structures and generally always seems to outperform the other methods. Quantitatively, it also distinguishes between familiar and unfamiliar features most accurately. This method achieves the highest overall accuracy, along with the highest TPR, TNR, PPV, and NPV. In terms of computation time, the *Gradients* method is fastest, but the *Masking*, *Perturbation*, and *Reconstruction* methods are also reasonably fast, obtaining their outputs in around 0.15 s or less.

Table 1. Quantitative comparison of regional competency explanation methods for a lunar environment. A true positive occurs when a method correctly identifies a pixel as one corresponding to an unfamiliar object (i.e., an astronaut or human-made structure), and a true negative occurs when a method identifies a pixel as one corresponding to a feature present in the training set (i.e., the sky, the lunar surface, or a shadow).

Method	Average Time (sec)	Average Accuracy (%)				
		Overall	TPR	TNR	PPV	NPV
Cropping	1.539	71.27	70.01	72.97	26.48	92.03
Masking	0.151	84.01	33.07	92.99	65.06	87.62
Perturbation	0.116	83.53	29.74	92.57	59.22	87.02
Gradients	**0.070**	82.16	51.32	87.59	64.44	91.01
Reconstruction	0.081	**96.16**	**88.87**	**98.83**	**93.53**	**95.95**

4.3 Dataset 2: Speed Limit Signs

The second dataset contains speed limit signs in Germany [63]. While the training dataset is composed of common speed limit signs (30 km/hr and higher), the test dataset set also contains an uncommon speed limit (20 km/hr). This dataset assesses the ability of the regional competency approaches to identify regions associated with unseen classes. A visual comparison of these approaches is shown in Fig. 7 and a quantitative comparison is given in Table 2. We also consider the performance of existing class activation mapping techniques for identifying regions associated with unseen classes in Fig. 12 and Table 5 of Appendix B.2.

Inspecting outputs across examples in the speed limit signs dataset, it appears that the *Cropping* method is not useful for this dataset, offering very little information through its outputs. The *Masking* method also does not appear very useful, as the outputs generally do not align with our expectations about the dependence of low model competency on elements of the unseen class (i.e., the number 2). Both of these methods have very low overall accuracy.

Compared to the first two methods, *Perturbation* and *Gradients* appear to offer more useful information. However, while both methods will sometimes assign higher dependency to the digit associated with the unseen class, the *Perturbation* method is not the most reliable and the *Gradients* method often places

Table 2. Quantitative comparison of regional competency explanation methods for a speed limit dataset. A true positive occurs when a method correctly identifies a pixel as one corresponding to a feature associated with an unseen class (i.e., the number 2), and a true negative occurs when a method identifies a pixel as one corresponding to a feature present in the training set (i.e., other parts of the traffic sign and background).

Method	Average Time (sec)	Average Accuracy (%)				
		Overall	TPR	TNR	PPV	NPV
Cropping	24.27	22.21	54.37	21.08	2.30	93.06
Masking	5.17	47.57	20.32	48.44	4.55	92.14
Perturbation	5.16	83.76	32.24	85.51	12.77	97.33
Gradients	0.319	83.98	**60.99**	84.85	12.76	98.41
Reconstruction	**0.109**	**92.19**	58.14	**93.43**	**26.20**	**98.44**

a higher dependency on the zero digit than it does on the two. While these two methods appear very similar in terms of their overall accuracy, the *Gradients* method has a much higher TPR, while maintaining a similar TNR.

Both visually and quantitatively, the *Reconstruction* method appears to perform the best. This method often indicates that low model competency is highly dependent on the digit 2, which is associated with an unseen class. It also achieves the highest overall accuracy, TNR, PPV, and NPV. In addition, this method has the lowest average computation time, maintaining a time of around 0.1 s.

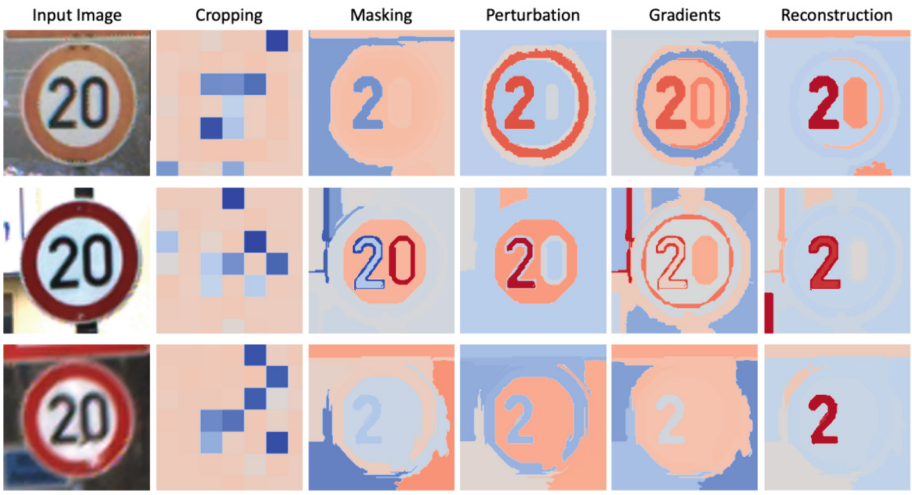

Fig. 7. Visual comparison of regional competency explanation methods for a speed limit dataset. Rows correspond to different input images and columns to different techniques. More red regions are more associated with low levels of model competency. (Color figure online)

4.4 Dataset 3: Outdoor Park

The third and final dataset contains images from regions in a park. While the training dataset only contains images from forested and grassy regions of the park, the test datset additionally includes images from around the park's pavilion. This dataset assesses the ability of the regional competency approaches to identify unexplored areas in an environment. A visual comparison of these approaches is shown in Fig. 8 and a quantitative one is given in Table 3. We also consider the performance of existing class activation mapping techniques for identifying unexplored regions in Fig. 13 and Table 6 of Appendix B.3.

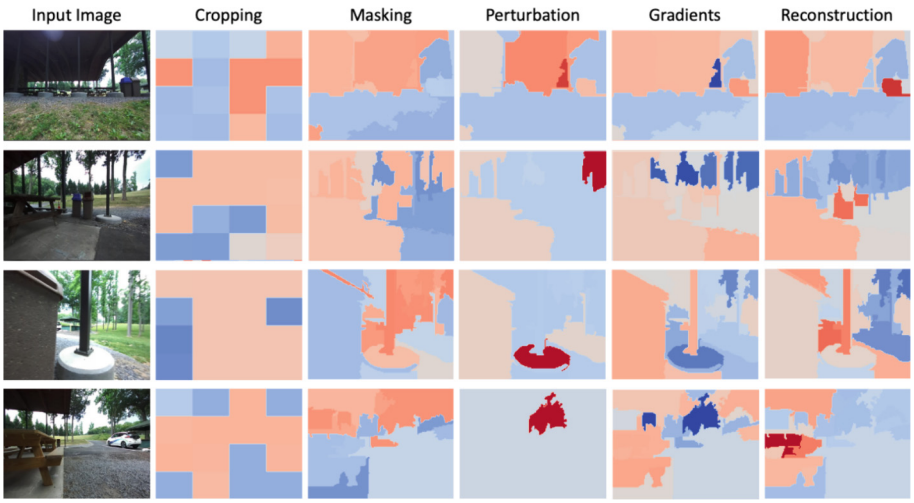

Fig. 8. Visual comparison of regional competency explanation methods for an outdoor park. Rows correspond to different input images and columns to different techniques. More red regions are more associated with low levels of model competency. (Color figure online)

As with the speed limit signs dataset, the *Cropping* method appears to offer little explanation for low model competency with the park dataset. The *Masking* and *Perturbation* methods sometimes identify unexplored regions as expected, but they do not consistently assign these regions higher incompetency dependency scores. These three methods all have low overall accuracy scores.

The *Gradients* and *Reconstruction* methods perform comparably, assigning high dependency scores to many of the image segments associated with unexplored areas and achieving similar levels of accuracy. Comparing these two methods, *Gradients* achieves a higher overall accuracy, TPR, and NPV, but *Reconstruction* achieves a higher TNR and PPV. The *Reconstruction* method is faster than *Gradients* in terms of computation time, but both methods are reasonably fast and could be used in most decision-making pipelines.

Because neither the *Gradients* nor *Reconstruction* method clearly outperforms the other, it is interesting to consider how they might be used in combination. The average accuracy achieved using the average of the outputs of these two methods is shown in the last row of Table 3. Combining these two methods increases the overall accuracy to over 80% and the TPR to over 90%. Using this technique, the NPV is also higher, but the TNR and PPV are lower. While using two methods would require more computational effort, the computation time is still within reason to be useful for decision-making.

Table 3. Quantitative comparison of regional competency methods for an outdoor park environment. A true positive occurs when a method correctly identifies a pixel as one corresponding to an unexplored region of the environment (i.e., areas around the pavilion), and a true negative occurs when a method identifies a pixel as one corresponding to a feature present in the training set (i.e., forested and grassy regions).

Method	Average Time (sec)	Average Accuracy (%)				
		Overall	TPR	TNR	PPV	NPV
Cropping	0.261	37.82	48.48	17.20	53.51	15.73
Masking	0.301	34.51	29.00	43.60	44.73	24.73
Perturbation	0.330	37.67	22.47	67.14	27.76	30.48
Gradients	0.130	77.13	81.13	69.52	81.99	71.86
Reconstruction	**0.053**	75.56	79.35	**73.81**	**84.53**	67.49
Grad. + Reco.	0.183	**80.09**	**90.26**	62.33	81.22	**79.73**

4.5 Analysis of Results

Comparing all of our results across the three datasets in Fig. 9, we see that the *Cropping* method may provide a rough idea of image regions contributing to low model competency for simple datasets, but this method is unlikely to provide utility for more complex datasets. The *Masking* and *Perturbation* methods may be useful for some datasets, but it is unlikely that they would provide more explanatory power than the *Gradients* and *Reconstruction* methods. Among all of the regional competency methods, the *Reconstruction* method appears the most promising for identifying regions contributing to low model competency, but the *Gradients* method also has merit. While the *Reconstruction* method more reliably informs us of regions in the image that are unfamiliar to the classifier, the *Gradients* method more directly tells us which pixels the competency score actually depends on. Because these two methods inform us of slightly different aspects of model confidence, there may be value in combining their outputs.

From Fig. 10, both the *Gradients* and *Reconstruction* methods can obtain their outputs in around 0.1 s or less on a 2.3 GHz Quad-Core Intel Core i7 processor. While these methods are fast enough to be usable in most decision-making

pipelines, their computation times are dependent on image size, classifier architecture, and segmentation algorithm parameters, so computation time can vary across systems. While the increase in computation time seen for the speed limit signs dataset was not terribly significant for the *Gradients* and *Reconstruction* methods, it was quite significant for the *Cropping*, *Masking*, and *Perturbation* methods. The *Cropping* method is unlikely to run fast enough for most systems, and the *Masking* and *Perturbation* methods may be too slow for systems with larger images and a greater numbers of image segments.

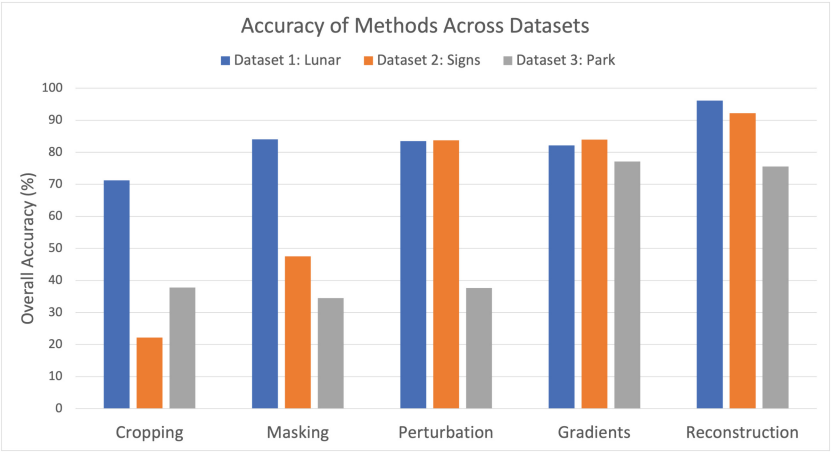

Fig. 9. Comparison of the overall accuracy of the five regional competency explanation methods in identifying regions of an image that are unfamiliar to the perception model.

From a visual comparison of the performance of the regional competency methods across the three datasets, it appears that these methods may be better at detecting unfamiliar objects and structures in an image (as with the lunar and park datasets), rather than identifying elements of unseen classes (as with the speed limit dataset). For the speed limit dataset, the '2' associated with the unseen class looks very similar to the '3' associated with a seen class. It is very possible that elements of the background appear more unfamiliar to the model than does the unseen digit. All five methods are much more precise when tasked with simply identifying features that are unfamiliar to the model, rather than those that the classifier should depend on but has not seen before.

Furthermore, these methods are generally more successful at detecting objects and structures that do not look very similar to elements in the training set. The *Gradients* and *Reconstruction* methods were generally effective for the park dataset but tended to assign low dependency scores to the gravel in unexplored areas of the park, which is unfamiliar to the model but looks similar to dirt and short grass that the classifier has seen. While the *Reconstruction* method

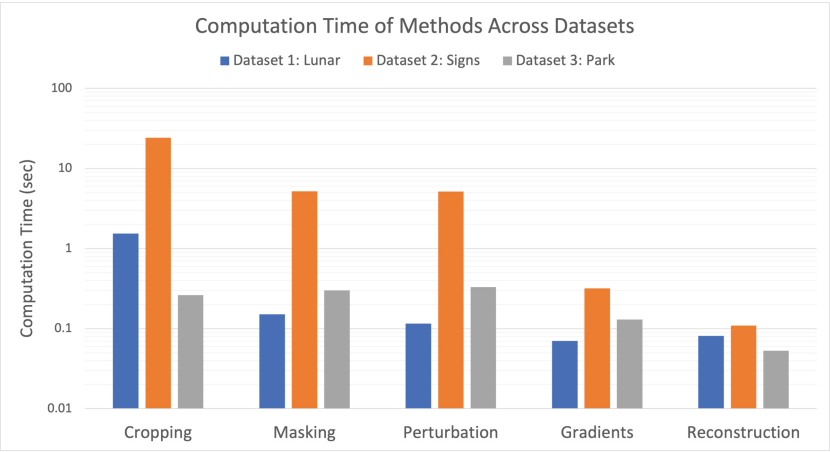

Fig. 10. Comparison of the computation time of the five regional competency methods.

was quite successful for the lunar dataset, the *Gradients* method struggled to identify dark and shadowy objects that blended into the lunar background.

In addition, these methods were generally most successful for the lunar dataset, which is relatively simple and can be more clearly segmented into a few distinct regions. The Felzenszwalb segmentation algorithm [61] is useful because it is fast and does not need to be trained on a specific dataset to work effectively. However, it often produced strange results, particularly for the speed limit and park datasets, affecting the performance of the four approaches that utilized this algorithm. These methods may be improved by using a different segmentation algorithm–especially one that is domain-specific.

Overall, the *Gradients* and *Reconstruction* methods show promise for identifying regions associated with low model competency, particularly when aspects of the image that are unfamiliar to the perception model cause this reduction in competency. These methods are generally accurate and fast enough to be used for decision-making, but additional work may be done to improve these methods.

5 Conclusions

In this paper, we sought to develop tools for understanding when DNN-based image classification models fail and why. We expand upon uncertainty quantification and OOD detection work to offer explanations for why a perception model lacks confidence in its prediction for a given input image by identifying specific regions in the image that lead to low model competency. Rather than offering explanations for classification decisions, as is typically done in explainable AI work, we aim to explain why a model is incompetent in its prediction.

With these goals in mind, we developed five distinct methods for identifying regions in an image contributing to low model competency, which we refer to as (1) *Image Cropping*, (2) *Segment Masking*, (3) *Pixel Perturbation*, (4) *Competency Gradients*, and (5) *Reconstruction Loss*. Each of these methods outputted an incompetency dependency score for each pixel in a set of input images. We evaluated each of these five approaches across three diverse datasets with the aim of determining which methods were most suitable for identifying unfamiliar objects, regions associated with unseen classes, and unexplored areas in an environment. We compared the dependency outputs of the five approaches visually, evaluated their ability to accurately identify image features that are unfamiliar to the classification model, and compared their computation times.

We found the *Reconstruction Loss* method to be the most effective and promising approach overall but also found merit in the *Competency Gradients* method. We believe both methods offer valuable information about why an image classification model is incompetent for a given input image, and their outputs can be obtained fast enough to be used in decision-making problems.

6 Limitations and Future Work

Additional work could be done to improve the performance of the regional competency estimation methods presented in this work. First, it may be worthwhile to explore different segmentation algorithms, particularly ones that are domain-specific or trained to work well with a particular dataset or environment of interest. For the *Reconstruction* method, it would be interesting to consider additional reconstruction loss functions, as well as ensembles of autoencoders. While the *Reconstruction* method is independent of the competency metric, the other four methods are directly dependent on the chosen metric. In this work, we only considered a single metric for competency, which incorporates different facets of uncertainty into a single score. It would be interesting to disentangle these sources of uncertainty or consider different metrics for competency. While the *Cropping*, *Masking*, and *Perturbation* methods do not have strict requirements for the chosen competency metric, the *Gradients* method is only useful if the competency score is differentiable with respect to the input image.

Beyond these more immediate next steps for improving the proposed methods, there are several other avenues for future work. First, this work focused on explaining low model competency for OOD images with regional features that are clearly unfamiliar to the perception model. Additional work would need to be done to understand low model competency for more nuanced cases, particularly with in-distribution images, and to understand high model competency. One could explore other methods for explaining model competency that are not focused on identifying particular regions contributing to low competency. It would be worthwhile to identify other types of image features that might contribute to low competency, such as image brightness. This work also seeks to understand perception model competency of trained models without modifying the underlying model architecture. It would interesting to also explore antehoc methods from the XAI literature to better understand model competency.

The approaches explored in this paper can be used to provide users a deeper understanding of how a model works and when it might fail. Additional work could be done with this goal in mind to combine visual explanations with language. Future work should also explore how the regional competency information presented in this paper should play into decision-making problems. While these methods provide perception-based systems with more information about why an image classification model is incompetent, it is still not clear how the system should respond. Additional work would need to be done to decide the appropriate response to model incompetency for different applications. Finally, the work presented in this paper focused on image classification problems. Additional work should be done to understand how these approaches could be adapted for regression, semantic segmentation, object detection, and other problems.

Disclosure of Interests. The authors have no competing interests to declare that are relevant to the content of this article.

A Data Sources and Algorithm Parameters

To reproduce this work, data is available on Drive, code is available on GitHub, and the chosen model architectures and training parameters are available on Drive. For more information on reproducing this work, please see this README.

B Comparison to Class Activation Maps

The goal of the work presented in this paper is to identify regions in an image contributing to low model competency and to display this incompetency dependency in a saliency map. This is closely related to work on developing class activation maps (CAMs), which are class-discriminative saliency maps that identify key regions in an image used by the model to make a particular class prediction. There are various methods for generating CAMs [64], but we choose to focus on a few representative approaches: Grad-CAM [41], Guided Grad-CAM [41], Grad-CAM++ [65], Integrated Gradients [42], SmoothGrad [66], DeepLIFT [48], Score-CAM [45], Ablation-CAM [67], Eigen-CAM [68], and LayerCAM [69]. These methods were implemented with the help of the PyTorch library for CAM methods [70] and the Captum model interpretability library for PyTorch [71].

The chosen mapping approaches are designed to explain model predictions, but we evaluate their ability to recognize unfamiliar objects (Sect. B.1), identify regions associated with unseen classes (Sect. B.2), and determine unexplored regions in an environment (Sect. B.3). We find that these approaches are not suited for generating competency maps, as is done in our work.

B.1 Dataset 1: Lunar Environment

Table 4. Quantitative comparison of CAM methods for a lunar environment.

Method	Average Time (sec)	Average Accuracy (%)				
		Overall	TPR	TNR	PPV	NPV
Grad-CAM	0.03	66.18	32.50	71.24	25.32	84.34
Guided Grad-CAM	0.04	76.24	30.97	83.88	35.94	88.89
Grad-CAM++	0.04	60.80	35.76	65.88	22.19	85.76
Integrated Gradients	0.46	71.88	21.26	79.34	20.47	87.58
SmoothGrad	2.68	77.51	18.19	85.88	19.32	87.92
DeepLIFT	0.05	73.36	24.01	80.58	20.67	87.96
Score-CAM	0.13	62.40	38.94	67.26	23.42	87.18
Ablation-CAM	0.23	62.39	35.44	67.20	21.52	85.76
Eigen-CAM	16.20	71.90	36.22	78.21	31.17	89.05
LayerCAM	0.11	73.81	19.87	82.31	15.68	87.25

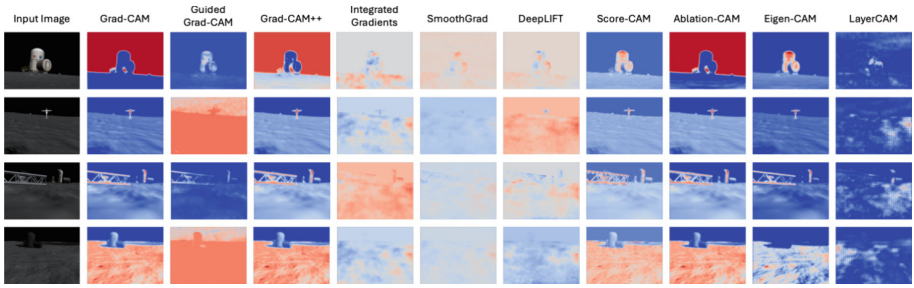

Fig. 11. Visual comparison of class activation maps (CAMs) for a lunar environment. Rows correspond to different input images and columns to different techniques for generating saliency maps. More red regions contribute more to the model's prediction. (Color figure online)

B.2 Dataset 2: Speed Limit Signs

Table 5. Quantitative comparison of CAM methods for a speed limit dataset.

Method	Average Time (sec)	Average Accuracy (%)				
		Overall	TPR	TNR	PPV	NPV
Grad-CAM	0.41	69.88	36.78	71.01	4.53	97.07
Guided Grad-CAM	0.60	83.15	17.39	85.39	5.65	96.59
Grad-CAM++	0.41	51.01	18.57	52.03	1.71	94.87
Integrated Gradients	18.06	51.43	49.46	51.50	3.33	96.78
SmoothGrad	80.19	50.30	49.89	50.31	3.28	96.78
DeepLIFT	0.87	52.90	48.20	53.06	3.35	96.80
Score-CAM	4.18	–	–	–	–	–
Ablation-CAM	4.11	51.72	25.95	52.54	2.17	95.30
Eigen-CAM	88.97	71.66	62.17	71.99	6.78	98.27
LayerCAM	0.62	87.59	54.00	88.78	14.09	98.21

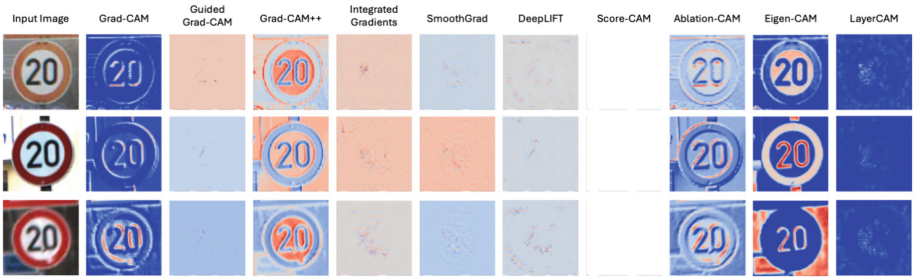

Fig. 12. Visual comparison of class activation maps (CAMs) for a speed limit dataset. Rows correspond to different input images and columns to different techniques for generating saliency maps. More red regions contribute more to the model's prediction. (Color figure online)

B.3 Dataset 3: Outdoor Park

Table 6. Quantitative comparison of CAM methods for an outdoor park environment.

Method	Average Time (sec)	Average Accuracy (%)				
		Overall	TPR	TNR	PPV	NPV
Grad-CAM	0.03	46.62	33.22	74.19	72.27	35.69
Guided Grad-CAM	0.04	34.43	13.30	78.22	60.28	30.69
Grad-CAM++	0.04	30.60	26.10	49.83	45.23	22.01
Integrated Gradients	0.46	30.69	13.25	67.67	34.40	28.66
SmoothGrad	2.65	33.98	22.23	59.38	43.78	28.68
DeepLIFT	0.05	31.57	18.97	57.25	45.10	26.82
Score-CAM	0.13	29.05	21.72	42.66	40.99	22.07
Ablation-CAM	0.22	32.16	20.99	55.19	46.16	25.71
Eigen-CAM	15.79	27.22	16.82	47.69	36.88	23.51
LayerCAM	0.10	31.27	14.96	63.28	44.79	27.66

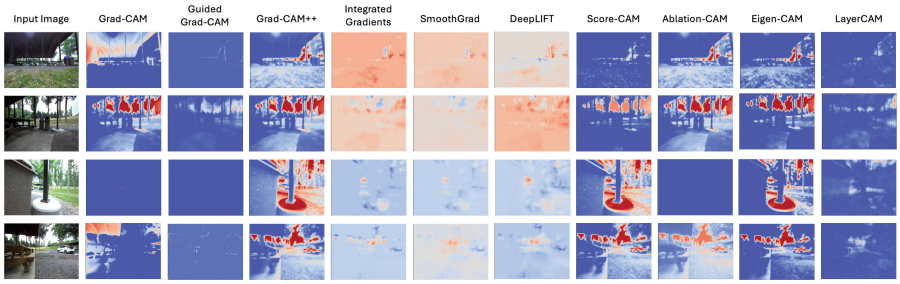

Fig. 13. Visual comparison of class activation maps (CAMs) for an outdoor park. Rows correspond to different input images and columns to different techniques for generating saliency maps. More red regions contribute more to the model's prediction. (Color figure online)

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A Guided Tour of Post-hoc XAI Techniques in Image Segmentation

Syed Nouman Hasany[1(✉)], Fabrice Mériaudeau[2], and Caroline Petitjean[1]

[1] Université de Rouen Normandie, Mont-Saint-Aignan, France
{syed-nouman,caroline.petitjean}@univ-rouen.fr
[2] Université de Bourgogne, Dijon, France
fabrice.meriaudeau@u-bourgogne.fr

Abstract. Deep learning models have shown tremendous gains in computer vision in the last decade. However, given their highly non-linear nature, they are often seen as black-boxes. This has led to the development of eXplainable Artificial Intelligence (XAI) as a parallel field with the aim of investigating the behavior of deep learning models. Research in XAI, however, has almost exclusively been focused on image classification models. Dense prediction tasks such as image segmentation have received little attention. The last few years have seen a shift in this trend with works focusing on exploring XAI in the context of image segmentation. A fair number of these works have borrowed from XAI techniques proposed in the context of image classification. It is safe to assume that going forward the number of XAI techniques focused on image segmentation are bound to increase. Reviewing the journey of XAI in image segmentation thus far would therefore be ideal, and is the goal of the present work. This review aims at presenting an overview of the XAI techniques proposed in the context of image segmentation. Another goal is to highlight the lack of interest in this field and its potential causes as well as to comment on potentially underexplored avenues. Given the relative nascency of the field, no review papers currently exist, a gap this work aims to fill.

Keywords: image segmentation · explainability · interpretability · XAI · review

1 Introduction

AlexNet's [47] breakthrough on ImageNet [13] in 2012 proved to be the pivotal moment in the evolution of computer vision. Going forward, the impact of classical computer vision declined, and that of deep learning increased. Since then, the field has been dominated by deep learning architectures, the most prominent of them being convolutional neural networks (CNNs). In recent years, owing to their competitive performance, transformer based architectures such as the vision transformer [15] have also started becoming popular. Whereas AlexNet was proposed to solve an image classification task, deep learning's influence has

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since broadened to include various other computer vision tasks as well such as image segmentation [54], object detection [91], pose estimation [92], image generation [87], etc.

Given the highly non-linear nature of deep learning algorithms, a parallel field of research soon came into existence, that of XAI (eXplainable Artificial Intelligence). Research in XAI is broadly concerned with understanding a deep learning model's behavior. For example, one sub-field concerns itself with trying to understand why a model arrived at a particular decision. The earliest techniques in this regard were proposed as occlusion analysis [88] and image-specific class saliency [73] in 2013 and 2014 respectively. These techniques were focused on identifying regions in an input image which were important for the model to arrive at its decision. This sub-field has received significant traction as is evident from the popularity of algorithms such as LIME [63], SHAP [52], RISE [61], Grad-CAM [70], etc. Another sub-field concerns itself with investigating the kind of representations learned by a model. An example is [73] in which representative examples of a given class are generated for a CNN. This allows researchers to identify patterns which a CNN has associated with each class.

While research in XAI commenced within an year of the deep learning revolution, the initial focus of the field remained almost exclusively on image classification. In recent years, however, dense prediction tasks, such as image segmentation, have also started receiving attention in the context of XAI. Due to the attention XAI in image classification has received, multiple survey papers detail the field's progression in this context [2,67]. No review paper, however, exists for XAI in image segmentation. We aim to fill this gap by providing an overview of the post-hoc techniques which have been proposed in the context of XAI in image segmentation.

One of the main reasons as to why image segmentation has received relatively little attention lies with the utility of a saliency map in the context of a dense prediction task. Given that the prediction is already in the spatial domain, it is arguable as to whether the end-user will find the saliency map to be of any use [17]. Nevertheless, XAI research in image segmentation is still dominated by saliency generation methods. Other than that, research has also focused on detecting biases within a segmentation model [38], identifying useful concepts learned by a segmentation model [17], generating insights about the segmentation model [58], etc.

2 Categorization of XAI Techniques for Image Segmentation

In order to search for relevant papers, we used two approaches. In the first approach, we searched for abstracts which mentioned **XAI**, **explainability**, or **interpretability**, along with **segmentation**[1] on Google Scholar. For the second approach, we considered some of the most widely cited papers relevant

[1] As well as morphological variants of these words.

to XAI in image segmentation [38,65,78,84]. Following a recursive approach, we exhaustively searched for works which had cited these papers followed by those which had cited the ones found in the previous step, and so forth. Ending up with **81** papers, we classified these into two categories. The first category consists of works in which techniques are proposed (**39** papers) whereas the second category consists of works in which techniques are utilized as tools (**42** papers). Barring passing mentions of the second category, in this review paper we have primarily focused on the first category. A complete list of techniques belonging to the first category can be found in Appendix A.

Techniques in XAI can generally be classified as belonging to local XAI or global XAI. Local XAI concerns itself with techniques focused on understanding the model's decisions for individual input samples. In this context the terms interpretability and explainability are often interchangeably used. We have opted for 'explainability', and consider algorithms falling under this category as those aiming at explaining the model's decision to an end user. These explanations generally take the form of saliency maps (also referred to as attribution maps or heatmaps). On the other hand, global XAI concerns itself with techniques focused on investigating the model's overall characteristics; an example of this is the feature space carved out by the model's internal representations. Another category which overlaps with both local and global XAI is that concerned with the concepts a model has learned. Figure 1 shows a taxonomy of the XAI algorithm categories discussed in this review.

3 Review of XAI Techniques for Image Segmentation

3.1 Local XAI

For local XAI in image classification, we are interested in explaining the model's output score (y^c) for a target class given an input image. However, due to image segmentation being a dense prediction task, there are three possibilities when it comes to what we are interested in explaining: (i) the entire segmentation map for a target class, (ii) a region of a segmentation map for a target class, or (iii) an individual pixel from the segmentation map for a target class. Going forward, we shall refer to the region to be explained as $\mathcal{M}$. In the discussion to follow, we categorize local XAI in image segmentation into three categories: (i) methods relying on the intermediate feature space, (ii) methods relying on the input space, and (iii) perturbation based methods.

Explainability Methods Relying on the Intermediate Feature Space. One of the most popular saliency generation methods in image classification has been Grad-CAM [71]. Grad-CAM generates saliency as the linear combination of the activation maps of a given model layer. This saliency for class c can be expressed as:

$$L^c_{Grad-CAM} = ReLU\left(\sum_k \alpha^c_k \cdot A^k\right) \tag{1}$$

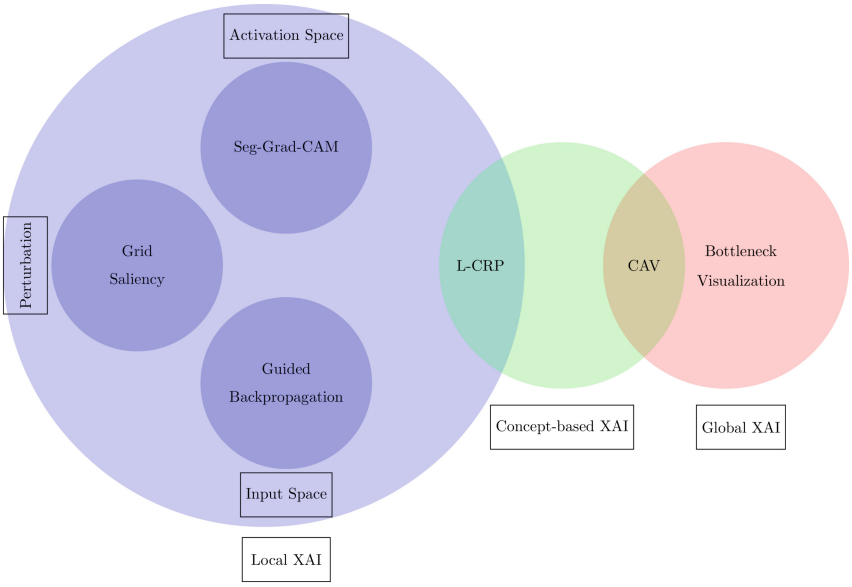

Fig. 1. Taxonomy of discussed XAI algorithms with a representative example for each category. Boxes indicates categories (and subcategories) whereas free text indicates instances of said categories.

A^k represents the kth activation map and α_k^c represents the kth coefficient. α_k^c is calculated in two steps. First, the gradient of the target class score with respect to the activation map (A^k) is computed, and this is followed by applying global average pooling (GAP) to this gradient matrix:

$$\alpha_k^c = GAP\left(\frac{\partial y^c}{\partial A^k}\right) \tag{2}$$

Since image segmentation returns a dense prediction instead of a single value, extending Grad-CAM to image segmentation required a necessary modification[2]. Primarily, two approaches can be tracked which aimed at extending Grad-CAM to image segmentation.

A pixel-wise application of Grad-CAM was proposed [80] leading to multiple saliency maps associated with individual pixels followed by an aggregation of these maps. Additionally, [80] also proposed the modification of Eq. (1) such that α_k^c is replaced by $\frac{\partial y_{i,j}^c}{\partial A^k}$ which is the gradient matrix, and the multiplication ($\cdot$) is replaced by the element-wise multiplication ($\odot$). This modification is proposed in order to avoid global average pooling (GAP) as GAP leads to a loss of spatial information owing to it collapsing the entire gradient matrix into a scalar. It is obvious that for a task such as image segmentation, this spatial information could have been crucial towards explaining the model's decision. This approach,

[2] Unless the goal is to explain the model's prediction for an individual output pixel.

named Grad-PAM, computes a map $L^c_{Grad-PAM}(i,j)$ for an individual pixel at location (i,j):

$$L^c_{Grad-PAM}(i,j) = ReLU\left(\sum_k \frac{\partial y^c_{i,j}}{\partial A^k} \odot A^k\right) \tag{3}$$

The most popular modification, however, is Seg-Grad-CAM [78] which works by replacing y^c in Eq. (2) with a sum of the target class' scores in the region of interest ($\mathcal{M}$). This leads us to α^c_k as being:

$$\alpha^c_k = GAP\left(\frac{\partial \sum_{(i,j)\in\mathcal{M}} y^c_{ij}}{\partial A^k}\right) \tag{4}$$

Grad-CAM has been considerably well-received in image classification and this has led to many derivatives such as Grad-CAM++ [1], Guided Grad-CAM [70], LayerCAM [42], etc. aimed at improving upon the original technique. Similarly, in image segmentation, Seg-Grad-CAM has led to techniques such as Seg-GradCAM++ [30,49,56], Seg-XGrad-CAM [30], Seg-Eigen-CAM [30], Seg-ScoreCAM [30,56], Seg-Ablation-CAM [27,30], and Seg-XRes-CAM [34]. As is obvious, these techniques are primarily extensions of their namesakes in image classification to image segmentation. It is worth noting that not all derivatives of Grad-CAM are gradient based; some such as Score-CAM, and Eigen-CAM bypass gradients in order to avoid the associated problems such as the noisy gradient problem [75].

Even though Seg-Grad-CAM (and its gradient-based derivatives) can be used to generate saliency maps when it comes to explaining an entire segmentation map for a target class, it is best avoided when it comes to the explanation of a region of interest within that segmentation map for a target class. This is attributable to the issue of spatial information collapsing due to the presence of global average pooling (GAP) [80]. If one wishes an explanation for the model's prediction of a region located near the top right of an image, it is unlikely that the bottom left region of the image would have made a significant contribution towards the model's decision. GAP, however, ensures that all spatial locations of an activation map (A^k) in Eq. (1) are multiplied by the same scalar α^c_k without any regard for their spatial relationship in the image.[3] Similar to the formulation of Eq. 3, Seg-XRes-CAM [34] and Seg-HiRes-Grad CAM [62] extend Seg-Grad-CAM such that the GAP is avoided when generating saliency maps for a region of interest given a target class.

Grad-CAM in image classification is generally applied to the final feature extraction convolutional layer. This layer is a good choice as its representations not only contain the effective summary of the original image, but these are also the sole convolutional representations responsible for the model's output decision. The situation in image segmentation, however, is not as straightforward.

[3] Non-gradient-based algorithms such as Seg-Ablation-CAM fail too since they are computing linear coefficients based on the complete spatial presence or absence of an activation map.

For a standard segmentation network such as a U-Net [64], the convolutional representations responsible for the final prediction (near the end of the decoder) do not contain a summary of the original image. A more likely candidate to contain the summary is the bottleneck layer (end of the encoder). The bottleneck's representations, however, are not directly responsible for the final segmentation output. These reservations notwithstanding, the bottleneck has been a common choice when it comes to applying Grad-CAM to image segmentation. Aiming to bypass this issue, [56] proposed an 'Adapted' alternative in which the technique is individually applied to all of the decoder's layers, and the obtained saliency maps are then aggregated by summing them up (appropriate transformations are applied in order to account for the differing spatial dimensions of the saliency maps). This adapted approach was utilized with Adapted Seg-Grad-CAM, Adapted Seg-GradCAM++, and Adapted Seg-ScoreCAM. Additionally, [56] also proposed a gradient-agnostic method of determining the linear coefficients such that the coefficient associated with each activation map is a product of the dependent and independent contribution of the activation map to the final segmentation. Dependence and independence, in this case, are defined such that they take into account the contribution of the activation map to the final prediction given its presence.

Explainability Methods Relying on the Input Image Space. Whereas Grad-CAM requires access to the intermediate activation space, many XAI techniques proposed in the context of image classification work directly with the input image space instead. Computing the saliency map as the derivative of the model's output score with respect to the input image was the earliest such method [73]. Owing to the noisy nature of the gradient signal in a deep network, techniques have been proposed which aim towards 'cleaning' this signal. An early technique proposed the inhibition of negative gradients in order to only highlight those gradient signals which contributed positively towards the model's decisions [76]. This method was one of the earliest XAI methods from image classification to be extended to image segmentation [84]. Other techniques proposed in the context of cleaning image gradient include SmoothGrad [75] which aims at generating saliency maps for multiple noisy versions of the input image followed by an averaging based aggregation. Similarly, Integrated Gradients [77] works by generating saliency maps for multiple versions of the input image each of which differs from the other in terms of the image brightness. This, too, is followed by an averaging based aggregation. Both of these method have been extended to image segmentation [33,89]. Similar to the problem of extending Grad-CAM to image segmentation, the necessary output modifications are applied to the final segmentation prediction before the application to these methods. Such modifications include applying global average pooling [39] to the final segmentation prediction or taking a summation over the target class' scores in the region of interest ($\mathcal{M}$).

Perturbation Based Methods. Another class of explainability methods are based on tracking the model's behavior as systematic modifications (perturbations) are applied to the input image. Perturbations which significantly impacted the model's output serve as indicators towards identifying salient regions in the original image. These techniques are categorized as perturbation based techniques. Given that they treat a model as a black box, they possess the distinct advantage of being model agnostic. This is advantageous because techniques belonging to the CAM family [93] are not guaranteed to work for non-convolutional architectures, a pressing example of which are architectures utilizing transformer blocks [8,35]. Grid saliency [38] was the earliest attempt of utilizing a perturbation based technique to image segmentation. This was an extension of the Meaningful Perturbation technique [21] originally proposed in the context of image classification. Grid saliency iteratively modifies an input image by removing irrelevant regions in order to finally arrive at the necessary context required by the segmentation model for its correct segmentation. This modification is guided by the gradient of a loss function, the aim of which is to simultaneously preserve the model's prediction as well as to get rid of unnecessary regions. Examining necessary contexts for various images allowed [38] to identify potential biases which the segmentation model might have learnt during its training.

Another popular perturbation based approach in image classification is SHAP [52] which works by dividing an image into superpixels, and then aims at distributing the overall contribution of each superpixel to the model's final decision. This is achieved by feeding the input image to the model multiple times such that on each occasion certain superpixels are masked and the remaining unmasked. Upon multiple iterations, this allows us to quantify the contribution of each superpixel towards the model's final decision. This approach was extended to image segmentation as well [12] with a slight modification. Instead of using algorithms such as SLIC [4] to identify superpixels, the authors reported better results if the input image is divided into a hexagonal grid structure, followed by treating each hexagon as a superpixel. The same work also extended RISE [61], another perturbation based approach, to image segmentation. For image classification, RISE starts by generating multiple random masks which are multiplied with the original image leading to multiple masked versions of the original image. The saliency map is then defined as the linear combination of these masks with the linear coefficients obtained using the model's classification scores associated with each masked image. Extending both RISE and SHAP to image segmentation is straightforward with the minor modification of averaging over the prediction scores of our target class in a region of interest ($\mathcal{M}$) in order to convert it to a scalar.

U-Noise [45] proposed a relatively novel approach in which an independent model is learned on top of the existing segmentation model (frozen) with the aim of directly predicting a saliency map given an input image. The goal of the independent model is to generate a noise mask, the addition of which to the original input image would not lead to a deterioration of the segmentation model's

prediction. The rationale is that the predicted noise mask will contain more noise for regions in the input image that are less important to the segmentation model and less noise for more important regions. An identical approach does not exist in image classification. Some similarities, however, can be seen in [11] in which a model is learned to predict a mask which is then element-wise multiplied with the input image before feeding it to the classification model. The goal in both cases [11,45] is to learn an additional model that can predict a mask (noise mask in the case of U-Noise) in real-time such that the application of this mask to the input image does not harm the original prediction. A modification to U-Noise's optimization procedure was recently proposed arguing for the inclusion of bilateral filtering to generate smooth noise masks [59].

Concept Based Methods. Instead of generating a single saliency map, concept based methods aim at identifying the useful concepts present in the image which might have contributed towards the final segmentation decision. [80] extended [79] to image segmentation in which the segmentation model's decision for a single pixel is decomposed into a decision tree. A saliency map can be generated for each decision node allowing us to track the conceptual journey undertaken by the model in order to arrive at its final decision. Another approach was proposed by [17] in which the explanation of a segmentation model's prediction is decomposed in terms of its concepts. Filters responsible for the detection of specific concepts are first identified in the segmentation model followed by a conditional application of Layer-wise Relevance Propagation (LRP) [6]. This allows us to highlight the contribution of individual filters leading to a disentangled concept-specific explanation.

3.2 Evaluation of Local XAI Methods

The results of local XAI methods are visual, and therefore relatively qualitative in nature, a consequence of which has been a lack of an agreed upon evaluation methodology. Occasionally, proposed techniques have found it sufficient to simply display the saliency maps without providing a quantitative qualification of the proposed technique. Quantitative evaluation, however, is not completely absent, and some evaluation metrics have been proposed. One popular evaluation technique is faithfulness [66,67] which tracks the model's performance as the most significant pixels from the input image - as identified from the local XAI technique - are removed.

Specifically in the context of image segmentation, a pair of metrics were proposed [56] in order to evaluate the generated saliency maps. The saliency map is first used to mask the input image such that the unimportant pixels - as per the saliency map - are removed. This masked image is then fed to the segmentation model. The first of our metrics, 'Prediction Preserved Score' (PPS), records the percentage of the prediction which is preserved given this masked image as compared to the model's prediction on the original image. The second metric, 'Image Preserved Score' (IPS), acting as a complement to

the first, records the percentage of the original image which is retained in the masked image. The idea behind this pair is that a good saliency map would score highly on the PPS and low on the IPS.

An evaluation strategy utilizing the U-Noise model was recently proposed [72]. The saliency map is first utilized to mask the input image. This masked image is then fed to the U-Noise model. The statistics of the generated noise mask serve as our evaluation metric with the rationale being that less noise would be added if the saliency map was correct in its identification of the important regions.

3.3 A Comparative Analysis of Local XAI Methods

Figure 2 shows the saliency maps generated using Seg-Grad-CAM, Seg-XRes-CAM, Seg-Eigen-CAM, Seg-AblationCAM, and RISE on a few samples from the COCO-2017 dataset [50]. The first four of these algorithms belong to the intermediate feature space category whereas RISE is a perturbation based method. Between the first four methods, Seg-Grad-CAM and Seg-XRes-CAM are gradient dependent whereas Seg-Eigen-CAM[4] and Seg-AblationCAM are gradient independent. The segmentation model was a pre-trained DeepLabv3[5], and the intermediate feature space methods were applied to the bottleneck layer.

From a qualitative point of view, a few things are observable. First, some parts of the object appear more salient as compared to other parts. For example, in the case of 'Cat', the ears and the nose appear more prominent compared to other regions of the cat. Similarly, for the 'Dog' case, the nose is comparatively brighter than the rest of the body in all saliency maps barring that of Seg-Eigen-CAM. Secondly, except for Seg-Eigen-CAM, the three intermediate feature space algorithms broadly agree with each other. For Seg-Eigen-CAM, in the cases of 'Dog', 'Bike', and 'Train', the saliency maps, interestingly, highlight the background instead of the object itself - almost a complement of the other saliency maps in the same row. Lastly, even though saliency maps for RISE broadly agree with the rest as far as general localization is concerned, the maps are much coarser as compared to the others. Where the intermediate feature space methods might highlight specific regions of interest in some detail, the same cannot be expected from RISE.

For the same dataset and segmentation model, Fig. 3 shows the saliency maps generated using RISE with varying number of masks. Unsurprisingly, the number of masks play an important role in the saliency map generation. It appears that between 100 masks and 2000 masks, the saliency map becomes more refined in the sense of being more concentrated around the object of interest. An interesting case presents itself in the third row where the segmentation model has predicted two cats whereas it is obvious that the left one is a dog. Saliency maps generated from RISE prominently highlight the correct cat whereas the incorrect dog is almost completely ignored.

[4] A gradient driven version of Seg-Eigen-CAM is also possible.
[5] https://pytorch.org/hub/pytorch_vision_deeplabv3_resnet101/.

For a quantitative comparison we utilize a slighlty modified version of the pair of metrics from [56]. The saliency map is first binarized, and this binarized version is then used to mask out the original image. This masked image is then passed on to the segmentation model. We define our first metric, Dice Explained, as the dice score between the segmentation model's prediction on this masked image and the segmentation model's prediction on the original image. Our second metric is defined as the ratio of the number of pixels in the binarized saliency map to the number of pixels in the prediction. Figure 4 summarizes our metric calculation process. For binarization purposes, thresholds of 0.05 and 0.1 are experimented with for intermediate feature space methods, and thresholds of 0.2 and 0.4 are used with RISE.

Table 1 summarizes the results of applying our methods on a subset of the COCO-2017 dataset. RISE with 500 masks reports the best dice explained of **0.948**, whereas for intermediate feature space methods, Seg-AblationCAM leads at **0.843**. The saliency ratios for the two are **10.59** and **6.68** respectively. The worst dice explained with RISE is **0.747** (2000 masks) with a saliency ratio of **2.196**, and for intermediate feature space methods, it is that of Seg-Eigen-CAM at **0.354** with a saliency ratio of **4.83**. While it might be tempting to decide in favor of RISE or Seg-AblationCAM when it comes to generating saliency maps for image segmentation, the computational time paints a disheartening picture with average times around **2.5** min for Seg-AblationCAM and **1.5** to **5.5** min for RISE. Both Seg-Grad-CAM and Seg-XRes-CAM, operate much more swiftly taking less than a second on average. This is easy to explain as both RISE (a perturbation based method) and Seg-AblationCAM (an intermediate feature space method) require multiple iterations before they can compute the necessary coefficients in order to generate a saliency map. Seg-Grad-CAM and Seg-XRes-CAM, on the other hand, only require a single forward and backward pass.

3.4 Global XAI

Similar to image classification, global XAI approaches in image segmentation are considerably sparse compared to local XAI approaches. [10] investigated the sensitivity of a segmentation model to image features using an approach borrowed from activation maximization. An image for which the segmentation model predicts the background class for a region of interest is used as the initial image. The goal is for this image to be modified such that the model predicts the desired foreground class for the region of interest. The derivative of the foreground class' activation with respect to the input image serves as the guiding signal which gradient ascent uses in order to modify the initial image. This can be formulated as follows:

$$X_{j+1} = X_j + \alpha \cdot \frac{\partial i}{\partial X_j} \tag{5}$$

X_j represents the image at iteration j, whereas i is the output activation one wishes to maximize. α is the learning rate.

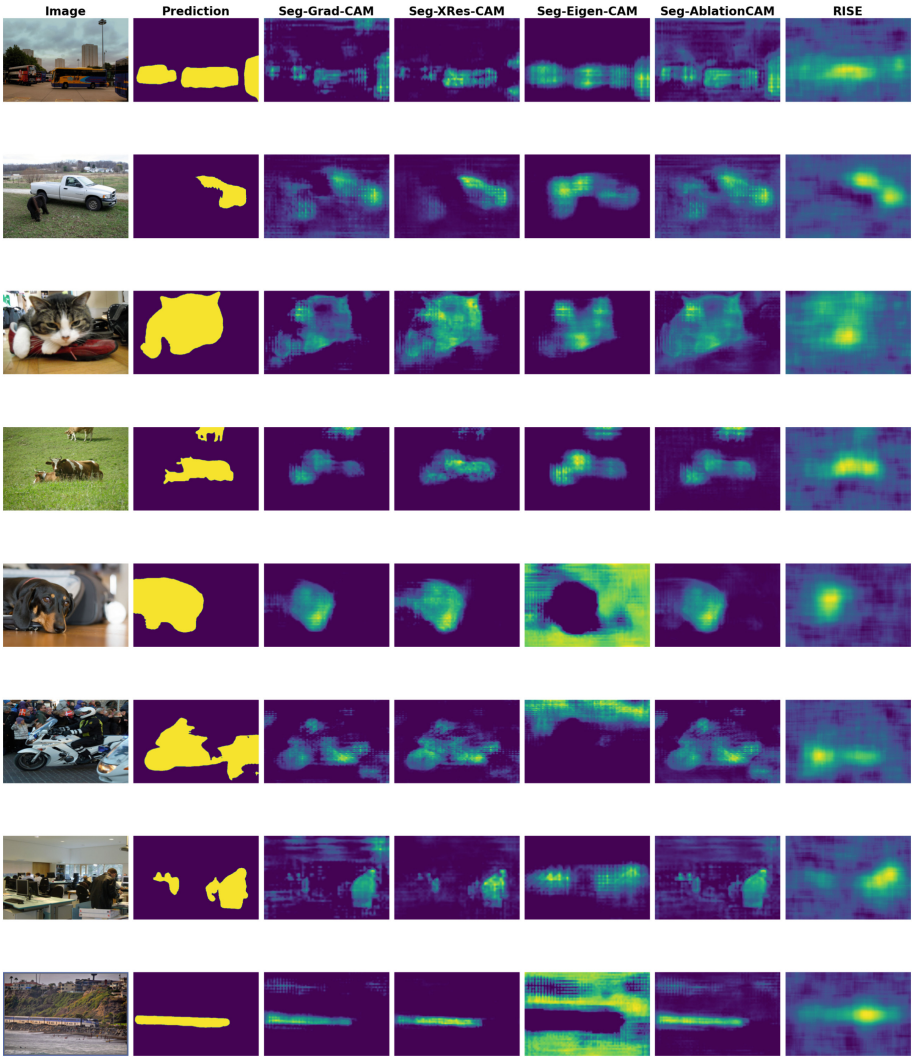

Fig. 2. Sample saliency maps generated from the application of Seg-Grad-CAM, Seg-XRes-CAM, Seg-Eigen-CAM, Seg-AblationCAM, and RISE on a few samples from the COCO-2017 dataset. For RISE, 2000 masks were utilized.

The path taken by the image is referred to as the DeepDream path with the idea being that by taking the steepest route, the image would have followed a path whereby only the most important features would have been modified. Accross multiple images, this analysis allows one to identify features to which a model is most or least sensitive to.

[58] too utilized activation maximization in order to investigate the kind of concepts being learned in various filters of an image segmentation model. An

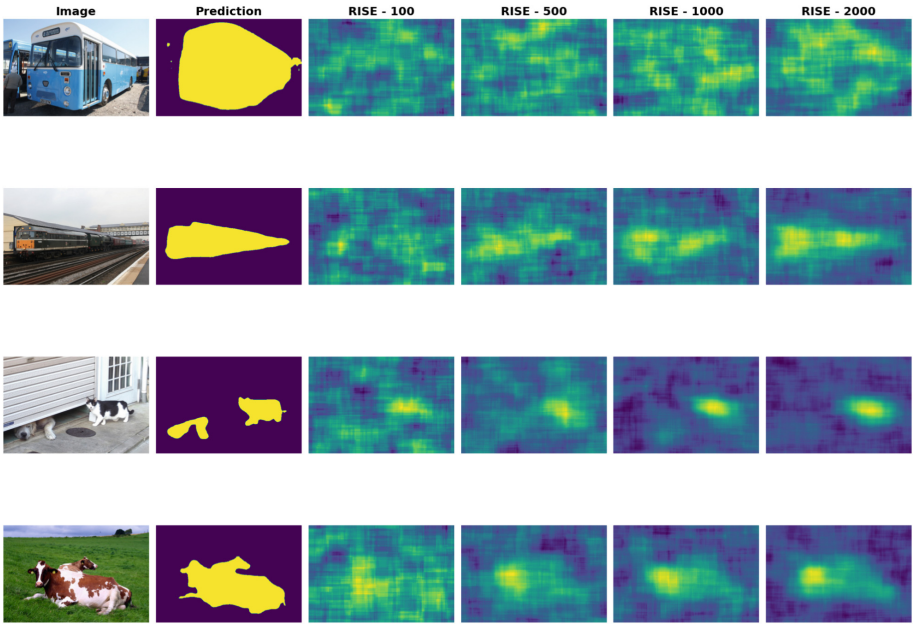

Fig. 3. Sample saliency maps generated from the application of RISE with various number of masks (100, 500, 1000, 2000) on a few samples from the COCO-2017 dataset.

Table 1. Comparison between Seg-Grad-CAM, Seg-XRes-CAM, Seg-Eigen-CAM, Seg-Ablation-CAM, and RISE on a subset of the COCO-2017 Dataset. The first four algorithms are representatives of the intermediate feature space methods whereas RISE represents the perturbation based methods. For dice explained, saliency ratio, and time, the mean value has been reported.

XAI Method	Binary Threshold	Gradient-based	No. of Masks	Dice Explained	Saliency Ratio	Time (s)
Seg-Grad-CAM	0.05	✓	-	0.606	3.47	0.25
Seg-Grad-CAM	0.1	✓	-	0.515	2.403	0.25
Seg-XRes-CAM	0.05	✓	-	0.671	1.788	0.26
Seg-XRes-CAM	0.1	✓	-	0.636	**1.25**	0.26
Seg-Eigen-CAM	0.05	✗	-	0.421	5.166	7.67
Seg-Eigen-CAM	0.1	✗	-	0.354	4.831	7.67
Seg-AblationCAM	0.05	✗	-	**0.843**	6.675	163.4
Seg-AblationCAM	0.1	✗	-	0.793	3.859	163.4
RISE	0.2	✗	100	0.927	11.155	16.1
RISE	0.4	✗	100	0.853	7.773	16.1
RISE	0.2	✗	500	**0.948**	10.59	80.4
RISE	0.4	✗	500	0.766	3.884	80.4
RISE	0.2	✗	1000	0.942	9.282	160.8
RISE	0.4	✗	1000	0.762	3.154	160.8
RISE	0.2	✗	2000	0.913	7.053	321.2
RISE	0.4	✗	2000	0.747	**2.196**	321.2

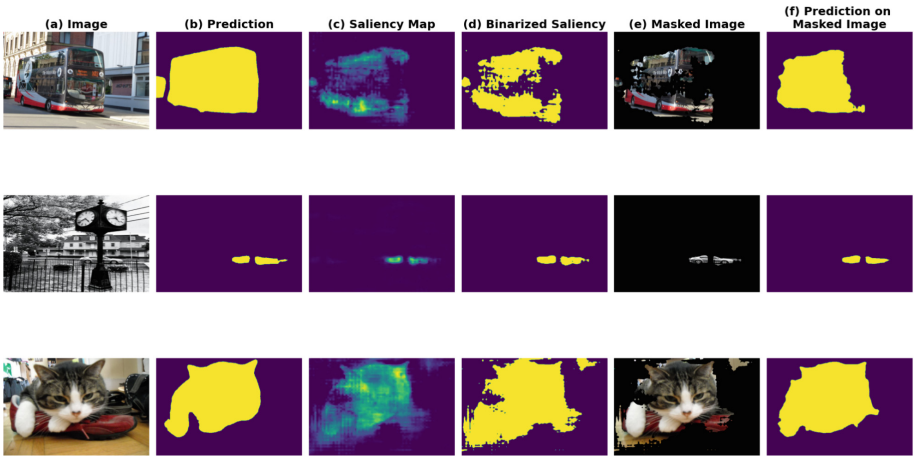

Fig. 4. Examples (from the COCO-2017 dataset) in order to explain the process of computing metrics for evaluating Saliency Maps for Image Segmentation. (a) Original Image, (b) Prediction obtained after passing (a) through the segmentation model, (c) Generated Saliency Map, (d) Binarizing the Saliency Map with a threshold (e) Masking the original image using the binarized Saliency Map (f) Obtaining the segmentation model's prediction on this Masked Image. The Dice Explained is then defined as the dice between (b) and (f). The Saliency Ratio is defined as the ratio of the number of non-zero pixels in (d) to the number of non-zero pixels in (b). For the current examples, the Dice Explained are: **0.81**, **0.86**, and **0.94** respectively. The Saliency Ratios are: **0.8**, **1.42**, and **1.45** respectively. For the current examples, the saliency maps were generated using Seg-XRes-CAM, and a threshold of **0.1** was utilized for binarization.

image is randomly initialized, and in order to determine the kind of concepts a filter might be looking for in an input image, gradient ascent is again utilized such that it is now guided by the derivative of an intermediate activation (that of the filter of interest) instead of the prediction space. Gradually, the random image transforms such that it is now dominated by the concept which the filter of interest is most activated by.

[41] explored the relationship of the segmentation model's bottleneck representations with the segmentation model's Intersection over Union (IoU) scores. Bottleneck representations of sample images followed by an application of dimensionality reduction serve as the input whereas the corresponding IoU scores for each of those images serves as the output. A simple regression model is then trained utilizing these input-output pairs allowing us to assess the segmentation model's predictions for unseen images whose ground truths are not available.

Another work [40] focused on the identification of useful concepts learned by an image segmentation model on a cardiac dataset. This was achieved with the help of Automated Concept-based Explanation (ACE) [25] which, in turn, is an extension of Concept Activation Vectors (CAV) [44]. Superpixels are first generated from individual images using SLIC [4]. These are then fed to the

segmentation model in order to obtain their bottleneck representations. Clustering is performed on these representations, and each cluster center is identified as a concept. A Concept Activation Vector (CAV) is then learned for each concept in order to determine its global importance for each segmentation category.

4 Tools for Practitioners

Existing tools for XAI in image segmentation are mostly focused on local methods. One of the earliest toolbox was developed by [69] which allows for the application of various local methods belonging to the intermediate feature space as well as the input image space. Another similar, but more recent, toolbox is [89] which provides seven local XAI algorithms. [26] provides a library which contains most of the CAM-based methods (intermediate feature space). The library is designed to work for both image classification as well as image segmentation[6]

5 Discussion

In terms of their application, the task for which XAI algorithms in image segmentation have been most utilized for is the generation of saliency maps, with Seg-Grad-CAM [78] being the favored choice of practitioners - see [32,85] for medical examples, and [9,43] for examples from natural images. However, another application of these saliency generation methods has been found in investigating the intermediate representations of segmentation models. The application of a saliency generation method to various layers of a segmentation model allows one to inspect the flow of information [48,53]. Additionally, if a custom layer is added to a segmentation model, these XAI methods can allow one to monitor whether the layer's representations follow the outcome one expected from the layer's design [51,82]. Another interesting application has been the identification of how much context is required by a segmentation model in order to successfully segment an object. An analysis of context information can allow one to identify whether the segmentation model has associated an object with spurious context [38]. Association with spurious context can be harmful as the segmentation model would fail to successfully segment the object in the absence of that context. Identifying such associations can help developers refine the training pipeline by introducing enough variation into the training data in order to minimize the possibility of a segmentation model learning irrelevant contexts.

While intermediate feature space dependent saliency generation methods have enjoyed considerable popularity for convolutional architectures, it is clear that with the advent of alternate architecture designs such as those incorporating transformer blocks [8,35], these methods would have to be accordingly adjusted or new methods be proposed. Given that the receptive field of transformer models is fundamentally different from convolutional models, it remains

[6] See also: https://jacobgil.github.io/pytorch-gradcam-book/Class%20Activation%20Maps%20for%20Semantic%20Segmentation.html.

an open question as to which layers need to be utilized in order to generate a saliency map.

With concept-based XAI, researchers have mostly been focused on identifying the various concepts which have been learned by the intermediate activations of the segmentation model [40,58]. However, a major issue in this regard, particularly for medical datasets, is concept identification. Concepts identified by automatic concept generation algorithms are hard to associate with tangible medical concepts, whereas labelled medical concept datasets using which a segmentation model can be probed are almost non-existent. For natural images, the problem is considerably alleviated due to the availability of a generous amount of online data regarding virtually any concept as well as due to the fact that one often doesn't require special training in order to interpret concepts associated with natural images.

6 Conclusion

The methods discussed above consistently reveal a common theme: XAI techniques used in image segmentation are derived from those employed in image classification. [78]'s Seg-Grad-CAM is an extension of Grad-CAM [12,71] being an extension of RISE [61] and SHAP [52], and [17] being an extension of [5], for example. Some exceptions to this rule include [41] and [45]. Despite its widespread use, the heavy dependence on XAI in classification is not straightforward to justify. Classification and segmentation are inherently distinct tasks: the former seeks to assign a single label to the entire image, while the latter involves a dense prediction task, aiming to assign a label to each pixel. Since the output of image segmentation is a spatial map, the effectiveness of generating saliency maps is diminished, as they often merely emphasize the object they were meant to clarify [17].

Instead of merely extending XAI approaches from image classification to image segmentation, it would be better if XAI in image segmentation was treated as an independent field by researchers. The initial focus should then be towards identifying potential applications of XAI in image segmentation as that would organically lead to the proposal of XAI methods fine-tuned to image segmentation. In the medical field, for example, saliency maps in image classification can aid a medical practitioner identify important regions in an image, whether the same role can be played by saliency maps of image segmentation is hardly obvious. Given the ubiquitous nature of image segmentation for medical images, the involvement of medical practitioners can help identify useful end goals for the field of XAI in medical image segmentation. Once potential applications have been identified, the design of XAI algorithms would then naturally be tailored towards fulfilling those roles allowing for informed borrowing from XAI algorithms in image classification. Taking inspiration from XAI in image classification is only natural, however, simple imitation should be discouraged owing to the fundamental difference between the two sets of computer vision problems.

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A Reviewed XAI Algorithms

Our search yielded **81** papers out of which **39** were works responsible for initially proposing a technique, and **42** were works which utilized these techniques

Table 2. Local XAI algorithms considered for this paper. Bold is used to indicate medical datasets. Additionally, if the algorithm is based on an existing algorithm from image classification, the predecessor is also identified. Key: **IFS** = Intermediate Feature Space.

Algorithm	Family	Gradient-based	Dataset	Precursor in Image Classification
Grad-PAM [80]	Local (IFS)	✓	Pascal-Context Cityscapes Look Into Person	Grad-CAM [71]
Seg-Grad-CAM [58,78]	Local (IFS)	✓	Cityscapes	Grad-CAM [71]
improved Grad-CAM [86]	Local (IFS)	✓	**Polyp (Colonoscopy)** **Liver Tumor (CT)** **Skin Lesion**	Grad-CAM [71]
Seg-GradCAM++ [49,56]	Local (IFS)	✓	Cityscapes **Brain Tumor (MRI)**	Grad-CAM++ [1]
Seg-ScoreCAM [56]	Local (IFS)	✗	**Brain Tumor (MRI)** Satellite Imaging	Score-CAM [81]
Seg-XGrad-CAM [30]	Local (IFS)	✓	Satellite Imaging	XGrad-CAM [22]
Seg-Eigen-CAM [30]	Local (IFS)	✗	Satellite Imaging	Eigen-CAM [55]
Seg-AblationCAM [27,30]	Local (IFS)	✗	Satellite Imaging Industrial Dataset	Ablation-CAM [14]
Seg-XRes-CAM [34]	Local (IFS)	✓	**Multi-organ (CT)** COCO-2017	HiResCAM [16]
Seg-Hires-Grad CAM [62]	Local (IFS)	✓	**Teeth (X-Ray)** **Kidney (CT)** Cityscapes	HiResCAM [16]
Seg-Sobol [72]	Local (IFS)	✗	Cityscapes Satellite Imaging	Sobol XAI [19]
Adapted Seg-Grad-CAM [56]	Local (IFS)	✓	**Brain Tumor (MRI)**	-
Adapted Seg-GradCAM++ [56]	Local (IFS)	✓	**Brain Tumor (MRI)**	-
Adapted Seg-ScoreCAM [56]	Local (IFS)	✗	**Brain Tumor (MRI)**	-
Kernel-Weighted [56,57]	Local (IFS)	✗	**Brain Tumor (MRI)**	-
Activations × Predictions [65]	Local (IFS)	✗	**Brain Tumor (MRI)**	-
Attribution-based Explanation [39]	Local (IFS)	-	Fungi	-
Score Maps [69]	Local (IFS)	✓	nuScenes	-

Table 3. Local XAI algorithms considered for this paper (cont.). Bold is used to indicate medical datasets. Additionally, if the algorithm is based on an existing algorithm from image classification, the predecessor is also identified. Key: **IS** = Input Space, **P** = Perturbation, **C** = Concept.

Algorithm	Family	Gradient-based	Dataset	Precursor in Image Classification
Guided backpropagation [83,84]	Local (IS)	✓	**Polyp (colonoscopy)**	Guided backpropagation [76]
Integrated Gradient [33]	Local (IS)	✓	**Lungs (CT)**	Integrated Gradient [77]
SmoothGrad [89]	Local (IS)	✓	**Brain Tumor (MRI)**	SmoothGrad [75]
Vanilla Saliency [3]	Local (IS)	✓	**Artery-Vein (OCT Angiography)**	Vanilla Saliency [73]
Grid Saliency [38]	Local (P)	✓	Cityscapes	Meaningful Perturbation [21]
SHAP [12]	Local (P)	✗	SAR Cityscapes	SHAP [52]
RISE [12]	Local (P)	✗	SAR	RISE [61]
Stable Explanation [23]	Local (P)	-	Cityscapes	-
U-Noise [45,59]	Local (P)	✓	**Pancreas (CT)**	-
Occlusion [29]	Local (P)	✗	COCO-2017	Occlusion [88]
SegNBDT [80]	Local (C)	✓	Pascal-Context Ctiyscapes Look Into Person	NBDT [79]
Concept Graphs [46]	Local (C)	✓	**Brain Tumor (MRI)**	-
L-CRP [18]	Local (C)	✓	Cityscapes	CRP
Example-based Explanation [37]	Local	✗	SemanticKITTI	-
Graph-based [24]	Local	-	**Liver Vessel (CT)**	-
Adversarial attacks on Saliency Maps [28]	Local	✓	Industrial	Adversarial Perturbation [31]
Counterfactual [74]	Local	✗	**Instrument (Endoscopy)** **Polyp (colonoscopy)**	-
Counterfactual - Generative [90]	Local	✓	BDD100k	-

as tools. Tables 2, 3, 4 list works belonging to the former category. This ends up being **44** proposed techniques (as some papers proposed more than one technique). Additionally, the datasets on which these techniques were applied in their parent papers are also mentioned with a symmetrical distribution of **31** datasets each for both medical as well as natural images.

Table 4. Global XAI algorithms considered for this paper. Bold is used to indicate medical datasets. Additionally, if the algorithm is based on an existing algorithm from image classification, the predecessor is also identified. Key: **C** = Concept.

Algorithm	Family	Gradient-based	Dataset	Precursor in Image Classification
Network Dissection [58]	Global (C)	✗	**Brain Tumor (MRI)**	Network Dissection [7]
Activation Maximization [58]	Global (C)	✓	**Brain Tumor (MRI)**	Activation Maximization [73]
CAV [40]	Global (C)	✓	**Heart (MRI)**	CAV [44]
DeepDream [12]	Global	✓	**Liver Tumor (CT)**	Activation Maximization [73]
Visualizing Bottleneck [41] Representations	Global	✗	Satellite Imaging	-
Emergent Language [68]	Global	✓	**Brain Tumor (MRI)**	-
Segmentation Ability [36]	Global	-	**Brain Tumor (MRI)** **COVID-19 (CT)** **Prostate (MRI)** **Pancreatic Mass (CT)**	-
Permutation Feature [60] Importance	Global	✗	Satellite Imaging	Permutation Feature [20] Importance

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Explainable Emotion Decoding for Human and Computer Vision

Alessio Borriero[2,4], Martina Milazzo[1], Matteo Diano[2], Davide Orsenigo[2,3], Maria Chiara Villa[2,4], Chiara DiFazio[2], Marco Tamietto[2,5], and Alan Perotti[3(✉)]

[1] La Sapienza University of Rome, Rome, Italy
[2] University of Turin, Turin, Italy
[3] Centai Institute, Turin, Italy
alan.perotti@centai.eu
[4] University of Camerino, Camerino, Italy
[5] Tilburg University, Tilburg, Netherlands

Abstract. Modern Machine Learning (ML) has significantly advanced various research fields, but the opaque nature of ML models hinders their adoption in several domains. Explainable AI (XAI) addresses this challenge by providing additional information to help users understand the internal decision-making process of ML models. In the field of neuroscience, enriching a ML model for brain decoding with attribution-based XAI techniques means being able to highlight which brain areas correlate with the task at hand, thus offering valuable insights to domain experts.

In this paper, we analyze human and Computer Vision (CV) systems in parallel, training and explaining two ML models based respectively on functional Magnetic Resonance Imaging (fMRI) and movie frames. We do so by leveraging the "StudyForrest" dataset, which includes functional Magnetic Resonance Imaging (fMRI) scans of subjects watching the "Forrest Gump" movie, emotion annotations, and eye-tracking data.

For human vision the ML task is to link fMRI data with emotional annotations, and the explanations highlight the brain regions strongly correlated with the label. On the other hand, for computer vision, the input data is movie frames, and the explanations are pixel-level heatmaps.

We cross-analyzed our results, linking human attention (obtained through eye-tracking) with XAI saliency on CV models and brain region activations. We show how a parallel analysis of human and computer vision can provide useful information for both the neuroscience community (allocation theory) and the ML community (biological plausibility of convolutional models).

Keywords: eXplainable Artificial Intelligence · NeuroImaging · Computer Vision · Emotion Recognition · Neuroscience

1 Introduction

Machine Learning (ML) algorithms have revolutionized data analysis by leveraging big data to extract valuable insights that might surpass and enhance human

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understanding. In the realm of emotion decoding, ML can be applied to process diverse data sources to unravel the complexities of human emotion, decoding the emotional content from various sensory sources like facial expressions, voice tones, and contextual cues [1]. By integrating high-level semantic information, ML goes beyond traditional methods, enabling the analysis of emotional cues within natural scenes with sufficient accuracy [43]. Indeed, the construction of emotional perception in the human brain is intrinsically complex and mediated by multiple hierarchical and parallel layers of processing, ranging from visual low-level features to the complex reconstruction of affective semantics. Each affective information is processed through feedback loops that extensively involve the entire brain, from subcortical to cortical areas and vice versa; the identification of the brain regions linked to emotion representation remains an unresolved issue [50,64]. The current debate features a clash between two opposing theories regarding the neural correlates involved in the creation and perception of emotional feelings: the *locationist* theory posits that discrete emotion categories consistently and specifically correspond to distinct brain regions, whereas the *constructionist* approach argues that discrete emotion categories are constructed of more general brain networks not specific to those categories [44]. In recent years, the application of ML techniques in neurophysiological studies regarding the decoding of emotional states from neuroimaging data (such as EEG and fMRI) has uncovered intriguing insights into the brain's processing of emotions [33,34]. The classification of emotional states has always been a challenge for computer vision as well, because of the complex nature of emotional information, which does not rely uniquely on low level spatial features, but mainly on high level semantic information. Numerous studies have focused on applying artificial intelligence to the task of decoding facial expressions [31,48], yet employing ML models to derive emotional information from natural scenes proves to be more challenging [55,66]. One of the central goals of this work is to examine side by side the processes by which the human brain and deep learning models decode emotional states from natural scenes, with the objective of properly comparing biological and artificial vision systems. The ambition to analyze the similiarities between human and artificial vision is motivated by many works, which have highlighted fundamental analogies between state-of-the-art computer vision algorithms and human brain visual system [5,35,58]. Specifically, it has been shown that the layers within Convolutional Neural Networks (CNNs) exhibit a similar structure to the brain's areas in the ventral stream visual pathways in terms of their internal representations [45,59,69]. However, little is known about the similarities and the differences regarding emotional decoding of human and machines. Does artificial intelligence generate similar inner representations of emotional visual input with respect to human beings? In order to pursue this research direction, we leveraged XAI techniques [4]. Since modern ML models (including CNNs) offer no human-understandable representation of their inner decision logic, XAI tackles this problem by providing additional information with respect to a ML model decision; in this work we rely on two canonical attribution methods, LIME [56] and SHAP [47]. When dealing

with brain decoding models, the explanation scores highlight the brain areas strongly correlated with the task at hand, while for CV models the explanation heatmaps provide saliencies at pixel level on movie frames. To explore the complexity of the human emotional spectrum, we exploited a dataset derived from fMRI acquisitions of subjects viewing the "Forrest Gump" movie, namely the StudyForrest project [23,60]. Although the dataset came with emotional annotations [39], we chose to use the annotations provided by Lettieri et al. in their work about emotionotopy in the human brain [41].

The present work aims to address the problem of the neural correlates related to emotion processing in the human brain exploiting machine learning models and XAI techniques. Moreover, merging eyetracking data, XAI results from brain decoding models and the explanation obtained from the emotion decoding on movie frames, we tried to bridge CV models with the human visual system, looking for neural patterns correlating with the degree of attention match between CNNs and human beings.

2 Related Works

2.1 Explainable Computer Vision

Since Alexnet [37], Deep Learning models have been the de-facto architecture for Computer Vision. However, these models offer no human-understandable representation of their inner decision processes, behaving like black boxes. The research field of XAI tackles this problem, developing approaches that help human understand the behaviour of black-box models. Since there is no mathematical definition of explainability and interpretability, it is important to keep in mind that this concept is linked only to the human understanding of neural network decision process: in [49] interpretability is explained as "the degree to which a human can understand the cause of a decision".

XAI is therefore a broadly defined concept which is implemented through a plethora of competing algorithms [8]. There are fundamental differences that help categorise these approaches: (i) *ante-hoc* vs *post-hoc*, (ii) *local* vs *global*, and (iii) *model-agnostic* vs *model-aware*. Ante-hoc approaches try to directly train explainable models, while post-hoc approaches aim at explaining an already existing opaque model. Local explanations pertain to a single data-point (e.g. *why was this image classified as deepfake?*), while global explanations try to extract a general description of the black-box model as a whole (e.g. *how does this deepfake detector work?*). Finally, model-aware explainers leverage the internals of a black-box, for instance computing scores based on gradients, while model-agnostic explainers only require to query a black-box at will. Most explainers are post-hoc and local, and there exist both model-aware and model-agnostic explainers for CV models, but the two most used ones, LIME [56] and SHAP [47], are model-agnostic.

LIME [56] (Local Interpretable Model-Agnostic Explanations) focuses on perturbing the single input sample, querying the model on the newly obtained synthetic point cloud in order to observe the impact on the output, and training

a white-box surrogate on the labelled point cloud. For computer vision tasks, LIME segments the input image into superpixels (patches of pixels) and exploits masking as a perturbation function.

SHAP (Shapley Additive exPlanations) [47] exploits the concept of Shapley value coming from cooperative game theory. The Shapley approach was to consider the power-set of players in a team, in order to measure the impact of each single player in the team outcome. SHAP keeps the same mindset, substituting players with features and team outcome with black-box output. Like LIME, there is a fundamental activity of perturbation/masking, but SHAP computes importance scores through marginalisation, without training a surrogate model.

2.2 Brain Decoding: Machine Learning on fMRI Data

Brain decoding refers to the process of interpreting both exogenous and endogenous brain states from observable brain activities, taking brain activity as input and brain states as output [32]. Concerning fMRI experiments, in which brain activity is measured through an indirect estimation of the metabolic changes in blood flow [25], decoding analysis has been traditionally performed through single-voxel univariate methods, like general linear model (GLM) [10]. This type of approach is referred to as 'univariate' because the corresponding statistical tests only consider the value of a single voxel or ROI (Region Of Interest) per condition at a time. Recently, an increasing number of researchers are adopting analyses that focus on patterns of responses across multiple voxels, known as multivariate pattern analysis (MVPA), instead of relying on values from single voxels or regions, in order to better assess the highly nonlinear information processing in the human brain [24,68]. MVPA techniques can be implemented using straightforward correlation analysis, linear classifiers, partial least squares algorithms [40] or by employing traditional machine learning algorithms, such as Support Vector Machines [38], with the most prominent examples of MVPA leveraging deep neural networks [17,32].

Compared to the CV case, less research work from the XAI field has focused on brain decoding tasks: except for a few valuable examples mostly regarding diagnostic brain imaging [15,54], fMRI and brain decoding studies lack of application of explainability approaches. With the present work we aimed to explore the possibility to exploit XAI to explore and study complex brain states.

2.3 Emotion Decoding for Human and Computer Vision

Numerous studies in CV have tackled the problem of emotion decoding, with a primary focus on decoding emotions from facial expressions [21]. While some studies categorize emotion through a limited set of basic emotional states (happiness, sadness, surprise, anger, fear, and disgust) [16], others decompose emotions into fundamental dimensions like valence and arousal [20]. State-of-the-art emotion decoding models exploit CNNs [46], while recent works analyze dynamic emotional facial expressions in videos [13].

Since fMRI data provide extensive insights into high-level cognitive processes in the human brain, numerous studies have focused on the task of decoding emotions through machine learning algorithms utilizing this type of neuroimaging data [33,34]. The state-of-the-art literature has demonstrated the possibility to predict at least information about emotional dimensions (valence and arousal) of emotional feelings perceived by human beings [7,26].

3 Experimental Setup

3.1 Frames, fMRI and Labels

The StudyForrest Project. The keystone of our study is the StudyForrest project [22,23]. This work is a scientific initiative that involves the collection and analysis of neuroimaging data, particularly functional magnetic resonance imaging (fMRI) data, related to dynamic visual and auditory stimuli, in order to study various aspects of human cognition and brain functions. "Forrest Gump" is a movie directed by Robert Zemeckis and released in 1994. Recognized as a classic of american cinema, the complexity of its narrative lay the foundation for our exploration into the emotional dynamics portrayed throughout its duration. It was shot at a standard cinematic frame rate of 23.97 frames per second (fps) The resolution, adhering to the standards of cinema, is 16:9. In total, the film comprises 204501 frames. For the ML+XAI pipelines described below, we adopted the movie version described of the original StudyForrest work: 8 movie segments, for a total of 172405 frames and an overall duration of 120 min, in order to divide the fMRI acquisition of each subject in 8 runs.

Emotion Annotation. An important component of this work is the inclusion of an emotional labeling, which we use as ground-truth labels for both the CV-ML model and the fMRI-based brain decoder. The StudyForrest project provides emotional labeling, but we decided to exploit the emotion annotations by Lettieri et al. [41] that has enriched the original dataset with the inclusion of labels related to the emotional responses associated with the fMRI. In this work 12 healthy Italian native speakers participants have been instructed to rate the movie with respect of 6 basic emotions (happiness, surprise, fear, sadness, anger and disgust) during the same reduced version of Forrest Gump used for the StudyForrest project. Annotators were allowed to report more than one emotion at the same time and ratings were continuously recorded. Therefore, six different times series for each of the twelve subjects has been produced. A temporal resampling has been implemented, in order to align the emotion data with the fMRI data in terms of temporal resolution. In this paper we focused on the four fundamental emotions of happiness, fear, sadness, and anger, as depicted in Fig. 1. In order to binarize our labels dataset and determine whether an emotion was dominant or not, a criterion was established. An emotion was considered dominant if at least one annotator assigned it a value greater than the other emotions at a given moment. If an emotion was dominant for one annotator at

a specific instant, it was considered positive; otherwise, it was considered negative. This binarization process allowed us to transform the continuous emotion intensity annotations into a binary representation, providing a clear distinction between dominant and non-dominant emotions. Regarding the brain decoding task, a further preprocessing step has been implemented before the binarization procedure. A sliding-windows smoothing has been applied over the emotion annotations time-series, with a windows size equal to 10 s and a stride equal to 2 s.

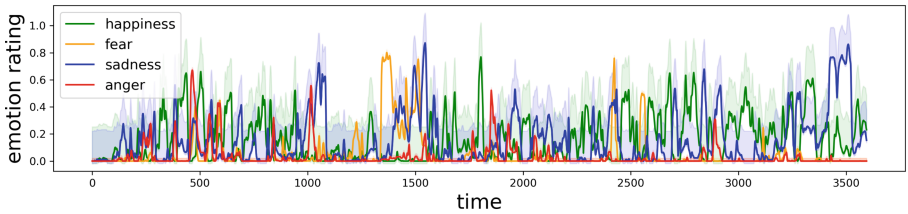

Fig. 1. Time series of emotion annotations. The work by Lettieri et al. provides the emotion annotation by 12 independent human annotators of the whole Forrest Gump movie; in our work we focused on happiness, fear, sadness and anger.

Through the procedure described above we built 5 different binary dataset, 4 for the emotions and 1 for the face, which were provided to 5 different ML models. The choice to have different models for different emotions lies on the theoretical prior of our work. One of our aims has been to investigate the dualism between constructionist and locationist theories of emotion processing in the human brain. By definition of the model itself, a single multi-class model would not allow us to see shared networks among different emotions, resulting in an imbalance towards the locationist vision.

Movie Frames Datasets. The first preprocessing step aimed to build the frames datasets for the emotion decoding task has been to resample the whole movie according to the fMRI repetition time (TR), namely 2 s. Thus, we obtained a set of 3599 frames. However, we observed that some subsequent frames were identical or nearly-identical, especially for long, static scenes. We wanted to remove duplicates in order to avoid a train-test leakage. We first tried to do so by measuring pixel-level difference between consecutive frames, but this approach did not capture semantical similarities: for instance, in scenes with crops blowing in the wind, we had high-pixel level differences - within a semantically monotone scene. In order to select significative frames, we resorted to exploit an Imagenet-pretrained CNN, specifically the EfficientNet B0 model [65]. For all frames, the EfficientNet was employed to derive labels indicative of the elements present in the input image. From these labels, the top 3 values among the 1000 provided by the network were selected. Subsequently, for each pair of consecutive frames,

the indices of the obtained labels were compared, leading to the elimination of frames that shared one or two labels. This process effectively removed frames that exhibited similarities in their content, enhancing the diversity and relevance of the dataset.

Alongside the emotion decoding task, in this work we aimed to asses the problem of facial recognition. To do that, we exploited DeepFace [61], an open-source library which embeds face recognition and facial expressions decoding. Through a specific function, we were able to extract bounding boxes around faces in individual frames. The binary labels were then selected based on the following criteria: frames without faces were labelled as negative cases, while frames with 1 or 2 faces with a bounding box's area greater than or equal to 4% of the whole frame were labelled as positive cases. In order to adapt our movie frames to the requirements of the CNN, which demands square images, we applied ad hoc cropping and padding to the whole collection of frames. Since our frames datasets were highly imbalanced, we performing an undersampling procedure. To conclude, after the preprocessing steps described above, we obtained 5 binary datasets of movie frames: 4 related to the 4 basic emotions and the last concerning the presence of faces within frames.

The fMRI Dataset. For the brain decoding task we exploited the rich fMRI dataset provided by StudyForrest, where 15 German native subjects participated to fMRI recording sessions, during which they were engaged in watching Forrest Gump. All subjects underwent 8 fMRI runs, for a total o 120 min of acquisition. The data preprocessing pipeline was prepared using AFNI [12] and FSL (FMRIB Software Library) [29] software. Structural images were resampled to 1mm3 (*3dAllineate*). Then they were brain extracted (*standard-spaceroi; bet2*), corrected for intensity bias (*3dUnifize*), and spatially normalized to the Montreal Neurological Institute (MNI) space with non-linear registration (*3dQwarp*). All functional volumes were slice timing corrected (*3dTshift*), spatially realigned to the first volume of the functional acquisition and corrected for scan motion (*3dvolreg*). All functional volumes were then spatially smoothed (*3dBlurToFWHM*) with a 6 mm full-width half-maximum isotropic Gaussian kernel (FWHM) and the signal was normalized (center: 0; variations in percentage). Each average EPIs were aligned to correspondent high resolution T1w and then re-sampled in the size of the functional acquisition using a weighted sinc-interpolation method. After the standard preprocessing pipeline described above, we introduced a lag of 2 s to account for the delay in hemodynamic activity. The resulting timeseries were then temporally smoothed using a moving average procedure (10 s window). This method allowed us to further account for the uncertainty of the temporal relationship between the actual onset of emotions and the time required to report the emotional state. As for the movie's frames dataset, we applied undersampling to our dataset in order to assess the imbalance in the class distribution, for each of the 5 datasets.

Eyetracking. In addition to the fMRI acquisition, the StudyForrest project also provides an eyetracking registration for each subject during each session. Such data enriches our analysis with an insightful information about the dynamics of the human attention of each subject with respect to the visual stimulus. In our study, we exploited the normalized version of the eye movement recordings, sampled uniformly at 1000 Hz, with a spatial resolution of 1280×546 pixels, the same as the original movie.

3.2 Machine Learning on Movie Frames

In order to decode emotional states from movie frames, we exploited a transfer learning approach [72]. We first chose a pretrained CNN, namely the EfficientNet B0 [65], then we added a customized layer on the top of the pre-trained network, allowing the model to learn features related to the binary emotion decoding and face/no-face tasks. The pre-training endowed the model with the ability to capture general features and patterns present in images, while the subsequent training phase allowed it to adapt to the variations of our domain-specific task. The additional layer was made by a global average pooling 2D layer, a dropout layer and dense layer (1024 neurons) and an output neuron implementing a sigmoid function for the classification. A key step of the fine-tuning procedure has been a grid-search over the hyper-parameters of the tailored layer. We explored the following space of hyperparameters: number of units in dense layers (128, 256, 512), dropout rate (0.1, 0.3, 0.5), learning rate (0.01, 0.001, 0.0001) and batch size (8, 16, 32). Moreover, we included in the grid-search two possible choices as the first layer, namely a global average pooling 2D layer or a flatten layer. To avoid overfitting and improve the model's generalizability, we relied on early stopping [70] in the fine-tuning procedure. Finally, to enrich the original datasets with the aim to further enhance the generalization capabilities of our models, we performed data augmentation [62], applying random flip, random rotation and random zoom to the frames. For what concerns the train-test splitting, we relied on a standard leave-one-out k-fold procedure, splitting each of the datasets in 5 folds. The results shown in the experimental results section are the average accuracy over the 5 splits.

3.3 Machine Learning on fMRI Data

In the context of the brain decoding task, the goal is infer information about dominant emotions and face occurrences from fMRI data. A key feature of our work has been to parcel each brain volume according to a brain atlas, namely the one by Glasser et al. [18]. Thus, each input of the ML models consists in a set of 394 features, representing the mean activity of each area of the brain in a given moment of the fMRI acquisition. We chose a within-subject approach, training and testing one model for each of the experimental subjects provided by the StudyForrest database. Our choice of training subject-specific models is due to the high variability, noisy nature and high dimensionality of the BOLD activity

from different brains, which makes the generalization of the fmri signal across-subject one of the main issues in brain decoding tasks [2]. Many solutions have been explored to address the possibility to obtain subject-independent models, such as hyperalignment [42] and roi-specific models [71]. In our work, we tried to apply a simple leave-one-subject-out approach, without the exploitation of more sophisticated techniques, achieving bad results in terms of generalization performance. Considering our goal of performing a whole-brain analysis, we then decided to implement a subject-wise paradigm, which was able to provide a sufficient amount of information about the structure of the processing of high-level semantic information within the human brain. We built a multi-layer perceptron [19] model, with relu as activation function, a single output neuron employing a sigmoid function and Adam algorithm as optimizer [30]. A grid-search procedure has been implemented in order to choose the best set of hyperparameters among the following: number of hidden layer (1 or 2) and the numbers of neurons for each layer (ranging from 40 to 300). In order to enforce generalization capabilities of the models, the learning algorithm has been equipped with a regularization parameters. As for the computer vision models, we performed a leave-one-out k-fold splitting, with $k = 5$, averaging the resulting accuracy values over the 5 splits.

3.4 XAI for Emotion Decoding

For both our set of emotion and face decoding models, namely the CV and the fMRI-based models, we exploited LIME and SHAP as XAI techniques, in order to unveil the decision-making processes behind the 'black-box'. An overview of these two ML+XAI pipelines is visualized in Fig. 2. Since exploring different explainability methods in brain decoding problems wasn't the aim of our work we do not exploit other XAI methods, such as gradient-based ones. Thus, we prefer to adopt the most used model-agnostic approaches.

For CV models, the XAI methods we applied generated heatmaps across frames, assigning significance scores to different areas of each frame. Specifically, for each frames of each dataset (regarding the four binary emotion plus face classification) we generated an explanatory heatmap, giving us an importance score for each pixel regarding the class discrimination task.

For what concerns the brain decoding models, we exploited XAI techniques in order to assign a score of importance to each area of the brain, namely the features for our ML problem. This approach produces a brain map of area importances for each sample of our dataset, i.e. the brain volumes of the fMRI acquisition.

In order to asses the statistical validity of the XAI brain maps that we obtained applying SHAP and LIME to our brain decoding models, we compute a null model. To do so, we trained the ML model with a shuffled set of binary labels, to assess chance-level performance. The set of feature importance values obtained by explaining the null model are used to build a null distribution, which in turn we will use to validate our explanation scores.

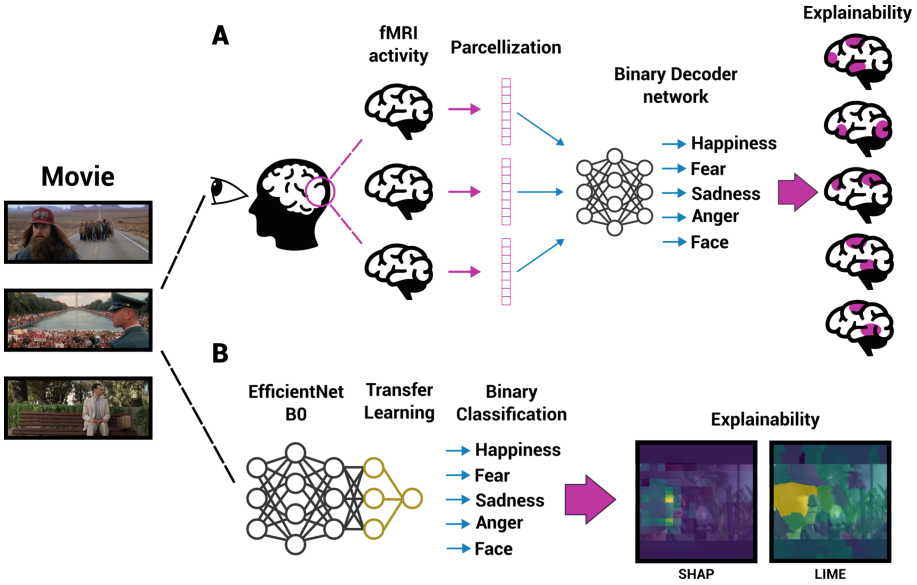

Fig. 2. Emotion decoding and XAI pipelines for brain data (A) and computer vision (B).

3.5 CNN-Humans Attentional Match: A Comparative Analysis

A crucial step of our work has been the cross-analysis between explainable CV and fMRI XAI results, alongside with the information provided by the eyetracking. Due to the temporal synchronization of all the different domains of our analysis, we could compare the visual attention of the CV model and of the human beings involved in the experiment. Our basic aim has been to identify which brain areas process the emotional content of the movie when the CNN and human look at the same spot within the visual stimulus. To do so, we first define a frame-wise score of overlap of visual attention. Exploiting the XAI heatmaps, we set this overlap score as the percentile of the heatmap's value for which the eyetracking recording fall within the pixels with importance values above that percentile. In order to provide more robustness to our analysis and to asses the timing of the human visual attention process, we consider a window of 1 s around the given frame when measuring the overlap score. Once we computed this measure for each of the movie's frames, we performed an area-wise correlation between overlap score of each frame and importance value of the given area at the time of each frame. This analysis provides a whole-brain map that describes which brain areas contain more information about the emotional content of the visual stimulus whenever the human subject and the computer vision model focus their attention on the same spot within the movie frame.

4 Experimental Results

4.1 Machine Learning on Movie Frames

Our investigation into emotion recognition through computer vision involved training distinct models for four key emotions (happiness, sadness, fear, and anger) and one for face detection. In order to optimize the performance of our emotion recognition models, we employed the grid search method, as mentioned in the experimental setup section. As reported in Table 1, each model shows good in-sample and out-of-sample performance. Specifically, the classifier designed to determine whether happiness is present in the frames exhibits higher accuracy compared to other emotion-related models.

For the face detection task, we obtained a remarkably higher performance with respect to the emotion classification models. This result is certainly due to the lower complexity of the face recognition problem, which relies on low-level visual features. On the contrary, the emotion decoding problem lives in a multi-faced semantic space making the identification of the visual features which define the presence of a certain emotion in a visual scene a non-trivial issue.

Table 1. Computer vision models performance

Model	***Happiness***	***Fear***	***Sadness***	***Anger***	***Face***
In-sample Accuracy	0.99	0.99	0.98	0.99	0.99
Out-of-Sample Accuracy	0.84	0.80	0.76	0.76	0.96

4.2 Machine Learning on fMRI Data

To assess the emotion and face decoding problem with fMRI data we trained 5 classifiers for each of the 15 experimental subjects. The overall out-of-sample performance we obtained, displayed in Table 2, is appreciably above the chance level. Moreover, the results seem stable across the subjects. Among the 5 models, the one related to face recognition shows the lowest performance. Such a result can be justified by a complex brain representation related to the face processing within the visual stimuli with respect to the emotional content of the movie. Moreover, the information about the faces rely only on visual stimuli, while the emotional nuances of the movie are carried by both visual and auditory stimuli. Thus, the emotion decoders can rely on a richer set of information throughout the brain with respect to the face decoder model. However, we analyzed in depth the brain representations related to each of the 5 decoder models through explainability techniques, as discussed in the following section.

Table 2. Brain decoding models performance

Model	*Happiness*	*Fear*	*Sadness*	*Anger*	*Face*
In-sample Accuracy	0.976	0.971	0.945	0.970	0.925
Out-of-Sample Accuracy	0.911	0.923	0.887	0.894	0.797

4.3 Explainability for fMRI-Based Models

One of the most intriguing aspects of our study has been the evaluation of the features' importance within the brain decoding models. The application of SHAP and LIME as explainers for our fMRI-based classifiers lead us to unveil the most important brain areas in terms of emotion and face processing, by defining a set of brain networks related to each of the 5 models. Figure 3 shows a representation of both SHAP and LIME brain maps related to the fear model. Table 3 summarizes the most significant areas for each of the four basic emotions and the face recognition, with respect to a null model.

At first, looking at the similarity among the two different explanatory maps obtained with SHAP and LIMES for each of the 5 models, we observed a good consistency in terms of resulting areas. As a stronger clue for this robustness with respect to XAI techniques, we compute pairwise spearman correlation corrected for a spin permutation test [3] among the couples of explanation for each classification task, resulting in a strong correlation for all the models (Spearman R values: Happiness - 0.81, Fear - 0.83, Sadness - 0.85, Anger - 0.82, Face - 0.81).

The feature importance maps we obtained provide interesting elements in context of the debate regarding how the brain processes different emotions. Looking at the overall pairwise Spearman correlation among the different maps corrected for a spin permutation test, (Fig. 4), we observe a high correlation level among all the emotions, including the face-related map. This result underlines the existence of a common brain network for the processing of all the basic emotions we studied through our analysis. In particular, the most significant region represented in all the 5 explanation maps is the Orbital and Polar Frontal Cortex (OPFC). Historically, OPFC plays a crucial role with respect to the locationist hypothesis regarding the brain representation of the anger [53,67]. Nevertheless, our results provide strong clues regarding a constructionist description of how emotional feelings and perceptions are represented in the human brain. From a psychological constructionist perspective, it is theorized that parts of the Orbital Frontal Cortex (OFC) contribute to core affect by serving as a hub for merging exteroceptive and interoceptive sensory data, thereby influencing behavior [44]. With the term core affect we describe the mental representation of bodily changes that are sometimes experienced as feelings of hedonic pleasure and displeasure with some degree of arousal [6]. Sensory information from the external environment and the body collectively direct an organism's reactions to its surroundings, allowing it to make actions that are appropriately adjusted to the context. Multiple connections involving the OFC to both various sensory systems and regions controlling visceral functions make this brain structure anatomically

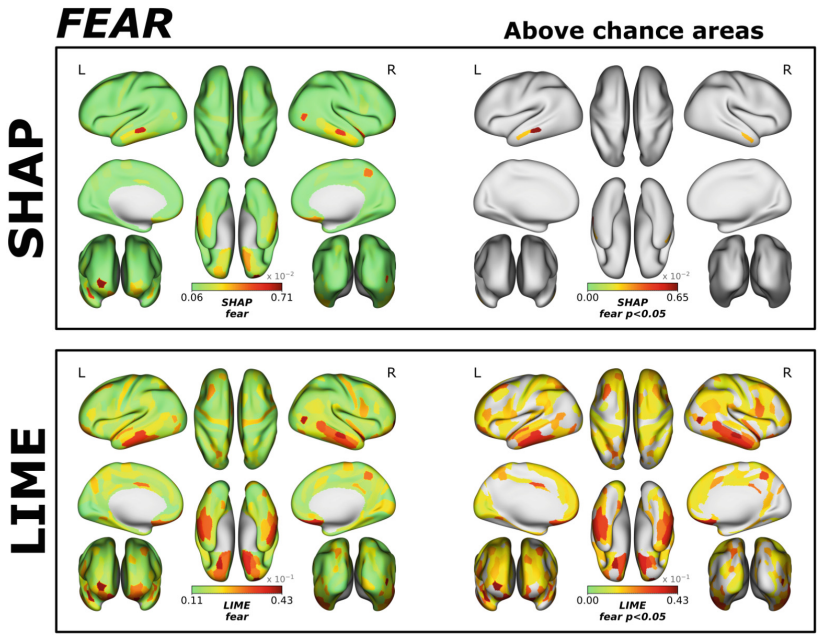

Fig. 3. Brain-wise feature importance maps obtained with SHAP and LIME. Through a null model we assess the significance of each area, obtaining a limited set of regions which process most information about the emotional content of the movie.

equipped to fulfill this function effectively [36]. Thus, OFC serves as an high-level decision making core, which produces behavioral and cognitive response to emotional stimuli.

Our results depict more complex pictures than a single brain network involved in the processing of all the emotions. In fact, beyond the OFPC, other areas emerge in more than one model. Looking at the feature importance maps that we obtained, the Insula Cortex seems to convey much information about the emotional processing in the human brain, specifically for happiness, fear and anger. From a locationist perspective, the Insular cortex, in particular the Anterior Insula, is more involved in processing of emotional content, indicating a transition from sensory to emotional processing within this area, and representing the core of the sensation of disgust [28]. The insula's intricate connections with other brain regions and its ability to integrate interoceptive inputs make it a key player in representing emotional experiences and bodily reactions to stimuli, as its activation has been linked to subjective feelings and cognitive factors, influencing decision-making processes and behavioral responses. Indeed, a more refined hypothesis, based on a constructionist approach, has identified a more complex role of this region in representing core affective feelings in awareness [9]. A core feature in the mental states labeled "disgust" is a representation of how an object will affect the body. In support of a psychological constructionist

Table 3. Most significant brain regions with respect to feature importance related to each brain decoding model.

	SHAP		LIME	
Model	**Macro Area**	**Value**	**Macro Area**	**Value**
Happiness	Early Auditory Cortex	0.0027	Insular Cortex	0.053
			Ventral Stream Visual Cortex	0.035
			Orbital and Polar Frontal Cortex	0.032
			Early Auditory Cortex	0.031
			Dorsal Stream Visual Cortex	0.030
Fear	Lateral Temporal Cortex	0.0065	Orbital and Polar Frontal Cortex	0.043
			Insular Cortex	0.041
			Lateral Temporal Cortex	0.040
			Early Auditory Cortex	0.028
			Superior Parietal Cortex	0.027
Sadness	Early Auditory Cortex	0.0044	Orbital and Polar Frontal Cortex	0.053
	Lateral Temporal Cortex	0.0035	Auditory Association Cortex	0.039
			Lateral Temporal Cortex	0.037
			Early Auditory Cortex	0.030
			Dorsolateral Prefrontal Cortex	0.019
Anger	Superior Parietal Cortex	0.0039	Insular Cortex	0.048
	Auditory Association Cortex	0.0038	Lateral Temporal Cortex	0.030
	Premotor Cortex	0.0038	Premotor Cortex	0.030
			Superior Parietal Cortex	0.027
			Orbital and Polar Frontal Cortex	0.024
Face	Lateral Temporal Cortex	0.0032	Orbital and Polar Frontal Cortex	0.034
	Anterior Cingulate Cortex	0.0022	Anterior Cingulate Cortex	0.019
			Auditory Association Cortex	0.017
			Superior Parietal Cortex	0.013

view, Insula activation emerges in a number of tasks that involve awareness of body states, shedding light on how specific aspects of insular activity can drive accurate emotional state predictions, but also on its potential inhibitory role in emotional processing.

Another notable brain area which emerges from our analysis is the Lateral Temporal Cortex, resulting as an important region for the classification models related to fear, sadness and anger. Looking with more details at our results, we find a peak of importance values in the Inferior Temporal Gyrus, namely the (bilateral) TE and TF areas. The state-of-the art literature describes these brain structures as a key step in the ventral visual stream implicated in object, face, and scene perception [11].

Alongside the Lateral Temporal Cortex's role in processing the visual aspects of emotions, it's important to highlight the significance of auditory characteristics in depicting emotional states. In this regard, it is crucial to mention the emergence of the Early Auditory Cortex as a significant area with respect to the

feature importance score assigned by the explainers. Such a brain region can be observed in the models related to happiness, fear and anger.

Thus, the results just discussed depict a well-defined emotion processing network which conveys most of the information about the emotional content of a complex visual and auditory stimulus. This set of brain regions seems to show a precise hierarchical structure. The OFPC represents a high-level cognitive stage, which integrate outer stimuli and inner representations influencing decision-making and modulating behavioral responses. At a mid-level, the Insular cortex acts as a relay for the aware response to sensory inputs, which can drive an emotional response. Finally, at the lower level, visual and auditory areas, namely the Lateral Temporal Cortex and the Early Auditory Cortex, represent a first integration step for the multisensory features conveying the emotional information.

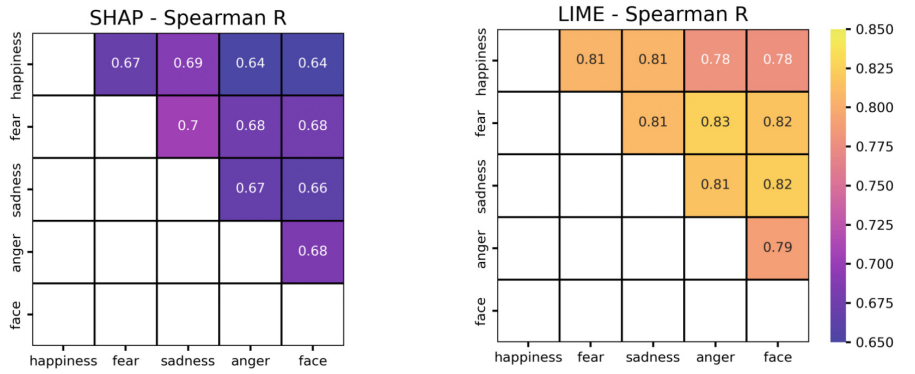

Fig. 4. Correlation among brain maps related to different models. The high correlation values we observed are due to the existence of a common brain network which processes information about the emotional content of a multisensory input. All the resulting correlations have a strong statistical significance, with p-values always below than 0.0002.

Finally, looking at the neural correlates obtained from our analysis associated with face-related visual stimuli, we basically observe the interplay of two brain regions. The OPFC seems to play an important role in the processing of human faces. As discussed before, this area represents a high-level stage in the processing of emotional stimuli, merging inner and outer signals and providing a behavioral output. Alongside this cognitive step in the processing pathway we are trying to discuss, the Anterior Cingulate Cortex (ACC) seems to process a significant amount of information regarding the presence of a face in a natural visual scene, as it is involved in the processing of various types of face stimuli, including social and self-related faces [51]. ACC represents a crucial hub in the human brain, integrating information at various levels through connection with both the "emotional" limbic system and the "cognitive" prefrontal cortex [63].

Interestingly, many works have highlighted the role of the ACC in processing face stimuli, both social or self-related [27,52]. In particular, this region plays a significant role in the rapid processing of emotionally salient facial expressions, such as fear, thus providing an immediate emotional assessment of faces; additionally, the ACC's connections with various brain regions, including the amygdala and temporal cortical areas, facilitate its role for interpreting facial expressions and understanding social cues [14]. The medial prefrontal network, which includes the ACC, is involved in processing emotional and social information from faces, indicating the ACC's role in social cognition and emotional regulation [57].

4.4 Comparative Analysis

Since the advent of Deep Learning models for effective CV, comparing how humans and artificial intelligence systems process visual information has been an open research question. In this paper we tried to address such a complex issue, exploiting eye-tracking registration, fMRI data and XAI techniques. In particular, our challenge has been to understand if a specific brain pattern emerges whereas the attention of the human visual system and the CNN focus on the same spot within the presented visual stimuli, i.e. the frame of the Forrest Gump movie. Our hypothesis is that, if some region in the brain activates more when human eye-tracking and explainability cover the same pixels over the frame, it may represent a region of high similarity with CNNs with respect to the way they process visual information. This local (in the sense that we tried to look at a brain-region level) analysis produced a brain map for each of our 5 decoding model, highlighting the areas with a feature importance which correlates more with the degree of overlap between computer vision explanation over the frames and eye-tracking (Fig. 5).

The brain maps we found exhibit an intricate set of regions, different among models and not robust with respect to the choice of the explainer, making the results hard to discuss. However, we can move at a global whole-brain level in order to get some insights from this kind of analysis. Looking at the distribution of the correlation values for each brain region, displayed in Figs. 6, we can observe interesting phenomena, mostly regarding the results with SHAP. Despite the distributions of the emotion-related models seem to basically overlap, with no appreciable difference with respect to a null distribution, the values associated to the face model displayed higher correlation values. Such a result tells us that, globally, there are some brain regions with a non-negligible tendency to activate the most when the computer vision and the human being focus on the same part of the visual stimulus. At a first level, this finding can be interpreted as a natural consequence of the multisensory nature of the emotional content. The face recognition task is a visual-only task, thus it is more probable to find correlations between visual attention and fMRI-based explanations. Then, another level of interpretation of our results regards the context-based nature of the emotional content of visual stimuli. The face recognition task is basically driven by local features of the visual input. The presence of a face within a frame is not

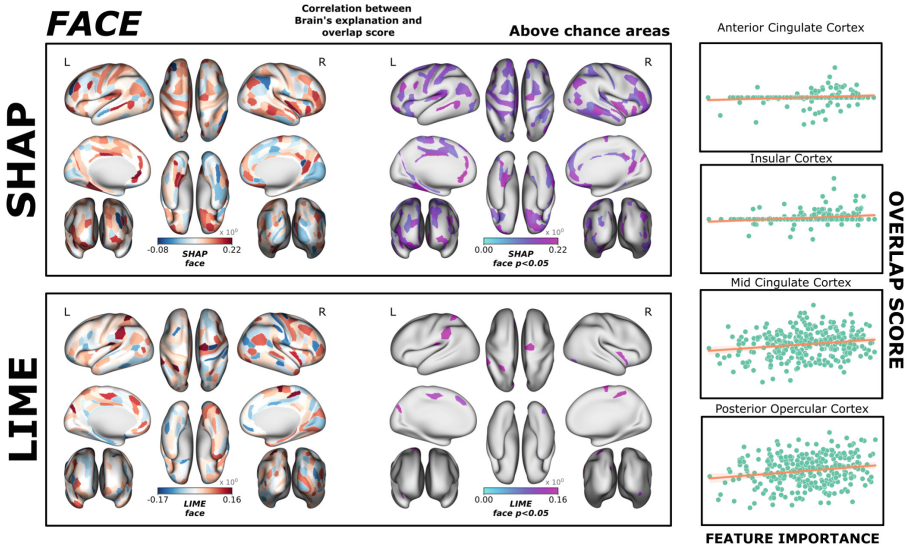

Fig. 5. Area-wise correlation between Brain's explanation and overlap score.

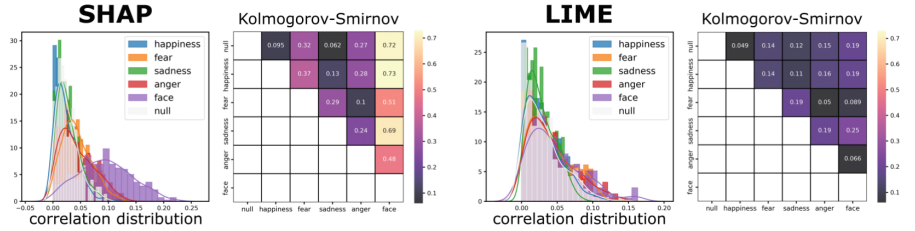

Fig. 6. Correlation distributions per model and Kolmogorov-Smirnov distance. For what concern SHAP results, the face-related distribution shows globally higher correlation values, highlighting the locality of the visual information about faces, in contrast with the context-based nature of the emotional content of a visual stimulus.

related to the context where the face is located. For what concerns emotions, it's clear that this kind of information about a stimulus is conveyed by the whole, not only by a limited detail. Under these assumptions, it's easier to understand why the information about the attentional match among humans and machines does not convey information about similarities among CNN and the brain visual pathways (Table 4).

Table 4. Most signigficative areas in terms of correlation among shap values and overlap between eyetracking and explainability over the frames.

	SHAP		LIME	
Model	**Macro Area**	**Correlation**	**Macro Area**	**Correlation**
Happiness	Early Auditory Cortex	0.119	Premotor Cortex	0.123
			Orbital and Polar Frontal Cortex	0.110
			Medial Temporal Cortex	0.103
			Mid Cingulate Cortex	0.098
Fear	TPOJ	0.119	Dorsal Stream Visual Cortex	0.134
	Insular Cortex	0.119	Early Auditory Cortex	0.109
	Superior Parietal Cortex	0.116	Nucleus Accumbens	0.108
	Neighboring Visual Areas	0.111	Premotor Cortex	0.107
	Ventral Stream Visual Cortex	0.107	Mid Cingulate Cortex	0.100
Sadness	Orbital and Polar Frontal Cortex	0.0966	Auditory Association Cortex	0.086
	Posterior Cingulate Cortex	0.0936		
	Dorsolateral Prefrontal Cortex	0.0836		
Anger	Dorsal Stream Visual Cortex	0.134	Insular Cortex	0.142
	Somatosensory and Motor Cortex	0.125	Anterior Cingulate Cortex	0.134
	Lateral Temporal Cortex	0.123	Premotor Cortex	0.132
	Posterior Cingulate Cortex	0.122	TPOJ	0.124
Face	Anterior Cingulate	0.223	Posterior Opercular Cortex	0.163
	Insular Cortex	0.218	Mid Cingulate Cortex	0.161
	Medial Temporal Cortex	0.217	Superior Parietal Cortex	0.161
	Posterior Cingulate Cortex	0.204	Dorsal Stream Visual Cortex	0.155
	Ventral Stream Visual Cortex	0.199		

5 Conclusions

To decode emotional states from visual stimuli in an ecological environment represents a challenging problem, in terms of understanding how the human brain processes such information and with respect to the attempt of building artificial intelligence systems able to replicate human perceptive and cognitive capabilities. In this paper we explored the possibility to retrieve information about emotional content of sensory input exploiting ML models. Moreover, we exploited XAI techniques in order to unveil how the brain processes emotional states. Through a computer vision-based emotion decoding analysis, we explored state-of-the-art techniques regarding the emotion classification within a set of complex visual stimuli, by showing that high-level semantic information can be successfully decoded by a CV model with high performance.

For the brain decoding problem, we obtained high performance results in terms of classification. Moreover, the XAI analysis we performed outlined the existence of a fundamental network which processes the emotional content of a complex sensory stimulus, with various levels of processing stages, from a lower-level sensory elaboration to a higher-level cognitive processing. These findings confirm a constructionist vision regarding how the brain creates emotional feelings, without the involvement of emotion-specific regions. However, the evidence put forth in this research is not compelling enough to invalidate the locationist theory. In fact, our machine learning-based analysis has been designed to

discriminate the presence or absence of a single emotion at times, not accounting for the discrimination among different emotions. The emotion-wise network that we observe tells us that an emotion-independent set of brain regions processes the whole spectrum of emotional content of a stimulus, but does not exclude that some specific areas can be involved in the elaboration of specific emotions.

Applying XAI techniques also to the computer vision models, we tried to compare the attentional mechanisms beyond the elaboration of emotional content of a stimulus in CNNs and human beings. This comparative analysis was aimed at eliciting specific patterns of activations in the brain whereas the human vision system and the artificial intelligence focus their attention in the same spot within the visual input. We found that it's possible to define some kind of coherent patterns of correlation with the attention match between humans and CNN only in the context of face recognition, because the evaluation of emotional context is intrinsically a context-based process. In other words, this kind of analysis aims to exploit an attentional similarity between CNN and human vision, which can be found in face recognition, but not in emotion decoding tasks, in order to define a relation about brain regions which show analogy with CNNs.

Through our analysis we addressed the emotion decoding problem from an XAI perspective, exploring innovative tools to explore the way the human brain and CNNs process emotional sensory stimuli. We also tried to look at the bridge between CNNs and the human visual system from a different perspective, exploiting the attentional mechanisms incorporated within these two systems in order to identify which portions of the brain show some kind of synchronized behavior. Our work has been focused on single CNN architecture, namely the EfficientNet B0 architecture. Further works could explore the relations between more complex models and the human brain. However, the transfer learning procedure we adopted provides to the CNN general visual features, ensuring a strong reliability of our results.

A further analysis that would extend our observations is the representation similarity analysis (RSA) between the fMRI brain activity and the emotion decoding with CNN [35]. RSA would be able to measure the similarity of processing between biological and artificial visual systems, as well as quantifying the differences of their inner representations.

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Explainable Concept Mappings of MRI: Revealing the Mechanisms Underlying Deep Learning-Based Brain Disease Classification

Christian Tinauer[1(✉)], Anna Damulina[1], Maximilian Sackl[1], Martin Soellradl[2], Reduan Achtibat[3], Maximilian Dreyer[3], Frederik Pahde[3], Sebastian Lapuschkin[3], Reinhold Schmidt[1], Stefan Ropele[1], Wojciech Samek[3,4,5], and Christian Langkammer[1(✉)]

[1] Department of Neurology, Medical University of Graz, 8036 Graz, Austria
{christian.tinauer,christian.langkammer}@medunigraz.at
[2] Department of Radiology and Radiological Sciences, Monash University, Melbourne, VIC 3800, Australia
[3] Fraunhofer Heinrich Hertz Institute, 10587 Berlin, Germany
[4] Technische Universität Berlin, 10623 Berlin, Germany
[5] BIFOLD - Berlin Institute for the Foundations of Learning and Data, 10587 Berlin, Germany
http://www.neuroimaging.at

Abstract. ***Motivation.*** While recent studies show high accuracy in the classification of Alzheimer's disease using deep neural networks, the underlying learned concepts have not been investigated.

Goals. To systematically identify changes in brain regions through concepts learned by the deep neural network for model validation.

Approach. Using quantitative R2* maps we separated Alzheimer's patients (n = 117) from normal controls (n = 219) by using a convolutional neural network and systematically investigated the learned concepts using Concept Relevance Propagation and compared these results to a conventional region of interest-based analysis.

Results. In line with established histological findings and the region of interest-based analyses, highly relevant concepts were primarily found in and adjacent to the basal ganglia.

Impact. The identification of concepts learned by deep neural networks for disease classification enables validation of the models and could potentially improve reliability.

Keywords: MRI · Medical diagnosis · Alzheimer's disease · Concepts identification · Histological validation

1 Introduction

Deep neural networks can learn features from magnetic resonance imaging (MRI) for the classification of Alzheimer's disease (AD) [25], but are generally seen as

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black boxes [7]. Medical applications are especially required to verify that the accuracy of those models is not the result of exploiting data artifacts. However, even when using conventional heat mapping approaches like Integrated Gradients [21], LIME [18] or Layer-wise Relevance Propagation (LRP) [3] it remains unclear whether atrophy, signal intensity changes or other contributors are relevant for the T1-based separation of patients with AD from normal controls (NC) [23]. Nevertheless, convolutional neural networks (CNNs) used for classification can be considered as feature extractors (e.g. channels of convolutional layers), which are later on classified by the fully connected layers. We make use of this network structure and the extracted features to explore the learned *concepts* and their importance to the classification results.

In this work, we utilized a quantitative MRI parameter, the effective relaxation rate R2* for AD classification by a relevance-regularized CNN. The relaxation rate R2* is defined as the inverse of the effective transversal relaxation time T2* ($R_2^* = \frac{1}{T_2^*}$), whereas T2* can be considered the time observed for the transverse magnetization decay in each voxel.

R2* is highly positively correlated with iron concentration in gray matter [14], and increased iron levels in the left and the right basal ganglia are frequent *concepts* found in AD [8]. We believe that CNNs implicitly learn such concepts and have recently implemented the tools to disentangle and visualize them individually.

The goal of this paper is to investigate the usage of concepts learned by the CNN for AD classification using Concept Relevance Propagation (CRP) [1]. Therefore, we compare *differences in concept maps* with the outcome of an established region of interest (ROI) based analysis and relate the results to histologically well-known areas of disease activity.

2 Methodology

2.1 Dataset

We retrospectively selected 226 MRI datasets from 117 patients with probable AD (mean age = 71.1 ± 8.2 years, male/female = 93/133) from our outpatient clinic and 226 MRIs from 219 propensity-logit-matched (covariates: age, sex) [12,19] normal controls (mean age = 69.6 ± 9.3 years, m/f = 101/125) from an ongoing community-dwelling study. Patients and controls were scanned using a consistent MRI protocol at 3 T (Siemens TimTrio) including a structural T1-weighted MPRAGE sequence with 1 mm isotropic resolution (TR/TE/TI/FA = 1900 ms/2.19 ms/900 ms/9°, matrix = 176 × 224 × 256) and a spoiled FLASH sequence ($0.9 \times 0.9 \times 2\,\text{mm}^3$, TR/TE = 35/4.92 ms, 6 echoes, 4.92 ms echo spacing, matrix = 208 × 256 × 64 × 6).

2.2 Preprocessing

Brain masks for each subject were obtained using FSL-SIENAX [20] and the structural T1-weighted image. Using the data acquired from the spoiled FLASH sequence, we solved the inverse problem given as

$$M_{xyz}[i] = M_{xyz}[0]e^{-t[i]R_2^*}, \tag{1}$$

for $M_{xyz}[0]$ and R_2^*, where $M_{xyz}[i]$ is the observed transversal magnetization in voxel xyz at echo i with echo time $t[i]$. The computations were executed for all voxels in the image volumes, yielding R2* maps (matrix $= 208 \times 256 \times 64$).

The obtained R2* maps were affinely registered to the subject's MPRAGE sequence using FSL-flirt and nonlinearly registered to the MNI152 standard-space brain template using FSL-fnirt.

2.3 Standard Classification Network

We utilize a classifier network, which uses the combination of a single convolutional layer (kernel $3 \times 3 \times 3$, 8 channels) followed by a down-convolutional layer (kernel $3 \times 3 \times 3$, 8 channels, stridding $2 \times 2 \times 2$) as the main building block. The overall network stacks four of those main building blocks before passing the data through two fully connected layers (16 and 2 units). Each layer is followed by a Rectified Linear Unit (ReLU) nonlinearity, except for the output layer where a Softmax activation is applied. For better reproducibility, we initially created 3 initializations of the network weights and used them as starting points for the network trainings.

2.4 Relevance-Guided Classification Network

To focus the network on relevant features, we proposed a relevance-guided network architecture, that extends the given classifier network with a relevance map generator [23], named $Graz^+$. In brief, to guide the training process, we extended the classifier's categorical cross entropy loss (loss_{CCE}) by a loss term to act as a regularizer:

$$\text{loss}_{\text{relevance}}(\mathbf{R}, \mathbf{M}) = -\mathbf{1}^T \text{vec}(\mathbf{R} \odot \mathbf{M}), \tag{2}$$

which enforces higher relevance values in $\mathbf{R}$ at positions marked by the focus region $\mathbf{M}$. Because the sum of relevance values in $\mathbf{R}$ is constant [15] this regularizer also automatically decreases the relevance values at positions not marked by $\mathbf{M}$. Consequently, this approach yields the total loss per data sample:

$$\begin{aligned} \text{loss}_{\text{Graz}^+} &= \text{loss}_{\text{relevance}} + \text{loss}_{\text{CCE}} \\ &= -\mathbf{1}^T \text{vec}(\mathbf{R} \odot \mathbf{M}) - \sum_{i=1}^{\text{outputs}} y_i \cdot \log(\hat{y}_i), \end{aligned} \tag{3}$$

where $\mathbf{R}$ denotes the relevance heat map (3D shape), $\mathbf{M}$ is the predefined binary attention mask obtained during image preprocessing (3D shape), $\text{vec}(\mathbf{A})$ denotes the row major vector representation of $\mathbf{A}$ resulting in a column vector (1D shape), and $\mathbf{1}$ is a column vector where all elements are set to 1 (1D shape). The inner product of the transposed vector $\mathbf{1}$ and the vector representation of $\mathbf{R} \odot \mathbf{M}$

gives the scalar value $\text{loss}_{\text{relevance}}$ (0D shape). The negative sign accounts for the maximization of the relevance inside the mask and $\odot$ denotes the element-wise product. For the categorical cross entropy y_i is the target value of the i-th output class and $\hat{y}_i$ its predicted value. For this study we used the FSL-SIENAX brain masks from image preprocessing as attention masks.

2.5 Training

We trained models on R2* maps in subject space using Adam [11] for 60 epochs with a batch size of 6. AD and NC data were separately sampled into training, validation, and test sets (ratio 70:15:15) while maintaining all scans from one person in the same set. To ensure the same class distribution in all setups, final sets were created by combining one set from each cohort. The sampling procedure was repeated 10 times to enable a bootstrapping analysis.

We reduced the learning by 0.3 when the validation loss did not decrease for 5 consecutive epochs and we also reset the model's weights to the state at the begin of the plateau phase. Initial learning rate was set to 10^{-3} and minimum learning rate was set to 10^{-6}.

2.6 Model Selection

The optimal models were selected from the last epoch of each run. The relevance inside the attention mask (brain mask) threshold was set to 95%, enforcing models where most of the relevance is inside the intracranial volume.

2.7 Bootstrapping Analysis

For each data sampling (10 data samplings) and for all network weights initializations (3 weights initializations), we repeated training of the network, creating overall 30 training sessions with the same input image configuration [4]. To identify concepts, we selected the best-performing run in terms of validation accuracy from the 30 training sessions.

2.8 Concepts Identification

Extending backpropagation-based heat mapping methods, CRP [1] enables conditioning on a concept encoded by a hidden-layer channel. Hence, for this analysis we computed the concept-conditional explanations for all eight channels of the last down-convolutional layer before the fully connected layers (cf. Fig. 1 for details). All concept maps were computed w.r.t. both output classes and the z^+-composite applied. In this study, the applied z^+-composite is made up from the z^+-rule for all convolutional- and all fully connected layers.

By applying RelMax [1], with the objective to maximize the relevance criterion, we ranked the test inputs for each concept channel. Furthermore, we computed the relative importance of the concept channels for the classification results and used it for ranking the concepts.

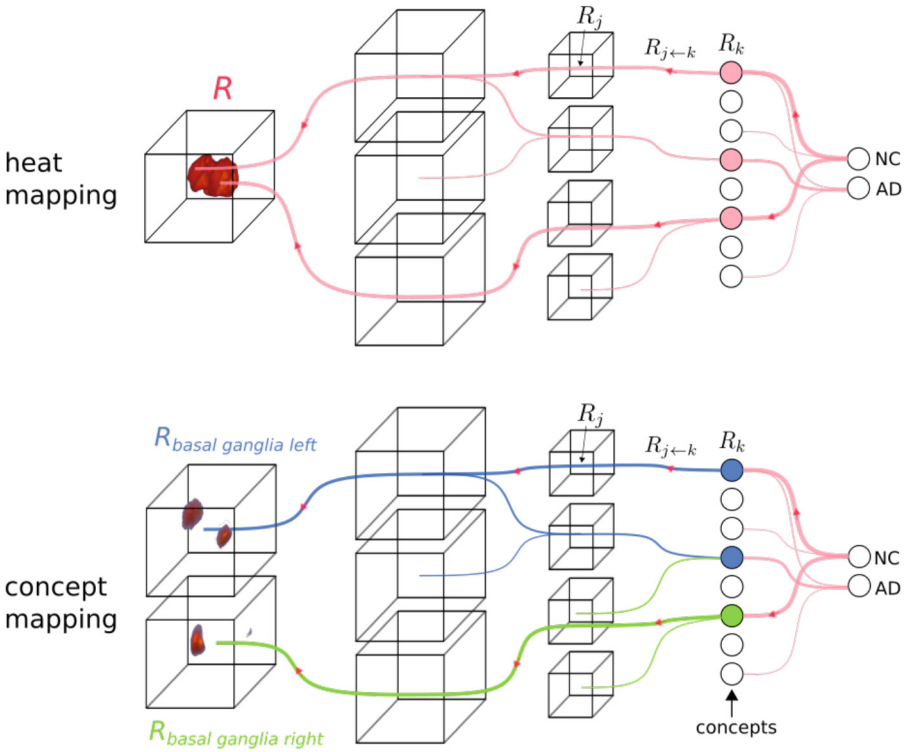

Fig. 1. Top row shows schematic overview of conventional back-propagation heat mapping propagating relevance scores backwards through the network creating a single heat map. Line thickness symbolizes the (relative) amount of relevance (in red) flowing through the connections. In contrast, shown in the bottom row, by conditioning on a concept encoded by a hidden-layer channel (highlighted in blue and green), Concept Relevance Propagation (CRP) and RelMax allow to compute concept-conditional explanations and provide semantic meaning for latent model structures, disentangling the learned and identified image features. *NC* normal control; *AD* Alzheimer's disease; *R* relevance. (Color figure online)

2.9 Relevance-Weighted Concept Map Representation

For local concept identification we present the top ranked test images in Fig. 3 in Appendix. Besides qualitatively investigating individual concept maps, we calculated mean concept maps for the classes NC and AD. As the concepts for both classes are visually very similar (cf. Figs. 4 and 5 in Appendix), we present the *class differences of the concepts* given as

$$\mathbf{R}_{diff} = \mathbf{R}_{NC,mean} - \mathbf{R}_{AD,mean}, \tag{4}$$

where $\mathbf{R}_{NC,mean}$ is the mean of the concept maps for the NC test images, and $\mathbf{R}_{AD,mean}$ is the mean of the concept maps for test AD images. Heat maps, concept maps, and differences of concept maps shown in this paper are overlaid on the MNI152 standard-space brain 1 mm template and windowed to present the top 50% of relevance. Windowing maps to the top 50% of relevance highlights voxel regions which receive high attributions of relevance, while suppressing less important regions. In mean maps five transversal slices are shown. In contrast, for individual concepts identification the same transversal slice for the five top ranked test images are presented.

2.10 R2* and Relevance Region of Interest Analysis

Anatomic structures were segmented on the T1-weighted MPRAGE sequence using FSL-FIRST [17]. Segmentation masks were applied to the R2* maps to calculate median R2* values in the basal ganglia (BG), hippocampi, and thalami [6]. As median R2* values were not normally distributed, Mann-Whitney U tests were applied to study R2* differences between AD and NC. For concepts identification and comparison, the significance of relevance sum differences between AD and NC in the same brain regions were computed using Mann-Whitney U tests.

3 Results

The bootstrapping setup yielded a mean balanced accuracy and standard deviation of $75.64\% \pm 5.16\%$, sensitivity of $69.67\% \pm 9.55\%$, specificity of $81.61\% \pm 5.45\%$ and an area under the curve of the receiver operating characteristics (AUC) of 0.76 ± 0.05.

The differences between the concept attribution for AD and NC classification of the best-performing bootstrapping run are shown in Fig. 2. Additionally, Table 1 shows the relevance sum in the basal ganglia for the four most important concepts and the results of the tests for differences in the classes. The base images for computing the differences in concepts between AD and NC are shown in Fig. 4 and Fig. 5 in the Appendix.

Qualitatively, concept maps differences show a substantial overlap with the basal ganglia regions. The median R2* values in selected anatomical brain regions are given in Table 2.

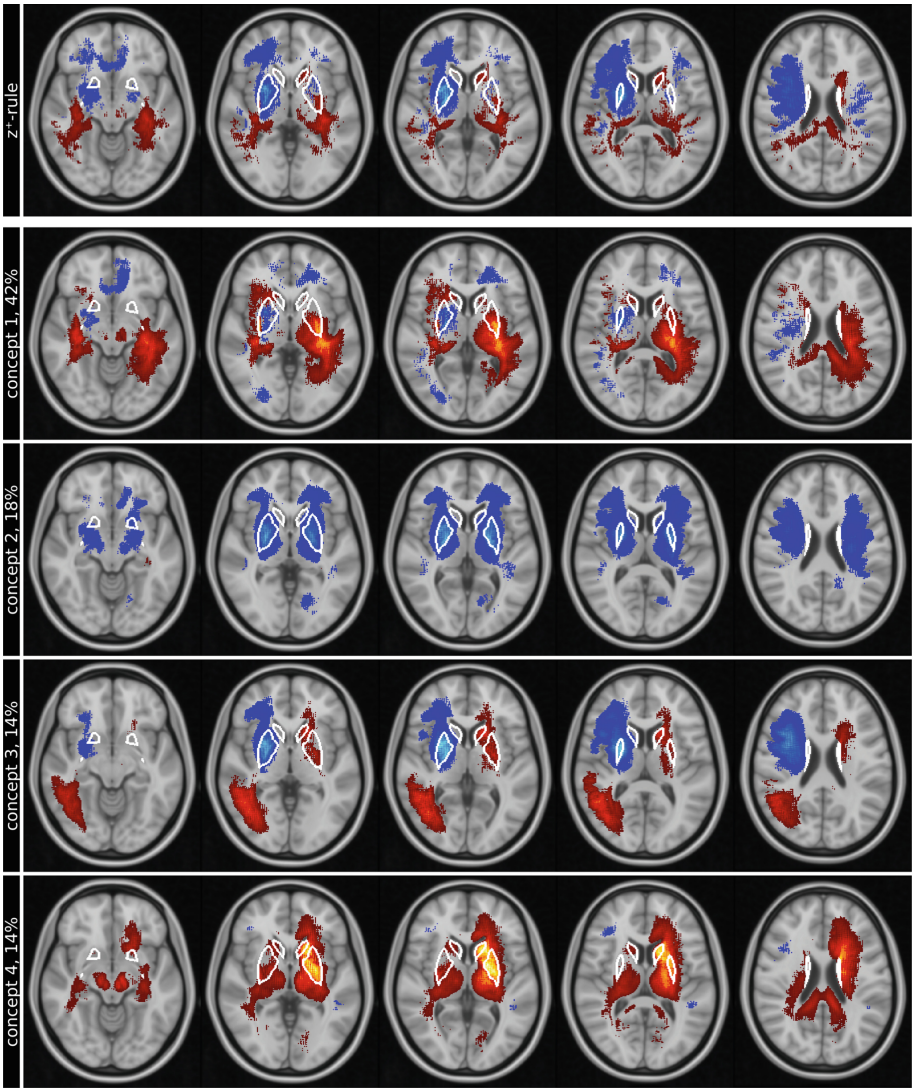

Fig. 2. Row (1) shows the difference of conventional mean global heat maps created for AD and NC using LRP-z^+-rule for five slices (columns), overlaid on the MNI152 standard-brain 1 mm template. In comparison, rows (2) to (5) show the differences of the four most important concepts on the same slices, ranked by their relative importance (percentage next to name) for the classification results. Negative differences are presented as blue-lightblue and indicate regions in concepts with more attribution in AD compared to NC, whereas positive differences, shown as red-yellow, highlight regions in concepts with more attribution in NC compared to AD. If the attribution in NC and AD is similar (e.g. concept is nearly equally used for both classes) the regions disappear. Images are shown in standard-radiological view, thus left and right is flipped and white lines outline the basal ganglia. (Color figure online)

Table 1. Sum of relevance values with interquartile range in parentheses in different brain regions in the four most important concepts. Mann-Whitney U test was used to calculate the p Value for study participants with AD and NC. Basal ganglia is defined as the sum of caudate nucleus, pallidum, putamen according to [10]. * indicates that p is less than 0.05.

	Region	Relevance sum NC	Relevance sum AD	p Value
concept 1	Left basal ganglia	0.0129 (0.0106–0.0148)	0.0077 (0.0060–0.0115)	**0.000***
	Left caudate nucleus	0.0020 (0.0016–0.0024)	0.0014 (0.0011–0.0021)	**0.008***
	Left pallidum	0.0036 (0.0028–0.0041)	0.0021 (0.0013–0.0030)	**0.001***
	Left putamen	0.0073 (0.0058–0.0082)	0.0042 (0.0032–0.0064)	**0.000***
	Right basal ganglia	0.0258 (0.0228–0.0291)	0.0271 (0.0174–0.0312)	**0.974**
	Right caudate nucleus	0.0032 (0.0023–0.0039)	0.0031 (0.0026–0.0044)	**0.673**
	Right pallidum	0.0061 (0.0051–0.0070)	0.0064 (0.0044–0.0079)	**0.940**
	Right putamen	0.0167 (0.0138–0.0185)	0.0161 (0.0105–0.0193)	**0.650**
concept 2	Left basal ganglia	0.0025 (0.0019–0.0043)	0.0072 (0.0047–0.0098)	**0.000***
	Left caudate nucleus	0.0004 (0.0003–0.0008)	0.0012 (0.0008–0.0020)	**0.000***
	Left pallidum	0.0006 (0.0005–0.0011)	0.0019 (0.0011–0.0024)	**0.000***
	Left putamen	0.0015 (0.0010–0.0025)	0.0040 (0.0027–0.0055)	**0.000***
	Right basal ganglia	0.0046 (0.0033–0.0057)	0.0081 (0.0052–0.0119)	**0.000***
	Right caudate nucleus	0.0006 (0.0003–0.0007)	0.0013 (0.0007–0.0017)	**0.000***
	Right pallidum	0.0013 (0.0007–0.0016)	0.0022 (0.0014–0.0030)	**0.000***
	Right putamen	0.0030 (0.0020–0.0037)	0.0051 (0.0034–0.0071)	**0.000***
concept 3	Left basal ganglia	0.0062 (0.0046–0.0073)	0.0040 (0.0025–0.0056)	**0.001***
	Left caudate nucleus	0.0015 (0.0011–0.0020)	0.0010 (0.0006–0.0014)	**0.001***
	Left pallidum	0.0014 (0.0011–0.0018)	0.0010 (0.0006–0.0013)	**0.004***
	Left putamen	0.0032 (0.0023–0.0039)	0.0020 (0.0014–0.0029)	**0.002***
	Right basal ganglia	0.0028 (0.0019–0.0043)	0.0059 (0.0045–0.0083)	**0.000***
	Right caudate nucleus	0.0004 (0.0003–0.0006)	0.0009 (0.0007–0.0013)	**0.000***
	Right pallidum	0.0007 (0.0005–0.0009)	0.0012 (0.0010–0.0017)	**0.000***
	Right putamen	0.0029 (0.0025–0.0033)	0.0022 (0.0016–0.0028)	**0.000***
concept 4	Left basal ganglia	0.0120 (0.0085–0.0144)	0.0068 (0.0034–0.0100)	**0.000***
	Left caudate nucleus	0.0029 (0.0021–0.0039)	0.0013 (0.0007–0.0024)	**0.000***
	Left pallidum	0.0028 (0.0018–0.0035)	0.0014 (0.0009–0.0024)	**0.000***
	Left putamen	0.0062 (0.0042–0.0070)	0.0028 (0.0018–0.0049)	**0.000***
	Right basal ganglia	0.0051 (0.0046–0.0060)	0.0037 (0.0026–0.0048)	**0.000***
	Right caudate nucleus	0.0009 (0.0006–0.0013)	0.0007 (0.0004–0.0009)	**0.012***
	Right pallidum	0.0013 (0.0011–0.0015)	0.0009 (0.0005–0.0012)	**0.000***
	Right putamen	0.0016 (0.0012–0.0027)	0.0036 (0.0028–0.0053)	**0.000***

Table 2. Median R2* values with interquartile range in parentheses in different brain regions in study participants with AD and NC. Mann-Whitney U test was used to calculate the p Value. Basal ganglia is defined as the mean of caudate nucleus, pallidum, putamen according to [10]. * indicates that p is less than 0.05.

Region	Median R_2^* NC (sec^{-1})	Median R_2^* AD (sec^{-1})	p Value
Left basal ganglia	29.00 (27.24–32.66)	30.76 (28.15–33.60)	**0.006***
Left caudate nucleus	23.37 (21.43–25.70)	24.28 (22.54–26.71)	**0.004***
Left pallidum	36.52 (33.17–40.58)	36.43 (33.61–41.30)	**0.401**
Left putamen	27.48 (25.15–31.75)	30.11 (27.27–33.94)	**0.000***
Left hippocampus	16.40 (15.34–17.49)	15.88 (14.78–17.06)	**0.003***
Left thalamus	19.70 (18.77–20.89)	19.71 (18.85–20.92)	**0.972**
Right basal ganglia	29.21 (26.74–32.14)	30.62 (28.09–32.91)	**0.002***
Right caudate nucleus	23.06 (21.35–25.17)	24.19 (22.11–25.94)	**0.002***
Right pallidum	36.25 (32.86–40.61)	36.76 (33.68–41.54)	**0.094**
Right putamen	28.08 (25.07–31.76)	30.17 (27.08–34.11)	**0.000***
Right hippocampus	16.44 (15.54–17.38)	16.02 (14.88–17.19)	**0.008***
Right thalamus	19.97 (18.82–20.71)	19.76 (18.90–20.62)	**0.497**

4 Discussion

Previous studies using T1-weighted MRIs showed that volumetric brain features are highly relevant for the CNN-based classification of AD [23], which has also been confirmed in recent related work on simulated aging [9]. To focus the classifier's attention within brain parenchyma, we applied a relevance-guided regularized network ($Graz^+$) [23]. With the availability of accessible large MRI databases from patients, such as the Alzheimer's Disease Neuroimaging Initiative (ADNI), AIBL or OASIS databases, various studies using CNN with explanation exploiting structural imaging data have been published [5,16,22,24]. However, from T1-weighted images alone it cannot be concluded whether volumetric features, signal intensity changes or other factors are relevant.

Therefore, this study identified the concepts of the CNN-based classification with R2* maps. Various studies have shown that brain iron increasingly accumulates in the deep gray matter of AD patients [6,13]. While the highest relevances in and adjacent to the basal ganglia are consistently involved in all concepts, differences in concept mapping identified complementary spatial patterns (e.g. concept 1 and concept 3, Fig. 2).

The differences in concept 1 show substantial positive differences in and adjacent to the left basal ganglia, whereas the right basal ganglia show nearly no differences. This was confirmed by the tests for relevance sum differences in Table 1 and was also found in susceptibility sensitive imaging in AD [2,6]. Concept 1 seems to activate more for NC images, favoring smaller R2* values in the left pallidum, left putamen and left white matter, whereas concept 2 shows

only negative differences, favoring higher R2* values in the left and right basal ganglia. Additionally, concept 3 shows positive differences in the right temporal and occipital lobes, which is in line with the analysis in [6]. Please note that all MR images in this paper are shown in standard-radiological view, causing the left and right side of the brain to be flipped.

Differences in the R2* values between AD and NC were additionally confirmed by the conventional ROI-based results in Table 2.

5 Conclusion

This study used quantitative MRI data (R2*) for deep learning classification and CRP in a clinical cohort of AD patients. Confirming histological and in-vivo iron mapping studies, this underlines that heat- and concept mapping can serve as an exploratory means to identify areas of pathological tissue changes and further reveal internal mechanisms of deep learning classification networks. Using the differences in concept maps increases the spatial sparsity of relevances and might indicate which changes in the input images are learned by the CNNs within each concept.

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Disclosure of Interests. The authors have no competing interests to declare that are relevant to the content of this article.

Appendix

Figure 3 shows the five top ranked test images for each concept on the same transversal slice, while Figs. 4 and 5 show the mean concept maps for NC and AD on five transversal slices.

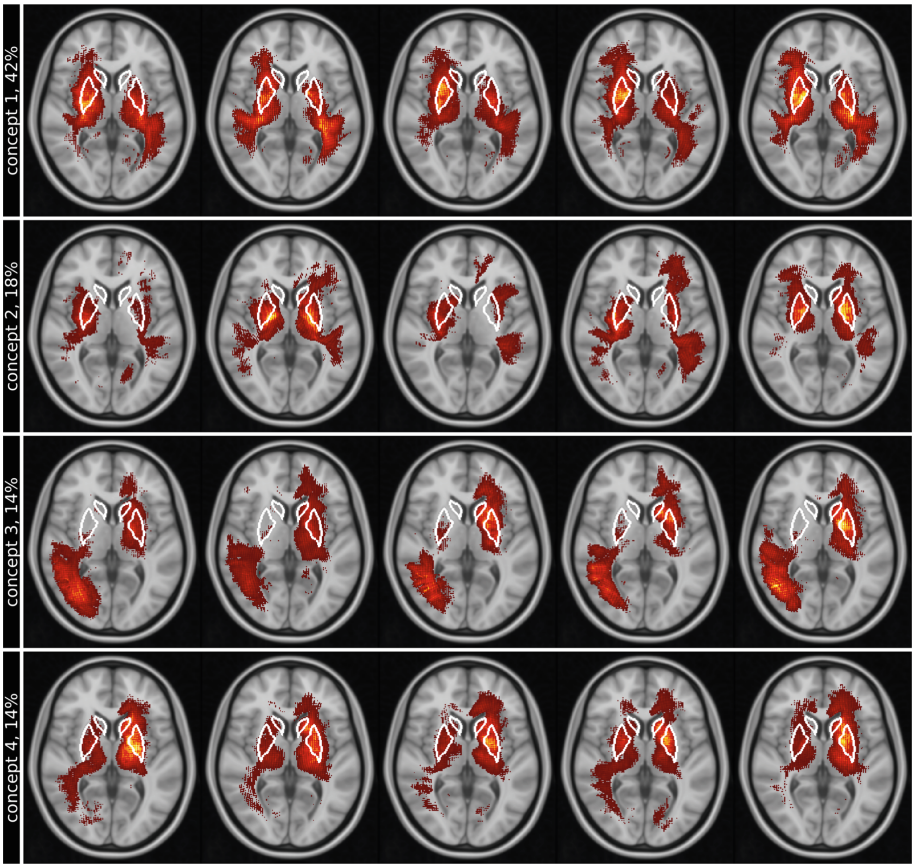

Fig. 3. Rows (1) to (4) show the 4 most important concepts, ranked by their relative importance (percentage next to name) for the classification results. Columns (1) to (5) show the 5 top ranked individual test images for each concept on the same transversal slice. Positive relevance, shown as red-yellow, highlight regions in concepts with higher attributions. White lines in slices outline the basal ganglia. Images are shown in standard-radiological view, causing the left and right side of the brain to be flipped. (Color figure online)

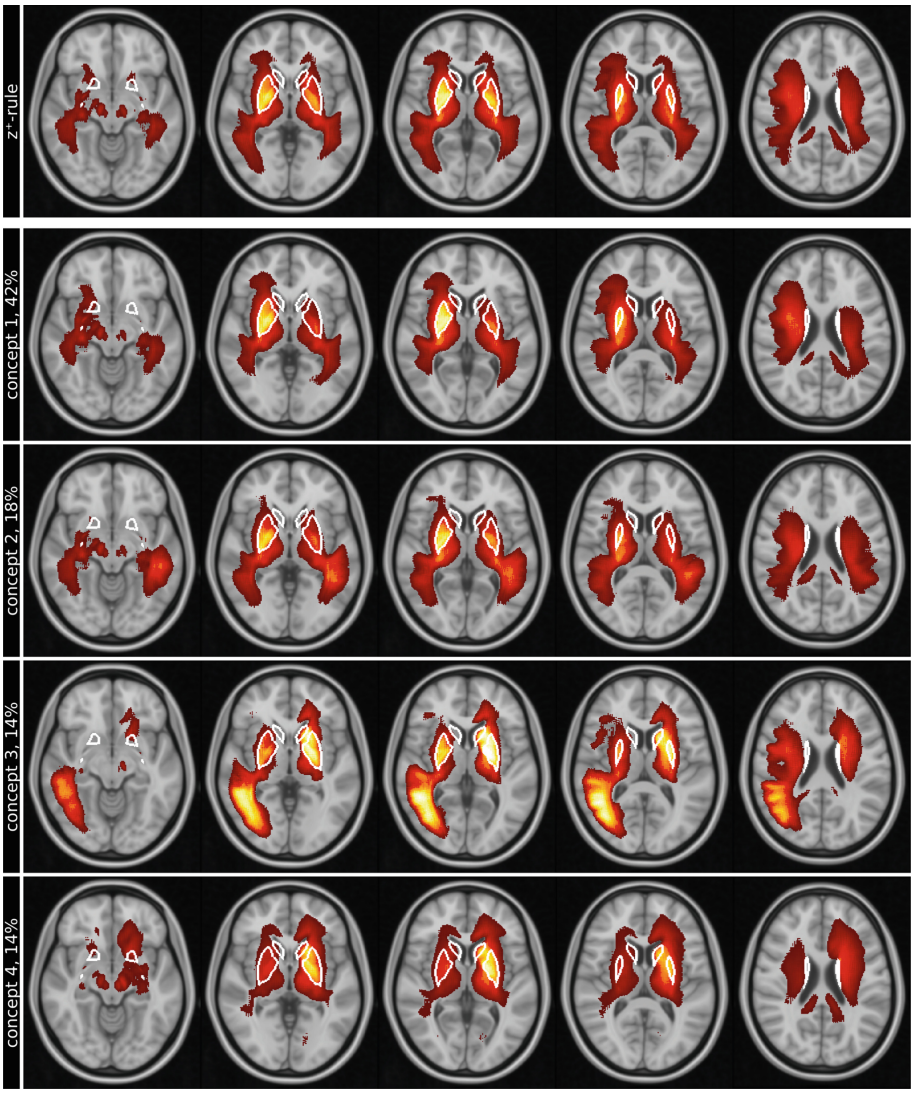

Fig. 4. Row (1) shows the conventional mean global heat maps created for NC using LRP-z^+-rule for 5 slices (columns), overlaid on the MNI152 standard-brain 1 mm template. In comparison, rows (2) to (5) show the 4 most important concepts on the same slices, ranked by their relative importance (percentage next to name) for the classification results. Positive relevance, shown as red-yellow, highlight regions in concepts with higher attributions. White lines in slices outline the basal ganglia. Images are shown in standard-radiological view, causing the left and right side of the brain to be flipped. (Color figure online)

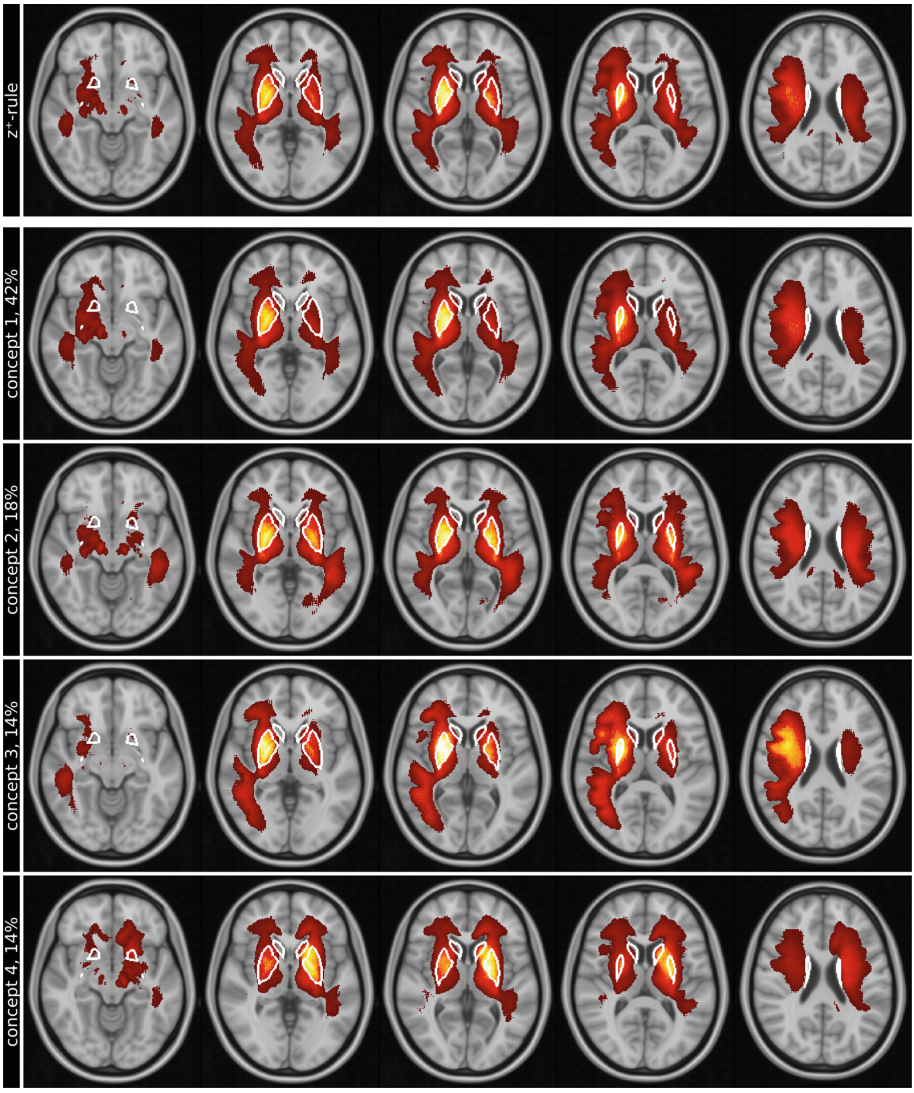

Fig. 5. Row (1) shows the conventional mean global heat maps created for AD using LRP-z^+-rule for 5 slices (columns), overlaid on the MNI152 standard-brain 1 mm template. In comparison, rows (2) to (5) show the 4 most important concepts on the same slices, ranked by their relative importance (percentage next to name) for the classification results. Positive relevance, shown as red-yellow, highlight regions in concepts with higher attributions. White lines in slices outline the basal ganglia. Images are shown in standard-radiological view, causing the left and right side of the brain to be flipped. (Color figure online)

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Logic, Reasoning, and Rule-Based Explainable AI

Template Decision Diagrams for Meta Control and Explainability

Clemens Dubslaff[1,2(✉)], Verena Klös[2(✉)], and Juliane Päßler[3(✉)]

[1] Eindhoven University of Technology, Eindhoven, The Netherlands
c.dubslaff@tue.nl
[2] Centre for Tactile Internet with Human-in-the-Loop (CeTI), Dresden, Germany
verena.kloes@tu-dresden.de
[3] University of Oslo, Oslo, Norway
julipas@ifi.uio.no

Abstract. *Decision tree classifiers (DTs)* provide an effective machine-learning model, well-known for its intuitive interpretability. However, they still miss opportunities well-established in software engineering that could further improve their explainability: separation of concerns, encapsulation, and reuse of behaviors. To enable these concepts, we introduce *templates* in *decision diagrams (DDs)* as an extension of multi-valued DDs. Templates allow to encapsulate and reuse common decision-making patterns. By a case study from the autonomous underwater robotics domain we illustrate the benefits of template DDs for modeling and explaining *meta controllers*, i.e., hierarchical control structures with underspecified entities. Further, we implement a template-generating refactoring method for DTs. Our evaluation on standard controller benchmarks shows that template DDs can improve explainability of controller DTs by reducing their sizes by more than one order of magnitude.

Keywords: Decision Diagrams · Decision Trees · Explainability · Control

1 Introduction

Cyber-physical systems are complex. Their behaviors may depend on different kinds of sensor data, have to deal with concurrent executions, and face heterogeneous components that all can be subject to uncertainty such as noise or partial observability. *Controllers* are usually used to steer their behaviors and ensure correct functioning, e.g., in safety-critical environments. Handcrafted engineering of controllers is error-prone and does not scale to large systems. Differently, provably correct controllers can be constructed from formal specifications by *automated controller synthesis* [47]. Due to recent demands by legal authorities [1–3,60] and to increase overall trustworthiness in systems, synthesized controllers should be *explainable*.

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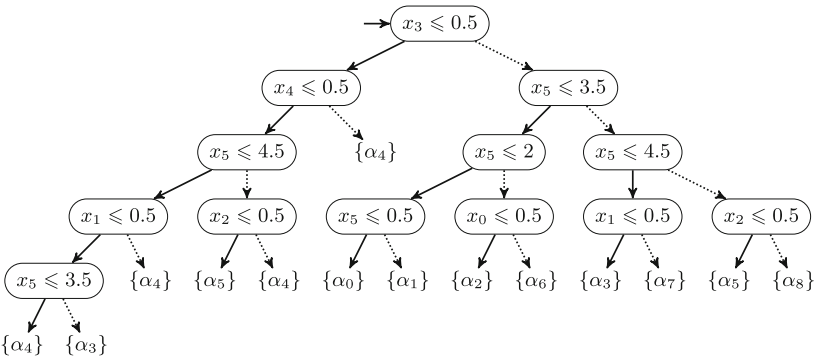

Fig. 1. DT representation of a control strategy for the TRIANGLE benchmark; solid edges are taken if the predicate is fulfilled, dotted edges if it is not.

Decision Trees for Strategy Explanation. *Decision trees (DTs)* [46,51] are an effective machine-learning model that also gained attention as explications of synthesized controllers due to their simplistic and interpretable nature [5,27]. Figure 1 shows an example DT of a controller for the probabilistic planning system TIREWORLD [41] that has been synthesized by the tool DTCONTROL [5].

In its basic form, DTs are binary trees where inner *decision nodes* are labeled by predicates over features such as observable system variables and where leafs are labeled with control actions. Each decision node has two outgoing edges for the predicate being fulfilled or not. By starting in the root of the DT and evaluating predicates according to the current state of the controlled system, DTs eventually lead to possible control actions of the controller. Depending on how the controller is synthesized, these actions all fulfill the controller specification, e.g., following its safety requirements.

Due to the intuitive evaluation of predicates in decision nodes, DTs are well-suited to represent and explain the step-wise reasoning in control policies. *Occam's razor* provides a major principle for explainability for a global perspective, according to which the best explanation for a phenomenon is the smallest one amongst all possible explanations [28]. For this, state-of-the-art learning algorithms minimize the size of DTs by choosing suitable decision predicates by entropy-based splitting and placing important decisions first. However, this minimization is only done within the class of DTs, which structurally hinders further improvements for explainability:

1. DTs are trees, duplicating shared decision making as isomorphic subtrees
2. DTs are monolithic, barely separating concerns [21]
3. DTs are deterministically evaluating predicates, only supporting nondeterminism in control actions of DT leafs

In this paper, we introduce the concept of *templates* in *decision diagrams (DDs)*, leading to a new data structure of *template DDs* that remedies the three drawbacks of DTs above. First, template DDs are diagrams instead of trees,

allowing for sharing of decision making by multiple incoming edges to decision nodes similar as within *binary decision diagrams (BDDs)* [12]. Therefore, common decision making but also common control actions can directly be deduced from the graphical representation as diagrams. Second, extending the first reasoning by merging isomorphic subtrees, we propose to even further reuse common decision making by introducing *box nodes*. Such nodes are labeled by a template that itself can stand for decision making modeled as template DDs, hence separating concerns similar to procedures in programming languages by encapsulating common behaviors. The hierarchical structure through boxes and assigned templates does not only improve explainability of control policies by allowing for more compact representation, they also enable *underspecification*, enhancing degrees of freedom. Decision making that is, e.g., irrelevant for ensuring safety but important for other requirements such as energy consumption, could hence be left out for explicating safety under *all possible implementations* of templates. To this end, template DDs are also suitable to specify *families of controllers*, enabling their explanation and family-based analysis [17,18,49]. Figure 3 shows a template DD for the DT controller of Fig. 1, reusing common decision making in a template used twice in the diagram.

Meta Control. Many cyber-physical systems are subject to uncertainty in both their environment and their internal structure. The reasons could be, e.g., partial observability, hardware or software failures, or noisy sensor data. To increase reliability, the systems can be implemented as *self-adaptive systems (SASs)* [62], enabling them to adapt their structure, task plan, control strategies, etc., mostly according to a predefined strategy. SASs that adapt their control strategy itself include a controller steering the adaptation in the system and can therefore be considered as a *meta control* system. One can distinguish between *internal* and *external* self-adaptation [54,62]. An internal self-adaptation mechanism is embedded in the application logic and realized by, e.g., exception handling or fault-tolerance mechanisms. External self-adaptation separates the application logic from the adaptation logic and introduces a two-layered system where the application logic, focussing on the domain concerns, is implemented in a *managed subsystem* (the control system), and the adaptation logic is implemented in a *managing subsystem* [35] (the meta control system) – see Fig. 2. The managed subsystem monitors the environment and may affect the environment with its actions. It can also contain internal self-adaptation like exception handling. The managing subsystem receives input from both the managed subsystem and the environment to determine if and which adaptation is needed, and adapts the managed subsystem if necessary.

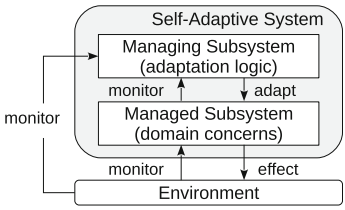

Fig. 2. A two-layered architecture for an SAS [49]

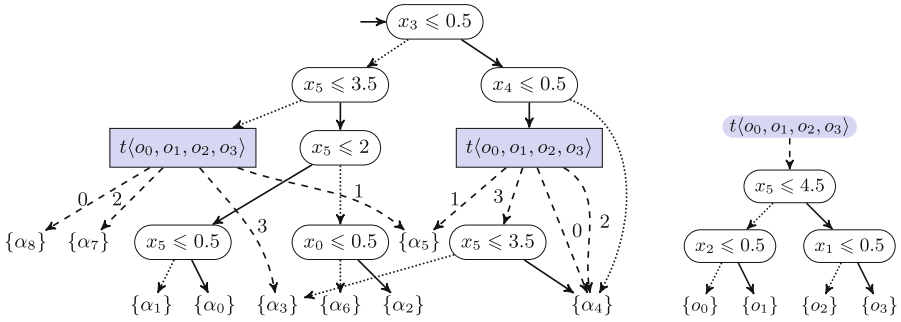

Fig. 3. Template DD for TRIANGLE control strategy

We focus here on easing the design and comprehension of meta control systems by distinguishing between high-level views for meta control and low-level means of adaptability. For this, we exploit the template DD's possibility of *underspecification*, realized by (i) leaving the specification of templates open and specifying families of controllers, and by (ii) introducing parameters in box types to steer template instantiations. The templates and parameter values can then be set by the meta controller to adapt the control strategy of the control system.

Contributions. We present three main contributions. First, we formally introduce our new data structure of template DDs as an extension of multi-valued DDs. They have dedicated box nodes which can themselves be identified with other template DDs, implementing box-associated decision making and leading to a hierarchical template DD structure. Second and third, we showcase the potential of template DDs by two applications:

1. Controller design with template DDs in the setting of *meta control*, and
2. synthesis of concise template DDs from DT controllers.

Here, we contribute to (1.) with an example of modeling a control and meta control system with template DDs for the case study of an autonomous underwater vehicle. We show that through the separation of concerns in the control structures, modeling can be eased through template DDs by reusing decision making and control underspecification. Further, we contribute to (2.) by refactoring DTs from standard DTCONTROL benchmarks towards template DDs and hierarchical DDs. We implement a simple refactoring heuristics that already shows huge potential in template DDs for explainability by reducing control strategy representations by more than one order of magnitude.

2 Related Work

Decision Trees (DTs) and Decision Diagrams (DDs). DT classifiers have a long history in artificial intelligence (AI) and statistical learning [11,46]. The explanatory power of Boolean DTs recently gains more and more attention in explainable AI [6]. Plambeck *et al.* [50] presented an approach to connect environment input to system behaviors and thus foster understanding which input caused what behavior. DTs are also used to generate *why*, *why-not*, *how-to*, and *what-if* explanations for context-aware intelligent systems [40].

While DTs enforce a tree structure, DDs are directed acyclic graphs and thus more general. Most prominently, binary DDs [12] are widely used in hardware design and verification to concisely represent Boolean functions [44,45]. The latter are in particular powerful when ordered and reduced, enabling a canonical representation and efficient operations. Extensions of binary DDs enable functions on more general co-domains (ADDs or MTBDDs, [26]), multi-valued decisions (multi-valued DDs, [58]), or nodes labeled by linear predicates instead of Boolean variables (linear DDs, [14]). MTBDDs for control strategy representation in synthesis have been considered in [63]. Recently, context-free DDs as hierarchical structures for efficient quantum circuit representation have been presented [57]. While this work also follows the idea of subdiagrams as components, advantages for modeling control strategies, explainability, and DT representations were not considered as we do with the use of template DDs.

Explainability of Self-adaptive Systems (SASs). Initially, research on SASs has been mainly focused on achieving self-adaptivity. However, due to increasing complexity of intelligent cyber-physical systems, explainability emerges as an important research direction [30,31]. Comprehensible adaptation rules for context-dependent effect expectations of adaptation options enable expressive, yet explainable, adaptation logics [36]. Li *et al.* proposed to use probabilistic model checking to determine when an explanation improves the overall system utility of an SAS with human-in-the-loop [39]. Camili *et al.* described different levels of explainability with corresponding meta-requirements, and demonstrated human-understandable explanations with robotic examples [13]. Autonomous agents can be also considered as adaptive systems. This research community has already a long tradition of investigating trust and explainability.

DTs have also been used as a specification formalism for SAS adaption rules, e.g., in the managing subsystem or meta controller for adapting control strategies [19,34,53]. Other works directly use DTs for self-adaptation: Chen *et al.* [15] model architectural design decisions available at runtime as DTs and combine them with goal models representing requirements for a new model-based self-adaptation approach. Cheng and Garlan [16] propose a language for describing adaptation strategies of architecture-based SASs where adaptation strategies are represented as DTs. None of these works uses a concept similar to template DDs.

Modularity in Decision Making. Behavior trees [20] are frequently used for robot controllers [33]. Their main advantage is a modular design where "individ-

ual behaviors can easily be reused in the context of another higher-level behavior, without needing to specify how they relate to the subsequent behaviors" [7]. There also exist approaches to parameterize behavior trees to improve reuse of subtrees [56] or reinforcement learning of parameters for complex robot control tasks [42]. Similar to these approaches, we introduce (parameterized) templates for DDs towards hierarchical template DDs that also benefit from modularity.

3 Foundations

We settle the foundations and notations used throughout the paper. Let us fix a set of *variables* Var on which we define *evaluations* as functions $\epsilon\colon \mathit{Var} \to \mathbb{Q}$ where $\mathbb{Q}$ are the rational numbers. The set of evaluations over Var is denoted by $\mathbb{E}$. A linear arithmetic *expression* e is defined by the grammar $e ::= c \mid c \cdot x \mid -e \mid (e + e)$ where x ranges over Var and c ranges over $\mathbb{Q}$. For an evaluation ϵ, the semantics $[\![e]\!]_\epsilon$ of an expression e is defined recursively as expected, e.g., $[\![x]\!]_\epsilon = \epsilon(x)$ and $[\![(e + e')]\!]_\epsilon = ([\![e]\!]_\epsilon + [\![e']\!]_\epsilon)$. We denote by $\mathit{Var}(e)$ the set of variables in the expression e. A linear *predicate* is of the form $(e \sim 0)$ where e is an expression and $\sim \in \{=, \geqslant, >\}$ is a comparison relation. The set of predicates is denoted by $\mathbb{P}$. A predicate $(e \sim 0) \in \mathbb{P}$ is *satisfied* by an evaluation ϵ, denoted $\epsilon \models (e \sim 0)$, iff $[\![e]\!]_\epsilon \sim 0$. We mainly consider *axis-aligned predicates*, i.e., predicates of the form $(c - x \geqslant 0)$ for $c \in \mathbb{Q}$ and $x \in \mathit{Var}$ and denote them by $x \leqslant c$. We also write x for the predicate $x - 1 \geqslant 0$, providing a Boolean interpretation of x.

Definition 1. *A* strategy *over states* $S \subseteq \mathbb{E}$ *and outcomes* Out *is a function* $\sigma\colon S \to 2^{\mathit{Out}}$. *We call* σ deterministic *if* $|\sigma(\epsilon)| = 1$ *for all* $\epsilon \in S$. *The strategy* σ *is* complete *if* $S = \mathbb{E}$.

To symbolically represent strategies through a decision diagram (DD) formalism that covers decision tree classifiers (DTs), we take inspiration from *linear DDs* [14] and *multi-terminal binary DDs* [26]. Recall that we fixed a set of variables Var used as domain of evaluations throughout the paper.

Definition 2. *A* decision diagram (DD) *is a tuple* $\mathcal{D} = (N, \mathit{Out}, \lambda, \mathtt{lo}, \mathtt{hi}, r)$ *where* $N = D \cup O$ *is a finite set of* nodes *comprising* decision nodes D *and* outcome nodes O *to which* $\lambda\colon N \to \mathbb{P} \cup 2^{\mathit{Out}}$ *labels either a predicate or a set of outcomes from* Out, *respectively,* $\mathtt{lo}, \mathtt{hi}\colon D \to N$ *are decision functions, and* $r \in N$ *is the root node of* $\mathcal{D}$.

A *path* in $\mathcal{D}$ is an alternating sequence $n_0, d_0, n_1, d_1, \ldots \in (D \times \{0,1\})^* \times N$ where for all i with $d_i = 1$ we have $n_{i+1} = \mathtt{hi}(n_i)$ and $n_{i+1} = \mathtt{lo}(n_i)$ if $d_i = 0$. Given a node $n \in N$, we denote by $\mathit{Reach}(n) \subseteq N$ the set of nodes reachable from n, i.e., the smallest set R where $n \in R$ and for all $n' \in R \cap D$ we have $\mathtt{lo}(n'), \mathtt{hi}(n') \in R$. We call $\mathcal{D}$ a *decision tree (DT)* if its underlying graph has a tree structure: for all $n, n' \in D$ with $n' \notin \mathit{Reach}(n)$ and $n \notin \mathit{Reach}(n')$ we have $\mathit{Reach}(n) \cap \mathit{Reach}(n') = \varnothing$. By $\mathcal{D}_n$ we denote the DD rooted at $n \in N$ that arises from $\mathcal{D}$ by replacing r by n. We assume that on all paths through $\mathcal{D}$, any

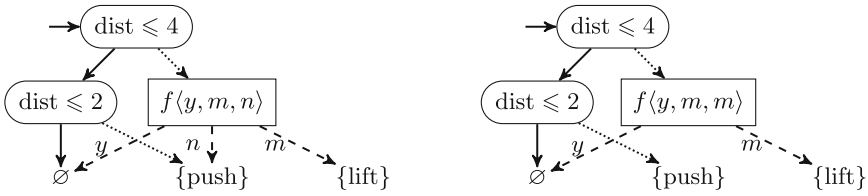

Fig. 4. Two simple example template DDs

predicate appears only once, i.e., $\lambda(n) = \lambda(n')$ implies $n = n'$ for all $n \in D$ and $n' \in \mathit{Reach}(n)$. We call $\mathcal{D}$ *deterministic* if $|\lambda(o)| = 1$ for all $o \in O$. Further, $\mathcal{D}$ is *reduced* if for all decision nodes $n, n' \in D$ we have $\mathtt{lo}(n) \neq \mathtt{hi}(n)$ (essentiality) and $\mathcal{D}_n \cong \mathcal{D}_{n'}$ implies $n = n'$ (merged isomorphic subdiagrams).

Figure 1 exemplifies how we graphically specify decision diagrams: the root node is specified by a sole incoming arrow; `hi`-edges are drawn with solid arrows, `lo`-edges with dotted ones; sets of outcomes are at the leafs of the diagram.

The semantics $[\![\mathcal{D}]\!]$ of $\mathcal{D} = \mathcal{D}_r$ is a complete strategy over $\mathbb{E}$ and Out, recursively defined for every evaluation $\epsilon \in \mathbb{E}$ as

$$[\![\mathcal{D}_n]\!](\epsilon) = \begin{cases} n & \text{if } n \subseteq \mathit{Out} \\ [\![\mathcal{D}_{\mathtt{lo}(n)}]\!](\epsilon) & \text{if } n \in D \text{ and } \epsilon \not\models \lambda(n) \\ [\![\mathcal{D}_{\mathtt{hi}(n)}]\!](\epsilon) & \text{if } n \in D \text{ and } \epsilon \models \lambda(n) \end{cases}$$

4 Template Decision Diagrams

In this section, we extend DDs towards *template DDs* where decisions are not exclusively made through completely specified predicates but can also be encapsulated in *boxes*. Formally, let $\mathbb{F}$ be a set of *function names* and define $\kappa(f)$ to denote the *arity* of the function $f \in \mathbb{F}$. Further, define a set $\mathbb{B}$ of *box types* that have the form $f\langle o_0, \ldots, o_{k-1}\rangle$ where $f \in \mathbb{F}$, $k = \kappa(f)$, and o_i for $0 \leqslant i < k$ are ordered *box outcomes*. The outcomes associated to a box type $\tau \in \mathbb{B}$, are collected in a set $\mathit{Out}(\tau) = \{o_0, \ldots, o_{k-1}\}$. Intuitively, a box of type τ as above encapsulates a strategy on $\mathit{Out}(\tau)$ by implementing function f. Enhancing DDs such that they can serve as templates of more complex decision making, we allow for *box nodes* that are non-binary as in *multi-valued DDs* [58]:

Definition 3. *A* template DD *is a tuple* $\mathcal{T} = (N, \mathit{Out}, \lambda, \mathtt{succ}, r)$ *where* $N = P \cup B \cup O$ *comprises* predicate nodes P, box nodes B, *and* outcome nodes O *to which* $\lambda\colon N \to \mathbb{P} \cup \mathbb{B} \cup 2^{\mathit{Out}}$ *labels either a predicate, a box type, or subsets from the set of outcomes Out, respectively. The decision function* `succ` *maps every predicate node* $p \in P$ *to a function* $\mathtt{succ}(p)\colon \{0,1\} \to N$ *and every box node* $b \in B$ *to a function* $\mathtt{succ}(b)\colon \mathit{Out}(\lambda(b)) \to N$, *and* $r \in N$ *is the root node of* $\mathcal{T}$.

Notations and definitions on DDs carry over to template DDs where we consider nodes $D = P \cup B$ as decision nodes. Similar to outcomes in box types,

we assume template DD outcomes to be ordered, i.e., $Out = \{o_0, \ldots, o_{k-1}\}$ where $k = |Out|$. If a box type is Boolean, we omit the outcome types and assume o_0 standing for "false", "no", or "0", and o_1 standing for "true", "yes", or "1".

Figure 4 shows example two template DDs to specify a strategy that performs actions in $Out = \{\text{push}, \text{lift}\}$ depending on the distance "dist" to some object. The left template DD contains a box typed as $f\langle y, m, n\rangle$, providing three box outcomes y ("yes"), m ("maybe"), and n ("no"). Already this simple example shows one of the core benefits of template DDs, namely encapsulation and separation of concerns. Here, the decision whether to lift the object is externalized to a strategy given by a function $f\langle y, m, n\rangle$ in case dist > 4. Otherwise, the decision to push iff dist > 2 is made. Similarly, the template DD on the right treats also the "no" outcome of function f as "maybe", leading to $Out(f\langle y, m, m\rangle) = \{y, m\}$ and hence a binary box.

Semantics of Template DDs. The definition of template DDs hints at decision making that is *underspecified*. For any box node $b \in B$ there is plenty of room which strategy $\sigma\colon \mathbb{E} \to 2^{Out(\lambda(b))}$ is actually represented by b. A natural approach to deal with such underspecification is to model outcomes nondeterministically, i.e., every box b represents a strategy σ where $\sigma(\epsilon) = Out(\lambda(b))$ for all evaluations $\epsilon \in \mathbb{E}$. Then, the semantics $[\![\mathcal{T}]\!]$ of a template DD $\mathcal{T} = \mathcal{T}_r$ as above is the complete strategy that is recursively defined over evaluations $\epsilon \in \mathbb{E}$:

$$[\![\mathcal{T}_n]\!](\epsilon) = \begin{cases} n & \text{if } n \subseteq Out \\ [\![\mathcal{T}_{\mathtt{succ}(n,0)}]\!](\epsilon) & \text{if } n \in P \text{ and } \epsilon \not\models \lambda(n) \\ [\![\mathcal{T}_{\mathtt{succ}(n,1)}]\!](\epsilon) & \text{if } n \in P \text{ and } \epsilon \models \lambda(n) \\ \bigcup_{o \in Out(\lambda(n))} [\![\mathcal{T}_{\mathtt{succ}(n,o)}]\!](\epsilon) & \text{if } n \in B \end{cases}$$

4.1 Hierarchical Decision Diagrams by Templates

Choosing box outcomes nondeterministically is usually too coarse when specifying strategies that have to fulfill certain requirements. A natural way to refine template DDs is by providing *function evaluations* that map boxes to more specific strategies over box outcomes. While our definitions allow for any representation of strategies, e.g., machine-learning classifiers, state-action lookup tables, etc., we especially have other template DDs as strategy representation for box evaluations in mind. Then, sets of template DDs provide a uniform formalism for strategy specification that follows a hierarchical structure similar to call graphs of procedural programs and context-free BDDs [57]:

Definition 4. *A* hierarchical DD *is a tuple* $\mathfrak{H} = (\mathcal{T}^0, \ldots, \mathcal{T}^{m-1}, \beta)$ *where* $\mathcal{T}^i = (N^i, Out^i, \lambda^i, \mathtt{succ}^i, r^i)$ *are template DDs for* $i \in \{0, \ldots, m-1\}$ *and* $\beta\colon \mathbb{F} \to \{0, \ldots, m-1\}$ *is a* function evaluation *that maps function names to template DDs such that for all* $f \in \mathbb{F}$*:*

- $\mathcal{T}^{\beta(f)}$ *is compatible with* f*, i.e.,* $|Out^{\beta(f)}| = \kappa(f)$*, and*

– $\mathcal{T}^{\beta(f)}$ *is decreasing, i.e.,* $\beta(\lambda^{\beta(f)}(b)) > \beta(f)$ *for all* $b \in B^{\beta(f)}$.

The latter two conditions on function evaluations ensure that evaluated boxes provide a decision on outcomes and that no recursion occurs, respectively. This leads to a well-defined semantics $[\![\mathfrak{H}]\!] = [\![\mathcal{T}^0_{r^0}]\!]_\beta$ of a hierarchical DD $\mathfrak{H}$ with function evaluation β, recursively defined over evaluations $\epsilon \in \mathbb{E}$:

$$[\![\mathcal{T}^i_n]\!]_\beta(\epsilon) = \begin{cases} n & \text{if } n \subseteq \mathit{Out}^i \\ [\![\mathcal{T}^i_{\mathtt{succ}^i(n,0)}]\!]_\beta(\epsilon) & \text{if } n \in P^i \text{ and } \epsilon \not\models \lambda^i(n) \\ [\![\mathcal{T}^i_{\mathtt{succ}^i(n,1)}]\!]_\beta(\epsilon) & \text{if } n \in P^i \text{ and } \epsilon \models \lambda^i(n) \\ \bigcup_{k<\kappa(f):o^j_k \in [\![\mathcal{T}^j_{r^j}]\!]_\beta(\epsilon)} [\![\mathcal{T}^i_{\mathtt{succ}^i(n,o_k)}]\!]_\beta(\epsilon) & \text{if } n \in B^i \text{ with } j{=}\beta(f) \text{ and} \\ & \lambda^i(n) = f\langle o_0, \ldots, o_{\kappa(f)-1}\rangle. \end{cases}$$

The last case of the above case distinction formalizes the evaluation of the jth template DD at its root r^j invoked from a box n in the ith template DD. For each outcome o^j_k the kth outcome o_k of the box type in the ith template DD is then chosen, continuing the evaluation within the ith template DD.

4.2 Standard Template Boxes

Templating through boxes can be done to specify common decision patterns that often occur in decision strategies or to encapsulate a sub-decision for reuse in a strategy-representing DD. The corresponding concept of procedures known from software engineering also provides inspiration for common structures how boxes are instantiated in most common applications.

Templates with Parameters. Often, recurring sub-decisions have the same structure but can differ in the values of constants used in the predicates of decision nodes. A common way to deal with that in software engineering is to introduce parameters. To enable the use of *parameterized templates*, we propose to introduce an additional variable in box types that serves as a parameter. For readability, we propose to list parameter names in square brackets behind the function name, e.g., parametrizing a box type $f\langle o_0, \ldots, o_{k-1}\rangle$ with parameters p_0 to p_{n-1} yields

$$f[p_0, \ldots, p_{n-1}]\langle o_0, \ldots, o_{k-1}\rangle$$

Standard Templates. In control decisions, we often have the same kind of checks that are performed on observable system variables. To provide explicit modeling solutions for these well-known patterns, we introduce standard templates and feasible implementations by (template) DDs. Next to simple checks like *less*, *equal*, or *greater*, variables often have to be within a certain interval and control actions depend on whether the variable is *smaller*, *within*, or *greater* than interval boundaries. We provide two templates for interval containment. First, templates with the box type "in_interval2$[x, \ell, u]\langle \text{in}, \text{out}\rangle$" decide whether

a variable x is evaluated within the interval $[\ell, u]$ or outside. It corresponds to the DD depicted in Fig. 5. This check is often used in control decisions to decide whether an action needs to be performed in order to keep the process variable within a reference interval. Second, control decision may also depend on whether the value is below or above the desired range. For this, we provide a template with box type "in_interval3$[x, \ell, u]\langle \text{in}, <, > \rangle$", which decides whether a variable x is evaluated within the interval $[\ell, u]$, smaller, or greater than the interval. The corresponding DT is depicted in Fig. 6.

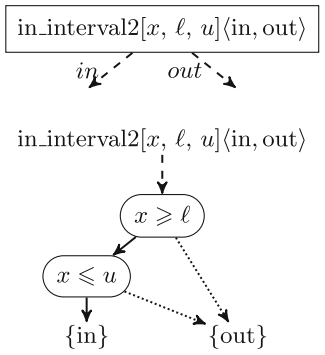

Fig. 5. Interval2 template (top) and a feasible implementation (bottom)

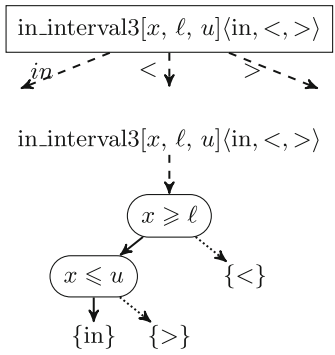

Fig. 6. Interval3 template (top) and a feasible implementation (bottom)

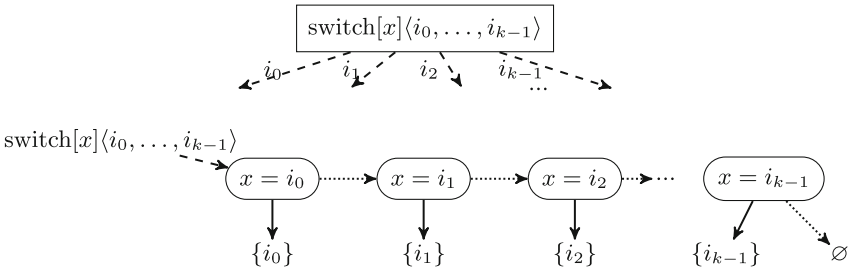

Fig. 7. Switch template (top) and a feasible implementation (bottom)

Another recurring pattern are *switches*, that compare the variable to several values and execute a corresponding control decision afterwards. The template of box type "switch$[x]\langle i_0, \ldots, i_{k-1} \rangle$" decides whether variable x is evaluated to i_0, i_1, ..., or i_{k-1}. This pattern is used in programming languages to avoid nested

if-statements. The corresponding DT is depicted in Fig. 7. Note that also alternative implementations of this template could be imagined, e.g., using binary search, from which one can abstract by using templates and leaving implementation details to future development steps or when further explanations on the implementation are required.

5 Templates for Self-adaptive Systems and Meta Control

Decision trees (DTs) as an instance of DDs are well-established to specify control policies. Using template DDs instead of DDs further provides two main advantages: (i) it accommodates reuse of decision policies and thereby introduces a separation of concerns by enclosing common behaviors into boxes, similar to procedures in programming, and (ii) it allows underspecification of certain parts of the control strategy, thereby enabling specification of families of controllers.

We illustrate the use of template DDs as modeling formalism for control and meta control policies with template DDs. As explained in Sect. 1, an SAS focussed on adapting control strategies can be seen as a special case of a two-layered system with a managed and a managing subsystem (see Fig. 2) where the managed subsystem is a control system and the managing subsystem is a meta control system. The meta control system adapts the control strategies of the control system according to input from the environment and the control system. The control system can be specified as a template DD, making common decision strategies explicit and thereby making it easier to understand the control system. Furthermore, template DDs can be left underspecified and contain parameters which are encoded as variables in the function name.

By instantiating template DDs and parameters of functions instead of adapting the control strategy of the control system in a different way, the adaptation strategy of the meta control system (i) is easier to specify because changing parameters of functions gives a good overview of the choices that can be taken, and the structure of the template DDs can be reused, (ii) has a compact structure which makes it easier to understand. The meta control system can itself be represented as a template DD.

We illustrate how interacting control and meta control systems can be modeled with template DDs on a case study inspired by the exemplar SUAVE [52]. To do so, the case study is first described in Sect. 5.1 and then modeled with template DDs in Sect. 5.2. As a first example for how to represent template DDs in a control system, Fig. 8 shows a template DD for an autonomous underwater vehicle (AUV). The template DD makes use of our standard template for interval checking, which can be implemented as shown in Sect. 4.2. When considering the AUV to be a two-layered system, the instantiation of the function parameters would be done by the meta control system, setting the values for *min* and *max* to adapt the minimal and maximal diving depth of the AUV.

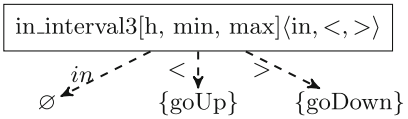

Fig. 8. Example template DD with standard interval template for an AUV

5.1 An Overview of the Case Study

In this section, we introduce our case study of an AUV used for searching for a pipeline located on a seabed and then following and inspecting it. The AUV has two kinds of vision sensors, a camera and a sonar, where the sonar consists of a side-scan sonar and a forward-facing sonar. Both kinds of vision sensors can be used for searching for the pipeline and for following it, but the sonar is preferred for searching since it can operate on a higher altitude, giving a bigger field of view, and the camera is preferred for inspecting the pipeline since it is easier to detect faults of the pipeline with the camera.

Depending on the current task of the AUV, i.e., searching for the pipeline or following it, a different camera resolution or sonar is required. When searching for the pipeline with the camera, the resolution does not need to be as high as for inspection since during inspection, faults need to be identified with the camera images. When searching for the pipeline with the sonar, the side-scan

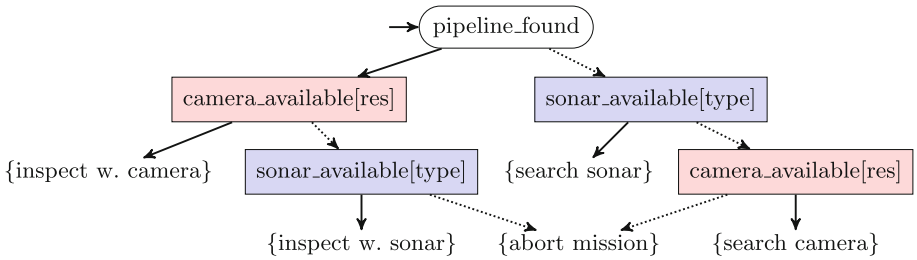

Fig. 9. The control system of the case study as a template DD

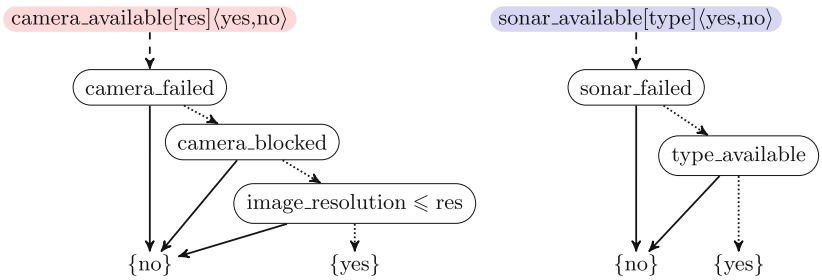

Fig. 10. The template DD instances for the control system

sonar is preferred since it provides a wider field of view than the forward-facing sonar. However, when following the pipeline, the forward-facing sonar is preferred. Both the sonar and the camera might fail during runtime, making it necessary to choose the other vision sensor for both tasks. Furthermore, the camera can get blocked temporarily, making a temporary switch to the sonar necessary for inspection. If both sensors fail, the mission should be aborted.

5.2 Modeling the Case Study with Template DDs

The AUV can be modeled as an SAS with a control system responsible for searching for and inspecting the pipeline, and a meta control system that adapts the control strategy of the control system depending on the current task of the AUV. Therefore, we model the control system of the AUV as a hierarchical template DD (see Figs. 9 and 10), and the meta control system as a (template) DD without box nodes that determines the values for the control system's function parameters depending on the state of the control system (see Fig. 11).

Control System. The control system consists of the hierarchical template DD representing how to choose the sensor for searching for the pipeline and following it, see Fig. 9. When searching for the pipeline (the pipeline has not been found, indicated by the Boolean value *pipeline_found*), the sonar is the preferred vision sensor so it is chosen if available. If only the camera is available, then the camera is chosen. Otherwise the mission should be aborted. When the pipeline has been found, the camera is preferred for following it and the sonar is only chosen if the camera is not available but the sonar is. In both cases, if the pipeline has been found and if it has not, the control system has to determine whether the camera and the sonar are available. These checks depend on the parameters res, the "resolution" of the camera, and type, the type of sonar used (side-scan and forward-facing), and have been summarized in the box types "camera_available[res]" and "sonar_available[type]", respectively. Recall that we simplify box types and outgoing edges of box types if they have only two outcomes, i.e., the box types "camera_available[res]$\langle o_0, o_1 \rangle$" and "sonar_available[type]$\langle o_0, o_1 \rangle$" have been simplified by omitting the box outcomes o_0 and o_1, and the outgoing edges of the box types are drawn as usual "yes/no-edges" for simplification.

Using the box types for checking if the camera and sonar are available makes understanding and adapting the control strategy of the control system much easier because (i) the box types summarize (template) DDs, increasing explainability and clarifying the structure of the DD; without them, this structure would be difficult to spot, (ii) the box nodes can be reordered easily to reflect a different preference for using camera and sonar; with the preferences described in Sect. 5.1, the two box types appear in a different order in the sub-template DDs, depending on whether the pipeline has been found.

The camera and sonar have to be available with different resolution and type, respectively, depending on the current task of the AUV, see Sect. 5.1. Therefore, the chosen instantiations of these box types depend on the parameters

res for "resolution" and type for the type of sonar to be used (side-scan or forward-facing). The template DDs implementing them are depicted in Fig. 10, where "camera_failed", "camera_blocked", "sonar_failed", and "type_available" are Boolean variables that are assumed to be set to the correct value by the system depending on sensor input on whether the camera failed, the camera is blocked, the sonar failed and the requested type is available, respectively. The variable "image_resolution" is an integer that is also set by the system depending on the current resolution of the camera images. In conclusion, the control system of the AUV is a hierarchical DD $\mathfrak{H} = (\mathcal{T}^0, \mathcal{T}^1, \mathcal{T}^2, \beta)$ where $\mathcal{T}^0, \mathcal{T}^1$, and $\mathcal{T}^2$ are the template DDs for the control system (Fig. 9), the camera and sonar available DDs (Fig. 10), respectively, and

$$\beta\colon \{\text{control}, \text{camera_available[res]}, \text{sonar_available[type]}\} \to \{0,1,2\}$$

$$\beta(f) = \begin{cases} 0 & \text{if } f = \text{control}, \\ 1 & \text{if } f = \text{camera_available[res]}, \text{ and} \\ 2 & \text{if } f = \text{sonar_available[type]}. \end{cases}$$

Meta Control System. The meta control system adapts the control strategy of the control system by instantiating the parameters res and type of the functions "camera_available[res]" and "sonar_available[type]" of the control system depending on whether the AUV is searching for the pipeline or following it, see Fig. 11. The variable "AUV_searching" is a Boolean variable that is assumed to be set by the system depending on the state of the control system, i.e., depending on whether the AUV is searching for the pipeline. This variable is different from the variable "pipeline_found" in the control system which captures whether the pipeline has been recognized on the sensor input.

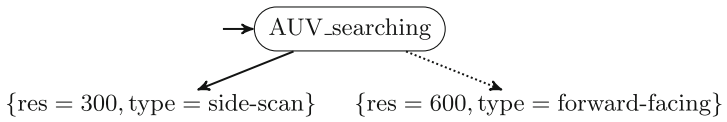

Fig. 11. The meta-control system of the case study

Enabling adaptation of the control system by instantiating parameters of functions enables reuse of box types and makes it easy to spot which parameters can be adapted and where they are used. With box types, the control strategy of the control system can easily be understood on a high level (i.e., without having to look at the implementation of the camera and sonar check) and the adaptation strategy of the meta control system is easy to understand and represent. Without box types, it would both be harder to see which parameters can be adapted and how the parameters in different parts of the DD relate to each other.

6 Improving Control Strategy Explanations

In software engineering, well-established concepts help to improve understandability and maintainability of software: separation of concerns, encapsulation, and reuse of behaviors. Their common intend is to reduce the amount of code that has to be captured and understood during software development. By encapsulating decisions and actions that belong together into subprograms, objects, and procedures, the overall program can be understood on a more abstract level. Encapsulated functionalities then only need to be evaluated once when inspecting software. Further, when changing functionalities or debugging, code modifications are reduced and mitigate negative side-effects of code duplication.

Our definition of *template DDs* and hence *hierarchical DDs* has been inspired by procedural software and thus can benefit from these concepts towards improved explainability of DTs and DDs. As boxes in template DDs usually encapsulate complex behavior including several decisions, a template DD is smaller than the equivalent DD without boxes, and, thus, easier to grasp. Furthermore, boxes allow for separation of concerns and can be examined individually.

6.1 Explainability Metrics for Template DDs

While these arguments are purely qualitative, the software engineering community also provides metrics that allow to quantify the understandability of code [64]. In the following, we describe the most important ones and transfer them to our template DD setting. Following Occam's razor, the best explanation for a phenomenon is the smallest one amongst all possible explanations [28]. Metrics to measure the psychological complexity [64] and hence understandability of source code can hence be used as metrics for explainability. Minimizing such metrics corresponds to increasing explainability. Most prominently used are the number of *lines of code (LOC)* and McCabe's *cyclomatic complexity (CC)* [43,61]. The latter measures the number of linearly independent control flows through a program. Formally, CC is defined as the sum of $|E| - |N| + 2$ for each procedure, where $|E|$ and $|N|$ are the number of edges and nodes of the control-flow graph, respectively. Note that with binary decisions and one terminal node only, this measure simplifies to $|N|$, i.e., the LOC measure.

Let us now establish corresponding metrics to measure the complexity of template DDs. For this, let $\mathcal{T} = (N, Out, \lambda, \texttt{succ}, r)$ be a template DD. The LOC measure on source code can directly be transferred to the setting of DDs as the number of decision nodes. This allows for a first impression of the size of information that needs to be understood. Formally,

$$LOC(\mathcal{T}) \;=\; |N|.$$

McCabe's cyclomatic complexity can also be interpreted on template DDs, where the DD is seen as control-flow graph in which each box corresponds to a procedure call and where each terminal node leads to an artificial control-flow sink node. Formally, we define the CC measure as

$$CC(\mathcal{T}) = |P| + \sum\nolimits_{b \in B} \big(|Out(\lambda(b))| - 1\big) + 1.$$

Note that each predicate node has two successors, while boxes might have higher arity. Both measures naturally extend to hierarchical DDs $\mathfrak{H} = (\mathcal{T}^0, \ldots, \mathcal{T}^{m-1}, \beta)$ as the sum of local measures, i.e.,

$$LOC(\mathfrak{H}) = \sum\nolimits_{i=0}^{m-1} LOC(\mathcal{T}^i) \quad \text{and} \quad CC(\mathfrak{H}) = \sum\nolimits_{i=0}^{m-1} CC(\mathcal{T}^i).$$

In the hierarchical DD of the TRIANGLE control example (see Fig. 3), we obtain $CC(\mathcal{T}^0) = 7 + 2 \cdot 3 + 1 = 14$ and $CC(\mathcal{T}^1) = 3 + 1 = 4$, leading to a cyclomatic complexity of $CC(\mathfrak{H}) = 18$. Note that $CC(\mathcal{D}) = 13 + 0 + 2 = 14$ in the DT representation of the very same control strategy (see Fig. 1). With respect to the LOC measure, we achieve better explainability for this example by $LOC(\mathfrak{H}) = 25$ compared to $LOC(\mathcal{D}) = 27$.

Note that the overall cyclomatic complexity $CC(\mathfrak{H})$ increased with refactoring due to the overhead of a second diagram where profit from reuse is not mitigated by the small size of the extracted subdiagram. However, when considering the meta control example in Fig. 9 and Fig. 10, we get $CC(\mathcal{D}) = 12$ for a non-refactored AUV and $CC(\mathcal{T}^0) = 6$, $CC(\mathcal{T}^1) = 4$ and $CC(\mathcal{T}^2) = 3$. Here, the overall cyclomatic complexity $CC(\mathfrak{H}) = 6+4+3 = 13$ is also slightly higher than $CC(\mathcal{D}) = 12$ but the individual diagrams that can be investigated in isolation have a much smaller cyclomatic complexity and have the benefit of explicitly showing code reuse.

6.2 Decision Diagram Refactoring

In Sect. 5, we showcased how template DDs can facilitate modeling of control systems with reuse of decision making. Legacy monolithic control systems, however, do not benefit from the opportunities of concise representation and reuse yet. *Refactoring* [25] is the standard approach in software engineering to improve the design of legacy code by encapsulating functionalities. This concept can also be used in the setting of DDs, factoring out isomorphic subdiagrams into template DDs towards hierarchical DDs that improve explainability measures.

Decision Diagram Refactoring Problem: Given a template DD $\mathcal{T}$ and an explainability measure η, synthesize a hierarchical DD $\mathfrak{H}_{\mathcal{T}} = (\mathcal{T}^0, \ldots, \mathcal{T}^{k-1}, \beta)$ such that $[\![\mathfrak{H}_{\mathcal{T}}]\!] = [\![\mathcal{T}]\!]$, $\bigcup_{i=0}^{k-1} P^i \subseteq P$, and $\eta(\mathfrak{H}_{\mathcal{T}})$ is minimal.

Note that we required the predicate nodes of the input template DD $\mathcal{T}$ to cover the predicate nodes of all template DDs $\mathcal{T}^i$ in the synthesized hierarchical DD $\mathfrak{H}_{\mathcal{T}}$. Intuitively, this corresponds to reusing nodes of $\mathcal{T}$ in template DDs and removing duplicates. Solving the refactoring problem clearly is intractable, since finding minimal refactoring involves to solve an NP-hard problem. To this end, heuristics for refactoring are favorable towards improving explainability of DD strategies. A naive greedy algorithm on a hierarchical DD $\mathfrak{H} = (\mathcal{T}^0, \ldots, \mathcal{T}^{k-1}, \beta)$

over predicate nodes $P = \bigcup_{i=0}^{k-1} P^i$ to factor out one promising template DD $\mathcal{T}^k$ would be to compute the maximal gain of explainability for isomorphic subdiagrams and opt for factoring out the maximal one:

1. Initialize the maximal gain $g_{\max} := 0$ and a template candidate $\mathcal{F} := \varnothing$.
2. For each pair $n \in P^i$ and $m \in P^j$ with $\lambda^i(n) = \lambda^j(m)$
   (a) determine the maximal subdiagram $\mathcal{S}$ of $\mathcal{T}_n^i$ such that
      – $\mathcal{S}$ is isomorphic to the top subdiagram of $\mathcal{T}_m^j$, and
      – $\mathcal{S}$ does not contain a node with multiple incoming edges;
   (b) compute the gain $g = \eta(\mathfrak{H}) - \eta(\mathfrak{H}')$ where $\mathfrak{H}'$ arises from $\mathfrak{H}$ by replacing every isomorphic occurrence of $\mathcal{S}$ in $\mathfrak{H}$ by a box pointing to a tentative fresh template DD $\mathcal{T}^k := \mathcal{S}$;
   (c) if $g > g_{\max}$, set $g_{\max} := g$ and set $\mathcal{F} := \mathcal{S}$.
3. Update $\mathfrak{H}$ with a fresh template DD $\mathcal{T}^k := \mathcal{F}$ and replace every isomorphic occurrence of $\mathcal{F}$ in $\mathfrak{H}$ by a box pointing to $\mathcal{T}^k$.

The above procedure can be initially applied to a hierarchical DD comprising a single DD only, and repeated until no gain in explainability is achieved anymore. Note that factoring out a template DD introduces new outcome nodes in the freshly generated template component, increasing explainability measures if there are only few template occurrences or small template sizes.

6.3 Implementation and Evaluation

We implemented template DDs and support for hierarchical DDs building on the Python-based DT learning framework dtControl [5]. For this, we first transfer DTs learned with dtControl into reduced DDs by merging outcome nodes and isomorphic subtrees. Then, we apply our refactoring heuristics from the last section and evaluate the LOC explainability measure, i.e., improving explainability through reducing DD sizes. Template DDs can be exported graphically towards explainable control strategies – Figs. 1 and 3 are generated by our implementation. Our implementation and experiments are publicly available[1].

Experiments. To investigate the explanatory potential of hierarchical DDs, we applied DD generation and refactoring through our heuristics to the standard benchmark set of dtControl, comprising different control strategies learned from cyber-physical systems, classical planning tasks, machine learning, as well as synthesized by probabilistic model checkers from examples of the Prism benchmark suite [38]. Note that for DD generation, we only applied reduction rules to the DT output, not changing the order of predicates. As explainability metrics we chose the LOC measure, since CC does not account well for reuse in templates (see Sect. 6.1). Table 1 shows the stepwise node-count reductions, where in the first half standard Prism benchmarks and in the second half cyber-physical systems benchmarks are separated through a horizontal line. The "all"

[1] https://anonymous.4open.science/r/tdd-97EC.

Table 1. Results for dtControl experiments

	DT		DD		factor(DD)		inner
Model	all	inner	all	inner	all	inner	reduction
beb	65	32	61	32	62	30	6%
blocksworld	1,235	617	800	616	855	572	7%
cdrive	2,401	1,200	2,152	1,200	2,049	422	65%
consensus	67	33	45	32	51	29	12%
csma	103	51	83	50	86	49	4%
eajs	167	83	112	79	115	78	6%
echoring	1,869	934	1,267	866	1,329	790	15%
elevators	16,327	8,163	6,325	6,260	4,696	4,348	47%
exploding-blocksworld	4,981	2,490	2,405	2,330	2,344	1,907	23%
firewire	307	153	232	153	237	132	14%
ij	1,291	645	510	500	525	418	35%
leader4	125	62	67	55	75	47	24%
mer30	137	68	89	68	91	61	10%
pacman	43	21	40	21	40	21	0%
philosophers-mdp	391	195	222	192	228	170	13%
pnueli-zuck	171,371	85,685	69,637	69,550	55,605	49,050	43%
rabin	111	55	67	55	70	47	15%
rectangle-tireworld	481	240	481	240	435	155	35%
triangle-tireworld	27	13	22	13	25	12	8%
wlan_dl	3,369	1,684	1,613	1,525	1,680	1,400	17%
zeroconf	381	190	187	178	197	165	13%
10rooms	17,297	8,648	748	723	534	313	96%
aircraft	915,877	457,938	332,424	332,408	307,395	264,419	42%
cartpole	253	126	211	126	218	116	8%
dcdc	271	135	138	135	138	135	0%
helicopter	6,339	3,169	3,079	2,841	3,049	2,312	27%
traffic_30m	12,573	6,286	2,888	2,876	2,950	2,281	64%
truck_trailer	338,283	169,141	142,915	141,995	138,245	111,975	34%
vehicle	13,235	6,617	5,878	5,860	6,261	5,256	21%

node counts are reflecting the standard LOC measure, while "inner" refers to the number of inner nodes, i.e., predicate and box nodes. The latter is more meaningful to mitigate the clear drawback of DTs in multiple outcome node duplications. Thus, we also provide reductions only w.r.t. inner nodes, leading to an inner reduction of $1-\frac{\textbf{factor(DD)}}{\textbf{DT}}$. Here, **DT** is the number of inner DT nodes and **factor(DD)** stands for applying our refactoring heuristic to the DD obtained from the original DT input from dtControl. We can observe clear reductions in size, even more than one order of magnitude in the case of the "10rooms" benchmark from the cyber-physical systems domain. There are diverse reasons

for reductions, which can be explained by the diverse benchmark set. Reductions are both, due to switching from DTs to DDs as representation formalism, and refactoring to hierarchical DDs. While refactoring has great impact in almost all cases, switching to DD formalism sometimes does not provide any reduction of inner nodes. For instance, in the case of the "cdrive" benchmark, refactoring provides the only source of more than 60% reduction.

7 Conclusion

We introduced templates in decision diagrams (DDs) as an extension of multi-valued DDs, which allow to encapsulate and reuse common decision-making patterns. Therefore, template DDs can yield more explainable representations than standard decision trees (DTs) due to their smaller size and by introducing different levels of abstraction in a hierarchical template DD. Furthermore, they enable underspecification which can be exploited for modeling meta control. We illustrated the benefits of template DDs for modeling and explaining meta controllers with a case study from the autonomous underwater robotics domain. To evaluate the potential benefits of template DDs for existing controllers given as DTs, we implemented a template-generating refactoring method for DTs. Our results on standard controller benchmarks showed template DDs to improve explainability of by reducing DT sizes by more than one order of magnitude.

Our new data structure for explainable AI and control strategy representation can be seen as a starting point for several future investigations. First, we did not opt for *ordered* template DDs, a common requirement usually present in binary DDs [12]. While this could lead to performance advantages for operations on DDs and allows for integration in state-of-the-art DD frameworks [32], this restriction would limit the modeling power: permuting templates as we illustrated in Fig. 9 would be not contained in the class of ordered template DDs. However, integration of template DDs and the impact of the ordering condition are an interesting future direction. Second, template DDs could be used as runtime models in self-explainable systems that autonomously decide when and how to explain themselves. Explanations could here be supported by what-if analyses for possible template instantiations or meta control decisions, importance of decisions, and causal reasoning [8,23,29]. Self-explanation can be also achieved by adding a meta-layer, similar to SASs, that monitors the system and constructs explanations from an abstract model of the system [10,22,24]. In [55], we proposed a high-level process to extract tree-like explanation models from system models with timing information. Here, template DDs could ease extraction and enable explanations at different levels of abstraction. Also on-the-fly adaptations of parameters and template instantiations could improve control performance and explainability, where in-depth considerations are future work. Third, learning algorithms that directly provide template DDs instead of DTs for control strategies could even further increase sharing and reuse of decision making, directly impacting existing methods in explainable AI. Our refactoring experiments conclusively demonstrated the potential of possible improvements

and provide a lower bound on how explainability could be increased with dedicated learning algorithms. In this vein, also the detection of appearances of standard templates as presented in Sect. 4.2 could be investigated, e.g., factoring out switch chains common in outputs from dtControl strategies. Fourth, the concept of templates in DD could be further investigated with applications in control theory [4] and reinforcement learning by templating and bootstrapping parts of the learning process [9,37,48,59]. Last but not least, further explainability measures accompanied with user studies to investigate the human understanding of control strategy explanations by template DDs would be important to achieve a profound integration of template DDs into engineering processes.

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A Logic of Weighted Reasons for Explainable Inference in AI

Stipe Pandžić(✉) and Joris Graff(✉)

Department of Philosophy and Religious Studies,
Utrecht University, Utrecht, The Netherlands
{s.pandzic,j.j.graff}@uu.nl

Abstract. We propose two methods that integrate justification logic, defeasible reasoning and numerical reasoning to lay the foundations for an explainable, reason-based neuro-symbolic architecture. The core idea behind the two methods is to model two different ways in which weighing default reasons can be formalized in justification logic. The two methods both assign weights to justification terms, i.e. modal-like terms that represent reasons for propositions. The first method obtains the values of these reasons solely on the basis of the extension-based operational semantics for default justification logic. This semantics handles default reasons in such a way that it extends consistent sets of reason-formula pairs as much as possible. The second method aims for a direct comparison of reasons, where the potential conflicts between default reasons are resolved by pooling together all the applicable reasons for or against propositions. Instead of applying default steps selectively in the fashion of the operational semantics, all available default reasons are applied simultaneously and interact directly with each other. We argue that the two methods show why combining justification logic, defeasible reasoning and numerical reasoning is an intuitive and promising logical approach to explainable neuro-symbolic integration.

Keywords: Justification logic · Defeasible reasoning · Weighted reasons · Neuro-symbolic integration

1 Introduction

Many logic-based approaches in artificial intelligence (AI) aim to create frameworks for *commonsense reasoning*. These frameworks depart from classical, deductive forms of logic in order to incorporate features that actual reasoning agents may be confronted with, such as defeasible inference steps and incomplete information. The advantages of such commonsense approaches are twofold. First, if AI systems are to perform in practical contexts, they will have to confront uncertain or conflicting sources of information, for which deductive models will not equip them. One approach is to use probabilistic models to estimate the probability of different states of affairs. Probabilistic models, however, generally require large amounts of quantitative inputs, which are often not available (see e.g. [18, p. 273]; [11, p. 165]). Non-probabilistic defeasible reasoning or argumentation models may thus be more appropriate for many realistic scenarios.

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Second, for humans, who similarly base most of their inferences on defeasible reasoning steps from uncertain or conflicting information sources, logical frameworks that mirror these conditions are likely to be easily interpretable. Compared to deductive proofs, on the one hand, and probabilistic models, on the other, step-by-step arguments for defeasible conclusions will appear intuitive to many users. Formalised commonsense reasoning may therefore contribute to the foundations of explainable AI (XAI).

The field of formalised commonsense reasoning is very wide, and includes nonmonotonic logics such as default logic [41] and certain forms of logic programming [22], Krause et al.'s logic of argumentation [28], Pollock's inference graphs [37,38], and abstract approaches to argumentation, most of which are inspired by the seminal work by Dung [15]. Recently, Pandžić has proposed to model commonsense arguments in justification logic [35,36]. The advantage of justification logic is that it explicitly models reasons for statements as modal terms called justification terms. In this way, the justification for a proposition can be observed from the justification term that precedes the proposition. Whereas standard justification logic is monotonic, and therefore unsuited for reasoning about uncertain information (indeed, it was originally proposed to reason about mathematical proofs; see [3]), Pandžić's version allows us to reason about potentially conflicting arguments.

Like most approaches in defeasible or commonsense reasoning, Pandžić's framework is qualitative; that is, propositions and arguments are evaluated according to non-numeric valuations or acceptability standards (Pandžić [35] uses classical binary valuations for propositions, and evaluates arguments using Dung's [15] criteria for extensions of argumentation frameworks, such as conflict-freeness and acceptability). This is appropriate for some contexts, where numeric estimations of the strength of reasons or arguments is unfeasible. Indeed, Fox, Glasspool and Bury [18] argue that non-probabilistic numeric logics (such as fuzzy logic) have the same shortcomings as probabilistic models, in that they "demand a great deal of quantitative input data" [18, p. 275]. However, non-probabilistic numeric approaches may still have distinct advantages. First, in many realistic scenarios, AI systems, like human agents, may be confronted with quantitative evidence that does not yet represent a probability estimate. For instance, evidence gathered from questionnaires often presents respondents' preferences or estimates on a numeric scale (e.g. 1 to 5), which does not have a straightforward probabilistic interpretation. Second, even evidence that is not given in a numeric format can often be given a numeric *estimate* that does not represent probability, but rather a more intuitive notion of *strength of reasons* or *strength of arguments*. For instance, I may estimate the credibility of a certain source – that is, my strength for believing what the source says – as 0.8, without committing myself to something as strong as a probability model according to which there is a 0.8 probability that the source's statements are true. Still, the numeric estimate may be useful since it allows for fine-grained comparisons of different sources. Of course, such estimates have a certain level of arbitrariness, but this is not necessarily a problem as long as the relative strengths of different

estimates reflects our non-numeric intuitions (e.g., as long as more believable sources are assigned a higher reliability scores than less believable ones).

There are, therefore, sufficient reasons to explore non-probabilistic gradual valuations for propositions and/or arguments in defeasible or commonsense logics. Such approaches are common within the subfield of Dung-style argumentation theory (e.g. [8–12,16,17,26,30,40]). These Dung-style approaches, however, do not model the internal structure of arguments, and are therefore less capable of capturing fine-grained patterns of reasoning. This paper, therefore, extends Pandžić's default justification logic, which can model such internal structure, with numeric valuations. Our aim hereby is to offer a reasoning system that is both able to handle reasons of different strengths, and to do so in a way that models reasons explicitly and is therefore in principle explainable. Thus, the results presented here are meant to contribute both to the formal theory of reasoning and to the foundations of XAI.

Below, we propose two methods that achieve the desired integration of default justification logic and numeric reasoning. The two methods both assign weights to justification terms, but compare weights in different ways. The first method (Sect. 2) obtains the values of reasons solely on the basis of the extension-based operational semantics for default justification logic. This semantics handles default reasons in such a way that it extends consistent sets of reason-formula pairs as much as possible, which is one of the standard semantics for Reiter's default logic [2]. The second method (Sect. 3) aims for a direct comparison of reasons. Instead of applying default steps selectively in the fashion of the operational semantics, all available default reasons are applied simultaneously and interact directly with each other. Conflicts between reasons are then directly resolved by addition of the weights of the corresponding reason terms. After presenting both methods, we argue that numeric default justification logic can serve as the basis for an explainable neuro-symbolic architecture (Sect. 4). We then discuss some related approaches (Sect. 5), before concluding.

In this paper, we provide foundational work for the development of the two methods and outline the syntax and semantics exemplified by a hypothetical application. Our main goal is to motivate the idea that numeric extensions of default justification logic are a promising avenue to model reasoning with weighted evidence in an intuitive way. Exploration of meta-theoretic properties of the systems, as well as how the systems differ in implementing weights in default justification logic, are therefore left to future work.

2 Weighted Default Justification Logic

In this section, we describe the theory of weighted defeasible reasons, where weights are a function both of making defeasible inference steps and of the interaction between defeasible reasons. This method relies on a numerical interpretation of the operations on default reasons that are inherited from the standard justification logic grammar of reason terms.

2.1 Justification Logic Preliminaries

The method described in this section takes a conservative approach with respect to the existing work in justification logic. That is, we want to keep the language and grammar of reason terms to correspond to Artemov's original idea of modeling reasons as terms in justification logic.[1] Our basic logic is a logic of non-defeasible reason terms **JT**, which will then be extended with defeasible weighted reasons.[2]

We first define the grammar of *reason terms* or *justification terms* in justification logic as follows:

$$t ::= x \mid c \mid (t \cdot t) \mid (t + t)$$

where x is a variable and c is a proof constant. We will use the function $Sub(t)$ whose output is the set of all subterms of t. The operators '$+$' (*sum*) and '$\cdot$' (*application*) are basic justification logic operations, where sum 'concatenates' reasons and application 'encodes' the *modus ponens* inference steps. Based on the grammar of reason terms, we define formulas of the justification logic **JT**:

$$F ::= \top \mid P \mid (F \to F) \mid (F \vee F) \mid (F \wedge F) \mid \neg F \mid t : F$$

where P is a propositional atom from the set of all propositional atoms Pr and t is a justification term from the set of all terms Tm. The axioms of the basic logic **JT** are:

A0 *All the instances of propositional logic tautologies from* Fm
A1 $t : (F \to G) \to (u : F \to (t \cdot u) : G)$ (Application)
A2 $t : F \to (t + u) : F$; $\quad u : F \to (t + u) : F$ (Sum)
A3 $t : F \to F$ (Factivity)

where Fm is the set of all **JT** formulas.

The logic **JT** includes *Modus ponens* and *Iterated axiom necessitation* as its inference rules. Axiom necessitation rules enable justification of logical tautologies. Each axiom instance of A0-A3 is justified by a justification constant $c \in CS$, where CS is the constant specification set, that is, the set of all justifications for logical axioms. By adopting the Iterated axiom necessitation rule, we assume that justified formulas are themselves justified by a justification constant.

The semantics of **JT** is an adaptation of Mkrtychev's [33] single-world model semantics for the original justification logic or the "logic of proofs". The interpretation $\mathcal{I}$ for **JT** relies on two functions. First, it relies on the standard *truth assignment* function for propositional formulas in $\mathcal{P}$. Secondly, $\mathcal{I}$ needs to answer

[1] The first justification logic was called "the logic of proofs" (LP) introduced in [3] and fully developed in [4]. The idea of modeling proofs already suggests monotonicity of reason terms. For more comprehensive introductions to justification logic systems, see [5] or [29].

[2] The abbreviation **JT** suggests that the axiom list includes the factivity schema **T**, but it does not include introspection axioms.

the question of which reason term justifies which formula. For this, $\mathcal{I}$ "forces" an answer by relying on the *reason assignment* function, $*(\cdot) : Tm \to 2^{Fm}$, which is mapping each term to a set of formulas from the set of all formulas Fm.

Based on the interpretation $\mathcal{I}$, we define the truth relation $\models$ on the set of formulas of **JT**. The truth relation contains standard clauses for the propositional connectives, together with the following clause for a justification assertion $t : F \in Fm$:

$$\mathcal{I} \models t : F \text{ iff } F \in *(t).$$

The **JT** consequence relation is then defined for all *reflexive* interpretations, which satisfy the condition that, if $F \in *(t)$, then $\mathcal{I} \models F$. We use this relation to define **JT** deductive closure of sets of formulas:

Definition 1 (JT closure). *JT closure is given by* $Th^{JT}(\Gamma) = \{F | \Gamma \models F\}$, *for a set of formulas* $\Gamma \subseteq Fm$ *and the JT consequence relation* $\models$.

We will also use a *non-monotonic* closure $Th^{JT^-}(\Gamma)$, which does not close sets of formulas under the sum operation.

The way in which a reason t justifies a formula F in the basic logic **JT** is reduced to the function reason assignment. But in most common uses of reasons, determining whether t justifies F or not requires a more intricate procedure of "testing" whether t can defeat all the available counter-reasons and whether its weight surpasses certain acceptance thresholds.

2.2 Weighted Default Justification Logic

To introduce weighted default justification logic (WDJL), we first introduce the basics of the default variant of justification logic. In defining weighted default justification logic, we follow default justification logic (as defined in [35]) and then we show how weights of reasons result from interactions among default reasons.

Default justification logic (DJL) is obtained by adding default rules to the syntax of **JT**, as done in [35]. The first type of default rules with defeasibly justified formulas is

$$\delta = \frac{t : F :: (u \cdot t) : G}{(u \cdot t) : G}$$

where u must be unique. Notice that the role of '::' is only to divide the formulas within the rule. Defaults of this type are called *normal*. Normal defaults codify a reasoning pattern that we call *default application* because of its connection to the Application axiom A2. To clarify their connection, notice that, for a default rule δ, if $t : F$ is in the knowledge base, then a consistent consequent $(u \cdot t) : G$ is added to the knowledge base, but not $u : (F \to G)$. Instead, the latter sentence serves as the *warrant* of δ. As Pandžić [35] explains, each default rule is *warranted* by an implicit justified conditional that is not (usually) known but can, in certain situations, be justifiedly assumed. Below, we will call u the *warrant justification*

term. Pandžić [35] introduces a *warrant assignment function* $\# : D \to Fm$, where D is the set of defaults and Fm is the set of all formulas, that maps each default to its warrant (e.g. $\#(\delta) = u : (F \to G)$).

For WDJL default rules, we introduce a new constraint on the weight function: for each default δ with *warrant justification term* u, it must be that $w(u) > 0$ and $w(u) < 1$. That is, each default rule gives us some reason to expect the conclusion to be true, but this reason is never perfect (else we would not have a default but a classical implication). The default value with which the reason "enters" the later process of weighing reasons can be altered, if there are reasons to attenuate its strength. We will refer to the initial, default weight of a reason t as $w(t^{\#})$. Determining the numeric value of default steps is a part of the new operational semantics for justification logic default theories.

The second type of default rules with justification assertions corresponds in structure to what are called *supernormal defaults* in Reiter's default logic:

$$\delta_{t:F} = \frac{\top :: t : F}{t : F},$$

where $w(t) > 0$ and $w(t) < 1$. Normal default rules with justification assertions encode inferential steps supported by warrants. However, rules such as $\delta_{t:F}$ can be considered as "givens" and they do not rely on warrants as their reason-based assumptions. Informally, the uncertainty in such defaults does not arise from making a default application step, but rather from uncertainty of a source of information. This type of uncertainty is explored in [36] and leads to the conflict of reasons that is often labelled as "undermining" in argumentation theory.

The semantics for default rules follows the operational semantics of [35], but the interaction of reasons is now determined by a combination of logical and numerical factors. The operational semantics is given for a default theory T, defined as a pair (W, D), where W is a finite set of **JT** formulas and D is a countable set of default rules. The set W only contains formulas $t : F$ such that $w(t) = 1$. All other formulas that represent starting sources of information, that is, for which it holds that $w(t) > 0$ and $w(t) < 1$, are "relegated" to supernormal defaults of the type $\delta_{t:F}$.

To define the operational semantics, we first determine when to consider a default rule applicable.

Definition 2 (Applicability). *For a set of* **JT***-closed formulas* Γ*, a default rule* $\delta = \frac{t:F::(u\cdot t):G}{(u\cdot t):G}$ *(or a default rule* $\delta_{t:F} = \frac{\top::t:F}{t:F}$*) is applicable to* Γ *iff* $t : F \in \Gamma$ *(or* $\top \in \Gamma$*) and* $\neg(u \cdot t) : G \notin \Gamma$ *(or* $\neg t : F \notin \Gamma$*).*

We postulate that supernormal default rules, as information sources, are always prioritized in application over normal default application rules. We can now define *evidence base* sets that extend an initial default theory by applying sequences of defaults $\Pi = (\delta_0, \delta_1, \ldots)$.

Definition 3 (Evidence base). $In(\Pi) = Th^{JT}(W \cup \{cons(\delta) \mid \delta \in \Pi\})$.

For the operational semantics, only some sequences of defaults are meaningful, namely, *default processes*. A sequence of default rules Π is considered to be a *process* of a default theory $T = (W, D)$ if and only if every k such that $\delta_k \in \Pi$ is *applicable* to the set $In(\Pi[k])$, where $\Pi[k]$ is the segment of the sequence Π with all the rules applied before δ_k. A process is *closed* when there are no applicable defaults left.

We are ready to define the basic type of attack between the reasons, namely, *undercut*:

Definition 4 (Undercut). *A reason u undercuts reason t being a reason for a formula F in a set of* **JT** *formulas $\Gamma \subseteq In(\Pi[k])$ iff*

$$\exists v \in Sub(t) \text{ such that } u : \neg[v : (G \rightarrow H)] \in Th^{JT}(\Gamma)$$

and $v : (G \rightarrow H) = \#(\delta)$ for a rule δ in a process Π' of T.

We say that a set of formulas Γ undercuts t if the **JT** closure of Γ contains an undercutter for t, and we use the notion of undercut to define conflict-free sets:

Definition 5 (Conflict-free sets). *A set of* **JT** *formulas Γ is conflict-free iff $Th^{JT}(\Gamma)$ does not undercut reason t for a formula F such that $t : F \in Th^{JT^-}(\Gamma)$.*

Before we define the ways in which we build extensions and weigh reasons against each other, we need to define the way in which we determine the weights of reasons. The semantics of WDJL is obtained by extending the semantics of default justification logic (as given in [35]) by adding numerical criteria to the existing conflict definitions. First, we introduce a *weight function* $w : Tm \rightarrow [0, 1]$ that assigns a number between 0 and 1 to each justification term. The function w is subject to the restriction that the weight of a complex justification term is a function of the weights of its subterms. Specifically:

Aggregation For a formula $(t + u) : F$, we define the reason weight $w(t + u) = max(tanh(w(t)+w(u)), max(w(t), w(u)))$.[3] The idea is that the weight of two aggregated reasons is a normalised function of the sum of the weights of both individual reasons. That is, the stronger (i.e. the closer to 1) the aggregated reasons are, the stronger the aggregation is. The aggregation $(t + u)$ respects the following two conditions:

- $\{t : F, u : F\} \in In(\Pi)$ and
- $\nexists v \in Sub(t) \bigcap Sub(u)$ such that $v : (G \rightarrow H) = \#(\delta)$ and $\delta \in \Pi$,

[3] This function normalises weights into the interval $[0, 1]$ and ensures that the aggregation of two reasons is at least as good as either of the reasons taken separately. For instance, if $w(t) = w(u) = 1$ – i.e. both reasons are certain – we would have $w(t + u) = max(tanh(2), max(1, 1)) = 1$. It is important that the aggregation of two certain reasons gives also a certain reason. For example, aggregating two separate logical proofs for a statement does not increase the uncertainty of the resulting reason.

for a process Π of T.

Default step For a formula $(t \cdot u) : F$, we define the reason weight $w(t \cdot u) = w(t) * w(u)$. The idea is that application of either an implication or a default (both of which result in a justification term of type $(t \cdot u)$) produces a reason the strength of which depends both on the strength of the antecedent/premise, and on the strength of the conditional/default step.

It is easily proven that the above functions preserve weights in the interval $[0, 1]$.

Building on the weights of complex terms, we define the fundamental operation that augments the operational semantics for default theories:

Definition 6 (Default attenuation). *For a warrant justification term t such that $t : (F \rightarrow G) = \#(\delta)$ for a rule δ in a process Π' of T, we define* reason attenuation *as $w(t) = w(t^{\#}) * (1 - w(u_1 + u_2 + \ldots + u_n))$, where each of the terms $u_1, u_2, \ldots, u_n$ is an undercutter in a set of* **JT** *formulas $\Gamma \subseteq In(\Pi[k])$ such that it holds that $u_k : \neg[t : (F \rightarrow G)] \in Th^{JT}(\Gamma)$, for each k such that $1 \leq k \leq n$.*

The notion of attenuation (derived from Dancy [13, p. 42]) indicates that a reason's strength is reduced, but not entirely removed. An attenuator can thus be seen as a gradual undercutter. The above definition is based on the idea that aggregating undercutting (or attenuating) reasons gives stronger grounds to disbelieve the warrant of a default inference.

We can now define the central notion of reason acceptability, which is in WDJL refined with the condition of an acceptability threshold:

Definition 7 (Acceptability). *For a process Π of a default theory $T = (W, D)$, a formula $t : F \in cons(\Pi)$ is acceptable w.r.t. a set of* **JT** *formulas $\Gamma \subseteq In(\Pi)$ iff for each undercutting reason u for t being a reason for F such that $u : G \in In(\Pi)$, $Th^{JT}(\Gamma)$ undercuts u being a reason for G and no warrant justification term $v \in Sub(t)$ is attenuated below the threshold weight value $w(v) = n$.*

By including the weights of reasons into the reason acceptability definition, we can now give a more fine-grained way of determining whether an argument or a justification assertion $t : F$ follows under a certain view on argument acceptance. These different ways to settle the question of the validity of an argument are defined in the following acceptance-based extensions in the style of [15]:

Definition 8 (WDJL Extensions). *The following reason acceptability-based extensions are defined for any default theory $T = (W, D)$ with the* **JT** *logic formulas (following [36, p. 308]):*

Admissible Extension *A potential extension set of* **JT** *formulas $\Gamma \subset In(\Pi)$ is an admissible extension of a default theory $T = (W, D)$ iff $Th^{JT}(\Gamma)$ is conflict-free, each formula $t : F \in \Gamma$ is acceptable with respect to Γ and Π is a closed process.*

Preferred Extension *A closure $Th^{JT}(\Gamma)$ of an admissible extension Γ is a preferred extension of T iff for any other admissible extension Γ', $\Gamma \not\subset \Gamma'$.*

Complete Extension *A closure $Th^{JT}(\Gamma)$ of an admissible extension Γ is a complete extension of T iff for each closed process Π of T such that there is a admissible extension Γ' in $In(\Pi)$ and $\Gamma \subset \Gamma'$, if a formula $t : F \in cons(D)$ is acceptable with respect to Γ in $In(\Pi)$, then $t : F$ belongs to Γ.*

Grounded Extension *A complete extension $Th^{JT}(\Gamma)$ is the unique grounded extension of T if Γ is the smallest potential extension with respect to set inclusion such that $Th^{JT}(\Gamma)$ is a complete extension of T.*

Notice that there are also many other options in which weights may interact with the way in which we define extensions, which are left for future work. We do not, for instance, discuss the basic principle of comparing two reasons, namely, if out of the two reasons for contradictory propositions one reason fares better, that extension should be favored. In such a way, we could easily adjudicate between, e.g., two preferred extensions of a default theory. Likewise, adjudicating between two or more competing extensions could be done by looking into reason aggregation, which gives a more exhaustive picture of reasons *pro* or *contra*. Moreover, if an attacker weighs more than the attacked reason, this fact alone could give grounds to include the attacker in the extension. Such an idea would lead to a simple "stable semantics" definition.

2.3 Example

As an example of reasoning with conflicting evidence, consider a hypothetical medical example. Imagine we are tasked with making medical diagnoses on the basis of two artificial neural networks that have been trained on separate data sets in order to predict diagnoses. For simplicity, say that both neural networks have only been trained to diagnose disease D, so that they have two output nodes (one giving the likelihood that a patient has disease D and one giving the likelihood that the patient does not have disease D), that have been normalised by a Softmax layer so that the outputs fall within the range $[0, 1]$ and add up to 1. Now, say we are tasked to predict whether a patient has disease D, and we have both neural networks at our disposal. It is possible to see the neural networks' output values as estimates of the conditional probabilities $P(D|S)$ and $P(\neg D|S)$ where D indicates the proposition (or state) that the patient has disease D and S indicates the data in the patient's file. However, when both neural networks give different outputs, we have no formal way of obtaining an 'aggregated' conditional probability estimation that takes into account both estimations. Thus, we will interpret both neural networks' outputs as roughly defined strengths of reasons to believe that the patient has disease D, rather than as formal probability estimates. More specifically, if the output of node D (call it n_D) is higher than the output of not-D (call it $n_{\neg D}$), we take the difference between n_D and $n_{\neg D}$ as the initial strength of a reason for D. The intuition is that if the difference between both nodes is very small – e.g. 0.51 vs. 0.49 – the network is very uncertain, and our reason for believing that D should be very low. *Vice versa*, if $n_{\neg D}$ is higher than n_D, then the difference is taken to be the initial strenth of a reason for $\neg D$.

The difference in outputs can be represented as a conflict of reasons that can be modeled in WDJL. Imagine that neural network A has an output of 0.9 for D and 0.1 for $\neg D$, and neural network B has an output of 0.75 for $\neg D$ and 0.25 for D, thus giving contradictory predictions. Take S to be the proposition that the patient's data are as in the file, but the certainty of such information is less-than-perfect, e.g. 0.9. This triggers the following default rule:

$$\delta_{s:S} = \frac{\top :: s : S}{s : S}.$$

Take D to be the proposition that the patient has disease D. Then, the two neural networks can be captured by the following two defaults:

$$\delta_a = \frac{s : S :: (t_a \cdot s) : D}{(t_a \cdot s) : D} \quad \text{and} \quad \delta_b = \frac{s : S :: (t_b \cdot s) : \neg D}{(t_b \cdot s) : \neg D}.$$

However, assume, next, that we have an independent reason to doubt the credibility of neural network A in this case. Assume for instance that we know that neural network A was trained on a data set that contained very few female patients, and is therefore less reliable for female patients, and assume that the patient in question is female. Then it seems that we have a third default:

$$\delta_c = \frac{u : F :: (v \cdot u) : \neg[t_a : (S \to D)]}{(v \cdot u) : \neg[t_a : (S \to D)]}.$$

Here, F is the proposition that the patient is female, u denotes whatever reason we have to believe that the patient is female, and v is our reason to

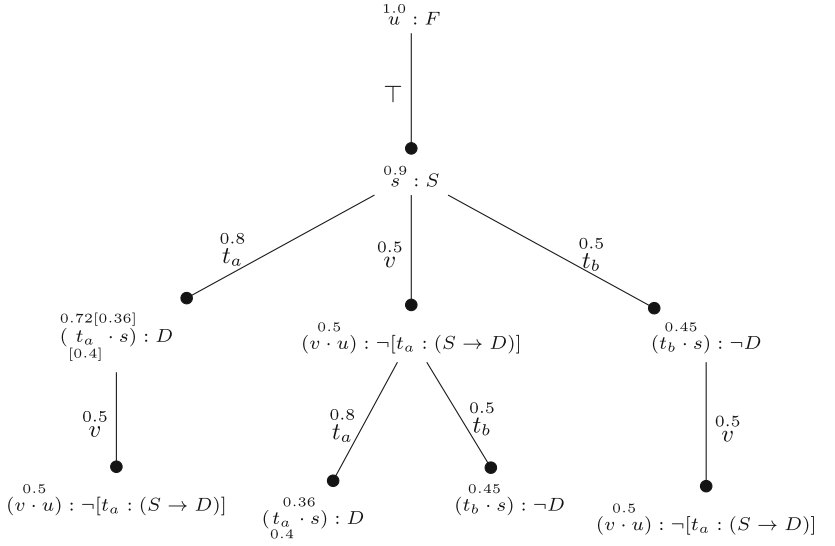

Fig. 1. Weighted process tree for the default theory T_1. Notice that calculating weights is a dynamic process that requires backtracking and attenuating the default weight for a reason term, as exemplified by the left-hand branch of the tree.

believe that the neural network is not reliable for female patients. We assume that the information about the patient's sex is a fact, thus the reason u has a weight of 1.

We can now define a default theory $T_1 = (W_1, D_1)$ with W_1 containing only the formula $u : F$ and D_1 containing the rules $\delta_{s:S}$, δ_a, δ_b and δ_c. The following weights are given as the starting values: $w(u) = 1.0$, $w(s) = 0.9$, $w(t_a) = 0.8$, $w(t_b) = 0.5$ and $w(v) = 0.5$. That is: we are certain that the patient is female (tracked by reason u), our credence in the file (tracked by s) is as given above, our *prima facie* credence in both neural networks (tracked by t_a and t_b) equals the difference between the values of both output nodes of these networks, and, for all we know, the credibility of neural network A is reduced by a factor 0.5. The output values for the above defined starting values can be visually represented by means of a process tree in Fig. 1.

We can easily check that the reason term $(t_a \cdot s)$ is attenuated by the information given in rule δ_c. Assume for the sake of an example that the acceptance threshold for T_1 is 0.4. In this case only $(t_b \cdot s)$ passes the threshold and a set containing the reason $(t_a \cdot s)$ also cannot undercut all the undercutters attacking its default step, which leaves us with the extension sets containing $(t_b \cdot s) : \neg D$ as the accepted argument, together with the argument $(v \cdot u) : \neg[t_a : (S \rightarrow D)]$. Thus, the combination of logical reasoning combined with the neural network outputs answers the example's question about the patient's disease in the negative. Notice that even in the situation in which the undercutter were much weaker, the fact that the attenuating strength is not bringing the undercut reason below the threshold would not necessarily save the undercut reason because of the condition of having to defend against an undercutter.

3 Strong Weighted Default Justification Logic

WDJL, as defined above, allows us to track the strength of reasons within arguments, allowing for more fine-grained definitions of extensions. However, it does not allow for weighing different reasons for or against the same proposition *within* a single argument. Recall that processes, in Pandžić [35], may only contain defaults the conclusions of which are consistent with the conclusions of earlier defaults in the process. Therefore, once a process contains a default with the conclusion $(t \cdot u) : G$, it may not contain any further defaults with the conclusion $(v \cdot w) : \neg G$, for any justification terms v and w. After all, given the factivity axiom, $(t \cdot u) : G$ and $(v \cdot w) : \neg G$ are not consistent. Of course, as we have seen, conflicting reasons can still be weighed *post hoc* in WDJL by comparing extensions. But intuitively, in everyday epistemic reasoning, we often *directly* weigh reasons for or against the same proposition. Therefore, in this section, we define a second version of WDJL, called *strong weighted default justification logic* (WDJL$^+$), that allows the weighing of reasons for and against the same proposition (based on [24]).

3.1 Preliminaries

The language of WDJL$^+$ is extended to contain two new operations on reason terms (next to the existing operations $\cdot$ and $+$), namely the operations $\sim$ and %. These operations are unary and binary, respectively. Thus the grammar of reason terms becomes:

$$t := x|c|(t \cdot t)|(t + t)| \sim t|(t\%t)$$

where x is a variable (an unspecified justification) and c is a proof constant.

The axioms of WDJL$^+$ are different from those of WDJL (and standard justification logic). First, Axiom 2 (the sum axiom), is adapted as follows:

A2$^+$ If $t \notin Sub(u)$ and $u \notin Sub(t)$: $t : F \rightarrow (u : F \rightarrow (t + u) : F)$ (Sum$^+$)

Thus a proposition with a summed reason term can be obtained only if both subterms are known to be reason terms for the proposition. This change reflects the idea that reasons are non-monotonic: if we have a reason for some proposition P, we cannot add any other reason and assume that the combined reason is also a reason for P. It may be, for instance, that the second reason interferes with the first reason, so that the combined reason is no longer a reason for, or even a reason against P. To use a classical example: my observation that Tweety is a bird gives me a reason to assume that Tweety can fly, which we may express as $o : F$, where F is the proposition that Tweety can fly and o is the reason provided by my observation. But if I would add my observation that Tweety's wings are clipped (represented by reason term p), I cannot conclude that $(o + p) : F$, since clipped birds generally cannot fly (although I may still be able to conclude that $(o + \sim p) : F$, which indicates that o is a reason for and p is a reason against F – see below).

The restriction that two summed reason terms may not be subterms of each other is meant to avoid 'double counting' of reasons. According to the axioms of standard justification logic, when we have a propositional tautology A, then there is a constant c such that $c : A$ is a theorem. Thus, for any sentence P, we have the theorem $c : (P \rightarrow P)$, for some constant c. Now, say that we have some defeasible reason for P, modelled as $t : P$. Using the application axiom, we can conclude $(c \cdot t) : P$. But then, without the condition on the Sum$^+$ axiom, we could conclude $[t + (c \cdot t)] : P$. But here the reason represented by reason term t seems to be counted double. This is not a problem in standard justification logic, but it is in WDJL$^+$, since the weight of the summed reason term would be an additive function of the weight of the individual terms – therefore, by adding together the weights of t and $(c \cdot t)$, we would seem to have more reason to conclude P than we really do. The restriction blocks this inference, since $t \in Sub(c \cdot t)$.

Moreover, Axiom 3 (the factivity axiom) is no longer valid in WDJL$^+$. Intuitively, this is because the sentence $t : P$ is now interpreted 't is *a* reason for P' – but from there being a reason for P it does not follow that P is true, since there may be stronger reasons against P. Instead, we introduce a new axiom to govern the behaviour of $\sim$:

A3$^+$ $t : F \rightarrow (\sim t) : \neg F$; $\quad (\sim t) : F \rightarrow t : \neg F$ (Inversion)

(There is no axiom for the % operation, since its behaviour is only defined in terms of the defeasible part of WDJL$^+$ presented below.)

3.2 Strong Weighted Default Justification Logic

Again, we define a weight function w. However, since we are now dealing with positive as well as negative reasons (as is clear from Axiom 3$^+$), w now has a codomain that also allows for negative numbers: $w : Tm \rightarrow [-1, 1]$. The functions determining the weights of complex reason terms are as follows:

Aggregation $w(t + u) = \tanh(w(t) + w(u))$. This is a simplified version of the aggregation function given in Sect. 2.2. The condition that the weight of the aggregated reason term may not be lower than the weight of each individual reason term is dropped, since it no longer makes sense when we also allow for negative weights (intuitively, two reasons against a proposition can give an even stronger reason against that proposition).

Default step $w(t \cdot u) = w(t) * w(u)$. This function is identical to the one given in Sect. 2.2.

Reason inversion $w(\sim t) = -w(t)$. If t is a reason for some proposition P, then $\sim t$ is the opposite reason for the opposite proposition $\neg P$. And intuitively, a reason to believe F is an equally strong reason to disbelieve $\neg F$. Therefore, the strength of a reason $\sim t$ should be the direct inverse of the corresponding positive reason t.

Attenuation $w(t\%u) = min(w(t), w(t) * (1 - w(u)))$. The operation % is used to denote attenuation of default reasons (see below). The idea here is that the stronger the attenuating reason u, the more the *prima facie* strength of default reason t is reduced to obtain the strength of the attenuated reason $(t\%u)$. In the extreme case where $w(u) = 1$ (i.e. there is maximum reason to believe that t is attenuated), the weight of $(t\%u)$ is reduced to 0, and in the extreme case where $w(u) \leq 0$ (i.e. there is no or even negative reason to believe that t is attenuated), the weight of $(t\%u)$ remains identical to the weight of t.

The new $\sim$ operation allows us to aggregate reasons for and against the same proposition. Consider a situation where we have the propositions $t : F$ and $u : \neg F$. Then, by Axiom 3$^+$, we also have $(\sim u) : F$ and $(\sim t) : \neg F$. And then, by Axiom 2$^+$, we also have $(t+ \sim u) : F$ and $(u+ \sim t) : \neg F$. Note also that $w(t+ \sim u) = -w(u+ \sim t)$.[4] This is indeed what we want: if the aggregated reasons count in favour of proposition F with a strength of n, then they should count against proposition $\neg F$ with a strength of $-n$, and *vice versa*.

The semantics of WDJL$^+$ introduce some significant changes with regards to WDJL (and standard DJL). Recall that in WDJL, propositions can only be

[4] Proof: $w(t+ \sim u) = tanh(w(t) + w(\sim u)) = tanh(-w(\sim t) - w(u)) = -tanh(w(\sim t) + w(u)) = -w(u+ \sim t)$.

true or false. However, now that we can aggregate reasons for and against the same proposition, it seems that credence in propositions should come in degrees. For instance, if I have two reasons t and u in favour of a proposition, it seems I should believe it more strongly than if I only have reason t. Thus, in WDJL$^+$, we define a valuation function as follows: $v : \mathcal{P} \to [-1, 1]$. This allows us to establish a correspondence between the weight function w and the valuation function v. That is, we constrain v such that the valuation of a proposition that starts with a reason term is equal to the weight of this reason term:

Definition 9 (Valuation). *For each proposition $t : P$ starting with reason term t, $v(t : P) = w(t)$.*

The valuation of atomic propositions that do not start with a reason term is unconstrained. The valuations of complex propositions depend on the valuations of their constituent propositions:

$$v(\neg P) = -v(P)$$
$$v(P \wedge Q) = min(v(P), v(Q))$$
$$v(P \vee Q) = max(v(P), v(Q))$$
$$v(P \to Q) = max(-v(P), v(Q))$$

Note that the valuation $v(P \to Q)$ is equal to the valuation $v(\neg P \vee Q)$.

Next, we consider application of defaults in WDJL$^+$. Defaults are defined in the same way as in WDJL. Below, we only focus on normal defaults. As defined in Sect. 2.2, the conclusion of a normal default is always of the form $(t \cdot u) : P$, where u is the reason term of the premise, and t is the warrant justification term. As before, the weight of $t \cdot u$ is equal to $w(t) * w(u)$. Thus, the valuation of the entire proposition $(t \cdot u) : P$ is also equal to $w(t) * w(u)$. This reflects the idea that uncertainty in the conclusion of a default can arise both from uncertainty in the justification of its premise, and from uncertainty in the justification of the warrant.

The next question is how to assess the interactions between multiple, possibly conflicting defaults. This is where the main difference between WDJL and WDJL$^+$ lies. Contrary to WDJL, WDJL$^+$ does not use processes. Instead, we simply apply all defaults simultaneously, provided that they are applicable (i.e. their premise is known with a non-zero credence). The reason we can apply multiple (potentially conflicting) defaults simultaneously is that the interactions between these defaults are automatically tracked in the weight of the resulting compound reason term.

Note that the defaults that are simultaneously applied may not only include defaults for or against the same conclusion (i.e. rebutters, in Pollock's [37]) terminology, but also defaults that attack the warrant of another default (i.e. undercutters, again following Pollock [37]). Thus, we need a way to record the effects of undercutters in justification terms. This is done with the % operator:

Definition 10 (Reason attenuation). *For any applicable default δ such that $conc(\delta) = (u \cdot t) : P$, where $u : (Q \to P)$ is a warrant of δ, if we have $v : \neg[u : (Q \to P)]$, then we can conclude $[(u \cdot t)\%v] : P$.*

Roughly speaking, the term $(u{\cdot}t)\%v$ means that the reason term $(u{\cdot}t)$ justifies the conclusion of a default that has been undercut by a proposition with reason term v. Given that undercutting reasons (like other reasons) come in degrees, we may say also that the proposition justified by $(u \cdot t)$ has been *attenuated* – in other words, it may still retain some of its strength, but its strength has been reduced by the proposition justified by v. We may also say that the proposition justified by v is an *attenuator* of the reasoning step leading to the conclusion justified by $(u \cdot t)$.

Thus, when we have a default theory – that is, a pair $\langle \mathcal{W}, \mathcal{D} \rangle$ where $\mathcal{W}$ is a knowledge base and $\mathcal{D}$ is a set of defaults – we can simply derive any conclusion that can be obtained by iteratively applying the rules of standard justification logic and any triggered defaults in $\mathcal{D}$. As long as $\mathcal{W}$ is consistent, this procedure will never lead to inconsistent sets of propositions. This is because the conclusion of any applied default δ has the form $(u \cdot t) : P$, where $(u \cdot t) : P$ must be consistent with the knowledge base (else δ could not have been applied), and where $(u \cdot t) : P$ is consistent with any reason against P (since we no longer have factivity), and therefore with the conclusion of any other default. The valuation of any proposition that can be derived is automatically tracked as the weight of the proposition's justification term. We give an example below.

3.3 Example

Recall the example of reasoning with neural networks A and B from Sect. 2.3. In order to assess our overall reason to believe that the patient has disease D, we can model this situation in WDJL$^+$. We still have the following three defaults (representing that neural network A predicts that the patient has disease D, neural network B predicts that the patient does not have D, and neural network A is less reliable since the patient is female):

$$\delta_a = \frac{s : S :: (t_a \cdot s) : D}{(t_a \cdot s) : D}, \qquad \delta_b = \frac{s : S :: (t_b \cdot s) : \neg D}{(t_b \cdot s) : \neg D}$$

$$\text{and} \quad \delta_c = \frac{u : F :: (v \cdot u) : \neg[t_a : (S \rightarrow D)]}{(v \cdot u) : \neg[t_a : (S \rightarrow D)]}.$$

The weights of the reason terms are still the same as in Sect. 2.3: $w(u) = 1.0$, $w(s) = 0.9$, $w(t_a) = 0.8$, $w(t_b) = 0.5$, and $w(v) = 0.5$ (again reflecting the facts that we are certain that the patient is female, that the information in the patient's file is quite but not fully reliable, that neural network A is more confident than neural network B, and that for all we know, the reliability of neural network A is halved for female patients).

Since in WDJL$^+$ we can apply all defaults simultaneously, we can conclude both $(t_a \cdot s) : D$ and $(t_b \cdot s) : \neg D$. By Definition 9, the valuation of both sentences depends on the weights of their reason terms, which are $w(t_a \cdot s) = w(t_a) * w(s) = 0.8 * 0.9 = 0.72$, and $w(t_b \cdot s) = w(t_b) * w(s) = 0.5 * 0.9 = 0.45$. Thus, the valuation of the sentence $(t_a \cdot s) : D$ is also equal to 0.72, and the valuation of the sentence $(t_b \cdot s) : \neg D$ is also equal to 0.45.

We can now aggregate both reasons directly, or we can first take into account the fact that δ_a is attenuated. Of course, the latter procedure is preferable if we want to take into account the full evidence. Thus we first apply δ_c, which leads to δ_a's conclusion being *attenuated* by reason term $(v \cdot u)$, which is written as $[(t_a \cdot s)\%(v \cdot u)] : D$ (see Definition 10). By Definition 9 and the attenuation function, the valuation of this attenuated sentence equals $w((t_a \cdot s)\%(v \cdot u)) = min(w(t_a \cdot s), w(t_a \cdot s) * (1 - w(v \cdot u))) = min(0.72, 0.72 * (1 - w(v) * w(u))) = min(0.72, 0.72 * (1 - 0.5 * 1)) = 0.36$.

Next, we want to aggregate the reasons provided by both neural networks, taking into account the fact that the reason relying on neural network A has been attenuated. To calculate the 'overall' reason in favour of D, we first use Axiom 3^+ (the inversion axiom) to conclude that $\sim (t_b \cdot s) : P$ – in words, $t_b \cdot s$ is a reason *against* D. By the inversion function, the weight of this inverted reason is $w(\sim (t_b \cdot s)) = -w(t_b \cdot s) = -0.45$. Next, using the Sum$^+$ axiom (Axiom 2^+), we can conclude $([(t_a \cdot s)\%(v \cdot u)]+ \sim [t_b \cdot s]) : D$. The valuation of this proposition equals $w(([[(t_a \cdot s)\%(v \cdot u)]+ \sim [t_b \cdot s])) = tanh(0.36 - 0.45) \approx -0.09$. Note that, equivalently, we can conclude that $(\sim [(t_a \cdot s)\%(v \cdot u)] + [t_b \cdot s]) : \neg D$ with a valuation of 0.09. Thus our overall reason counts against proposition D as strongly as it counts in favour of proposition $\neg D$ (which is what we want, since intuitively, any reason to believe a proposition is also a reason to disbelieve its negation, and *vice versa*). In other words, due to neural network A's reduced reliability, we now have a slight reason to disbelieve that the patient has the disease. This outcome is analogous to the one in Sect. 2.3.

3.4 Justification Default Graphs

As the above example makes clear, WDJL$^+$ has the feature that justification terms exactly track the way in which a conclusion has been reached. This is desirable if we wish to explicitly represent arguments in object language, but the implication is that justification terms can become very long and therefore difficult to understand. From an XAI perspective, it is desirable to have a more perspicuous representation of arguments. For this aim, we can represent the derivation of conclusions in WDJL$^+$ in a form of a graph. However, contrary to the previous approach, these graphs do not represent different processes (since WDJL$^+$ does not have any processes). Rather, the graph structure reflects the construction of the complex justification term of the main conclusion. Since this justification term tracks the reasoning steps used to arrive at the conclusion, the structure of the graph represents this reasoning as well. The resulting types of graphs are structurally similar to the default graphs introduced in Graff [24], which also represent different, potentially conflicting defaults in the same structure. Sticking to the example above, we can represent the reasoning steps leading to the conclusion $([(t_a \cdot s)\%(v \cdot u)]+ \sim [t_b \cdot s]) : D$ (with a valuation of -0.09) as in Fig. 2. Each node here represents a proposition (rectangular nodes) or default rule (circular nodes), and each edge represents application of a default, or a non-defeasible derivation. Next to each proposition node, we show the proposition's valuation (i.e. the weight of its justification term), as well as the equation by

which this valuation is obtained (except for input nodes, the valuation of which is primitive). Next to each default node, we show the weight of the default's warrant justification term.

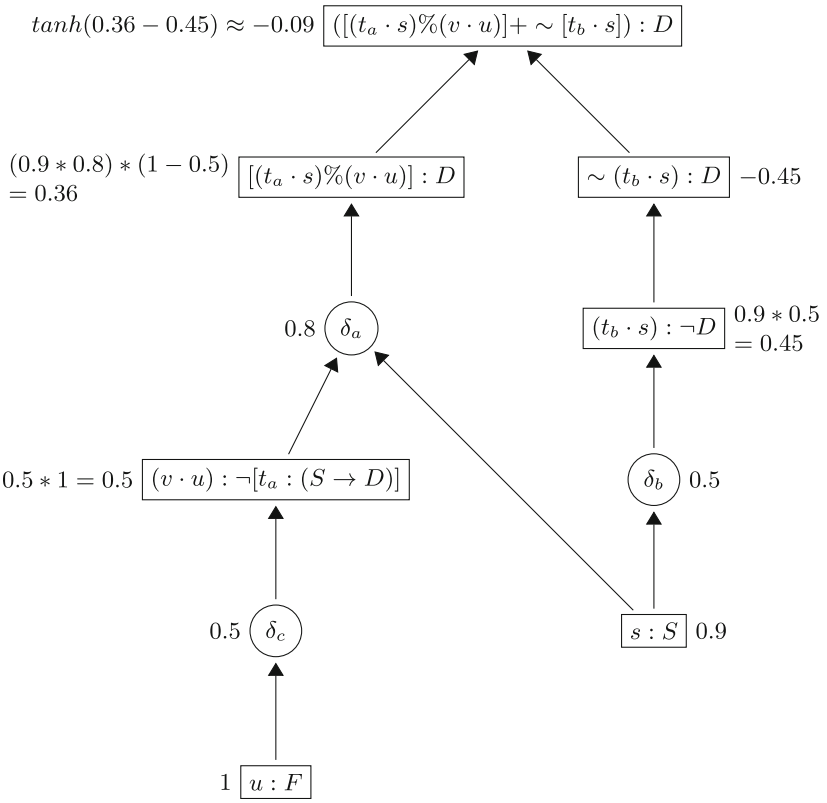

Fig. 2. Example 3.3, represented as a default graph

4 WDJL and WDJL$^+$ as Explainable Neuro-Symbolic Architectures

By combining numeric weights with a symbolic semantics, WDJL and WDJL$^+$ can be seen as neuro-symbolic architectures. WDJL process trees, e.g. Figure 1, can be compared to artificial neural networks, in that both are based on the propagation of numeric values within a graph structure. As a result, numeric process trees could be used to instantiate different arguments, by varying the values of the reason terms, just like artificial neural networks can be used to classify different items by varying the input activations. For instance, if the weights of reason terms in Fig. 1 would be different, different formulas would be acceptable, and therefore the semantics would yield different extensions. One

difference between WDJL and neural network-based approaches, however, lies in WDJL's operational semantics. That is, process trees delay some aspects of numeric reasoning and prioritize the rule-based reasoning component via the notion of consistency.

The numeric reasoning is more prioritized in WDJL$^+$. As the graph in Fig. 2 shows, arguments in WDJL$^+$ are structurally closer to artificial neural networks. That is to say, each argument in WDJL$^+$ can be instantiated as a graph, where the activations of input nodes (i.e. the strength of reasons for premises and for default warrants) are given, and the activation of each non-input node is a function of the activations of its input nodes. The example in Sect. 3.3 was of course very simple, but in principle, more complex patterns of argumentation (including many reasoning steps, nested attenuators, etc.) could be modelled in a single graph. The same graph could then again be used to instantiate different arguments, by varying the input values. For instance, the graph in Fig. 2 could be used for any patient to estimate the strength of our reason to believe that patient has disease D, simply by varying the strength of the input and default nodes. Such numeric argumentation models are not necessarily limited to reasoning about the outputs of different neural networks, as in our running example. They could be used to reason about any form of evidence that can be given a numeric valuation. Examples are data sampled from questionnaires using numeric scales, or numeric knowledge bases where numeric values represent the credibility or importance of sources (see e.g. [32]).

Graff [24, p. 253] speculates that default graphs could be trained like neural networks. That is to say, these graphs could be presented with different input-output pairs, learning an optimal input-output mapping by adjusting the connection weights through backpropagation. A similar approach is followed by D'Avila-Garcez et al. ([20, Ch. 3], [21, Ch. 4]), who represent extended logic programs as neural networks, and show how these neural networks can be trained on the basis of input data. However, D'Avila-Garcez et al.'s neural networks are of a simpler structure than the WDJL$^+$ models presented here, since they do not model aggregation of reasons or undercutting/attenuating defaults.

The connections in Fig. 2 do not have weights, but these could easily be added (see [24] for details). If represented as default graphs, WDJL$^+$ models could in principle allow for machine learning. At the same time, the models would remain inherently explainable, since the nodes correspond to interpretable symbols (propositions and default rules). If successful, this approach would integrate the flexibility afforded by neural machine learning with the explainability of symbolic systems. However, exploring this possibility falls outside of the scope of the current paper and is left for future work.

A more recent approach by Giordano and Dupré interprets a standard type of pre-trained deep neural networks (multilayer perceptrons) within a weighted, non-monotonic multipreference semantics of fuzzy description logics [23]. They note that such interpretations could help to explain deep neural networks and therefore contribute to the field of XAI. In further work, we aim to explore whether deep neural networks could similarly be interpreted according to the semantics of WDJL or WDJL$^+$.

5 Related Work

There are several existing proposals to model some aspects of commonsense reasoning in justification logic. We will mention some of the most prominent ones. The logic of approximate conditional probabilities developed by [34] introduces a way to model default reasoning by combining conditional probabilities with justification assertions. Their proposal is based on defining operators for approximate probabilities of a justified formula, given some condition formula. However, their approach does not focus on encoding the uncertainty in the internal structure of reasons, nor on weighing justifications by directly assigning numeric values to terms. Instead, the numeric aspect applies to justification assertions. Likewise, the probabilistic justification logic presented in [27] focuses on exploring statements of the following type: "the probability of $t : F$ is greater than or equal to s" and does not present a mechanism for weighing reasons nor for modelling defeat. Moreover, Baltag, Renne and Smets [6,7] define extensions of justification logic in which it is possible to receive new information that defeats currently accepted reasons. The nature of defeat in their system can be seen as a result of prioritizing new information, and they do not aim to encode defeasibility in reason terms or express the conflicts between reasons numerically. Instead, their logics use belief revision and dynamic epistemic logic techniques to model how agents consolidate beliefs in response to changes to their evidence.

Apart from this work in justification logic, and the work in nonmonotonic and neuro-symbolic reasoning already mentioned, there have been some strands of work in numeric argumentation theory that bear some closer similarities to the current approach. In this section, we mention two strands of this work, namely Pollock's inference graphs with variable degrees of justification, and numeric Dung-style argumentation frameworks.

5.1 Numeric Inference Graphs

Pollock has developed inference graphs as a means of formalising defeasible reasoning [37–39]. Such graphs contain inference steps, representing propositions, that are connected by two types of edges, representing support and defeat relations, respectively. Pollock's original semantics for inference graphs made use of 'defeated' and 'undefeated' status assignments [37,38]. However, in his [39], Pollock introduced a numeric semantics. In this new approach, *justification strengths* are assigned to inference steps, and *reason strengths* are assigned to both support and defeat relations. A proposition's justification strength is then defined to be a function of a) the strengths of its supporting propositions and of the support edges, and b) the strengths of its defeaters.

The current approach towards modelling arguments is inspired by Pollock's inference graphs, and the reasoning behind these, in some important aspects. Most importantly, we have taken the distinction between undercutting and rebutting defeaters from Pollock's work (e.g. [37]). At the same time, we go further than Pollock did in applying this distinction to the numeric valuation

of reasons. In Pollock's gradual inference graphs, rebutters and undercutters affect the attacked proposition's justification strength identically. That is, an undercutting defeater of an argument from A to B affects B's justification strength just in the way that a direct rebutting defeater would, namely by directly reducing B's strength proportionally to the strength of the defeater. This seems implausible, however. Undercutters (or attenuators) seem not to provide direct reasons to disbelieve a conclusion. Rather, they make some other reason for the conclusion less strong. This is more naturally modelled by having attenuators multiply the strength of an argument by a value lower than 1, rather than providing an independent counterweight to the argument's conclusion.

Another difference between Pollock and our second method specifically is that Pollock's inference graphs do not allow for aggregation of reasons, while in our second method, different reasons for or against a conclusion can be added (via the normalised addition function). Pollock offers some arguments why he believes aggregation to be implausible in domains of theoretical reasoning [39, pp. 245–246]. But this seems to not always be the case. Intuitively, if we have two independent sources confirming proposition A, our overall reason to believe A is higher than if only one source would have been present. Our second method is meant to capture this intuition.

5.2 Numeric Argumentation Frameworks

Argumentation frameworks (AFs) were first proposed by Dung [15] to model arguments in a maximally abstract way. An AF is nothing more than a pair $\langle \mathcal{A}, \mathcal{D} \rangle$, where $\mathcal{A}$ is a set of arguments and $\mathcal{D} \subseteq \mathcal{A} \times \mathcal{A}$ is an attack relation between arguments. The internal structure of arguments is thus entirely ignored. Despite this simplicity, Dung showed how some powerful semantics can be defined to establish *extensions*, roughly, subsets of arguments that are mutually acceptable given the attack structure. That being said, Dung's framework does not provide for comparison of strengths of arguments, and is therefore less suitable to reason about sources with varying levels of reliability. Partially for this reason, many authors have proposed extensions of Dung's framework that assign numeric weights to arguments, attack relations, or both. Table 1 contains an overview of these approaches, along with some of their properties, which we shortly explain below.

First, some of the numeric AF-based approaches are *probabilistic* in that the weights of arguments or attack relations explicitly represent (relative) probabilities. In other approaches, the interpretation of the weights is left unspecified, or specified in a non-probabilistic way. As mentioned above, both our approaches are non-probabilistic. We interpret weights in an abstract way as 'strength of reasons'. In specific domains, this abstract notion may be specified so that the weights obtain a clearer interpretation (such as the reliability of sources [11,26]).

Second, some of the AF-based approaches assign fixed strengths to arguments or attack relations. In other words, the strength assignment is *external* to the structure of the AF. In other approaches, the strength of arguments or attack relations depends fully [10] or – more often – partially on the structure of the

Table 1. Numeric AF–based approaches

Approach	Probabilistic?	Strength source?	Semantics?
Cayrol and Lagasquie-Schiex [10]	No	Internal	Argument–based
Dunne et al. [16]	No	External	Extension–based
Li et al. [31]	Yes	External	Extension–based
Da Costa Pereira et al. [11]	No	Mixed	Argument–based
Leite and Martins [30]	No	Mixed	Argument–based
Thimm [43]	Yes	External	Argument–based
Coste-Marquis et al. [12]	No	External	Extension–based
Hunter [25], first approach	Yes	External	Argument–based
Hunter [25], second approach	Yes	External	Extension–based
Eğilmez et al. [17]	No	Mixed	Argument–based
Kaci and Labreuche [26]	No	External	Extension–based
Dondio [14]	Yes	Mixed	Extension–based
Baroni et al. [8] and Rago et al. [40]	No	Mixed	Argument–based
Bistarelli et al. [9]	No	External	Extension–based

AF itself. Our own approach is most similar to the latter approaches, since the weights of individual justification terms are assigned externally, but the weights of complex justification terms (and thus of the propositions justified by these terms) are functions of their sub-terms, and therefore of the argument structure.

Third, the semantics of numeric AF-based approaches mostly fall into two classes. Some of these approaches follow Dung's [15] original semantics by applying gradual valuations in first instance to extensions or labellings, i.e. groups of arguments, and valuating individual arguments indirectly on the basis of their occurrence within these groups. Other approaches apply gradual valuations directly to individual arguments – i.e., each argument is valuated directly on the basis of its own strength and/or the strengths of its attacking arguments or the incoming attack relations themselves. As should be clear from the foregoing, our first method is extension-based (the numeric weights affect the acceptability of extensions first and foremost), while our second method is argument-based (the numeric weights directly determine the valuation of individual arguments – or, more concretely, propositions). In this sense, our approach is similar to that of Hunter [25], who also considers both an extension-based valuation and a direct valuation of arguments. (In other ways, though, our approach differs from Hunter's, most notably in the fact that valuations of Hunter's arguments are independent of the structure of the AF itself.)

The main difference between our approaches and any of the AF-based approaches (regardless of the above-mentioned subdivisions) is, of course, that the language we use allows for fine-grained representation of the internal structure of arguments and attacks between arguments that cannot be represented in AFs. (Some of the AF-based approaches in Table 1 contain support as well as

attack relations between arguments. But these still do not represent the reasoning *within* arguments.) For this reason, we can also distinguish between different ways in which arguments can be attacked – specifically, as we have seen, we can distinguish between rebuttals and undercutters (or attenuators). This allows for more fine-grained interactions between different arguments' strengths, since a rebutting proposition affects the strength of the rebutted proposition in a different way than a proposition that attacks a proposition's warrant. At the same time, representation of the internal structure of arguments is desirable from the perspective of XAI. Showing a user how exactly a conclusion is derived from certain premises is more informative than stating that there exists *some* argument for a conclusion.[5]

6 Conclusions and Future Work

In the present paper, we focused on laying the groundwork for a unique method that combines qualitative and quantitative reasoning about uncertainty in justification logic. We have offered two separate approaches for integrating default justification logic (as presented in [35]) with numeric reasoning. The first approach stays closer to standard justification logic by following an extension-based operational semantics, while the second approach moves closer to a fully neuro-symbolic (but still explainable) architecture, by comparing different, possibly conflicting reasons directly.

Since the paper is meant to be foundational, it leaves open several important questions for future work. First, we still need to prove that the system behaves correctly and that it delivers intuitive outputs beyond 'hand-picked' examples. To establish that the systems are 'well-behaved', we still need a clear formulation of the desirable properties that the systems need to satisfy. Another question beyond the scope of this paper is how feasible implementations based on WDJL and WDJL$^+$ are, given the complexity problems that have been established for standard default logic [42]. Nonetheless, we hope to have shown that the approaches offer a promising framework for reasoning with uncertain and conflicting evidence, in a way that simultaneously allows for fine-grained numeric evaluations and explicit modeling of the structure of arguments in an intuitive way.

Disclosure of Interest. The authors have no competing interests to declare.

[5] Our overview of numeric approaches is not meant to be exhaustive. We could, e.g., follow [19] in translating an argumentation system into classical logic and then use probabilistic classical logic to determine the probabilities via the attack relation. In [1], numerical strength is integrated in structured argumentation by extending a possibilistic logic with argumentation methods based on defeasible logic programming.

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On Explaining and Reasoning About Optical Fiber Link Problems

George Theodorou[1(✉)], Sophia Karagiorgou[1], Annamaria Fulignoli[2], and Roberto Magri[2]

[1] UBITECH Ltd., Thessalias 8 & Etolias 10, 15231 Chalandri, Athens, Greece
{gtheodorou,skaragiorgou}@ubitech.eu
[2] Ericsson Telecomunicazioni S.p.A., Via Anagnina 203, 00118 Roma, Italy
{annamaria.fulignoli,roberto.magri}@ericsson.com

Abstract. Optical fiber links are known for their high bandwidth and reliable data transmission. However, problems may still arise, affecting signal quality and network performance. These problems are usually happening due to external physical extrusion or excessive bending, insufficient transmission power, damaged connectors causing signal loss; or failures of splice tray connector. In response to increasing optical fiber link problems transparency and interpetability, various attempts have been made to bring explainability in Artificial Intelligence (AI) decision-making and reasoning processes. This paper tackles a crucial and timely topic, i.e., understand the various factors contributing to optical link problems by explaining opaque AI models with two goals: (i) either providing instance explanations for a given decision by using a local and model agnostic approach; or (ii) providing global explanations able to describe the overall logic assuming knowledge of the black box model or its internals. The scientific contribution of this paper entails novel explainable AI (XAI) models harvesting data from optical fiber link events to first derive local explanations, and then apply a hierarchical approach to educe global explanations from the local ones. The proposed approach shows that we can efficiently tackle both explanation complexity and fidelity to reason about the causes that have resulted in optical fiber link problems.

Keywords: Explainable AI · SHAP · Reasoning · Optical Fiber Links · Fault Interpretation

1 Introduction

Optical fibers have emerged as the most commonly used transmission medium in telecommunications. Service Providers depend on optical fibers to reliably transport a continuously growing amount of data traffic, thanks to the medium huge bandwidth and low attenuation. These key attributes make fiber the predominant choice for multiple communication applications ranging from telecommunication backbone infrastructure to metro aggregation segments down to the

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access network serving fixed users and mobile access infrastructure (e.g. 5G Radio Access Network).

The electronic process miniaturisation paired with the advent of photonic integration has allowed the manufacturing of high-rate optical transceivers (currently up to 800 Gbps) in small pluggable formats (e.g. Small Form-factor Pluggable (SFP) or Quad Small Form-Factor Pluggable (QSFP) [2] standardised by the SFF committee [15]) directly interconnected by optical fibers. Many of these transceivers can be housed in a single telecommunication or data centre equipment allowing scaling of capacity provisioning as requested by the traffic demand trends. The huge capacity transported by these optical links requires great attention to the telecommunication infrastructure monitoring to prevent faults when possible and provide fast and reliable diagnostics when an optical link goes down. Long traffic disruptions and cumbersome troubleshooting and repair procedures would cause high operational costs for a Service Provider. Optical cables damage is one of the major issues: underground and overhead cables are exposed to numerous threats, as they traverse urban environments or travel long distances across remote sites. Construction works is, for example, a relevant cause for fiber cuts. Fiber issues, however, are not the only ones that can impact an optical link connection. For instance, optical link loss can be caused by optical disconnections at the optical distributor frames by operation personnel or by malicious actions in a cabinet, optical transceiver disconnections in a central office, or power failures either in the transceiver itself or in the hosting equipment or on-site. The classification of the root cause for a fiber link fault due to a Loss-of-Signal (LOS) is the first step in the troubleshooting of a network outage. The more rapid is the classification the faster the network operator can fix the issue or put in place further investigation actions. Current level of information to the operator is a binary information, i.e., loss-of-signal reports an alarm on/off. At current state-of-the-art, different tools exist in order to check the root-cause, but they are complex and require dedicated and expensive solutions at the central office, as well as at remote sides, i.e., Remote Fiber Monitoring System (RFTS). However, a data-driven approach can offer simpler solutions providing more precise information for a first estimation of the root cause together with the basic alarm information. To this respect, transceivers support standard Digital Diagnostics Monitoring (DDM) functions [1]. This capability allows monitoring of the transceiver operating parameters in real-time. Parameters include optical output power, optical input power, temperature, laser bias current, and transceiver supply voltage, and many other. This data is sampled by the transceiver internal sensors and made available through standardised digital interfaces to the host equipment and then, to any software application that can exploit them for diagnostic purposes. The refresh rate of this parameters is typically of several milliseconds, allowing for fast variation detections. Magri et al. [12] have shown that the received optical power from a transceiver is a precious source of data to help the root cause analysis of an optical link fault. In particular, they have shown that different types of fault induce a different transient behavior of the received power from its steady state normal operation value to its loss due to the

link disruption. In other words, each root cause leaves a different fingerprint in terms of optical power drop transient than can be captured in the form of a time series when the onset of a disruption is detected. The time series of samples can be fed to a Machine Learning (ML) classifier trained to distinguish between the different root causes. Figure 1 depicts the different optical link fault root causes and the relevant operational actions. The classification performance has been proven to be very satisfactory in distinguishing between four classes of issues (but the concept can be further generalised to other source of disruptions):

1. Fiber stress or cut: in this case the problem is on the deployed fiber and proper personnel must be sent to check the fiber with proper dedicated investigation instrumentation like Optical Time Domain Reflectometer (OTDR);
2. Connector disconnection: this may highlight that some personnel or malicious intruder is operating in part of the installation where distributor frames or patch panels are present;
3. SFP connector: this is the case when someone is disconnecting the fiber directly on the optical transceiver on the host equipment; and
4. Shutdown: in this case the issue is cause by the loss of power supply to the host equipment or within the host equipment causing the transceiver to shutdown. Dedicated personnel should then check directly in the equipment room to verify the power supply.

Summarising, each identified root cause would trigger a different intervention to solve the issue. Leveraging ML simplifies the personnel trouble-shooting task allowing for less skilled personnel to take a proper action much faster with great cost saving for the infrastructure owner. A typical scenario where this ML-driven fault classification is valuable is the radio access network for mobile communication (e.g. 5G). In this scenario, for example troubleshooting may require climbing an antenna tower with very high costs and this should be avoided if the problem is elsewhere. Besides the fault classification itself induced by ML, it is indeed important to add explanatory information to support the decision-making process and select the proper fixing or investigating actions since the classification explanation can complement the root cause feedback with the necessary reliability, trust and transparency required to reduce the risk of misinterpretation. In the following section, we investigate how to add Artificial Intelligence derived explanations (XAI) and feedback to the optical link fault classification problem. The proposed approach shows how XAI can be interpreted to highlight relevant features of the time series or time dependent behavior of the receiver power samples that had contributed to drive an ML model towards the determined class.

The reference dataset has been collected in a dedicated laboratory experimental setup. An optical link based on SFP pluggable transceivers is hosted on a commercial transponder equipment. The link includes fiber patch-cords and optical connectors to provide means to emulate real optical link faults. Figure 1 illustrates the laboratory experimental setup for the collection of the dataset.

Four possible faults have been reproduced and repeated hundreds of times in various manners. The received optical power time series have been sampled by

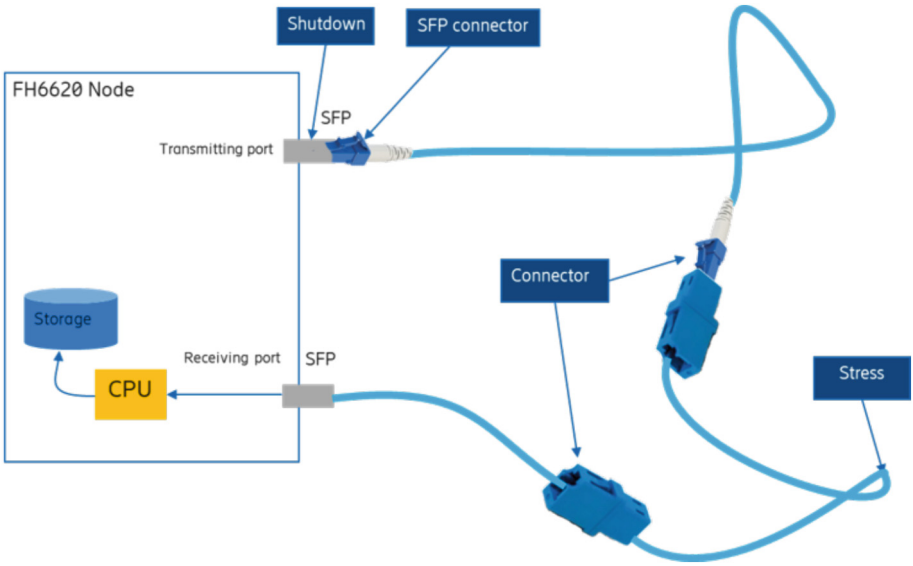

Fig. 1. Laboratory Setup for Dataset Collection.

the host node micro-controller reading the relevant information from the module memory map, stored and labelled with the corresponding fault class:

- Connector: the loss of signal is caused by the disconnection of one of the connectors between patch-cords;
- SFPConnector: the loss of signal is caused by unplugging the optical connector of the SFP module on the transmitting side;
- Shutdown: the SFP module on the transmitting side has powered off; and
- Stress: the fiber patch-cord manually stressed and broken in random ways by different persons.

This paper investigates how data collected from the laboratory experimental setup can be used to explain and reason about different aspects that cause problems to optical fiber links. The goal of this work is to harvest data from optical fiber link events to first derive local explanations, and then apply a hierarchical approach to derive global explanations from the local ones. The contributions of this paper are, as follows:

- Harvests information from events occurring within fiber optic links. These events indicate Loss-of-Signal (LOS) occurrences caused by disconnections, unplugging, normal shutdowns and stress;
- Analyse the collected data to derive local explanations for each event to identify the root cause of the issue;
- Apply a hierarchical approach towards deriving global explanations after interpreting individual events to identify broader patterns and trends across end-to-end network cases; and

– Tackle both explanation complexity and fidelity via XAI methods within an ML pipeline to gain insights into how the ML models arrive at their conclusion and reason about the causes that have resulted in optical fiber link problems.

The rest of the paper is organised, as follows: Sect. 2 provides the current literature review around innovative applications of AI for proactive and precise optical link fault detection in optical communications. Section 3 offers a detailed overview of the dataset. Section 4 presents the technical architecture of the explanations pipeline. Section 5 discusses our first experimental results, and Sect. 6 concludes the paper and provides insights to be pursued in the near future.

2 Literature Review

Numerous studies have delved into innovative applications of AI for fault detection in optical communications. Among the notable contributions, researchers have explored advanced AI techniques such as ML algorithms, Deep Learning (DL) models, and hybrid approaches combining AI with signal processing methods. These efforts have resulted in significant advancements in fault detection accuracy, response time, and robustness. The breadth of research underpins the importance of leveraging AI technologies to mitigate failures or errors in optical communication systems and improve the reliability and resilience of optical networks.

Fan, Z. et al. [6] focus on the crucial role of optical networks in modern information society, where approximately 50% of optical service faults originate from errors in the optical transmission system links, monitored through frame overheads. Current maintenance methods often fail to accurately predict these errors, leading to potential disruptions in network services. To tackle this challenge, the study introduces a Gini importance-based feature selection approach and a maximum likelihood-based logistic regression algorithm for predicting performance degradation in optical transmission system links. Moreover, the study proposes a Network AI (NAI) three-level intelligent architecture to facilitate real-world application, culminating in the development of an optical transmission performance degradation prediction system based on this architecture. Experimental results from network experiments validate the effectiveness of the proposed performance degradation model and the NAI-based system.

Li X. et al. [9] address the critical need for fiber fault detection in optical communication systems, particularly within established optical access networks facing challenges such as increased insertion loss. Traditional fault-detection methods struggle to capture fault echoes effectively under these circumstances. To overcome these limitations, the paper introduces a modulation-enhanced external-cavity-resonant-frequency method utilising a laser for fault echo reception, offering superior sensitivity compared to photo-detector-based methods. While previous studies focused on sensitivity and spatial resolution, this research develops a practical model based on Lang-Kobayashi rate equations and validates it through experiments, identifying optimal detection performance associated

with specific modulation parameters, including a modulation depth of 0.048, a frequency sweeping range of 0.6 times the laser relaxation oscillation frequency, and a sweeping step of 0.1 times the external cavity resonant frequency. The conclusion highlights the practical analysis of this high-sensitive fiber fault detection method, emphasising the optimal frequency-sweeping approach while considering various practical factors, recommending specific parameter settings crucial for high-sensitivity fiber fault detection and serving as a foundation for practical prototype development.

Liu, P. et al. [10] introduce a customized AI module integrated into an Optical Time Domain Reflectometer (OTDR) device, along with an optical power monitoring module, to establish an AI-assisted optical network fault location mechanism tailored for high-density data center interconnections. This mechanism optimally utilizes optical link data by leveraging the AI module to predict potential link failures, subsequently monitored by the optical power module for rapid fault localization and response. Experimental testing demonstrates a significant improvement in link fault detection efficiency, with the AI model enhancing average fault detection efficiency by 98.41%, which is a significant increase in failure detection and localization within data centers.

Karandin, Oleg et al. [8] demonstrate the application of XAI in fault localization within optical networks, showcasing how it enhances the comprehensibility of ML-driven decisions. Specifically, it compares the performance and interpretability of ML models trained on Optical Signal-to-Noise Ratio (OSNR) measurements collected from a single receiver monitor versus multiple monitors distributed along the network path. The conclusion drawn from the analysis highlights the stark difference in explainability between the two scenarios. While the model utilizing telemetry from a single monitor successfully identifies faults, its reasoning remains opaque due to the intricate nature of OSNR statistics. Conversely, deploying additional monitors improves both fault-localization accuracy and model explainability, albeit at increased cost. The study suggests that while comprehensive monitoring may be impractical for expansive networks, a judicious deployment of additional monitors enhances both the trustworthiness and utility of ML-based fault management tools.

Ayoub, Omran et al. [4] focus on the application of XAI, particularly Shapley Additive Explanations (SHAP), in the context of light-path Quality-of-Transmission (QoT) estimation. It provides an overview of XAI and SHAP, discusses their benefits in networking, and surveys existing studies employing XAI in networking tasks. The study formulates the QoT estimation problem as a supervised binary classification task and develops an ML model using eXtreme Gradient Boosting (XGB). Through the application of SHAP, the paper demonstrates how insights into the model's behavior can be extracted and misclassifications inspected, thereby enhancing transparency and trustworthiness. The conclusion highlights the efficacy of SHAP in providing explanations for model decisions and its utility in feature selection to improve model efficiency. Additionally, it underscores the importance of scrutinizing dataset biases to mitigate

inconsistencies in model behavior. Overall, the study underscores the significance of XAI in enhancing the interpretability and trustworthiness of ML models.

However, the practical application of AI on a large scale necessitates the ability to understand the predictions and decisions made by ML or DL models. This requirement underpins the pivotal role of explainable AI (XAI) in fostering trust between AI systems and human actors. By providing transparency and clarity in the decision-making process, XAI bridges the gap between the complex algorithms employed by AI and the understanding of end users. This transparency cultivates trust and confidence to AI systems, thereby facilitating their widespread adoption and integration into various domains with high and safe guarantees. In essence, the ability of AI to offer interpretable explanations for its decisions is paramount for realising its full potential and ensuring harmonious collaboration between humans and machines.

Jacovi et al. [7] address the critical role of trust in human-AI interactions, particularly within the XAI contexts, aiming to provide clarity and structure to the concept of trust. They introduce a model of trust inspired by interpersonal trust but tailored for AI interactions, emphasising the user's vulnerability and the anticipation of AI decisions. The study advocates for explicit contracts between users and AI models to foster warranted trust, highlighting the importance of assessing risk and distinguishing between warranted and unwarranted trust. It asserts the necessity of evaluating whether trust is warranted and identifies explanation as a key factor in fostering intrinsic trust. They conclude by offering guidelines for designing trustworthy AI and emphasise the need for further research in evaluating warranted trust in human-AI interactions.

Adadi, Amina et al. [3] explore the emerging field of XAI in the context of the rapidly evolving landscape of AI applications. Recognising the critical need for transparency and trust in AI systems, the study provides a comprehensive overview of existing research on XAI, addressing fundamental questions such as what, who, when, why, where, and how. Through an interdisciplinary lens, the survey examines various approaches to XAI and identifies key trends and challenges. While acknowledging the significant impact of XAI across diverse domains, the paper also highlights the lack of formalism in problem formulation and the need for more thorough exploration of human involvement in explainability approaches.

Ribeiro et al. [14] introduce the Local Interpretable Model-Agnostic Explanations (LIME), a method for explaining the predictions of ML models in an interpretable and trustworthy manner. By learning an interpretable model locally around each prediction, LIME provides insights into the reasons behind model predictions, crucial for assessing trust in decision-making scenarios. Additionally, the paper proposes SP-LIME, a method for selecting representative and non-redundant predictions to offer a global view of the model's behavior. Through experiments in text and image classification domains, involving both expert and non-expert users, they demonstrate the utility of explanations in various trust-related tasks, such as choosing between models, assessing trust, and improving untrustworthy models.

Lundberg et al. [11] introduce SHapley Additive exPlanations (SHAP), a unified framework for interpreting predictions, aiming to address the tension between accuracy and interpretability in complex models like ensemble or DL models. SHAP assigns importance values to each feature for a given prediction, offering a novel class of additive feature importance measures with desirable properties. By unifying six existing methods, SHAP provides insights for improved computational performance and better consistency with human intuition. The framework demonstrates a promising step towards reconciling the accuracy-interpretability trade-off in model predictions, offering a foundation for future development of interpretation methods.

Existing works are limited either by only addressing the supervised or unsupervised learning aspects of optical links failure detection, or by not leveraging the strengths of XAI to derive interpretable cause-effect insights. Compared to the above-mentioned approaches, the scientific contribution of this work is the combinatorial approach towards optical links fault detection and XAI-fuelled interpretation of the detected faults to derive more informed decisions. The differentiation of the proposed approach results in faster root cause identification, by understanding the underlying reasons behind the optical link fault detection. It also contributes to reduced downtime and minimises service disruptions, specifically in critical applications. Last, by analysing trends and patterns in explanations across different events, recurring problems within the optical network infrastructure can support proactive maintenance and reasoning about potential causes to be considered before failures occur.

3 Dataset Overview

The dataset utilized for the analysis in this paper stems from a dedicated test setup, owned by Ericsson, designed to investigate LOS events within optical networks. At the receiving side of the setup, the CPU samples the received optical power at a rate of approximately 5 ms. These sampled readings are then stored in a memory buffer, capable of holding up to 1000 samples. As a consequence, each data instance within the dataset constitutes a time series of the received optical power with a maximum length of 1000 timestamps.

The time series stored within the dataset underwent a rigorous filtering process to ensure data quality and relevance for subsequent analyses. Among a larger pool of files, specific selection criteria were applied within the optical domain to identify and retain time series that meet stringent validation standards.

Firstly, the mean time delta, represented by the first column of the file, was scrutinized to ensure adherence to the nominal value of 5 ms, with a tolerance of $\pm 50\%$. This criterion ensures that the temporal resolution of the time series aligns closely with the intended sampling rate, maintaining consistency and accuracy in capturing temporal dynamics.

Furthermore, additional criteria were applied to the optical power readings within the time series. The minimum value of the time series was required to fall below the low threshold of -30 dBm, indicative of sufficient sensitivity to

detect low-power signals. Conversely, the maximum value of the time series was mandated to surpass the high threshold of −17 dBm, ensuring that the dataset encompasses a diverse range of power levels encountered within optical networks.

Moreover, each retained time series was required to exhibit at least one recorded power drop, signifying a transition from a maximum value to a minimum value. This criterion ensures that the dataset captures instances of significant signal attenuation, characteristic of loss of signal events within optical networks.

The received optical power readings are expressed in decibels relative to one milliwatt (dBm), offering a standardized metric for power levels. To facilitate diverse analyses and interpretations, the dBm readings can be readily converted to milliwatts (mW) utilizing the formula illustrated below. This conversion allows for a seamless integration of the dataset with various analytical methodologies and tools, ensuring its usability across different research contexts.

$$P_{\mathrm{mW}} = 10^{\left(\frac{P_{\mathrm{dBm}}}{10}\right)} \tag{1}$$

During a LOS event within optical networks, the received optical power undergoes a significant drop from its maximum level to a "no-power" state. The precise detection of LOS events is governed by the specific module datasheet, which outlines parameters such as assert and de-assert thresholds, as well as the relevant hysteresis range. For instance, the Ericsson SFP+ specification provides an example of such thresholds and ranges.

It is important to note that the LOS detection logic embedded within the SFP module may introduce a certain latency with respect to the actual diagnostic reading of received power. Consequently, in this project, a deliberate decision was made to utilize diagnostic values that are not synchronized by the LOS alarm. This approach ensures that the dataset captures the raw received power readings, unaffected by any potential detection delays or logic within the SFP module.

The storage of the received power file is initiated by the sampling software itself, triggered by the sampling interval configured within the experimental setup. This ensures that the dataset accurately reflects the temporal dynamics of received power, enabling detailed analyses of LOS events and their associated characteristics.

For a visual representation of the described process, refer to Fig. 2, which provides an illustrative explanation of the sequence of events involved in LOS detection and dataset generation. This figure aids in understanding how the received power readings are captured, stored, and utilized for subsequent analyses within the project framework.

Overall, the dataset comprises a total of 3235 individual time series, each documenting the received power readings from an SFP module during a LOS event. Furthermore, each time series is associated with one of four fault categories under examination as shown in Fig. 1. For a detailed breakdown of the dataset regarding the specific fault categories, refer to Table 1.

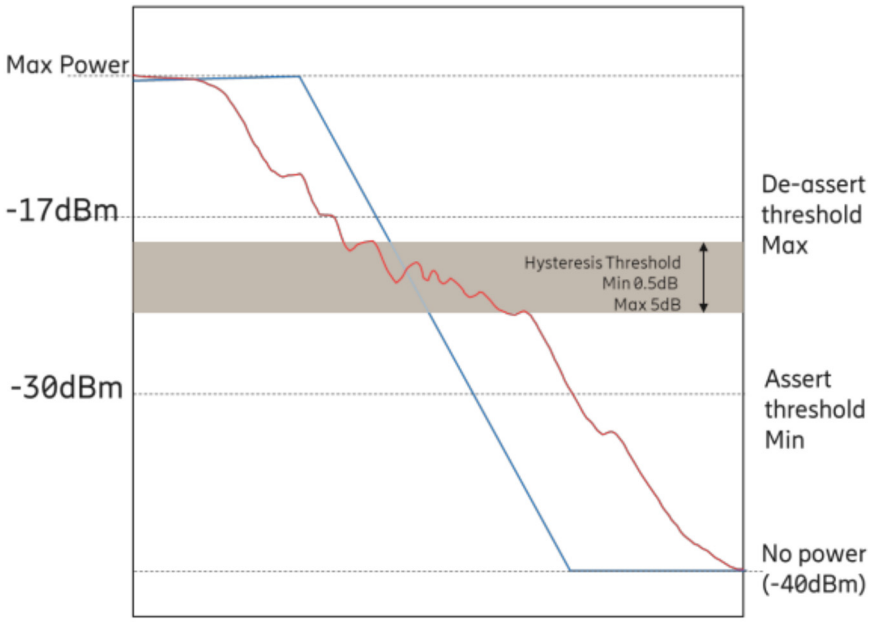

Fig. 2. LOS Event.

Table 1. Time Series Breakdown per Fault Category.

Fault category	Occurrences	% of Occurrences
Connector	906	28%
SFPConnector	833	26%
Shutdown	1190	37%
Stress	306	9%

This dataset serves as a valuable resource for analyzing LOS events within optical networks, providing insights into the temporal patterns and characteristics associated with different fault categories. Through leveraging this dataset, researchers can gain a deeper understanding of LOS phenomena and develop more effective strategies for network monitoring, fault detection, and mitigation.

4 Explanations Pipeline Architecture

An essential part of learning over the collected data is the Explanations Pipeline Architecture. In this section, we provide the pipeline and the data preparation steps followed before deriving the explanations. The explanations pipeline serves two purposes: (i) harvest the optical power loss time-series data to predict the type of fault that caused it; and (ii) generate explanations that justify the model's decision regarding the reason and type of fault.

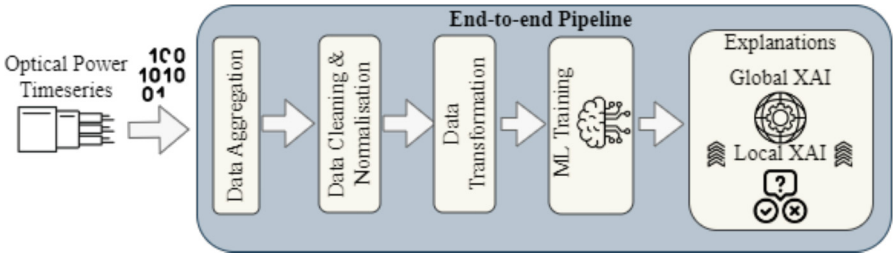

Fig. 3. Explanations Pipeline.

The explanations cultivate trust and transparency to humans and allow for their widespread adoption. As it is depicted in Fig. 3, we harvest the optical power loss time-series data by first cleansing, then extracting window-based features in time and finally feeding the data to various ML models to learn patterns about each fault category. Last but not least, SHAP is employed through the Explainer Dashboard [5] to provide meaningful explanations about how models have made their classifications and thus decisions. As shown in the Algorithm 1, the pipeline first performs Data Aggregation (line 3) to create a unified dataset of all optical power time-series. Then, Data Cleansing and Normalization takes place (lines 4–5), followed by Data Transformation (line 6). ML Training is the next thing in the pipeline (line 7). Finally, local and global explanations are derived to explain the model (lines 9–10).

Algorithm 1: Explainable AI Pipeline

Input: Optical power values OP in time-series T
Output: Local and Global Explanations on AI Model about Faults XAI_{F_L} and XAI_{F_G}

1 **foreach** $OP \in T_i$ **do**
2 $\quad OP \leftarrow$ Aggregation(OP)
3 $\quad OP \leftarrow$ Cleansing(OP)
4 $\quad OP \leftarrow$ Normalization(OP)
5 $\quad OP \leftarrow$ Transformation(OP)
6 $\quad OP \leftarrow$ ML_Training(OP)
7 **end**
8 $XAI_{F_L} \leftarrow$ ExplainLocal(OP)
9 $XAI_{F_G} \leftarrow$ ExplainGlobal(OP)

4.1 Data Aggregation

For data aggregation, we consolidate all time-series data pertaining to various fault categories into a unified data structure. This consolidation process enables

seamless integration and organisation of diverse fault-related information, ensuring a comprehensive analysis of the system's behaviour. Additionally, we introduce a new column within this unified dataset to denote the specific type of fault associated with each observation. This additional attribute serves as a crucial label for a set of analyses conducted in Sect. 5, as part of our experimentation. By categorising faults within a structured framework, our pipeline facilitates systematic and targeted investigations into the distinct characteristics and behaviors of different fault types, ultimately enhancing the interpretability and efficacy of the proposed explanability approach.

4.2 Data Cleansing and Normalisation

In the data cleansing phase, a crucial step involves pruning each optical power time-series up to the initial point in time when optical power loss is detected. This approach is adopted because only the data preceding the optical power loss event holds relevance in analysing the underlying reasons leading to the outage. By focusing exclusively on the data preceding the optical power loss, the explanations pipeline ensures that the subsequent analysis is centered on identifying the root causes and contributing factors associated with signal power disruptions. This meticulous data pruning strategy not only streamlines the dataset but also facilitates a more targeted and insightful examination of the events leading up to optical power losses, thereby enhancing the overall effectiveness of the proposed architecture.

4.3 Data Transformation

In this step, we adopt a systematic approach where each generated feature encapsulates the average optical received power value within a predefined time window. These time windows are constructed from the precise moment when the optical power loss event is triggered, extending backward over a specified number of time steps. Overall, 15 features are constructed ranging from a window of just the last 5 timestamps to the last 500 timestamps, see the Table 5 at the Appendix. By adopting this strategy, our pipeline effectively captures the dynamic evolution of optical power values across short-term, medium-term, and long-term perspectives towards a signal disruption. In essence, our feature extraction process is tailored to encode the nuanced impact of optical power fluctuations leading up to and during the optical power loss event, thereby providing a comprehensive representation of the underlying dynamics within the system. This feature engineering not only enhances the granularity of our analysis but also empowers our AI models to discern intricate patterns and correlations embedded within the data, facilitating more accurate and insightful predictions.

In addition to facilitating predictive accuracy, our AI pipeline prioritises the interpretability of the generated models. To achieve this, the features representing the average optical power values within specified time windows are transformed into categorical features, each assigned one of six possible values based on their quantile representation. This feature encoding scheme, see Table 2,

enhances the interpretability of the features by simplifying their representation while preserving essential information about their distribution. By categorising the features in this manner, our pipeline enhances the transparency of the AI models' decision-making process, enabling end users to better understand the insights provided.

Table 2. Feature Encoding Scheme.

Quantile Value	Encoding
0%–25%	0
25%–50%	1
50%–75%	2
75%–85%	3
85%–95%	4
95%–100%	5

4.4 ML Training

During ML training, we adhere to a standard methodology by partitioning the dataset into distinct training (80%) and test sets (20%) to rigorously assess the performance accuracy of our models. Our objective is to detect the type of optical power fault based on the received power time-series data. To accomplish this task, we deploy several conventional ML models, including Random Forest, AdaBoost, KNN, and Light Gradient Boost. The selection of standard ML models is deliberate, as we prioritise interpretability and explainability, necessitating the use of ML algorithms for which features are directly provided and not inherently constructed as it happens with Deep Learning (DL) algorithms. In that way, clear understanding of how the features impact the model's decisions both at a local and global level is achieved. By employing conventional ML techniques, we ensure that our models remain transparent and interpretable, enabling stakeholders to derive actionable insights from their outputs. Through comprehensive evaluation and analysis, we achieve to develop robust and effective AI solutions capable of accurately detecting the 4 distinct optical fault types while providing meaningful explanations for their predictions.

4.5 AI Explainability

Towards AI explainability, our focus shifts to leveraging the Explainer Dashboard to extract valuable insights and understand the predictions made by these models. Powered by the SHAP TreeExplainer, the Explainer Dashboard facilitates an in-depth analysis of model behavior by offering a comprehensive view of how each feature contributes to the model predictions, enabling stakeholders to interpret model decisions with clarity and confidence both on a per case scenario as well as on a global level. Through this transparent and interpretable

approach towards AI explainability, our pipeline ensures that stakeholders can extract actionable insights while fostering trust in the predictive capabilities of our models.

5 Experimental Results

This section presents the experimental results on the ML models performance regarding their precision, recall, f1 score and PR-AUC. We also evaluate the explainability of the ML models using local and then progressing to derive global explanations.

5.1 Model Performance

We conducted experiments with a variety of ML models to compare between models performance and explainability insights. Specifically, we trained four distinct traditional ML models using the time-series dataset. To evaluate the performance of these models, we computed key metrics such as precision, recall, F1-score, and PR-AUC [13]. The results, presented in Table 3 below, provide insights into the effectiveness of each model in capturing the nuances of the data and achieving the desired classification outcomes. Since, we deal with a four-class classification problem, the weighted average has been used to calculate the performance metrics.

Table 3. Performance Metrics of Model Trained with Feature Encoding.

Model	Precision	Recall	F1 Score	PRAUC
Random Forest	74%	72%	71%	75%
Ada Boost	49%	53%	49%	45%
LGBM	75%	73%	72%	75%
KNN	74%	72%	71%	71%

Upon analysing the results presented in the table above, Light Gradient Boosting Machine (LGBM), a powerful and efficient ensemble learning technique, stands out as the top-performing model among the trained models. LGBM leverages gradient boosting to construct decision trees in a sequential manner, optimising model performance by focusing on misclassified instances. Its ability to handle large datasets efficiently and its high scalability make it a preferred choice for complex classification tasks. Given its exceptional performance in terms of precision, recall, and F1-score, LGBM emerges as a promising candidate for exploring the models' decision-making processes through Explainable AI techniques, providing valuable insights into the underlying mechanisms driving predictive outcomes.

It is worth noting, as mentioned in Sect. 4 regarding data transformation, that we made a deliberate decision to discretize the average power values into

six distinct categories based on quantile values to aid in the interpretation of the results.

In case that we were using the average optical received power without encoding then the corresponding ML models performance would be as shown in Table 4.

Table 4. Performance Metrics of Model Trained without Feature Encoding.

Model	Precision	Recall	F1 Score	PRAUC
Random Forest	85%	85%	85%	91%
Ada Boost	54%	33%	23%	55%
LGBM	85%	85%	85%	89%
KNN	83%	83%	83%	83%

5.2 Model Explainability

We are going to perform the AI Explainability exercise by using the explainer dashboard. Explainerdashboard is a powerful library designed to swiftly construct interactive dashboards tailored for analyzing and clarifying the predictions and operations of machine learning models compatible with scikit-learn, including xgboost, catboost, and lightgbm. Explainerdashboard offers access to a plethora of features, including SHAP values, permutation importances, interaction effects, partial dependence plots, various performance metrics, and the ability to explore individual decision trees within random forests.

In the pursuit of generating comprehensive explanations, the dataset utilized encompasses both accurately classified time series and those that have been misclassified. This inclusive approach ensures a thorough examination of the model's performance across various scenarios, facilitating a deeper understanding of its predictive capabilities and potential areas for improvement.

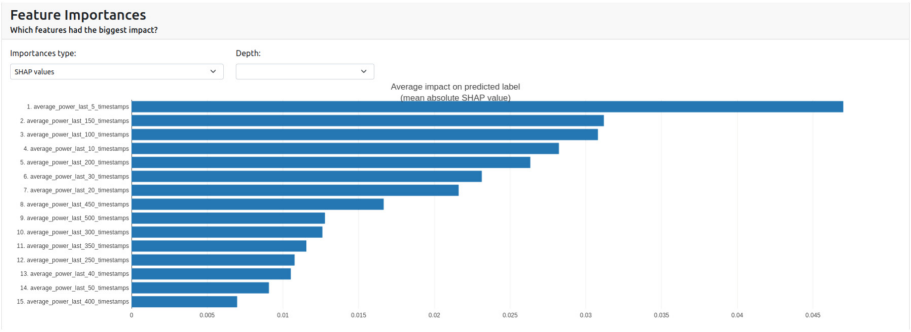

Fig. 4. SHAP Global Features Importance.

Model Global Explainability. In Fig. 4, the SHAP global features importance are depicted, revealing crucial insights into the factors influencing optical power loss prediction. Notably, the analysis highlights that events occurring within the last 5 time-steps hold the highest significance, underscoring the substantial impact of short-term power drops on the model's decision-making process. Additionally, the second and third most important features correspond to events occurring approximately 150 and 100 time-steps before the optical power loss, indicating that both short-term and longer-term events play pivotal roles as predictors of future optical power loss occurrences. This finding underscores the importance of considering both short-term fluctuations and longer-term trends in understanding and predicting power system dynamics effectively. Furthermore, our analysis delved into the SHAP global feature importance categorised by fault type within our dataset.

For the Connector fault category, see Fig. 5, caused by the disconnection of one of the connectors between patch-cords, the most critical feature is events within the last 5 timestamps. Hence, this finding suggests that these faults manifest abruptly and are less influenced by longer-term patterns.

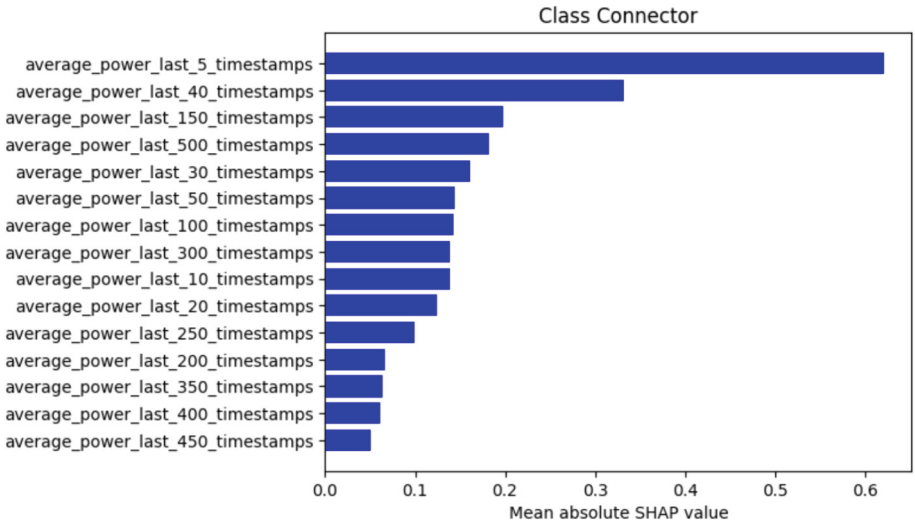

Fig. 5. SHAP Global Features Importance for the Connector Fault Type.

Conversely, the SFP-Connector fault category, see Fig. 6, caused by the direct fiber disconnection on the optical transceiver on the host equipment, exhibits a balanced dependence on both short-term (5 timestamps) and longer-term (100 timestamps before the power-loss event) features, indicating a more nuanced relationship between fault occurrence and temporal context.

In contrast, the Stress fault category, see Fig. 7, caused by problems on the deployed fiber, showcases a unique pattern where a mixture of short-term and longer-term features contributes to its detection. Notably, while the SFP-Connector category prioritises short-term events within the last 5 timestamps,

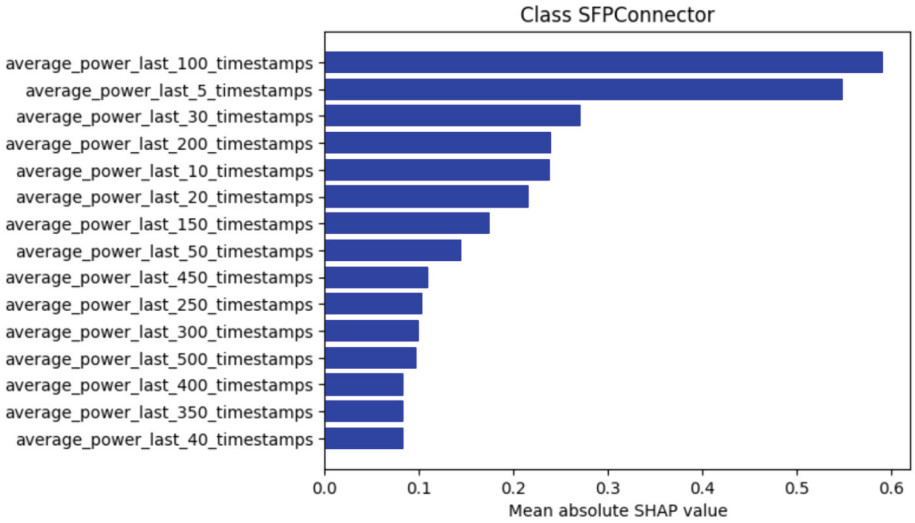

Fig. 6. SHAP Global Features Importance for the SFP-Connector Fault Type.

features spanning the last 20 to 30 timestamps are more informative for detecting fiber stress related issues.

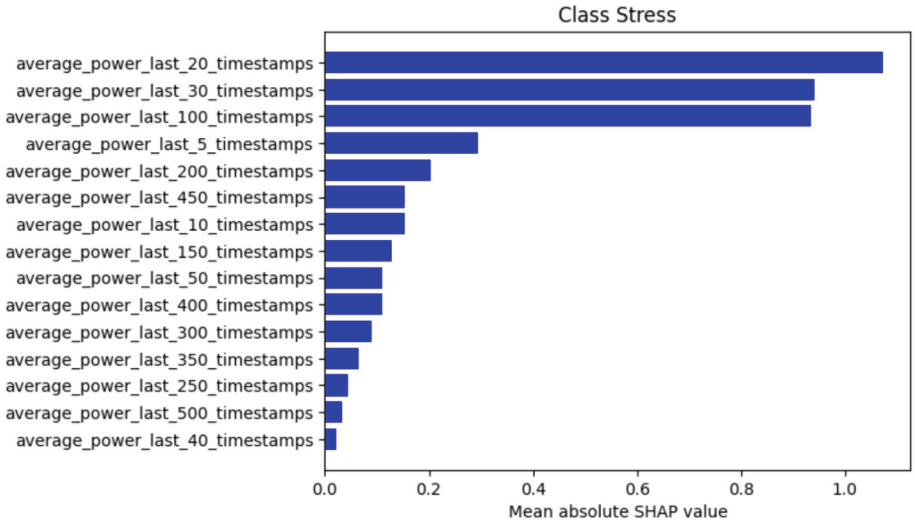

Fig. 7. SHAP-driven Global Features Importance for the Stress Fault Type.

Last but not least, the Shutdown fault category, see Fig. 8, caused by the loss of power supply, the most predictive features are primarily concentrated between 5 and 30 timestamps before the power-loss event, highlighting the significance of short-term events in this context.

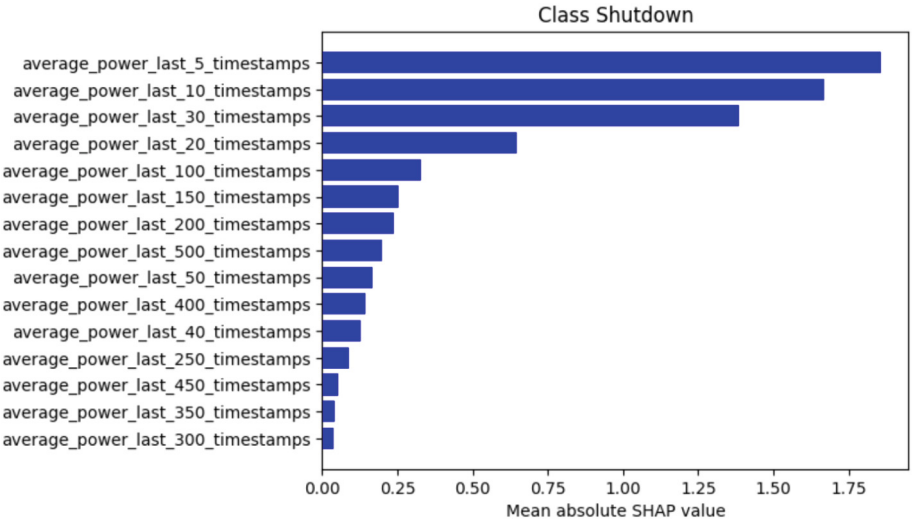

Fig. 8. SHAP-driven Global Features Importance for the Shutdown Fault Type.

Our findings suggest that certain fault types exhibit a higher predictability when considering events immediately preceding the optical power loss event, emphasising the significance of short-term contextual factors. Conversely, other fault categories demonstrate a stronger correlation with longer-term optical power level information, highlighting the importance of incorporating longer-term data to capture underlying trends and patterns. By understanding these distinctions in global feature importance across fault categories, we can enhance the accuracy and reliability of our fault prediction models tailored to specific scenarios.

Model Local Explainability. Furthermore, in addition to analysing the SHAP global features importance, an equally crucial aspect of explainability in AI is justifying individual predictions, often referred to as local feature importance. The explainer dashboard offers a valuable tool to inspect the most influential features for individual predictions.

For instance, in the Fig. 9 the model confidently predicts the class as SFP-Connector with 67.5% probability. Upon examination Fig. 10, we find that this figure depicts the model's features contributions, showing green the positive contributors and red the negative ones. The yellow bar on the left shows the

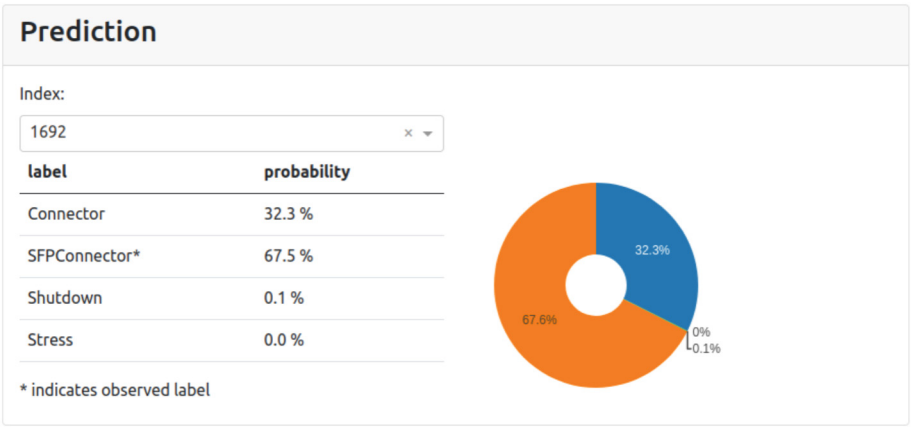

Fig. 9. Model's Individual Prediction.

averaged out probability out of all classes in the dataset and the blue bar on the right is the final contribution of all the features to the model's decision. However, even more interestingly, the most significant contributing features to the specific prediction are the events occurring within the last 30, 20 and 10 time-steps. This level of granularity in understanding the model's decision-making process enhances transparency and trust in AI systems, enabling stakeholders to gain deeper insights into the rationale behind specific predictions and improve the model interpretability.

The utilization of Explainable AI (XAI) techniques, such as the Explainer Dashboard and SHAP (SHapley Additive exPlanations), holds immense potential for enhancing the practical application of artificial intelligence in optical networks. By providing transparent insights into the predictive behavior of machine learning models, stakeholders within the optical networking domain can make informed decisions with greater confidence and precision. The ability to interpret model decisions at both a global and local level empowers network operators and engineers to understand the underlying factors driving predictions, thereby facilitating proactive maintenance, optimization, and troubleshooting of optical network infrastructure.

Furthermore, the adoption of XAI methodologies can significantly improve the reliability and efficiency of fault management systems in optical networks. By leveraging advanced visualization techniques and interpretable feature analysis, network operators can identify potential fault patterns, anomalies, and degradation trends with greater accuracy and timeliness. This proactive approach enables preemptive measures to be taken to mitigate network disruptions, optimize resource allocation, and enhance overall network resilience. Ultimately, the integration of XAI into optical network management processes can lead to a more robust and adaptive infrastructure, capable of meeting the evolving demands of modern telecommunications while minimizing downtime and maximizing operational efficiency.

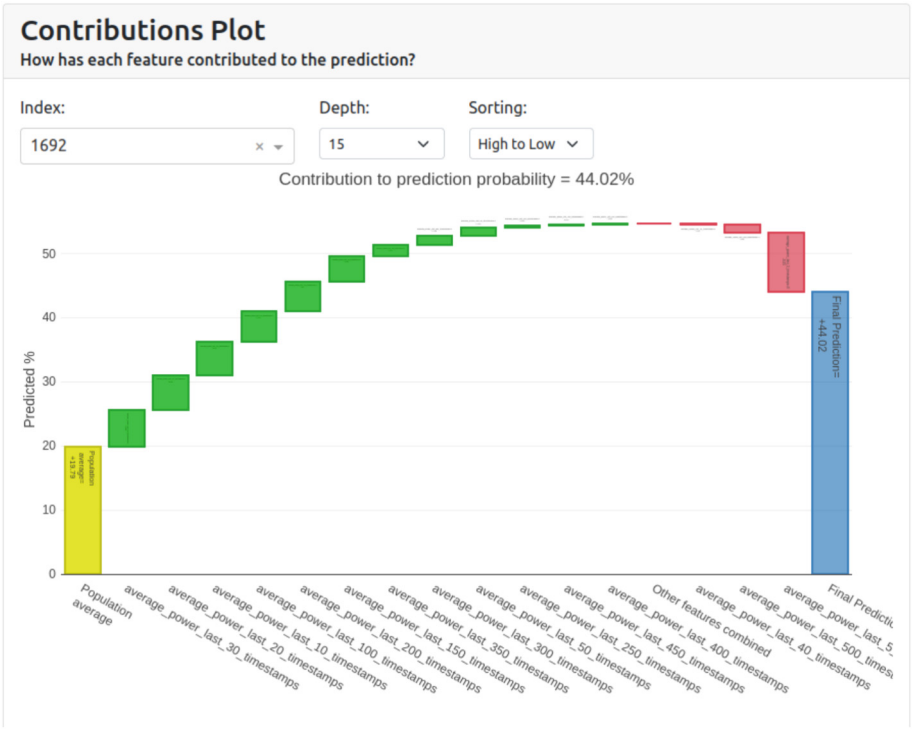

Fig. 10. Features Contributing to the Prediction. (Color figure online)

6 Conclusion

This paper presents an innovative approach to leveraging state-of-the-art explainable AI techniques for understanding the predictions generated by an ML model designed to detect faults in optical fiber links. We described a comprehensive comparison of four different traditional ML models and found that Light Gradient Boosting Machine (LGBM) outperforms the others. Furthermore, we explored the application of explainable AI in enhancing the model's interpretability, revealing that each type of fault is characterised by a distinct set of features that significantly contribute to explaining the occurrence of the fault. This research sheds light on the importance of transparency and interpretability of AI models deployed in critical infrastructures monitoring and fault detection systems related to optical fiber links.

In the future, we plan to extend the proposed Explanations Pipeline Architecture towards simulating how XAI outputs would change under different input conditions, aiding in better understanding the reasoning of ML models and deriving counterfactual explanations.

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Disclosure of Interest. The authors have no competing interests to declare that are relevant to the content of this article.

A Appendix A

Table 5. List of features used to represent the average optical received power over the different time windows

Feature	Description
average_power_last_5_timestamps	Last 5 timestamps prior to the LOS event
average_power_last_10_timestamps	Last 10 timestamps prior to the LOS event
average_power_last_20_timestamps	Last 20 timestamps prior to the LOS event
average_power_last_30_timestamps	Last 30 timestamps prior to the LOS event
average_power_last_40_timestamps	Last 40 timestamps prior to the LOS event
average_power_last_50_timestamps	Last 50 timestamps prior to the LOS event
average_power_last_100_timestamps	Last 100 timestamps prior to the LOS event
average_power_last_150_timestamps	Last 150 timestamps prior to the LOS event
average_power_last_200_timestamps	Last 200 timestamps prior to the LOS event
average_power_last_250_timestamps	Last 250 timestamps prior to the LOS event
average_power_last_300_timestamps	Last 300 timestamps prior to the LOS event
average_power_last_350_timestamps	Last 350 timestamps prior to the LOS event
average_power_last_400_timestamps	Last 400 timestamps prior to the LOS event
average_power_last_450_timestamps	Last 450 timestamps prior to the LOS event
average_power_last_500_timestamps	Last 500 timestamps prior to the LOS event

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Construction of Artificial Most Representative Trees by Minimizing Tree-Based Distance Measures

Björn-Hergen Laabs(✉), Lea L. Kronziel, Inke R. König, and Silke Szymczak

Institute of Medical Biometry and Statistics, University of Lübeck, Lübeck, Germany
b.laabs@uni-luebeck.de

Abstract. The random forest (RF) algorithm is known for its predictive performance but has been criticized for its lack of interpretability due to its complex ensemble nature. To address the issue of explainability our study questions the traditional approach of using most representative trees (MRTs) to simplify RF interpretation, highlighting the potential for misinterpretation due to non-informative early splits. To overcome these limitations, we propose a new method involving the construction of artificial representative trees (ARTs) through a greedy algorithm that iteratively builds a tree to minimize the distance to the RF ensemble, thereby preserving the predictive performance of the RF. We give a detailed description of the methodological framework for ART construction, including strategies for reducing computational complexity through variable preselection and quantile-based splitting. Results from extensive simulations demonstrate that ARTs provide a more accurate reflection of the RF's predictive performance and substantially reduce the false discovery rate, thus offering a more reliable interpretative model. The findings suggest that ARTs represent an advance in addressing the interpretation of RF models.

Keywords: Machine learning explainability · random forests · most representative trees

1 Introduction

While machine learning (ML) methods are now well established in the context of data science in general, their lack of explainability is still often criticized. For example, most ML algorithms outperform classical statistical approaches such as regression models with regard to the prediction performance, but they also tend to overfit and lack interpretability making them black-box methods. Random forest (RF) is one of these black box methods. They use bootstrapping to generate a large ensemble of decision trees based on a given training data set. While the interpretation of a single tree is quite intuitive, the interpretation of several hundreds of trees is not.

Variable importance measures (VIMs) [1–3] are a valuable tool for interpretation but only give an estimate which variables are important for the prediction, not to which

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extent they influence the outcome. To overcome this limitation, most representative trees (MRTs) have been proposed to summarize the structural information of all trees of the ensemble into a single tree [4]. Briefly, pairwise distances are calculated between all pairs of trees, and an average dissimilarity score is computed for each tree. Then, the tree with the lowest dissimilarity score, i.e., one that is most similar to all other trees, is selected as the MRT. Recent studies have shown that this method is able to summarize most of the information within an RF and consistently outperforms a separately trained decision tree. Furthermore, the choice of an appropriate distance metric is crucial for the performance of the MRT [5].

While MRTs show that complex ML models can in general be reduced to smaller representative models, their interpretation is still problematic. When training an RF, the hyperparameter `mtry` defines the number of randomly chosen independent variables presented to the algorithm at each node. As a result, the single trees in the forest are decorrelated, so that even two trees that are trained on the same bootstrap sample are different from each other avoiding overfitting [1]. As a consequence, not all splits made by a single decision tree within the forest have to be meaningful. As an example, consider a scenario in which only one of 100 independent variables in a training data set influences the studied outcome, and all other variables can be considered as noise. With an `mtry` of 10 (10% of the number of all independent variables), the probability of having only noise variables to choose from at a given node is about 91%. Thus, it is likely that even the first split in one of the decision trees is not meaningful. If such a tree is selected as MRT, the interpretation of this tree will be biased because one would assume that the first split is the most important although it is in fact meaningless. As another consequence, the single decision tree in the RF will include more irrelevant variables, because by `mtry` it is sometimes forced to select from unimportant variables although the overall prediction could be improved by adding other splits.

Here, we propose an alternative approach to generate MRTs from the RF by creating a new artificial tree that is not already included in the ensemble. By this, we aim to reduce the number of irrelevant variables in the representative tree and the overall size of the surrogate model while preserving the prediction performance of the forest. A greedy algorithm is used to iteratively build this tree by adding new possible nodes and re-estimating the distance to the forest. This procedure minimizes the dissimilarity score which measures the average distance to all other trees.

Our proposed method is similar to rule extraction methods like *Node Harvest* [6], *inTrees* [7], or *SIRUS* [8] which try to identify the relevant rules (path from root to leave in a tree) in an RF to generate smaller models while preserving the prediction performance of the ensemble. Here, we aim to build a new model that is as close as possible to the inner structure of the RF by minimizing the tree distances instead of directly optimizing the similarity to the predictions. It also follows the same idea as *CMM* [9] and *mimetic classifiers* [10] to train a new interpretable model based on the ensemble. But instead of generating new training data from the ensemble [9, 10], we aim to directly optimize the similarity to the ensemble.

By constructing an artificial most representative tree (ART) based on the optimization of a complex distance measure, we aim to solve the issue of `mtry` in MRTs and gain more insight into the common tree structures of RFs.

In the next section, we will first briefly summarize the basic ideas of RFs and MRTs. Afterwards, we will describe the construction of ARTs for an arbitrary distance measure and describe simulation study and empirical data benchmarking. The results of the simulated and benchmarking data are shown in Sect. 3. Finally, we discuss our results and the limitations and implications of our approach in Sect. 4.

2 Methods

2.1 Random Forests

RFs [1] are a method of supervised ML that combines classification and regression trees (CART) [11] with the concept of bagging [12]. Given an initial data set D the RF algorithm first uses bootstrapping or subsampling to generate multiple training data sets D_b followed by the generation of a single decision tree T_b on each of the training data sets D_b. Each decision tree starts with the complete training data set D_b at the root node and iteratively searches for the best binary split that optimally reduces impurity in the two child nodes. Usually, impurity is measured by the Gini index for classification and the variance for regression. Unlike CART, the RF algorithm restricts the search for an optimal split to a subset of the predictor variables of size `mtry` which is randomly selected for each split. In addition to the bootstrapped training data, this is the second level of randomness in order to decorrelate the individual trees in a forest, thus avoiding overfitting. Tree growth is stopped when one of the following three stopping criteria is met: 1. All terminal nodes are pure, 2. All observations per terminal node have identical values in their predictor variables, and 3. All terminal nodes are smaller than a pre-specified threshold (minimal node size). To predict the class (or quantitative value for regression) of a new observation, this is dropped down all trees, and the majority vote or the mean over all trees for regression is calculated.

2.2 Variable Importance Measures (VIMs)

Consisting of several hundred and up to several thousand trees, RFs are considered black box models. However, to gain insight into the mechanisms that lead to a particular prediction, several variable importance measures (VIMs) have been proposed [1–3]. The idea of VIMs is to summarize the contribution of each variable to the prediction of an RF into a single value. For example, the permutation VIM measures the change in prediction performance of an RF when the values of a single predictor variable are permuted. The permutation removes the association between predictor and outcome as well as all interactions between the permuted predictor and any other independent variable. For a more important variable, the decrease in prediction accuracy should be greater than for less important variables. The advantage of VIMs is that they are relatively easy to compute, but they only measure how a single variable affects the prediction of an RF. While some recent approaches [13, 14] try to additionally incorporate the underlying tree structures, they can only partially capture interactions between variables.

2.3 Selection of Most Representative Trees (MRTs)

A potential alternative to VIMs for interpreting RFs is to extract the MRTs from the ensemble [4, 5]. The goal of this method is to identify a surrogate model that optimally preserves the predictive performance and the generalized tree structures of the RF. Therefore, some arbitrary distance measure $d\left(T_i, T_j\right)$ is estimated for each pair of trees T_i, T_j Afterwards, the tree with the minimal mean distance to all other trees is selected as the MRT. The performance of these MRTs depends on the selection of a suitable distance measure. We previously introduced a new tree-based weighted splitting variables (WSV) distance measure [5]

$$d_{WSV}\left(T_i, T_j\right) = \sum_{v \in V} \left(U_i(v) - U_j(v)\right)^2, \tag{1}$$

where V is the set of all features and $U_i(v)$ a usage score measuring how much tree T_i relies on variable v. This usage score is defined by

$$U_i(v) = \frac{\sum_k \frac{1_{[s_i(k)=v]}}{2^{l(k)-1}}}{\max\limits_k(l(k))}, \tag{2}$$

where k indicates the nodes of T_i, $l(k)$ represents the level of the tree T_i in which k occurs and $1_{[s_i(k)=v]}$ indicates whether the splitting variable selected at node k is the variable of interest. We showed that this measure is a good and robust choice for selecting the MRTs. It is as good as other measures in preserving the prediction performance of the complete ensemble and, unlike others, does not rely on an independent data set to calculate the distances.

While these MRTs, selected from the ensemble, have been shown to retain the predictive performance of the RFs, they should be interpreted with caution due to the nature of their construction. As described above, when training each decision tree in the ensemble, only a fraction of the predictor variables (`mtry`) is considered for splitting. Thus, it is possible that in the first split of a tree, no causal variable is presented to the algorithm, so that the tree randomly chooses a noise variable to start with. This is not a disadvantage in the context of RFs, as it will be canceled out by aggregating several hundred trees. However, it can lead to misleading interpretations, if such a tree is selected as MRT.

2.4 Construction of Artificial Most Representative Trees (ARTs)

Here we propose an algorithm for the construction of an ART that is no part of the ensemble to avoid the problem of non-meaningful splits in the first splits of an MRT. A greedy algorithm (Fig. 1, Algorithm 1) is implemented for the complex optimization process of finding a new tree that minimizes a given tree-based distance metric. First, a tree consisting only of the root node is initialized. Then, all possible binary splits are built (resulting in a stump) and the pairwise distances to all trees of the RF are calculated. Now, the tree with the minimal mean distance to all trees of the RF is selected and all other stumps are discarded (first iteration in Fig. 1). The same procedure is then repeated for the next levels of child nodes. Here, all possible next extensions at this level

are considered, i.e., at the second level, splitting right or left child node or both (second iteration in Fig. 1). Tree growth is continued while at least one possible ART in iteration *i* is closer or equally close but with a better prediction accuracy to the RF than the ART from iteration *i-1*. If in any iteration two or more possible splits lead to the same smallest mean distance to the RF the one is chosen that has the highest prediction performance on an independent test data set.

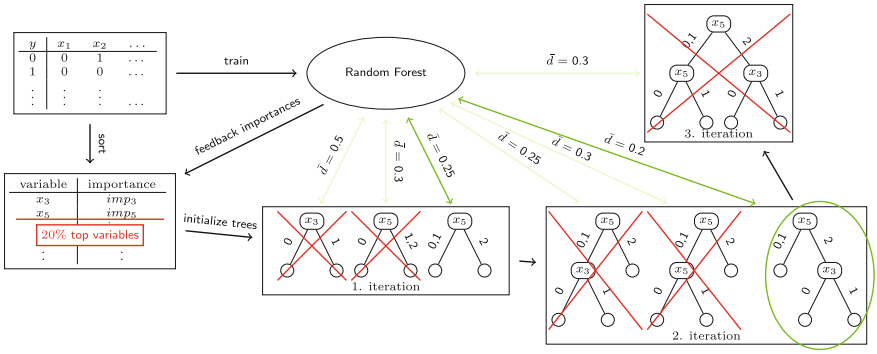

Fig. 1. Example for the construction of an artificial most representative tree (ART). Based on a given data set an RF is trained and variable importances are estimated. For the construction of the ART, only the most important variables are selected, and all possible trees with only one split are built (1. Iteration). Afterwards, the mean distance of all newly constructed trees to the original RF is calculated. The third tree splitting on x_5 is selected because it minimizes the distance, and all other trees are discarded. In the next step (2. Iteration) all possible next splits are considered, and again the tree that minimizes the distance to the RF is kept, while all other trees are discarded. In this example, the algorithm ends after the second iteration because all trees in the third iteration have a higher mean distance to the RF.

Especially for high-dimensional data sets the search for the optimal next split will be time-consuming. Therefore, we have added two further constraints that limit the number of split points that are tested at each step of the tree-generation process. Note that each split point refers to a combination of predictor variable and threshold value used in at least one split in the original RF. First, we consider only a subset of the predictor variables. All variables are ranked based on a VIM, and only the *imp.num.var* most important variables are used for the construction of the ART. Second, we restrict the number of splits for an important variable. This is especially important for continuous variables for which a large number of splits are possible. Otherwise, these could lead to a substantial selection bias favoring continuous over categorical variables, especially with a low number of categories [2]. For each important variable, the available splits in the original RF are interpreted as a continuous variable and only specific quantiles *probs_quantiles* (e.g., 10%, 25%, or 50% quantiles) of the splits are considered in the tree generation. Of note, based on the design of the method as a greedy algorithm and the fact that the tree-based distance measure is not necessarily monotonically decreasing with the depth of the ART it cannot be guaranteed that the algorithm will find the optimal ART but only the first local optimum. However, this reduces computation time and model complexity.

Alg 1. Pseudo code for the construction of an artificial representative tree (ART):

```
Require: RF, similarity metric metric, dataset with training data train_data, number of preselected variables based on importance values imp.num.var, quantiles of split points to use for variables with many split points probs_quantiles

Function    generate_tree(rf, metric, train_data,
            imp.num.var, probs_quantiles)

  Extract   all split_points from rf
  Reduce    split_points based on imp.num.var variables
            and probs_quantiles
  Build     all possible stumps using split_points
  Estimate  similarities of all stumps to rf using metric
  Select    all stumps with maximum similarity
  Estimate  accuracies of all remaining stumps on
            train_data
  Select    stump with best accuracy → ART
  Remove    all other stumps

  Repeat:
    Build     all possible trees with one additional split
              using split_points
    Estimate  similarities of all new trees to rf using
              metric
    Select    all new trees with maximum similarity
    Estimate  accuracies of all remaining new trees on
              train_data
    Select    tree with best accuracy → ART_candidate
    Remove    all other trees from this iteration

    If similarity(ART_candidate) < similarity(ART):
      Break
    Else if   similarity(ART_candidate) == similarity(ART)
              & accuracy(ART_candidate) <= accuracy(ART)
      Break
    Else
        ART <- ART_candidate

    Return ART
```

2.5 Simulation Design

We evaluated our newly developed method in an extensive simulation study on a high-performance cluster using R version 4.3.0. The code for reproducing our simulations can

be found on GitHub (https://github.com/imbs-hl/ART_paper). The following description of the simulation study is based on the recommended ADEMP scheme [15].

Aim: The simulation study is conducted to compare MRTs and ARTs in terms of their prediction performance and ability to detect relevant predictor variables.

Data Generating Mechanism: The 5 different settings from [5] are simulated. Specifically, in each setting, $p = 100$ variables are generated for training and validation data each with 1000 observations and a continuous outcome. Except for settings 3 and 5, the independent variables x_j are all drawn independently from a Bernoulli distribution $B(0.5)$ with $j = 1,\ldots,p$.

1. *Scenario (few large effects):* The outcome is simulated based on 5 effect variables with large effect sizes of $\beta = 2$ using:

$$y = x_1 \cdot \beta + \ldots + x_5 \cdot \beta + N(0, 1), \tag{3}$$

where $N(0,1)$ denotes the standard normal distribution.

2. *Scenario (many small effects):* The outcome is simulated based on 50 effect variables with small effect sizes of $\beta = 0.2$ using:

$$y = x_1 \cdot \beta + \ldots + x_{50} \cdot \beta + N(0, 1) \tag{4}$$

3. *Scenario (correlated variables):* The first 10 variables are drawn independently from $B(0.5)$ and are directly used to simulate the outcome. Five of them are used to create correlated variables. For each of the 5 variables, 4 correlated variables are generated without a direct influence on the outcome. The function `rmvbin` from the R package `bindata` (version 4.1.19) is used with a correlation of 0.3 between the effect variable and each of the 4 correlated variables. The remaining 70 variables are drawn independently from $B(0.5)$.

$$y = x_1 \cdot \beta + \ldots + x_{10} \cdot \beta + N(0, 1) \tag{5}$$

4. *Scenario (interaction effects):* The outcome is simulated by 5 variables with an effect size of $\beta = 2$ in combination with pairwise interactions of 10 other variables with the same effect size:

$$y = x_1 \cdot \beta + \ldots + x_5 \cdot \beta + x_6 \cdot x_7 \cdot \beta + \ldots + x_{14} \cdot x_{15} \cdot \beta + N(0, 1) \tag{6}$$

5. *Scenario (binary and continuous variables):* The outcome is simulated based on 5 Bernoulli distributed variables and 5 continuous variables from a standard normal distribution with effect sizes of $\beta = 2$. The remaining variables are drawn independently from $B(0.5)$.

$$y = x_1 \cdot \beta + \ldots + x_{10} \cdot \beta + N(0, 1) \tag{7}$$

Estimands: The similarity of surrogate model predictions to the original forest predictions in the independent validation data is estimated and the variables included in the MRTs and ARTs are recorded.

Methods: MRTs and ARTs are used to construct interpretable trees from RFs.

Random Forests: RFs are trained using the `ranger` package (version 0.15.1) [16], with 500 trees, a minimal node size of 100, and an `mtry` of 10. Variable importances are calculated using the permutation feature importance [1].

Most Representative Trees: MRTs are created using the weighted splitting variable distance measure [5] and the `timbR` package (version 2.0.3, https://github.com/imbs-hl/timbR).

Artificial Most Representative Trees: The ARTs are also created using `timbR` with the weighted splitting variable distance measure [5]. Using the variable importance calculated for the RF, ARTs are constructed based on the 5, 10, 15, 20, 25, and 100 most important variables. Limitation of the number of splits for continuous variables was performed based on 50%, 25%, and 10% quantiles, respectively.

Performance Measure: All performance measures are estimated using 100 repetitions and the validation data set. All variables that are directly used to simulate the outcome are defined as effect variables and the remaining ones as noise variables.

Deviation from RF: To estimate the deviation of the predictions of the surrogate models from the RF predictions, the mean squared difference between the predictions of each created tree and the original forest predictions is calculated on the validation data set.

Included Variables: The relative frequencies indicate how many of the effect or noise variables are used at least once as split variables in a tree in relation to the total number of effect or noise variables.

False Discovery Rate: The false discovery rate (FDR) indicates how often a noise variable was used as a split variable in a tree in relation to the total number of split variables.

Runtime: For all simulations, the runtime is measured in minutes using a high-performance cluster and parallelization with the package `batchtools` (version 0.9.17) [17].

2.6 Benchmarking Experiment

To compare MRTs and ARTs beyond simulation studies on real data, we used the AutoML Benchmark Regression collection from OpenML (https://openml.org/search?type=benchmark&sort=tasks_included&study_type=task&id=269) consisting of 33 different data sets with a continuous outcome. We filtered each data set for outliers which we defined as observations with an outcome value that is more than five times the IQR from the median. Additionally, we omitted all observations with missing values and all constant variables. Afterwards, we trained RFs using default parameters in a 7-fold cross-validation. Data sets where the RFs showed an R^2 of less than 0.3 were excluded from further analyses resulting in a total of 20 data sets. Finally, we built MRT and ART for each RF and estimated the fraction of variance in random forest predictions that can

be explained by the respective surrogate model

$$R^2 = \frac{\sum_i \left(\hat{g}(x) - \hat{\bar{f}}(x)\right)^2}{\sum_i \left(\hat{f}(x) - \hat{\bar{f}}(x)\right)^2}, \tag{8}$$

where $\hat{g}(x)$ is the prediction of the surrogate model and $\hat{f}(x)$ the original forest prediction. Additionally, we extracted the number of splits per tree and estimated the Akaike information criterion (AIC)

$$AIC = 2k - 2ln\left(\hat{L}\right), \tag{9}$$

where k is the number of splits in each tree and $\hat{L}$ is the estimated loss of each tree, measured by its MSE on a test data set.

3 Results

3.1 Prediction Performance

The first aspect for evaluating representative trees as a surrogate model for RFs is the similarity of their predictions to the original forest predictions. Therefore, we compared the predictions of both types of surrogate models (ART and MRT) with the predictions of the original model (RF). The results for all five scenarios are depicted in Fig. 2.

For all scenarios the mean deviation from the forest predictions is lower for the ARTs, indicating that our newly developed method leads to predictions that are closer to the predictions of the complex model. While this effect is relatively small in the simpler scenarios 1 and 2 it gets larger for the more complex scenarios including correlated variables (scenario 3), interactions (scenario 4), and continuous variables (scenario 5).

3.2 Included Variables

Another aspect for the evaluation of the proposed surrogate model algorithm is that it should include all the relevant variables but none of the noise variables (Fig. 3). The first row of boxplots illustrates the proportion of covered effect variables, showing that ARTs consistently cover fewer effect variables compared to MRTs in all scenarios.

The second row of boxplots displays the proportion of covered noise variables. Here, ARTs generally include nearly no noise variables and thus remarkably less than MRTs.

The third row depicts the FDR and demonstrates that ARTs have a substantially lower FDR across all settings, with medians close to 0. This is in strong contrast to MRTs, which show median FDRs larger than 25% in all scenarios indicating that more than 25% of splits in each MRT use noise variables as splitting variables.

In summary, the results suggest that ARTs are superior in terms of excluding noise variables and minimizing the FDR while being more conservative in the inclusion of effect variables compared to MRTs. This could imply that the ART is more specific, but it may lack some sensitivity in identifying effect variables. However, its lower FDR suggests that when it does identify variables, they are more likely to be truly influential, thereby providing a more reliable model for interpretation.

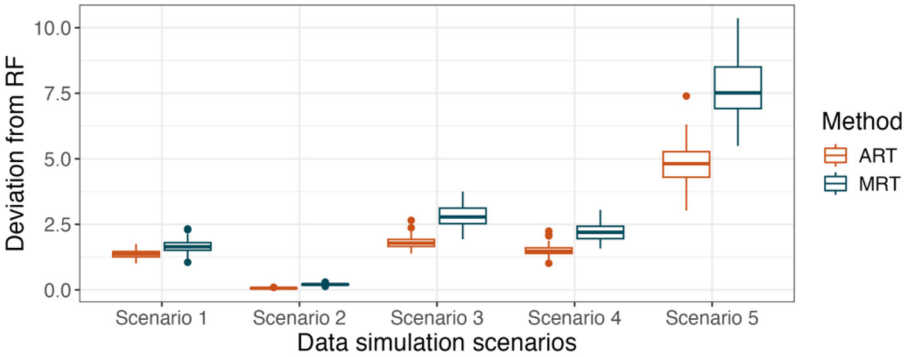

Fig. 2. Comparison of artificial (ART, orange) and selected most representative trees (MRT, blue) with regard to their deviation in predictions from the original forest predictions for all five simulation scenarios in 100 repetitions. ARTs were generated using the five most important variables and 25% quantiles to limit the number of split points for continuous variables. Deviation was measured by the mean squared difference of the predictions made by the surrogate model and the original RF on an independent validation data set. (Color figure online)

3.3 Computation Time

With regard to computation time the traditional selection of an existing tree from the ensemble outperforms ARTs. While MRTs have a constant computation time of about two seconds across all scenarios, ARTs complete in under three minutes for the first four scenarios. Only in the last scenario, they take ten times to compute. This is caused by the continuous variables that were simplified by using 25% quantiles as splitting variables. These still result in three times more possible splits than the binary variables. Computation time could be reduced further by using only one possible split point (Table 1).

3.4 Influence of Tuning Parameters

In our simulation study, we analyzed the influence of both possible tuning parameters (variable preselection and split point limitations for continuous variables). Increasing the number of preselected variables based on variable importance measures had no large effect on the deviation of the original random forest prediction (Suppl. Fig. 1), increases relative frequencies of selected noise variables (Suppl. Fig. 3), FDR (Suppl. Fig. 4), and runtime (Suppl. Fig. 5) while it decreases the relative frequency of selected effect variables if the threshold is larger than the number of simulated effect variables (Suppl. Fig. 2). Thus, the number of preselected variables should be reasonably small but close to the number of expected effect variables in the data set.

A smaller number of split points for continuous variables showed only a slight influence on all performance measures (Suppl. Figs. 6, 7, 8 and 9) but a substantial decrease in computation time (Suppl. Fig. 10).

3.5 Tree Depth

In general, we observed that the ART algorithm leads to smaller trees than MRTs (see Fig. 4 for a representative example). In the example from the first simulation scenario the ART is remarkably smaller (eight splits vs. 16 splits) and uses all of the effect variables and none of the noise variables, while the classical MRT also uses all effect variables but additionally eight noise variables. On average, the ARTs in this scenario use 36.6% fewer splits. Overall, we observed that ARTs are smaller in scenarios 1 – 4, and only in some instances of scenario 5 the ARTs were larger than the MRTs (Suppl. Fig. 11). In

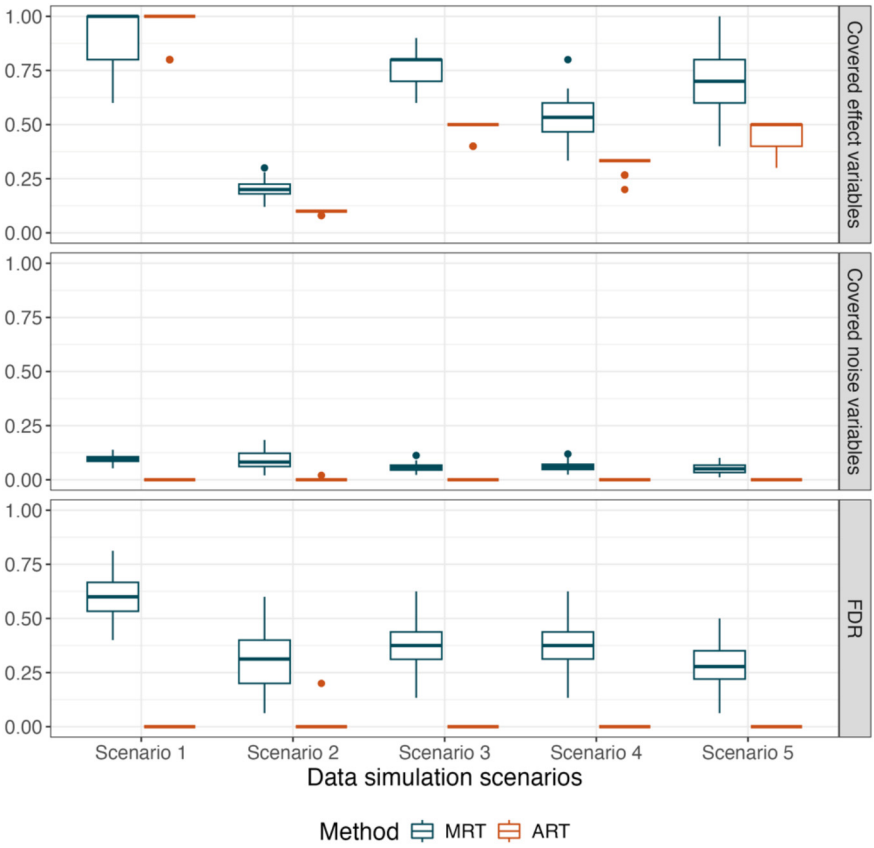

Fig. 3. Comparison of artificial (ART, orange) and selected most representative trees (MRT, blue) with regard to included variables in the surrogate model for all five simulation scenarios in 100 repetitions. ARTs were generated using the five most important variables and 25% quantiles to limit the number of split points for continuous variables. The fractions of covered effect and noise variables were calculated by the number of effect (noise) variables that occur at least once in the surrogate model divided by the number of all simulated effect (noise) variables. False discovery rate is represented by the number of splits in the surrogate model using one of the noise variables divided by the total number of splits in the surrogate model. (Color figure online)

Table 1. Comparison of computation time in minutes for artificial (ART) and selected most representative trees (MRT).

	Scenario 1	Scenario 2	Scenario 3	Scenario 4	Scenario 5
MRT	0.03	0.03	0.03	0.03	0.03
ART	2.71	2.36	2.28	2.56	31.38

general, we also observed that larger representative trees lead to a larger deviation from the original random forest.

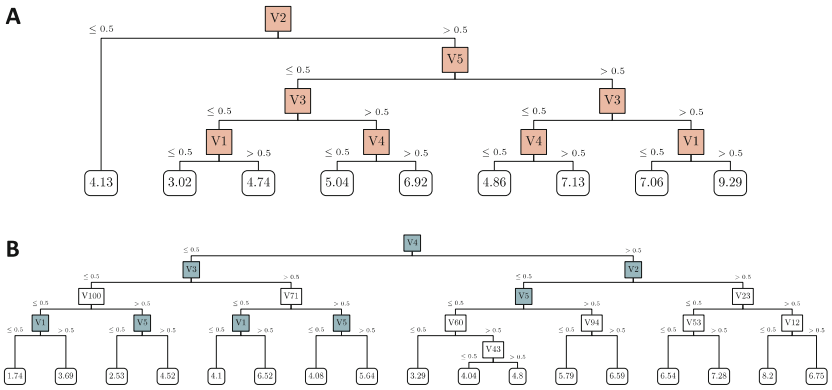

Fig. 4. Comparison of ART (**A**) and MRT (**B**) for one run of the first simulation scenario using simulated effect variables (blue and orange) and noise variables (white). (Color figure online)

3.6 Benchmarking

For the R^2 MRTs are in the majority of data sets (13 out of 20) better than ARTs (see Table 2). However, the ARTs are always substantially smaller than the MRTs. For example, the MRTs in the Mercedes example have a mean of 40 splits per tree and reach an R^2 of 0.876. In contrast, the corresponding ARTs on the same data set need only four splits to reach an R^2 of 0.830, meaning reducing the complexity by 90% reduces the similarity to the RF by only 4.6%. This is also reflected in the AIC combining prediction performance and model size showing an advantage of ARTs in all 20 data sets.

Table 2. Results for a benchmark experiment using 20 regression tasks and a 7-fold cross-validation comparing MRT and ART with regard to the variability of the random forest prediction that can be explained by the respective surrogate model (R^2), model size (mean number of splits), and prediction performance relative to model size (AIC). NA refers to an instance where at least one of the cross-validation folds did not finish in less than 180 h.

Datasets		R2		Mean number of splits		AIC	
OpenML Task ID	Name	MRT	ART	MRT	ART	MRT	ART
233212	Allstate_Claims_ Severity	0.596	**0.694**	27.4	**10.9**	24.3	**-8,9**
359938	Brazilian_houses	**0.899**	0.782	15.6	**7.3**	1.9	**-16,6**
233213	Buzzinsocial-media_Twitter	**0.933**	NA	**16.9**	NA	**17.3**	NA
233215	Mercedes_Benz_ Greener_Manufacturing	**0.876**	0.830	39.9	**3.9**	71.2	**-1.0**
167210	Moneyball	0.590	**0.678**	19.9	**8.7**	24.6	**2.3**
359948	SAT11_HAND_ runtime_regression	0.484	**0.589**	25.6	**8.3**	20.9	**-13,8**
359944	abalone	**0.791**	0.737	20	**13.3**	36.4	**22.9**
359937	black_friday	**0.746**	0.360	19.1	**12**	4.9	**-9,8**
359950	boston	0.610	**0.646**	22.3	**12.4**	37.6	**17.5**
233211	diamonds	**0.980**	0.854	18.6	**6.3**	8.3	**-17.8**
359936	elevators	0.752	**0.806**	20.1	**6.1**	62.0	**33.7**
359952	house_16H	**0.612**	0.414	22	**16**	3.1	**-11.1**
359949	house_sales	0.699	**0.705**	20.7	**11.9**	-7.2	**-25.2**
359943	nyc-taxi-green-dec-2016	**0.736**	0.536	24.7	**17.4**	47.9	**32.9**
359946	pol	**0.900**	0.889	34.6	**15.6**	58.0	**19.7**
359932	socmob	**0.865**	0.566	19.3	**11.4**	29.7	**13.4**
359933	space_ga	**0.773**	0.594	20.3	**7.3**	48.5	**22**
359934	tecator	**0.697**	0.583	23.6	**9.7**	40.1	**11.4**
359945	us_crime	0.338	**0.758**	20.6	**7.3**	47.4	**21.6**
359935	wine_quality	**0.584**	0.579	23.9	**12.4**	48.8	**25.8**

4 Discussion

In this work, we have shown that the previously proposed method for selecting MRTs from an RF can lead to misinterpretations of the resulting MRT when early splitting variables are uninformative. We solved this problem by developing an algorithm that iteratively builds a new artificial representative tree by minimizing the tree-based distance

to the original RF. In our extensive simulations, we showed that ARTs lead to similar results as MRTs with regard to the similarity of their prediction compared to the RF prediction, meaning that they are able to preserve the RF's predictive performance. Furthermore, ARTs can substantially reduce the false discovery rate, meaning that a variable that is included in such a representative tree is likely to be an effect variable.

The iterative generation of an ART by testing all possible splits in each step for finding the best approximation to the complete RF indeed leads to a substantial increase in computation time. Therefore, we introduced a preselection of the variables used in the generation step based on VIMs. This limitation to already known important variables substantially reduces computation time without drastically deteriorating the performance of the ART.

However, the number of preselected variables is a new hyperparameter that needs to be tuned when training ARTs. We observed that increasing this hyperparameter increases relative frequencies of selected noise variables, FDR, and runtime. However, it decreases prediction performance and the relative frequency of selected effect variables if the threshold is larger than the number of simulated effect variables. Thus, we recommend using reasonable small values close to the number of expected effect variables. To give a more precise recommendation more simulations (especially for high dimensional data settings) are needed.

While this algorithm is a promising step towards improving the explainability of RFs, there are several directions for further research. First, we utilized a relatively simple approach to preselect variables for the algorithm, using only the most important variables. More sophisticated variable selection methods such as Vita [18] or Boruta [19] are available which are able to directly classify variables as important and unimportant. Using these methods could further improve the performance and especially the probability to include all important variables in the ART. Second, the ART building algorithm itself only considers a single variable at a time. Comparing not only single splits but whole subtrees would improve the ability of ARTs to also include interaction effects, but again will increase the computation time. Third, RFs are by design limited to binary splits. While this is a reasonable limitation for RFs, ARTs might benefit from using other forms of splits. Fourth, we only compared ARTs with the previously proposed MRTs and not with other methods like *Node Harvest* [6], *inTrees* [7], *SIRUS* [8], *CMM* [9], or *mimetic classifiers* [10] which have similar objectives. A detailed comparison study including these methods and more diverse complexity metrics for decision trees is needed. Finally, the whole concept of projecting an RF onto a single tree might be flawed in some cases, where the trees in the RF are too diverse to be summarized in a single ART.

In conclusion, we have shown that ARTs are a good tool for the interpretation of RFs and lead to substantially smaller trees including fewer noise variables than the previously proposed MRTs while preserving the prediction performance. This new method will help to further improve the applicability of RF prediction modeling in practice.

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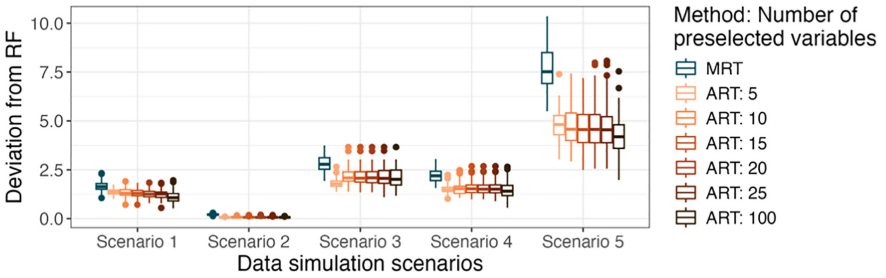
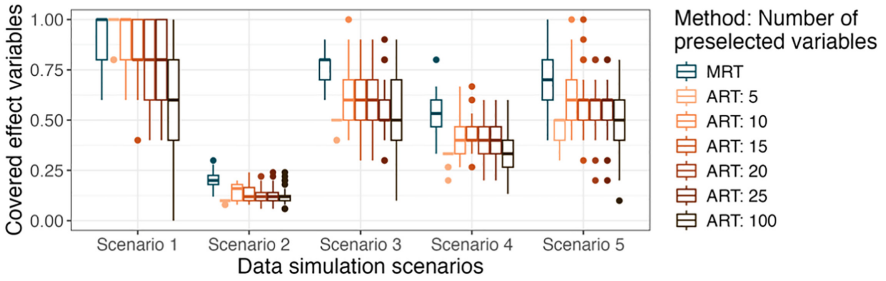
Disclosure of Interests. All authors declare that they have no conflict of interest.

Data Availability. All methods developed in this manuscript are publicly available within the R package `timbR` on GitHub (https://github.com/imbs-hl/timbR). Additionally, precalculated results of the simulations can be accessed in a shiny app (https://bioweb2.imbs.uksh.de/rsconnect/ART_paper/) and code for reproducing the simulation study and regenerating figures is publicly available on GitHub (https://github.com/imbs-hl/ART_paper).

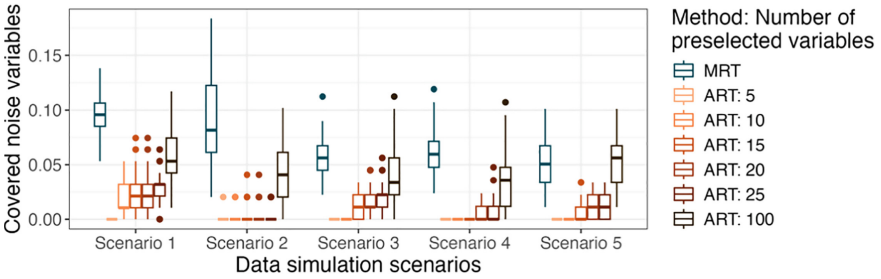
Appendix



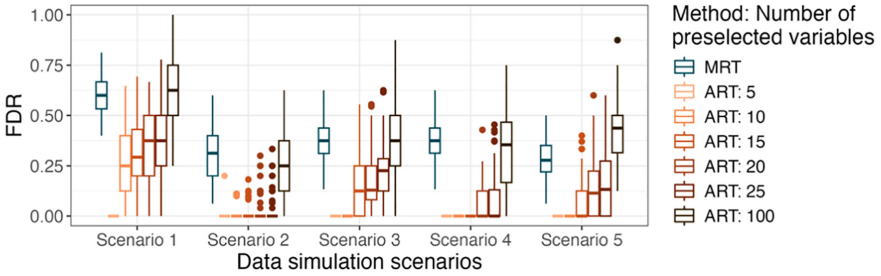
Suppl. Fig. 1. Comparison of artificial (ART, orange) and selected most representative trees (MRT, blue) with regard to their deviation in predictions from the original forest predictions for all five simulation scenarios in 100 repetitions. ARTs were generated using the five (10, 15, 20, 25, and 100) most important variables and 25% quantiles to limit the number of split points for continuous variables. Deviation was measured by the mean squared difference of the predictions made by the surrogate model and the original RF on an independent validation data set. (Color figure online)



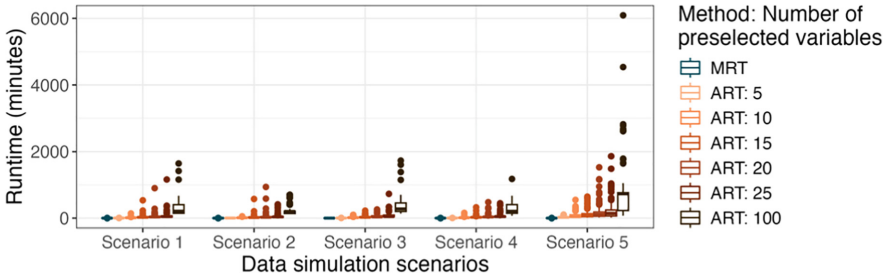
Suppl. Fig. 2. Comparison of artificial (ART, orange) and selected most representative trees (MRT, blue) with regard to included variables in the surrogate model for all five simulation scenarios in 100 repetitions. ARTs were generated using the five (10, 15, 20, 25, and 100) most important variables and 25% quantiles to limit the number of split points for continuous variables. The fractions of covered effect variables were calculated by the number of effect variables that occur at least once in the surrogate model divided by the number of all simulated effect variables. (Color figure online)



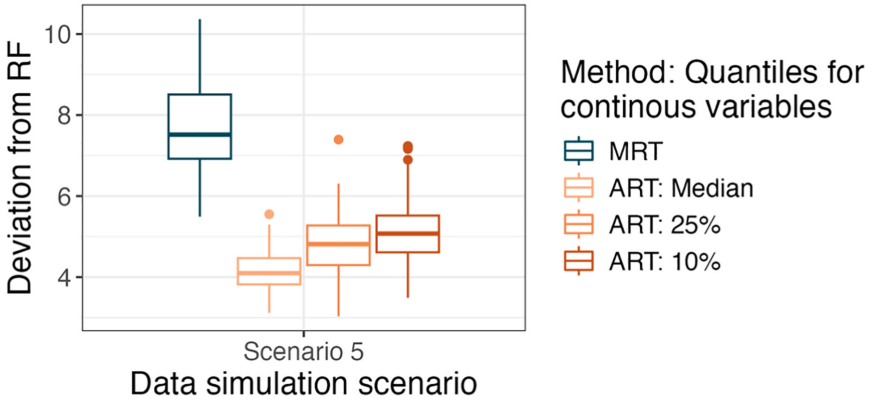
Suppl. Fig. 3. Comparison of artificial (ART, orange) and selected most representative trees (MRT, blue) with regard to included variables in the surrogate model for all five simulation scenarios in 100 repetitions. ARTs were generated using the five (10, 15, 20, 25, and 100) most important variables and 25% quantiles to limit the number of split points for continuous variables. The fractions of covered noise variables were calculated by the number of noise variables that occur at least once in the surrogate model divided by the number of all simulated noise variables. (Color figure online)



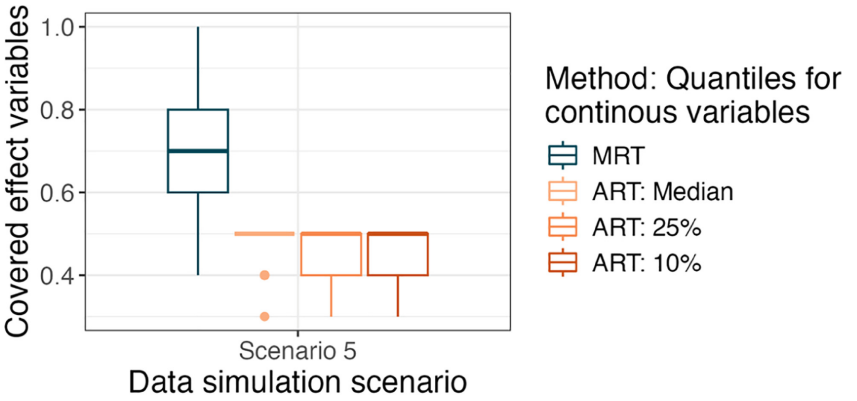
Suppl. Fig. 4. Comparison of artificial (ART, orange) and selected most representative trees (MRT, blue) with regard to included variables in the surrogate model for all five simulation scenarios in 100 repetitions. ARTs were generated using the five (10, 15, 20, 25, and 100) most important variables and 25% quantiles to limit the number of split points for continuous variables. False discovery rate is represented by the number of splits in the surrogate model using one of the noise variables divided by the total number of splits in the surrogate model. (Color figure online)



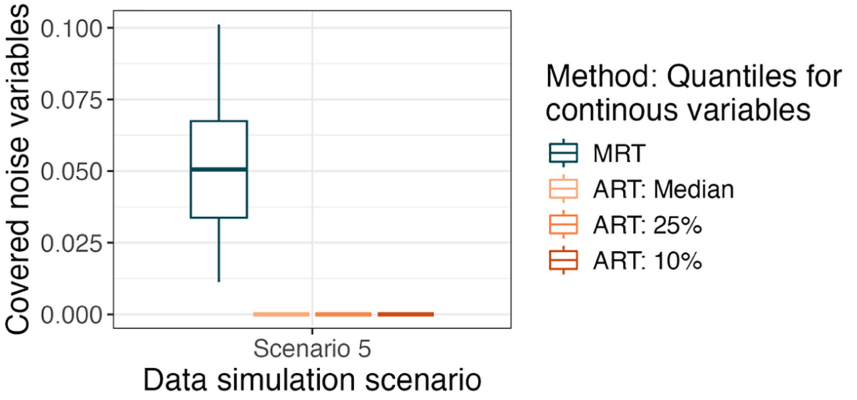
Suppl. Fig. 5. Comparison of artificial (ART, orange) and selected most representative trees (MRT, blue) with regard to runtime for all five simulation scenarios in 100 repetitions. ARTs were generated using the five (10, 15, 20, 25, and 100) most important variables and 25% quantiles to limit the number of split points for continuous variables. (Color figure online)



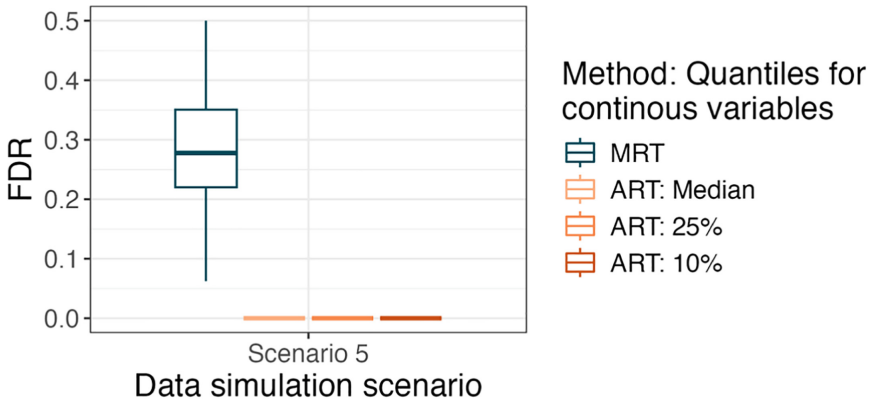
Suppl. Fig. 6. Comparison of artificial (ART, orange) and selected most representative trees (MRT, blue) with regard to their deviation in predictions from the original forest predictions for all five simulation scenarios in 100 repetitions. ARTs were generated using the five most important variables and 50%, 25%, and 10% quantiles to limit the number of split points for continuous variables. Deviation was measured by the mean squared difference of the predictions made by the surrogate model and the original RF on an independent validation data set. (Color figure online)



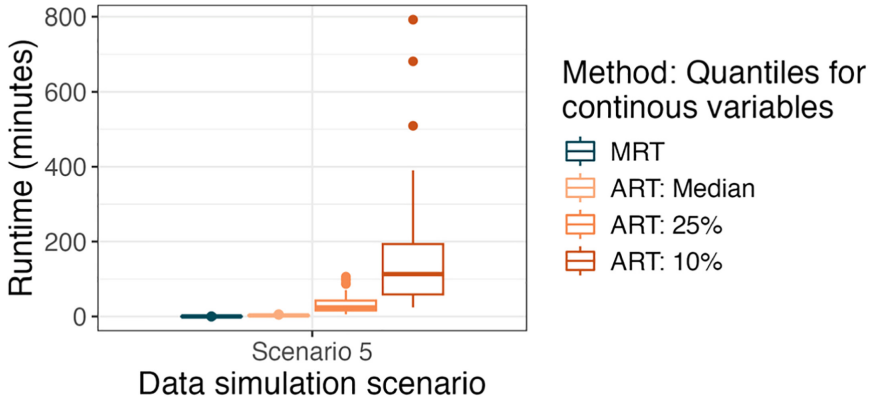
Suppl. Fig. 7. Comparison of artificial (ART, orange) and selected most representative trees (MRT, blue) with regard to included variables in the surrogate model for all five simulation scenarios in 100 repetitions. ARTs were generated using the five most important variables and 50%, 25%, and 10% quantiles to limit the number of split points for continuous variables. The fractions of covered effect variables were calculated by the number of effect variables that occur at least once in the surrogate model divided by the number of all simulated effect variables. (Color figure online)



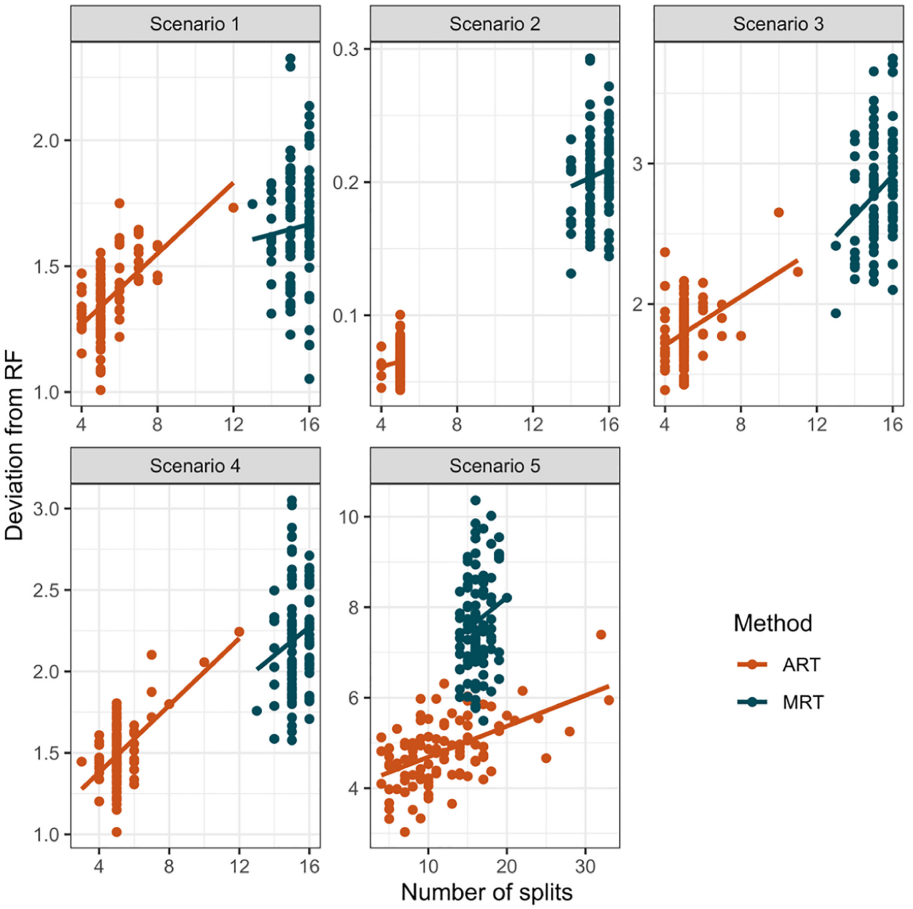
Suppl. Fig. 8. Comparison of artificial (ART, orange) and selected most representative trees (MRT, blue) with regard to included variables in the surrogate model for all five simulation scenarios in 100 repetitions. ARTs were generated using the five most important variables and 50%, 25%, and 10% quantiles to limit the number of split points for continuous variables. The fractions of covered noise variables were calculated by the number of noise variables that occur at least once in the surrogate model divided by the number of all simulated noise variables. (Color figure online)



Suppl. Fig. 9. Comparison of artificial (ART, orange) and selected most representative trees (MRT, blue) with regard to included variables in the surrogate model for all five simulation scenarios in 100 repetitions. ARTs were generated using the five most important variables and 50%, 25%, and 10% quantiles to limit the number of split points for continuous variables. False discovery rate is represented by the number of splits in the surrogate model using one of the noise variables divided by the total number of splits in the surrogate model. (Color figure online)



Suppl. Fig. 10. Comparison of artificial (ART, orange) and selected most representative trees (MRT, blue) with regard to runtime for all five simulation scenarios in 100 repetitions. ARTs were generated using the five most important variables and 50%, 25%, and 10% quantiles to limit the number of split points for continuous variables. (Color figure online)



Suppl. Fig. 11. Relationship of number of splits (simplicity) for artificial (ART, orange) and selected most representative trees (MRT, blue) with deviation of the surrogate model predictions from the original random forest predictions. (Color figure online)

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Decision Predicate Graphs: Enhancing Interpretability in Tree Ensembles

Leonardo Arrighi[1(✉)], Luca Pennella[2(✉)], Gabriel Marques Tavares[3,4], and Sylvio Barbon Junior[5]

[1] Department of Mathematics and Geosciences, University of Trieste, Trieste, Italy
leonardo.arrighi@phd.units.it
[2] Department of Economics, Business, Mathematics and Statistics, University of Trieste, Trieste, Italy
luca.pennella@phd.units.it
[3] LMU Munich, Munich, Germany
tavares@dbs.ifi.lmu.de
[4] Munich Center for Machine Learning (MCML), Munich, Germany
[5] Department of Engineering and Architecture, University of Trieste, Trieste, Italy
sylvio.barbonjunior@units.it

Abstract. Understanding the decisions of tree-based ensembles and their relationships is pivotal for machine learning model interpretation. Recent attempts to mitigate the human-in-the-loop interpretation challenge have explored the extraction of the decision structure underlying the model taking advantage of graph simplification and path emphasis. However, while these efforts enhance the visualisation experience, they may either result in a visually complex representation or compromise the interpretability of the original ensemble model. In addressing this challenge, especially in complex scenarios, we introduce the Decision Predicate Graph (DPG) as a model-specific tool to provide a global interpretation of the model. DPG is a graph structure that captures the tree-based ensemble model and learned dataset details, preserving the relations among features, logical decisions, and predictions towards emphasising insightful points. Leveraging well-known graph theory concepts, such as the notions of centrality and community, DPG offers additional quantitative insights into the model, complementing visualisation techniques, expanding the problem space descriptions, and offering diverse possibilities for extensions. Empirical experiments demonstrate the potential of DPG in addressing traditional benchmarks and complex classification scenarios.

Keywords: Ensemble Learning · Explainable Artificial Intelligence · Interpretability · Explainability · Tree-based Ensemble Method · Graph · Random Forest

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1 Introduction

Artificial intelligence, although still under strong development, is now a consolidated and widely used tool. This is thanks to the continuous growth of computing power, which allows the use of increasingly complex and computationally expensive machine learning (ML) methods. The challenges presented by modern-world problems are growing in complexity as well as the proposed solutions.

Dealing with large quantities of data and frequently encountering unbalanced datasets are still significant obstacles in addressing many real-world issues; however, tree-based ensemble algorithms offer several advantages in overcoming these challenges, including robustness to noise and outliers, scalability to large datasets, automatic handling of missing values, and the ability to capture complex relationships and interactions within the data [24,26].

Without delving into the intricacies of various algorithms, exhaustively described by Hastie et al. [19], the process of learning tree-based ensemble models involves training multiple decision trees to combine their predictions, optimising performance, and enhancing generalisation capabilities. While these models indeed offer concrete solutions to a diverse array of problems, developers and users are confronted with new challenges. The training outcome yields an exceptionally complex model, commonly referred to as an opaque-box model (also known as the black-box problem), whose internal workings are not transparent or easily interpretable [5]. In response to this context, the Decision Predicate Graph (DPG) is proposed in this paper.

Drawing inspiration from the expanding theme of eXplainable Artificial Intelligence (XAI), we designed a graph structure to tackle transparency and explainability challenges inherent in tree-based ensemble models. This facilitates a better understanding of the intricate choices underlying these ML models. DPG is created with inspiration from the concept of aggregating random forests (RF) [4,21], as introduced by Gossen and Steffen [13]. The approach proposed in [13] suggests visualising the decisions within the RF by combining the branches of the tree base learners into a single and compact decision diagram. The concept behind DPG is to convert a generic tree-based ensemble model for classification into a graph, a defined and studied structure with known properties. In this graph, nodes represent predicates, i.e., the feature-value associations present in each node of every tree, while edges denote the frequency with which these predicates are satisfied during the model training phase by the samples of the dataset. The DPG structure enables comprehending the choices made by the model, enhancing transparency and understandability. Moreover, it allows the exploitation of graph properties to develop metrics and algorithms facilitating the analysis of the ensemble model. This, in turn, aids in understanding the decisions it makes, easing the task of visualising the graph which can be vast and complicated for larger models with numerous tree base learners.

DPG serves as a model-specific tool offering a comprehensive interpretation of tree-based ensemble models. It provides descriptive metrics that enhance the understanding of the decisions inherent in the model, offering valuable insights.

This tool proves particularly useful for models that are *a priori* considered satisfactory in terms of performance.

Our work contributes in the following ways:

- we introduce DPG, a novel interpretability structure that transforms an opaque-box tree-based ensemble model into an enriched graph;
- we present the algorithm used to create DPG, accompanied by pseudo-code to enhance understanding and facilitate replication, complete with its asymptotic complexity;
- we provide the interpretation of three metrics from graph theory, enriching the model comprehension and gaining insights;
- we demonstrate the use of the proposed method through two case studies: the application of DPG to two RF models, respectively, on the Iris dataset [10] and a challenging dataset.

It is important to highlight that these results are achieved in a generic fashion, utilising a standard classifier on well-established datasets. Significantly, we intentionally avoided incorporating scenario-specific heuristics. Therefore, we posit that our aggregation approach has the potential for widespread application across a diverse range of related scenarios.

2 Literature Review

Complex yet effective models have become increasingly prevalent, especially in fields where the outcomes bear significant importance and require a heightened level of sensitivity. In these scenarios, there is a growing demand for models to exhibit transparency, accountability, and a comprehensive understanding. Consequently, the discussion on XAI has experienced significant growth, leading to an extensive body of literature [1,14,17].

In XAI, as suggested by Dwivedi et al. [9], classifications are often based on scope, distinguishing between global interpretability, which reveals overall data trends and provides insights into the entire model, and local interpretability, which elucidates the reasoning behind specific predictions for individual instances. Another classification criterion involves how interpretability is achieved: intrinsic interpretability relies on straightforward model structures (e.g., concise decision trees or linear models), while post hoc interpretability involves methods applied after model training [1]. Interpretation methods are further categorised as model-specific or model-agnostic. Model-agnostic tools are versatile, capable of being employed with any ML model, and are applied post-training (e.g., SHAP [23], LIME [35], PDP [12], and Anchors [36]). In contrast, model-specific interpretation tools are tailored to specific model classes, such as our proposal centred around tree-based ensemble models.

In this context, XAI tools, especially those providing global interpretations, become valuable instruments for understanding tree-based ensemble models. These models are widely used in addressing diverse problem domains, as highlighted in several surveys [2,6,16]. As a result, the number of studies delving into model-specific techniques tailored for ensembles of trees has also increased.

The first significant study is proposed by Mashayekhi and Gras [25]. The authors introduced *RF+HC*, an approach that employs a hill climbing algorithm in RF to search for a decision set. This rule set reduces the number of decisions dramatically, which significantly improves the comprehensibility of the underlying model built by RF. Similarly, Hara and Hayashi [18] exploit Bayesian model selection to extract the decision set. These approaches share similarities with our method; however, our proposal extends beyond the extraction of decisions from the RF. Visualising the decisions of tree-based ensemble models and simultaneously complementing them with metrics developed by graph theory makes DPG more adaptable and holistic. This approach allows for obtaining insights into the model beyond just decision information.

Zhao et al. [41] proposed *iForest*, a visual analysis system specifically designed to interpret RF models and their predictions. They built a feature view to illustrate the relationships between input features and outcome predictions and proposed a design that summarises multiple decision paths based on feature occurrences and ranges, allowing users to explore and understand the partitioning logic of these paths. The iForest, like numerous visualisation systems, faces significant challenges related to scalability and interpretability when dealing with large ensembles. To overcome this challenge, our approach does not prioritise visualisation but instead strives for a comprehensive understanding of the ensemble's logic.

Hatwell et al. [20] contributed with *Collection of High Importance Random Path Snippets* (CHIRPS), a method that incorporates the explanation of RF classification for each data instance and extracts a decision path from each tree in the forest, resulting in a set of decisions that elucidate the classification process. However, this method is limited to rules extraction and lacks insight into the model's structure. Additionally, it does not incorporate metrics to explain the logic of the tree-based ensemble model.

Another technique focused on visualising decisions underlying RF models is introduced by Neto and Paulovich [31]. *Explainable Matrix* (ExMatrix) employs a matrix-like visual metaphor, where rows represent decisions, columns denote features, and cells encapsulate decision predicates, thereby facilitating the scrutiny of models and the audit of classification outcomes. The visualisation capability of ExMatrix for global visualisation has limitations in terms of scalability because the number of decisions increases significantly with the number of trees in large ensembles. Moreover, ExMatrix layouts can rapidly become challenging to explore, while the complexity of the model increases. As mentioned earlier, our approach enables us to retrieve information without relying solely on visualisation.

Dedja et al. [7] introduced an approach denoted as *Building Explanations through a LocalLy AccuraTe Rule EXtractor* (BELLATREX), which is designed to explain the forest predictions for a given test instance by a set of logical rules based on the features of the dataset. However, a potential limitation lies in the computational complexity of the approach. While explaining a single prediction is quick, applying the method to a complete dataset becomes computationally

expensive. Furthermore, BELLATREX focuses on instance-level explanations rather than providing a global perspective. In addition, BELLATREX uses clustering techniques to simplify the representation and decision logic, whereas our approach uses graphs to avoid simplifications that can lead to loss of information.

Various studies [8,15,39,42] proposed several tree similarity metrics through the process of clustering based on tree representations. However, these approaches, while beneficial for interpretability, require simplifying the model through techniques such as selection, pruning, and frequency analysis, which can result in information loss.

A number of works [13,28,29,37] established a connection between tree-based ensemble models and graph theory. The foundational concept of these techniques has been explored by [11,22,30,32,38,40,43]. These works established the theory that introduces the transformation of decision trees into graphs, aiming for more efficient and non-redundant tree structures. Nakahara et al. [29] and Silva et al. [37] works focus on performance optimisation, with Gossen and Steffen [13] and Murtovi et al. [28] being the sole contributors that employed these techniques for interpretability purposes.

In particular, Gossen and Steffen [13] introduced the *Algebraic Decision Diagram* (ADD), aiming to transform tree-based ensemble models into bipartite graphs. The ADD serves as an alternative construction to RF, providing an additional predictive model that functions as a surrogate. ADD proves valuable for specific aspects of interpretability, such as outcome explanation, logic of *majority vote*, and visualising the path. Acting as a surrogate model, their primary contributions lie in the ability to provide a simple, optimised model. In comparison, our objective is to extend their ideas by fully leveraging graph theory. This goes beyond the transposition of tree-based ensemble models into a graph; rather, it involves using graph theory and its associated metrics to gain insights into the functionality of the models.

ForestGUMP, an online tool developed by Murtovi et al. [28], is designed for generating ADDs. This tool provides valuable information such as graph visualisation, hypothetical sample path display, and logic of majority vote. However, it has some limitations in terms of the number of usable trees (only 20) and in visualising problems that involve multiple features and classification choices, making graph navigation complex. In our work, while we enable visualisation, our focus is on utilising graph-related metrics, and we do not incorporate the simplification of the analysed models.

3 Decision Predicate Graphs

DPGs capture the details of a tree-based ensemble model and learned dataset specifics, emphasising predicate paths while preserving crucial decision points. Essentially, DPG converts complex ensemble models into a graph structure, where nodes represent predicates made by the model and edges denote the occurrence of these predicates during model training.

In this segment, we present the formalisation of DPG, elucidate the algorithm employed in its development through pseudo-code and its asymptotic complexity,

and provide a detailed exposition of various metrics and properties essential for comprehending the tree-based ensemble model. Moreover, we meticulously outline the advantages of metrics and articulate why they can serve as a valuable complementary aid to graph visualisation, particularly in overcoming its inherent limitations.

As previously mentioned in Sect. 1, DPG is tailored for tree-based ensemble models designed specifically for classification tasks.

3.1 Definition

Let $\mathcal{M}_n$ be the tree-based ensemble model consisting of n tree base learners $T(x; \Theta_b)$, where x is a generic sample, and Θ_b characterizes the bth learner in terms of split variables, cutpoints at each node, and terminal-node values. More specifically, Θ_b includes:

- all the splitting conditions associated with each jth internal node n_{bj} based on a specific feature f_{bj} and a threshold (for numerical features) or a set of possible values (for categorical features) t_{bj};
- the values assigned to leaf nodes c_b.

Let $\mathcal{D}$ be the training set on which $\mathcal{M}_n$ is trained. Every base learner is trained on a dataset $\mathcal{D}_b$, where $\mathcal{D}_b$ is a subset of $\mathcal{D}$. We define $\mathcal{O}$ as the set of logical operations: $\mathcal{O} = \{\leq, >, =, \neq\}$.

The predicate set $\mathcal{P}(\mathcal{M}_n)$, for an ensemble method $\mathcal{M}_n$ is the set obtained by the union of the set of all the triples $p = (f_{bj}, o, t_{bj})$, where $o \in \mathcal{O}$, and f_{bj}, $t_{bj} \in \Theta_b$, and the set of all leaf nodes c_b, for all n tree base learners of $\mathcal{M}_n$. The triples p are called decisions, while the elements of $\mathcal{P}(\mathcal{M}_n)$ are called predicates.

A Decision Predicate Graph (DPG($\mathcal{M}_n$)) for a model $\mathcal{M}_n$ is a directed weighted graph $(\mathcal{P}, E)$ where:

- $\mathcal{P}$ is the set of nodes, which corresponds to the predicate set $\mathcal{P}(\mathcal{M}_n)$;
- E is the set of edges, where each directed edge connects two predicates if and only if there exists an element in the training set $\mathcal{D}$ that satisfies both conditions specified by the predicates in an immediately consecutive manner. The weight of the edge corresponds to the number of training set elements satisfying these consecutive conditions.

For conciseness, from this point onward, we will use the acronym DPG to refer to the graph, indicating that it is constructed based on a model.

3.2 From Ensemble to a DPG

We introduce an algorithm, outlined in Algorithm 1, for constructing the DPG based on a tree-based ensemble model, by traversing all tree base learners with the training samples.

Algorithm 1: Construct DPG from Ensemble Tree Model

Input: Ensemble tree model $\mathcal{M}_n$, Training set $\mathcal{D}$
Output: DPG($\mathcal{M}_n$)

Initialise empty set DPG($\mathcal{M}_n$);
foreach T *(learner)* ***in*** $\mathcal{M}_n$ **do**
    Initialise empty predicate set $\mathcal{P}$ and edge set E;
    foreach x *(sample)* ***in*** $\mathcal{D}$ **do**
        Initialise empty predicate set $\mathcal{P}_x$ and edge set E_x;
        `// To obtain the predicates path for` x `on the tree` T
        ($\mathcal{P}_x$, E_x) ← `TRAVERSING(`T, x`);`
        Add ($\mathcal{P}_x$, E_x) to $\mathcal{P}$ and E;
DPG($\mathcal{M}_n$) ← `AGGREGATING(`$\mathcal{P}$, E`);`
return *DPG*($\mathcal{M}_n$);

The algorithm iterates over each base learner in the ensemble tree model $\mathcal{M}_n$ and each training sample x in the training set $\mathcal{D}$.

To clarify, the `TRAVERSING` function follows the predicate path of a particular input sample x through the decision tree T, starting from the root node and navigating to the appropriate leaf node based on the feature values of x. Meanwhile, the `AGGREGATING` function processes the predicates and edges obtained from `TRAVERSING` the decision trees for all samples into a single graph representation, DPG($\mathcal{M}_n$), by taking the union of $\mathcal{P}$ and computing the frequency of elements of E.

The algorithm presents a systematic methodology for constructing DPG. The overall asymptotic complexity can be formally expressed as follows:

$$O(b \times s \times (k + k^2)) = O(b \times s \times k^2)$$

where:

- b is the number of learners in the ensemble,
- s is the number of samples in the training set, $|\mathcal{D}|$, and
- k represents the size of the ($\mathcal{P}_x$, E_x) processed by the `TRAVERSING` and `AGGREGATING` functions.

This analysis takes into account the linear time complexity $O(k)$ for the `TRAVERSING` function and the quadratic time complexity $O(k^2)$ for the `AGGREGATING` function. Our *Python* 3.10 implementation is accessible here[1].

3.3 DPG Interpretability

In this section, we enumerate and elucidate some of the advantages that DPG can offer. The graph-based nature of DPG provides significant enhancements in the direction of a complete mapping of the ensemble structure. Weighted directed

[1] https://github.com/LeonardoArrighi/DPG.

graphs, such as DPGs, are studied structures with well-established properties that enable the identification or construction of useful metrics and algorithms. It is crucial to emphasise that all the observations presented in this section are valuable for comprehending and analysing the obtained model.

Visualisation. DPG provides an immediate advantage by allowing the visualisation of the entire tree-based ensemble models through a single comprehensive graph. Similar to the idea proposed by Gossen and Steffen [13], consolidating all individual basic learners within the model into a unified graph provides a holistic representation of the decision-making process. This visualisation not only elucidates the decisions made by the learners but also reveals the intricate relationships between them. Consequently, it facilitates a comprehensive understanding of the utilised features and, more importantly, the associations between features and their values that enable the model to accurately classify a sample into a specific class.

Another noteworthy aspect of DPG lies in the representation of weights for pair-wise nodes. This feature enables a discerning analysis of the most significant path through predicates, shedding light on decisions consistently employed by numerous learners or across multiple samples. This insight not only highlights the prevalence of certain decisions but also opens avenues for targeted enhancement strategies, focusing on those influential aspects within the model.

Moreover, by traversing all the possible paths between predicates in reverse, starting from one of the classes, we can discern the essential characteristics that a sample must possess to be classified into a specific class. This capability facilitates the *a priori* elimination of certain elements from the dataset when considering the particular class.

Nevertheless, we acknowledge that while visualisation is a valuable tool, its effectiveness diminishes with an increasing number of tree base learners. A multitude of tree base learners implies an increase in decisions and, consequently, an abundance of predicates. As a result, the size of the graph, in terms of nodes, grows proportionally with the model's scale. This expansion can render the graph illegible or impractical to visualise due to its intricate complexity. To address this challenge, we provide additional tools that complement the visualisation, aiding in the extraction of model properties and facilitating a more comprehensible understanding.

One approach is based on the desire and feasibility of determining the specific characteristics a sample must exhibit to be assigned to a particular class. Taking inspiration from the *outcome explanation problem* introduced by Gossen and Steffen [13], to enhance the immediacy and effectiveness of this analysis, we provide an aggregation of predicates, referred to as *constraint*, which represent intervals associated with the features of each class. The constraints are defined as follows: for a given class identified in the DPG, we list all nodes connected by a path originating from the node itself and culminating in the class. For each feature within the node predicates, we delineate the most extensive possible interval using the values associated with the features. This interval is defined

by two endpoints. The minor endpoint is the smallest value within the set of values less than the feature, while the major endpoint is the largest value within the set of values greater than the feature. If either of these two sets is empty, the interval is deemed infinite. Each class has its constraints for every feature contributing to the classification of the samples.

It is important to note that constraints are not a substitute for visualisation; instead, they offer insights resembling those promoted by it. It is anticipated that additional graph measures could complement insights from constraints, providing similar interpretations as visualisation.

Centrality. The centrality of a node is defined as a number or rank corresponding to the node position within the network. By observing centrality, we can make considerations that allow us to better understand the process hidden in the ensemble method. The notion of centrality encompasses a wide range of metrics. In this section, we explore those metrics that offer the most insightful information.

According to Brandes [3], we define the *betweenness centrality* (BC) of a node as the fraction of all the shortest paths between every pair of nodes of the graph passing through the considered node. Let DPG $= (\mathcal{P}, E)$ be the graph and $s, t, v \in \mathcal{P}$ three vertexes of DPG, we can denote with $\sigma(s, t)$ the number of shortest paths between s and t and with $\sigma(s, t|v)$ the number of shortest paths between s and t passing through v. Then, the BC of the node $v \in \mathcal{P}$ is defined as:

$$BC(v) = \sum_{s,t \in \mathcal{P}} \frac{\sigma(s, t|v)}{\sigma(s, t)}.$$

All details and observations about BC can be read in [3]. BC serves as a relevant metric for gaining a deeper insight into the significance of decisions within the ensemble model. We can observe that a node with a higher BC value has a more significant influence on the flow of information within the graph; nodes with high BC can be considered potential bottleneck nodes because they play a crucial role in facilitating interactions between different parts of the DPG. For this reason, we can assert that these nodes are meaningful to understanding the tree-based ensemble models: in all tree base learners, the decision contained in the node is essential to classify the elements of the dataset. We highlight that the significance extends beyond the characteristic itself; it encompasses the value associated with it.

According to Mones et al. [27], we define the *local reaching centrality* (LRC) of a node v of the DPG as the proportion of other nodes reachable from node v via outgoing edges. LRC can be generalised to weighted graphs by measuring the average weight of a given directed path starting from node v (more details are available on [27]). The LRC serves as a metric for assessing the importance of DPG's nodes. It gauges the extent to which decisions contained in these nodes are employed by diverse tree base learners for classifying samples in the training set. This, in turn, reflects the importance of these decisions in the classification of new samples. The LRC offers a comprehensive perspective on the concept

of feature importance (FI) by extending its definition to encompass the values associated with features across various decisions. Additionally, the prominence of paths between highlighted predicates indicates their frequent utilisation, providing insights into how new samples can be classified with fewer decisions.

Community. While there is no single definition of a community, we can observe structures similar to communities in the DPG. According to Radicchi et al. [33], we define a *community* as a subset of nodes of the DPG characterised by dense interconnections between its elements and sparse connections with the other nodes of the DPG that do not belong to the community. Based on the properties of DPG, we employed *asynchronous label propagation* algorithm, proposed by Raghavan et al. [34], to detect communities within the graph. The core concept of the algorithm involves each graph node determining its community membership based on the majority of its neighbours. Without delving into details, which can be appreciated in [34], the algorithm comprises a series of steps: each node initially possesses a unique label. As these labels diffuse through the graph, closely connected groups of nodes converge on a common label. These consensus groups then expand outward until further expansion becomes impractical. After this label propagation process, nodes sharing the same labels are identified as belonging to the same community. This process is iterated until each node in the network aligns its label with the community that includes the maximum number of its neighbouring nodes. The algorithm is defined asynchronous, as each node receives updates without waiting for updates on the remaining nodes. Identifying communities in the DPG provides insights into the ensemble model: visualising them allows us to discover groups of nodes that similarly contribute to the classification of samples.

By employing the asynchronous label propagation algorithm, we observe that each formed community is associated with a class. Within these communities, the features utilised by the ensemble model to classify samples belonging to the community's class are emphasised. Once again, the association between features and values plays a key role, highlighting the specific decisions made by the learners. To quote Raghavan et al. [34]:

> Communities in social networks can provide insights about common characteristics or beliefs among people that make them different from other communities.

Similarly, we observe that communities within the DPG offer a valuable understanding of the characteristics for samples to be assigned to a particular community class. This intuition extends to identifying predominant features and those that play a marginal role in the classification process. Moreover, it is noteworthy that communities also provide insights into the entire dataset and the complexity of the problem. A community comprising a substantial number of nodes, each associated with different predicates often involving distinct features, indicates that the model makes diverse decisions to assign samples to the community class. This implies that the model encounters challenges in classifying samples for this particular class, and data from different classes are not easily distinguishable within the dataset.

Finally, communities, functioning as sub-graphs, can be used to visualise the decisions made in the ensemble model, enabling the identification of a specific class. This replaces the complex illustration of the DPG, especially when we are focused on visualising a single class and dealing with many tree base learners. We summarised the utility of discussed properties and metrics in Table 1.

Table 1. Summary of Constraints, Betweenness Centrality (BC), Local Reaching Centrality (LRC), and Community, featuring provided definitions and their utility in offering insights into tree-based ensemble models.

Property	Definition	Utility
Constraints	The intervals of values for each feature obtained from all predicates connected by a path that culminates in a given class.	Calculate the classification boundary values of each feature associated with each class.
BC	Quantifies the fraction of all the shortest paths between every pair of nodes of the graph passing through the considered node.	Identify potential bottleneck nodes that correspond to crucial decisions.
LRC	Quantifies the proportion of other nodes reachable from the local node through its outgoing edges.	Assess the importance of nodes similarly to feature importance, but enrich the information by encompassing the values associated with features across all decisions.
Community	A subset of nodes of the DPG which is characterised by dense interconnections between its elements and sparse connections with the other nodes of the DPG that do not belong to the community.	Understanding the characteristics of nodes to be assigned to a particular community class, identifying predominant predicates, and those that play a marginal role in the classification process.

4 Empirical Results and Discussion

In this section, we demonstrate the effectiveness of DPG to the well-known Iris dataset [10] and a synthetic multiclass dataset[2]. Each experiment in this section was conducted using the RF classifier, with variations limited to the number of tree base learners. The implementation is available here[3]. Finally, we discuss potential enhancements to DPG and explore further development opportunities.

[2] https://github.com/LeonardoArrighi/DPG/datasets.
[3] https://github.com/LeonardoArrighi/DPG.

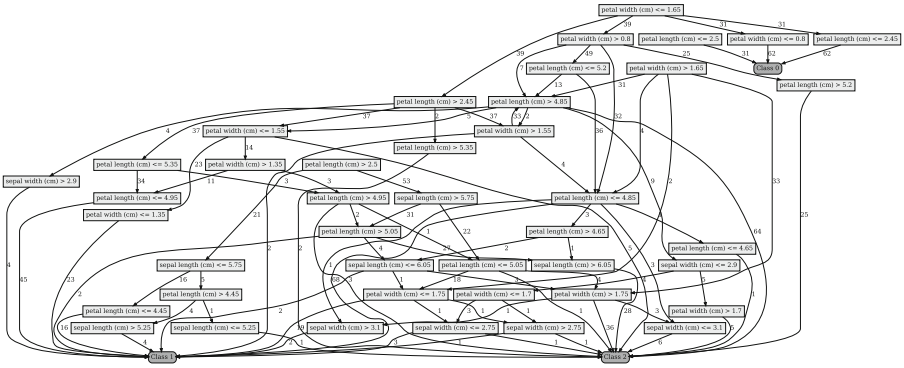

Fig. 1. DPG of the RF composed of 5 tree base learners trained on Iris dataset.

4.1 DPG: Iris Insights

The first case study concerns the classification of the Iris dataset [10]. The simplicity, manageability, versatility, and relevance of this dataset make it an interesting and relevant resource for discussions and demonstrations of interpretability in ML. The dataset comprises measurements of sepals and petals for iris flowers encompassing three distinct species with a total of four features and three classes. The RF was selected as the tree-based ensemble model due to its well-established reputation and high-performance capabilities. To conduct the classification, we partitioned the dataset into training and test sets, following a 80-20% proportion, respectively. A seed value of 42 was established for randomness control, and the number of base tree learners was set to 5. The RF performances, evaluated on the test set, are summarised in the confusion matrix in Table 2. The model demonstrates 100% accuracy.

Table 2. Confusion matrix depicting the performance evaluation of the RF model with 5 base tree learners on the test set.

	Prediction		
Ground truth	Class 0	Class 1	Class 2
Class 0	19	0	0
Class 1	0	13	0
Class 2	0	0	13

After training the model, we applied the algorithm outlined in Sect. 3.2 to obtain the DPG, which can be visualised in Fig. 1. Then, we can analyse the obtained graph using the metrics and algorithms proposed in Sect. 3.3, verifying their utility and effectiveness in gaining insights into the model. It is important

to note that the DPG leads to the calculation of both global metrics, referring to the overall graph, and metrics at the level of individual nodes.

To illustrate the effectiveness and one of the advantages of employing DPG, we highlight the constraints for the different classes in the Table 3. The class-specific constraints delineate the necessary characteristics a sample must exhibit to be assigned to that particular class by the tree-based ensemble model. This insight contributes to a better understanding of how the model utilises features for effective classification.

Table 3. Constraints for each class based on the DPG for an RF model within 5 tree base learners.

Class	Constraints
0	`petal width (cm)` ≤ 1.65
	`petal length (cm)` ≤ 2.50
1	$5.25 <$ `sepal length (cm)` ≤ 6.05
	$0.80 <$ `petal width (cm)` ≤ 1.75
	$2.45 <$ `petal length (cm)` ≤ 5.35
	$2.75 <$ `sepal width (cm)` ≤ 2.90
2	$5.75 <$ `sepal length (cm)` ≤ 6.05
	$0.80 <$ `petal width (cm)` ≤ 1.75
	$2.45 <$ `petal length (cm)` ≤ 5.35
	$2.75 <$ `sepal width (cm)` ≤ 3.10

The first metric under discussion is the BC of the nodes, as depicted in Table 4, where we identify potential bottleneck nodes. These nodes encapsulate significant information, particularly representing decisions made by numerous tree base learners. We quickly discern that the decision associated with `petal length (cm)` and the value 4.85 is pivotal, as it is frequently relied upon by multiple basic learners and is essential for successful classification.

Furthermore, the LRC metric provides additional information. Examining Table 5a, we can emphasise the most crucial predicates influencing the decision-making process of the tree-based ensemble model. This includes not only identifying the most frequently used features but also recognising the associated values that lead to significant and divisive splits in the dataset across various basic learners. As observed in Table 5, a comparison between the LRC of the nodes and the FI, calculated on the same model on which DPG is based, suggests that the metric may rank the predicates similarly. FI is calculated using the *Mean Decrease Impurity* (MDI) algorithm introduced by Breiman [4]. This comparison also provides additional information about the values used in the decisions and the frequency of paths extending the concept of FI.

By employing the global metric community, we identified the presence of three distinct communities. Table 6 illustrates the association between each

Table 4. Top eight predicates by evaluating their BC obtained from the DPG based on an RF model consisting of 5 tree base learners.

Predicate	BC
`petal length (cm) > 4.85`	0.053
`petal length (cm) <= 4.85`	0.036
`petal width (cm) > 1.55`	0.034
`sepal length (cm) <= 6.05`	0.032
`petal length (cm) > 4.95`	0.028
`petal length (cm) > 4.65`	0.022
`petal width (cm) <= 1.75`	0.022
`petal width (cm) <= 1.55`	0.021

Table 5. Comparison of the top eight predicates by evaluating their LRC obtained from the DPG based on an RF model consisting of 5 tree base learners (Table 5a), alongside the FI of the same model (Table 5b) calculated exploiting MDI algorithm.

(a) LRC evaluation

Predicate	LRC
`petal width (cm) <= 1.65`	1.531
`petal length (cm) > 2.45`	0.919
`petal width (cm) > 0.80`	0.874
`petal length (cm) > 2.50`	0.699
`petal width (cm) > 1.65`	0.618
`petal length (cm) <= 5.20`	0.565
`petal width (cm) > 1.55`	0.540
`sepal length (cm) > 5.75`	0.332

(b) FI evaluation

Feature	FI
`petal length (cm)`	0.550
`petal width (cm)`	0.373
`sepal length (cm)`	0.054
`sepal width (cm)`	0.023

community and a distinct class obtained by applying the asynchronous label propagation algorithm to the DPG. We can affirm that each node within a community contains decisions that significantly contribute to the accurate classification of samples belonging to a specific class. For instance, when applying the predicates of Community 3 (comprising two features and two predicates) to the test set and traversing from the root node, it achieved 100% accuracy for Class 0, the class delineated in the mentioned community.

Finally, upon comparing the Table 6 and the Fig. 2, we can state that the communities facilitate the comprehension of how the model addresses the classification problem. Examining the number of decisions and features utilised in each community reveals that differentiating between Class 1 (Community 1) and Class 2 (Community 2) poses a greater challenge for the model. This indicates the difficulty in effectively separating samples within the dataset into their respective classes. This difficulty becomes apparent when visualising the dataset across the

Table 6. Communities obtained from the DPG based on an RF model composed of 5 tree base learners. The table shows the number of predicates belonging to each community, the number of features in the community nodes, and the class involved in each community.

Community	# Predicates	# Features	Class
Community 1	23	4	1
Community 2	18	4	2
Community 3	4	2	0

features, as in Fig. 2. Conversely, the community encompassing Class 0 (Community 3) consists of fewer predicates, signifying that it is more distinguishable from other classes, as confirmed in the Fig. 2.

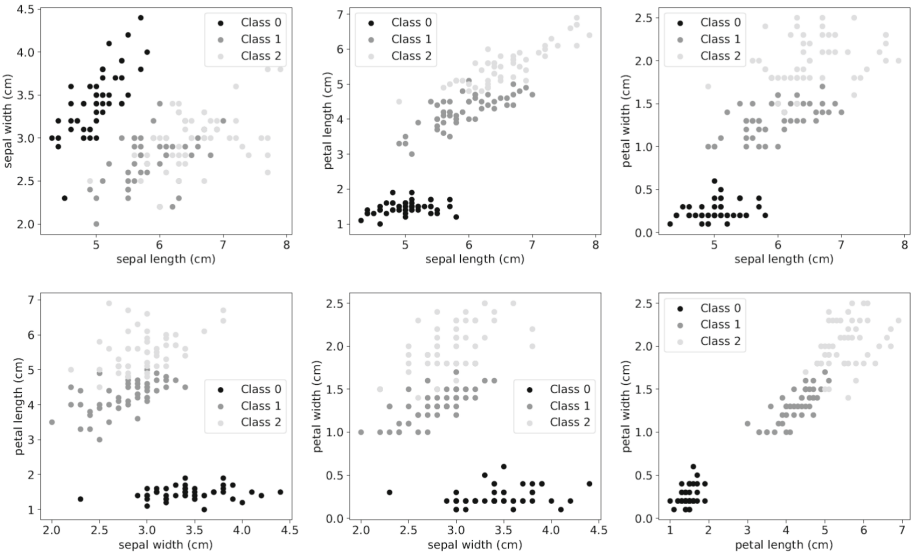

Fig. 2. Two-dimensional depiction of the Iris dataset, employing feature pairs in each graph for visual representation.

4.2 Comparing to the Graph-Based Solutions

As outlined in the Sect. 2, Gossen and Steffen [13] and Murtovi et al. [28] conducted studies on the interpretability of tree-based ensemble models exploiting graph structures. We compared DPG with their proposed method by examining their outcomes and potential insights. Using the same tree-based ensemble model

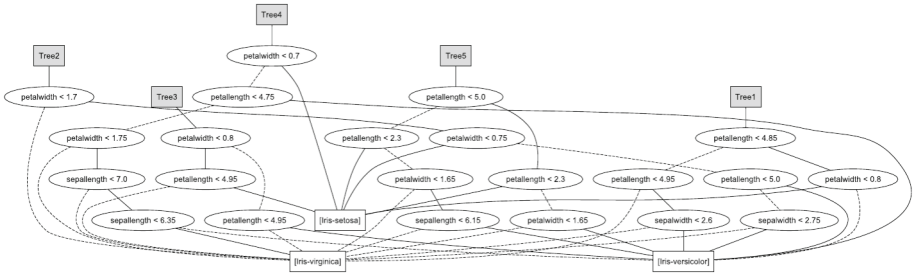

Fig. 3. ADD of an RF model with 5 tree base learners induced for Iris dataset.

we studied in Sect. 4.1 as input, we generated the ADD displayed in Fig. 3. The first noticeable distinction from DPG lies in the ADD structure, as it forms a bipartite graph. We can observe that the ADD is generated from the trained model, albeit without fully leveraging the training dataset. Consequently, the evaluation of connections between nodes and the assessment of the significance of decisions made by different tree base learners are not fully exploited. This implies that each branch carries equal weight and impact in the diagram, and a classification error significantly influences the structure by introducing an incorrect path. Another crucial difference is that ADD does not provide graph metrics, leaving the user the interpretation of the diagram and potentially missing out on relevant information. Moreover, an additional limitation, as indicated in the studies by [13,28], involves the challenge that emerges when generating ADDs and dealing with large ensembles. Visualisation becomes intricate, even with a modest count of 20 tree base learners. In contrast, DPG allows the computation of both global and local metrics, even with a higher number of tree base learners.

To further examine these aspects, we use two RF models, one with 20 tree learners and the other with 100, to analyse a complex multiclass problem with a dataset comprising 4 classes, 1000 samples, and 16 features. This introduces a four-class problem that was randomly generated. The dataset was created using the `make_classification`[4] function from `scikit-learn`. The following setup has been maintained for training both models. We divided the dataset into training and test sets, with an 80-20% ratio. We fixed a seed value of 42 for randomness control to ensure reproducibility. The RF performances, assessed on the test set, are summarised in the confusion matrices presented in Table 7. The model with 100 tree base learners shows better performance in every parameter, reaching an overall accuracy of 66%, outperforming the model with 20 tree base learners, which barely reaches 58%.

We emphasise that even in this context, DPG is a useful tool. Both DPG and ADD present intricate visualisations with 20 tree base learners. However, DPG overcomes this obstacle by providing metrics that can still offer valid insights into

[4] https://scikit-learn.org/stable/modules/generated/sklearn.datasets.make_classification.html.

Table 7. Confusion matrices of the RF models with 20 tree base learners (RF 20) and with 100 tree base learners (RF 100) tested on the synthetic dataset.

	Prediction (RF 20)				Prediction (RF 100)			
Ground truth	Class 0	Class 1	Class 2	Class 3	Class 0	Class 1	Class 2	Class 3
Class 0	38	4	12	6	38	4	14	4
Class 1	5	31	3	5	4	32	2	6
Class 2	11	2	29	2	5	2	33	4
Class 3	10	13	10	19	9	9	5	29

the model. The first insight is displayed in Table 8, where we provide constraints for the four classes of the dataset. Constraints, even in this complex scenario, allow the visualisation of intervals where sample features should be situated for precise classification into their respective classes.

Table 8. Constraints for each class based on the DPG for an RF model within 100 tree base learners.

Class 0	Class 1	Class 2	Class 3
$-5.87 < \texttt{F1} \leq 5.74$	$-5.79 < \texttt{F1} \leq 5.72$	$-5.76 < \texttt{F1} \leq 5.72$	$-5.76 < \texttt{F1} \leq 5.72$
$-2.64 < \texttt{F2} \leq 2.63$	$-2.61 < \texttt{F2} \leq 2.61$	$-2.61 < \texttt{F2} \leq 2.61$	$-2.61 < \texttt{F2} \leq 2.63$
$-5.24 < \texttt{F3} \leq 3.75$	$-5.38 < \texttt{F3} \leq 3.75$	$-5.24 < \texttt{F3} \leq 3.75$	$-5.24 < \texttt{F3} \leq 3.75$
$-4.80 < \texttt{F4} \leq 4.37$	$-5.15 < \texttt{F4} \leq 4.37$	$-4.80 < \texttt{F4} \leq 4.37$	$-4.80 < \texttt{F4} \leq 4.37$
$-2.61 < \texttt{F5} \leq 2.44$	$-2.61 < \texttt{F5} \leq 2.71$	$-2.61 < \texttt{F5} \leq 2.44$	$-2.61 < \texttt{F5} \leq 2.44$
$-2.29 < \texttt{F6} \leq 2.58$	$-2.29 < \texttt{F6} \leq 2.58$	$-2.29 < \texttt{F6} \leq 2.58$	$-2.29 < \texttt{F6} \leq 2.58$
$-2.82 < \texttt{F7} \leq 2.47$	$-2.82 < \texttt{F7} \leq 2.47$	$-2.82 < \texttt{F7} \leq 2.47$	$-2.82 < \texttt{F7} \leq 2.86$
$-4.62 < \texttt{F8} \leq 4.74$	$-4.62 < \texttt{F8} \leq 4.74$	$-4.62 < \texttt{F8} \leq 4.74$	$-4.62 < \texttt{F8} \leq 4.74$
$-2.40 < \texttt{F9} \leq 2.59$	$-2.40 < \texttt{F9} \leq 2.59$	$-2.40 < \texttt{F9} \leq 2.71$	$-2.40 < \texttt{F9} \leq 2.59$
$-4.71 < \texttt{F10} \leq 4.32$	$-4.71 < \texttt{F10} \leq 4.32$	$-4.71 < \texttt{F10} \leq 5.43$	$-4.71 < \texttt{F10} \leq 4.32$
$-2.87 < \texttt{F11} \leq 2.77$	$-2.87 < \texttt{F11} \leq 2.77$	$-2.87 < \texttt{F11} \leq 2.86$	$-2.87 < \texttt{F11} \leq 2.77$
$-2.42 < \texttt{F12} \leq 2.37$	$-2.42 < \texttt{F12} \leq 2.37$	$-2.42 < \texttt{F12} \leq 2.37$	$-2.42 < \texttt{F12} \leq 2.37$
$-4.28 < \texttt{F13} \leq 5.01$	$-4.28 < \texttt{F13} \leq 5.01$	$-4.28 < \texttt{F13} \leq 5.01$	$-4.28 < \texttt{F13} \leq 5.01$
$-7.31 < \texttt{F14} \leq 8.33$	$-7.31 < \texttt{F14} \leq 8.33$	$-7.31 < \texttt{F14} \leq 8.33$	$-7.31 < \texttt{F14} \leq 8.33$
$-4.46 < \texttt{F15} \leq 4.19$	$-4.46 < \texttt{F15} \leq 4.19$	$-4.46 < \texttt{F15} \leq 4.19$	$-4.97 < \texttt{F15} \leq 4.5$
$-4.70 < \texttt{F16} \leq 4.21$	$-4.70 < \texttt{F16} \leq 4.21$	$-4.70 < \texttt{F16} \leq 4.52$	$-4.70 < \texttt{F16} \leq 4.21$

The BC metric helps identifying potential bottleneck nodes. Upon observing Table 9a, we can see that there is not a large difference between the BC values associated with the predicates. Therefore, we can assume that there are no bottleneck nodes.

Examining the information provided in Table 9b, the LRC underscores which predicates significantly impact the decision-making process of the ensemble model.

Table 9. Top eight predicates by evaluating their BC (Table 9a), and top eight predicates by evaluating their LRC (Table 9b), both obtained from the DPG based on an RF model consisting of 100 tree base learners.

(a) BC evaluation

Predicate	BC
`F15 > 1.17`	0.018
`F15 <= 1.61`	0.015
`F12 > 0.20`	0.015
`F12 > 0.41`	0.014
`F4 <= 0.33`	0.014
`F8 > 0.36`	0.014
`F1 <= -1.10`	0.013
`F11 <= 1.16`	0.013

(b) LRC evaluation

Predicate	LRC
`F7 <= 1.62`	15.812
`F1 <= 3.10`	14.475
`F14 > -1.78`	13.313
`F4 > -2.97`	13.158
`F5 > -1.92`	13.065
`F4 > -1.36`	12.989
`F1 <= 2.49`	12.986
`F13 > 1.98`	12.920

Another insight can be obtained by employing the global metric community. In this scenario, we identified the presence of four distinct communities, displayed in Table 10. We note that each community contains a distinct class. Furthermore, upon observing the table, we can conclude that each community exhibits a high number of involved features and predicates, confirming the complexity of the classification problem.

Table 10. Communities obtained from an RF model composed of 100 tree base learners. The table shows the number of predicates belonging to each community, the number of features in the community nodes, and the class involved in each community.

Community	# Predicates	# Features	Class
Community 1	7767	16	2
Community 2	2149	16	0
Community 3	2351	16	3
Community 3	2100	16	1

4.3 Potential Improvements

Several avenues await exploration in the future. The primary aim is to reduce the computational cost of DPG, as many real-world problems involve large datasets that do not scale well with the current implementation of DPG. Expanding the application scope of DPG is another key goal, including its utility in explaining models relevant to regression-type problems. Given DPG's applicability to any model and dataset, we aim to introduce new tests and use cases to delve deeper into the method. This includes proposing applications to novel datasets

and exploring their compatibility with other tree-based ensemble models. Furthermore, while this paper introduces certain metrics and algorithms derived from graph theory, the field offers extensive possibilities for future exploration. In the future, we plan to introduce new tools associated with DPG to enhance the interpretation of tree-based ensemble models.

5 Conclusion

In this paper, we introduced Decision Predicate Graphs (DPG) as a novel model-specific tool for tree-based ensemble interpretability. DPG is obtained from a trained model and data, ensuring the maintenance of its performance. The concept behind DPG is to convert an opaque-box tree-based ensemble model into an enriched graph. DPG enables graph-based evaluations and the identification of model decisions towards facilitating comparisons between features and their associated values while offering insights into the entire model. In particular, we introduced Betweenness Centrality, Local Reaching Centrality, Community and Constraints as useful metrics and properties towards improving and extending the XAI interpretability approaches. While DPG is still considered an evolving work, its potential is substantial, given the robust underlying theory and the versatility of the tool. Furthermore, the effervescent research on graphs, knowledge graphs, and complex networks might strengthen the possibilities grounded in DPG. As the next step of our current research, we expect to apply DPG to provide local interpretability and to enhance the scalability of the current implementation.

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Model-Agnostic and Statistical Methods for eXplainable AI

Observation-Specific Explanations Through Scattered Data Approximation

Valentina Ghidini[1], Michael Multerer[1], Jacopo Quizi[1(✉)], and Rohan Sen[1,2]

[1] Euler Institute, USI Lugano, Lugano, Switzerland
{valentina.ghidini,michael.multerer,jacopo.quizi,rohan.sen}@usi.ch
[2] Institute of Finance, USI Lugano, Lugano, Switzerland

Abstract. This work introduces the definition of observation-specific explanations to assign a score to each data point proportional to its importance in the definition of the prediction process. Such explanations involve the identification of the most influential observations for the black-box model of interest. The proposed method involves estimating these explanations by constructing a surrogate model through scattered data approximation utilizing the orthogonal matching pursuit algorithm. The proposed approach is validated on both simulated and real-world datasets.

Keywords: Observation-specific explanations · functional reconstruction · surrogate model · reproducing kernel Hilbert space

1 Introduction

In the rapidly evolving landscape of machine learning, the need for understanding and interpreting complex models has become increasingly important. As models get more sophisticated and are applied to critical decision-making processes across various domains, the demand for transparency and interpretability has never been higher. This necessity has led to the development of explainable artificial intelligence (XAI), which aims to shed light on the inner workings of black-box algorithms.

Most of the conventional XAI techniques found in the current literature, such as the ones in [5,8,9,11,12,16,17], can be broadly classified as measures of variable importance. The common rationale of all these techniques is to assign a score to the different input features according to their respective importance in the inner mechanisms of the black-box model to explain. In fact, the higher the score, the more significant the variable. This approach is very general and versatile, as these methods can be applied to different data formats, including images and texts, and are compatible with diverse types of black-box models.

In this work, however, we propose to shift the perspective from explanatory variables to individual data points. Specifically, our objective is to understand the importance of each data instance in estimating the decision-making process. We

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anticipate that some data points may have a greater impact than others on the definition of the predictions. This change of focus offers an opportunity to better understand the internal operations of a model. By detecting the most important observations, we can analyze what are the influential data instances driving the estimation of the black box and understand whether the model accurately captures the "typical" data points or is ill-fit to represent outliers or unexpected instances. In essence, this approach enables us to identify the data points that have the greatest influence on the decision-making process and draw pertinent conclusions accordingly.

The idea proposed in this paper is closely related to influential observations in statistical models [3,6]. Influential observations are conceptually different from outliers: in the literature, they are broadly defined as data points whose impact on the model estimation significantly outweighs that of others. In linear models, influential data points are well-defined and diagnosed through metrics like leverage and Cook's distance [7]. However, identifying influential observations in black boxes is still an open problem.

We begin by providing a proper definition of observation-specific explanations. These explanations are designed to measure the influence of each data point on the decision-making process of the black box. Then, we propose to estimate such explanations using the scattered data approximation [19], effectively defining an interpretable surrogate model for the black box. If the surrogate lives in a reproducing kernel Hilbert space, it can be represented with respect to an orthonormal basis. This allows us to use the resulting coefficients in the expansion to define normalized observation-specific explanations. The surrogate model is estimated using the orthogonal matching pursuit algorithm [13–15] that selects an optimal subset of data points to represent the black-box model of interest. The resulting predictions of the black box at the selected data instances are perfectly reproduced by the surrogate, and the approximation error at all data points is rigorously bounded. This even enables an observation-wise diagnostic of the fidelity of the surrogate model [12]. Then, the approach is validated on synthetic data and on a data set concerning anatomical measurements of animals.

The remainder of the paper is organized as follows: Sect. 2 introduces the definition of observation-specific explanations and proposes an estimation procedure leveraging orthogonal matching pursuit in reproducing kernel Hilbert spaces. Section 3 presents the application of the proposed method on both simulated scenarios and real datasets. Finally, Sect. 4 summarizes the findings and discusses future research directions.

2 Methodology

We adopt a standard machine-learning notation. Consider a sample set of n observations, each containing p features, denoted as $X = \{\mathbf{x}_1, \ldots, \mathbf{x}_n\} \subset \mathcal{X} \subseteq \mathbb{R}^p$, where $\mathcal{X}$ is the sample space. We denote by f the black-box model of interest which outputs the predictions $f(\mathbf{x}_1), \ldots, f(\mathbf{x}_n)$ at the sample points.

Given this setting, we begin by defining observation-specific explanations that offer instance-wise significance measures proportional to the importance of each data point in the definition of the prediction process. Following this idea, we introduce a framework for computing such explanations using a surrogate model based on scattered data approximation techniques.

2.1 Observation-Specific Explanations

Definition 1. *Given a model f, we define* observation-specific explanations *as*

$$\gamma = \{\gamma_1, \ldots, \gamma_n\},$$

where $\gamma_i \in [0, \infty)$, $1 \leq i \leq n$, is a positive index proportional to the importance of the i^{th} observation for the model f.

To the best of our knowledge, this definition introduces a novel class of explanations in the literature. It proposes a family of importance metrics, indicating how much each observation influences the predictions made by the black-box model under examination. The viewpoint is then shifted with respect to standard explainability techniques: indeed, traditionally, importance is assessed for each explanatory variable regarding the prediction process, typically by averaging across dataset observations. However, our proposal reverses the approach: the aim is to understand the impact of each data point on the process of generating the predictions. As mentioned above, this concept draws inspiration from influential observations in statistical models, which are intuitively defined as instances whose removal or addition significantly impacts the model estimation [3].

Definition 1 is kept general since it can encompass any specific algorithm that computes the observation-specific explanations. To address the latter, we propose a procedure based on the definition of a surrogate model using scattered data approximation in the next subsection. However, future research directions could explore the development of alternative algorithms to provide observation-specific explanations.

2.2 Surrogate Models Using Scattered Data Approximation

To estimate observation-specific explanations for a particular black-box function f, we define a surrogate model f^* such that $f^*(\mathbf{x}) \approx f(\mathbf{x})$ for $\mathbf{x} \in \mathcal{X}$. Recall that a surrogate model is a function designed to closely mimic the behavior of the black-box model under consideration [12,17].

Suppose that the surrogate model f^* is contained in a reproducing kernel Hilbert space (RKHS) $\mathcal{H}$ that contains real-valued functions on the sample space $\mathcal{X}$ with reproducing kernel $\mathcal{K}\colon \mathcal{X} \times \mathcal{X} \to \mathbb{R}$, see [4]. Using the canonical feature map $\mathbf{x} \mapsto \mathcal{K}(\mathbf{x}, \cdot) \in \mathcal{H}$, the sample set $\{\mathbf{x}_1, \ldots, \mathbf{x}_n\}$ can be embedded into $\mathcal{H}$. We posit that the black-box model f can be well approximated by the surrogate model f^*, that is

$$f(\mathbf{x}) \approx f^*(\mathbf{x}) = \sum_{i=1}^{n} c_i \, \mathcal{K}(\mathbf{x}_i, \mathbf{x}) \quad \text{for all } \mathbf{x} \in \mathcal{X}. \tag{1}$$

This assumption implies that the surrogate is defined as a linear model as in Eq. (1) in the RKHS $\mathcal{H}$. Here, the c_i's are the coefficients corresponding to the basis functions $\mathcal{K}(\mathbf{x}_i, \cdot)$ parametrizing the linear representation of the surrogate model. From (1), it is possible to derive a class of surrogate models that exactly match the values of the predictions $f(\mathbf{x}_{i_1}), \ldots, f(\mathbf{x}_{i_{n^*}})$ at a selected subsample $\{\mathbf{x}_{i_1}, \ldots, \mathbf{x}_{i_{n^*}}\} \subset \{\mathbf{x}_1, \ldots, \mathbf{x}_n\}$. To compute this subsample, we adaptively compute an orthonormal basis $\{\mathcal{N}_1, \ldots, \mathcal{N}_{n^*}\} \subset \mathcal{H}$ by means of *orthogonal matching pursuit* [14,15] (OMP). This approach amounts to representing the surrogate model with respect to this new basis:

$$f^*(\mathbf{x}) = \sum_{j=1}^{n^*} a_j^* \, \mathcal{N}_j(\mathbf{x}). \tag{2}$$

Herein, the $a_j^*, j = 1, \ldots, n^*$, are the coefficients with respect to the new (orthonormal) basis functions. However, we cannot use these coefficients directly for observation-specific explanations since any particular basis function $\mathcal{N}_j(\mathbf{x})$ is not uniquely linked to a single observation. Nevertheless, the orthonormal basis functions $\mathcal{N}_1, \ldots, \mathcal{N}_{n^*}$ can be computed by a Gram-Schmidt procedure with respect to a certain reordering of the basis of kernel translates $\{\mathcal{K}(\mathbf{x}_1, \cdot), \ldots, \mathcal{K}(\mathbf{x}_n, \cdot)\}$. Therefore, we obtain

$$\mathcal{N}_j(\mathbf{x}) = \sum_{k=1}^{j} d_{k,j} \mathcal{K}(\mathbf{x}_{i_k}, \mathbf{x}) \tag{3}$$

for certain coefficients $d_{k,j}$. Thus, Eq. (3) can be used to represent the surrogate in terms of the basis functions induced by the kernel translates, which can subsequently be used for observation-specific explanations. Notice that the indices of the selected observations can be obtained in an iterative manner. Indeed, having computed $m < n^*$ of these basis functions, the next index i_{m+1} is chosen such that the error

$$\max_{i=1,\ldots,n} \left| f(\mathbf{x}_i) - \sum_{j=1}^{m} a_j^* \, \mathcal{N}_j(\mathbf{x}_i) \right| \tag{4}$$

is minimized.

A surrogate model such as the above is particularly advantageous for two main reasons. First, using (3), we can express f^* in terms of $\mathcal{K}(\mathbf{x}_1, \cdot), \ldots, \mathcal{K}(\mathbf{x}_n, \cdot)$ to define quantities that can be interpreted as observation-specific explanations. In particular, we can use the respective coefficients of $\mathcal{K}(\mathbf{x}_i, \cdot)$ to assign an appropriate weight to each observation proportional to its significance. It is worth noting that each $\mathcal{K}(\mathbf{x}_i, \cdot)$ corresponds to a particular sample point $\mathbf{x}_i$ for $1 \leq i \leq n$. Thus, the OMP [14] effectively selects a subsample of n^* data points for the reconstruction process, which are then targeted as the most influential observations. Second, it offers analytical bounds on the reconstruction of $f(\mathbf{x}_i)$ for $i = 1, \ldots, n$. This implies that within a standard mathematical framework, there are theoretical guarantees that f^* accurately replicates the behavior of the black-box model f on X. Furthermore, if we assume that $f \in \mathcal{H}$, then we can also have

the guarantees of a high fidelity on the entire sample space $\mathcal{X}$. It is worth noting that such an approach allows us to obtain observation-specific errors, defined as

$$\text{err}_i := |f(\mathbf{x}_i) - f^*(\mathbf{x}_i)|, \qquad \text{for } i = 1, \ldots, n. \tag{5}$$

In general, it can also be proven that using a tolerance ε^2 in the OMP algorithm proposed in [14], there exists the uniform error bound

$$|f(\mathbf{x}_i) - f^*(\mathbf{x}_i)| \leq \varepsilon \, \|f\|_{\mathcal{H}}, \quad \text{for } i = 1, \ldots, n. \tag{6}$$

Notice that the tolerance can be interpreted as the extent to which we want to reconstruct the black-box model using the surrogate model and gives us uniform error bounds as in (6). In general, given a tolerance ε, the number of selected data points n^* depends on the overall distribution of the observations in the support as well as the type of kernel used for the surrogate model. As a result, the choice of the kernel for the reconstruction process is important. Future research work should assess the sensitivity and study the nature of such a choice.

Analyzing such errors (and their normalized versions), we can discern for which observations the prediction process is accurately represented by the surrogate, i.e., where the error is close or equal to zero and where this is not the case. Thus, we can also sketch a tentative analysis of the surrogate model to see whether there are some systematic areas of the point cloud bound for errors.

2.3 Estimation of the Observation-Specific Explanations

A representation as in (2) lends itself to offering observation-specific explanations. Note that the expression of the surrogate model f^* in terms of an orthonormal basis implies that each coefficient a_j^* is simply the projection of the black-box model f onto the corresponding basis function pertaining to a particular observation from the set of samples. We wish to use this characterization of the coefficients to identify the most influential observations. Using (3) in (2), we have that

$$f^*(\mathbf{x}) = \sum_{j=1}^{n^*} a_j^* \sum_{k=1}^{j} d_{k,j} \mathcal{K}(\mathbf{x}_{i_k}, \mathbf{x}) = \sum_{k=1}^{n^*} c_{i_k} \mathcal{K}(\mathbf{x}_{i_k}, \mathbf{x}), \tag{7}$$

where

$$c_{i_k} = \sum_{j=k}^{n^*} d_{k,j} a_j^*. \tag{8}$$

Each coefficient c_{i_k} is then uniquely associated with a selected observation. Denoting the set $S := \{i_1, \ldots, i_{n^*}\}$, we set $c_i = 0$ if $i \notin S$ for all $1 \leq i \leq n$. Finally, we define the observation-specific explanations as

$$\gamma_i = \frac{|c_i|}{\max\limits_{1 \leq i \leq n} |c_i|}. \tag{9}$$

Thus, the data points not deemed as important are associated with a null explanation, effectively annihilating their impact on the estimation of the surrogate

model, and consequently on the prediction process provided by the black box. The resulting values of γ_i serve as normalized measures of the importance of each observation: a higher γ_i indicates greater importance of the i^{th} data point for the prediction process of the black box, replicated by the surrogate model.

3 Application

In this section, we showcase the application of the proposed method. In particular, Subsect. 3.1 displays results in simulated scenarios, using well-known data-generating processes. Then, Subsect. 3.2 presents an application on a real dataset.

3.1 Simulated Studies

In this subsection, we will explain two known data-generating processes that are suitably defined, rather than dealing with an actual black-box model. This framework facilitates assessing the suitability of the surrogate model and understanding the features of the resulting observation-specific explanations in a well-understood context.

The first example concerns a simple, bidimensional, quadratic function. We simulate $n = 1000$ data points according to

$$f(X_1, X_2) = X_1^2 + X_2^2 + 1, \tag{10}$$

where X_1 and X_2 are random, independent, standard Gaussian explanatory variables. We construct the surrogate model in (2) by employing a Gaussian kernel, see [18], with a length scale parameter equal to $\sqrt{7}$. We discuss an alternative method for determining this crucial hyperparameter using the real dataset in Subsect. 3.2. The tolerance is set to $\varepsilon = 10^{-3}$, and the procedure selects $n^* = 36$ data points for the definition of the surrogate model. In particular, the reconstruction error in (6) is of the order of 10^{-4}, indicating that we are achieving a nearly perfect reconstruction of the real function using only 3.6% of the data size: this means that the surrogate model has a provable high fidelity in replicating the generating model of interest. Figure 1 reports the results on the bidimensional, quadratic simulated dataset: the left plot displays the data points with colors proportional to the observation-specific errors in Eq. (5) — darker hues indicate higher errors and consequently a lower fidelity of the surrogate model. This visualization offers insight into the data points best represented by the surrogate model. The surrogate reproduces better the instances at the data cloud's boundaries where the points' density is lower. These boundary points are also where we anticipate having statistically more uncertainty in the functional reconstruction, but also perfect candidates for interpolation. In the right plot, the colored points represent those selected for defining the surrogate in Eq. (2), with the intensity of red and the point size proportional to the observation-specific explanations in (9). As expected, such instances shape the actual point cloud, focusing on more representative areas of the support.

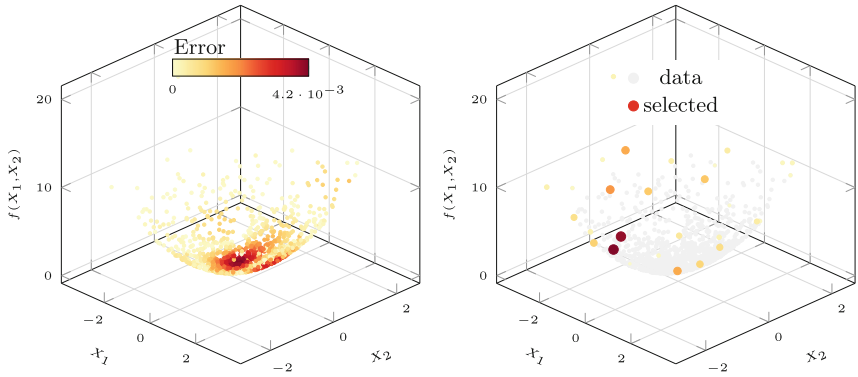

Fig. 1. First simulated scenario: the data generating process is a quadratic function. In the left plot, the colors correspond to the observation-specific relative error in the surrogate model reconstruction: darker shades represent higher errors. In the right plot, selected data points are colored and sized according to the magnitude of their explanations: Darker and larger points indicate higher values.

In the second simulated scenario, we employ the well-known Ackley function [1], commonly used as a benchmark for optimization studies due to its highly oscillatory nature and the presence of multiple local optimal points. We opt for this function to explore how the surrogate model in (2) behaves in the presence of a smooth yet intricate generating model, producing peculiarly shaped point clouds. To estimate the surrogate model, we run the OMP algorithm choosing a tolerance of $\varepsilon = 10^{-3}$, which results in $n^* = 370$ selected points out of the $n = 1000$ simulated ones. We simulate X_1, X_2 from independent standard Gaussian distributions and employ the Matérn kernel $\mathcal{K}_{3/2}$, see [2]. The resulting reproduction error is of the order of 10^{-4}. Figure 2 reports the results for this scenario. As before, the plot on the left illustrates the cloud of the simulated data points, with colors indicating the observation-specific relative error in (5). Lighter points denote a better approximation (higher fidelity) by the surrogate model. Consequently, the boundaries of the point cloud reveal the lower fidelity of the surrogate, as these areas are statistically the most challenging to reproduce. However, even in these low-fidelity regions, we still select representative subsamples that can be used for effective reconstruction. In the right plot, the points chosen by the OMP algorithm to construct the surrogate model are displayed. Their colors and sizes are proportional to the observation-specific explanation in (9), with darker shades and bigger points representing higher values. Notably, these selected data instances mostly correspond to the local optima of the Ackley function, heuristically selected for their informativeness. Furthermore, the points with higher values of explanations are strategically positioned in less dense areas of the point cloud.

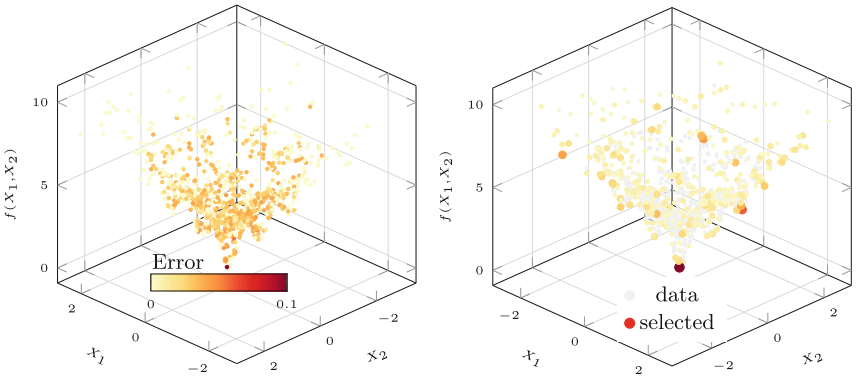

Fig. 2. Second simulated scenario: the data generating process is the Ackley function. In the left plot, the colors correspond to the observation-specific relative error in the surrogate model reconstruction: darker shades represent higher errors. In the right plot, selected data points are colored and sized according to the magnitude of their explanations: Darker and larger points indicate higher values.

3.2 Real-World Application

In this section, we present an application of the proposed technique on a publicly available dataset[1], whose task is to predict the height of $n = 101$ possums given their anatomical features, including the skull size, foot length, and others, standardized. We consider a black-box model by training an AdaBoost regressor [10], achieving a mean square error of 3.64 centimeters on the predicted length of the possum. This error is deemed acceptable since the mean length of the animals in this dataset is roughly 87 centimeters, with a standard deviation of 13.23 centimeters. We run the OMP algorithm, setting the tolerance $\varepsilon = 10^{-2}$ that gives us the number of selected data points to be equal to $n^* = 20$. Here, we choose the length scale parameter for the Exponential kernel, see [2], with K-fold cross-validation [10], where the parameter is set to the minimizer of the cross-validation error, which is found to be equal to $9.32 \cdot 10^{-4}$. The final, reconstruction error is of the order of 10^{-3}.

Even though visually exploring multidimensional patterns can be challenging, Fig. 3 depicts the bivariate dependence of possum length with the corresponding head and skull structure, with the foot length and the size of the ear conch, and with the chest and belly dimensions. Notice that we have arranged the bidimensional relationships based on pairwise correlations, with the displayed pairs of explanatory variables being the most correlated ones. The results confirm that data points with the highest observation-specific explanations tend to be located in less dense areas and near the boundaries of the point cloud. It is also possible to infer that the most important possums for the prediction process are the numbers 24 and 48. The first animal is characterized by a narrow head (skull width =

[1] kaggle.com/datasets/abrambeyer/openintro-possum.

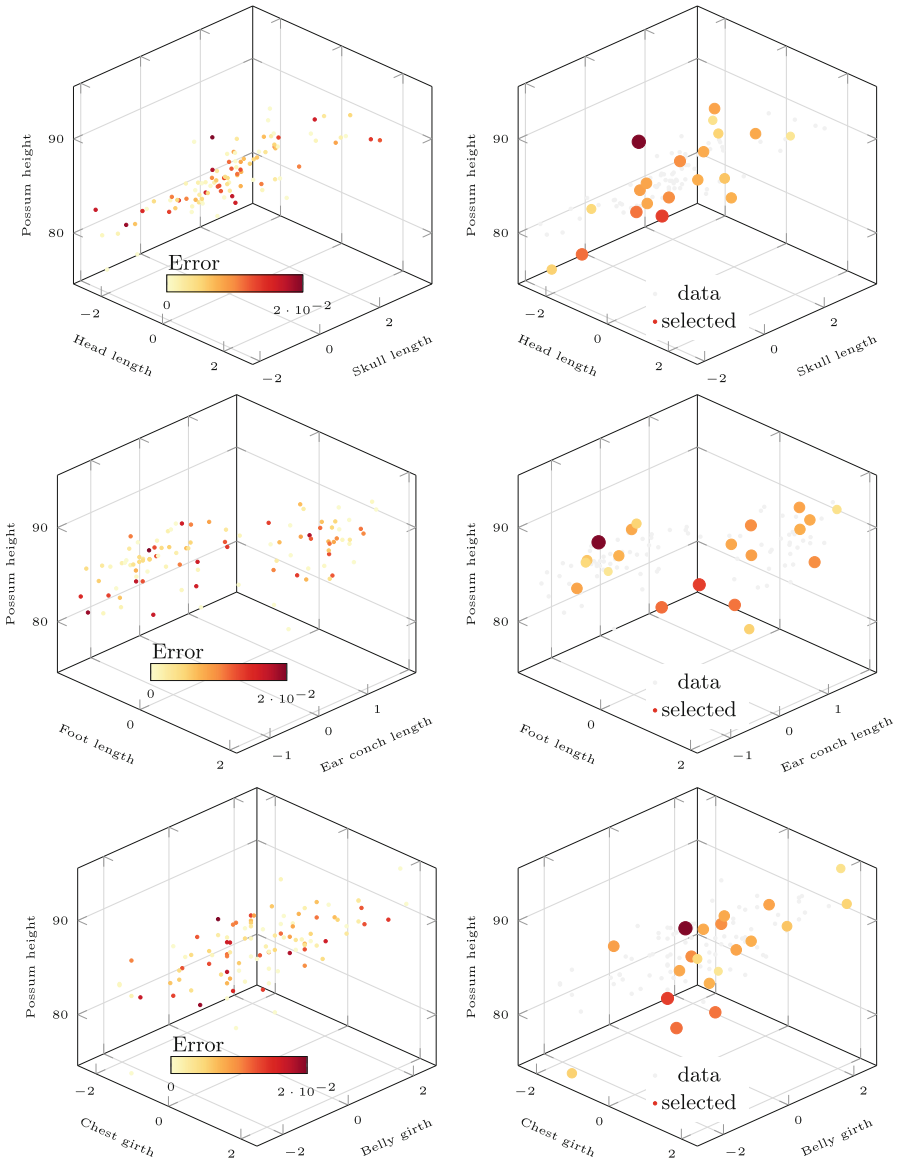

Fig. 3. Possum dataset: scatter plots of the height of the animal versus pairs of correlated, standardized anatomical features. In the left plot, point colors are with respect to the observation-specific relative error: darker shades represent higher errors. In the right plot, selected data points are colored and sized based on the magnitude of their explanations: darker and bigger points indicate higher values.

54.9), but longer feet (foot length = 75) and ear conch (ear conch length = 53.5). On the other hand, the second important possum has a wider head size (skull

length = 59.6), but shorter feet(foot length = 64), and the smaller ear conch (ear conch length = 43.9). A similar analysis can be conducted for all selected instances in the surrogate model. Therefore, we can conclude that observation-specific explanations highlight the most representative data instances (possums) for the reconstruction of the prediction process, discerning any redundancy of information.

4 Discussion

In this work, we shift the perspective of traditional explainability techniques from the importance of features to the importance of observations. We introduce the definition of observation-specific explanations, with the rationale of understanding the impact of each data point in the prediction process of a black-box model. Then, we propose a technique to obtain such observation-specific explanations. We employ scattered data approximation to define a surrogate model with provable high fidelity to the black box, using the orthogonal matching pursuit algorithm to estimate it. This procedure selects a subset of points to optimally define the surrogate model, which is then assigned individual coefficients. The observation-specific explanations are then obtained from these coefficients. Note that the proposed technique is completely model-agnostic: it can be applied to any black box of interest. Future research endeavors could explore the development of novel techniques for computing observation-specific explanations. This may involve alternative approaches to defining surrogate models or bypassing their definition altogether. Furthermore, there is potential for investigating the applicability of our approach to different and unstructured data types, such as images or texts, or in different fields (finance, healthcare, and so forth). Additionally, studying the efficiency of the method in scenarios with high dimensionality or a large number of data points could provide valuable insights.

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CNN-Based Explanation Ensembling for Dataset, Representation and Explanations Evaluation

Weronika Hryniewska-Guzik[1(✉)], Luca Longo[2], and Przemysław Biecek[1]

[1] Warsaw University of Technology, Warsaw, Poland
{weronika.hryniewska.dokt,przemyslaw.biecek}@pw.edu.pl
[2] Technological University Dublin, Dublin, Ireland
luca.longo@tudublin.ie

Abstract. Explainable Artificial Intelligence has gained significant attention due to the widespread use of complex deep learning models in high-stake domains such as medicine, finance, and autonomous cars. However, different explanations often present different aspects of the model's behavior. In this research manuscript, we explore the potential of ensembling explanations generated by deep classification models using convolutional model. Through experimentation and analysis, we aim to investigate the implications of combining explanations to uncover a more coherent and reliable patterns of the model's behavior, leading to the possibility of evaluating the representation learned by the model. With our method, we can uncover problems of under-representation of images in a certain class. Moreover, we discuss other side benefits like features' reduction by replacing the original image with its explanations resulting in the removal of some sensitive information. Through the use of carefully selected evaluation metrics from the Quantus library, we demonstrated the method's superior performance in terms of Localisation and Faithfulness, compared to individual explanations.

Keywords: Explainable Artificial Intelligence · Convolutional Neural Network · model evaluation · data evaluation · representation learning · ensemble

1 Introduction

Deep learning models, despite their unprecedented success [70,75], lack full transparency and interpretability in their decision-making processes [39,52]. This has led to growing concerns about the use of "black box" models and the need for explanations to better understand their inferential process [47]. Using examples of specific cases from a dataset, generated explanations might reveal which elements are most important in a model's prediction [40,49,61].

Currently, explanations generated for a trained deep learning models often are presented as individual insights that need to be investigated separately

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and then compared [3]. Each explanation provides a limited view of the model's decision, as it tends to focuse on specific aspects, making it challenging for a human to obtain a comprehensive understanding. This approach hinders the ability to discern the reasons behind a model's predictions.

There has been an emerging trend in explanation ensembling, which is derived from model ensembling, which involves combining multiple predictive models to reduce variation of predictions which often leads to higher overall performance. Examples of such predictive techniques are random forests [11] and gradient boosting [21]. This tendency shows that it is plausible that individual explanations possess unique pieces of information that, when combined, might form a more comprehensive and accurate understanding of a model's inferential process. [10,76]

However, despite the presence of metrics to evaluate the explanations' [32,63] and model's [64] quality, quantitative metrics that assess the representation learned by the model are not well-defined [71]. Instead, still many research manuscripts [53,56] use various visualisation methods, such as those visualising the components of Principal Component Analysis (PCA) [35] and t-Distributed Stochastic Neighbor Embedding (t-SNE) [41], which are challenging for quantitative evaluation.

Hence, models continue to be benchmarked on extensive datasets, like ImageNet [18]. Nevertheless, relying solely on single datasets is no longer adequate for effectively distinguishing between models in terms of quality [8]. Evaluating models on benchmarks comprising multiple datasets is recognized as both time and resource-intensive. However, it is crucial to note that even such comprehensive evaluations do not ensure an optimal performance of a chosen pretrained model when fine-tuning is applied to a specific target dataset [65].

For this reason, our method introduce metrics to evaluate three important aspects when training representation-based models, namely the quality of a dataset, the representations learned, and the completeness (exhaustiveness) of explanations. We apply novel ensembling explanations with the use of Convolutional Neural Networks (CNNs). One of ensembling explanations advantages is that they can predict the possibility of bias in both learned representation and data. This happens when the results of training our explanation ensembling method will perform much better on the training set than on the testing set.

In addition, our approach may be useful for techniques that reduce unnecessary information contained in the image data, for example by reducing the number of features. The removal of sensitive information presented in images holds particular significance in critical fields such as medicine to ensure patient privacy [5,17,68]. Furthermore, the ability to reduce unnecessary information and noise from the original image, potentially might improve training results and speed up it by focusing on relevant features, improving signal-to-noise ratio and reducing risk of overfitting [29,62].

Furthemore, our method can be particularly important for classes that might involve sensitive attributes such as race or ethnicity [69]. This will help identify the need to increase the number of images related to a specific class to support

their discrimination through model's learning. Another advantage of our method is the possibility to check the impact of the component explanations on the final explanation.

Understanding the explanations becomes crucial in this context, especially when they may conflict. This common problem can be overcome with our approach, which can be learned from explanations. It might learn to prioritize one explanation method over another, considering it more reliable. Alternatively, conflicting parts of explanations may lead to a cancellation effect, where our XAI Ensembler decides that certain areas are not relevant.

This manuscript addresses the research question of whether CNN-based explanations aids in dataset, representation, and explanation assessment. In the following sections, we present our approach, introduce metrics and conduct experiments providing insights into the suitability of our method for evaluation purposes.

2 Related Works

Explanations for deep learning models have gained considerable attention in the field of computer vision, particularly in the context of post-hoc analysis. Various approaches have been proposed to generate meaningful explanations for deep learning models [32]. Perturbation-based methods, such as Lime [49], Occlusion [72], and Shapley Value Sampling [13], involve modifying an input image to evaluate the impact on a model's output and identify significant regions or features. On the other hand, gradient-based methods, including Integrated Gradients [61], DeepLift [57], Guided GradCAM [55], GradientShap [40], or Guided Backpropagation [59], utilize gradients to highlight the salient regions of an input image that contribute most to the model's inferential process. Additionally, Noise Tunnel [27] is an approach that introduces noise into the input image and analyzes its impact on the model's output.

The ensemble of explanations has not yet received significant attention in the field of Explainable Artificial Intelligence (XAI). Zou et al. [76] proposed an approach for XAI ensembling using a Kernel Ridge regression [43] to combine the normalized Grad-CAM++ technique [55] and the normalized positive SHAP values. In an alternative method for XAI ensembling based on metrics (consistency, stability, and area under the loss) [10], the authors calculated the weighted importance of three different metrics and created a combined vector of explanations by taking a weighted sum of ensemble scores and their corresponding original explanations. Another study explored the use of mean to combine explanations generated by multiple explanation methods [50], aiming to provide a more noise-resistant ensembled explanation. Additionally, Bhatt et al. [9] proposed a XAI ensembling method based on optimization around the points of interest, minimizing explanation sensitivity and bridging the gap between AI decision-making system and human interpretability, particularly in the context of medical applications. These research works contribute to the growing body of research on XAI ensembling, offering various techniques and perspectives to enhance the interpretability and trustworthiness of AI systems. It is

worth mentioning that ensembles of explanations also appear in the literature under the title of aggregations of explanations [9,50].

Furthermore, NoiseGrad [12] and SmoothGrad [58] explanation methods can be considered as ensembles of explanations. NoiseGrad injects noise into the model's weights and then averages over perturbed models. At the same time, SmoothGrad adds noise to the input data and then averages the resulting heatmaps for each of the input images.

Evaluation of explanations using pixel-wise masks plays a crucial role to assess the quality and reliability of computer vision models. Several works have proposed metrics and approaches to evaluate explanations using pixel-wise masks. One of such contributions is the benchmark dataset CLEVR-XAI [6], which was specifically designed for the evaluation of neural network explanations with ground truth. The authors proposed two novel quantitative metrics, relevance mass accuracy and relevance rank accuracy, which are suitable for evaluating visual explanations in reference to ground-truth masks. They emphasized the importance of establishing a well-defined common ground for evaluating XAI methods, which can significantly advance XAI research. To address challenges associated with the assumption that humans can accurately recognize explanations [51,66], the Quantus library [24] was developed. This library offers a versatile and comprehensive toolkit for collecting, organizing, and explaining various evaluation metrics proposed for explanation methods. It aims to automate the process of explanation quality quantification, assisting researchers in the evaluation of explanation techniques, including localization metrics.

In the work [73], Excitation Backprop was introduced as a technique for evaluating explanations. The evaluation process involved verifying whether the single pixel with the highest relevance lies within the object corresponding to a model's predicted class, referred to as the pointing game accuracy. This metric assesses the ability of an explanation method to accurately highlight the salient regions associated with the predicted class by the underlying trained model. These notable research contributions highlight the importance of developing rigorous evaluation methodologies for explanation techniques in the context of pixel-wise masks. The proposed metrics and tools provide valuable resources to the XAI community, enabling researchers to objectively assess and compare the performance of different explanation methods in terms of their localization accuracy and relevance to ground truth masks.

Representation learning aims to distill complex data into low-dimensional, reusable structures that exhibit specific qualities [45]. A good representation is characterized by several key attributes. These include low dimensionality for efficiency, spatial coherence for proximity of related aspects, and disentanglement to isolate specific features [37]. Additionally, maintaining hierarchy and meaningfulness aids interpretability.

The utilization of models that have informative representations facilitates the generation of segmentation masks through explanations. This approach has gained popularity in weakly-supervised semantic segmentation problems

[15,16,38]. Typically, these studies employ the Class Activation Mapping (CAM) explanation method to provide explanations and then to generate segmentation masks. However, recent work [54] demonstrates that other explanation methods can also be used to obtain pixel-wise masks.

In summary, while post-hoc explanations for deep learning models have seen significant advancements, the potential of generating masks using explanations ensemble methods remains unexplored. To the best of our knowledge, there are no metrics for the evaluation of explanations that are based on the relevance of the component explanations in the ensemble of explanations. In the upcoming section, we delve into details of the method and conduct experiments to explore the effectiveness of our CNN-based explanations ensembling.

Evaluation of representation goes beyond individual image analysis using post-hoc explainability methods. Moreover, relying solely on the accuracy on ImageNet may be misleading, as a high accuracy metric does not guarantee effective representation learning or generalization. Beyer et al. [8] emphasize that performance metrics on large datasets may not capture a model's ability to handle diverse data. Beyond accuracy, evaluating representations through interpretability methods or transfer learning tasks provides a more comprehensive measure of a model's capabilities, ensuring it generalizes effectively in real-world scenarios.

While t-SNE [41] is commonly chosen to compare learned representations visually [2,44], it has limitations. Firstly, t-SNE effectively preserves local structure but may not accurately represent global relationships in the data. It tends to focus on preserving pairwise similarities between nearby data points, which may not capture the overall structure of the data. Secondly, t-SNE has hyperparameters, such as the perplexity, that need to be set, which can significantly impact the resulting visualization. Selecting appropriate values for these hyperparameters can be subjective and may lead to different interpretations of the data. Lastly, t-SNE can be sensitive to noisy or sparse data, potentially producing misleading visualizations. In the context of evaluating deep learning representations, noisy or irrelevant features may influence the t-SNE visualization in ways that do not accurately reflect the quality of the learned representations.

To conclude, t-SNE can be a powerful tool for visualization, but it fails in quantitative assessment of data representation quality. Additionally, precision, recall, f1 score and auc roc are good metrics for assessing the quality of classification on a given dataset. However, they are not suitable for assessing the quality of representation, because they do not check the image features learned by a model.

3 The Concept of CNN-Based Ensembled Explanations

In our proposed methodology, presented in Fig. 1 and Algorithm 1, we introduce a novel approach for ensembling explanations to enhance the interpretability of classification models. The process begins by generating explanations using

established techniques for the trained classification model. Consider a set of p explanation methods. Let $\mathcal{E}_{ij}$ represent the explanation created by j-th method for the i-th image (I_i) in the dataset. For each image, we aim to generate a set of explanations:

$$\mathcal{E}_i = \{\mathcal{E}_{i1}, \mathcal{E}_{i2}, \ldots, \mathcal{E}_{ip}\}. \tag{1}$$

To ensemble these explanations, we employ two strategies depending on the availability of ground truth masks. Let $\mathcal{M}_i$ denote the pixel-wise annotated mask corresponding to the i-th image. In the first strategy, we utilize pixel-wise annotated masks for a selected class of images, which serve as a reference for generating ensembles. Alternatively, in cases where annotated masks are not readily available, a domain specialist can manually annotate a small subset of representative images.

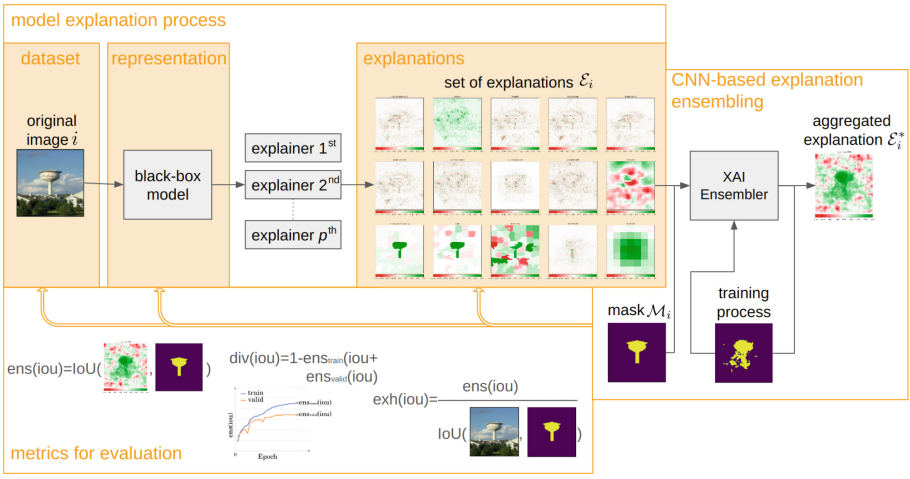


Fig. 1. Overview of CNN-based ensembling of explanations. The diagram illustrates the process that involves an original image being processed by a black-box classification model, followed by an explainer generating explanations. These explanations, along with masks depicting objects in the original image, are then used to train XAI Ensembler, which architecture is presented in Fig. 2. The ensembler's training process results in an aggregated image. CNN-based ensembling of explanations with metrics we defined can be used to assess dataset, representation and explanations.

Once the explanations are generated, our objective is to train a CNN-based explanation ensembling model (XAI Ensembler), shown in Fig. 2, to predict masks given explanations. The training procedure involves minimizing the segmentation loss $\mathcal{L}_{\text{seg}}$ as follows:

$$\mathcal{L}_{\text{seg}} = \sum_{i=1}^{n} \text{SoftDice}(\mathcal{M}_i, \text{XAI_Ensembler}(\mathcal{E}_i)), \tag{2}$$

where n represents the total number of images. The Soft Dice coefficient [42], refers to a similarity metric that quantifies the overlap between predicted and ground-truth segmentation masks and is utilized to assess the segmentation accuracy of the *XAI Ensembler* model. The explanation aggregation obtained by XAI Ensembler is denoted as $\mathcal{E}_i^*$.

The detailed training procedure for this segmentation model is outlined in detail in Sect. 3.1. It provides comprehensive insights into the specific steps, techniques, and parameters employed during the training process. We aim to train a segmentation model that effectively captures and interprets the underlying image features by incorporating the generated explanations and their corresponding masks.

After training the segmentation model using the explanations and masks, we progress to evaluating its performance with unseen images. This evaluation serves as a crucial step in assessing the generalization capabilities and effectiveness of the trained segmentation model. By applying the model to unseen data, we can gain valuable insights into its ability to accurately delineate the regions of interest and align with human perception.

Overall, our proposed method of ensembling explanations offers a unique perspective on interpreting and understanding the predictions of a trained classification model. Through the integration of explanations and the subsequent training of a segmentation model, we strive to enhance the interpretability and trustworthiness of the underlying classification model. The evaluation of the trained segmentation model with unseen images serves as a crucial testing step, providing valuable insights into the complex classification model.

3.1 Experimental Setup for Training

Our experiments employed a segmentation model based on CE-Net [22], which has shown promising performance in various image analysis tasks [28]. The dataset used for training and evaluation was ImageNet-S50 [19]. We split the data 80/20 for training and testing. The input images were resized to 224×224 pixels. During training, we utilized a learning rate of 0.0002, and the learning rate was halved when no improvement was observed, with a minimum interval of 20 epochs. The training process was stopped if the learning rate dropped below 10^{-9}, or after 200 epochs.

To optimize the segmentation model, we employed the Soft dice loss [42] and utilized the Adam optimizer [31]. The Soft dice loss has been proven effective in handling classes with lesser spatial representation in an image, and is commonly used in medical image segmentation tasks [20].

To enhance the robustness of the model and minimize overfitting, we applied various image transformations, which are standard for ImageNet. These included a center crop to focus on the central region of the image, Imagenet normalization to standardize pixel values, random adjustments of hue and saturation

Algorithm 1. CNN-based Ensembled Explanations training

if pixel-wise masks available **then**
  Utilize pixel-wise annotated masks $\mathcal{M}$ for ensembling.
else
  Manually annotate a small subset of representative images.
end if
Select the classification model for which the ensemble of explanations will be generated.
Choose the number of epochs e.
Count total number of images n.
Select p explanation methods for ensembling.
Define the architecture of the CNN-based explanation ensembling model
for $d = 1$ to e **do**
  for $i = 1$ to n **do**
    for $j = 1$ to p **do**
      Generate explanation $\mathcal{E}_{ij}$ for the trained classification model.
    end for
    Train the CNN-based explanation ensembling model using the explanations $\mathcal{E}_i$ and masks $\mathcal{M}_i$ (see Section 3.1).
  end for
end for

values, random shifts, scaling, and rotations, random flips to introduce horizontal and vertical symmetry variations, as well as random rotations by 90°.

The code written using PyTorch library is available on Github: https://github.com/Hryniewska/CNNbasedXAIensembling.

3.2 Ablation Studies

The objective of our experiments is to select the best ensemble model's architecture. We created three different model's architectures, presented in Fig. 2. The first and the second architecture used many encoders. Each encoder takes one explanation method as an input; therefore the number of encoders is linearly dependent on the number of explanations that are going to be ensembled.

In the first architecture, we used concatenation as the method for combining inputs from multiple encoders. Concatenation involves stacking encoders' outputs along a new channel dimension. This type of stacking is useful when you want to retain all the information from each input and allow the network to learn how to use each input separately. It is particularly suitable for problems where the inputs represent different aspects of the problem. It is worth noting that the dimensionality of the input space increases when using concatenation, potentially leading to a large number of model parameters. This architecture's total number of parameters is 17 675 256 multiplied by the number of encoders squared.

The second approach employed summation as the combining method. It entails an element-wise addition of the encoder outputs, channel by channel. Summation is used when the inputs are expected to be complementary, and it emphasizes their joint influence on the output. This method helps keep the model's number of parameters low (17 675 256 parameters) and can be beneficial when the inputs are additive.

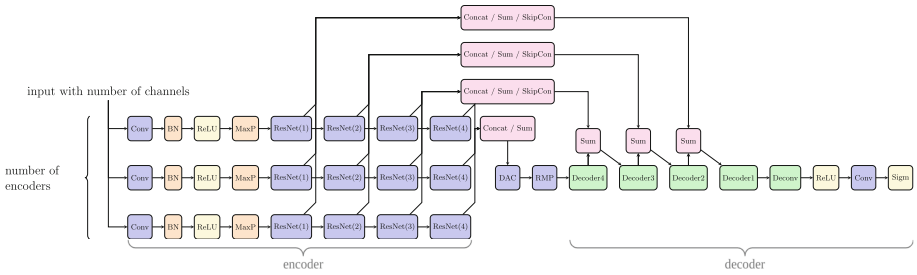


Fig. 2. Proposed three architectures of CNN-based explanation ensembling: concatenation, summation and multi-channel approach. Key architectural components are abbreviated as follows: BN (BatchNorm2D), MaxP (MaxPool2D), Conv (Conv2D), Concat (Concatenation), Sum (Summation), and SkipCon (Skip connection). The number in brackets after "ResNet" denotes the specific number of layers in the ResNet architecture.

The last architecture consists of a single encoder that takes explanations in channels, resulting in the number of channels being a product of three and the number of explanations for ensembling. The Captum library [32], employed for generating explanations, provides output with three channels for each explanation. The output values fall within the range of [–1, 1] or [0, 1], depending on the explanation method. Additional channels in convolutional layers are appropriate when we need to leverage the spatial relationships in the inputs while allowing the network to learn features from each input independently. It is commonly used in tasks like multi-modal image processing, where different inputs (RGB image, depth map, infrared image) are processed jointly to improve feature extraction [36]. The number of parameters is equal to 17 675 256.

Typically, the choice among these methods depends on empirical experimentation and the specific requirements of a machine learning or deep learning task. It is important to consider the unique characteristics of the data and the problem to determine which input combination method is most appropriate.

For this reason, we trained all three architectures using the same training procedure, as described in Sect. 3.1. The proposed model architectures were trained on several dozen models with the same configurations encompassing various explained deep learning classification models, explanation methods, and specific classes. These configurations included explained deep learning classification models, such as SqueezeNet1.1, MobileNet v2, VGG16, DenseNet121, ResNet50,

and EfficientNet B0. An effort was made to select diverse classes, e.g. objects that are large or small, popular or rare, always look the same or always differently. The target classes comprised digital watch, dining table, gibbon (monkey), grand piano, kuvasz (dog), ladybird, purse, umbrella, beach wagon, water bottle, and water tower. The experimental setup for sets of explanation methods consisted of the following configurations: zero explanations (the original image is displayed instead of the explanation), three local explanations that are based on different approaches (Shapley Value Sampling, NoiseTunnel, Guided Backprop), four explanations based on the most frequently cited XAI methods (GradientShap, Lime, Occlusion, Guided GradCAM) and seven explanations that show differentiation between XAI methods (DeepLift or Integrated Gradients for ResNet, GradientShap, Guided Backprop, Guided GradCAM, Lime, Occlusion, Shapley Value Sampling).

The obtained results were averaged and are presented in Table 1. The detailed interpretation of the used metrics is presented in Sect. 4. However, it can be seen that the best results are obtained for the architecture with the concatenation operation, so all further experiments will only be performed using this architecture.

Table 1. Performance metrics of different ensembling model architectures on training and testing sets.

Ensembling method	Training				Testing			
	ens(acc)	ens(f1)	ens(IoU)	loss	ens(acc)	ens(f1)	ens(IoU)	loss
Concatenation	0.930	0.828	0.734	0.314	0.865	0.723	0.599	0.405
Sum	0.898	0.615	0.504	0.233	0.809	0.511	0.407	0.379
Channel	0.881	0.546	0.440	0.271	0.792	0.425	0.332	0.407

3.3 Method Evaluation

The decision to employ the Quantus library for the evaluation of our CNN-based XAI explanation ensembling model is rooted in its capability to offer a comprehensive and systematic comparison of various XAI methods. Quantus [24] provides a library for evaluating the performance of XAI techniques, ensuring a thorough examination of key aspects.

The chosen metrics for this evaluation were thoughtfully selected to optimize the meta-consistency score [25]. It is pivotal as it indicates the stability and reliability of an XAI method across different contexts and datasets. Prioritizing metrics contributing to a higher meta-consistency score ensures that our method not only explains individual instances effectively but also maintains consistency across diverse scenarios.

The radar plot in Fig. 3 visually represents the comparative performance of our CNN-based XAI ensembling method against individual non-ensembled XAI methods. The inclusion of key evaluation metrics-Faithfulness, Robustness,

Localisation, Complexity, and Randomisation-offers a holistic view of a method's capabilities. The subsequent paragraphs delve into detailed explanations of each of these five metrics.

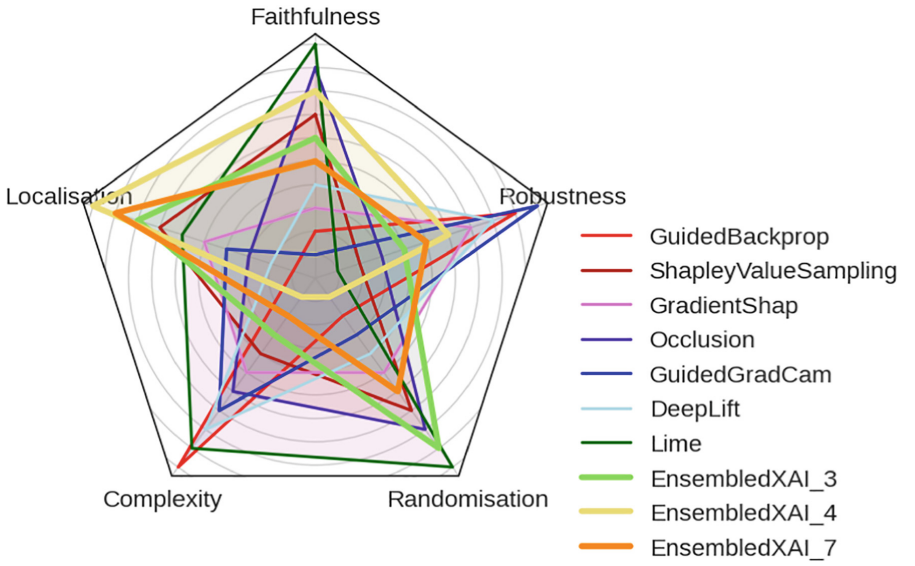


Fig. 3. Radar plot comparing the performance of CNN-based explanation ensembling with individual non-ensembled XAI methods. The plot displays the ranking of explanations in five key evaluation metrics: Faithfulness, Robustness, Localisation, Complexity, and Randomisation. Higher values indicate better performance.

Localisation (Pointing-Game [73]*).* This metric helps to evaluate whether the highest-scored attribution in the explanation is accurately localized within the targeted object or region of interest. It provides valuable insights into the accuracy and precision of the explanations in pinpointing relevant features. In terms of Localisation, CNN-based explanation ensembling method exhibited the best performance over all explanations.

Faithfulness (Pixel-Flipping [7]*).* This metric assesses the faithfulness of the explanations by perturbing pixels in descending order of their attributed values and observing the impact on the classification score. The results are aggregated using mean. Our CNN-based explanation ensembling method achieved competitive scores in the Faithfulness metric, ranking near the top.

Robustness (Local Lipschitz Estimate [4]*).* This metric measures the consistency of explanations between adjacent examples. It quantifies how much the explanations change when the input data points are slightly perturbed, providing insights into the stability and reliability of the explanations. Our CNN-based explanation ensembling method demonstrated relatively good robustness, ranking in the middle of the other selected XAI methods.

Complexity (Sparseness [14]*).* This metric utilizes the Gini index to measure the sparseness of the explanations. It helps assess whether an explanation predominantly highlights highly attributed features that are truly predictive of a model's output. The complexity metric favors explanations that fill a small part of the segmentation mask. The principle of our method does not allow for high scores on this metric. In our approach, the explanation should contain as much of the mask as possible. However, during the XAI Ensembler model's inference, the cut-off can be increased to get smaller areas marked.

Randomisation (Model Parameter Randomisation [1]*).* While MetaQuantus [25] recommends Random Logit for randomizing model parameters, in evaluation of our XAI ensembling models, we used an alternative approach due to the impossibility of carrying out calculations for a random other class. The ModelParameterRandomisation metric involves randomizing the parameters of single model layers in a cascading or independent manner. It then measures the distance between the original and randomized explanations, providing insights into the robustness of the original explanations. The results are aggregated using mean. In the Randomisation metric, our CNN-based explanation ensembling method achieved relatively competitive scores, particularly when ensembled with the three XAI methods, outperforming GuidedBackprop and GuidedGradCam. However, suprisingly, the scores associated to our ensembling with 4 XAI methods were low, which may be related to the component explanations chosen.

By utilizing the Quantus library and the metrics following MetaQuantus [25] recommendations, we were able to systematically evaluate and compare the performance of our CNN-based explanation ensembling. Higher values on the radar plot signify superior performance in these metrics. Our XAI method may excel in faithfully representing the underlying model, demonstrating potential robustness across diverse scenarios. Moreover, it may accurately localise influential features, manage complexity effectively, and exhibit controlled randomization. However, our approach results in lower scores in the complexity metric.

This evaluation approach emphasizes the qualitative aspects of the proposed solution, going beyond a mere quantitative analysis. By aligning with best practices in XAI evaluation, the use of Quantus metrics may help enhancing the credibility and reliability of our XAI ensemble method.

4 Metrics for Representation, Dataset and Explanation Evaluation

While post-hoc explainability methods provide insights into individual images, they lack a holistic view. These methods fail to comprehensively assess factors like dataset representativeness, learned representations, and explanation quality. Furthermore, relying solely on ImageNet accuracy can be misleading. Even visualization techniques like t-SNE fail in quantitative assessment of data representation quality.

Representation-Oriented Ensembling Performance Metric
To address these challenges, a novel metric is proposed for quantitatively assessing data representation quality through CNN-based XAI ensembling. The objective of explanations is to reveal the elements within an image that guide the model's inferential process. Evaluating these explanations becomes challenging when multiple methods generate diverse interpretations [34]. This is where CNN-based XAI ensembling could prove valuable. The model endeavors to learn to segment the object by exclusively considering explanations and the ground-truth mask of the classified object. Success in learning indicates that explanations might effectively represent the classified object, minimizing the impact of noise for an accurate interpretation.

The matrix $\mathcal{E}_i$, defined in values within the range $[-1, 1]$. To compute the evaluation metrics for *representation-oriented ensembling performance (ens)*, we discretize $\mathcal{E}_i$ into a binary version denoted as $\mathcal{E}^*_{bin}$, where values above a selected cut-off are set to 1, and values below are set to 0. In this research work, we use cut-off equal to 0.5, which is typical for segmentation tasks. The evaluation metric ens(iou), which measures the overlap between the one values of $\mathcal{E}^*_{bin}$ and any overlapping ground-truth instances $\mathcal{M}$, is calculated as follows:

$$ens(iou) = IoU(\mathcal{E}^*_{bin}, \mathcal{M}). \tag{3}$$

Other metrics for CNN-based XAI ensembling, such as ens(f1) or ens(acc), can be calculated analogously by comparing the binary ensemble of explanations ($\mathcal{E}^*_{bin}$) with the ground truth ($\mathcal{M}$). True positives, true negatives, false positives and false negatives are determined based on the overlap between the binarised explanations and the ground truth.

Data-Oriented Diverseness Metric
In scenarios, when an image contains multiple objects (for example, "cat", "dog", "mouse", "cottage") and the class assigned to the image is "animals", traditional classification metrics may not show how well the model has actually learned. This limitation poses challenges for the manual analysis, making it difficult to discover potential issues during model training. Our proposed method shows that the XAI Ensembler can have the capability to explain what the underlying model has learned from the explanations and highlight disparities from the expected outcomes. Through the analysis of these distinctions, it allows for a precise understanding of rare or peculiar images, different from the average content of most of the other images, and the reasons behind their challenges.

Furthermore, to address the performance disparities between test and training data, we introduce the *data-oriented diverseness (div)* metric. This metric serves as an indicator of a dataset's quality, determining whether a sufficiently diverse set of images is present to train a model with a robust representation. Denoted as *div(m)*, it quantifies the similarity between validation and training performance metrics. Specifically, for IoU, the formula is expressed as:

$$div(iou) = 1 - ens_{train}(iou) + ens_{valid}(iou). \tag{4}$$

Low values of *div(iou)* suggest that there are substantial differences between the model's ensembler performance on training and testing datasets. This can be an indication of potential issues in the dataset, such as insufficient image diversity, leading to challenges in training a model that generalizes well across various scenarios.

Explanation-Oriented Exhaustiveness Metric

The *explanation-oriented exhaustiveness (exh)* metric helps to assess the completeness of an explanation relative to an entire input image, measuring the difficulty of extracting the object of interest in explanations compared to segmenting it in the original image. Specifically, for the Intersection over Union (IoU) metric, the formula is given by:

$$exh(iou) = \frac{ens(iou)}{IoU(I_i, \mathcal{M})}. \tag{5}$$

This metric provides insights into how well the generated explanations capture the salient features of an object within the broader context of the segmenting original image. A higher *exh* value indicates a more exhaustive and reliable explanation, emphasizing a model's ability to highlight important elements.

For each of the above metrics, the visuals in Fig. 4 highlight the actual problem and what they actually are aimed to measured.

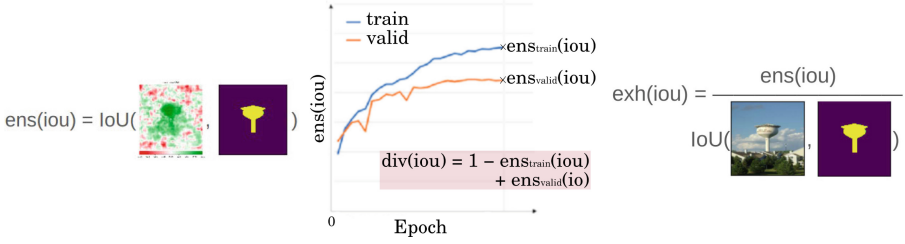


Fig. 4. A simplified illustration of introduced metrics: ensembling performance, diverseness and exhaustiveness.

4.1 Representation Evaluation

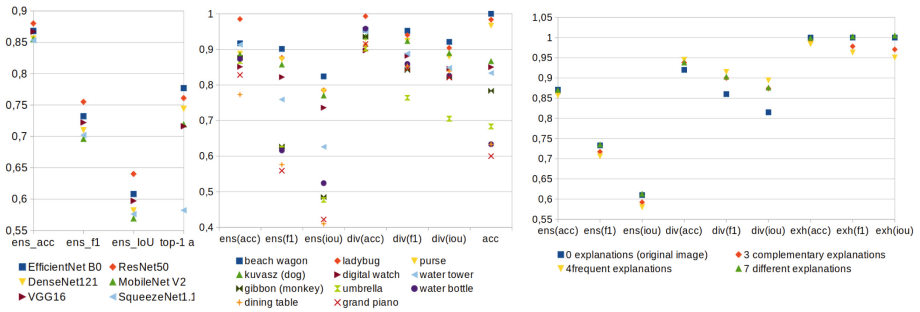
High accuracy on datasets like ImageNet may not be enough for thorough model evaluation [8]. The experiment focuses on adding an interpretability layer to enable a model's explanation, allowing a deeper understanding of the learned representations beyond accuracy metrics. This aligns with the perspective that assessing model performance requires considering not only predictive accuracy but also interpretability and the meaningfulness of the learned representations.

This experiment aims to determine which classification model from SqueezeNet1.1, MobileNet v2, VGG16, DenseNet121, ResNet50, and EfficientNet B0 learned the best representation. To achieve this, we trained XAI ensembling models using explanations generated by mentioned models. We decided

to average the results for eleven classes and three different sets of explanations, listed in legends to Figs. 5b and 5c, respectively. The results are shown in Fig. 5a. Moreover, in Fig. 5a a performance comparison of various architectures on ImageNet dataset is presented, illustrating their top-1 accuracy.

From Fig. 5a, several key observations can be made. Namely, the ensembling accuracy metric is the least differentiating. This is in line with assumptions, because segmentation models are evaluated at the pixel level and often there is a significant class imbalance between the object area and the background. ResNet50 outperforms all other models in all ensembling performance metrics. However, in classification metrics, it has lower accuracy than EfficientNet B0, which took second place in ensembling performance metrics. The order of models according to ensembling f1 and IoU metrics is the same. This is because both metrics are similar to each other, but it is worth noting that they are very suitable for evaluating the segmentation task. VGG proved better at learning representation than the very deep DenseNet121 model and two small models: MobileNet and SqueezeNet. What may seem surprising is the result of SqueezeNet, which according to ensembling performance metrics, positions itself higher than MobileNet, which contradicts the accuracy metric, of which it is more than 10% lower.

To sum up, our experiment highlights the strong performance of ResNet50, a trend consistent with its frequent use as a backbone in various computer vision tasks, including semantic segmentation (as seen in Mask R-CNN [23]) and object detection (as seen in Faster R-CNN [48]). Researchers and practitioners often leverage its capabilities as a feature extractor [26], utilizing pre-trained ResNet weights to boost the performance of their models on specific tasks.



(a) Ensembling performance metric for different models.

(b) Ensembling performance, and diverseness metric for different classes.

(c) Ensembling performance, diverseness, and exhaustiveness metric for different sets of explanations.

Fig. 5. Averaged results of introduced metrics on the testset.

4.2 Dataset Evaluation

In this experiment, our objective was to investigate whether the difficulty of an image class from the ImageNet dataset could be evaluated through explanation ensembling. We trained CNN-based explanation ensembling on eleven various image classes using 5-fold cross-validation, explanations generated by ResNet50, and using three sets of explanation methods. Cross-validation is important especially for dataset evaluation, as it reduces the impact of splitting the dataset into training and test data. The averaged across cross-validations performance metrics are shown in Fig. 5b.

By analyzing the performance metrics, we can gain insights into the difficulty of each image class. Firstly, identifying classes with the lowest *ens(f1)* and *ens(IoU)* highlights challenging classes, such as "dining table" and "grand piano". Addressing the difficulties in recognizing these classes may be crucial for applications like training models for object recognition in autonomous vacuum cleaners. Particularly, in some cases, it is important to ensure fairness by preventing subjects such as "monkeys" from being misidentified as "humans". Secondly, it is important to examine the cases where the greatest difference in the *ens(f1)* and *ens(IoU)* between the training and testing sets occurs. Taking the "water bottle" as an example, presented in Fig. 6, understanding such disparities is essential, especially if the context of the images in the dataset significantly differs between themselves, as observed in the varying poses and contexts of "water bottles".

The last column in Fig. 5b presents the accuracy of the classification task. The correlation can be seen between the individual class score in the introduced metrics (ensembling performance and diverseness) and the accuracy score. From the obtained results, it can be seen that classes that get high scores in the classification usually get high scores in our metrics. Our metrics have the advantage over the simple classification metrics, because their results can be easily visualised as masks or heatmaps, which help to find the reason for a potential misclassification error.

By leveraging explanation ensembling, we can gain a holistic understanding of the difficulty level associated with different image classes. This information can be valuable for various applications, such as prioritizing resources for

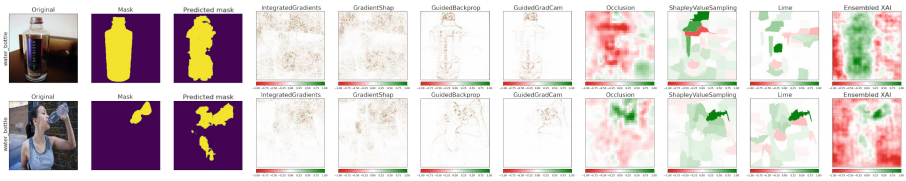


Fig. 6. Two examples from the ImageNet dataset belonging to the "water bottle" class. The first example is commonly found in the dataset, while the second example is challenging and infrequently encountered. Each example consists of an image, a mask, a predicted mask, and individual component explanations. The final component is the ensembled explanation.

challenging image classes, improving model performance on difficult classes, or tailoring training strategies to address class-specific difficulties.

4.3 Explanations Evaluation

In this section, we present the experimental results of our study, which aimed to answer the research question of how well the explanations succeeded in extracting relevant features. We also determine the influence of each input explanation on the final result.

The experimental setting is the same as in the previous subsection. To understand the significance of the number of XAI methods incorporated in the CNN-based explanation ensembling, we visualised the results in Fig. 5c. The measure *exh(iou)* determines how much of the important features are left in the picture. Our analysis revealed that the results of segmenting an original image and applying XAI ensembling on seven different explanations obtained almost the same results.

These findings indicate that, based on the selected performance metric, seven XAI methods have the potential to deliver the same predictive performance as the original images. The original image contains many more features than its explanations. Demonstrating that explanations can serve as substitutes for original images, this approach holds promise for facilitating reduction of redundant information or the anonymization process of images. Anonymization involves transforming data into a format that does not disclose individuals' identities. In this context, one possible application could be modifying existing images [67], or alternatively, employing synthetic data generation as an alternative method for data anonymization.

Several studies report that no single explanability method performs best in all metrics evaluating the quality of an explanation [30,33,74]. However, it is hard to clearly state how explanability methods should be compared among themselves. It is possible to choose a single metric, but according to each metric, the result may be different. Faced with this, comes the problem of evaluating the validity of metrics for evaluating explanations among themselves.

In our solution, we do not have such a problem. Our explanation ensembling solution allows, after training the XAI Ensembler, to disable, in other words set the attribution values to zero, individual input explanations. Disabling one component explanation can measure its validity in the aggregated explanation. For this reason, we check the impact of different explanability methods on the ensembling result.

In Table 2, the "difference" columns show the average difference in *ens(acc)*, *ens(f1)* and *ens(IoU)* between the model with all XAI methods and the model with the specified XAI method removed. The "quotient" columns show the average ratio of the performance of the model with all the XAI methods to the performance of the model with the specified XAI method removed.

As Table 2 shows, the GuidedBackprop explanation contributes the most. What is worth noting is that the explanations are complementary. After removing one of them, there is no significant drop in the quality of the generation of the

corresponding mask (the difference in the resulting mask when giving all explanations in the input and when giving all of them without one explanation). It is important to note that these results are based on a ResNet50 model and may not generalize to other models or datasets.

Table 2. Impact of removing XAI methods on ensemble model performance. The "difference" columns show the average decrease in performance metrics, while the "quotient" columns express the remaining performance as a ratio compared to the model with all XAI methods.

Removed XAI	Difference			Quotient		
	ens(acc)	ens(f1)	ens(IoU)	ens(acc)	ens(f1)	ens(IoU)
GuidedBackprop	0.0478	0.0770	0.0964	0.9465	0.8937	0.8501
GuidedGradCam	0.0208	0.0413	0.0482	0.9763	0.9389	0.9156
ShapleyValueSampling	0.0086	0.0113	0.0135	0.9899	0.9667	0.9586
Occlusion	0.0072	0.0110	0.0116	0.9917	0.9666	0.9600
GradientShap	0.0037	0.0037	0.0035	0.9963	0.9874	0.9853
Lime	0.0028	0.0004	0.0003	0.9972	0.9906	0.9901
IntegratedGradients	0.0025	-0.0005	-0.0012	0.9974	0.9911	0.9912

5 Conclusions and Future Works

The novel aspect of our CNN-based explanation ensembling lies in using explanations to evaluate the learned representation. Our approach provides a unique way to assess the difficulty of each class and the quality of trained models. It aids in the selection of a pretrained backbone for tasks like detection, providing quantitative evaluation and dataset analysis. Noteworthy is the possibility of ensembling explanations from trained vision transformers, as their explanations are also commonly presented as heatmaps [60].

We gain insights into where the model excels and where it struggles by analyzing the performance metrics of the generated explanations for many class. Through the study on the elements that should or should not be present in the ensembled explanation, our method helps to predict the possibility of bias, which is particularly crucial for sensitive attributes, and suggests strategies to address imbalances in the dataset. This valuable information can guide further improvements in dataset preparation and model training. Our approach finds unnecessary information in data, which may help in designing algorithms that enhance anonymity.

Morever, our CNN-based explanation ensembling presents a novel approach to address the challenge of generating comprehensive explanations, in which we do not have to worry about the importance of each of the aggregated explanation methods. Through the use of carefully selected evaluation metrics

from the Quantus library, we demonstrated the method's superior performance in terms of Localisation and Faithfulness, compared to individual XAI methods.

While we have highlighted the key strengths of our approach, it is essential to acknowledge its limitations. The performance of our method is highly dependent on the quality and diversity of the training data, as limited or biased training data can result in suboptimal explanations and potentially undermine a model's reliability. Furthermore, ensembling explanations using CNNs can be computationally intensive, which may limit its applicability in resource-constrained environments. Interpretability remains an ongoing challenge, as even the best explanations might not fully capture the complex inferential processes of deep learning models. It is crucial to recognize these limitations to ensure the responsible and informed use of the CNN-based explanation ensembling.

In light of these limitations, future research can explore several directions. First of all, we see a growing area of universal, robust feature extractors, such as DINOv2 [46]. We believe that the CNN-based explanation ensembling might be available as pretrained backbone and work on many different classes. Additionally, an idea of the trainable XAI Ensembler can be generalised to other modalities. The code for our CNN-based explanation ensembling is available at https://github.com/Hryniewska/CNNbasedXAIensembling.

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Local List-Wise Explanations of LambdaMART

Amir Hossein Akhavan Rahnama(✉), Judith Bütepage, and Henrik Boström

KTH Royal Institute of Technology, Stockholm, Sweden
{amiakh,butepage,bostromh}@kth.se

Abstract. LambdaMART, a potent black-box Learning-to-Rank (LTR) model, has been shown to outperform neural network models across tabular ranking benchmark datasets. However, its lack of transparency challenges its application in many real-world domains. Local list-wise explanation techniques provide scores that explain the importance of the features in a list of documents associated with a query to the prediction of black-box LTR models. This study investigates which list-wise explanation techniques provide the most faithful explanations for LambdaMART models. Several local explanation techniques are evaluated for this, i.e., Greedy Score, RankLIME, EXS, LIRME, LIME, and SHAP. Moreover, a non-LTR explanation technique is applied, called Permutation Importance (PMI) to obtain list-wise explanations of LambdaMART. The techniques are compared based on eight evaluation metrics, i.e., Consistency, Completeness, Validity, Fidelity, ExplainNCDG@10, (In)fidelity, Ground Truth, and Feature Frequency similarity. The evaluation is performed on three benchmark datasets: Yahoo, Microsoft Bing Search (MSLR-WEB10K), and LETOR 4 (MQ2008), along with a synthetic dataset. The experimental results show that no single explanation technique is faithful across all datasets and evaluation metrics. Moreover, the explanation techniques tend to be faithful for different subsets of the evaluation metrics; for example, RankLIME out-performs other explanation techniques with respect to Fidelity and ExplainNCDG, while PMI provides the most faithful explanations with respect to Validity and Completeness. Moreover, we show that explanation sample size and the normalization of feature importance scores in explanations can largely affect the faithfulness of explanation techniques across all datasets.

Keywords: Explainability for Learning to Rank · Local list-wise explanations · Local explanations · Explainable Artificial Intelligence · Explainable Machine Learning

1 Introduction

Learning-to-Rank (LTR) Models are omnipresent in our daily lives; whether we search for a song on our music streaming service or search for a product on e-commerce websites, they provide us with relevant documents (items) [21].

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However, the size and complexity of ranking datasets are growing, and therefore, Learning-To-Rank (LTR) models are becoming increasingly complex [7,8].

LambdaMART is an LTR boosted tree model, widely popular in the LTR research community, which has achieved excellent predictive performance across numerous LTR benchmarks, often outperforming neural networks [28]. Even though a single decision tree can be interpretable under certain circumstances [14], LambdaMART is considered to result in black-box models, often consisting up to several hundred or thousand trees [8]. Understanding such models is clearly too challenging in most cases. Moreover, studies have shown that such complex tree models, such as those generated by LambdaMART, can fail silently, e.g., in the case of adversarial examples [9,39]. Therefore, there is an apparent need to better understand the logic behind the predictions of such models.

Local explanations of LTR models aim to provide insightful information about the predicted score or ranking for the document(s) for a given query. These explanations are obtained post-hoc, i.e., after training the LTR model. One of the most popular forms of local explanations is feature attributions, where each feature is assigned an importance score that represents its contribution to the predicted output [16,25][1]. These explanations can be model-based or model-agnostic. Model-agnostic explanations are widely popular since, unlike their model-based counterparts, they make no assumption on the internal workings of the explained model [16].

In Learning-to-Rank (LTR), Local model-agnostic explanations are further divided into two categories: point-wise, where an explanation is obtained for a single document that has been associated with a given query [32,35], or list-wise, where a single explanation is provided for explaining the prediction of a set (list) of documents for a given query [12,33,34] (see Fig. 1).

Point-wise and List-wise explanations have different uses [12,33]. Consider the case where a single document has received a surprisingly low (high) predicted relevance score (rank) by an LTR Model; for example, a search query for the song "Drake Love" allocates a low relevance score to the song "Love All" by Drake. We can understand what features contributed to this surprising prediction by obtaining a point-wise explanation. In some cases, the predicted relevance scores for a list of documents associated with a query have surprisingly low (high) predicted scores (ranks); for example, the search query "Hotel Malta" may allocate low predicted scores to a list of hotels in the outskirts of Valletta (the capital of Malta). We can understand the underlying contributing features for those surprising relevance scores by obtaining a list-wise explanation. To tackle these cases, model users and developers can use the explanations to adjust feature scaling, de-biasing the features, add interaction terms between features, or even re-train the model [4,19].

There are two main approaches to obtain list-wise explanations. In the first approach, point-wise explanations of individual documents are aggregated into a list-wise explanation, e.g., using a normalized sum function [32,35]. In the

[1] Other types of local explanations include rules, counterfactual explanations, and examples, which, however, fall outside the focus of our study; see [25] for more details on such types of explanation.

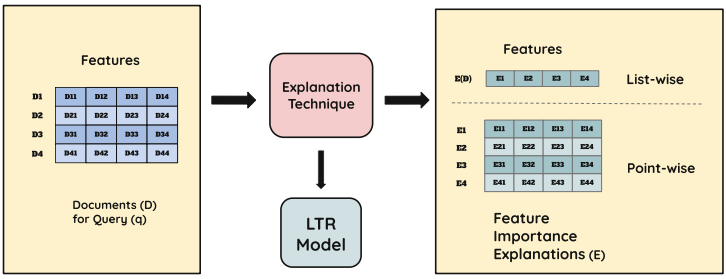

Fig. 1. List-wise $E(D)$ and Point-wise Explanations E_i for D_i $(i = 1, ..., 4)$ have different interpretations and utility. Point-wise explanations show us the importance of features for the change in the predicted rank of each document D_i. In contrast, list-wise explanations show us the importance of features for the list of documents D. It is possible to aggregate the point-wise explanations for all documents D_i to obtain list-wise explanations.

second approach, the explanation technique form the explanation directly from the joint list of documents. In [12,34], it is argued that the first approach has limitations since point-wise explanation techniques consider the feature values of other documents to be fixed. By supporting their arguments using *synthetic examples*, they showed that aggregating importance scores in this way can cause some importance to cancel out one another even though they are important for all documents.

Local explanations come with a caveat: their evaluation. The challenge is that the ground truth importance scores cannot be directly extracted from the complex black-box models [2,17,18,29]. However, several measures for evaluating local explanations have been proposed in the literature [1,12,36].

We have noticed several gaps in the literature on the list-wise explanations of LTR models. Firstly, no open code or benchmarks are available for explanation techniques of LTR models, similar to [1,17] in the context of supervised learning models. Secondly, in the current studies, LTR explanation techniques are compared and evaluated with respect to only a small subset of the available explanation techniques and proposed evaluation measures (See 2 for a detailed discussion). Thirdly, previous studies have focused mainly on evaluating neural network approaches to LTR models trained on a few text datasets [12,32,35] even though studies have shown that LambdaMART provides state-of-the-art performance also for numerous tabular data LTR tasks [28]. Moreover, tabular data is extensively used in high-stakes decision-making domains where explanations are needed the most, such as clinical search [3] or risk assessment [38]; the current studies have not focused on such tabular benchmarks. We argue that the gap above hinders the rapid development and comparison of explainability techniques for such crucial tasks.

Our study aims to tackle these gaps as follows. Firstly, we provide the implementations of an inclusive set of LambdaMAT's available explanation techniques

openly available at our GitHub repository[2]. Secondly, we focus on evaluating list-wise explanations of LambdaMART on an inclusive set of three benchmark datasets, namely Yahoo [7], Microsoft Bing Search (MSLR-WEB10K), and LETOR 4 (MQ2008) [27], along with a synthetic use case. Thirdly, Permutation Importance (PMI) explanations [13] are for the first time evaluated in the context of list-wise expansion of LTR models. Unlike LIME and similar techniques that require surrogate models, PMI follows a rather straightforward logic; see Sect. 3.2.1 for details on this. We evaluate the faithfulness of PMI along with other list-wise explanation techniques: Greedy-Score [33] and RankLIME [12]. To investigate whether the aggregation of point-wise explanations is a suitable alternative for obtaining list-wise explanations, we compare all the techniques mentioned above with the aggregated point-wise LTR-based explanations of LIRME [35], EXS [32], and non-LTR-based point-wise explanations of LIME [30] and KernelSHAP[3] [23]. The aggregation is performed over all documents for a given query. In this study, we aim to thoroughly investigate the faithfulness of explanations across several measures. These are Consistency, Validity, Completeness, (In)fidelity, Fidelity, and ExplainNDCG@10, Feature Frequency (FF), and Ground Truth (GT) similarity; see Sect. 3.7 for definitions of these metrics.

The key findings from the presented empirical investigation are: 1) for the synthetic dataset, unlike the real-world tabular datasets, the aggregation of point-wise explanations provide the most faithful list-wise explanations with respect to several of the measures; GT, FF, Fidelity, and ExplainNDCG. 2) for the non-synthetic datasets, the results indicate that the explanation techniques are faithful toward different evaluation measures; PMI is the most faithful technique with respect to Validity and Completeness, while RankLIME is the most faithful technique with respect to Fidelity and ExplainNDCG. 3) The faithfulness of explanation techniques based on measures Fidelity, Infidelity, and ExplainNCDG can drastically change by simply normalizing the feature importance scores output from the explanation techniques. 4) The faithfulness of LIME list-wise explanations can largely depend on their sample size hyper-parameter; this is the case in particular for the Yahoo dataset. 5) We show that explanation techniques are most faithful with respect to Feature Frequency measures when they can equally include features that are used in nodes with larger and lower depths in the trees in the LambdaMART model. 6) The faithfulness of point-wise explanations does not significantly improve in tabular datasets when we limit their aggregation to the explanations of the top-rank document associated with a query.

2 Related Work

A comprehensive study of the evaluation of list-wise explanations of LTR models in general and LambdaMART, in particular, is not available in the literature on explainability. However, the available studies have included only a subset of

[2] https://github.com/amir-rahnama/listwise_lambdamart_explanations.

[3] For brevity, we refer to KernelSHAP as SHAP.

explanation techniques, evaluation metrics, or benchmark LTR datasets. In [33], Greedy-Score provides the more optimal performance compared to non-LTR explanations of LIME and SHAP concerning Completeness and Validity when explaining the LambdaMART and RankNet models on the MQ2008 dataset. In [12], RankLIME outperforms EXS explanations of neural ranking models on the MSMARCO dataset based on fidelity and ExplainNDCG@10 metrics. In [35], the authors show that LIRME explanations are correct when explaining a neural language model on the TREC-8 dataset. In their study, correctness is the agreement between the feature's importance scores and the relevant terms in the text. No systematic evaluation is present in the study of [32] for EXS explanations.

Apart from the explanation we have included in our study, which we discuss in Sect. 3, other list-wise explanation techniques are proposed for explaining neural networks trained on text datasets. In [24] and [37], the authors propose list-wise explanation techniques Generalized Preference Coverage (GPC) and LiEGe for explaining neural ranking models trained on text datasets. These techniques rely strongly on 1) optimization based on the relationship between each query token with tokens in their associated documents and 2) bag of words representation of text datasets. Even though the second limitation can be adjusted to a tabular setting similar to RankLIME, we could not find an alternative to the first criteria for the tabular setting. Therefore, these techniques are excluded from our study.

3 Background

In this section, we first formalize the problem of listwise explanations of LTR models, then provide an overview of local list-wise (Sect. 3.2) and point-wise explanation techniques (3.3), along with point-wise non-LTR explanations (Sect. 3.4 of LTR models describe the evaluation measures employed in this study. We finish this section by providing an overview of the evaluation measures used in our study.

3.1 Problem Statement

Let $X = (q, D)$ where $D \in \mathbb{R}^m$ the list of m documents d for a query q and $d_i \in x \in \mathbb{R}^n$ the i-th document in that list. A ranking function $f : X^m \rightarrow \Pi^m$ produces a permutation or rank where π_i is the rank of the i-th document. Learning-to-rank models learn the function f from the training data using point-wise, pairwise, or list-wise loss approaches [11]. The pointwise approach treats

the ranking problem as a regression problem and optimizes the LTR models based on the relevance of individual documents. Pairwise loss functions treat the ranking problem as a classification problem. Therefore, the model is trained to learn which documents among pairs in a query have a larger relevance. Lastly, the list-wise approach aims to optimize the model directly on the set of relevance scores for all documents in a given query. LambdaMART is a pairwise LTR model shown to approximate list-wise objective functions [6].

After training, the function f outputs $f(q, D)$ where R is the predicted relevance score. Local list-wise feature attribution techniques output their explanations in the form of $\phi \in \mathbb{R}^n$ in which ϕ_j is the importance scores of feature j for the predicted output of $f(q, D)$ where $j = 1, ..., N$ and f is the trained black-box LTR model.

3.2 List-Wise Explanations

3.2.1 PerMutation Importance (PMI)

PerMutation Importance was originally proposed by [5] for random forests, and in [13], the authors proposed its model-agnostic variation for global explanations. PMI is widely used in supervised learning; however, to our knowledge, we are the first to evaluate its explanations for list-wise explanations. Unlike other model-surrogate explanations, PMI has a key advantage in that it does not need to train a surrogate interpretable model[4]. One of the key problems in training a surrogate model is that it is more difficult to troubleshoot the explanations as the surrogate model is subject to under-fitting or over-fitting, similar to any other machine learning model [31].

The logic of PMI is shown in Algorithm 1. Let $D \in \mathbb{R}^{M \times N}$ be the dataset of documents D for query q and f a black-box model. We permute each feature's value in the D one at a time. We then measure the absolute difference in the average predicted score of the black box before and after the permutation of features. We repeat this process for T trials; T is considered a hyper-parameter, and these values are averaged across the trials. The importance scores of feature j, i.e., FI_j, can be interpreted as the magnitude that permuting feature j changes the absolute predicted score of model f given all other features are fixed.

In our implementation, the PMI algorithm has only access to the documents D associated with a single query q and not the entire dataset as considered in [13]. We chose this design because we focus on obtaining list-wise explanations only at the query level, and we do not want the data from other queries to leak into the explanation process.

3.2.2 Greedy-Score

Greedy-cover is a list-wise iterative greedy explanation technique that aims to find the set of features that can cause the largest change in the predicted rank of documents associated with a query [33]. At time step 0, the algorithm starts with an empty set of important features $E = \emptyset$. In each step, the algorithm

[4] We discuss how LIME-based explanation techniques train their surrogate models in Sect. 3.2.3 and 3.3).

Algorithm 1. PMI algorithm

Require: $D \in \mathbb{R}^{M \times N}$ documents for query q, f: black-box model,
procedure PMI(D, f, R, T)
  $t \leftarrow 1$
  for feature j in $j = 1, ..., N$ **do**,
    while $t \leq T$ **do**
      $D' \leftarrow$ permute values of feature j in D
      $FI_j^t = \sum_M \left| f(D') - f(D) \right| / M$
      $t \leftarrow t + 1$
    end while
    $FI_j = \sum_T FI_j^t / T$ ▷ Importance for Feature j
  end for
  return FI_j
end procedure

finds a potential candidate feature to be added to E iteratively by nullifying the features $E \cup f$ and measuring the following objective function:

$$z_p^f = (R(d_i, E \cup f) - R(d_j, E \cup f)) \cdot w_p \tag{1}$$

where R is the predicted rank of model f, and w_p is the predicted rank differences between d_i and d_j where $i, j \subset P$ and P are the set of concordant pairs[5] between the originally predicted rank and the new predicted rank after the nullification of features at each step. The feature that maximizes the sum of the z_p^f values across all concordant pairs in P is added to E. See [33] for more details.

3.2.3 RankLIME

RankLIME explanation technique is an extension of LIME explanations [30], and it was initially developed for obtaining explanations of neural models trained on text [12]. RankLIME and other LIME-based explanation techniques, such as LIRME, EXS, LIME, and SHAP, assume feature independence in their explanation process. RankLIME starts with a perturbation step. In this step, uniform Gaussian noise is added to randomly selected features of **all** documents D for query q. The standard deviation of this noise is equal to the standard deviation of the selected feature. After repeating this process T times to create a new set of documents D', the samples D' are weighted by a kernel function. The sample size T is a hyper-parameter.

An interpretable linear surrogate model g with list-wise loss $\mathcal{L}$ is trained on pairs of $(D', f(D'))$:

$$\mathcal{L} = \sum_i \frac{2^{f(D'_i)} - 1}{\log_2(1 + \text{APrank}(s_i))}$$

$$\text{APRank}(s_i) = 1 + \sum_j \frac{1}{1 + \exp\left(-\frac{(f(D'_j) - f(D'_i))}{t}\right)}$$

[5] Concordant pairs are the pairs of documents that maintain the same relative rank between two ranked lists.

where t is the temperature and is usually set to 0.1 [11]. The idea is that by generating the sample of different variations of the explained list and recording the predicted output $f(D')$, the surrogate model can model the relationship between feature changes and the predicted output. The local explanations of RankLIME are the weights of the interpretable surrogate model g and indicate the importance of features in the list of documents to the predicted list-wise scores of the documents, i.e., $f(D)$.

3.3 Poin-Twise Explanations

LIRME [35] and EXS [32] are point-wise explanation techniques that are also extensions of LIME but developed for explaining LTR models. Since they are point-wise explanation techniques, the explanation technique starts with a single document $d \in D$ in their perturbation step. In their original studies, LIRME and EXS sample interpretable bag-of-words representations of the explained text document. We have adapted them to include the tabular interpretable sampling of TabularLIME [15,30]. In this process, LIRME and EXS create binary interpretable samples by binning the features of d into quantiles. For each newly generated document d', we randomly sample a feature value, and if the feature value lies in the same bin as the feature value in d, the label is equal to one; otherwise, it is zero[6]. After repeating this process T times, the samples of D' are created. A kernel function weights these samples based on their distance to d.

LIRME trains a Ridge surrogate model on pairs of $(D', f(D'))$ with the following loss function:

$$\mathcal{L}(D'_i, fD'_i, k) = \sum_{j=1}^{T} k(d'_{i,j}, d_i)(g(d'_{i,j}) - f_{d'_{i,j}})^2 + \alpha|\Theta| \tag{2}$$

EXS, On the other hand, uses a Linear SVM surrogate and has three labeling processes built for $f(D')$ in pairs $(D, f(D')$: Score-based (S), top-K binary (B), and rank-based (R). In score-based, label equals $1 - \frac{R(d')-R(d_1)}{R(d_1)}$ where $R(d_1)$ is the rank of the top-1 document in that query. Top-K binary generates a label one for sample d' if its predicted rank is larger than the rank of the Top-K document for the query (the anchor). In Rank-based, the label of d' is zero if its rank is less than the top-K document in the query (the anchor). Otherwise, the label equals $1 - \frac{R(d')}{k}$. EXS uses a hinge square loss or epsilon-insensitive loss function to train its surrogate, depending on the type of labeling used. We set the anchor document to be the document that is ranked 10-th in the list of documents since the anchor needs to be positioned relatively high on the list of documents as suggested in [32].

The explanations of LIRME and EXS are the weights of their surrogate models. We aggregated the point-wise explanations across all documents in a query using a normalized sum function, as proposed in their representative studies, to obtain list-wise explanations for these techniques. For more details, see [32,35].

[6] For a detailed description of the sampling algorithm, see [15].

Table 1. Key Differences between the explanation techniques in our study. RankLIME, Greedy-Score, and PMI are list-wise, LIRME and EXS point-wise, and LIME and SHAP are non-LTR explanation techniques.

Name	Sampling	Kernel	Labeling	Surrogate	Objective
RankLIME	Gaussian	exp	Scores	Ridge	Listwise Loss
Greedy Score	Greedy	None	Scores	None	Weighted MSE
PMI	Permutation	None	Scores	None	MSE
LIRME	Quantile	exp	Scores	Ridge	Weighted MSE
EXS	Quantile	exp	Anchor	SVM	Squared Hinge
LIME	Gaussian	exp	Scores	Ridge	Weighted MSE & Larspath
SHAP	Gaussian	discrete	Scores	Ridge	Weighted MSE & Larspath

3.4 Non-LTR Explanations

Even though LIME [30] and KernelSHAP [23] are not developed for explaining LTR models, they can provide point-wise explanations of LTR models by casting the problem as a regression problem.

There are some key differences between LIME and SHAP. The most significant difference is the choice of kernel function that weights the generated samples. LIME uses an exponential kernel, while SHAP uses a discrete combinatorics kernel. Moreover, unlike other techniques, LIME and SHAP use Larspath feature selection after training their surrogate. Similar to the process in Section. We obtain list-wise explanations of LIME and SHAP by aggregating the explanations of all documents in a query using a normalized sum function. LIME and SHAP use Gaussian sampling, similar to RankLIME, the main difference between the Gaussian sampling in LIME and SHAP versus RankLIME is that LIME and SHAP add a Gaussian noise with a standard deviation of 0.1. For more details, see their respective studies.

3.5 Key Similarities and Differences

In Table 1, we summarize and clarify the difference between the explanation techniques. The table helps us in analyzing the reasons behind the empirical success and failures of these techniques on our studied datasets later in Sect. 4 and Sect. 5. The explanation techniques generally differ in the way they generate samples, their kernel function, the labeling technique they use, the surrogate models, and their objective functions.

3.6 Explanation Representation

For tabular datasets, we often divide explanations into K important and $N - K$ unimportant features by sorting (ranking) the features based on their absolute importance scores in descending order. The value of K is a hyper-parameter, where N is the total number of features in the dataset. Features are sorted based on absolute importance scores to prioritize features with the largest effect on the predicted score regardless of their sign of scores [25].

3.7 Evaluation of Local Explanations

Numerous studies have proposed several systematic evaluation measures for local explanation techniques [11,12,33,35]. We selected evaluation measures that can be applied to explaining LTR models that are not based on neural networks and trained on tabular datasets.

A subset of our evaluation measures work by direct comparison with a baseline either from the data generation process or the model (Sect. 3.7.1 and 3.7.2). Other evaluation measures evaluate the explanations by removing important (unimportant) features of explained instance based on its explanations (Sect. 3.7.3 or the dot product of query document pairs and their associated explanations (Sect. 3.7.4). Infidelity measure combines the ideas from the two previously mentioned evaluations (Sect. 3.7.5). Explanation Consistency measures the change in the explanations as their hyper-parameters of explanation techniques change. (Sect. 3.7.6).

3.7.1 Ground Truth

In [22], the authors proposed using synthetic datasets to evaluate local explanations. A set of document vectors in $\mathbb{R}^m$ is generated for which polynomial functions such as $Y = W_1X_1 + W_2X_2 + ... + W_NX_N$ are employed, after which the documents are sorted based on the value of Y to obtain relevance. This process is repeated for a set of N queries, after which an accurate LTR model is trained on all the query-document pairs. In Ground Truth similarity, we measure $\rho(\lambda, \phi)$ where ϕ is local list-wise explanations, $\lambda = [W_1, ..., W_N]$ is the ground truth vector of the data generation process, and ρ is the Kendall rank correlation coefficient. The larger the similarity to Λ, the more faithful the explanations are with respect to this measure.

3.7.2 Feature Frequency

In our study, we have included one model-specific evaluation measure. In tree models, features used on the decision paths play a significant role in predicting the relevant outcome of each document. Feature frequency importance scores show the frequency at which features are used along the decision paths of documents [10,20]. In our study, we calculate the normalized feature frequency after filtering the decision paths to include the ones used for predicting the list of explained documents associated with a query. We use the Kendal Tau correlation function as the similarity function as similarly proposed for the Fidelity measure by [12]. List-wise explanations that show large similarity to the Feature Frequency baseline are considered optimal.

3.7.3 Validity and Completeness

Validity and Completeness [33] are two measures to evaluate the quality of local explanations in LTR models. Validity (Completeness) measures the change in the predicted score of the black-box LTR models for documents D after top-K unimportant (important) features from the explanations are nullified in the explained

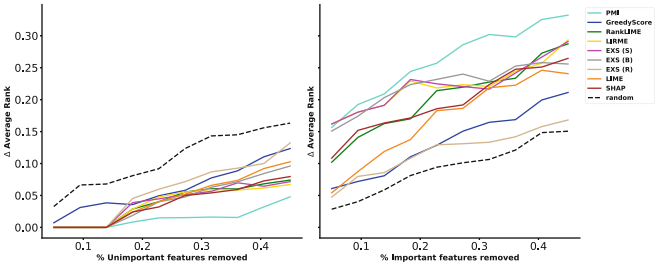

Fig. 2. MQ2008 dataset: Validity (Left) and Completeness (Right) of top-K important features in the list-wise explanations of LambdaMART for progressively increasing the values of K. The charts show the average values of these measures across all query document pairs in the test dataset. Across both measures, PMI is the most optimal explanation. As expected, all explanations perform better than the random baseline across both measures.

documents D. Formally, we define these measures following the notation of [33]. Validity is defined as $V_{\phi,K} = \rho(f(D), f(D'_k))$ where ρ is the Kendal Tau, $f(q, D)$ is the predicted score of the list of documents D in query q we explain, and D'_k is the vectors of documents where the K least important features from their list-wise explanations ϕ are nullified [32]. In practice, nullification is performed by replacing the feature value with the average value of that feature in the dataset. Valid explanations minimize the value of $V_{\phi,K}$. Completeness is defined similarly, but in that case, D'_k is now the vector of documents where the top-K important features are nullified. To study the sensitivity of the Validity and Completeness of explanations to the values of K, we progressively increase the values of K and calculate the Area Under Curve for these measures after averaging the values for all query document pairs, similar to the proposition in [18] (see Fig. 2). Faithful explanations need to have large values of Completeness and small values of Validity.

3.7.4 Fidelity and ExplainNDCG

In [12], the authors proposed Fidelity and ExplainNDCG@10 for evaluating explanations of LTR models. Given a local explanation ϕ and a list of documents D and a black-box model f, the fidelity is calculated as $\rho(D \cdot \phi, f(D))$ where ρ is the Kendall rank correlation coefficient. The ExplainNDCG@10 [12] is similar to fidelity, but instead of Kendal Tau correlation, we uses the NDCG@K function. We use the values of $K = 10$ in our tabular dataset, as suggested in the original study. For our synthetic case, we use $K = 3$ given that the number of documents associated with queries is smaller.

3.7.5 Infidelity

In [36], the authors propose the measure Infidelity that evaluates explanations by calculating the Mean Squared Error (MSE) between the significant perturbed

variations of documents D (performed by adding Gaussian noise) and the explanations ϕ with the change in predicted output after perturbing the explanations. The significant perturbation is considered to be the nullifying the top-20/‰ of important features in query document pairs. Moreover, we have replaced the MSE function with the Kendall rank correlation coefficient to adapt the measure to the learning-to-rank context. Optimal explanations have small values of Infidelity.

3.7.6 Explanation Consistency

Explanation consistency [35] is one of the desired properties of local explanations that have hyper-parameters. For example, as mentioned in Sect. 3, LIME-based explanation techniques such as LIME, SHAP, EXS, LIRME, and RankLIME have a hyper-parameter called sample size. In [35], the authors propose Explanation consistency that measures the change in the top-K ($k \ll M$) important features as the hyper-parameters in an explanation technique increases. We use Jacacrd Similarity to measure Explanation Consistency. Consistent explanations show minimal changes in their set of top important features as these hyper-parameters increase. Explanation Consistency can be considered an approach in which we can automatically tune the sample size hyper-parameters.

4 Empirical Investigation

In this section, we first describe the experimental setup. After that, we provide our evaluation of list-wise explanations using a synthetic dataset. The aim is to have a baseline to benchmark the explanations on simpler datasets. After that, we continue our evaluation of list-wise explanations of the LambdaMART model on the benchmark tabular datasets.

4.1 Setup

The total number of available queries in MQ2008 [27], Web10k [27], and Yahoo [7] is 800, 10000, and 29921. There are 46, 137, and 699 features in the document pairs of the queries[7]. All datasets include their previously defined validation and test set. The validation set is used for hyper-parameter optimization of the explanations and LambdaMART models. We used the test dataset to evaluate explanations. The implementation of LambdaMART is based on the LightGBM package [20]. We used the default parameters of LambdaMART models that achieved respective NDCG@10 scores of 0.79. 0.76 and 0.54 on the test dataset, nearly on par with the state-of-the-art performance results reported in [28].

We use Explanation consistency as proposed by [35] for tuning the hyper-parameters of explanation techniques. In measures based on the Kendall rank correlation coefficient, the correlation is performed by comparing absolute importance and GT baseline scores. Based on this, we focus on the correct ranking of

[7] For more details about the datasets, see the corresponding studies.

important features regardless of their sign, as it is common in evaluating tabular explanations [25,26,29].

To create list-wise explanations from the point-wise explanations, we used the sum aggregation of all feature importance scores of documents normalized by the number of documents in a given query as suggested in [32,35]. Our Random baseline allocates uniformly random importance scores to all documents. For more details, see the code of our experiments.

4.2 Synthetic Dataset

In this section, we evaluate the list-wise explanations of LambdaMART on our synthetic dataset. This example allows us to evaluate these explanations in simpler settings with datasets with fewer features. The data generation function of Gaussian input $X \in \mathbf{R}^{10\times 5}$ with the data generation function $Y = 0.2X_1 - 0.5X_3 - 0.4X_4$ (second and fifth feature have zero coefficients) and we sort the values of Y to obtain relevance. We stack 1000 queries and train a LambdaMART model on top of the dataset with default parameters that achieve an NDCG@10 score of 0.98 on the test dataset. We select a random subset of 30% of instances as the test held-out dataset for obtaining the explanations. The Ground Truth (GT) baseline vector $\lambda = [0.2, 0, 0.5, -0.3, 0]$, which are the coefficients of the data generation process. The Feature Frequency (FF) baseline from the model obtained is $[0.28, 0.1, 0.25, 0.27, 0.1]$. This baseline also indicates that the second and fifth features were used in fewer splits than the other features.

In Fig. 3, we see an example of list-wise explanations for a test query-documents pair of our dataset. The Ground Truth (GT) and Feature Frequency (FF) baseline vectors are also visualized for clarification. Note that EXS (R) and Greedy Score explanations (wrongfully) have allocated large importance scores to the fifth feature in the dataset.

In Table 2, the average values of each evaluation measure are reported for the test query document pairs in the synthetic dataset. The bold values indicate the most optimal explanations for each measure. LIRME and EXS (S) are the top explanations concerning GT similarity, Fidelity, and ExplainNDCG@K (here $K = 5$) measures. Moreover, RankLIME is the optimal measure with respect to completeness and is also on par with LIRME and EXS for ExplainNDCG. PMI outperforms other techniques with respect to validity, and to our surprise, all explanations fail to provide a more faithful explanation than our random baseline when measuring Infidelity. We provide a possible explanation of this in Sect. 4.3.3. On an overall inspection of the table, we can see that the aggregation of point-wise explanations does lead to faithful explanations in GT, FF, Fidelity, and ExplainNDCG.

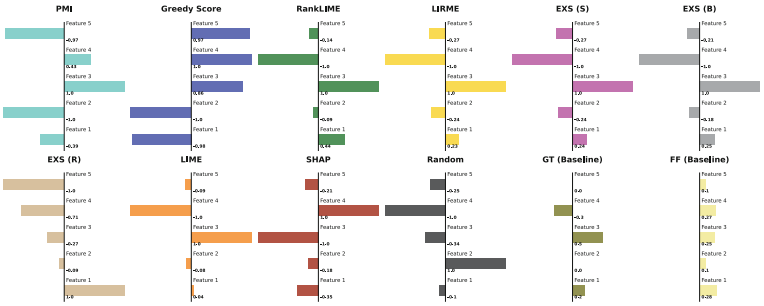

Fig. 3. Local Listwise explanations for a test query document pairs of the synthetic dataset. In addition, we visualized the Ground Truth (GT) from the data generation and Feature Frequency (FF) vectors in the bottom right side of the visualization. Scores are normalized between –1 and 1 for visualization purposes.

Table 2. Synthetic Dataset: Average values of evaluation measures for test query documents. Bold values indicate the most optimal values for each measure. Arrows indicate whether faithful explanations need to maximize or minimize the measure.

	PMI	Greedy Score	RankLIME	LIRME	EXS (S)	EXS (B)	EXS (R)	LIME	SHAP	Random
GT ↑	0.91	0.04	0.91	**0.95**	**0.95**	0.94	–0.02	0.66	0.74	–0.03
FF ↑	0.35	–0.27	0.32	0.32	0.32	0.32	–0.03	0.44	**0.48**	0.05
Fidelity ↑	–0.33	0.11	0.93	**0.94**	0.92	0.92	0.00	–0.04	0.02	–0.14
ExplainNCDG ↑	0.70	0.82	**1.00**	**1.00**	**1.00**	**1.00**	0.78	0.78	0.80	0.75
Completeness ↑	40.20	15.15	**41.55**	39.60	39.60	39.60	19.50	29.40	30.15	14.10
Validity ↓	**1.20**	23.70	1.65	1.35	1.65	1.35	12.15	4.95	6.45	25.80
Infidelity ↓	–0.05	0.02	0.08	0.05	0.05	0.05	–0.03	–0.00	–0.03	**–0.06**

4.3 Benchmark Datasets

In this section, we evaluate local list-wise explanations of LambdaMART on our benchmark datasets, namely MQ2008, Web10k, and Yahoo. To have more intuition about the datasets, we have provided the global feature importance scores (based on feature splits) obtained from our trained LambdaMART models (Fig. 4). This way, we have an intuition of the histogram of normalized feature importance used by our LambdaMART models. Notice that unlike Web10K and Yahoo datasets, the importance scores of features in MQ2008 are more evenly distributed.

4.3.1 Explanation Consistency

We investigate the Explanation Consistency measure with respect to the explanation sample size. This also helps us choose a threshold for the sample size hyper-parameter of LIME, SHAP, LIRME, EXS, and RankLIME. As suggested in [35], consistent explanations tend to include fewer changes in their most important features as their sample size increases. We measure the Explanation Consistency with Jaccard Similarity of the Top-50% of features in each dataset between explanations obtained at sample size T and $T+1$. We change the values of T to

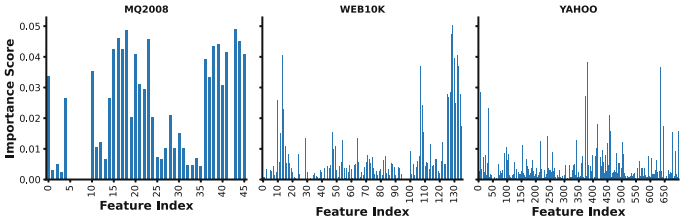

Fig. 4. Global Feature Importance Scores of LambdaMART

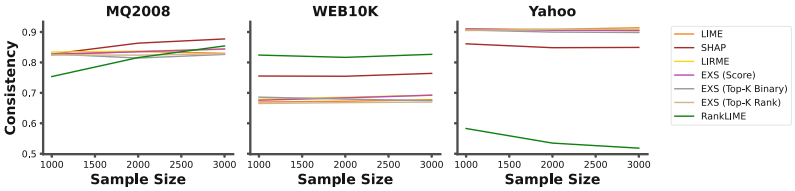

Fig. 5. The explanation consistency point-wise explanations. We expect the LIME-based explanations to provide larger consistency as their sample size increases.

be $1000, 1500, 2000, 2500, 3000$. The explanation consistency of the sample size of 1000 is compared with the sample size of 500.

In Fig. 5, we can see that most point-wise explanations provide nearly consistent list-wise explanations except for RankLIME and SHAP in Yahoo datasets. Our results show that in most cases, our explanation techniques do not change their top 50 % important features starting from the sample size value of 1000. However, based on this result, we set the cutoff threshold and selected 5000, 3000, and 3000 as the sample sizes for these explanations for MQ2008, Web10k, and Yahoo datasets, respectively. The idea is to allocate the same sample size value to all LIME-based explanations for a fair comparison since, in this case, they receive an equal computational budget. However, we still show the effect of sample size on different evaluation metrics later in Sect. 4.3.4.

4.3.2 Fidelity and Other Measures

After Explanation Consistency, we investigate teh faithfulness of explanations with respect to FF, Fidelity, ExplainNCDG@10, Completeness, Validity, Infidelity, and Feature Frequency. In Table 3, the average values of each evaluation measure are reported for the explanations of test query document pairs in the MQ2008 dataset. PMI outperforms other explanation techniques for Completeness, Validity, and FF similarity in this dataset. On the other hand, RankLIME continues to be the most optimal explanation concerning Fidelity, ExplainNDCG@10, and Infidelity measures. In this case, we can see that the aggregated point-wise explanations of LIRME and EXS, along with LIME and SHAP, show poor faithfulness across all measures.

Table 3. MQ2008: Average values of evaluation measures for test query documents. Bold values indicate the most optimal values for each measure. Arrows indicate whether faithful explanations need to maximize or minimize the measure.

	PMI	Greedy Score	RankLIME	LIRME	EXS (S)	EXS (B)	EXS (R)	LIME	SHAP	Random
FF ↑	**0.7**	0.07	0.42	0.48	0.48	0.43	0.27	0.3	0.5	–0.03
Fidelity ↑	0.66	0.02	**0.99**	0.69	0.67	0.66	0.15	0.03	–0.09	0.52
ExplainNDCG ↑	0.96	0.8	**1**	0.97	0.96	0.96	0.85	0.79	0.76	0.93
Completeness ↑	**46.13**	24.34	38.6	40.2	40.84	40.34	20.89	33.19	39.79	13.41
Validity ↓	**4.3**	13.83	7.7	6.82	7.37	8.02	10.65	8.37	8.36	20.8
Infidelity ↓	–0.13	–0.01	**–0.21**	–0.09	–0.05	–0.13	–0.04	0	0.06	0.17

Table 4. Web10k: Average values of evaluation measures for test query documents. Bold values indicate the most optimal values for each measure. Arrows indicate whether faithful explanations need to maximize or minimize the measure.

	PMI	Greedy Score	RankLIME	LIRME	EXS (S)	EXS (B)	EXS (R)	LIME	SHAP	Random
FF ↑	0.55	0.13	0.05	**0.57**	**0.57**	0.43	0.01	0.37	0.56	0.01
Fidelity ↑	–0.02	–0.01	**0.88**	0.03	0.05	0.09	0.02	0.08	0.15	0
ExplainNDCG ↑	0.52	0.51	**0.99**	0.56	0.55	0.61	0.52	0.57	0.62	0.53
Completeness ↑	**41.78**	16.43	17.73	38.13	37.64	38.05	11.5	33.5	41.72	19.09
Validity ↓	**0.06**	14.32	10.93	2.38	2.55	3.89	15.67	4.83	2.84	12.43
Infidelity ↓	0.03	0.05	–0.06	**–0.16**	–0.05	–0.08	–0.03	–0.01	–0.02	0.04

Table 4 shows the same results for the Web10K. New trends have emerged. LIRME and EXS (S) are most faithful concerning FF similarity, while LIRME is also most faithful concerning Infidelity. As before, PMI and RankLIME continue to outperform other techniques with respect to Completeness and Validity and Fidelity and ExplainNDCG, respectively.

In the Yahoo dataset (Table 5), the trends change for FF and Infidelity measures. Here, SHAP and EXS (R) are the most faithful with respect to FF similarity and Infidelity. In Sect. 4.3.5, we provide an intuition about the underlying reasons behind the faithfulness of PMI, LIRME, EXS (S), and SHAP with respect to FF similarity in these datasets.

Before finishing this section, we provide an overview of the ranks of each explanation technique across all datasets and measures in Fig. 6. The ranks are adjusted so that more faithful explanations have lower ranks for all measures. Overall, we see that specific explanations such as PMI and RankLIME can be faithful across datasets; however, they can only be applied to a subset of the measures we have studied. We can also see that overall, Greedy-Score, EXS (R), and LIME are not faithful across numerous measures and datasets. Among the point-wise explanations, LIRME explanations in all datasets and SHAP for the Web10k dataset provide relatively low fidelity ranks.

Table 5. Yahoo: Average values of evaluation measures for test query documents. Bold values indicate the most optimal values for each measure. Arrows indicate whether faithful explanations need to maximize or minimize the measure.

	PMI	Greedy Score	RankLIME	LIRME	EXS (S)	EXS (B)	EXS (R)	LIME	SHAP	Random
FF ↑	0.55	0	0.46	0.55	0.55	0.45	0.33	0.53	**0.58**	–0
Fidelity ↑	0.61	–0.04	**0.99**	0.61	0.47	0.59	0.01	0.29	–0	0.48
Explain NDCG ↑	0.95	0.69	**1**	0.96	0.9	0.94	0.73	0.81	0.69	0.9
Completeness ↑	**49.99**	8.83	44.97	38.71	42.15	33.68	14.8	29.65	36.77	16.39
Validity ↓	**0**	18.03	44.97	1.41	0.91	2.43	3.83	0.43	0.53	10.08
Infidelity ↓	0.22	0.02	0.12	0.12	0.24	0.08	**–0.03**	0.1	–0	0.08

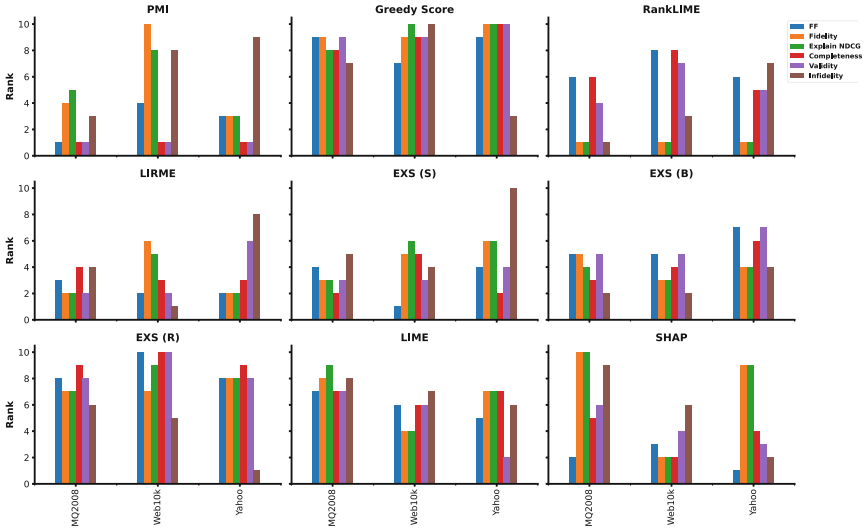

Fig. 6. The average rank of all explanations across all measures in our benchmark datasets. Lower ranks indicate more faithful explanations for all measures.

4.3.3 The Effect of Normalization

As we mentioned in Sect. 3.7, Fidelity, Infidelity, and ExplainNDCG measures include the dot product between predicted scores for query document pairs and their associated explanations. In this section, we show that normalization of explanations, **after** the explanations are obtained from the respective techniques, can cause significant change to the faithfulness of explanations based on these measures. This is surprising as normalization does not change the relative rank of scores in the feature importance scores.

In Table 6, we have re-evaluated the faithfulness of explanations based on Fidelity, Infidelity, and ExplainNDCG after min-max normalization of their importance scores in the explanations. We can see that in this case, for example, in Web10k, SHAP and GreedyScore can outperform RankLIME. Similarly, PMI and EXS (R) can outperform other techniques for the Yahoo dataset, including RankLIME.

Table 6. The effect of normalization on Fidelity, Infidelity, and ExplainNDCG where the dot product of explanations and predicted score of their associated documents are calculated.

Dataset	Measure	PMI	GreedyScore	RankLIME	LIRME	EXS (S)	EXS (B)	EXS (R)	LIME	SHAP	Random
MQ2008	Fidelity ↑	0.66	0.58	**0.69**	0.64	0.65	0.66	0.58	0.58	0.57	0.52
	ExplainNDCG ↑	0.96	0.94	**0.97**	0.96	0.96	0.97	0.95	0.94	0.94	0.93
	Infidelity ↓	–0.13	–0.09	**–0.15**	–0.09	–0.08	–0.14	0.03	0.15	0.08	0.17
Web10k	Fidelity	–0.02	0.02	0.02	0.02	0.01	0.02	0.02	0.02	**0.03**	0
	ExplainNDCG	0.52	**0.56**	0.52	0.55	0.55	0.55	0.54	0.52	0.55	0.53
	Infidelity ↓	0.03	0.05	0.05	**–0.02**	–0	–0.01	0.04	0.03	0.01	0.04
Yahoo	Fidelity ↑	**0.61**	0.47	0.59	0.47	0.48	0.49	0.47	0.5	0.47	0.48
	Explain NDCG ↑	**0.95**	0.9	0.94	0.91	0.91	0.92	0.91	0.92	0.9	0.9
	Infidelity ↓	0.22	0.1	0.14	0.26	0.28	0.21	**0.04**	0.13	0.27	0.08

We find this surprising as measures such as FF, Completeness, and Validity are robust with the normalization of explanations as long as they do not cause any change to the relative rank of features based on their importance scores. We discuss this further in Sect. 5.

4.3.4 The Effect of Sample Size

In Sect. 4.3.1, we measured the effect of explanation sample size with respect to Explanation Consistency. In this section, we investigate whether sample size can cause significant differences to measures evaluated in Sect. 4.3.

The results for MQ2008, Web10k and Yahoo are respectively shown in Figs. 7, 8 and 9. There are cases where sample size can change the relative rank of explanations. In MQ2008, LIME and RankLIME for Validity, EXS (R), RankLIME, and LIME for Completeness, to name a few. In Web10k, EXS (R) for Validity and ExplainNDCG and EXS (B) for Completeness are other examples of this phenomenon. In Yahoo datasets, the effect of sample size on the relative rank of explanations is more significant for measures such as FF, Completeness, and Infidelity. For example, EXS (R) or SHAP can be considered most faithful based on Infidelity and the sample size we choose.

Even though we could select sample sizes that optimize each explanation technique individually, we would like to reiterate that in our study and for a fair comparison, we allocated the same sample size to all LIME-based explanation techniques [35].

4.3.5 The Effect of Depth

In the structure of decision trees in LambdaMART, features utilized for splitting in nodes with shallower depths are regarded as more significant [5]. This is because a larger number of documents are likely to traverse through these nodes along the decision paths of the tree. Let's compute the median depth of features among the top-K important features in each explanation. As we elevate the value of K, it can be expected that the median rank of the feature set will also rise, given that the median depth increases with the inclusion of less significant features that are used in nodes with greater depths.

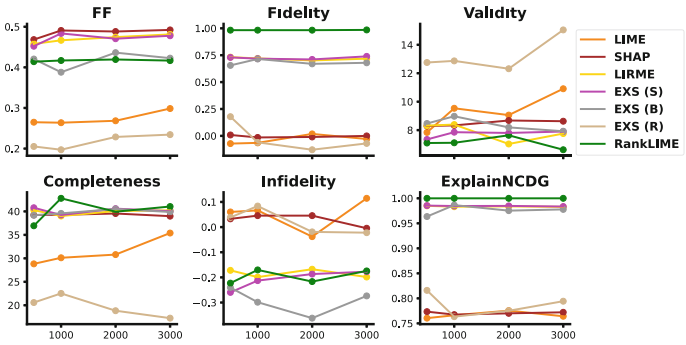

Fig. 7. MQ2008: Effect of sample size on evaluation measures. The values are reported averages across all test query document pairs.

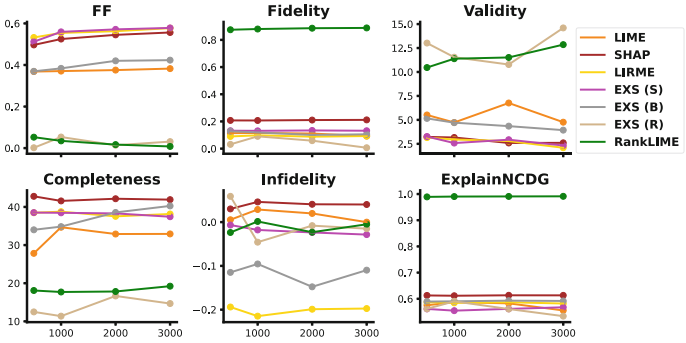

Fig. 8. Web10K: Effect of sample size on evaluation measures. The values are reported averages across all test query document pairs.

Figure 10 shows the result of our analysis, averaged over all query document pairs in the test dataset. For example, PMI can successfully increase the median depth of features in MQ2008 but not in Web10k and Yahoo. By comparing this figure and the results in Table 3, 4 and 5, we can now realize the reasons PMI is not the most faithful explanation with respect to FF for Web10k and Yahoo datasets. By examining other explanation techniques, we can see that the positive trend between increasing values of K and the median depth of features in explanation is a proxy to the faithfulness of explanations with respect to Feature Frequency (FF) similarity.

4.3.6 Aggregation Effect

In this section, we investigate whether the poor performance of point-wise explanations in tabular datasets can be due to the way in which we aggregate the explanations. In our study, we aggregated the point-wise explanations for explanations for all the documents associated with queries. We want to investigate whether aggregating the top-rank documents associated with a query *instead of*

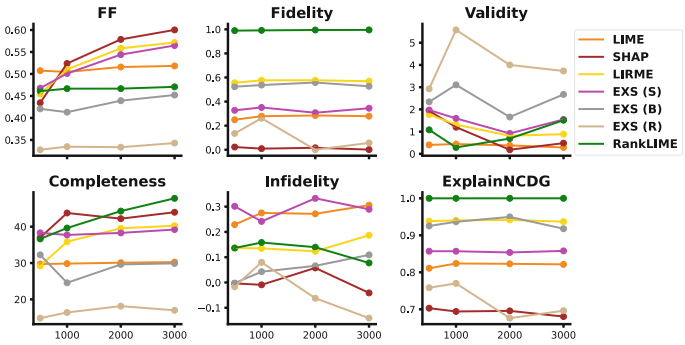

Fig. 9. Yahoo: Effect of sample size on evaluation measures. The values are reported averages across all test query document pairs.

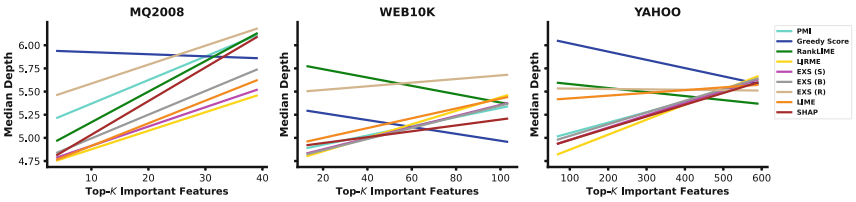

Fig. 10. The relationship between the median depth of features in the top-K important features in each explanation. The results are averaged across all test query document pairs. We expect the median rank of features will increase as the values of K increase.

all documents can improve the faithfulness of explanations with respect to some measures.

For this, we report the fidelity of point-wise and non-LTR techniques by aggregating the explanations of the top-10 and 20% documents ranked by their predicted relevance scores in the Web10k dataset. Table 7 shows that except for minimal improvements in Completeness and ExplainNDCG, the aggregations from the larger set of explanations provide more faithful explanations. By comparing the results with that of Table 4, we can see that our original aggregation, across the explanations of all documents for an associated query, provides the most faithful explanations. As a result, the aggregation cannot directly explain the low fidelity of aggregated point-wise explanations in our tabular datasets.

Table 7. Web10k: Evaluation of List-wise explanations obtained from aggregating only the top-10% and top-20% of documents (ranked in descending order) by their predicted relevance scores. Bold values indicate the explanation with the most optimal value.

	Top 10-%						Top 20-%					
	LIRME	EXS (S)	EXS (B)	EXS (R)	LIME	SHAP	LIRME	EXS (S)	EXS (B)	EXS (R)	LIME	SHAP
FF ↑	0.49	0.51	0.35	0	0.32	0.52	**0.53**	**0.53**	0.38	0	0.34	**0.53**
Fidelity ↑	0.03	0.04	0.02	–0	0.06	0.09	0.02	0.05	0.03	–0.04	0.08	0.11
Explain NDCG ↑	0.56	0.54	0.57	0.54	0.55	**0.6**	0.56	0.55	0.57	0.53	0.57	0.59
Completeness ↑	37.32	37.92	38.15	14.21	36.94	**38.63**	37.58	36.66	37.98	12.15	33.33	35.02
Validity ↓	2.64	2.67	3.91	10.69	5.7	2.75	2.73	**2.44**	3.69	14.29	5.32	2.94
Infidelity ↓	**–0.15**	–0.06	–0.07	0.05	0.01	–0.07	**–0.15**	–0.06	–0.1	0.03	0.02	–0.06

5 Concluding Remarks

In this study, we have systematically evaluated local list-wise explanations of LambdaMART across the datasets MQ2008, Yahoo, and Web10k, along with a synthetic dataset. The study has considered seven explanation techniques; PMI, RankLIME, Greedy-Score, LIRME, EXS, LIME, and SHAP, as well as seven evaluation measures; Completeness, Validity, (In)fidelity, Fidelity, ExplainNDCG@10, Feature Frequency Similarity, and Ground Truth similarity.

The presented results show that the explanation techniques tend to be faithful toward a subset of the evaluation measures. For example, PMI is the most faithful technique regarding Validity and completeness, while RankLIME is the most faithful technique regarding Fidelity and ExplainNDCG across all datasets. The aggregation of point-wise explanation techniques led to more faithfulness for the synthetic dataset than our tabular benchmark datasets across different measures. The underlying reason behind this can be the fact that our synthetic dataset was limited to only five features. This suggests that the aggregation of point-wise explanations to obtain list-wise explanations is not a reliable solution to achieve faithful explanations for tabular datasets with a larger number of features.

We also investigated two factors that may cause a significant change in the faithfulness of explanations. Firstly, we showed that min-max normalization of the feature importance scores could largely affect the relative rank of documents with respect to measures such as Fidelity, Infidelity, and ExplainNDCG. These are the measures in which the dot product of explained documents and their associated explanations are calculated. Secondly, we saw empirical evidence that the sample size hyper-parameter of LIME-based explanations can cause the faithfulness of explanations to change across numerous measures. This effect is largely visible for list-wise explanations of LambdaMART in Yahoo datasets.

One important question that we raise from our study is whether evaluation measures such as Fidelity, Infidelity, and ExplainNDCG also need to be robust to such post-processing techniques. This is because Completeness, Validity, GT and FF similarity are all robust to post-processing of explanations since the relative rank of features based on their importance scores are unchanged. This question has not been addressed in [12] or [36], where these measures were originally proposed.

Another important question that we raise from our study is: "What is the relationship between the sample size hyper-parameter and the faithfulness of LIME-based explanations across our studied evaluation measures"?. Moreover, we used the proxy measure of Explanation Consistency for tuning the sample size of these techniques. Are there better alternative measures than Explanation Consistency to tune the sample size without the possibility of over-fitting to specific measures?

We have identified some limitations with our study. Firstly, we have evaluated local list-wise explanations without using human subjects. Before their use in high-stakes decision-making cases, explanation techniques must be further evaluated with human subjects. Secondly, the performance of explanation techniques in our study does not necessarily generalize to other datasets and models. We postulate that the results of our study are generalizable, to some extent, to tabular LTR datasets with similar characteristics (number of features, distributions of values of features). However, the empirical investigation could be extended to include a larger selection of datasets for a thorough analysis. Lastly, the study evaluates model-agnostic explanations of LTR models that provide feature attributions. For future studies, similar evaluations can be conducted on counterfactual or instance-based explanations of LambdaMART or neural ranking models.

Disclosure of Interests. The authors have no competing interests to declare that are relevant to the content of this article.

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Sparseness-Optimized Feature Importance

Isel Grau[1,2(✉)] and Gonzalo Nápoles[3]

[1] Information Systems, Eindhoven University of Technology, Eindhoven, The Netherlands
i.d.c.grau.garcia@tue.nl
[2] Eindhoven Artificial Intelligence Systems Institute, Eindhoven University of Technology, Eindhoven, The Netherlands
[3] Department of Cognitive Science and Artificial Intelligence, Tilburg University, Tilburg, The Netherlands
g.r.napoles@uvt.nl

Abstract. In this paper, we propose a model-agnostic post-hoc explanation procedure devoted to computing feature attribution. The proposed method, termed Sparseness-Optimized Feature Importance (SOFI), entails solving an optimization problem related to the sparseness of feature importance explanations. The intuition behind this property is that the model's performance is severely affected after marginalizing the most important features while remaining largely unaffected after marginalizing the least important ones. Existing post-hoc feature attribution methods do not optimize this property directly but rather implement proxies to obtain this behavior. Numerical simulations using both structured (tabular) and unstructured (image) classification datasets show the superiority of our proposal compared with state-of-the-art feature attribution explanation methods. The implementation of the method is available on https://github.com/igraugar/sofi.

Keywords: model-agnostic explainability · feature importance · sparse explanations

1 Introduction

Explainable Artificial Intelligence (XAI) has emerged as a crucial paradigm to enhance the transparency and interpretability of machine learning models, particularly in domains where the decision-making impact is significant. Understanding the rationale behind model predictions is essential for building trust, ensuring fairness, and facilitating decision-making processes. Feature attribution analysis plays a fundamental role in XAI by identifying the key factors driving model predictions, thereby enabling stakeholders to gain insights into the underlying mechanisms of complex models.

G. Nápoles—Equal contribution.

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Several methods have been proposed to compute feature attribution in tabular and unstructured prediction tasks. The results of these methods are usually presented as feature importance rakings in the case of structure data, or saliency maps or heatmaps in the case of unstructured data such as images or text. Approaches such as Permutation Feature Importance (PFI) [6], SHAP [13], LIME [20] or RISE [19] rely on perturbing the input data and observe the changes that this perturbation causes in the prediction of the underlying black box model. Other model-specific methods, such as gradient-based approaches, rely on the structures of neural networks to backpropagate a gradient or a relevance signal, revealing the influence of the input features in the output of the model. In order to evaluate the quality of the explanations obtained through feature attribution methods, the most common approach is incrementally deleting the features in the order of importance and measuring the change in the black box model predictions [18]. A smaller area under the performance curve and a steep curve indicate that the explanations obtained are correct, faithful to the black box, and also sparse.

In this paper, we propose a novel model-agnostic post-hoc explanation method, termed Sparseness-Optimized Feature Importance (SOFI), specifically designed for computing sparse feature importance rakings. Unlike existing methods that rely on perturbations or gradient propagation for computing attributions, SOFI directly addresses the sparseness of the explanations by defining the problem of computing the feature importance ranking as an optimization problem. This approach is grounded on the intuition that marginalizing the most important features significantly impacts model performance while marginalizing the least important ones has minimal effect. Our method offers both local and global feature attribution explanations and demonstrates superior performance compared to state-of-the-art methods, as evidenced by numerical simulations on 16 structured (tabular) and one unstructured (image) classification datasets.

The remainder of this paper is organized as follows. Section 2 provides a comprehensive review of related work, including several approaches for computing feature attribution. Section 3 introduces the proposed Sparseness-Optimized Feature Importance (SOFI) method, detailing its formulation, optimization approach, and implementation. In Sect. 4, we present experimental results demonstrating the effectiveness and superiority of SOFI compared to state-of-the-art methods across diverse datasets and model architectures. Finally, Sect. 5 summarizes the key contributions of this work.

2 Related Work

Various methods have been proposed in the literature to compute feature attribution. Based on whether they depend on the structures of the underlying black box models or only the pairs of input and output, they are grouped into model-specific or model-agnostic, respectively. For example, the model-agnostic method permutation feature importance (PFI) [6] measures the increase in the black box model error when destroying the information about a feature by shuffling its values in the data. This process effectively breaks the relation of each feature with

the rest. The higher the error, the more important the feature. This method offers simplicity and computational efficiency in prediction tasks with tabular data. However, the results may be biased by unrealistic data instances that result from the permutation process [15]. Furthermore, the feature importance ranking obtained by PFI can only be analyzed at a global level (i.e., for the entire black box model).

An alternative is the use of model-agnostic methods that rely on studying the variance in the prediction of the model itself instead of the error incurred. A well-known example of these methods is SHAP [13], which approximates Shapley values for obtaining local explanations signaling the positive or negative contribution of each feature to a given prediction with respect to the average prediction for the dataset. SHAP is also able to provide global explanations by aggregating the aggregating the absolute value of the feature contributions for the entire dataset.

Another well-known method that outputs feature attribution explanations is LIME [20]. Similarly to PFI and SHAP, LIME relies on the perturbation of data points to compute the feature attribution. The main difference with the previous approaches is that LIME creates this perturbation locally, around the data point of interest, and then creates a surrogate interpretable model (e.g., a linear model or decision tree) that approximates the behavior of the black box. However, several studies have signaled that LIME is sensitive to the sampling process, leading to a lack of robustness in the explanations [8,24]. This means that for similar instances, we might obtain very dissimilar explanations.

Another example of a perturbation-based explanation method is RISE [19], which creates smaller random binary masks that occlude parts of an input image. The resulting images multiplied by each mask are predicted by the black box model, and the scores for each prediction are computed. The saliency map can be computed as a weighted sum of random masks, where the weights are the probability scores. This method outperformed LIME and Grad-CAM [21] in terms of area under the performance curve after the deletion of important pixels.

Model-specific methods, on the other hand, rely on the structures or computations of the underlying black box model to compute a measure of feature attribution. Tree-based models, such as Random Forests and other ensembles, can make use of the accumulated impurity decrease associated with each feature in the trees to compute a measure of feature importance [3], also known as Gini importance. This method is convenient for tabular classification or prediction tasks, where tree-based ensembles have proven to offer generally very good performance. However, this method might attribute more importance to features with numerous unique values.

The majority of model-specific methods are conceived for different architectures of neural networks. These methods rely either on the computation of a gradient or the propagation of compositions. For example, gradient-based methods such as the simple gradient saliency method [23] compute the derivative of the model output with respect to the input of interest. Therefore, each pixel receives an attribution that can be interpreted as the relevance of the pixel to

the prediction. The larger the absolute value of the gradient, the stronger the influence of that pixel on the prediction. However, when backpropagating the gradient, this method discards the negative influence of features when non-linear activation functions such as ReLU are used. Other variants of this method, such as Integrated Gradients [26], Gradient × Input [22], or SmoothGrad [25] attempt to increase the quality of the saliency map by reducing the noise. Layer-wise relevance propagation method [16] backpropagates a relevance score starting from the activation of the predicted class neuron, taking the input and their sign into consideration. Several LRP propagation rules are available for different types of neural network layers, however, the resulting heatmaps can be sensitive to its hyperparameters [2].

Grad-CAM [21] is a more elaborated gradient-based method, where the gradients corresponding to a class of interest are backpropagated until the last convolutional layer and then multiplied by its aggregated feature maps. The resulting feature maps are rescaled and visualized as a saliency map over the original image. This method also applies the ReLU function to the averaged feature map, effectively ignoring the negative gradients. Nonetheless, Grad-CAM is perhaps the most applied method from this category in the literature.

A different class of feature attribution methods that is less explored than gradient-based methods is the compositional explanations. For example, Network Dissection (ND) [1] is a method that evaluates convolutional neural networks by analyzing their latent representations through labeled image datasets. It views each unit in the network as a concept detector, offering a detailed understanding of the model at the neuron level. Compositional Explanations (CE), a variation of ND introduced in [17], extends this approach by employing compositional operators on concepts to generate more intricate explanations through beam search. While CE outperforms ND in providing higher-quality explanations, it relies on a dataset with a comprehensive inventory of predefined concepts, posing a limitation if a concept is absent from the inventory. In contrast, Forward Composition Propagation [9] describes each neuron with a composition vector expressed in terms of features describing the structured classification or regression task. Such composition vectors provide semantics to the hidden components of the network and are propagated through the whole network from the input layer to the last one. This method does not make use of an extra dataset of predefined concepts but is limited to tabular classification or regression tasks.

With the increase in the number of feature attribution methods proposed in the last years comes the need for methods for evaluating their results. This is a challenging aspect of XAI in general due to the lack of ground truth for explanations or agnostic measures of the explanation quality. A recent and detailed survey on the different approaches for evaluating XAI methods can be found in [18]. In the case of feature attribution methods that are usually represented as heatmaps or rankings of feature importance, the most common technique for evaluating the correctness of explanations is the incremental deletion of features. This method consists of sequentially modifying the input by removing or altering features according to the ranking of importance provided by the expla-

nation. For each modified input, the method assesses the corresponding change in the output of the predictive model and calculates the change in terms of area under the performance curve. The results of the method under evaluation are then compared with a random ranking or other baseline methods. Obtaining a smaller area with the steepest curve can be interpreted as a sparse explanation, which is also faithful to the black box model prediction since removing the most important features accounts for the biggest loss in the prediction performance of the black box.

3 Proposed Method

In this section, we explain the rationale of the SOFI method as an optimization framework. The goal of the algorithm is to find a ranking of features $\mathbf{\Pi} = (\pi_1 \preceq \pi_2 \preceq, \ldots, \preceq \pi_n)$, such that $\pi_i \in \mathbf{F}$ is the feature ranked in the i-th ranking position, and $\mathbf{F}$ is the set of all problem features. In the feature importance problem, it holds that $\pi_i \preceq \pi_{i+1} \iff g(\pi_i) \geq g(\pi_{i+1})$, such that $g(\pi_i)$ is a function that returns model's performance after marginalizing up to the i-th feature in the ranking. The base case, where no feature is marginalized, is denoted as π_0 to maintain consistency with the notation.

The proposed algorithm seeks to solve a two-term optimization problem to find $\mathbf{\Pi}$ in such a way that $g(\pi_1) \geq g(\pi_2) \geq \cdots \geq g(\pi_n)$. More explicitly, the first term is devoted to minimizing the area under the performance curve given by the points $(0, g(\pi_0)), (1, g(\pi_1)), \ldots, (n, g(\pi_n))$ so that the curve is monotonically decreasing. The curve should not have peaks since it is desired to observe a degradation in the model's performance as more features are marginalized, although plateaus may occur. This behavior is enforced through the second term, which penalizes the function when peaks are detected. The severity with which the function is penalized depends on the importance ranking of features involved in the peak (the more important the feature, the harsher the penalization). By minimizing this function, we obtain rakings of feature importance with sparseness, which is a desired property of explanations. Additionally, we find a permutation that represents the raking of features that affect the most the prediction of the model for a given instance. This is a property that signals the correctness of the explanations and faithfulness to the behavior of the black box model [18].

Equation (1) displays the two-term objective function to be minimized in order to obtain a feature importance ranking associated with the SOFI explanation method,

$$f(\mathbf{\Pi}) = \sum_{i=0}^{n} w(i) g(\pi_i) + \lambda \sum_{i=0}^{n-1} \max\{0, \Delta_i\} \tag{1}$$

such that

$$w(i) = 1 - \frac{i}{n+1} \tag{2}$$

and

$$\Delta_i = g(\pi_{i+1}) - g(\pi_i). \tag{3}$$

Note that the penalization factor $\lambda > 0$ is a hyperparameter utilized to prevent peaks in the performance curve. The interpretation of this parameter is akin to the regularization parameter introduced in loss functions during neural network learning. In most cases, a small λ value is sufficient to enforce the desired decreasing behavior.

Due to the non-convex combinatorial nature of the optimization problem, Genetic Algorithms (GA) will be used to optimize the objective function in Eq. (1) and produce the feature importance ranking. More specifically, we use the non-dominated sorting genetic algorithm II (NSGA-II) [4] as implemented in the `pygad` library [7]. This implementation facilitates extending our proposed algorithm to a multi-objective setting that considers other explanation desiderata besides sparseness. Moreover, other evolutionary or gradient-based algorithms could be considered as optimizers in the framework of our SOFI method. Figure 1 summarizes the overview of the proposed SOFI method.

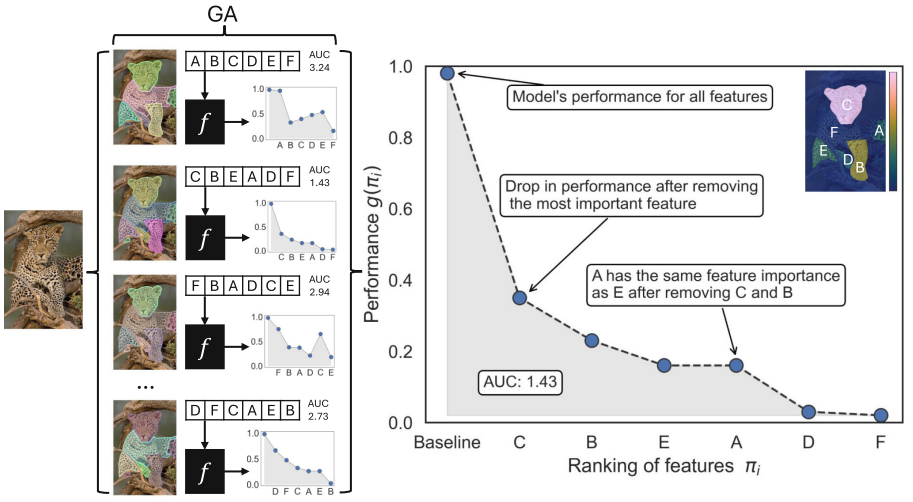


Fig. 1. Overview of SOFI method: Based on the input data, the ranking of features is optimized using a GA and evaluating a fitness function based on the area under the performance curve and a penalization factor for avoiding picks in the curve. The best permutation is returned as the resulting ranking of feature importance and optionally mapped to the instance for visualization purposes.

In our implementation using genetic algorithms, the genome will encode a permutation of the first n integers that index the problem features. Evaluating the objective function with a given black box and a chromosome involves (a) computing the model's performance (e.g., probability of correct class, accuracy, F1-score, precision, recall, etc.) using all features, and then (b) recalculating the performance scores after incrementally marginalizing the problem features in the same order as indicated by the chromosome under evaluation. In our implementation it is guaranteed that only feasible permutations are obtained after

applying the crossover and mutation operators. The optimization process outputs the best-evaluated feature importance ranking after a number of iterations.

4 Experiments and Discussion

In this section, we assess the effectiveness of the proposed SOFI post-hoc method and empirically demonstrate its superiority when contrasted with state-of-the-art feature attribution methods. To accomplish this, we will utilize both structured and unstructured data for tabular and image classification tasks, respectively.

Throughout all experiments conducted in this section, we will arbitrarily employ the following hyperparameters for the GA optimizer of the SOFI explanation method. The number of generations is set to 20, the population size is set to 100, the number of chromosomes manipulated to obtain the next population is set to 20, while the mutation and crossover probabilities are set to 0.01 and 0.9, respectively. It is important to note that the genomes cannot contain duplicated genes, as the solution represents a permutation. Finally, the penalization factor in Eq. (1) is set as $\lambda = 0.1$ to prevent the occurrence of peaks in the performance curve.

4.1 Experiments Using Structured Data

As the first experiment, we will assess the performance of SOFI, Shapley Additive Explanations (SHAP), and Permutation Feature Importance (PFI) using 16 tabular classification problems. The number of features in the datasets ranges from 7 to 29, the number of instances from 106 to 4898, and the number of decision classes from 2 to 15. In the case of SHAP, we build the explainer using 20 prototypes of the training data due to its demanding computational complexity. For the PFI method, no extra hyperparameters are needed. The area under the performance curve after incrementally marginalizing features is used as an evaluation metric for comparing the methods. The smaller the area, the sparser the explanation based on the feature importance raking.

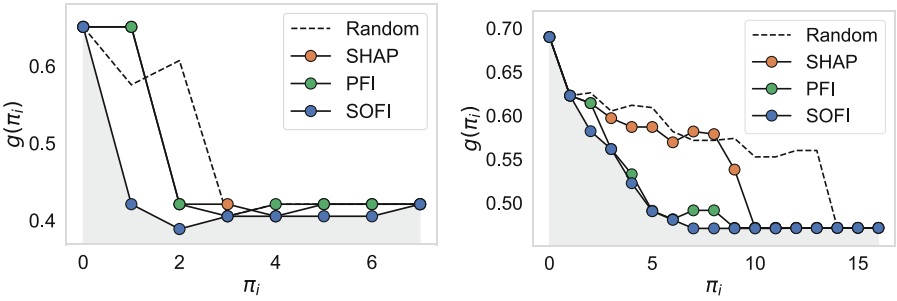
As for the black-box model, we used a gradient boosting classifier with 5-fold cross-validation and hyperparameter optimization for each dataset. The hyperparameter grid includes the following values: 100, 200, and 300 base estimators, maximum depth of the trees set to 3, 5, and 7, and learning rate values of 0.1, 0.01, and 0.001. The choice of algorithm for the black box is based on the generally good performance expected from this technique in tabular classification. The macro F1-score was selected as the model's performance metric due to being better suited for imbalanced settings, which is the case of some of the datasets included in the experiment. However, our method is completely model-agnostic, and any other performance measure could be selected. In the case of using a loss function as a performance metric for the curve, then the optimization problem should be modeled as a maximization problem, but the rationale would be analogous.

Figure 2 displays the performance degradation curves associated with all post-hoc methods for each dataset. A baseline curve representing a random feature importance ranking is also included with the goal of showing whether the post-hoc methods discover meaningful rankings. In these plots, the x-axis depicts the feature ranking indexes (which might represent different features since the rankings often differ among methods). The y-axis represents the average model's performance on the test set after marginalizing up to the i-th problem feature as indicated by each post-hoc method.

A closer inspection of sub-figures clearly reveals that SOFI produces smaller areas under the performance curve when compared to the other two post-hoc methods. This means that SOFI rankings are sparser explanations than the ones from SHAP or PFI. In 11 out of the 16 datasets analyzed, our method produces monotonically-decreasing performance degradation curves (i.e. without peaks), as opposed to SHAP and PFI, which avoid peaks in only five datasets each. The presence of peaks in the curve indicates that removing a particular feature after the previous ones have been removed increases the performance of the algorithm. This, in turn, suggests that the features that were removed already might not be the most important ones since the model is still performing better by relying on the remaining features, and therefore, the feature importance ranking is not optimally sparse. In the curves, some plateaus can also be observed, especially towards the last features of the ranking (e.g., see datasets *bank*, *ecoli*, *heart-statlog*, etc.). In our method, this means that after removing previous features, the remaining features are not important for the model since the performance equals the expected base performance with all features marginalized (i.e. similar to the base performance in the SHAP method). If the plateau is observed in the middle of the curve, this means that all features in the plateau carry the same importance. Therefore, the resulting SOFI ranking for an individual feature, is conditioned to the features that have been already removed. In this way we partially consider feature interactions.

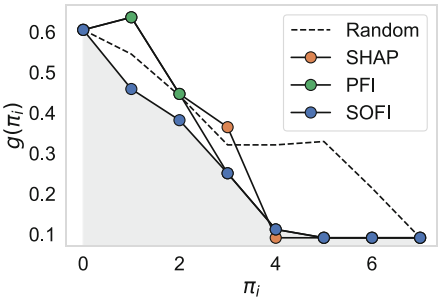
Table 1 summarizes the simulation results and includes the basic statistics of each dataset, such as the number of instances (k), the number of features (n), and the number of decision classes (m). The table reports the area under the performance curve (AUPC) and the execution time (in seconds) associated with each explanation method. It is worth mentioning that the experiments were run on an Intel Core i5-8265U CPU @ 1.60 GHz processor with 16 GB of RAM available.

The simulation results are straightforward: the proposed SOFI algorithm always reports the smallest area under the performance curve while being, on average, the second-fastest algorithm after the PFI method. The differences are statistically significant with p-values = 6.10E−05 after applying the Wilcoxon pair-wise test with Holm's correction. It is noticeable that SOFI outperformed the remaining methods using a relatively simple setting and without tuning its hyperparameters. In case SOFI produces noticeable peaks, we simply recommend increasing the λ parameter or the number of generations together with the population size. Such flexibility is absent in most post-hoc feature attribu-

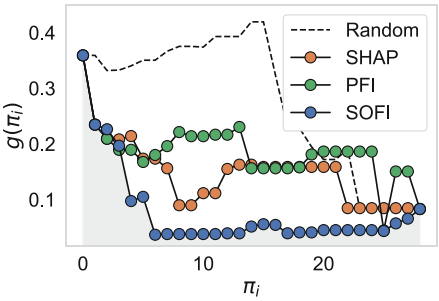


(a) appendicitis

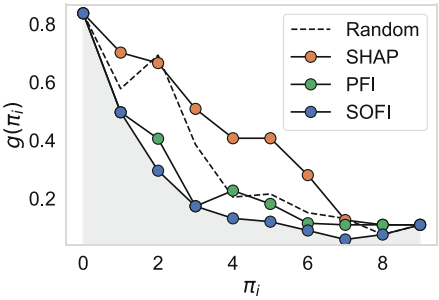
(b) bank



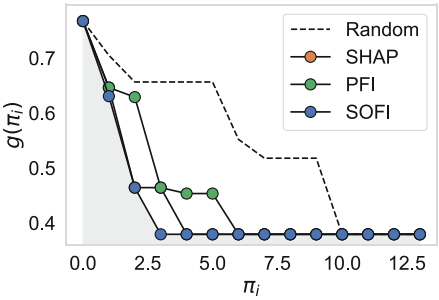
(c) ecoli



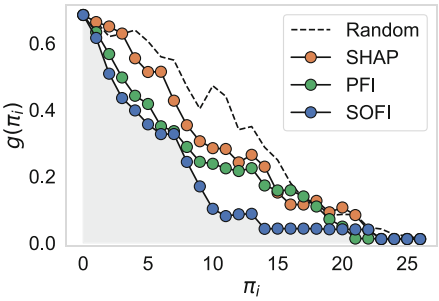
(d) flags



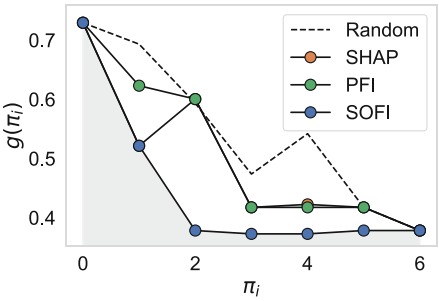
(e) glass



(f) heart-statlog

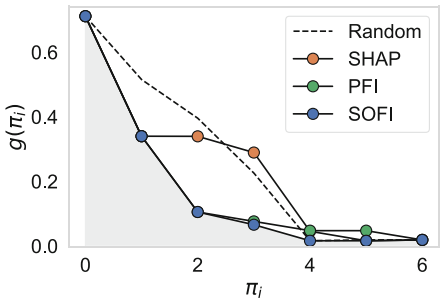


(g) libras

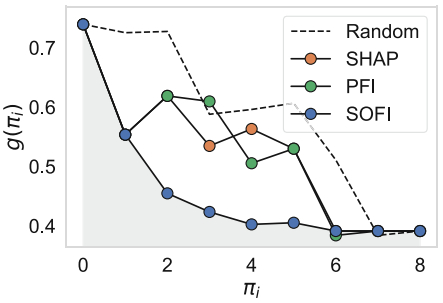


(h) liver-disorders

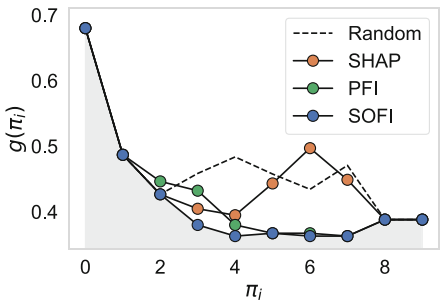
Fig. 2. Performance curves reported by feature attribution methods for selected tabular datasets



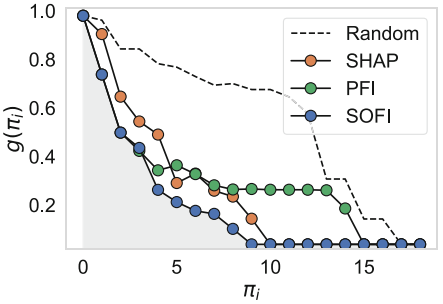
(i) mfeat-morphological



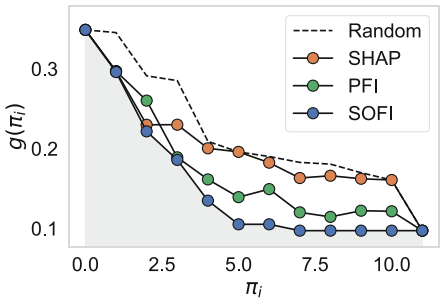
(j) pima



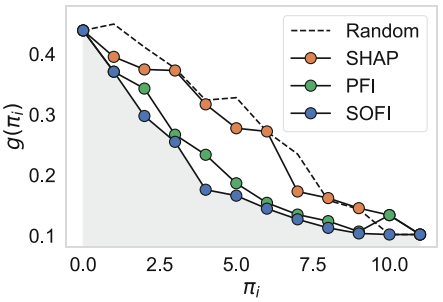
(k) saheart



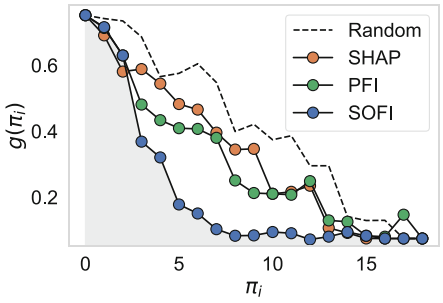
(l) segment



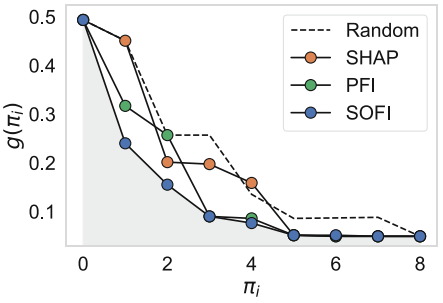
(m) wine-quality-red



(n) wine-quality-white



(o) vehicle



(p) yeast

Fig. 2. (*continued*)

Table 1. Area under the curve reported by PFI, SHAP, and SOFI for each structured dataset used in this study.

Dataset	Statistics			AUPC			Time		
	k	n	m	SOFI	SHAP	PFI	SOFI	SHAP	PFI
appendicitis	106	8	2	2.07	2.32	2.32	12.25	0.98	1.72
bank	4521	17	2	4.83	5.22	4.89	115.58	433.3	5.85
ecoli	336	8	8	1.57	1.84	1.78	12.82	6.37	2.68
flags	194	29	8	1.55	2.59	3.05	202.62	69.7	11.8
glass	214	10	6	1.86	2.99	2.07	28.47	11.44	2.59
heart-statlog	270	14	2	3.54	3.62	3.86	41.02	22.15	2.74
libras	360	27	15	3.89	5.56	4.74	165.79	198.07	16.19
liver-disorders	345	7	2	1.98	2.2	2.28	6.59	1.29	1.1
mfeat-morphological	2000	7	10	1.13	1.44	1.16	38.3	35.87	10.33
pima	768	9	2	2.53	2.88	2.89	19.87	11.71	2.42
saheart	462	10	2	2.5	2.65	2.56	18.97	9.67	1.99
segment	2310	19	7	3.25	4.17	4.2	150.9	407.31	13.06
vehicle	846	19	4	3.15	4.53	4.22	92.84	137.03	6.83
wine-quality-red	310	7	3	1.27	1.52	1.39	114.2	362.41	9.37
wine-quality-white	4898	12	7	1.65	2.1	1.76	106.33	823.99	19.03
yeast	1484	9	10	0.99	1.33	1.14	33.16	49.06	7.01

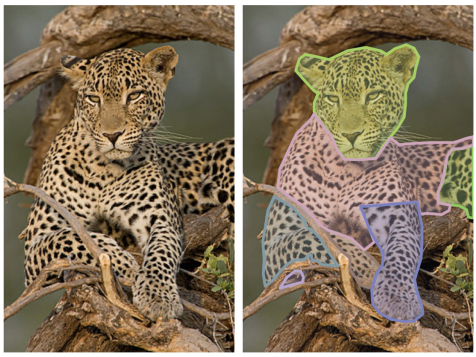
tion methods such as SHAP and PFI. Finally, in order to compare fairly with PFI in this experiment, we tested SOFI as a global explanation method where the feature importance was computed for all instances in the test set. However, as we will show in the next section, SOFI can also be used as a local explanation method, providing feature importance at the instance level.

4.2 Experiments Using Unstructured Data

As a second experiment, we test the usability of our method in unstructured data, more specifically, in an image classification task. For this experiment, we define the features as super-pixels or segments in an image. Our motivation for this approach is to provide explanations at a meaningful level of granularity in the image, represented by its parts, rather than proposing yet another post-hoc method for computing a saliency map. This representation can also be more efficient towards validating our method in user studies, where humans prefer to handle a limited number of concepts [14].

We test our approach by using the PartImageNet dataset [10], which has 24000 images annotated with 158 ImageNet [5] classes. More importantly, this dataset provides detailed part-level segmentation for a diverse range of classes. For example, the category of quadruped animals contains annotations about the head, tail, body, and leg parts. The dataset contains 11 categories of images and 40 types of annotations. As a black box for this experiment, we use ResNet50 [11] architecture, initialized with the weights after training on the ImageNet dataset.

In Fig. 3, we show an example instance of the PartImageNet dataset with its corresponding segment annotation. For our experiment, we additionally denote the background as another segment and, therefore, is also represented as a feature. Analogous to the marginalization of features in the previous experiment, we should choose a perturbation technique for images. In this case, we use a Gaussian filter with a standard deviation of 5.0, which produces a significant blurring of the segment.



(a) Original image (b) Segmentation

Fig. 3. Example image from PartImageNet dataset and its segment annotations. The top three classes predicted by ResNet50 for this instance are *leopard* (0.99), *cheetah* (0.01), and *jaguar* (0.01).

Figure 4 shows the effect of blurring segments in the example image. Here, the order of the blurring process corresponds to a random feature importance ranking created as a baseline. On the top of each image, we show the change in the probability of the correct class (*leopard*) predicted by the black box. From this image, it is evident that the baseline ranking is far from being optimally sparse since removing the background of the image as the most important feature only reduces the probability of recognizing the class in 0.04.

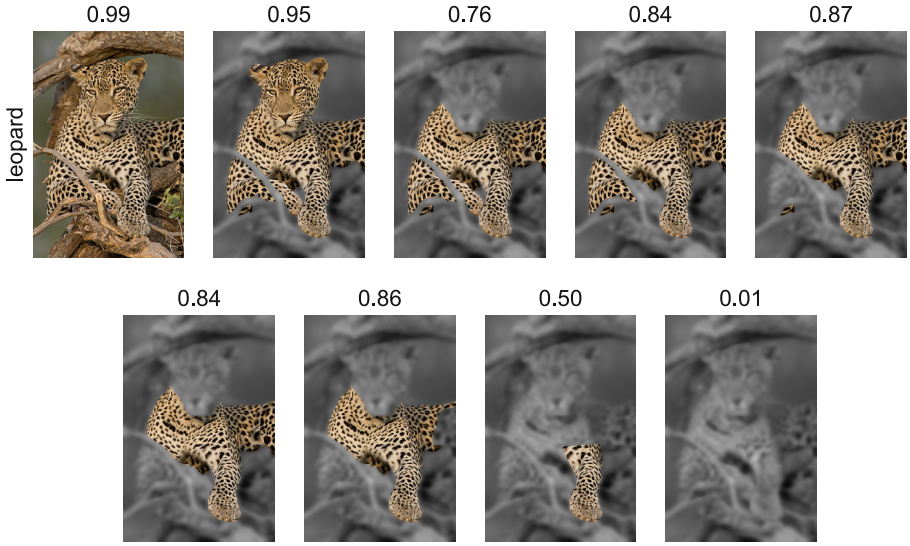


Fig. 4. Incremental blurring of segments and the effect on the probability of detecting the correct class for a random baseline ranking of feature importance.

Next, we compute the SOFI permutation for this instance while keeping the hyperparameters reported in the previous experiment. The result of SOFI is a ranking of the segments which the method considers the most important ones. For comparison, we also compute the saliency maps resulting from the state-of-the-art methods SHAP, Grad-CAM, and RISE. In the case of SHAP, we use a maximum number of evaluations of the underlying model equal to 1000 and a batch size of 50. For RISE, we use the number of masks set to 2000, and Grad-CAM does not require any hyperparameter. For the compared methods, after obtaining their saliency maps in differing scales, we aggregate their absolute values on each PartImageNet segment, creating a segmented feature importance mask for each method that can be compared to SOFI results. Figure 5 shows the original and the segmented saliency maps for each method.

From Fig. 5, we can observe that the saliency maps obtained differ in what they consider to be the most important part of the image for the underlying neural network. While Grad-CAM emphasizes the head, SHAP focuses its feature contribution on the leg of the leopard and RISE in other parts. Interestingly, SOFI highlights the head in first place and the leg in second place, with other parts being less relevant. In Fig. 6, we show the effect of blurring the image according to the resulting feature importance rakings associated with the four methods tested. The associated probability of predicting the correct class is shown at the top of the image.

Figure 6 shows that both Grad-CAM and SOFI are able to detect the segment of the image (i.e., the head of the leopard), which reduces the most the probability of predicting the correct class, after removal. However, Grad-CAM attributes

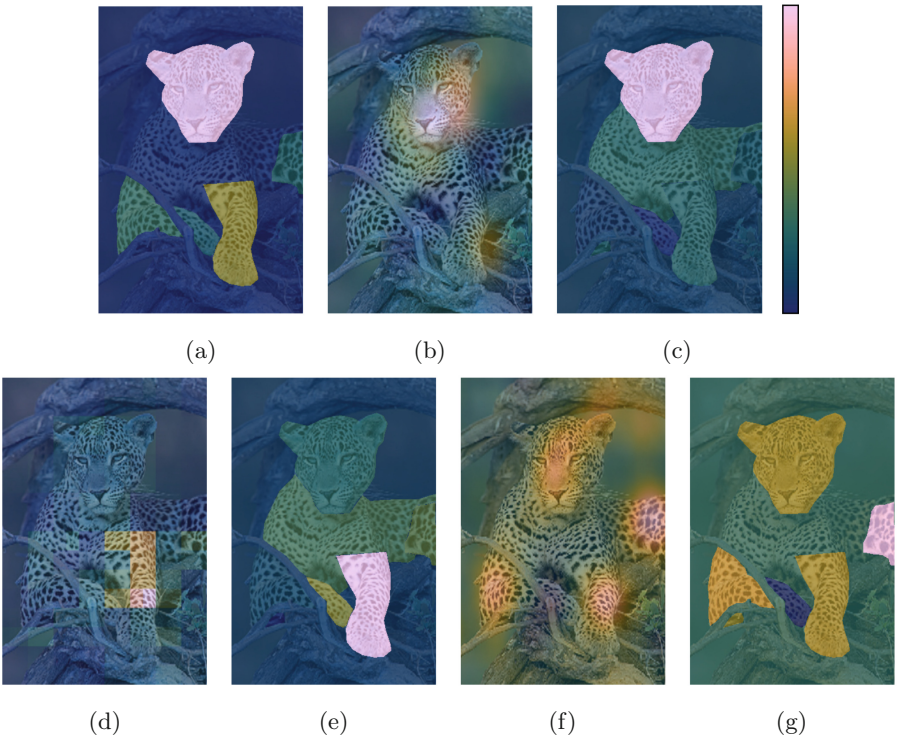
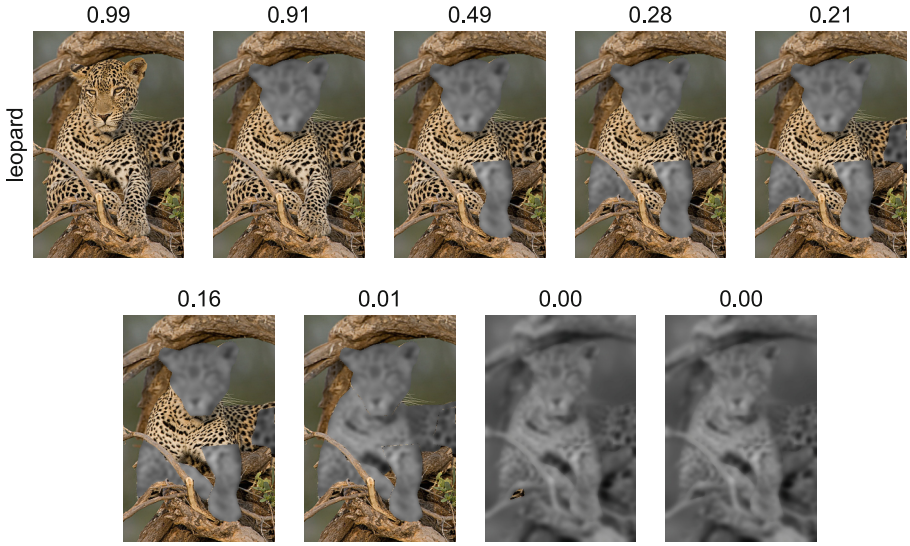
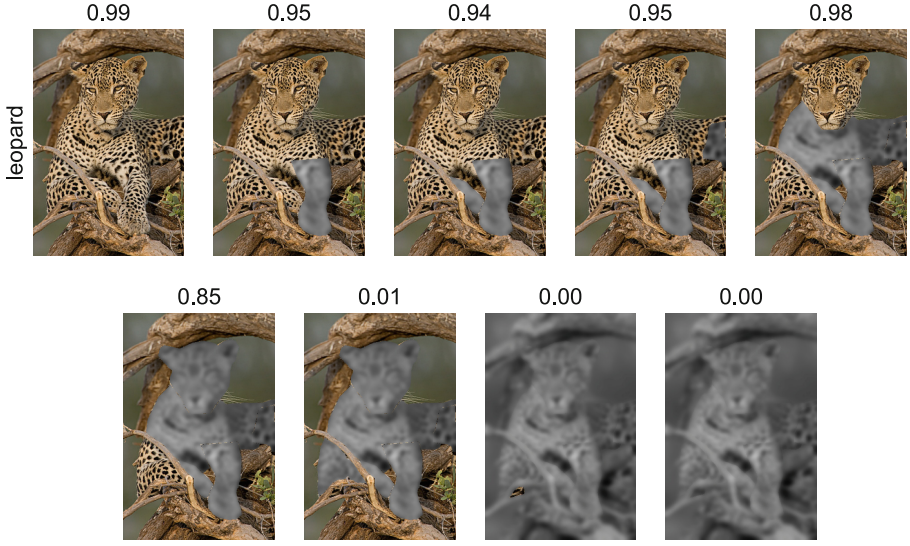


Fig. 5. Original and segmented feature importance masks for each method: (a) SOFI segmented mask, (b) Grad-CAM original saliency map, (c) Grad-CAM segmented saliency map, (d) SHAP original saliency map, (e) SHAP segmented saliency map, (f) RISE saliency map, (g) RISE segmented saliency map. The segmented saliency maps for Grad-CAM, SHAP, and RISE are computed as the mean absolute pixel or superpixel importance of the PartImageNet segments.

importance to other regions of the body, which do not harm the performance of the model when occluded. In contrast, SHAP attributes the highest importance to the segment representing the front leg, which incurs a smaller reduction of the probability (0.04) than SOFI and Grad-CAM while giving importance to other parts of the body. In the case of RISE, the first two attributions focus on parts of the body that do not reduce the model's capacity to predict the leopard significantly. However, for all models, it is observed that after blurring the head, a significant drop in performance is observed, reinforcing the role of this segment in the prediction. We consider this instance particularly challenging since the animal class has a clear pattern that can be associated with any part of the body. Nevertheless, it is evident that the features of the head and front leg are relevant to the model prediction.

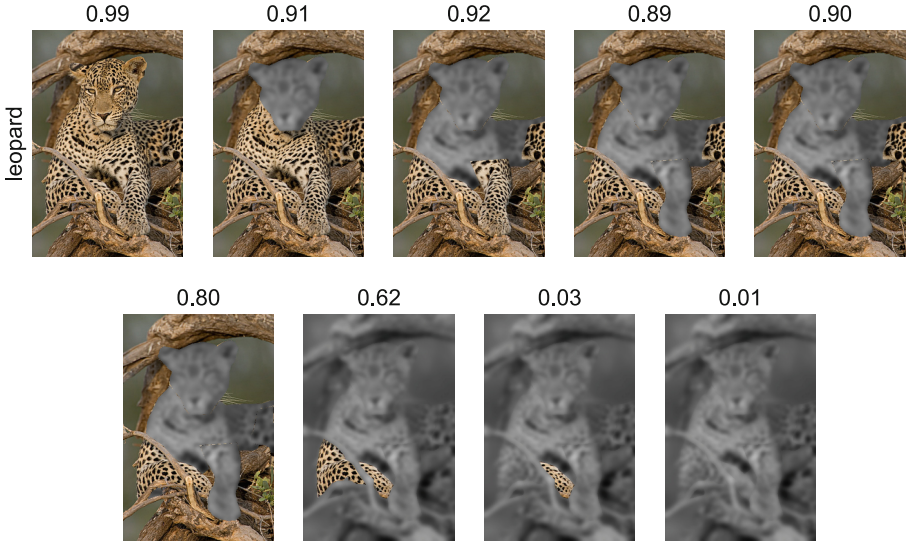


(a) SOFI

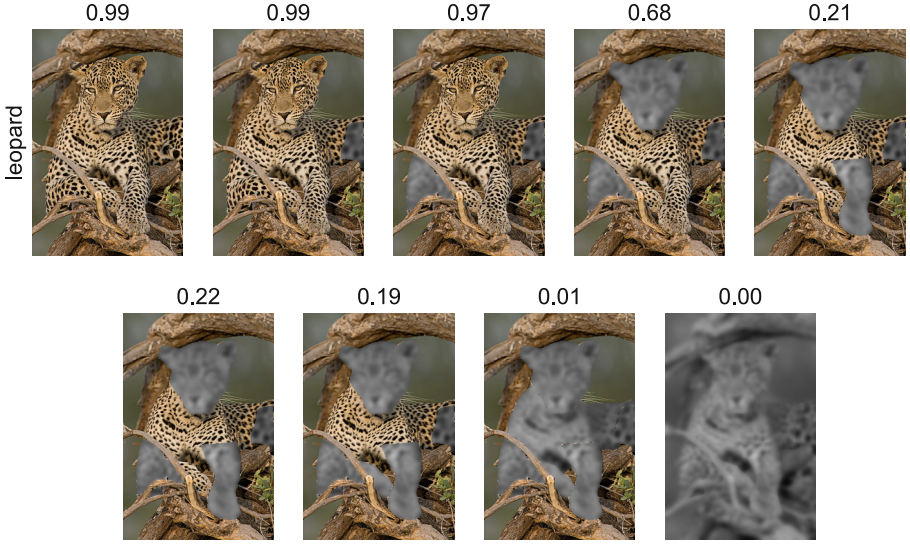


(b) SHAP

Fig. 6. Incremental blurring of segments and the effect on the probability of detecting the correct class for different feature rakings: a) SOFI, b) SHAP, c) Grad-CAM, and d) RISE saliency maps.



(c) Grad-CAM



(d) RISE

Fig. 6. (*continued*)

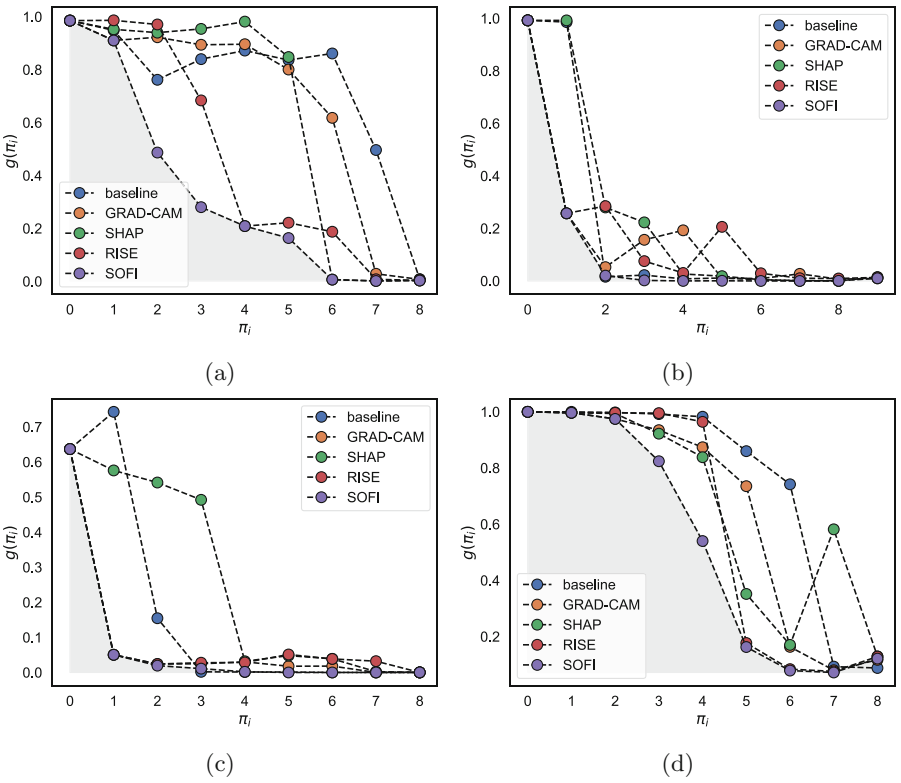


Fig. 7. Performance degradation curves for the ResNet50 architecture when predicting four example instances, according to the feature importance rankings computed by the four methods in the study. The classes corresponding to each instance are: (a) *leopard*, (b) *chow*, (c) *mink*, and (d) *gordon setter*. The incremental blurring plots for instances b, c, and d are provided in the Appendix.

Finally, Fig. 7 shows the performance degradation curves for all methods studied on the instance of reference (a) and another three instances of the dataset with a different number of segments. We randomly choose three instances that correspond to different classes and categories in the segmentation annotations in order to show the performance of our algorithm in different settings. The incremental blurring plots associated with the other three inspected instances can be found in Figs. 9, 10, and 8, in the Appendix.

From these plots, we can observe that SOFI produces the rankings of segments that cause the biggest reduction in the ability of the model to recognize the true class. This means that SOFI produces explanations in the form of feature importance rankings that are the sparsest (in terms of the number of concepts that they involve) compared to those produced by SHAP, Grad-CAM, and RISE.

5 Concluding Remarks

This paper presented SOFI, an agnostic post-hoc method for generating either local or global feature attribution explanations, which solves a two-term combinatorial problem. The algorithm produces an explanation in the form of feature importance ranking, with optimal sparseness in terms of the number of features considered relevant. The rationale of our algorithm is that marginalizing the most important features will cause a sharper drop in the model's performance compared to marginalizing the least important ones. As a result, the feature importance rankings obtained by SOFI are also deemed more correct and faithful to the behavior of the black box model since they show the steepest performance degradation curves in the experiments. In the literature, the sparseness and correctness properties derived from incrementally deleting features continue to be the mainstream approach to compare feature attribution post-hoc methods, regardless of their inner working principles.

When tested on structured classification tasks, the SOFI method continuously reported the smallest area under the performance curve compared to PFI and SHAP methods while being the second fastest algorithm after PFI. Similarly, SOFI reported sparser explanations than SHAP, GradCam, and RISE methods when tested on image datasets instances. In the image classification task, segmentation annotations were used as super-pixels. This approach is not perceived as a relevant limitation since there are efficient algorithms for image segmentation [12] that allow obtaining meaningful super-pixels, thus conveniently reducing the granularity of explanations. Future work will be focused on extending our approach to multi-objective optimization settings where more than one desiderata for explanations can be optimized, especially after the automated semantic segmentation of unstructured data such as images.

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Disclosure of Interests. All authors declare that they have no conflicts of interest.

Appendix

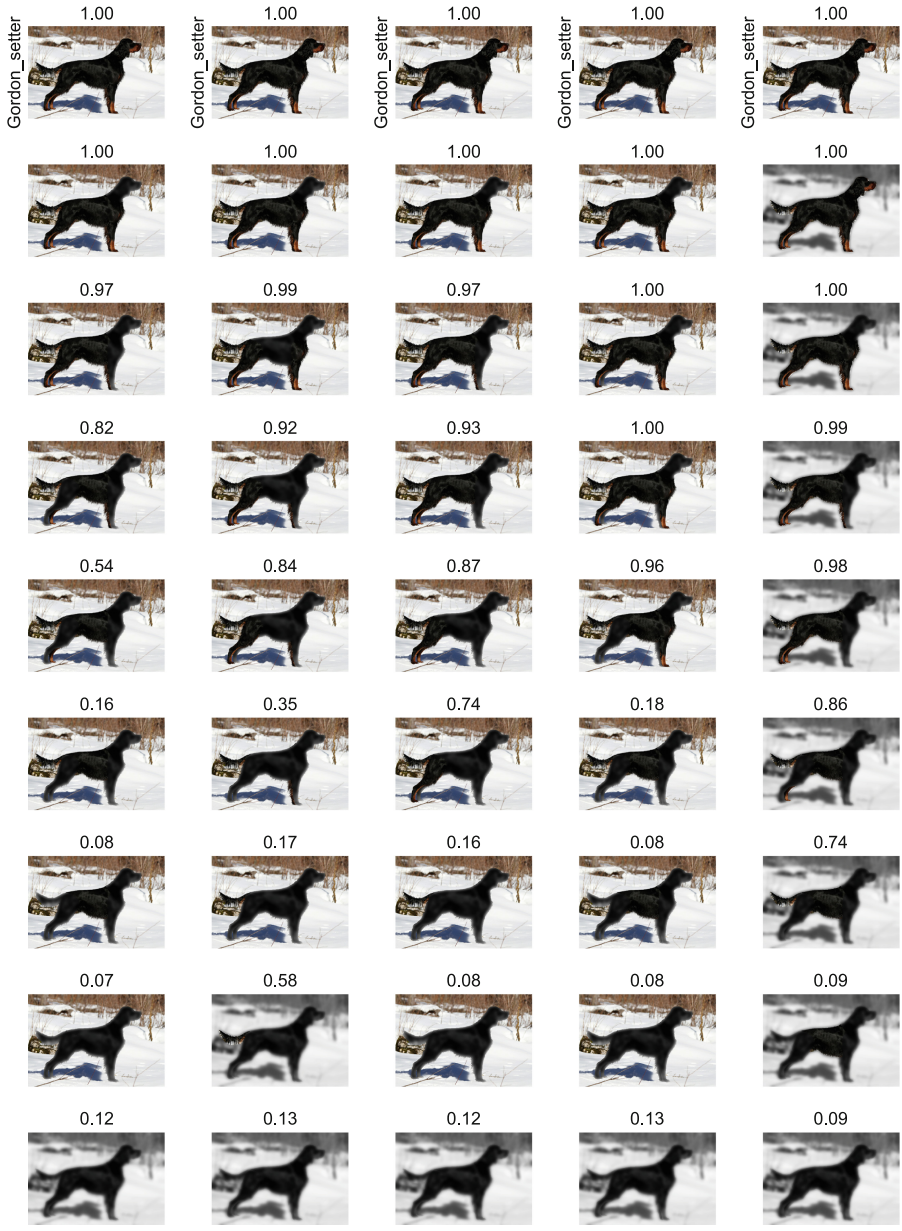

Fig. 8. Incremental blurring of segments and the effect on the probability of detecting the correct class *gordon setter*, for different feature rakings: a) SOFI, b) SHAP, c) Grad-CAM, d) RISE and e) a random baseline. The incremental blurring process is represented vertically from the top to the bottom of the figure.

(a) SOFI (b) SHAP (c) Grad-CAM (d) RISE (e) Baseline

Fig. 9. Incremental blurring of segments and the effect on the probability of detecting the correct class *chow*, for different feature rakings: a) SOFI, b) SHAP, c) Grad-CAM, d) RISE and e) a random baseline. The incremental blurring process is represented vertically from the top to the bottom of the figure.

(a) SOFI (b) SHAP (c) Grad-CAM (d) RISE (e) Baseline

Fig. 10. Incremental blurring of segments and the effect on the probability of detecting the correct class *mink*, for different feature rakings: a) SOFI, b) SHAP, c) Grad-CAM, d) RISE and e) a random baseline. The incremental blurring process is represented vertically from the top to the bottom of the figure.

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Stabilizing Estimates of Shapley Values with Control Variates

Jeremy Goldwasser[1(✉)] and Giles Hooker[2]

[1] Department of Statistics, UC Berkeley, 367 Evans Hall, Berkeley, CA 94720, USA
jeremy_goldwasser@berkeley.edu
[2] Department of Statistics and Data Science, University of Pennsylvania, 265 South 37th Street, Philadelphia, PA 19104, USA
ghooker@wharton.upenn.edu

Abstract. Shapley values are among the most popular tools for explaining predictions of black-box machine learning models. However, their high computational cost motivates the use of sampling approximations, inducing a considerable degree of uncertainty. To stabilize these model explanations, we propose ControlSHAP, an approach based on the Monte Carlo technique of control variates. Our methodology is applicable to any machine learning model and requires virtually no extra computation or modeling effort. On several high-dimensional datasets, we find it can produce dramatic reductions in the variability of Shapley estimates.

Keywords: SHAP · Shapley Values · Interpretability · Monte Carlo

1 Introduction

Machine learning models have become widely deployed in high-stakes domains such as healthcare, finance, and criminal justice [11,13,24]. Use of algorithmic decision-making in these areas has profound human consequences, but the predictive models themselves may be opaque. This motivates the development of methods to understand the reasoning behind the predictions of a black-box model [2].

To that end, feature importance methods attribute a value to each feature passed into the model, indicating its predictive importance [4,10,23,29,38]. In this paper, we focus on Shapley values, a seminal concept in game theory which have emerged as one of the most popular tools for model interpretability [23,31]. In layman's terms, Shapley values quantify how much information is gained from being told the value of each feature, averaged over orderings in which the features are revealed.

Shapley values are rarely computed exactly, as the computational cost is exponential in the number of input features. Rather, they are typically estimated using the Shapley Sampling or KernelSHAP algorithm [23,33,34]. These algorithms, however, are subject to sampling variability; as a result, running the same procedure twice may yield different estimated Shapley values, including

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different estimated orderings of features. This instability raises questions about the trustworthiness of insights gleaned from Shapley values.

We seek to mitigate this issue by employing Monte Carlo variance reduction techniques. In particular, we use control variates, a method that adjusts one random estimator based on the known error of another. Here, the related estimator approximates the Shapley values of a first or second order Taylor expansion to the original model, depending on whether the value function assumes features are correlated or independent. In the independent case, these estimates entail essentially no additional computation; otherwise we must put some effort into pre-computing terms which can then be used for any query point for which we need to calculate Shapley values. While variations on our methods are possible, from tuning parameters to more complex Monte Carlo schemes [28], our goal is to provide a default scheme that requires minimal computational or intellectual effort.

Estimates of Shapley values are significantly more stable with control variates. In some cases, variance reductions reach as high as 90% for dozens of features, along with noticeably more consistent rankings of Shapley values among features.[1]

2 Shapley Values in Machine Learning

2.1 Shapley Values

In the context of machine learning, the Shapley values for an input sample $x \in \mathbb{R}^d$ present the change in prediction - i.e. the information gained - from learning the value of each feature. They do so by averaging the marginal contribution of a feature to every subset, or "coalition," of other features.

Formally, let $f : \mathcal{X} \to \mathbb{R}$ be a machine learning model trained on $X \in \mathbb{R}^{n \times d}$. Further, let $S \subseteq [d] := \{1, \ldots, d\}$ denote a subset of the d features. Following SHAP [23], the value function for input x maps a subset S to an estimate of the mean prediction, conditioned on the known features x_S:

$$v_x(S) := \hat{\mathbb{E}}[f(x_S, X_{[d]\setminus S})] = \hat{\mathbb{E}}[f(X)|X_S = x_S]. \tag{1}$$

In some settings - e.g. if the features are multivariate Gaussian or mutually independent - the conditional mean can be computed exactly. Otherwise, when the conditional distribution is intractable, the true mean is impossible to compute. Hence, in this case the value function $v_x(S)$ estimates $\mathbb{E}[f(x_S, X_{[d]\setminus S})]$ using an approximation of the conditional distribution $X|X_S$.

The j$^{\text{th}}$ Shapley value for input x, which we denote $\phi_j(x)$, averages the change in value function across all subsets of $[d]$, with and without the j^{th} feature.

$$\phi_j(x) = \frac{1}{d} \sum_{S \subseteq [d]\setminus\{j\}} \binom{d-1}{|S|}^{-1} \big(v_x(S \cup \{j\}) - v_x(S)\big). \tag{2}$$

[1] https://github.com/jeremy-goldwasser/ControlSHAP contains our code and experimental results.

Each Shapley value is a function of $\mathcal{O}(2^d)$ subsets. Since this is computationally prohibitive in high dimensions, we consider sampling approximations.

2.2 Shapley Estimation

Shapley sampling [33] is the most straightforward algorithm to approximate Shapley Values. For each iteration $m \in \{1, \ldots, M\}$, randomly permute the d features, and select the subset $S^m \subseteq [d] \backslash \{j\}$ that appears before j. Shapley sampling averages the change in value function from adding feature j to each subset.

$$\hat{\phi}_j(x) = \frac{1}{M} \sum_{m=1}^{M} v_x(S^m \cup \{j\}) - v_x(S^m). \tag{3}$$

In contrast, the KernelSHAP method [23] estimates all d features at once. Again, in each iteration m we select a coalition S^m and compute its value function $v_x(S^m)$. Rather than sampling coalitions via permutations, however, S^m is sampled from the distribution

$$p(S) \propto \frac{d-1}{\binom{d}{|S|}(|S|)(d-|S|)}. \tag{4}$$

Each coalition S can be equivalently expressed as a binary vector $z \in \{0,1\}^d$ by defining $z_i = \mathbb{1}\{i \in S\}$ for all $i \in [d]$. Accordingly, z vectors $\mathbf{0}$ and $\mathbf{1}$ correspond to $S = \emptyset$ and $[d]$, respectively.

Abusing notation, let $v_x(z) := v_x(S)$, where z is the binary representation of S. Consequently, we can express $v_x(\mathbf{0}) = \mathbb{E}[f(X)]$ and $v_x(\mathbf{1}) = f(x)$. The KernelSHAP estimator is defined as

$$\begin{aligned} \hat{\phi}(x) &= \underset{\phi \in \mathbb{R}^d}{\operatorname{argmin}} \frac{1}{M} \sum_{m=1}^{M} \left(\phi^T z^m - (v_x(z^m) - v_x(\mathbf{0}))\right)^2 \\ &\quad \text{s.t.} \quad \mathbf{1}^T \phi = v_x(\mathbf{1}) - v_x(\mathbf{0}) \\ &= (Z^T Z)^{-1} \left(I - \frac{\mathbf{1}\mathbf{1}^T (Z^T Z)^{-1}}{\mathbf{1}^T (Z^T Z)^{-1} \mathbf{1}} \right) Z^T (V - v_x(\mathbf{0})) \\ &\qquad + \left(\frac{v_x(\mathbf{1}) - v_x(\mathbf{0})}{\mathbf{1}^T (Z^T Z)^{-1} \mathbf{1}} \right) (Z^T Z)^{-1} \mathbf{1}, \end{aligned} \tag{5}$$

where $Z \in \mathbb{R}^{M \times d}$ and $V \in \mathbb{R}^M$ contain the coalitions z^m and values $v_x(z^m)$.

Drawing from [6,23] proved that the weighting kernel in 4 produces linear regression coefficients that are equal to the Shapley values.

2.3 Related Work

A number of works seek to reduce the sample variability of Shapley values. Mitchell et al. propose several Monte Carlo techniques for choosing subsets [25];

independently, Covert and Lee advocate antithetic sampling for KernelSHAP [9]. Song et al. accelerate computation by storing pre-computed subsets [32]. Our method can be applied on top of these methods, leveraging their contributions for added benefit.

Alternatives to sampling-based methods have also been proposed. Chen et al. derive expressions for Shapley values in settings for which the features contribute to predictions with a well-understood graphical structure [8]. FastSHAP directly predicts Shapley values using a neural network [18]; it generalizes well when the neural network is trained on a large, diverse dataset. Other works more efficiently compute Shapley values given specific predictive models, e.g. Tree SHAP and Deep SHAP [22,23]. These methods can quickly estimate Shapley values or obtain them exactly. However, they are limited to settings that satisfy their preconditions on model types, data structures, and number of inputs to explain.

3 Control Variates for Shapley Values

3.1 Control Variates

The method of control variates is a variance reduction technique for Monte Carlo sampling problems. To reduce the error of an estimator, it leverages information about the known error of another.

Suppose $\hat{A}$ and $\hat{B}$ are unbiased estimators of A^* and B^*, where only B^* is known. Define the control variates estimator as

$$\tilde{A} := \hat{A} - c(\hat{B} - B^*). \tag{6}$$

For any constant c, $\tilde{A}$ is an unbiased estimator of A^*. Consequently, its mean squared error (MSE) is equivalent to its variance. The optimal c is selected with the following lemma, using plug-in estimates of the variance and covariance.

Lemma 1. *Suppose $\hat{A}, \hat{B}$ are unbiased estimators and $\rho_{\hat{A},\hat{B}} > 0$, where ρ is Pearson's correlation coefficient. Then the control variates estimator (6) has minimal variance $Var(\tilde{A}) = (1 - \rho^2_{\hat{A},\hat{B}})\, Var(\hat{A})$ when $c^* = \frac{Cov(\hat{A},\hat{B})}{Var(\hat{B})}$.*

3.2 ControlSHAP

Our method, ControlSHAP, applies the method of control variates to the Shapley values for input x. The estimands A^* and B^* in Sect. 3.1 are represented by $\phi_j^{\text{model}}(x)$ and $\phi_j^{\text{approx}}(x)$; these are the j^{th} Shapley values of f and its Taylor approximation. In the following subsections, we will elaborate on the approximate model and its true Shapley values $\phi^{\text{approx}}(x)$.

Either Shapley sampling or KernelSHAP can be used to compute $\hat{\phi}_j^{\text{model}}(x)$ and $\hat{\phi}_j^{\text{approx}}(x)$. These are both appropriate for control variates because they are essentially unbiased for the true Shapley value (Eq. 2), whose value function is the approximate conditional mean (Eq. 1). Trivially, Shapley sampling (Eq. 3)

is unbiased, and [9] showed that KernelSHAP has negligible bias (Eq. 5). Prior works have established that these methods may be biased for the Shapley value whose value function is the true conditional mean, $v_x(S) = \mathbb{E}[f(X)|X_s]$; however, by definition this Shapley value is not the estimand for this problem, as these means are intractable [1,7,35]. By virtue of (6), control variate corrections to these estimands do not change their bias.

Whichever method we choose, we must use the same subsets S in computing $\hat{\phi}_j^{\text{model}}(x)$ and $\hat{\phi}_j^{\text{approx}}(x)$ to ensure they are correlated. Following Eq. 6 and Lemma 1, the ControlSHAP estimator is

$$\hat{\phi}_j^{\text{CV}}(x) = \hat{\phi}_j^{\text{model}}(x) - \hat{\alpha}\big(\hat{\phi}_j^{\text{approx}}(x) - \phi_j^{\text{approx}}(x)\big) \tag{7}$$
$$\text{where } \hat{\alpha} = \frac{\widehat{\text{Cov}}(\hat{\phi}_j^{\text{model}}(x), \hat{\phi}_j^{\text{approx}}(x))}{\widehat{\text{Var}}(\hat{\phi}_j^{\text{approx}}(x))}.$$

If α is obtained using the oracle covariance and variance, then this estimator reduces variance and MSE by a factor of $\rho^2(\hat{\phi}_j(x)^{\text{model}}, \hat{\phi}_j(x)^{\text{approx}})$.

3.3 Independent Features

To compute Shapley values at x, we must estimate the conditional mean for each sampled subset, $\mathbb{E}[f(X)|X_S = x_S]$. To do so, we generate a number of samples from $X|X_S$, compute $f(X)$ on each, and take a sample mean. The conditional distribution $X|X_S$ may be difficult or impossible to characterize. Because of this, the standard approach popularized by SHAP samples the features in S^C from their marginal distributions [23,33]. Doing so makes the implicit assumption that the features are independent for the sake of computing $\mathbb{E}[f(X)|X_S = x_S]$.

Consider the second-order Taylor approximation of f around x. If the value function is computed sampling features from their marginal distributions, then the Shapley values $\phi^{\text{approx}}(x)$ of this approximate model can be computed exactly. While we refer to this as the "independent features case," the following theorem holds *regardless* of whether the features are actually independent.

Theorem 1. *Define $\mu := \mathbb{E}[X]$ and $\Sigma_{jk} := Cov(X_j, X_k)$. Let the value function $v_x(S)$ compute conditional means by sampling each feature from its marginal distribution. The j^{th} Shapley value of the second-order Taylor approximation of f around x is*

$$\begin{aligned}\phi_j^{approx}(x) = &\frac{\partial f}{\partial x_j}(x_j - \mu_j) \\ &- \frac{1}{2}\left[\sum_{k=1}^{d}(x_k - \mu_k)\frac{\partial^2 f}{\partial x_j x_k}\right](x_j - \mu_j) \\ &- \frac{1}{2}\sum_{k=1}^{d}\Sigma_{jk}\frac{\partial^2 f}{\partial x_j x_k}.\end{aligned}$$

Proof of Theorem 1 is in the Sect. A.1 of the appendix.

3.4 Correlated Features

Input features are often correlated with one another. When this is the case, it may be imprudent to falsely assume independence for the sake of computational convenience. This may produce Shapley values that misrepresent the relationship between inputs and predictions, and rely on the values that a machine learning function makes in areas where it has very little data [17].

Unfortunately, respecting feature correlations requires that we sample from $X|X_S$ for all sampled subsets S. Nonparametric density estimation in high dimensions is challenging due to the curse of dimensionality, and most approaches typically require considerable modeling effort.

To circumvent this, a common alternative estimates the conditional mean by characterizing $X|X_S$ as a multivariate normal distribution [1,7]. This makes the value function easy to compute, without having to ignore dependencies between the features. One can even faithfully represent binary data under this categorization by simulating Gaussian random variables and thresholding accordingly [21].

Moreover, using this value function enables us to compute the true Shapley value of a first-order Taylor expansion of f. As in Theorem 1, the formula in Theorem 2 holds whether or not the features are indeed normally distributed.

Theorem 2. *Define the value function to take the conditional expectation of $f(X)|X_S$ over a multivariate normal distribution. Let $S^m \subseteq [d]\backslash\{j\}$ be the subset of features appearing before j in the m^{th} permutation of $[d]$, and let Q_S and R_S be matrices defined in Sect. A.2 of the supplementary material that do not depend on x. Define*

$$D_j := \frac{1}{d!}\sum_{m=1}^{d!}([Q_{S_j^m \cup j} + R_{S_j^m \cup j}] - [Q_{S_j^m} + R_{S_j^m}]).$$

The j^{th} Shapley value of the first-order Taylor approximation of f around x is

$$\phi_j^{approx}(x) = \nabla f(x)^T D_j (x - \mu).$$

Evaluating $\phi_j^{\text{approx}}(x)$ still requires computing an exponential number of terms for D_j. However, because D_j does not depend on x, we can reuse it for other inputs x. That is, we can front-load computational work into accurately estimating D_j; then, we can instantly obtain close approximations to $\phi_j^{\text{approx}}(x)$ for as many inputs x as we want. This is valuable if we want to estimate Shapley values at many inputs, either for future data or to provide a global notion of feature importance.

The normality condition in Theorem 2 can be relaxed somewhat. It is needed insofar as it produces conditional means of the form

$$E[X_1|X_2 = x_2] = \mu_1 + \Sigma_{12}\Sigma_{22}^{-1}(x_2 - \mu_2),$$

which occur whenever the variance of the conditional distribution is independent of the values of the conditioning set. This property holds for all elliptically-symmetric distributions, including the multivariate t-, Laplace, and logistic distributions [12].

3.5 Non-differentiable Models

The discussion so far assumes that we have access to first and sometimes second derivatives of f. This excludes models based on trees that result in piecewise constant functions, as well as neural networks with ReLU activation functions, which have first but not second derivatives. While there are specific Shapley methods for trees [5], these only apply to models with axis-oriented splits (counter to e.g. [37]).

For our purposes, we use a Taylor series only to provide linear and quadratic approximations to f. Thus, we can replace $\partial f/\partial x_i$ and $\partial^2 f/\partial x_i \partial x_j$ with finite-difference approximations based on large step sizes, so as to approximate model behavior across substantial changes in the features.

In our experiments below, we used finite differencing on random forest predictors. We chose the step size of each numeric feature to be its marginal standard deviation; for categorical features, we swapped the observed values. Alternative choices would include basing step sizes on conditional distributions, or tuning them for agreement with f. Our goal has been to examine a straightforward control variate method with minimal costs both computationally and in human effort, so we have not pursued these variations here. We speculate that additional tuning would provide further improvements, which may be worthwhile in particular applications.

4 Variance Estimation

To use the method of control variates, we need to estimate the variance and covariance of our two estimators, as in Eq. 7. This section outlines strategies for doing so.

4.1 Shapley Sampling

Let $G_S := v_x(S \cup j) - v_x(S)$ for subset S. Shapley sampling selects M subsets S via independent permutations, then takes a sample mean of their G_S values. Therefore the variance of Shapley sampling values is

$$\mathrm{Var}(\hat{\phi}_j) = \frac{1}{M}\mathrm{Var}(G_S).$$

We estimate $\mathrm{Var}(G_S)$ empirically across the M observations of G_S. Estimating the covariance between the Shapley estimates is similarly straightfoward:

$$\mathrm{Cov}(\hat{\phi}_j^{\mathrm{model}}, \hat{\phi}_j^{\mathrm{approx}}) = \frac{1}{M}\mathrm{Cov}(G_S^{\mathrm{model}}, G_S^{\mathrm{approx}}),$$

which again we estimate empirically.

4.2 KernelSHAP

We present three approaches to estimate the KernelSHAP variance and covariance. These are unrelated to the choice of independent or correlated features.

1. **Bootstrapping**. For many iterations, take M samples with replacement from the observed subsets S. Compute the KernelSHAP estimates $\hat{\phi}_{\text{boot}}^{\text{model}}$ and $\hat{\phi}_{\text{boot}}^{\text{approx}}$ on the pairs $\{(S_{\text{boot}}, v_{\text{boot}}^{\text{model}}(S))\}_{i=1}^{M}$ and $\{(S_{\text{boot}}, v_{\text{boot}}^{\text{approx}}(S))\}_{i=1}^{M}$. Repeat this process to produce many bootstrapped estimates of $\hat{\phi}_{\text{boot}}^{\text{model}}$ and $\hat{\phi}_{\text{boot}}^{\text{approx}}$, with which we compute empirical variance and covariance.
2. **Least Squares**. KernelSHAP learns $\hat{\phi}$ by fitting a constrained least squares (Eq. 5). Writing $A(Z) = (Z^T Z)^{-1}\left[I - (\mathbf{1}\mathbf{1}^T(Z^T Z)^{-1})/(\mathbf{1}^T(Z^T Z)^{-1}\mathbf{1})\right] Z^T$ we make the approximation

$$\widehat{\text{Var}}(\hat{\phi}) = A(Z)\text{Var}(v_x(S))A(Z)^T.$$

   The diagonals of $Var(v_x(S))$ are computed via sample variance, provided more than one sample is used to estimate each $v_x(S)$. Analogously,

$$\widehat{\text{Cov}}(\hat{\phi}^{\text{model}}, \hat{\phi}^{\text{approx}}) = A(Z)\widetilde{\Sigma}A(Z)^T$$

   where $\widetilde{\Sigma} = \text{Cov}(v_x(S)^{\text{model}}, v_x(S)^{\text{approx}})$.
   Here we note that this formula ostensibly accounts only for the sampling variability in $v_x(S)$ rather than the Monte Carlo choice of coalitions Z. However, by the total covariance formula writing $\hat{\phi} = B(Z, V)$ we have that

$$\begin{aligned}&\text{Cov}(\hat{\phi}^{\text{model}}, \hat{\phi}^{\text{approx}})\\ &\quad= \mathbb{E}_Z\text{Cov}(B(Z, V^{\text{model}}), B(Z, V^{\text{approx}})|Z)\\ &\qquad+ \text{Cov}_Z(\mathbb{E}_V B(Z, V^{\text{model}}), \mathbb{E}_V B(Z, V^{\text{approx}}))\end{aligned}$$

   [9] demonstrated that kernelSHAP has small bias, making the second term small, and this calculation provides a consistent estimate of the first term. $\text{Var}(\hat{\phi}^{\text{model}})$ is estimated with an equivalent calculation.
3. **Grouped**. [9] introduced a method based on their finding that the variance evolves at a $O(\frac{1}{M})$ rate. First, split the $\{(S, v_x(S))\}_{i=1}^{M}$ pairs into disjoint groups, and fit $\hat{\phi}^{(k)}$ in each. Then, take the empirical variance and covariance of these $\hat{\phi}^{(k)}$ values, and scale linearly.

Empirically, we observed that the bootstrapped and least squares methods yield roughly the same variances and covariances. Section D of the appendix shows that ControlSHAP with these two methods has nearly identical performance, across multiple datasets and machine learning models. Our experimental results all use the least squares method, since bootstrapping has a moderately higher computational cost.

In our experiments, we were not able to get the grouped method to reliably work. We observed that it occasionally produced highly erroneous estimates of variance and covariance. While this may be remedied by fitting a larger number of groups, the group size must be large enough to avoid singularity issues in fitting $\hat{\phi}^{(k)}$. More details are in Sect. D of the appendix.

5 Experiments

We demonstrated the efficacy of our ControlSHAP methodology on five datasets. A simulated dataset contained ten normally distributed features with block covariance structure; each feature had a correlation of 0.5 with one feature and 0.25 with another, and the response given by a logistic regression. We also used four public datasets from the UCI ML repository: Portuguese bank [26], German credit [16], census income [19], and breast cancer subtyping [3].

These real-world datasets are higher-dimensional, containing roughly 20 features on average. Many features are categorical with more than two levels, e.g. Month. We represent each of these features as a set of one-hot encodings. These transformations are necessary to use models like logistic regression and neural networks; they are also required for ControlSHAP, which takes gradients with respect to the features. The Shapley value of a categorical feature is the sum across the individual levels [23].

In the following analyses, we refer to the two approaches for the value function as assuming features are independent versus correlated (dependent). This terminology only serves to distinguish the method of sampling and Taylor expansion. It does not entail making actual assumptions about the data, as detailed in Sect. 3. The ControlSHAP estimator's properties such as unbiasedness do not require feature independence or Gaussianity to hold.

For each data set, and each model, we sampled 40 data points from the test set at random and estimated Shapley values 50 times for each, with and without control variates and using both correlated and independent sampling. For each estimate, we sampled 1000 coalitions. For KernelSHAP, we generated 10 points per coalition to estimate pointwise variances. We ran 200 bootstrap samples and 20 groups to estimate variances and covariances for control variates for bootstrap and grouped estimators. For Shapley sampling, which re-runs the Monte Carlo calculation for each feature, we used 1 sample per coalition. Computations were run on an internal cluster.

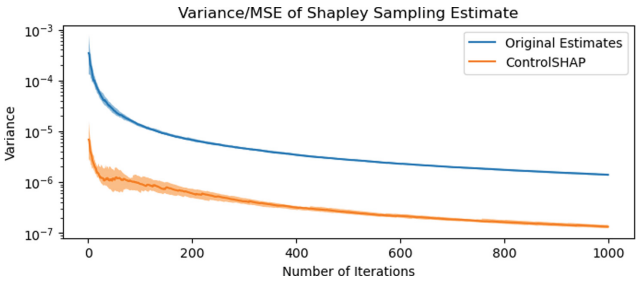


Fig. 1. Variance (or MSE) of Shapley estimates. Computed independent-features Shapley sampling estimates on "Prev Days" feature, using logistic regression model on Bank dataset. Bands show inner quartiles across 50 iterations.

As Fig. 1 demonstrates, ControlSHAP significantly improves the variability of Shapley estimates, regardless of the number of samples drawn. For this feature - chosen because it has the highest Shapley value - adjusting the Shapley estimate consistently reduces its variance by at least 90%.

Next, we consider the improvements in variance - equivalently, in MSE - across all 5 datasets, using a variety of machine learning predictors. We calculated the variance reduction of our ControlSHAP methods by taking the sample variance of each feature's Shapley value across 50 iterations.

$$\text{VarReduc}_j(x) = \frac{\left[\widehat{\text{Var}}(\hat{\phi}_j^{\text{model}}(x)) - \widehat{\text{Var}}(\hat{\phi}_j^{\text{CV}}(x))\right]}{\widehat{\text{Var}}(\hat{\phi}_j^{\text{model}}(x))}. \tag{8}$$

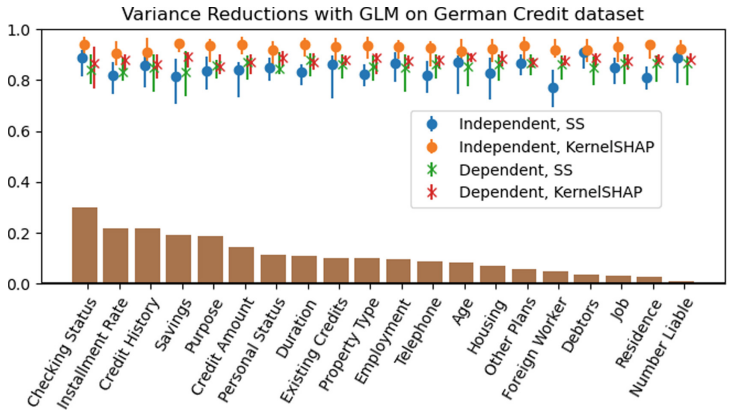


Fig. 2. Variance reductions of ControlSHAP methods. Sorted features by absolute Shapley values (unscaled in plot) with logistic regression predictor. Confidence bands denote lower and upper quartiles of Eq. 8 across 40 held-out values of x.

Figure 2 displays results for all features of the German Credit dataset, using logistic regression as a predictor. Our four ControlSHAP methods produce large variance reductions on all features, whose Shapley values span several orders of magnitude. On average, our final Shapley estimates have roughly 80–95% lower variability than the original estimates.

A second stability metric we used was the number of ranking changes. This metric is motivated by the fact that often practitioners care more about identifying features with high Shapley values than the values themselves [14,15,27]. Formally, let $\hat{r}_j(x)$ be the ranking of feature j within a set of d Shapley estimates. For each method, we compute the average number of pairwise ranking changes across 50 resamplings.

$$\text{RankChgs}(x) = \binom{50}{2}^{-1} \sum_{a \neq b} \left[\sum_{j=1}^{d} |\hat{r}_j^a(x) - \hat{r}_j^b(x)| \right]. \tag{9}$$

Figure 3 shows the average number of ranking changes for 40 sample inputs. We observe that the rankings are significantly more stable with the control variate. ControlSHAP averages around 20 rank changes, compared to 60 for the original estimates. Note that this statistic highly depends on the stability of Shapley sampling prior to using control variates: if the original estimates exhibit no ranking changes between simulations, this statistic will not show an improvement even if they variance of the estimates is reduced. We believe our simulations to be representative of current practice where a noticeable reduction is evident.

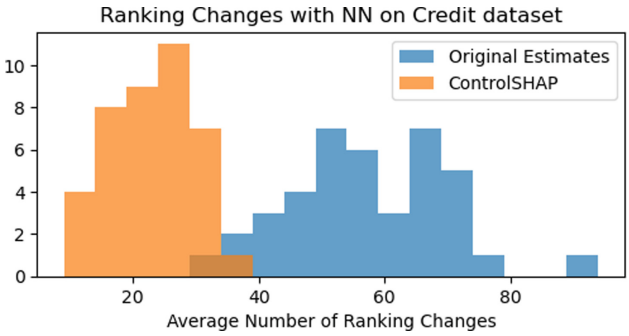


Fig. 3. Average number of Ranking Changes (Eq. 9) across 50 iterations, with and without ControlSHAP correction. Shapley values estimated on 40 inputs via KernelSHAP with correlated features. Neural network trained on Credit dataset.

Table 1 contains results across all 5 datasets with logistic regression, neural network, and random forest predictors. In addition to reductions in variance, we also present reductions in the number of rank changes, averaged over 40 sample inputs. Full sets of visualizations for variance reductions and rank changes can be found in Sect. B of the appendix.

In every setting, ControlSHAP reduces the variability of Shapley estimates, often by a wide margin. For both logistic regression and neural networks, at least one ControlSHAP method produces over a 50% reduction in variance and 30% reduction in number of rank changes on all five datasets. On the BRCA and German credit datasets, these figures rise to 85% and 55% for most ControlSHAP methods. Every ControlSHAP method reduces variance by over 67% on three of the four real-world datasets with logistic regression.

While ControlSHAP is generally strong for neural networks, it tends to work better when in the dependent features case. On the simulated, bank, and census datasets, variance reductions are typically below 25% assuming independence and above 50% otherwise. Presumably, this owes to the fact that neural networks are less smooth. The quadratic model from the independent features case (Theorem 1) is more prone to neural networks' steep gradients and hessians, and may poorly approximate model behavior as a result.

Table 1. ControlSHAP Percent Reductions in Variance (left) and Ranking Changes (right). Shapley Sampling (SS) and KernelSHAP (KS), with and without independence assumption. Averaging variance and rank-change reductions over 40 sample inputs. Taking median variance reduction of features with 5 highest Shapley values.

Logistic Reg.	**SIM**	**BANK**	**BRCA**	**CENSUS**	**CREDIT**
Indep., SS	10% | 6%	76% | 57%	75% | 57%	33% | 13%	83% | 60%
Indep., KS	24% | 17%	84% | 43%	87% | 54%	3% | 2%	94% | 67%
Correlated, SS	59% | 32%	67% | 42%	92% | 68%	71% | 58%	85% | 64%
Correlated, KS	51% | 27%	69% | 29%	94% | 71%	59% | 30%	87% | 59%

Neural Net	**SIM**	**BANK**	**BRCA**	**CENSUS**	**CREDIT**
Indep., SS	9% | 6%	8% | 2%	76% | 52%	4% | 3%	55% | 36%
Indep., KS	22% | 15%	25% | 8%	86% | 55%	-1%| 1%	80% | 45%
Correlated, SS	58% | 31%	58% | 37%	92% | 65%	61% | 42%	84% | 61%
Correlated, KS	51% | 28%	40% | 13%	92% | 68%	51% | 24%	87% | 60%

RF	**SIM**	**BANK**	**BRCA**	**CENSUS**	**CREDIT**
Indep., SS	31% | 15	18% | 3%	19% | 7%	4%| 2%	15% | 6%
Indep., KS	37% | 19%	4% | 3%	14% | 6%	7%| 4%	14% | 6%
Correlated, SS	56% | 31%	6% | 5%	22% | 11%	2% | 5%	24% | 17%
Correlated, KS	54% | 31%	1% | -2%	17% | 9%	3% | -1%	22% | 10%

When the machine learning model is not differentiable, gradients and hessians can be taken via finite differencing. These coarse estimates use a smooth approximation to step-wise constant model behavior, which also results in a lower correlation between our target and control variate.

Nevertheless, ControlSHAP increases stability on random forests, again particularly in the correlated features case. Figure 4 showcases its solid performance on the simulated dataset, reducing variability by over 50% with dependent features. Table 1 shows its reductions in variance and rank changes on all datasets and ControlSHAP methods.

Finally, we analyzed the ControlSHAP estimates to confirm they are unbiased. While the true SHAP values are computationally prohibitive to compute, their sum is known to be $f(x) - E[f(X)]$ by the efficiency property [23,31]. To search for bias, we compared the sum of Shapley estimates from this target, across multiple ML predictors. Figure 5 demonstrates that ControlSHAP exhibits no evidence of bias; on the contrary, the sum of its Shapley estimates is typically closer to the target.

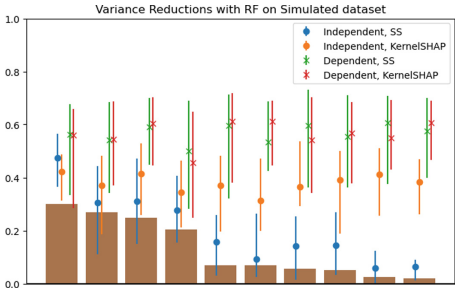


Fig. 4. Variance Reductions of Random Forest on simulated dataset with 10 features. Upper and lower quantiles of reductions provided across 40 inputs. Computed gradients via finite differencing.

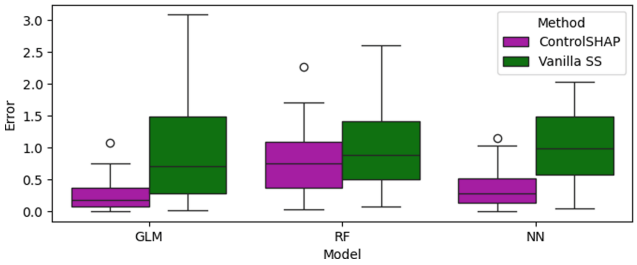


Fig. 5. Normalized absolute difference from sum of Shapely estimates to $f(x) - Ef(X)$. Ran for 10 iterations on credit dataset with independent features.

6 Discussion

While Shapley values are capable of providing useful model explanations, they fall short in one critical way: stability. Stability is necessary to instill trust in our interpretations of black-box machine learning algorithms. Practitioners cannot use such a system with confidence if rerunning the same procedure yields different explanations.

This work proposes a method, ControlSHAP, to address this. ControlSHAP can be applied in conjunction with other attempts to mitigate sampling variability. Further, while designed for differentiable models, it can help explain other predictive models via finite differencing. Specifically, it adjusts Shapley estimates using the method of control variates, which track the extent to which the random samples over- or underestimate on a correlated estimator.

ControlSHAP stabilizes Shapley estimates across a wide range of datasets and algorithms, in some cases reducing variability by as high as 90%. Moreover, we can easily obtain accurate estimates of the added benefit from performing ControlSHAP adjustments. Recall that ControlSHAP reduces variance by approximately $\rho^2(\hat{\phi}_j(x)^{\text{model}}, \hat{\phi}_j(x)^{\text{approx}})$ (Eq. 7). We can estimate this correlation using the variance and covariance used to compute $\widehat{\alpha}$; this enables us to

estimate the variance reduction. As Sect. C in the appendix demonstrates, these anticipated reductions closely mirror the empirical quantities.

Alternatively, ControlSHAP can be used to accelerate convergence. [9], for example, suggest terminating when the variability falls below a certain threshold. Using the control variate here could lead to a dramatic speed-up. Note in Fig. 1 that the ControlSHAP estimates have comparable variability after 100 iterations as the original values after 1000.

ControlSHAP can be employed as a relatively "off-the-shelf" tool, in the sense that it stabilizes Shapley estimates with close to no extra computational or modeling work. The only model insight required is the gradient, as well as the hessian in the independent features case. Computationally, the single substantial cost is in the correlated features case, which requires accurate estimation of pre-compute matrices. Otherwise, ControlSHAP only necessitates passing each $X|X_S$ through the Taylor approximation, and computing the Shapley values' covariance - both of which are extremely quick tasks.

It is worth noting that other explanation algorithms also suffer from various forms of instability. [40] and [39] found that the explanations from LIME and distillation trees are sensitive to the synthetic data used to generate them. In addition, counterfactual explanations for adjacent points may be subject to vary widely [20].

These drawbacks motivate future work stabilizing model explanations. Shapley computations may benefit from tuning finite differences, more sophisticated Monte Carlo schemes, or alternative control variates. In some contexts it may be preferable to use inherently interpretable models like decision trees or the LASSO [30,36]. We intend ControlSHAP to complement these models, enabling more trustworthy use of black-box methods.

Disclosure of Interests. We have no competing interests to disclose.

Appendix

A Exact Shapley Values

A.1 Shapley Value, Assuming Feature Independence

Proof. Assume features are sampled independently in the value function. Consider a second-order Taylor approximation to f at x:

$$g(x') := f(x) + (x' - x)^T \nabla f(x) + \frac{1}{2}(x' - x)^T \nabla^2 f(x)(x' - x).$$

Let $J = \nabla f(x)$, $H = \nabla^2 f(x)$, $\Sigma = \text{Cov}(X)$, and $\sigma^2_{k\ell} = \text{Cov}(X_k, X_\ell)$. The Shapley value function is

$$\begin{aligned} v_x(S) &= E[g(X')|X'_S = x_S] \\ &= f(x) + E[(x' - x)|X'_S = x_S]^T \nabla f(x) \\ &\qquad + \frac{1}{2} E\Big[(x' - x)^T \nabla^2 f(x)(x' - x)|X'_S = x_S\Big] \\ &= f(x) + (\mu_{S^C} - x_{S^C})^T J_{S^C} \\ &\qquad + \frac{1}{2} E\Big[(x'_{S^C} - x_{S^C})^T \nabla^2 f(x)_{S^C S^C}(x'_{S^C} - x_{S^C})\Big], \end{aligned}$$

where the quadratic term is equal to

$$\begin{aligned} &tr(H_{S^C S^C} \Sigma_{S^C S^C}) + (\mu_{S^C} - x_{S^C})^T H_{S^C S^C}(\mu_{S^C} - x_{S^C}) \\ =& \sum_{k \in S^C} \sum_{\ell \in S^C} H_{k\ell}\Big(\sigma^2_{k\ell} + (\mu_k - x_k)(\mu_\ell - x_\ell)\Big) \\ =& H_{jj}(\sigma^2_{jj} + (\mu_j - x_j)^2). \end{aligned}$$

Recall $\phi_j(x) := \dfrac{1}{d} \displaystyle\sum_{S \subseteq [d] \setminus \{j\}} \binom{d-1}{|S|}^{-1} \big(v_x(S \cup \{j\}) - v_x(S)\big)$. The difference in value functions is

$$v_x(S \cup j) - v_x(S) = \underbrace{(x_j - \mu_j)J_j - \frac{1}{2} H_{jj}\Big(\sigma^2_j + (\mu_j - x_j)^2\Big)}_{\text{(a)}} - \underbrace{\sum_{k \in (S \cup j)^C} \underbrace{H_{jk}\Big[\sigma^2_{jk} + (\mu_j - x_j)(\mu_k - x_k))\Big]}_{\text{(c)}}}_{\text{(b)}}.$$

Define the Shapley weight $w_S = \frac{1}{d}\binom{d-1}{|S|}^{-1}$.

$$\begin{aligned}
\phi_j(x) &= \sum_{S\subseteq[d]\setminus\{j\}} w_S(\textcircled{a} - \textcircled{b}) \\
&= \textcircled{a} \sum_{S\subseteq[d]\setminus\{j\}} w_S - \sum_{S\subseteq[d]\setminus\{j\}} w_S \textcircled{b} \\
\sum_{S\subseteq[d]\setminus\{j\}} w_S \textcircled{b} &= \sum_{S\subseteq[d]\setminus\{j\}} w_S \sum_{k\in\{S\cup j\}^C} \textcircled{c} \\
&= \sum_{S\subseteq[d]\setminus\{j\}} \sum_{k\in\{S\cup j\}^C} w_S \textcircled{c} \\
&= \sum_{k\neq j} \sum_{S\subseteq[d]\setminus\{j,k\}} w_S \textcircled{c} \\
&= \sum_{k\neq j} \textcircled{c} \sum_{S\subseteq[d]\setminus\{j,k\}} w_S.
\end{aligned} \tag{10}$$

Noting subsets of equal size have the same Shapley weight, we can easily show $\sum_{S\subseteq[d]\setminus\{j\}} w_S = 1$.

$$\begin{aligned}
\sum_{S\subseteq[d]\setminus\{j\}} w_S &= \sum_{S\subseteq[d]\setminus\{j\}} \frac{1}{d}\binom{d-1}{|S|}^{-1} \\
&= \sum_{a=0}^{d-1} \binom{d-1}{a} \frac{1}{d} \binom{d-1}{a}^{-1} = \sum_{a=0}^{d-1} \frac{1}{d} = 1
\end{aligned} \tag{11}$$

With a bit more arithmetic, we can show $\sum_{S\subseteq[d]\setminus\{j,k\}} w_S = \frac{1}{2}$.

$$\begin{aligned}
\sum_{S\subseteq[d]\setminus\{j,k\}} w_S &= \sum_{a=0}^{d-2} \binom{d-2}{a} \frac{1}{d} \binom{d-1}{a}^{-1} \\
&= \sum_{a=0}^{d-2} \frac{(d-2)!}{a!(d-a-2)!} \frac{a!(d-a-1)!}{d!} \\
&= \sum_{a=0}^{d-2} \frac{d-a-1}{d(d-1)} \\
&= \frac{1}{d(d-1)} \Big[d(d-1) - \sum_{a=0}^{d-2} (a+1) \Big] \\
&= \frac{1}{d(d-1)} \Big[d(d-1) - \frac{d(d-1)}{2} \Big] = \frac{1}{2}.
\end{aligned} \tag{12}$$

Plugging the results of 11 and 12 into 10 yields the final expression for the Shapley value:

$$\begin{aligned}
\phi_j(x) &= \textcircled{a} \sum_{S \subseteq [d] \setminus \{j\}} w_S - \sum_{S \subseteq [d] \setminus \{j\}} w_S \textcircled{b} \\
&= \textcircled{a} - \frac{1}{2} \sum_{k \neq j} \textcircled{c} \\
&= (x_j - \mu_j) J_j - \frac{1}{2} H_{jj} \Big(\sigma_j^2 + (\mu_j - x_j) \Big) \\
&\qquad - \frac{1}{2} \sum_{k \neq j} H_{jk} \Big[\sigma_{jk}^2 + (\mu_j - x_j)(\mu_k - x_k)) \Big] \\
&= (x_j - \mu_j) J_j - \frac{1}{2} \Big[\sum_{k=1}^{d} (x_k - \mu_k) H_{jk}) \Big] (x_j - \mu_j) \\
&\qquad - \frac{1}{2} \sum_{k=1}^{d} \sigma_{jk}^2 H_{jk}.
\end{aligned}$$

□

A.2 Shapley Value, Correlated Features

Consider the linear model $g(x') = \beta^T x' + b$, where $x' \sim \mathcal{N}(\mu, \Sigma)$. (For our problem, $\beta = \nabla f(x)$ and $b = f(x)$.)

Let P_S be the projection matrix selecting set S; define $R_S = P_{\bar{S}}^T P_{\bar{S}} \Sigma P_S^T (P_S \Sigma P_S^T)^{-1} P_S$ and $Q_S = P_S^T P_S$. We consider the set of permutations of $[d]$, each of which indexes a subset S as the features that appear before j. Averaging over all such permutations, the Shapley values of $g(x)$ [1,7,23] are

$$\begin{aligned}
\phi_j(x') = \beta \underbrace{\left[\frac{1}{d!} \sum_m ([Q_{\{S^m \cup j\}^C} - R_{S^m \cup j}] - [Q_{\{S^m\}^C} - R_{S^m}]) \right]}_{C_j} \mu \\
+ \beta \underbrace{\left[\frac{1}{d!} \sum_m ([Q_{S^m \cup j} + R_{S^m \cup j}] - [Q_{S^m} + R_{S^m}]) \right]}_{D_j} x'.
\end{aligned}$$

Lastly, we observe that $C_j = -D_j$. For each subset S, the R terms cancel out; $Q_{S^C} + Q_S = I_d$, so $Q_{\{S \cup j\}^C} - Q_{S^C} = -(Q_{S \cup j} - Q_S)$. This yields our expression for the dependent Shapley value:

$$\phi_j^g(x) = \beta^T D_j (x - \mu).$$

B Comprehensive Results

We display results across all 5 datasets and 3 machine learning predictors. All datasets were binary classification problems, so we used the same models to fit them. For logistic regression and random forest, we used the sklearn implementation with default hyperparameters. For the neural network, we fit a two-layer MLP in Pytorch with 50 neurons in the hidden layer and hyperbolic tangent activation functions.

Figure 6 displays the variance reductions of our ControlSHAP methods in the four settings: Independent vs Dependent Features, and Shapley Sampling vs KernelSHAP. The error bars span the 2^{th} to 7^{th} percentiles of the variance reductions for 40 held-out samples.

Figure 7 compares the average number of changes in rankings between the original and ControlSHAP Shapley estimates. Specifically, we look the Shapley estimates obtained via KernelSHAP, assuming correlated features.

C Anticipated Correlation

Figure 8 compares the observed and *anticipated* variance reductions, across two combinations of dataset and predictor. Recall that the variance reduction of the control variate estimate for $\phi_j(x)$ is $\rho^2(\hat{\phi}_j(x)^{\text{model}}, \hat{\phi}_j(x)^{\text{approx}})$, where ρ is the Pearson's correlation coefficient. We compute the sample correlation coefficient of the Shapley values on sample x as follows:

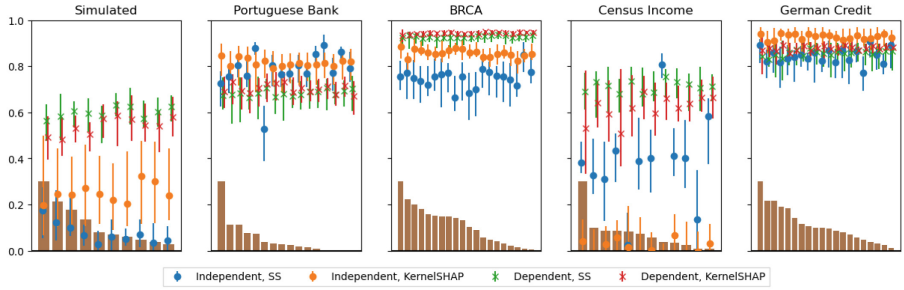
$$\hat{\rho} := \frac{\widehat{\text{Cov}}(\hat{\phi}_j(x)^{\text{model}}, \hat{\phi}_j(x)^{\text{approx}})}{\sqrt{\widehat{\text{Var}}(\hat{\phi}_j(x)^{\text{model}})\widehat{\text{Var}}(\hat{\phi}_j(x)^{\text{approx}})}}$$

We average this across the 50 iterations to obtain a single estimate for the correlation $\hat{\rho}$. The plots display the median and error bars for $\hat{\rho}^2(\hat{\phi}_j(x)^{\text{model}}, \hat{\phi}_j(x)^{\text{approx}})$ across 40 samples. Once again, the error bars span the 2^{th} to 7^{th} percentiles.

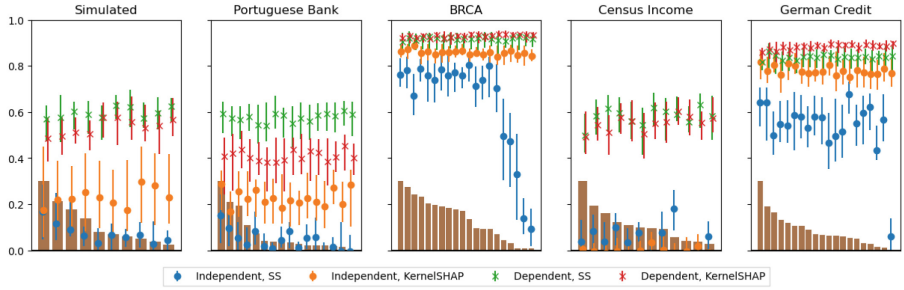
D Comparing KernelSHAP Variance Estimators

Section 4.2 of the paper details methods for computing the variance and covariance between KernelSHAP estimates. Bootstrapping and the least squares covariance produce extremely similar estimates. In turn, these produce ControlSHAP estimates that reduce variance by roughly the same amount, as shown in Fig. 9. This indicates that both methods are appropriate choices.

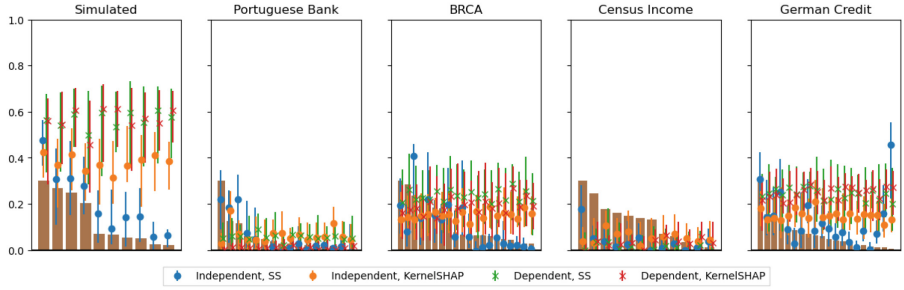
In contrast, we were not able to get the grouped method to reliably work. Its variance estimates were somewhat erratic, as they are drawn from a heavy-tailed χ^2 distribution (Fig. 10). As a result, its ControlSHAP estimates were occasionally *more* variable than the original KernelSHAP values themselves.



(a) Logistic Regression

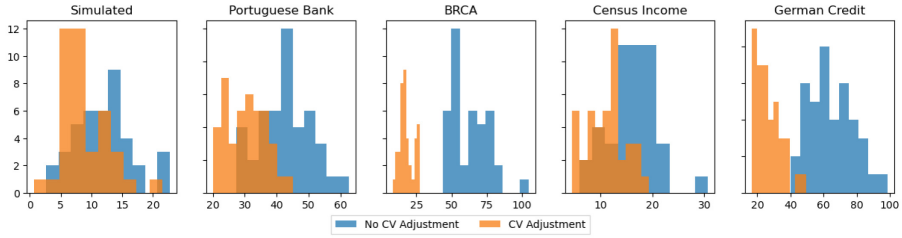


(b) Neural Network

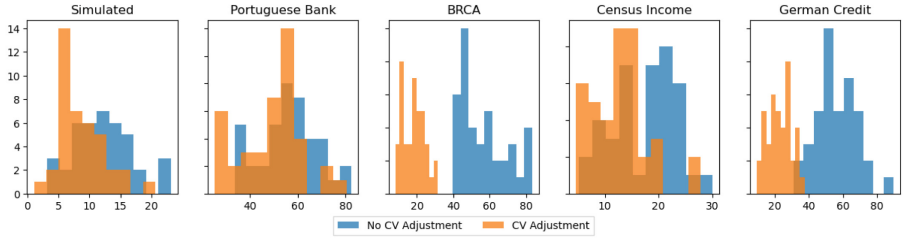


(c) Random Forest

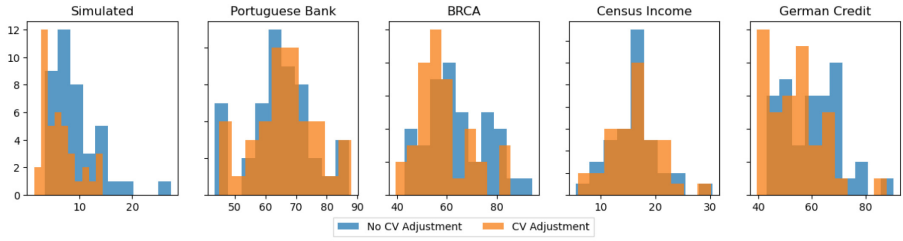
Fig. 6. Variance Reductions



(a) Logistic Regression

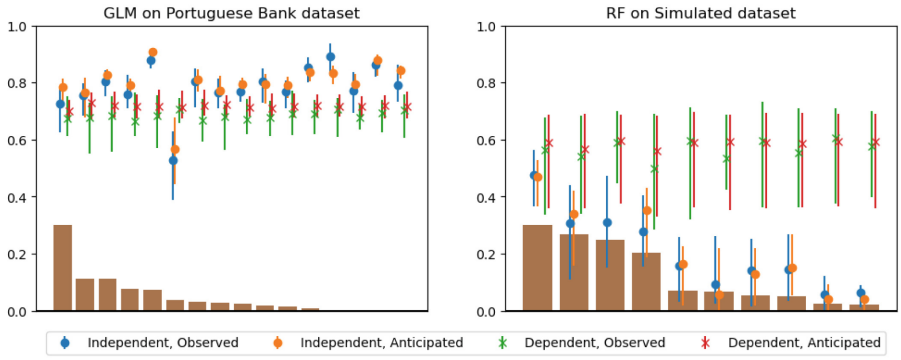


(b) Neural Network

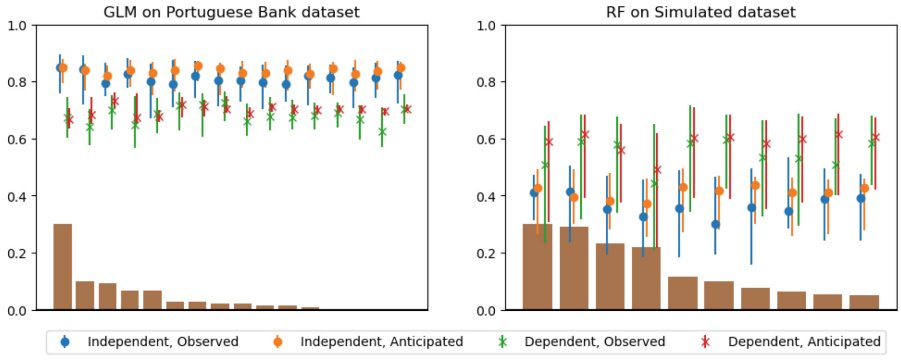


(c) Random Forest

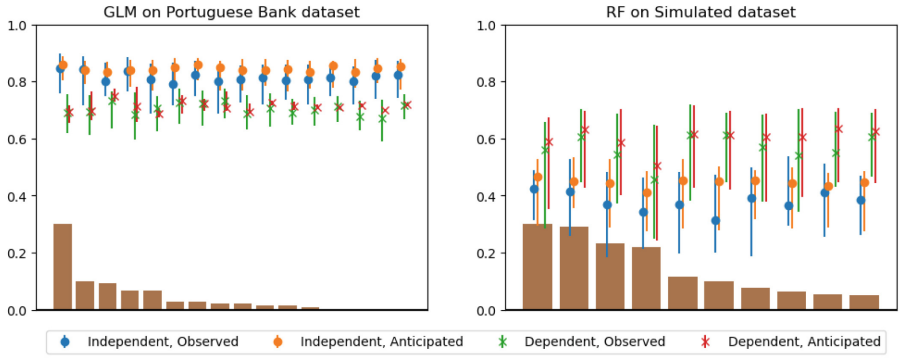
Fig. 7. Average Number of Rank Changes, with and without ControlSHAP's adjustment via Control Variates (CV).



(a) Shapley Sampling



(b) KernelSHAP, Least Squares Variance Estimation



(c) KernelSHAP, Bootstrapped Variance Estimation

Fig. 8. Anticipated vs Observed Variance Reduction

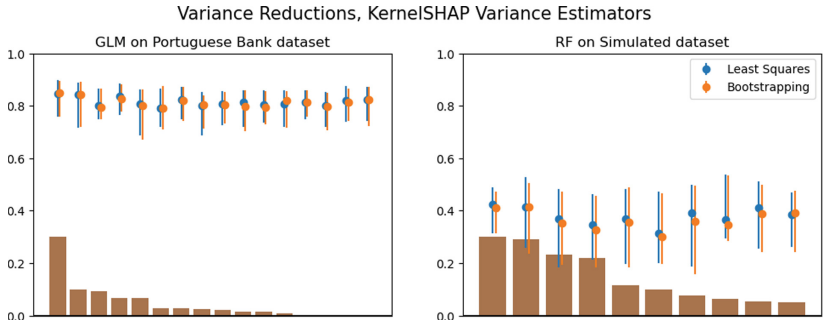


Fig. 9. Variance reductions of ControlSHAP across 40 samples with bootstrapped and least-squares estimates of KernelSHAP variance and covariance. Shapley estimates computed assuming independent features.

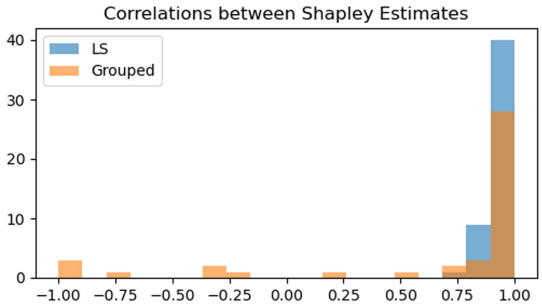


Fig. 10. Correlation between model and Taylor approximation, grouped and least squares. Same input and feature across 50 iterations, with logistic regression on the bank dataset.

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A Guide to Feature Importance Methods for Scientific Inference

Fiona Katharina Ewald[1,2], Ludwig Bothmann[1,2], Marvin N. Wright[4,5,6], Bernd Bischl[1,2], Giuseppe Casalicchio[1,2(✉)], and Gunnar König[3]

[1] Department of Statistics, LMU Munich, Munich, Germany
Giuseppe.Casalicchio@stat.uni-muenchen.de
[2] Munich Center for Machine Learning (MCML), Munich, Germany
[3] Department of Computer Science and Tübingen AI Center, University of Tübingen, Tübingen, Germany
[4] Leibniz Institute for Prevention Research and Epidemiology – BIPS, Bremen, Germany
[5] University of Bremen, Bremen, Germany
[6] University of Copenhagen, Copenhagen, Denmark

Abstract. While machine learning (ML) models are increasingly used due to their high predictive power, their use in understanding the data-generating process (DGP) is limited. Understanding the DGP requires insights into feature-target associations, which many ML models cannot directly provide due to their opaque internal mechanisms. Feature importance (FI) methods provide useful insights into the DGP under certain conditions. Since the results of different FI methods have different interpretations, selecting the correct FI method for a concrete use case is crucial and still requires expert knowledge. This paper serves as a comprehensive guide to help understand the different interpretations of global FI methods. Through an extensive review of FI methods and providing new proofs regarding their interpretation, we facilitate a thorough understanding of these methods and formulate concrete recommendations for scientific inference. We conclude by discussing options for FI uncertainty estimation and point to directions for future research aiming at full statistical inference from black-box ML models.

Keywords: Feature Importance · Model-agnostic Interpretability · Interpretable ML

1 Introduction

Machine learning (ML) models have gained widespread adoption, demonstrating their ability to model complex dependencies and make accurate predictions [32]. Besides accurate predictions, practitioners and scientists are often equally interested in understanding the data-generating process (DGP) to gain insights into

G. Casalicchio and G. König—Equal contribution as senior authors.

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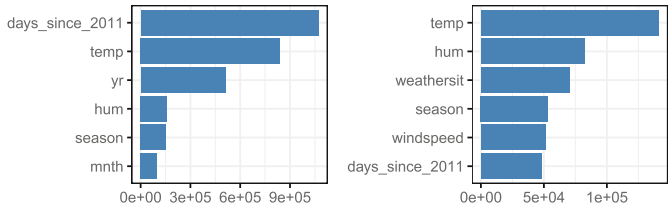
the underlying relationships and mechanisms that drive the observed phenomena [53]. Since analytical information regarding the DGP is mostly unavailable, one way is to analyze a predictive model as a surrogate. Although this approach has potential pitfalls, it can serve as a viable alternative for gaining insights into the inherent patterns and relationships within the observed data, particularly when the generalization error of the ML model is small [43]. Regrettably, the complex and often non-linear nature of certain ML models renders them opaque, presenting a significant challenge in understanding them.

A broad range of interpretable ML (IML) methods have been proposed in the last decades [11,25]. These include *local* techniques that only explain one specific prediction as well as *global* techniques that aim to explain the whole ML model or the DGP; *model-specific* techniques that require access to model internals (e.g., gradients) as well as *model-agnostic* techniques that can be applied to any model; and *feature effects* methods, which reflect the change in the prediction depending on the value of the feature of interest (FOI), as well as *feature importance* (FI) methods, which assign an importance value to each feature depending on its influence on the prediction performance. We argue that in many scenarios, analysts are interested in reliable statistical, population-level inference regarding the underlying DGP [41,60], instead of "simply" explaining the model's internal mechanisms or heuristic computations whose exact meaning regarding the DGP is at the very least unclear or not explicitly stated at all. If an IML technique is used for such a purpose, it should ideally be clear, what property of the DGP is computed, and, as we nearly always compute on stochastic and finite data, how variance and uncertainty are handled. The relevance of IML in the context of scientific inference has been recognized in general [53] as well as in specific subfields, e.g., in medicine [8] or law [15]. Krishna et al. [34] illustrate the disorientation of practitioners when choosing an IML method. In their study, practitioners from both industry and science were asked to choose between different IML methods and explain their choices. The participants predominantly based their choice on superficial criteria such as publication year or whether the method's outputs align with their prior intuition, highlighting the absence of clear guidelines and selection criteria for IML techniques.

Motivating Example. The well-known "bike sharing" data set [17] includes 731 observations and 12 features corresponding to, e.g., weather, temperature, wind speed, season, and day of the week. Suppose a data scientist is not only interested in achieving accurate predictions of the number of bike rentals per day but also in learning about the DGP to identify how the features are associated with the target. She trains a default random forest (RF, test-RMSE: 623, test-R^2: 0.90), and for analyzing the DGP, she decides to use two FI methods: permutation feature importance (PFI) and leave-one-covariate-out (LOCO) with L2 loss (details on these follow in Sects. 5 and 7). Unfortunately, she obtains somewhat contradictory results – shown in Fig. 1. The methods produce results that agree on using temperature (`temp`), season (`season`), the number of days elapsed since the start of data collection in 2011 (`days_since_2011`), and humidity (`hum`) as part of the top 6 most important features, but the rankings of these features



(a) PFI (b) LOCO

Fig. 1. Six most important features following (a) PFI and (b) LOCO.

differ across different methods. She is unsure which feature in the DGP is the most important one, what the disagreement of the FI methods means, and, most importantly, what she can confidently infer from the results about the underlying DGP. We will address her questions in the following sections.

Contributions and Outline. This paper assesses the usefulness of several FI methods for gaining insight into associations between features and the prediction target in the DGP. Our work is the first concrete and encompassing guide for global, loss-based, model-agnostic FI methods directed toward researchers who aim to make informed decisions on the choice of FI methods for (in)dependence relations in the data. The literature review in Sect. 3 highlights the current state-of-the-art and identifies a notable absence of guidelines. Section 4 determines the type of feature-target associations within the DGP that shall be analyzed with the FI methods. In Sect. 5, we discuss methods that remove features by perturbing them; in Sect. 6 methods that remove features by marginalizing them out; and in Sect. 7 methods that remove features by refitting the model without the respective features. In each of the three sections, we first briefly introduce the FI methods, followed by an interpretation guideline according to the association types introduced in Sect. 4. At the end of each section, our results are stated mathematically, with some proofs provided in Appendix A. We return to our motivational example and additionally illustrate our theoretical results in a simulation study in Sect. 8 and formulate recommendations and practical advice in Sect. 9. We mainly analyze the *estimands* of the considered FI, but it should be noted that the interpretation of the *estimates* comes with additional challenges. Hence, we briefly discuss approaches to measure and handle their uncertainty in Sect. 10 and conclude in Sect. 11 with open challenges.

2 General Notation

Let $\mathcal{D} = \left(\left(\mathbf{x}^{(1)}, y^{(1)}\right), \ldots, \left(\mathbf{x}^{(n)}, y^{(n)}\right)\right)$ be a data set of n observations, which are sampled i.i.d. from a p-dimensional feature space $\mathcal{X} = \mathcal{X}_1 \times \ldots \times \mathcal{X}_p$ and a target space $\mathcal{Y}$. The set of all features is denoted by $P = \{1, \ldots, p\}$. The realized feature vector is $\mathbf{x}^{(i)} = (x_1^{(i)}, \ldots, x_p^{(i)})^\top$, $i \in \{1, \ldots, n\}$, where $\mathbf{y} = \left(y^{(1)}, \ldots, y^{(n)}\right)^\top$

are the realized labels. The associated random variables are $X = (X_1, \ldots, X_p)^\top$ and Y, respectively. Marginal random variables for a subset of features $S \subseteq P$ are denoted by X_S. The complement of S is denoted by $-S = P \setminus S$. Single features and their complements are denoted by j and $-j$, respectively. Probability distributions are denoted by F, e.g., $F_Y(Y)$ is the marginal distribution of Y. If two random vectors, e.g., feature sets X_J and X_K, are unconditionally independent, we write $X_J \perp\!\!\!\perp X_K$; if they are unconditionally dependent, which we also call unconditionally associated, we write $X_J \not\perp\!\!\!\perp X_K$.

We assume an underlying true functional relationship $f_{\text{true}} : \mathcal{X} \to \mathcal{Y}$ that implicitly defines the DGP by $Y = f_{\text{true}}(X) + \epsilon$. It is approximated by an ML model $\hat{f} : \mathcal{X} \to \mathbb{R}^g$, estimated on training data $\mathcal{D}$. In the case of a regression model, $\mathcal{Y} = \mathbb{R}$, and $g = 1$. If $\hat{f}$ represents a classification model, g is greater or equal to 1: for binary classification (e.g., $\mathcal{Y} = \{0, 1\}$), g is 1; for multi-class classification, it represents the g decision values or probabilities for each possible outcome class. The ML model $\hat{f}$ is determined by the so-called learner or inducer $\mathcal{I} : \mathcal{D} \times \lambda \mapsto \hat{f}$ that uses hyperparameters λ to map a data set $\mathcal{D}$ to a model in the hypothesis space $\hat{f} \in \mathcal{H}$. Given a loss function, defined by $L : \mathcal{Y} \times \mathbb{R}^g \to \mathbb{R}_0^+$, the risk function of a model $\hat{f}$ is defined as the expected loss $\mathcal{R}(\hat{f}) = \mathbb{E}[L(Y, \hat{f}(X))]$.

3 Related Work

Several papers aim to provide a general overview of existing IML methods [11,12,25,26], but they all have a very broad scope and do not discuss scientific inference. Freiesleben et al. [19] propose a general procedure to design interpretations for scientific inference and provide a broad overview of suitable methods. In contrast, we provide concrete interpretation rules for FI methods. Hooker et al. [30] analyze FI methods based on the reduction of performance accuracy when the FOI is unknown. We examine FI techniques and provide recommendations depending on different types of feature-target associations.

This paper builds on a range of work that assesses how FI methods can be interpreted: Strobl et al. [56] extended PFI [7] for random forests by using the conditional distribution instead of the marginal distribution when permuting the FOI, resulting in the conditional feature importance (CFI); Molnar et al. [42] modified CFI to a model-agnostic version where the dependence structure is estimated by trees; König et al. [33] generalize PFI and CFI to a more general family of FI techniques called relative feature importance (RFI) and assess what insight into the dependence structure of the data they provide; Covert et al. [10] derive theoretical links between Shapley additive global importance (SAGE) values and properties of the DGP; Watson and Wright [58] propose a CFI based conditional independence test; Lei et al. [35] introduce LOCO and are among the first to base FI on hypothesis testing; Williamson et al. [60] present a framework for loss-based FI methods based on model refits, including hypothesis testing; and Au et al. [4] focus on FI methods for groups of features instead of individual features, such as leave-one-group-out importance (LOGO).

In addition to the interpretation methods discussed in this paper, other FI approaches exist. Another branch of IML deals with variance-based FI methods aimed at the FI of an ML model and not necessarily regarding the DGP, as they only use the prediction function of an ML model without considering the ground truth. For example, the feature importance ranking measure (FIRM) [63] uses a feature effect function and defines the standard deviation as an importance method. A similar method by [23] uses the standard deviation of the partial dependence (PD) function [20] as an FI measure. The Sobol index [55] is a more general variance-based method based on a decomposition of the prediction function into main effects and high-order effects (i.e., interactions) and estimates the variance of each component to quantify their importance [45]. Lundberg et al. [38] introduced the SHAP summary plot as a global FI measure based on aggregating local SHAP values [39], which are defined only regarding the prediction function without considering the ground truth.

4 Feature-Target Associations

When analyzing the FI methods, we focus on whether they provide insight into (conditional) (in)dependencies between a feature X_j and the prediction target Y. More specifically, we are interested in understanding whether they provide insight into the following relations:

(A1) Unconditional association ($X_j \not\perp\!\!\!\perp Y$).
(A2) Conditional association ...
  (A2a) ... given all remaining features X_{-j} ($X_j \not\perp\!\!\!\perp Y \mid X_{-j}$).
  (A2b) ... given any user-specified set X_G, G $\subset P\backslash\{j\}$ ($X_j \not\perp\!\!\!\perp Y \mid X_G$).

An unconditional association (A1) indicates that a feature X_j provides information about Y, i.e., knowing the feature on its own allows us to predict Y better; if X_j and Y are independent, this is not the case. On the other hand, a conditional association (A2) with respect to (w.r.t.) a set $S \subseteq P\backslash\{j\}$ indicates that X_j provides information about Y, even if *we already know* X_S. When analyzing the suitability of the FI methods to gain insight into (A1)–(A2b), it is important to consider that no FI score can simultaneously provide insight into more than one type of association. In supervised ML, we are often interested in the conditional association between X_j and Y, given X_{-j} (A2a), i.e., predicting Y better if we are given information regarding all other features.

For example, given measurements of several biomarkers and a disease outcome, a doctor may not only be interested in a well-performing black-box prediction model based on all biomarkers but also in understanding which biomarkers are associated with the disease (A1). Furthermore, the doctor may want to understand whether measuring a biomarker is strictly necessary for achieving optimal predictive performance (A2a) and to understand whether a set of other biomarkers G can replace the respective biomarker (A2b).

Example 1 shows that conditional association does not imply unconditional association ((A2) $\not\Rightarrow$ (A1)). Additionally, unconditional association does not imply conditional association, as Example 2 demonstrates ((A1) $\not\Rightarrow$ (A2)).

Example 1. Let $X_1, X_2 \sim Bern(0.5)$ be independent features and $Y := X_1 \oplus X_2$ (where $\oplus$ is the XOR operation). Then, all three features are pairwise independent, but X_1 and X_2 together allow us to predict Y perfectly.

Example 2. Let $Y := X_1$ with $X_1 \sim N(0,1)$ and $X_2 := X_1 + \epsilon_2$ with $\epsilon_2 \sim N(0, 0.1)$. Although X_2 provides information about Y, all of this information is also contained in X_1. Thus, X_2 is unconditionally associated with Y but conditionally independent from Y given X_1.

Furthermore, conditional (in)dependence w.r.t. one feature set does not imply (in)dependence w.r.t. another, e.g., (A2a) $\not\Leftrightarrow$ (A2b). This is demonstrated by adding unrelated features to the DGP and the conditioning set, as shown in Examples 1 and 2.

5 Methods Based on Univariate Perturbations

Methods based on univariate perturbations quantify the importance of a feature of interest (FOI) by comparing the model's performance before and after replacing the FOI X_j with a perturbed version $\tilde{X}_j$ (permuted observations):

$$\text{FI}_j = \mathbb{E}\left[L\left(Y, \hat{f}(\tilde{X}_j, X_{-j})\right)\right] - \mathbb{E}\left[L\left(Y, \hat{f}(X)\right)\right]. \tag{1}$$

The idea behind this approach is that if perturbing the feature increases the prediction error, the feature should be important for Y. Below, we discuss the three methods PFI (Sect. 5.1), CFI (Sect. 5.2), and RFI (Sect. 5.3) differing in their perturbation scheme: Perturbation in PFI [7,18] preserves the feature's *marginal distribution* while destroying all dependencies with other features X_{-j} and the target Y, i.e.,

$$\tilde{X}_j \sim F_{X_j}(X_j) \text{ and } \tilde{X}_j \perp\!\!\!\perp (X_{-j}, Y); \tag{2}$$

CFI [56] perturbs the FOI while preserving its dependencies with *the remaining features*, i.e.,

$$\tilde{X}_j \sim F_{X_j \mid X_{-j}}(X_j \mid X_{-j}) \text{ and } \tilde{X}_j \perp\!\!\!\perp Y \mid X_{-j}; \tag{3}$$

RFI [33] is a generalization of PFI and CFI since the perturbations preserve the dependencies with *any user-specified set* G, i.e.,

$$\tilde{X}_j \sim F_{X_j \mid X_G}(X_j \mid X_G) \text{ and } \tilde{X}_j \perp\!\!\!\perp Y, X_{P \setminus (G \cup \{j\})} \mid X_G. \tag{4}$$

To indicate on which set G the perturbation of j is conditioned, we denote RFI_j^G. We obtain PFI by setting $G = \emptyset$ and CFI by setting $G = -j$. As will be shown, the type of perturbation strongly affects which features are considered relevant.

5.1 Permutation Feature Importance (PFI)

Insight into $X_j \not\perp\!\!\!\perp Y$ (A1): Non-zero PFI does not imply an unconditional association with Y (Negative Result 5.12). In the proof of Negative Result 5.12, we construct an example where the PFI is non-zero because the perturbation breaks the dependence between the features (and not because of an unconditional association with Y). Based on this, one may conjecture that unconditional feature independence is a sufficient assumption for non-zero PFI to imply an unconditional association with Y; however, this is not the case, as Negative Result 5.13 demonstrates. For non-zero PFI to imply an unconditional association with Y, the features must be independent *conditional on* Y instead (Result 5.11).

Zero PFI does not imply independence between the FOI and the target (Negative Result 5.14). Suppose the model did not detect the association, e.g., because it is a suboptimal fit or because the loss does not incentivize the model to learn the dependence. PFI may be zero in that case, although the FOI is associated with Y. In the proof of Negative Result 5.14, we demonstrate the problem for L2 loss, where the optimal prediction is the conditional expectation (and thus neglects dependencies in higher moments). For cross-entropy optimal predictors and given feature independence (both with and without conditioning on Y), zero PFI implies unconditional independence with Y (Result 5.11).

Insight into $X_j \not\perp\!\!\!\perp Y$ Conditional on X_G or X_{-j} (A2): PFI relates to unconditional (in)dependence and, thus, is not suitable for insight into conditional (in)dependence (see Sect. 4).

Result 5.11 (PFI Interpretation). *For non-zero PFI, it holds that*

$$(X_j \perp\!\!\!\perp X_{-j} \,|\, Y) \wedge (PFI_j \neq 0) \quad \Rightarrow \quad X_j \not\perp\!\!\!\perp Y. \tag{5}$$

For cross-entropy loss and the respective optimal model,

$$(X_j \perp\!\!\!\perp X_{-j}) \wedge (X_j \perp\!\!\!\perp X_{-j} \,|\, Y) \wedge (PFI_j = 0) \quad \Rightarrow \quad X_j \perp\!\!\!\perp Y. \tag{6}$$

Proof. The first implication directly follows from Theorem 1 in [33]. The second follows from the more general Result 5.31. □

Negative Result 5.12. $PFI_j \neq 0 \not\Rightarrow X_j \not\perp\!\!\!\perp Y$.

Proof (Counterexample). Let $Y, X_1 \sim N(0,1)$ be two independent random variables, $X_2 := X_1$, and the prediction model $\hat{f}(x) = x_1 - x_2$. It is simple to calculate that this model has expected L2 loss of 1, as $\mathbb{E}[L(Y, X_1 - X_2)] = \mathbb{E}[Y^2] = 1$. Now let $\tilde{X}_1$ be the perturbed version of X_1 ($\tilde{X}_1 \sim F_{X_j}(X_1)$), and $\tilde{X}_1 \perp\!\!\!\perp (Y, X_2)$. The expected L2 loss under perturbation now is $\mathbb{E}[(Y - (\tilde{X}_1 - X_2))^2] = \text{Var}(Y - \tilde{X}_1 + X_2) = 3$, which implies $\text{PFI}_1 = 2$. So PFI_1 is non-zero, but $X_1 \perp\!\!\!\perp Y$. □

Negative Result 5.13. $(X_j \perp\!\!\!\perp X_{-j}) \wedge (PFI_j \neq 0) \not\Rightarrow X_j \not\perp\!\!\!\perp Y$.

Proof (Counterexample). Let $X_1, X_2 \sim Bern(0.5)$ with $X_1 \perp\!\!\!\perp X_2$, and $Y := X_1 \oplus X_2$, where $\oplus$ is XOR. Consider a perfect prediction model $\hat{f}(X) = x_1 \oplus x_2$, and $\hat{f}$ encodes the posterior probability for $Y = 1$ (here, Y can be only 0 or 1). This model has a cross-entropy loss of 0, since $Y = \hat{f}(X)$. Furthermore, it holds that $X_1 \perp\!\!\!\perp Y$. Again, let $\tilde{X}_1$ be the perturbed version of X_1. One can easily verify that $Y = (X_1 \oplus X_2) \perp\!\!\!\perp (\tilde{X}_1 \oplus X_2) = \tilde{\hat{Y}}$ and $Y, \tilde{\hat{Y}} \sim Bern(0.5)$. Thus, the prediction $\tilde{\hat{Y}}$ using the perturbed feature $\tilde{X}_1$ assigns probability 1 to the correct and wrong class with probability 0.5 each. Thus, the cross-entropy loss for the perturbed prediction is non-zero (actually, positive infinity), and $\text{PFI}_j \neq 0$. □

Negative Result 5.14. $FI_j = 0 \not\Rightarrow X_j \perp\!\!\!\perp Y \mid X_G$ for any $G \subseteq P \backslash \{j\}$, even if the model is L2-optimal.
NB: This result holds not only for PFI but also for any FI method based on univariate perturbations, including PFI, CFI, and RFI (Eq. 1).

Proof (Counterexample). If a model does not rely on a feature X_j, $\text{FI}_j = 0$. We construct an example where $\hat{f}$ is L2-optimal but does not rely on the feature X_1, which is dependent with Y conditional on any set $G \subseteq P \backslash \{j\}$. Let $Y \mid X_1, X_2 \sim N(X_2, X_1)$ with $X_1, X_2 \sim N(0, 1)$ and $X_1 \perp\!\!\!\perp X_2$. Then, Y is dependent with X_1 conditional on any set $G \subseteq P \backslash \{1\}$: Here, G could either be $G = \emptyset$ or $G = \{2\}$. Now, for small X_1, extreme values of Y are less likely than for $X_1 = 100$, irrespective of whether we know X_2. Now consider $\hat{f}(x) = x_2$. $\hat{f}$ is L2-optimal since $\mathbb{E}[Y|X] = X_2$, but $\hat{f}$ does not depend on X_1. □

5.2 Conditional Feature Importance (CFI)

Insight into $X_j \not\perp\!\!\!\perp Y \mid X_{-j}$ (A2a): Since CFI preserves associations between features, non-zero CFI implies a conditional dependence on Y, even if the features are dependent (Result 5.21). The converse generally does not hold, so Negative Result 5.14 also applies to CFI. However, for cross-entropy optimal models, zero CFI implies conditional independence (Result 5.21).

Insight into $X_j \not\perp\!\!\!\perp Y$ (A1) and $X_j \not\perp\!\!\!\perp Y \mid X_G$ (A2b): Since CFI provides insight into conditional dependence (A2a), it follows from Sect. 4 that CFI is not suitable to gain insight into (A1) and (A2b).

Result 5.21 (CFI interpretation). *For CFI, it holds that*

$$CFI_j \neq 0 \quad \Rightarrow \quad X_j \not\perp\!\!\!\perp Y \mid X_{-j} \tag{7}$$

For cross-entropy optimal models, the converse holds as well.

Proof. The first equation follows from Theorem 1 in [33]. The second follows from the more general Result 5.31. □

5.3 Relative Feature Importance (RFI)

Insight into $X_j \not\perp\!\!\!\perp Y \mid X_G$ (A2b): Result 5.31 generalizes Results 5.11 and 5.21. While PFI and CFI are sensitive to dependencies conditional on *no* or *all* remaining features, RFI is sensitive to conditional dependencies w.r.t. *a user-specified feature set* G. Nevertheless, we must be careful with our interpretation if features are dependent. RFI may be non-zero even if the FOI is not associated with the target (Negative Result 5.32). In general, zero RFI does not imply independence (Negative Result 5.14). Still, for cross-entropy optimal models and under independence assumptions, insight into conditional independence w.r.t. G can be gained (Result 5.31).

Insight into $X_j \not\perp\!\!\!\perp Y$ (A1) and $X_j \not\perp\!\!\!\perp Y \mid X_{-j}$ (A2a): If features are conditionally independent given Y, setting G to $\emptyset$ (yielding PFI) enables insight into unconditional dependence. Setting G to $-j$ (yielding CFI) enables insight into the conditional association given all other features.

Result 5.31 (RFI interpretation). *For $R = P \backslash (G \cup \{j\})$, it holds that*

$$(X_j \perp\!\!\!\perp X_R \mid X_G, Y) \wedge (RFI_j^G \neq 0) \quad \Rightarrow \quad X_j \not\perp\!\!\!\perp Y \mid X_G. \tag{8}$$

For cross-entropy optimal predictors and $G \subseteq P \backslash \{j\}$, it holds that

$$(X_j \perp\!\!\!\perp X_R \mid X_G, Y) \wedge (X_j \perp\!\!\!\perp X_R \mid X_G) \wedge (RFI_j^G = 0) \Rightarrow X_j \perp\!\!\!\perp Y \mid X_G. \tag{9}$$

Proof. The first implication follows directly from Theorem 1 in [33]. The proof of the second implication can be found in Appendix A.1.

Negative Result 5.32. $RFI_j^G \neq 0 \not\Rightarrow X_j \not\perp\!\!\!\perp Y \mid X_G$.

Proof (Counterexample). Let $G = \emptyset$. Then, $RFI_j^G = PFI_j$ and $X_j \not\perp\!\!\!\perp Y \mid X_G \Leftrightarrow X_j \not\perp\!\!\!\perp Y$. Thus, the result directly follows from 5.12. □

6 Methods Based on Marginalization

In this section, we assess SAGE value functions (SAGEvf) and SAGE values [10]. The methods remove features by marginalizing them out of the prediction function. The marginalization [39] is performed using either the conditional or marginal expectation. These so-called reduced models are defined as

$$\begin{aligned} \hat{f}_S^m(x_S) &= \mathbb{E}_{X_{-S}}\left[\hat{f}(x_S, X_{-S})\right], \quad \text{and} \\ \hat{f}_S^c(x_S) &= \mathbb{E}_{X_{-S} \mid X_S}\left[\hat{f}(x_S, X_{-S}) \mid X_S\right], \end{aligned} \tag{10}$$

where $\hat{f}^m$ is the marginal and $\hat{f}^c$ is the conditional-sampling-based version and $\hat{f}_\emptyset^m = \hat{f}_\emptyset^c$ the average model prediction, e.g., $\mathbb{E}[Y]$ for an L2 loss optimal model

and $\mathbb{P}(Y)$ for a cross-entropy loss optimal model. Based on these, SAGEvf quantify the change in performance that the model restricted to the FOIs achieves over the average prediction:

$$v^{m/c}(S) = \mathbb{E}\left[L\left(Y, \hat{f}_{\emptyset}^{m/c}\right)\right] - \mathbb{E}\left[L\left(Y, \hat{f}_{S}^{m/c}(X_S)\right)\right] \tag{11}$$

We abbreviate SAGEvf depending on the distribution used for the restricted prediction function (i.e., $\hat{f}^m$ or $\hat{f}^c$) with mSAGEvf (v^m) and cSAGEvf (v^c).

SAGE values [10] regard FI quantification as a cooperative game, where the features are the players, and the overall performance is the payoff. The surplus performance (surplus payoff) enabled by adding a feature to the model depends on which other features the model can already access (coalition). To account for the collaborative nature of FI, SAGE values use Shapley values [52] to divide the payoff for the collaborative effort (the model's performance) among the players (features). SAGE values are calculated as the weighted average of the surplus evaluations over all possible coalitions $S \subseteq P \setminus \{j\}$:

$$\phi_j^{m/c}(v) = \frac{1}{p} \sum_{S \subseteq P \setminus \{j\}} \binom{p-1}{|S|}^{-1} \left(v^{m/c}(S \cup \{j\}) - v^{m/c}(S)\right), \tag{12}$$

where the superscript in ϕ_j denotes whether the marginal $v^m(S)$ or conditional $v^c(S)$ value function is used.

6.1 Marginal SAGE Value Functions (mSAGEvf)

Insight into $X_j \not\perp\!\!\!\perp Y$ (A1): Like PFI, mSAGE value functions use marginal sampling and break feature dependencies. mSAGEvf may be non-zero ($v^m(\{j\}) \neq 0$), although the respective feature is not associated with Y (Negative Result 6.12). While an assumption about feature independence was sufficient for PFI for insight into pairwise independence, this is generally not the case for mSAGEvf. The feature marginalization step may lead to non-zero importance for non-optimal models (Negative Result 6.13). Given feature independence and L2 or cross-entropy optimal models, a non-zero mSAGEvf implies unconditional association; the converse only holds for CE optimal models (Result 6.11).

Insight into $X_j \not\perp\!\!\!\perp Y$ Conditional on X_G or X_{-j} (A2): The method mSAGEvf does not provide insight into the dependence between the FOI and Y (Negative Result 6.12) unless the features are independent and the model is optimal w.r.t. L2 or cross-entropy loss (Result 6.11). Then, mSAGEvf can be linked to (A1) and, thus, is not suitable for (A2) (Sect. 4).

Result 6.11 (mSAGEvf interpretation). *For L2 loss or cross-entropy loss-optimal models (and the respective loss) and $(X_j \perp\!\!\!\perp X_{-j})$, it holds that*

$$v^m(\{j\}) \neq 0 \quad \Rightarrow \quad X_j \not\perp\!\!\!\perp Y \tag{13}$$

For cross-entropy optimal predictors, the converse holds as well.

Proof. The proof can be found in Appendix A.2.

Negative Result 6.12. $v^m(\{j\}) \neq 0 \not\Rightarrow \exists G \subseteq P \backslash \{j\} : X_j \not\perp\!\!\!\perp Y \mid X_G$

Proof (Counterexample). Let us assume the same DGP and model as in the proof of Negative Result 5.12. In the setting, both the full model $\hat{f}_1^m(x) = x_1 - x_2 = 0$ and $\hat{f}_\emptyset^m = 0$ are optimal, but $\hat{f}_1^m(x_1) = x_1 \neq 0$ is sub-optimal. Thus, $v^m(\{1\}) \neq 0$ (although $X_1 \perp\!\!\!\perp Y$ for any $G \subseteq P \backslash \{j\}$). □

Negative Result 6.13. $(v(\{j\}) \neq 0) \wedge (X_j \perp\!\!\!\perp X_{-j}) \not\Rightarrow X_j \not\perp\!\!\!\perp Y$

Proof (Counterexample). Let $X_1, Y \sim N(0,1)$, and let X_{-1} be some (potentially multivariate) random variable, with $X_{-1} \perp\!\!\!\perp Y$ and $X_{-1} \perp\!\!\!\perp X_1$. Let $\hat{f}(x) = x_1$ be the prediction model. Then, $\hat{f}_\emptyset^m = \hat{f}_\emptyset^c = \mathbb{E}_X[X_1] = 0$ and $\hat{f}_1^m(x_1) = \hat{f}_1^c(x_1) = x_1$. Since the optimal prediction $\hat{Y}^* = \mathbb{E}[Y|X] = \mathbb{E}[Y] = 0$, the average prediction $\hat{f}_\emptyset^m = \hat{f}_\emptyset^c = 0$ is loss-optimal and $\hat{f}_1^m(x_1) = \hat{f}_1^c(x_1) = x_1$ is not loss-optimal. Consequently, $v(\{1\}) \neq 0$ (although X_1 is independent of target and features). Notably, the example works both for v^m and v^c. □

6.2 Conditional SAGE Value Functions (cSAGEvf)

Insight into $X_j \not\perp\!\!\!\perp Y$ (A1): Like for mSAGEvf, model optimality w.r.t. L2 or cross-entropy loss is needed to gain insight into the dependencies in the data (Negative result 6.13). However, since cSAGEvf preserves associations between features, the assumption of independent features is not required to gain insight into unconditional dependencies (Result 6.21).

Insight into $X_j \not\perp\!\!\!\perp Y$ Conditional on X_G or X_{-j} (A2): Since cSAGEvf provide insight into (A1), they are unsuitable for gaining insight into (A2) (see Sect. 4). However, the difference between cSAGEvf for different sets, called surplus cSAGEvf ($\text{scSAGEvf}_j^G := v^c(G \cup \{j\}) - v^c(G)$, where $G \subseteq P \backslash \{j\}$ is user-specified), provides insights into conditional associations (Result 6.21).

Result 6.21 (cSAGEvf interpretation). *For L2 loss or cross-entropy loss optimal models, it holds that:*

$$v^c(\{j\}) \neq 0 \quad \Rightarrow \quad X_j \not\perp\!\!\!\perp Y \tag{14}$$

$$scSAGEvf_j^G \neq 0 \quad \Rightarrow \quad X_j \not\perp\!\!\!\perp Y \mid X_G \tag{15}$$

For cross-entropy loss, the respective converse holds as well.

Proof. The first implication (and the respective converse) follows from the second (and the respective converse) by setting $G = \emptyset$. The second implication was proven in Theorem 1 in [40]. For the converse, [10] show that for cross-entropy optimal models $v^c(G \cup \{j\}) - v^c(G) = I(Y, X_j | X_G)$; it holds that $I(Y, X_j | X_G) = 0 \Leftrightarrow X_j \perp\!\!\!\perp Y \mid X_G$. □

6.3 SAGE Values

Since non-zero cSAGEvf imply (conditional) dependence and cSAGE values are based on scSAGEvf of different coalitions, cSAGE values are only non-zero if a conditional dependence w.r.t. some conditioning set is present (see Result 6.31).

Result 6.31. *Assuming an L2 or cross-entropy optimal model, the following interpretation rule for cSAGE values holds for a feature X_j:*

$$\phi_j^c(v) \neq 0 \quad \Rightarrow \quad \exists\, S \subseteq P \backslash \{j\} : X_j \not\perp\!\!\!\perp Y \mid X_S. \tag{16}$$

For cross-entropy optimal models, the converse holds as well.

Proof. The Proof can be found in Appendix A.3.

We cannot give clear guidance on the implications of mSAGE for (A1)–(A2b) and leave a detailed investigation for future work.

7 Methods Based on Model Refitting

This section addresses FI methods that quantify importance by removing features from the data and refitting the ML model. For LOCO [35], the difference in risk of the original model and a refitted model $\hat{f}^r_{-j}$ relying on every feature but the FOI X_j is computed:

$$\text{LOCO}_j = \mathbb{E}\left[L\left(Y, \hat{f}^r_{-j}(X_{-j})\right)\right] - \mathbb{E}\left[L\left(Y, \hat{f}(X)\right)\right], \tag{17}$$

where $\hat{f}^r_{-j}$ keeps the learner $\mathcal{I}(\mathcal{D}, \lambda)$ fixed.[1]

Williamson et al. [60] generalize LOCO, as they are interested in not only one FOI but also in a feature set $S \subseteq P$. As they do not assign an acronym, we from here on call it Williamson's Variable Importance Measure (WVIM):

$$\text{WVIM}_S = \mathbb{E}\left[L\left(Y, \hat{f}^r_{-S}(X_{-S})\right)\right] - \mathbb{E}\left[L\left(Y, \hat{f}(X)\right)\right]. \tag{18}$$

Obviously, WVIM, also known as LOGO [4], equals LOCO for $S = j$. For $S = P$, the optimal refit reduces to the optimal constant prediction, e.g., for an L2-optimal model $\hat{f}^r_{-S}(X_{-S}) = \hat{f}^r_{\emptyset}(x_{\emptyset}) = \mathbb{E}[Y]$ and for a cross-entropy optimal model $\hat{f}^r_{\emptyset}(x_{\emptyset}) = \mathbb{P}(Y)$.

7.1 Leave-One-Covariate-Out (LOCO)

For L2 and cross-entropy optimal models, LOCO is similar to $v^c(-j \cup j) - v^c(-j)$, with the difference that we do not obtain the reduced model by marginalizing out one of the features, but rather by refitting the model. As such, the interpretation is similar to the one of cSAGEvf (Result 7.11).

[1] In Eq. (10), we tagged the reduced models $\hat{f}^m$ and $\hat{f}^c$, indicating the type of marginalization. For refitting-based methods, we use the superscript r.

Result 7.11. *For an L2 or cross-entropy optimal model and the respective optimal reduced model $\hat{f}^r_{-j}$, it holds that $LOCO_j \neq 0 \Rightarrow X_j \not\perp\!\!\!\perp Y \mid X_{-j}$. For cross-entropy loss, the converse holds as well.*

Proof. For cross-entropy and L2-optimal fits, the reduced model that we obtain from conditional marginalization behaves the same as the optimal refit (for cross-entropy loss $\hat{f}^r_S = \hat{f}^c_S = \mathbb{P}(Y|X_S)$, for L2 loss $\hat{f}^r_S = \hat{f}^c_S = \mathbb{E}[Y|X_S]$) [10, Appendix B] and thus $\text{LOCO}_j = v^c(j \cup -j) - v^c(-j)$. As such, the result follows directly from Result 6.21.

7.2 WVIM as Relative FI and Leave-One-Covariate-In (LOCI)

For $S = j$, the interpretation is the same as for LOCO. Another approach to analyzing the relative importance of the FOI is investigating the surplus WVIM (sWVIM_j^{-G}) for a group $G \subseteq P \backslash \{j\}$:

$$\text{sWVIM}_j^{-G} = \mathbb{E}\left[L\left(Y, \hat{f}^r_G(X_G)\right)\right] - \mathbb{E}\left[L\left(Y, \hat{f}^r_{G\cup\{j\}}(X_{G\cup\{j\}})\right)\right]. \tag{19}$$

It holds that sWVIM_j^{-G} equals scSAGEvf_j^G, only differing in the way features are removed, so the interpretation is similar to the one of scSAGEvf. A special case results for $G = \emptyset$, i.e., the difference in risk between the optimal constant prediction and a model relying on the FOI only. We refer to this (leaving-one-covariate-in) as LOCI_j. For cross-entropy or L2-optimal models, the interpretation is the same as for cSAGEvf, since $\text{LOCI}_j = v^c(\{j\})$ (Result 7.21).

Result 7.21. *For L2 or cross-entropy optimal learners, it holds that*

$$LOCI_j \neq 0 \quad \Rightarrow \quad X_j \not\perp\!\!\!\perp Y, \qquad \textit{and} \tag{20}$$

$$sWVIM_j^{-G} \neq 0 \quad \Rightarrow \quad X_j \not\perp\!\!\!\perp Y \mid X_G. \tag{21}$$

For cross-entropy, the converse holds as well.

Proof. For L2-optimal models, $\hat{f}^c_\emptyset = \mathbb{E}[Y] = \hat{f}^r_\emptyset$ and $\hat{f}^c_G = \mathbb{E}[Y|X_G] = \hat{f}^r_G$. For cross-entropy optimal models, $\hat{f}^c_\emptyset = \mathbb{P}(Y) = \hat{f}^r_\emptyset$ and $\hat{f}^c_G = \mathbb{P}(Y|X_G) = \hat{f}^r_G$. Thus, the interpretation is the same as for cSAGEvf (Result 6.21). □

8 Examples

We can now answer the open questions of the motivational example from the introduction (Sect. 1). To illustrate our recommendations (summarized in Table 1), we additionally apply the FI methods to a simplified setting where the DGP and the model's mechanism are known and intelligible, including features with different roles.

Table 1. Summary of our results. The abbreviation "CE" stands for cross-entropy loss and "L2" for L2-loss, each with the respective optimal model.

Outcome	Assumptions	Implication
$\text{PFI}_j \neq 0$	$X_j \perp\!\!\!\perp X_{-j} \mid Y$	$\Rightarrow \quad X_j \not\perp\!\!\!\perp Y$
$\text{PFI}_j = 0$	$\text{CE} \wedge (X_j \perp\!\!\!\perp X_{-j}) \wedge (X_j \perp\!\!\!\perp X_{-j} \mid Y)$	$\Rightarrow \quad X_j \perp\!\!\!\perp Y$
$\text{mSAGEvf}_j \neq 0$	$(\text{L2} \vee \text{CE}) \wedge (X_j \perp\!\!\!\perp X_{-j})$	$\Rightarrow \quad X_j \not\perp\!\!\!\perp Y$
$\text{mSAGEvf}_j = 0$	$\text{CE} \wedge (X_j \perp\!\!\!\perp X_{-j})$	$\Rightarrow \quad X_j \perp\!\!\!\perp Y$
$\text{cSAGEvf}_j \neq 0$	$\text{L2} \vee \text{CE}$	$\Rightarrow \quad X_j \not\perp\!\!\!\perp Y$
$\text{cSAGEvf}_j = 0$	CE	$\Rightarrow \quad X_j \perp\!\!\!\perp Y$
$\text{LOCI}_j \neq 0$	$\text{L2} \vee \text{CE}$	$\Rightarrow \quad X_j \not\perp\!\!\!\perp Y$
$\text{LOCI}_j = 0$	CE	$\Rightarrow \quad X_j \perp\!\!\!\perp Y$
$\text{CFI}_j \neq 0$	-	$\Rightarrow \quad X_j \not\perp\!\!\!\perp Y \mid X_{-j}$
$\text{CFI}_j = 0$	CE	$\Rightarrow \quad X_j \perp\!\!\!\perp Y \mid X_{-j}$
$\text{scSAGEvf}_j^{-j} \neq 0$	$\text{L2} \vee \text{CE}$	$\Rightarrow \quad X_j \not\perp\!\!\!\perp Y \mid X_{-j}$
$\text{scSAGEvf}_j^{-j} = 0$	CE	$\Rightarrow \quad X_j \perp\!\!\!\perp Y \mid X_{-j}$
$\text{LOCO}_j \neq 0$	$\text{L2} \vee \text{CE}$	$\Rightarrow \quad X_j \not\perp\!\!\!\perp Y \mid X_{-j}$
$\text{LOCO}_j = 0$	CE	$\Rightarrow \quad X_j \perp\!\!\!\perp Y \mid X_{-j}$
$\text{RFI}_j^G \neq 0$	$X_j \perp\!\!\!\perp X_R \mid X_G, Y$	$\Rightarrow \quad X_j \not\perp\!\!\!\perp Y \mid X_G$
$\text{RFI}_j^G = 0$	$\text{CE} \wedge (X_j \perp\!\!\!\perp X_R \mid X_G, Y) \wedge (X_j \perp\!\!\!\perp X_R \mid X_G)$	$\Rightarrow \quad X_j \perp\!\!\!\perp Y \mid X_G$
$\text{scSAGEvf}_j^G \neq 0$	$\text{L2} \vee \text{CE}$	$\Rightarrow \quad X_j \not\perp\!\!\!\perp Y \mid X_G$
$\text{scSAGEvf}_j^G = 0$	CE	$\Rightarrow \quad X_j \perp\!\!\!\perp Y \mid X_G$
$\text{sWVIM}_j^{-G} \neq 0$	$\text{L2} \vee \text{CE}$	$\Rightarrow \quad X_j \not\perp\!\!\!\perp Y \mid X_G$
$\text{sWVIM}_j^{-G} = 0$	CE	$\Rightarrow \quad X_j \perp\!\!\!\perp Y \mid X_G$

Returning to Our Motivating Example. Using Result 5.11, we know that PFI can assign high FI values to features even if they are not associated with the target but with other features that are associated with the target. Conversely, LOCO only assigns non-zero values to features conditionally associated with the target (here: bike rentals per day, see Result 7.11). We can therefore conclude that at least the features `weathersit, season, temp, mnth, windspeed` and `weekday` are conditionally associated with the target, and the TOP 5 most important features, according to PFI, tend to share information with other features or may not be associated with `bike rentals per day` at all.

Illustrative Example with Known Ground-Truth. This example includes five features $X_1, \ldots, X_5$ and a target Y with the following dependence structure (visualized in Fig. 2, left plot):

- X_1, X_3 and X_5 are independent and standard normal: $X_j \sim N(0, 1)$,
- X_2 is a noisy copy of X_1: $X_2 := X_1 + \epsilon_2, \epsilon_2 \sim N(0, 0.001)$,
- X_4 is a (more) noisy copy of X_3: $X_4 := X_3 + \epsilon_4, \epsilon_4 \sim N(0, 0.1)$,
- Y depends on X_4 and X_5 via linear effects and a bivariate interaction: $Y := X_4 + X_5 + X_4 * X_5 + \epsilon_Y$, $\epsilon_Y \sim N(0, 0.1)$.

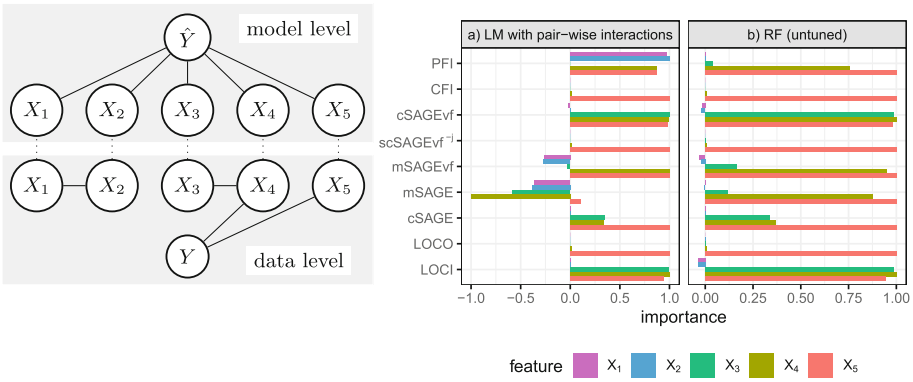


Fig. 2. Left: Graph illustrating the model and data level associations. Right: Results of FI methods for the LM in panel (a) and the RF in panel (b); importance values are relative to the most important feature.

Regarding (A1), features X_3, X_4 and X_5 are unconditionally associated with Y, while only X_5 is conditionally associated with Y given all other features (A2a).

We sample $n = 10,000$ observations from the DGP and use 70% of the observations to train two models: A linear model (LM) with additional pair-wise interactions between all features (test-MSE = 0.0103, test-R^2 = 0.9966), and a random forest (RF) using default hyperparameters (test-MSE = 0.0189, test-R^2 = 0.9937). We apply the FI methods on 30% test data with L2 loss to both models using 50 repetitions for methods that marginalize or perturb features. We present the results in Fig. 2.[2] The right plot shows each feature's FI value relative to the most important feature (which is scaled to 1).

(A1): *LOCI* and *cSAGEvf* correctly identify X_3, X_4 and X_5 as unconditionally associated. *PFI* correctly identifies X_4 and X_5 to be relevant, but it misses X_3, presumably since the model predominantly relies on X_4. For the LM, *PFI* additionally considers X_1 and X_2 to be relevant, although they are fully independent of Y; due to correlation in the feature sets, the trained model includes the term $0.36x_1 - 0.36x_2$, which cancels out in the unperturbed, original distribution, but causes performance drops when the dependence between X_1 and X_2 is broken via perturbation. For *mSAGEvf*, similar observations can be made, with the difference that X_1 and X_2 receive negative importance. The reason is that for *mSAGEvf*, the performance of the average prediction is compared to the prediction where all but one feature are marginalized out; we would expect that adding a feature improves the performance, but for X_1 and X_2, the performance worsens if adding the feature breaks the dependence between X_1 and X_2.

[2] All FI methods and reproducible scripts for the experiments are available online via https://github.com/slds-lmu/paper_2024_guide_fi.git. Most FI methods were computed with the Python package `fippy` (https://github.com/gcskoenig/fippy.git).

(A2): *CFI*, *LOCO*, and $scSAGEvf^{-j}$ correctly identify X_5 as conditionally associated, as expected. *cSAGE* correctly identifies features that are dependent with Y conditional on any set S, specifically, X_3, X_4 and X_5. The results of *mSAGE* for the RF are similar to those for *cSAGE*; on the LM, the results are quite inconclusive – most features have a negative importance.

Overall, the example empirically illustrates the differences between the methods as theoretically shown in Sects. 5 to 7.

9 Summary and Practical Considerations

In Sects. 5 to 7, we presented three different classes of FI techniques: Techniques based on univariate perturbations, techniques based on marginalization, and techniques based on model refitting. In principle, each approach can be used to gain partial insights into questions (A1) to (A2b). However, the practicality of the methods depends on the specific application. As follows, we discuss some aspects that may be relevant to the practitioner.

For (A1), PFI, mSAGEvf, cSAGEvf, and LOCI are – in theory – suitable. However, PFI and mSAGEvf require assumptions about feature independence, which are typically unrealistic. cSAGEvf require marginalizing out features using a multivariate conditional distribution $P(X_{-j}|X_j)$, which can be challenging since not only the dependencies between X_j and X_{-j} but also the ones between X_{-j} have to be considered. LOCI requires fitting a univariate model, which is computationally much less demanding than the cSAGEvf computation.

For (A2a), a comparatively more challenging task, CFI, scSAGEvf and LOCO are suitable, but it is unclear which of the methods is preferable in practice. While CFI and scSAGEvf require a model of the univariate conditional $P(X_j|X_{-j})$, LOCO requires fitting a model to predict Y from X_{-j}. For (A2b), the practical requirements depend on the size of the conditioning set. The closer the conditioning set is to $-j$, the fewer features have to be marginalized out for scSAGEvf, and the fewer feature dependencies may lead to extrapolation for RFI. For sWVIM, larger relative feature sets imply more expensive model fits.

Importantly, all three questions (A1) to (A2b) could also be assessed with direct or conditional independence tests, e.g., mutual information [9], partial correlation tests [5], kernel-based measures such as the Hilbert-Schmidt independence criterion [24,62], or the generalized covariance [51]. This seems particularly appropriate for question (A1), where we simply model the association structure of a bivariate distribution. Methods like mSAGEvf can arguably be considered overly complex and computationally expensive for such a task.

10 Statistical Inference for FI Methods

So far, we have described how the presented FI methods should behave in theory or as point estimators. However, the estimation of FI values is inherently subject to various sources of uncertainty introduced during the FI estimation procedure, model training, or model selection [41,60]. This section reviews available

techniques to account for uncertainty in FI by applying methods of statistical inference, e.g., statistical tests and the estimation of confidence intervals (CIs).

All FI methods in this paper measure the expected loss. To prevent biased or misleading estimates due to overfitting, it is crucial to calculate FI values on independent test data not seen during training, aligning with best practices in ML performance assessment [37,54]. Computing FI values on training data may lead to wrong conclusions. For example, Molnar et al. [43] demonstrated that even if features are random noise and not associated with the target, some features are incorrectly deemed important when FI values are computed using training data instead of test data. If no large dedicated test set is available, or the data set is not large in general to facilitate simple holdout splitting, resampling techniques such as cross-validation or bootstrap provide practical solutions [54].

In the following, we will first provide an overview of method-specific approaches and then summarize further ideas about more general ones.

PFI and CFI. Molnar et al. [41] address the uncertainty of model-specific PFI and CFI values caused by estimating expected values using Monte Carlo integration on a fixed test data set and model. To address the variance of the learning algorithm, they introduce the learner-PFI, computed using resampling techniques such as bootstrapping or subsampling on a held-out test set within each resampling iteration. They also propose variance-corrected Wald-type CIs to compensate for the underestimation of variance caused by partially sharing training data between the models fitted in each resampling iteration. For CFI, Watson and Wright [58] address sampling uncertainty by comparing instance-wise loss values. They use Fisher's exact (permutation) tests and paired t-tests for hypothesis testing. The latter, based on the central limit theorem, is applied to all decomposable loss functions calculated by averaging instance-wise losses.

SAGE. The original paper of SAGE [10] introduced an efficient algorithm to approximate SAGE values, since the exact calculation of SAGE values is computationally expensive. They show that, according to the central limit theorem, the approximation algorithm convergences to the correct values and that the variance reduces with the number of iterations at a linear rate. They briefly mention that the variance of the approximation can be estimated at a specific iteration and can be used to construct CIs (which corresponds to the same underlying idea of the Wald-type CI for the model-specific PFI mentioned earlier).

WVIM including LOCO. Lei et al. [35] introduced statistical inference for LOCO by splitting the data into two parts: one for model fitting and one for estimating LOCO. They further employed hypothesis testing and constructing CIs using sign tests or the Wilcoxon signed-rank test. The results' interpretation is limited to the importance of the FOI to an ML algorithm's estimated model on a fixed training data set. Williamson et al. [60] construct Wald-type CI intervals for LOCO and WVIM, based on k-fold cross-validation and sample-splitting[3]. Compared to LOCO, it provides a more general interpretation of the results as it considers the FI of an ML algorithm trained on samples of a particular size,

[3] This involves dividing the k-folds into two parts to serve distinct purposes, allowing for separate estimation and testing procedures.

i.e., due to cross-validation, the results are not tied to a single training data set. The approach is related to [41] but removes features via refitting instead of sampling and does not consider any variance correction. The authors note that, while sample-splitting helps to address issues related to zero-importance features having an incorrect type I error or coverage of their CIs, it may not fully leverage all available information in the data set to train a model.

PIMP. The PIMP heuristic [2] is based on model refits and was initially developed to address bias in FI measures such as PFI within random forests. However, PIMP is a general procedure and has broader applicability across various FI methods [36,43]. PIMP involves repeatedly permuting the target to disrupt its associations with features while preserving feature dependencies, training a model on the data with the permuted target, and computing PFI values. This leads to a collection of PFI values (called null importances) under the assumption of no association between the FOI and the target. The PFI value of the model trained on the original data is then compared with the distribution of null importances to identify significant features.

Methods Based on the Rashomon Set. The Rashomon set refers to a collection of models that perform equally well but may differ in how they construct the prediction function and the features they rely on. Fisher et al. [18] consider the Rashomon set of a specific model class (e.g., decision trees) defined based on a performance threshold and propose a method to measure the FI within this set. For each model in the Rashomon set, the FI of a FOI is computed, and its range across all models is reported. Other works include the Variable Importance Cloud (VIC) [13], providing a visual representation of FI values over different model types; the Rashomon Importance Distribution (RID) [14], providing the FI distribution across the set and CIs to characterize uncertainty around FI point estimates; and ShapleyVIC [44], extending VIC to SAGE values and using a variance estimator for constructing CIs. The main idea is to address uncertainty in model selection by analyzing a Rashomon set, hoping that some of these models reflect the underlying DGP and assign similar FI values to features.

Multiple Comparisons. Testing multiple FI values simultaneously poses a challenge known as multiple comparisons. The risk of falsely rejecting true null hypotheses increases with the number of comparisons. Practitioners can mitigate it, e.g., by controlling the family-wise error rate or the false discovery rate [43,49].

11 Open Challenges and Further Research

Feature Interactions. FI computations are usually complicated by the presence of strong and higher-order interactions [43]. Such interactions typically have to be manually specified in (semi-)parametric statistical models. However, complex non-parametric ML models, to which we usually apply our model-agnostic IML

techniques, automatically include higher-order interaction effects. While recent advances have been made in visualizing the effect of feature interactions and quantifying their contribution regarding the prediction function [3,23,27], we feel that this topic is somewhat underexplored in the context of loss-based FI methods, i.e., how much an interaction contributes to the predictive performance. A notable exception is SAGE, which, however, does not explicitly quantify the contribution of interactions towards the predictive performance but rather distributes interaction importance evenly among all interacting features. In future work, this could be extended by combining ideas from functional decomposition [3,27], FI based on those [29] and loss-based methods as in SAGE.

Model Selection and AutoML. As a subtle but important point: it seems somewhat unclear to which model class or learning algorithms the covered techniques can or should be applied to, if DGP inference is the goal. From a mechanistic perspective, these model-agnostic FI approaches can be applied to basically any model class, which seems to be the case in current applications. Considering what Williamson et al. [60] noted in and, following our results, many statements in the Sects. 5 to 7 only hold under a "loss-optimal model". First of all, in practice, the construction of a loss-optimal model with certainty is virtually impossible. Does this imply we should try to squeeze out as much predictive performance as possible, regardless of the incurred extra model complexity? Williamson et al. [60] use the "super learner" in their definition and implementation of WVIM [59]. Modern AutoML systems like AutoGluon [16] are based on the same principle. While we perfectly understand that choice, and find the combination of AutoML and IML techniques very exciting, we are unsure about the trade-off costs. Certainly, this is a computationally expensive technique. But we rather also worry about the underlying implications for FI methods (or more generally IML techniques), when models of basically the highest order of complexity are now used, which usually contain nearly unconstrained higher-order interactions. We think that this issue needs to be more analyzed.

Rashomon Sets and Model Diagnosis. Expanding on the previous issue: In classical statistical modeling, models are usually not exclusively validated by checking predictive performance metrics only. The Rashomon effect tells us that in quite a few scenarios, very similarly performing models exist, which give rise to different response surfaces and different IML interpretations. This hints at the effect that ML researchers and data scientists might likely have to expand their model validation toolbox, in order to have better options to exclude misspecified models.

Empirical Performance Comparisons. We have tried to compile a succinct list of results to describe what can be derived from various FI methods regarding the DGP. However, we would also like to note that such theoretical analysis often considerably simplifies the complexity of real-world scenarios to which we apply these techniques. For that reason, it is usually a good idea to complement such mathematical analysis with informative, detailed, and carefully constructed empirical benchmarks. Unfortunately, not a lot of work on empirical benchmarks

exists in this area. Admittedly, this is not easy in FI, as ground truths are often only available in simulations, which, in turn, lack the complexity found in real-world data sets. Moreover, even in simulations, concrete "importance ground truth numbers" might be debatable. So far, there are no extensive benchmarks in the literature on FI methods. Many compare local importance methods [1,26], but few global methods: E.g., Blesch et al. [6] and Covert et al. [10] compare FI methods for different data sets, metrics, and ML models. However, the comparisons are not applied with regard to different association types, as the methods are not differentiated in this respect as in our paper.

Causality. Beyond association, scientific practitioners are often interested in causation (see, e.g., [21,22,50,57,61]). In our example from Sect. 4, the doctor may not only want to predict the disease but may also want to treat it. Knowing which features are associated with the disease is insufficient for that purpose – association remains on rung 1 of the so-called ladder of causation [47]: Although the symptoms are associated with the disease, treating them does not affect the disease. To gain insight into the effects of interventions (rung 2), experiments and/or causal knowledge and specialized tools are required [28,31,46,48].

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Appendix

A Additional Proofs

A.1 Proof of Result 5.31

Proof. We show that $X_j \not\perp\!\!\!\perp Y \mid X_G \Rightarrow RFI_j^G \neq 0$: For cross-entropy loss,

$$\begin{aligned} RFI_j &= (\mathbb{E}_X[D_{KL}(p(y|x)||f(y|x_{-j}, \tilde{x}_j))] - H(Y|X)) \\ &\quad - (\mathbb{E}_X[D_{KL}(p(y|x)||f(y|x))] - H(Y|X)) \\ &\overset{f=p}{=} \mathbb{E}_X[D_{KL}(p(y|x)||f(y|x_{-j}, \tilde{x}_j)] \end{aligned}$$

It remains to show that KL-divergence for $f(y, x_{-j}, \tilde{x}_j)$ is non-zero:

$$\begin{aligned} p(x_j, y, x_G, x_R) &= p(x_j|y, x_G, x_R) \\ &= p(x_j|y, x_G)p(y, x_G, x_R) && (X_j \perp\!\!\!\perp X_R|X_G, Y) \\ &\neq p(x_j|x_G)p(y, x_G, x_R) && (X_j \not\perp\!\!\!\perp Y \mid X_G) \\ &= p(\tilde{x_j}|x_G)p(y, x_G, x_R) && (\text{def. of } \tilde{X}_j) \\ &= p(\tilde{x_j}, y, x_G, x_R) \end{aligned}$$

Since $X_j \perp\!\!\!\perp X_R | X_G$ it holds that $p(\tilde{x}_j, x_R, x_G) = p(x_j, x_R, x_G)$ and, thus, $p(y|\tilde{x}_j, x_R, x_G) \neq p(y|x_j, x_R, x_G)$. With model optimality, $p(y|x) \neq f(y|\tilde{x}_j, x_{-j})$. Since KL divergence > 0 for $p \neq f$ it holds that $RFI_j > 0$. □

A.2 Proof of Result 6.11: mSAGEvf Interpretation

Proof. The implication is shown by proving the counterposition:

$$X_j \perp\!\!\!\perp (Y, X_{-j}) \quad \Rightarrow \quad v^m(\{j\}) = 0.$$

Since $X_j \perp\!\!\!\perp (Y, X_{-j}) \Rightarrow f_j^{*,m}(x_j) = f_j^{*,c}(x_j)$ it holds that $v^m(\{j\}) = v^c(\{j\})$. $X_j \perp\!\!\!\perp (Y, X_{-j}) \Rightarrow X_j \perp\!\!\!\perp Y$ and thus $v^m(\{j\}) = v^c(\{j\}) = 0$ (Result 6.21). □

A.3 Proof of Result 6.31: cSAGE Interpretation

Proof. The equation is shown by proving the contraposition

$$\forall S \subseteq P \backslash \{j\} : X_j \perp\!\!\!\perp Y \,|\, X_S \quad \Rightarrow \quad \phi_j^c(v) = 0.$$

From Result 6.21 we know that $X_j \perp\!\!\!\perp Y \,|\, X_G \Rightarrow v^c(G \cup \{j\}) - v^c(G) = 0$ for L2 and cross-entropy optimal predictors. If $\forall S \subseteq P \backslash \{j\} : X_j \perp\!\!\!\perp Y \,|\, X_S$, all summands of the SAGE value are zero, and thus $\phi_j^c = 0$.

Converse for cross-entropy loss: We prove the converse by counterposition

$$\phi_j^c(v) = 0 \quad \Rightarrow \quad \forall S \subseteq P \backslash \{j\} : X_j \perp\!\!\!\perp Y \,|\, X_S.$$

If L is the cross-entropy loss and f^* the Bayes model, using [10, Appendix C.1]

$$\phi_j^c(v) = \frac{1}{p} \sum_{S \subseteq P \backslash \{j\}} \binom{p-1}{|S|}^{-1} I(Y; X_j \,|\, X_S) = 0,$$

where the mutual information I and the coefficients are always non-negative. Thus, we add non-negative terms so the sum can only be zero if $\forall S \subseteq P \backslash \{j\} : I(Y; X_j \,|\, X_S) = 0$ and, thus, $\forall S \subseteq P \backslash \{j\} : X_j \perp\!\!\!\perp Y \,|\, X_S$. □

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Interpretable Machine Learning for TabPFN

David Rundel[1(✉)], Julius Kobialka[1], Constantin von Crailsheim[1], Matthias Feurer[1,2], Thomas Nagler[1,2], and David Rügamer[1,2]

[1] Department of Statistics, LMU Munich, Munich, Germany
{david.rundel,julius.kobialka,constantin.crailsheim,matthias.feurer,thomas.nagler,david.ruegamer}@stat.uni-muenchen.de
[2] Munich Center for Machine Learning, Munich, Germany

Abstract. The recently developed Prior-Data Fitted Networks (PFNs) have shown very promising results for applications in low-data regimes. The TabPFN model, a special case of PFNs for tabular data, is able to achieve state-of-the-art performance on a variety of classification tasks while producing posterior predictive distributions in mere seconds by in-context learning without the need for learning parameters or hyperparameter tuning. This makes TabPFN a very attractive option for a wide range of domain applications. However, a major drawback of the method is its lack of interpretability. Therefore, we propose several adaptations of popular interpretability methods that we specifically design for TabPFN. By taking advantage of the unique properties of the model, our adaptations allow for more efficient computations than existing implementations. In particular, we show how in-context learning facilitates the estimation of Shapley values by avoiding approximate retraining and enables the use of Leave-One-Covariate-Out (LOCO) even when working with large-scale Transformers. In addition, we demonstrate how data valuation methods can be used to address scalability challenges of TabPFN. Our proposed methods are implemented in a package `tabpfn_iml` and made available at https://github.com/david-rundel/tabpfn_iml.

Keywords: In-Context Learning · Prior-Data Fitted Networks · Tabular Data · SHAP

1 Introduction

Interpretable machine learning (IML) has become increasingly vital in the field of machine learning (ML), especially when applied to real-world tasks requiring an understanding of the model's predictions. IML serves both as a debugging tool to enhance fairness and ethics, as well as a means to improve social acceptance by fostering trust through interpretability [20].

Recently, Hollmann et al. [14] introduced TabPFN, a prior-data fitted network (PFN) for tabular data that achieves state-of-the-art performance on a diverse range of classification tasks in mere seconds. It approximates probabilistic inference by in-context learning with a Transformer. For small tabular tasks,

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TabPFN shows great potential due to its straightforward use without requiring hyperparameter tuning. This renders TabPFN an appealing choice across diverse application domains. TabPFN, like other deep neural networks (DNNs), is however a black box and its missing interpretability makes it less attractive for applications in some research fields, where understanding the predictions can have significant ethical and practical implications [14,22]. IML implementations have so far focused on classical supervised ML algorithms and are mostly model-agnostic. While there already exist model-specific implementations for certain deep learning (DL) and tree-based models that take advantage of their architectural properties [20], it is not always clear how to apply IML methods to PFNs and TabPFN in particular in a meaningful and efficient manner.

Our Contributions: In this work, we close this gap by modifying existing IML methods to take advantage of the unique properties of TabPFN and thereby enabling efficient computation of IML methods. In particular, these adaptations allow for a more accurate calculation of Shapley values through exact retraining and the use of LOCO, which is usually not a viable option for Transformers [28]. Also, we employ data valuation measures for context optimization, thereby mitigating scalability challenges of TabPFNs. Our methodological contribution is accompanied by a comprehensive, easy-to-use IML toolbox.

2 Supervised Machine Learning and TabPFN

Consider n_{train} independent and identically distributed (*iid*) training observations, gathered in a training data set $\mathcal{D}_P = \{\boldsymbol{x}^{(i)}, y^{(i)}\}_{i=1}^{n_{\text{train}}}$ where $\boldsymbol{x}^{(i)} \in \mathcal{X} \subseteq \mathbb{R}^p$ are p-dimensional feature vectors and $y^{(i)} \in \mathcal{Y} \subseteq \mathbb{R}^g$ are categorical or continuous targets. In the classical supervised ML setting, the goal is to learn the functional relationship between features and targets with a model $\hat{f} : \mathcal{X} \to \mathcal{Y}$ from a hypothesis space $\mathcal{H}$ that is fully determined by its parameters $\boldsymbol{\theta}$. For most ML algorithms, given the dataset and a hyperparameter configuration $\boldsymbol{\lambda} \in \Lambda$, a learner or inducer $\mathcal{I}(\mathcal{D}_P, \boldsymbol{\lambda})$ yields such a model by minimizing the empirical risk $\mathcal{R}_{\text{emp}}\big(\hat{f}(\{\boldsymbol{x}^{(i)}\}_{i=1}^{n_{\text{train}}}), \{y^{(i)}\}_{i=1}^{n_{\text{train}}}\big)$. This is accomplished by optimizing the parameters $\boldsymbol{\theta}$ that were either randomly initialized or pre-trained. The resulting model can be used to perform inference $\hat{f}(\{\boldsymbol{x}^{*(i)}\}_{i=1}^{n_{\text{inf}}})$ on an unseen inference dataset $\{\boldsymbol{x}^{*(i)}\}_{i=1}^{n_{\text{inf}}}$ of size n_{inf} [3,17]. In the following, we focus on the binary classification task, i.e., $\mathcal{Y} = \{0, 1\}$, and a probabilistic model $\hat{f}(\cdot) \in [0, 1]$ predicting the probability of the positive class given features and parameters $\boldsymbol{\theta}$. To highlight that $\hat{f}$ is fitted on the set of all available features $P = \{1, \ldots, p\}$, we use $\hat{f}_P$ with index P. When the model is fitted on a feature subset $S \subseteq P$, giving rise to a reduced dataset $\mathcal{D}_S = \{\boldsymbol{x}_S^{(i)}, y^{(i)}\}_{i=1}^{n_{\text{train}}}$, where $\boldsymbol{x}^{(i)} = \{\boldsymbol{x}_S^{(i)}, \boldsymbol{x}_{SC}^{(i)}\}$ for all $i = 1, \ldots, n_{\text{train}}$, we denote the model as $\hat{f}_S$.

For PFNs and TabPFNs [14,21], the training and inference procedure, however, differs. As opposed to classical supervised ML, where a model is trained

from scratch or fine-tuned on the training subset of a new dataset, PFNs perform inference in a single forward pass. In this approach, a pre-trained neural architecture with fixed parameters is fed with a training set $\mathcal{D}_P$ and inference features $\{\boldsymbol{x}^{*(i)}\}_{i=1}^{n_{\text{inf}}}$ from a previously unseen task. The model leverages the training set, also known as *context*, to conduct inference on the inference observations (so-called *in-context learning*). We denote the PFN model function as $\hat{f}_{\text{PFN}}(\{\boldsymbol{x}^{*(i)}\}_{i=1}^{n_{\text{inf}}}, \mathcal{D}_P)$, emphasizing that both the inference and training data are supplied during inference, and to highlight that the parameters of PFNs remain unchanged with the training dataset. To achieve this, the underlying model has been pre-trained on artificially generated tasks sampled from a prior distribution. Hence, PFNs are trained in two steps: The first step is the offline training phase, where the model learns to approximate the posterior predictive distribution (PPD) for an inference observation on many synthetic training datasets from diverse tasks (about 10 million datasets in the case of TabPFN). In detail, for every synthetic dataset $\mathcal{D}_{\text{syn}}$, it is trained to replicate the PPD on a query tuple $(Y_{\text{syn}}^{(n_{\text{syn}}+1)}, \boldsymbol{X}_{\text{syn}}^{(n_{\text{syn}}+1)})$, using $\mathcal{D}_{\text{syn}} \cup (Y_{\text{syn}}^{(n_{\text{syn}}+1)}, \boldsymbol{X}_{\text{syn}}^{(n_{\text{syn}}+1)})$ as training input. The second step is the inference phase, where a new training set and inference features are fed into the pre-trained network with fixed parameters [14,22].

TabPFN specifically uses a Transformer model and a prior over tabular classification tasks. The tokens in the Transformer represent the observations, with training observations attending to each other and inference observations only attending to training observations. In particular, there are $\binom{n_{\text{train}}}{2} + n_{\text{train}} \cdot n_{\text{inf}}$ token connections and the computational cost for a forward pass scales as $\mathcal{O}(n_{\text{train}}^2 + n_{\text{train}} \cdot n_{\text{inf}})$. Due to the quadratic complexity in the number of training observations, the recommended maximum size of the training set is 1024 observations [14,22]. TabPFN eliminates the need for parameter training or hyperparameter tuning during the inference stage, enabling almost instantaneous predictions [22]. Moreover, its unique characteristics include the fact that loss gradients w.r.t. the input data are available in the backward pass, as well as its attention mechanism. Several studies have demonstrated the model's effectiveness, achieving state-of-the-art performance across various benchmark datasets, specifically for moderately large tabular classification tasks [14,19].

3 Interpretable Machine Learning for TabPFN

Existing IML implementations have primarily focused on traditional supervised ML algorithms or work for specific DNN architectures not meaningful or applicable to PFNs. In the following, we describe how to adapt existing IML methods to exploit the unique characteristics of TabPFN. This will render the methods more efficient while also preserving their conventional usage paradigms. Our methodological contributions and implementations include local and global methods for assessing feature effects (FEs), feature importance (FI), and data valuation (DV). We have selected the methods according to their popularity, and such that they complement each other w.r.t. their limitations as well as benefit from

TabPFN's computational efficiency. Where sensible, both continuous and categorical features are supported. In our package, we have not included model-specific Transformer IML methods [2,5], because the available methods either produce unsuitable (relevance) scores or cannot be applied to TabPFN due to their architecture. Table 1 summarizes the methods and modifications made in this paper and implemented in the package `tabpfn_iml`.

Table 1. Overview of methods adapted and modified for TabPFN. In addition to FE, FI, and DV, we also include methods for clinical value assessment (CVA), uncertainty quantification (UQ), and local explanations (LE).

Method	Type	Scope	Modifications
ICE (Sect. 3.1)	FE	local	Improved efficiency
PD (Sect. 3.1)	FE	global	Improved efficiency
ALE (Sect. 3.1)	FE	global	Improved efficiency
Kernel SHAP (Sect. 3.3)	FE	local & global	Exact retraining
SA (Sect. 3.5)	FE	local & global	–
LOCO (Sect. 3.2)	FI	global	Rendered tractable
SAGE (Sect. 3.6)	FI	global	Exact retraining
LOO (Sect. 3.4)	DV	global	Rendered tractable
Data Shapley (Sect. 3.4)	DV	global	Exact retraining
SA (Sect. 3.5)	DV	local & global	–
DCA (Sect. 3.6)	CVA	global	–
CP (Sect. 3.6)	UQ	local	–
MOC (Sect. 3.6)	LE	local	–

3.1 ICE, PD and ALE

Individual conditional expectation (ICE) curves [13,20], partial dependence (PD) plots [10,20] and accumulated local effect (ALE) plots [1,20] show how the predictions of a model change along a feature space of interest by systematically varying the respective input values. This necessitates prediction on artificial observations, where the feature of interest is varied according to a specified grid generated by discretizing the feature space into G grid points. PD approximates the expected target response, while ALE addresses the extrapolation issue of ICE and PD [20].

Each TabPFN forward pass involves $n_{\text{train}}((n_{\text{train}}-1)/2+n_{\text{inf}})$ token connections (cf. Sect. 2) and thus the computational cost per inference sample scales as $\mathcal{O}(n_{\text{train}}^2/n_{\text{inf}})$. Consequently, TabPFN yields runtime improvements when processing as many inference samples as possible in a single forward pass. For the PD plot, widely used implementations such as the scikit-learn `partial_dependence`

function, execute a prediction step for every point of the grid. Instead, we propose a more efficient way to compute the PD plot in just a single forward pass by constructing a large inference array that contains all artificial observations. Compared to the scikit-learn implementation which has computational complexity $\mathcal{O}((n^2_{\text{train}} + n_{\text{train}} \cdot n_{\text{inf}}) \cdot G)$, our implementation then scales as $\mathcal{O}(n^2_{\text{train}} + n_{\text{train}} \cdot n_{\text{inf}} \cdot G)$. Thus, our PD plot implementation implies substantially lower runtimes. The modifications and implementations of ICE and ALE follow the same logic.

3.2 LOCO

Leave-One-Covariate-Out (LOCO) estimates the effect of features on the predictive performance of a learner by retraining it on a subset of features without the feature of interest [16,20]. The starting point is a model $\hat{f}_P$ trained with all features. Then, for each feature of interest j, the feature is removed from the dataset ($\mathcal{D}_{P\setminus\{j\}}$), the model is retrained without it ($\hat{f}_{P\setminus\{j\}}$) and the resulting change in empirical risk is tracked. LOCO approximately indicates whether a feature contains unique information that no other feature contains [16,20].

For many DL models, particularly extensive Transformer architectures, calculating LOCO scores is not feasible due to the high computational cost of repeatedly retraining models on different feature subsets. However, we can easily compute LOCO for TabPFN. Due to in-context learning, TabPFN enables inference on an updated training set through a single forward pass. In particular, we propose to calculate LOCO for the j-th feature as

$$\begin{aligned}\text{LOCO}_j = \mathcal{R}_{\text{emp}}\big(\hat{f}_{\text{PFN}}(\{\boldsymbol{x}^{*(i)}_{P\setminus\{j\}}\}_{i=1}^{n_{\text{inf}}}, \mathcal{D}_{P\setminus\{j\}}), \{y^{*(i)}\}_{i=1}^{n_{\text{inf}}}\big) \\ -\mathcal{R}_{\text{emp}}\big(\hat{f}_{\text{PFN}}(\{\boldsymbol{x}^{*(i)}_{P}\}_{i=1}^{n_{\text{inf}}}, \mathcal{D}_{P}), \{y^{*(i)}\}_{i=1}^{n_{\text{inf}}}\big).\end{aligned}$$

3.3 Kernel SHAP

Shapley values [24] have their origin in game theory and aim to fairly attribute payouts. In ML, they estimate the local FE for an observation $\boldsymbol{x}^{*(i)}$ in a model $\hat{f}_P$ [20]. In detail, the Shapley value for the j-th feature considers the (weighted) average marginal contribution of the feature to the model prediction across all feature subsets. While Shapley values provide a theoretically sound approach to attribution considering feature interactions, their calculation can be costly since the learner needs to be retrained for each feature subset S. As retraining ML models is in general resource-intensive and given that there are 2^p distinct feature subsets, this becomes impractical for most learners and datasets.

To address these challenges, Kernel SHapley Additive exPlanations (Kernel SHAP) [17] seeks to provide tractable approximations to Shapley values for a more efficient estimation. It employs an additive feature attribution measure by fitting a weighted linear model as local surrogate model that regresses model predictions on a binary representation of feature subsets, where only the features that appear in the subset are used for prediction [17,20]. To make the computations tractable, it incorporates the following additional approximations:

- M feature subsets are randomly sampled, as opposed to exhaustive consideration of all 2^p subsets. M serves as a hyperparameter that balances computational cost and approximation quality [17,26].
- To avoid retraining the learner on every sampled feature subset S, retraining is approximated by using the model trained on all features $\hat{f}_P$. This approach involves marginalizing over the features S^C that are absent in a subset while assuming that all features are independent and approximating an integral by Monte Carlo.[1] Specifically, the approximation can be expressed as

$$\hat{f}_S(\cdot) \approx L^{-1} \sum_{l=1}^{L} \hat{f}_P\big((\cdot, \boldsymbol{x}_{S^C}^{\kappa(l)})\big). \tag{1}$$

  The indices $\kappa(l) \in \{1, \ldots, n_{\text{train}}\}$ for $l \in \{1, \ldots, L\}$ are randomly sampled and referred to as *imputation samples*, used to impute S^C [17,26].

We can exploit the unique characteristics of TabPFN to enhance Kernel SHAP by obtaining exact solutions, rather than approximations, for certain terms in the Shapley value estimation. First, consider a naïve implementation of Kernel SHAP coupled with TabPFN. Using Eq. 1, the approximation for retraining TabPFN on a feature subset S is given by

$$\hat{f}_{\text{PFN}}(\{\boldsymbol{x}_S^{*(i)}\}_{i=1}^{n_{\text{inf}}}, \mathcal{D}_S) \approx L^{-1} \sum_{l=1}^{L} \hat{f}_{\text{PFN}}(\{(\boldsymbol{x}_S^{*(i)}, \boldsymbol{x}_{S^C}^{\kappa(l)})\}_{i=1}^{n_{\text{inf}}}, \mathcal{D}_P).$$

An *approximate retraining* implementation could process all n_{inf} inference samples and M feature subsets in parallel using L imputation samples, scaling as

$$\mathcal{O}(n_{\text{train}}^2 + n_{\text{train}} \cdot n_{\text{inf}} \cdot M \cdot L). \tag{2}$$

However, since training and prediction are done jointly in TabPFN's forward pass, $\hat{f}_{\text{PFN}}(\{\boldsymbol{x}_S^{*(i)}\}_{i=1}^{n_{inf}}, \mathcal{D}_S)$ does not need to be approximated. Instead, we iterate over M feature subsets, for each restricting the features of both the training and inference set accordingly, and evaluate the model for all n_{inf} inference samples. We refer to this approach as *exact retraining*, scaling as

$$\mathcal{O}((n_{\text{train}}^2 + n_{\text{train}} \cdot n_{\text{inf}}) \cdot M). \tag{3}$$

Assuming $n_{train} \approx n_{inf}$ and $L \geq 2$, our method has thus lower computational cost. In Eq. 2, we assume that all calculations can be executed in a single forward pass. As this is mostly not possible due to memory constraints on commonly used hardware, we also provide an *approximate retraining* implementation, where we iterate over M and L, predicting sequentially.

3.4 Data Valuation for Context Optimization

One of the primary limitations of TabPFN is its scalability in n_{train}. As a consequence, its context is restricted to approximately 1000 observations. This limits the widespread adoption of TabPFN, especially for larger datasets [9].

[1] In this work, we chose not to consider the optional model linearity assumption as it is overly simplistic.

A straightforward way to use TabPFN with larger datasets involves random sketching. This entails selecting a random subset of size $n_{\text{sub}} \leq 1024$ from the complete training set as context. In contrast, context optimization involves using more advanced strategies to determine appropriate contexts when dealing with large datasets. Recently, researchers proposed prompt-tuning strategies to learn an optimal context via gradient descent [9,18]. In our work, we pursue a different approach to context optimization by selecting a representative training subset based on IML methods. Various data valuation techniques, e.g., Data Shapley, which quantify the contribution of training observations to the predictive performance, have been suggested in the literature. Empirical evidence further suggests that incorporating training samples based on data values can enhance performance of ML models [12].

In contrast to Shapley values for FEs, Data Shapley considers the marginal contributions of training observations – as opposed to features – to the model performance across all training subsets. As retraining is impractical for complex learners and datasets, several approximations such as Gradient Shapley [12] have been proposed. Gradient Shapley employs simplistic approximations to estimate the marginal contributions, such as utilizing stochastic gradient descent with a batch size of one. Some advancements exist, but most of these involve additional approximations and yet require training multiple models, limiting their applicability [15,23,29]. We can, however, implement Data Shapley as a modification of Kernel SHAP with exact retraining by making use of in-context learning. This eliminates the reliance on such simplistic approximations. Consider a training dataset of size $n_{\text{train}} > n_{\text{sub}}$ (typically $n_{\text{sub}} = 1024$) where our objective is to identify an optimal subset of size n_{sub}. To achieve this, we evaluate M randomly sampled training subsets, ranging in size from 256 (for sufficient context) to n_{sub}, on a validation set of size n_{val}. The resulting empirical risk values are then attributed to the training set observations by fitting a weighted linear model as a local surrogate. Subsequently, we select the n_{sub} training observations with lowest coefficients in the linear model as the context, expecting that these will contribute the least to the increase in empirical risk. Kernel SHAP forgoes the need to examine all $2^{n_{\text{train}}}$ training subsets. In order to ensure estimation of the surrogate model, we, however, require $M \geq n_{\text{train}}$, leading to substantial computational expenses.

Furthermore, TabPFNs architecture facilitates the incorporation of additional DV techniques, including Leave-one-out (LOO) testing [6]. Similar to LOCO, LOO removes a single training observation from the training procedure and computes the change in predictive performance [23,29].

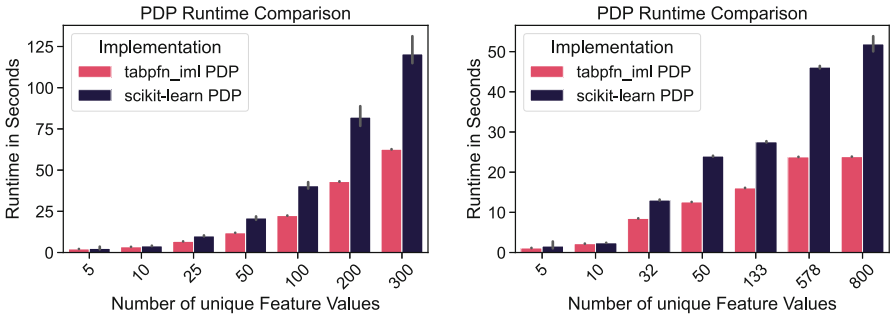
3.5 Sensitivity Analysis

Sensitivity Analysis (SA) is a model-specific IML method that leverages gradients computed in a backward pass to extract relevance scores for features [11]. Computing SA for FE is straightforward for TabPFN and also included in our implementation. Leveraging the differentiability of TabPFN w.r.t. training

observations, we follow a similar approach to adapt SA for data values. For the data value of the j-th training observation, we compute the gradient

$$R_j = \left\| \nabla_{(\boldsymbol{x}^{(j)}, y^{(j)})} \mathcal{R}_{\text{emp}} \left(\hat{f}_{\text{PFN}}(\{\boldsymbol{x}^{*(i)}\}_{i=1}^{n_{\text{inf}}}, \mathcal{D}_P), \{y^{*(i)}\}_{i=1}^{n_{\text{inf}}} \right) \right\|_2$$

of the empirical risk w.r.t. that observation and aggregate it to a scalar using the L2-norm. This allows identifying training observations for which the empirical risk is highly sensitive.



(a) Synthetic datasets **(b)** OpenML-CC18 suite datasets

Fig. 1. Runtime comparison of the implementations of `tabpfn_iml` PD vs. scikit-learn `partial_dependence()`. (Color figure online)

3.6 Further Methods

In order to complement our toolbox to provide a comprehensive IML tool, we include Shapley Additive Global Importances (SAGE) [7] as an additional FI method. Our implementation involves utilizing Kernel SHAP with the empirical risk as the function to be explained and also benefits from exact retraining. We also offer wrappers for decision curve analysis (DCA), a CVA method as implemented in [25], and conformal prediction (CP), an UQ method as implemented in [27]. Last but not least, we implement multi-objective counterfactual explanations (MOC) from the realm of LE using evolutionary multi-objective algorithms (EMOAs) [8]. For more information, we refer to the tabpfn_iml repository.

4 Numerical Experiments[2]

In the following, we provide numerical evidence of the improvement in computational efficiency and approximation error obtained with our modifications and implementation. Furthermore, we show how DV methods can be employed to improve classification performance by context optimization.

[2] All experiments were run on a Nvidia GTX A6000 GPU. Furthermore, our experiment code in the repository enables the replication of all results.

4.1 ICE and PD

To show the improved efficiency of our ICE and PD implementation, we conduct a runtime comparison across 25 synthetic datasets with 10 features and 1000 observations in Fig. 1a and across 7 binary classification datasets with increasing size from the OpenML-CC18 suite in Fig. 1b. For the synthetic data, we define a complex non-linear relationship between 10 features and a binary target, add a random noise term, and discretize the feature of interest to the number of intervals indicated on the x-axis, as detailed in our GitHub repository. We compare our implementation to scikit-learn's `partial_dependence()` across five different seeds per feature level, setting the `grid_size` equal to the number of unique feature values. In Fig. 1b, we set the `grid_size` to the default value of 100 to depict runtimes for different real-world datasets. For both Figures, n_{train} comprises 80% of all observations, and n_{inf} the remaining 20%. Our modifications to ICE and PD result in much better runtime, with our implementation being substantially faster as the grid size or data set size increase.

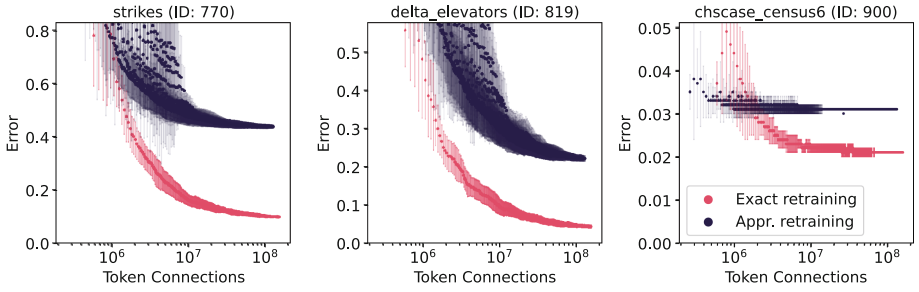


Fig. 2. Average errors (points) and corresponding standard deviations (error bars) across all runs plotted against the token connections for various configurations and three datasets (different plots). Exact retraining configurations are red, while approximate retraining configurations are black. (Color figure online)

4.2 Kernel SHAP

Next, we empirically compare our proposed exact retraining method to the alternative approximate method. The comparison of the two Kernel SHAP approaches is done by defining a (stochastic) error metric, which for each calculates the sum of the absolute deviations from the exactly computed Shapley values $\phi_j(\hat{f}_{\text{PFN}}, \boldsymbol{x}^{*(i)})$ across features j, averaged over inference observations i:

$$\frac{1}{n_{\text{inf}}} \sum_{i=1}^{n_{\text{inf}}} \sum_{j=1}^{P} \quad |\phi_j(\hat{f}_{\text{PFN}}, \boldsymbol{x}^{*(i)}) - \hat{\phi}_{j,(M,L)}(\hat{f}_{\text{PFN}}, \boldsymbol{x}^{*(i)})|.$$

$\hat{\phi}_{j,(M,L)}(\hat{f}_{\text{PFN}}, \boldsymbol{x}^{*(i)})$ indicates the estimated Shapley value for the j-th feature and i-th inference observation. We further specify $M \in \{1, 2, \dots\}$ and $L \in$

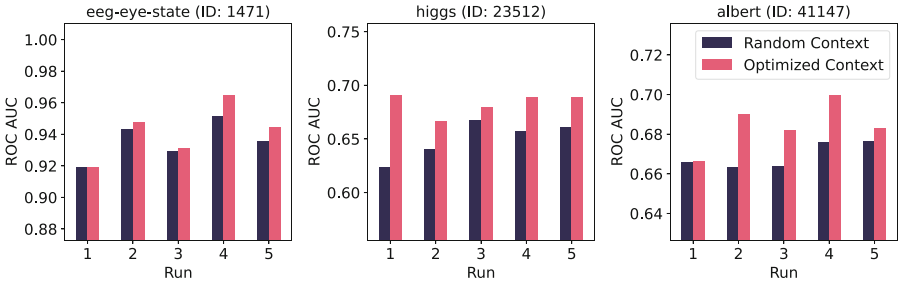


Fig. 3. Comparison of the ROC AUC of randomly selected TabPFN contexts (black) and contexts optimized through Data Shapley coupled with Kernel SHAP (red) across five runs and three datasets. (Color figure online)

$\{-1, 1, 2, \dots\}$ as the number of feature subsets and imputation samples, where we denote exact retraining as $L = -1$. Since the exact computation of Shapley values scales exponentially in the number of features, we select three binary classification tasks from the OpenML-CC18 suite [4] with six numerical features, aligning with the optimal application scenario for TabPFN [14]. For each dataset, we randomly sample a training set of size $n_{\text{train}} = 256$ and an inference set of size $n_{\text{inf}} = 128$ and evaluate a fine-grained grid across all values of $M \in \{6, \dots, 160\}$ and $L \in \{1, \dots, 25\}$ for approximate retraining, and across values of $M \in \{6, \dots, 1600\}$ for exact retraining. We repeat all experiments 25 times for each dataset, with feature subsets and imputation samples randomly sampled in each iteration.

In Fig. 2, we present the mean and one standard deviation of the error across all runs, contrasted with the corresponding number of token connections for each (M, L) configuration according to Eqs. 2 and 3. The plot shows that exact retraining almost exclusively achieves lower approximation error for a fixed computational budget. In contrast, achieving comparable error values with approximate retraining often demands a larger number of feature subsets and multiple imputation samples. Moreover, exact retraining converges at a lower error and the standard deviations indicate that its Shapley estimates have lower variance. On top, it is important to stress that the number of token connections for approximate retraining is a lower bound and approximate retraining converges even slower in practice. Notably, for the chscase_census6 dataset, there are configurations with a small number of token connections where approximate retraining outperforms the exact equivalent, which might be attributed to the cost-effectiveness of approximate retraining for low values of L.

4.3 Data Valuation for Context Optimization

We conduct an empirical evaluation of our Data Shapley implementation for context optimization by comparing it to random sketching. Our benchmark consists of three datasets from the OpenML-CC18 suite [4], chosen in accordance with

[9], suggesting that TabPFN's performance on these datasets can be enhanced through context optimization. For each dataset, we create a training set of size $n_{\text{train}} = 3072$ and aim to identify an optimal context of size $n_{\text{sub}} = 512$ thereof. We choose $M = 3 \cdot n_{\text{train}}$, $n_{\text{val}} = 512$ and ultimately assess the optimized context's performance on a previously unseen test set of size $n_{\text{test}} = 1024$, comparing it to the performance of randomly selected contexts of the same size. This procedure is repeated 5 times with differently sampled subsets.

In Fig. 3, we compare the ROC AUC on the test set of both random and optimized contexts across all runs and three datasets. The plot indicates that our method consistently outperforms random sketching. On the eeg-eye-state and albert datasets, we observe an average increase in ROC AUC of 0.57% and 1.52%, respectively. An even more substantial improvement is obtained for the higgs dataset, yielding a difference of 3.3% in ROC AUC. The results suggest that our implementation of Data Shapley produces sensible results, indicating its potentially meaningful application in domains beyond context optimization. However, note that this procedure requires nearly 10,000 TabPFN forward passes and is – in contrast to our other proposed methods – notably more expensive.

5 Discussion and Conclusion

By exploiting the architecture underlying TabPFN and in-context learning, we propose tailored modifications and adaptations of IML methods for TabPFN. Specifically, we provide efficient implementations of ICE, PD, and ALE. Furthermore, using similar insights, we make the computation of LOCO and LOO possible despite TabPFN's Transformer nature. Moreover, we have shown that our exact retraining approach to Kernel SHAP dominates current alternatives with respect to both approximation error and runtime. Finally, our implementation of Data Shapley yields meaningful results that can be used for context optimization. Together with further methods in IML, we provide a comprehensive and easy-to-use IML package that allows to shed light on TabPFN's predictions. The suggested modifications to IML methods also translate to the broader realm of in-context learning, especially for tabular data, and exploring their potential is a promising direction for future research.

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Statistics and Explainability: A Fruitful Alliance

Valentina Ghidini(✉)

Euler Institute, Università della Svizzera Italiana, Lugano, Switzerland
valentina.ghidini@usi.ch

Abstract. In this paper, we propose standard statistical tools as a solution to commonly highlighted problems in the explainability literature. Indeed, leveraging statistical estimators allows for a proper definition of explanations, enabling theoretical guarantees and the formulation of evaluation metrics to quantitatively assess the quality of explanations. This approach circumvents, among other things, the subjective human assessment currently prevalent in the literature. Moreover, we argue that uncertainty quantification is essential for providing robust and trustworthy explanations, and it can be achieved in this framework through classical statistical procedures such as the bootstrap. However, it is crucial to note that while Statistics offers valuable contributions, it is not a panacea for resolving all the challenges. Future research avenues could focus on open problems, such as defining a purpose for the explanations or establishing a statistical framework for counterfactual or adversarial scenarios.

Keywords: Explainability · Statistics

1 Introduction

This paper takes an unconventional approach, for either computer scientists or statisticians. It will not delve into technicalities, present groundbreaking methodologies or introduce new explainability techniques. It will however explore a reflection on the use of statistics in the explainability context to argue that a closer relationship between standard statistical procedures and explainability techniques is needed to tackle some foundational challenges.

Considerable debate has revolved around the desirability of employing black-box models, especially in highly sensitive applications [20]. However, in this work, we acknowledge the impracticality of completely eliminating the use of black-box models in real-world applications. Consequently, we start from the assumption that there is a necessity for explainability techniques, recognizing the need to provide explanations for black boxes somehow, since such inscrutable models are not going to disappear anytime soon.

The main question at the core of this work is whether we can address certain challenges currently undermining the foundations of eXplainable Artificial

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Intelligence (XAI) through the application of statistical methods, analyzing the corresponding advantages and disadvantages. The aim is to persuade practitioners that statistics can serve as a valuable tool, offering a robust ground for the introduction of novel methods and techniques. Specifically, we try to leverage statistics to systematically tackle existing challenges in the realm of explainability, employing a formal mathematical methodology to address these open issues.

1.1 Related Literature

Numerous works in the literature have brought attention to foundational issues and ongoing challenges in XAI, but, to our knowledge, no tentative solution has been outlined yet. This paper draws inspiration from various sources, including but not limited to [9,12–15,18,20,21]. As previously mentioned, our objective is to show how statistics can alleviate (if not entirely resolve) various complaints and challenges posed by researchers. Standard statistical procedures can indeed offer partial or comprehensive solutions to the principal issues, as outlined in a detailed point-by-point analysis in the next sections.

The remainder of this paper is structured as follows: Sect. 2 lists some of the issues currently contained in the literature that are the target of our analysis. Section 3 addresses such issues point-by-point, illustrating how statistical procedure can be used. Section 4 highlights some additional perks of using statistical explanations, while Sect. 5 addresses some problems which can not be solved merely applying standard statistical procedures. Then, Sect. 6 contains some conclusive thoughts about future and desirable research directions.

2 Foundational Issues in Explainability

In this section, we enumerate foundational issues in explainability as identified in various works within the literature. In particular, we list the challenges we aim to tackle using standard statistical procedures.

2.1 Issue 1: Lack of Proper Definitions

First, let us address the biggest elephant in the room: the absence of a clear, unique and precise definition for the term *explanation*. Numerous scholars have already pointed out this issue in works such as [6,12,15,22] and references therein. While certain referenced works offer definitions of the term *explanation*, these definitions often lack practical applicability and, as a result, they are not actionable. In other words, translating the English meaning into a precise mathematical definition of explanations is usually difficult, if not impossible. For example, citing [12], the authors define an explanation as "*an "interface" between humans and a decision maker that is at the same time both an accurate proxy of the decision maker and comprehensible to humans*". However, such definition is still ambiguous, as it remains unclear what, e.g., an interface is

from either a mathematical or an algorithmic viewpoint, or how to obtain one. Another definition of explanation provided in [7] is the following: "*we define the act of "interpreting" some object X as the activity performed by an agent A ... assigning a subjective meaning to X. Such meaning is what we call interpretation. We define "explaining" as the activity of producing a more interpretable object X' out of a less interpretable one, namely X, performed by agent A*". While this definition introduces a notion reminiscent of algebraic concepts, it still falls short in providing concrete information about what an explanation is, its form, or the methodology for obtaining one. As of our current understanding, it appears that there is a recurrent, ephemeral illusion in the literature, according to which the English definition of explanation is perceived to seamlessly convert into an actionable technical definition, which, in reality, is often not the case. The lack of a proper definition for explanations is one of the root causes of the issues discussed in the next subsections, namely the lack of theoretical guarantees, simple quantitative assessments of the goodness of explanations, and uncertainty quantification.

Arguably, a class of more actionable explanations implicitly defines what in statistics is known as variable importance measures. For example, the work in [8] gives the following definition: "*An explanation is additional meta information, generated by an external algorithm or by the machine learning model itself, to describe the feature importance or relevance of an input instance towards a particular output classification*". Even though not explicitly mentioned, it is natural for statisticians to think of variable importance measures, such as the world-famous ones introduced by Breiman in [5]. In the next section, we argue that, among other things, statistics can be seen as the bridging gap enabling the interpretation of traditional variable importance measures as explanations for black-box models.

2.2 Issue 2: Lack of Theoretical Guarantees

Another problem is the lack of theoretical guarantees about the explanations. Consider standard XAI approaches, such as GradCAM [23], LIME [19], RISE [17] among others. These works offer empirical algorithms that yield explanations. Their appeal lies in the ease of interpreting the explanations and the overall sensibility of the entire procedure, contributing to their popularity and widespread adoption. However, these approaches usually lack a theoretical analysis. For instance, when considering a probabilistic modeling framework where data are treated as realizations of random variables living on probability spaces, it remains unclear whether explanations exist in general as well-defined quantities (*existence result*) and, if so, how they can be precisely defined in a mathematical sense. Moreover, as of our current understanding, it remains uncertain whether the algorithms provide correct explanations, at least in some defined sense (*correctness result*). Essentially, we argue that there is a need to precisely determine the quantity being estimated as explanation for the model. Furthermore, a desirable property of XAI algorithms would be that the estimation of the explanations improves as the number of collected data instances grows (*convergence result*).

Notice however that these theoretical properties can only be verified if and only if we can establish a proper mathematical definition of explanations. All these results are standard in statistical analysis and we think they would be significant for the theoretical understanding and enhancement of current XAI techniques.

2.3 Issue 3: Lack of (Simple) Evaluating Metrics

A significant challenge in explainability lies in the absence of simple, quantitative evaluation metrics to assess the goodness of explanations [11]. This difficulty is partially due to the lack of a clear and actionable definition of explanation, as previously mentioned in Subsect. 2.1. This makes it difficult to establish a ground truth with respect to which the evaluation of explanations can be conducted. Indeed, if the quantity defined as explanation is ill-defined or ambiguous, assessing the correctness of the output from a proposed technique is impossible. Furthermore, the absence of a clear and actionable definition for explanations makes it difficult to establish benchmarks where the true explanation is analytically known. Consequently, defining simulated scenarios to assess the correctness of XAI algorithms (i.e. whether they return the correct explanations) is unfeasible.

Since the lack of precise quantitative metrics inhibits a quantitative and objective evaluation, the literature often resorts to sanity checks performed by humans, as highlighted by [9]. These assessments however come at a cost—they are time-consuming, require financial resources, and are subject to individual judgment. In fact, in this context, explanations are often considered correct if and only if they align with human judgment regarding the importance of features. For instance, in a cat-dog image classifier, an explanation is deemed correct if it identifies the pixels representing the actual animal as the most crucial part of the input image according to the model. However, this enables a vicious cycle: certain explanations may be considered inadequate or, worse, incorrect simply because they do not align with human expectations. Nevertheless, it is plausible that the machine learning model in question operates in an unexpected manner, and explanations could be valuable in detecting discrepancies between the human symbolic representation of features and the models' sub-symbolic system.

2.4 Issue 4: Lack of Uncertainty and Robustness Evaluation

Last but not least, one crucial aspect often missing in many XAI techniques is the uncertainty quantification of the resulting explanations. Relying solely on point estimates of the explanations is not the best practice; it is essential to gain an understanding of how much these estimates vary across different data instances or realizations of the training set. Bayesian statisticians are well aware of this, and they typically provide estimates of the entire posterior distribution for the quantities they are interested in inferring. Nevertheless, uncertainty quantification is not yet a common practice in the XAI community. Generally, only single-shot outputs of explanatory algorithms are provided, offering no insight into the robustness of the proposed explanations. The variability of the proposed

explanations is usually not thoroughly studied or understood. For instance, it is often unclear how the explanations would change with slight alterations to the model, data points, or when using a different, randomly selected dataset. If the explanations exhibit significant variability, one may hesitate to trust them, or alternatively, one might consider collecting additional data for a more reliable estimation.

3 A (Tentative) Answer: Statistics

In this section, we aim to offer point-by-point solutions to the challenges outlined in Sect. 2. We argue that standard statistical tools and techniques can be leveraged to sketch several solutions.

3.1 Solution 1: Mathematical Measures and Definitions

First, we can define explanations in a statistical sense, and in this framework the most natural choice is to employ the notion of variable importance measures, as in the seminal works in [4,5]. For instance, given a set of explanatory variables $X = (X_1, \ldots, X_p)$, we can define explanations as expected value of variations in some output of the model f with respect to changes in a specific variable X_j (possibly multidimensional). An easy, intuitive and general approach is to define a difference $d(\cdot, \cdot)$ of summary statistics $s_f(\cdot)$ pertaining to the model of interest and consider its expected value $\mathbb{E}_{X_{\text{interest}}}[d\big(s_f(X_j), s_f(\widetilde{X}_j)\big)]$, where $\widetilde{X}_j$ is a perturbed version of X_j. Classical examples of perturbations of inputs are summation of white noise [17], or random permutation of the values [4]. The expected value is taken with respect to $X_{\text{interest}} = X_j$ (as in sensitivity analysis [3]) or $X_{\text{interest}} = X_{(-j)}$, that is with respect to all the other covariates. To summarize, in this example, the explanations are defined as expected values of differences in summary statistics of the model outputs, whenever X_j is perturbed. This formulation yields explanations which can be written as:

$$I_f(X_j) = \mathbb{E}_{X_{\text{interest}}}[d\big(s_f(X_j), s_f(\widetilde{X}_j)\big)]. \tag{1}$$

Given such expected values, explanations can be estimated by means of proper estimators, endowed with estimates [3,10]. As previously mentioned, this approach is not novel: it aligns precisely with the concept of variable importance measures. In the end, both procedures assign a score to each covariate (possibly multidimensional) which is proportional to its importance for the model. Even though variable importance measures have been around for decades, their flexibility is endless: according to the choice of $d(\cdot, \cdot)$ and of $s_f(\cdot)$ we can measure the impact of each covariate (or each group of explanatory variables) in infinite ways. It is then clear that such definition bridges the gap between explanations and variable importance measures, exploiting a proper statistical quantity, which can be endowed with an estimator, as explanation.

With this example, we have delved into the natural interpretation of statistical variable importance measures as a general way of providing explanations

which can be endowed with proper statistical estimators. However, the possibilities are endless: one can use statistical quantities such as variance, quantiles, probability distributions computed either on differences on the data or the model outputs to define explanations suited for the application at hand. Indeed, notice that explanations defined as in Eq. (1) are just one (flexible) possibility showcased as example. In general, the aim of this subsection is to advocate for providing explanations using statistical quantities to accurately capture, through their definition, specific characteristics of the model to be explained. Another example of well-known tools in XAI are partial dependence plots [1] or accumulated local effects plots [2]. In both cases, the explanations are provided by effects or measures of dependences defined through probabilistic expected values. As such, those kind of explanations (and many others) can be analyzed with a statistical viewpoint in mind, employing all the techniques discussed in the next sections to obtain proper statistical explanations.

3.2 Solution 2: Studies of Convergence as Theoretical Guarantees

With a proper statistical definition of explanation in place, it is natural to leverage standard statistical tools to establish convergence results. For example, consider the previously defined explanations $I_f(X_j)$ as a running example and suppose we observe n independent and identically distributed $p-$dimensional data points $\mathbf{x}_1, \ldots, \mathbf{x}_n$, which are commonly arranged row-wise into a design matrix X. Then, it is natural to estimate the explanations $I_f(X_j)$ using statistical estimators, such as the empirical mean. By employing such a procedure and assuming standard regularity constraints on the summary statistic $s_f(\cdot)$ and the difference $d(\cdot, \cdot)$, it is possible to leverage well-known theorems like the law of large numbers and the central limit theorem to establish convergence results. The Xi method [3,10] provides an example illustrating such a pipeline. The referenced works offer theoretical convergence results for the explanations, demonstrating that as the number of data points increases, the estimated explanations converge to their true, existing values. Future endeavors could also address distributional results: it would be interesting to prove the asymptotic Gaussianity of the estimators of the explanations. This could be utilized to construct proper asymptotic confidence intervals.

3.3 Solution 3: Definition of Evaluating Metrics

If the explanations are well-defined to sensibly capture interesting features of the model under scrutiny, it becomes straightforward to establish appropriate and quantitative evaluation metrics. Indeed, it is easier to evaluate the disparity between the estimated explanations and the true ones. A natural metric is a simple difference between the true values of the explanations and the estimated ones in, e.g., simulated data, but various other discrepancy measures can also be employed. Indeed, besides the quantitative assessment, it is now easy to simulate benchmarks and scenarios where the true explanations are designed. This allows for the evaluation of whether a proposed XAI technique effectively captures the

true explanations underlying the data-generating process. This approach completely eradicates the need for human studies, as the evaluation is now entirely objective, quantitative, and conducted by means of a properly defined metric. It also avoids the subjectivity associated with determining what constitutes a good explanation based on different users' perspectives, as human evaluations are inherently influenced by varying levels of expertise, education, and other factors. Furtherly, this approach clarifies and eliminates any ambiguity regarding what constitutes a wrong explanation: an explanation (or its estimator/estimate) is considered wrong if and only if it fails to converge or is not "close enough" (in the newly-defined metric sense) to the true value. In such cases, one should reconsider either the algorithm or the estimator used in the procedure.

Notice however that the previous discussion assumes that a good explanation, at least in this statistical context, is one that is estimated correctly with respect to the true quantity underlying the data. However, since interpretability is closely bounded to the human perception of data, one could also argue that a good explanation is a *useful* one, that is it can be easily employed by the end users. Indeed, explanations are of little use if they are too complex to be understood or too simple to be faithful to the black box. Evaluation metrics in this regard are trickier to define, and may involve a human in the loop at some point, but more objective and quantitative alternatives employing statistical techniques could be explored in future research avenues.

3.4 Solution 4: Uncertainty Quantification

A significant challenge is the absence of a comprehensive framework for uncertainty quantification in explanations. In general, few papers discuss the robustness of the outputs generated by proposed XAI techniques [3,10]. We argue that the gold standard should be to provide uncertainty quantification for the explanations, but as of the current state of the field at the time of writing, this is not yet a common practice. Indeed, reliability and robustness of explanations are always among the essential requirements for the deployment in real-world scenarios. Intuitively, when either the data instances to be explained or the prediction process provided by the model are slightly perturbed, the explanations should not undergo significant changes. By incorporating statistics into the process, we gain access to standard tools that can be used to provide not only point estimates of explanations but also confidence intervals, whether asymptotical or empirical. In cases where we have an asymptotic distribution of the estimators for the explanations, obtaining confidence intervals becomes a relatively straightforward process. For example, if one can establish that the explanations are asymptotically normal, it becomes possible to derive closed-form asymptotic confidence intervals. Even if verifying asymptotic normality is not feasible, one can resort to standard statistical resampling techniques, such as the bootstrap. In this context, the data are resampled B times to obtain the B different estimates of the explanations allowing for the computation of the corresponding empirical distribution, variability measures, and/or confidence interval. As a final note, notice that this section only analyzes explanation uncertainty, but there exist

many other sources of variability, such as model or sampling uncertainty. However, we argue that model uncertainty should be addressed prior to explaining it (in the definition/training part of the pipeline). As for sampling uncertainty, it should be analyzed in the algorithmic part: for example, if the estimation of the explanations involves some Monte Carlo algorithm, then a proper diagnostic procedure should be carried out, computing effective sample sizes and examining suitable trace plots.

4 Other Statistical Perks

There are additional significant benefits to utilizing statistics to define explanations. We list some advantages below, but such list is not comprehensive and could be updated in the future.

4.1 Bonus 1: Trustworthy Explanations

Using a coherent procedure to have well-defined explanations has the undeniable advantage of having assurances about the accuracy of the quantities that we are estimating. If convergence results can also be established, there are theoretical guarantees regarding the correctness of the explanation estimates when the number of data points is sufficiently large. This establishes a solid foundation for *trust* in the provided explanations.

Consequently, there are two potential outcomes in the analysis of explanations: first, the explanations of the black-box models align with our human expectations, highlighting the features in the prediction process that we expect. In this case, we can infer that the model of interest is effectively capturing expected meaningful dependence structures in the data. In a second scenario, the explanations may emphasize features that we, as human beings, do not consider important. In some current studies, such explanations are categorically considered incorrect. However, with theoretical assurances regarding the correctness of the estimated explanations, a more nuanced interpretation can be gained. Generally speaking, a mismatch between the most important features according to the explanations and the researcher's expectations indicates a divergence in the black-box sub-symbolic representation of the data from what was anticipated. Hence, it is plausible that the researcher's perception of the importance of certain features is misplaced, or there may be multicollinearity issues causing the model to favor other variables. In any case, the explanations can also serve as a diagnostic tool for the black-box model.

4.2 Bonus 2: Statistics Offers Interpretable Models

In this section, we draw inspiration from prominent works that assert post hoc interpretability is not always the optimal choice and advocate against the use of black-box models in high-stakes decisions [20,21]. Much of the support for these arguments stems from the observation that some XAI procedures generate

explanations that lack stability, trustworthiness, and robustness. We argue that, by incorporating statistical methods, many of these issues can be alleviated, if not entirely resolved, as discussed in the previous sections.

Even though we believe black-box models will not fall in disgrace anytime soon, we acknowledge that in certain sensitive contexts, like medical or financial applications, it is preferable to use transparent models. A straightforward approach to define interpretable models is to resort to classic inferential statistics. For instance, one can utilize linear regression, generalized linear models, generalized additive models, trees, survival models, and so forth. These models are inherently interpretable and well-performing, if defined correctly. The main drawback, however, is that they necessitate a deeper understanding of statistics, as well as more meticulous prior design and feature engineering, which could be challenging or require a higher level of domain expertise. Notice that we are not claiming that *all* statistical models are interpretable: for example, nonparametric models may bee too complex to be white boxes. Nonetheless, classical statistics provides a multitude of choices to model various types of data using inherently interpretable models.

4.3 Bonus 3: Easy-to-Define Fairness

Statistics also provides the means to comprehend and rigorously assess fairness in black-box models. One straightforward approach involves introducing tests for explanations concerning sensitive attributes in the design matrix, such as gender or age. Indeed, a natural procedure for a statistician would be to define a statistical test where the null hypothesis examines the significance of an explanation, e.g. a variable importance measure, of a sensitive attribute being null. This can be achieved by leveraging asymptotic and distributional results of estimators of explanations. Of course, this represents the more straightforward option; alternatively, one could test differences in certain outputs of the model before and after perturbing specific sensitive factors, offering actual p-values to evaluate the fairness of the model. Such methods are fairly naive and limited, but they serve just as a starting point for a statistical definition of fairness: future research avenues should be concerned with more sophisticated statistical structures to capture complex characterization of fairness.

4.4 Bonus 4: Explanations for the Data Generating Process

A common topic of discussion in the field of explainability is whether explanations should solely focus on black boxes or also encompass the data generating process (DGP) [9,10]. It is often argued that explanations concerning the DGP can establish the ground truth for their counterpart interpreting the black box of interest [16]. Although we may not entirely endorse this assertion, we acknowledge the importance of understanding the data generating process first and the black box model second. Analyzing the DGP beforehand can be crucial for identifying potential disparities between the explanations of the DGP and

those of the model. Some works in the literature have defined pre hoc explanations [3,10], that are explanations specifically devised for the data generating process of interest. Future research endeavors may explore various techniques to establish proper definitions for pre hoc explanations.

5 Open Problems Not Solved by Statistics

In this section, we list some issues highlighted by works in the literature [9], for which it is currently unclear how to address them using statistics.

5.1 Definition of a Purpose

The work in [9] advocates for the establishment of a purpose in the development of an explainability technique. This aspect cannot be resolved by statistics alone, as the purpose typically varies depending on the field and heavily relies on the specific application of interest. Is the purpose to ensure a fair model? Or to verify whether the model behaves as expected? Or perhaps to examine the joint dependence structure of the covariates on the target? Or even to increase the trust in the model in practitioners?

5.2 Simplicity of Explanations

Some authors [12,15] also argue for the simplicity of explanations to guarantee the usability of XAI techniques for a broad audience of users. This notion is already prevalent in the literature. An illustrative example of simple explanations is provided by heatmaps explaining models for images: it is intuitively clear that highlighted pixels in the pictures are also the most crucial for the black-box model's explanation. Generally, it is not that immediate to convey such intuitive explanations using statistical measures of differences in the output, or other statistical quantities. Additional work should ensure that explanations are easily comprehensible to a diverse audience.

5.3 Need of Researchers Trained in Statistics

Another unavoidable drawback of using statistics to formulate explainability techniques is that one necessitates statistical training to appropriately define and operate in such a context. While gaining proficiency in statistics is fairly accessible for individuals with a scientific background (e.g., computer scientists, engineers, etc.), it still requires time and dedication to delve into a (potentially) new literature and engage with a distinct community, albeit not significantly distant from the ones from which XAI stemmed as a research field.

6 Discussion

In this paper, we have outlined some foundational issues of explainability highlighted in the literature by different authors. We focus on the absence of proper definitions for explanations and of theoretical guarantees, on the challenges in defining evaluation metrics, and on the doubts regarding whether the proposed explanations are robust or exhibit excessive variability, thereby raising concerns about their trustworthiness. We argue that a solution to all these issues can be found through the application of statistical techniques. By utilizing statistical estimators for explanations, we can leverage well-established results such as the law of large numbers or the central limit theorem to furnish theoretical guarantees for explanations. Furthermore, it is natural to incorporate uncertainty quantification into the resulting estimates, as is customary in any statistical analysis. Statistical explanations may also enhance the trust in either the models or the outcome of the XAI procedures. It is also natural for a statistician to employ hypothesis testing to assess the fairness of a model. However, there is still much work to be done in this regard: statistical explanations should be comprehensible to users with various backgrounds, not necessarily statisticians by training. Additionally, each technique should be crafted with a specific purpose in mind. Future research should also define the statistical counterparts of adversarial and counterfactual explanations, which is neither straightforward nor immediate. This considerations can have implications for adversarial attacks and generative models as well.

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How Much Can Stratification Improve the Approximation of Shapley Values?

Patrick Kolpaczki[1(✉)], Georg Haselbeck[2], and Eyke Hüllermeier[2,3]

[1] Paderborn University, Paderborn, Germany
patrick.kolpaczki@upb.de
[2] University of Munich (LMU), Munich, Germany
[3] Munich Center for Machine Learning, Munich, Germany

Abstract. Over the last decade, the Shapley value has become one of the most widely applied tools to provide post-hoc explanations for black box models. However, its theoretically justified solution to the problem of dividing a collective benefit to the members of a group, such as features or data points, comes at a price. Without strong assumptions, the exponential number of member subsets excludes an exact calculation of the Shapley value. In search for a remedy, recent works have demonstrated the efficacy of approximations based on sampling with stratification, in which the sample space is partitioned into smaller subpopulations. The effectiveness of this technique mainly depends on the degree to which the allocation of available samples over the formed strata mirrors their unknown variances. To uncover the hypothetical potential of stratification, we investigate the gap in approximation quality caused by the lack of knowledge of the optimal allocation. Moreover, we combine recent advances to propose two state-of-the-art algorithms *Adaptive SVARM* and *Continuous Adaptive SVARM* that adjust the sample allocation on-the-fly. The potential of our approach is assessed in an empirical evaluation.

Keywords: Shapley Value · Cooperative Games · Explainable Artificial Intelligence · Feature Importance

1 Introduction

Over the last decade, machine learning models exhibited a significant increase in complexity, turning them eventually into non-transparent black boxes that seemingly resist any attempt to transfer their inner workings to a level of human comprehension. Meanwhile, developers are confronted with a recent rise in societal and legal pressure to ease understanding of their decision-making and thus provide trustworthiness, as for example the EU AI Act [1]. A common approach to deal with this rising demand is by providing post-hoc feature explanations. Additive feature explanations divide an observed numerical effect among the available features used by the applied model. This allows for interpreting the

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part assigned to a feature as its individual contribution to the effect of interest. Here, one commonly distinguishes between two feature explanation types. On the one hand, local explanations consider the model's prediction outcome for a single data point of interest as the effect to be split up, called feature attribution [20]. On the other hand, global explanations quantify the features' individual contributions to the generalization performance of the model on a chosen set of data points, also known as feature importance [8].

An established way to additively decompose an effect is by adopting a game-theoretical view. *Cooperative games* capture the spirit that features are agents or *players* which can form groups, called *coalitions*, and perform a task together, for which the group receives a numerical *reward*. Constructing this reward mechanism fittingly as the prediction value or the generalization performance elicits local, respectively global explanations. This in turn reduces the explanation task to finding an appropriate partition of the collective benefit obtained by all players cooperating together. The so far most popular solution to the problem of assigning fair payoffs is the Shapley value [29], since it is the only solution to provably fulfill desirable axioms that one would demand from such a partition. The Shapley value of a player can be understood as the weighted average of its *marginal contributions* to all coalitions, with the marginal contribution of a player to a coalition simply being the increase in received reward that the inclusion of that player causes. Unfortunately, it entails an inherent drawback. As its formula contains the rewards of all possible coalitions (feature subsets), the Shapley value's computational complexity is exponential w.r.t. the number of involved players in the game, turning quickly intractable in practice.

Current research on tackling this problem goes in two directions. By assuming a tightly restricted model type and reward mechanism, one stream of works reduces its computation to polynomial time for feature attribution [19,20]. However, these approaches are neither model-agnostic nor fruitful for feature importance. Even more importantly, the Shapley value found its way into numerous more areas, spanning from data valuation [12,16] to quantifying the contribution of base learners in ensembles [27] and neurons in deep networks [13], and well beyond machine learning such as for example economics [3]. For a broader overview in machine learning we refer to [28]. Hence it is of vital importance to tackle the problem on a more abstract and domain-independent level that allows to transfer solutions. The second stream does justice to this assessment and proposes to approximate the Shapley value with barely any underlying assumptions (see Sect. 2). Most approximation methods strike a balance between precision and approximation time by returning sample-based mean-estimates.

Among these, the technique of stratification has been employed by a number of algorithms. Stratification takes advantage of the observation that coalitions of the same size may tend to obtain similar rewards. Grouping them by size creates subpopulations, called *strata*, of higher homogeneity w.r.t. the coalitions' rewards than then population of the whole power set of players. The increase in homogeneity, or the reduction in variance in other words, speeds up sample-based mean estimation, as subestimates for each *stratum* converge faster. The key to

exploit this technique to its fullest is an allocation of available samples that prioritizes strata with higher variance. However, this comes with two hurdles. First, the optimal allocation has to be derived analytically after investigating the algorithm's precision depending on each stratum's variance and the number of samples spent on it. And second, the stratum variances are a priori unknown and can only be estimated. We call approaches oblivious to these variances with a fixed sample allocation *static*, and those that adjust their allocation during the approximation process by learning from the observed samples *adaptive*.

Contribution. Despite first refinements offered in the literature, the hypothetical potential of stratification using the optimal allocation is left unexplored. Assessing it would not only shed light on the gap that current stratifying methods have to close, but also reveal what performance improvements are to be expected at most. Moreover, adaptive methods have only been proposed for the class of approaches that sample marginal contributions, while the more recent class of sampling coalition-reward pairs *Stratified SVARM* [18] is being left untouched so far. Hence our contribution is divided into multiple parts:

- First, we reflect upon stratification for Shapley approximations and establish guiding terminology in Sects. 4 and 6.
- We derive the theoretically optimal sample allocation for the state-of-the-art *Stratified SVARM* algorithm in Sect. 5.
- By transferring and improving adaptive techniques, we propose the enhanced model-agnostic algorithms *(Continuous) Adaptive SVARM* in Sect. 7.
- Finally, we conduct an empirical evaluation of the benefit of adaptive compared to static stratification for both sampling approaches in Sect. 8.

2 Related Work

Interpreting the Shapley value as a weighted average of marginal contributions allows to also view it as an expectation of those and thus *ApproShapley* [6] approximates it by sampling marginal contributions, with further theoretical guarantees provided in [21]. Following, [21] introduced with *Stratified Sampling* the stratification by size, employing a static sample allocation over the strata. *Structured Sampling* [30] falls within this class too as it distributes the samples uniformly over all strata. The first algorithms to consider adaptive stratification were *Standard Deviation Sampling* [25] and *St-ApproShapley-opt* [5]. While the former, represents a multi-armed bandit-based philosophy, the latter chases the optimal allocation by estimating the strata's variances. A more sophisticated mechanism is employed by the *Stratified Empirical Bernstein Method* [4], which evaluates for each sample to be drawn next the most promising stratum. Further model-agnostic methods relying on the notion of marginal contributions are given in [15,22,26]. Shifting to sampling coalition-reward pairs, [7] divide the Shapley value into two sums, for which [18] propose and theoretically analyze *Stratified*

SVARM. More outstanding, yet popular is *KernelSHAP* [20]. It exploits the correspondence of the Shapley value to the solution of a weighted least square optimization problem, which it approximately constructs with each observed coalition, but does not allow to apply stratification.

3 The Shapley Value and Its Approximation

Cooperative games are formally represented by a tuple $(\mathcal{N}, \nu)$ comprising the set of players $\mathcal{N} = \{1, \ldots, n\}$, which we identify by natural numbers, and a value function $\nu : \mathcal{P}(\mathcal{N}) \to \mathbb{R}$ that assigns to each *coalition* $S \subseteq \mathcal{N}$ a real-valued *worth* $\nu(S)$. Despite its simplicity, this formalism possesses the capability of modelling countless scenarios in which human or possibly nonhuman entities can form groups to attain a collective benefit. Given the availability of the players in $\mathcal{N}$, the question of how to divide the worth $\nu(\mathcal{N})$ of the grand coalition $\mathcal{N}$ among all members in equitable manner arises. The Shapley value [29] provides a solution based on the notion of *marginal contributions*. We call the increase in worth $\Delta_i(S) := \nu(S \cup \{i\}) - \nu(S)$ caused by the inclusion of player i in presence of a coalition S its marginal contribution to S. The Shapley value of any player $i \in \mathcal{N}$ is given by a weighted average of its marginal contributions:

$$\phi_i := \sum_{S \subseteq \mathcal{N} \setminus \{i\}} \frac{1}{n \cdot \binom{n-1}{|S|}} \cdot \Delta_i(S)\,. \tag{1}$$

One can derive this formula by imposing the four axioms efficiency, symmetry, additivity, and the dummy-property (see [28] for further explanations), which capture a widely accepted intuition of fairness in the context of profit allocation. The Shapley value is provably the only solution to fulfill all of these axioms simultaneously [29]. This uniqueness arguably constitutes the key driver for its popularity in and outside of XAI. However, it comes with the major drawback of computational complexity. Without strong assumptions on the structure of ν, the exact computation of the Shapley value is NP-hard [10] because the number of subsets grows exponentially fast with the number of players n. Approximations are therefore needed to make the computation practically feasible.

We consider the *fixed-budget setting* in which all of the latent but unknown Shapley values $\phi_1, \ldots, \phi_n$ are to be approximated by estimates $\hat{\phi}_1, \ldots, \hat{\phi}_n$. The approximation algorithm is aware of $\mathcal{N}$, but has only restricted access to ν in the sense that the number of times it can evaluate the worth $\nu(S)$ of any S is limited by a fixed budget T. We judge the approximation quality by the *mean squared error* (MSE) averaged over all players:

$$\text{MSE} := \frac{1}{n} \sum_{i=1}^{n} \mathbb{E}\left[\left(\hat{\phi}_i - \phi_i\right)^2\right]. \tag{2}$$

The minimization of this measure is a widely demanded goal in the literature given its accessibility in theory as well as for empirical measurements.

4 Static Stratification

Since the Shapley value is a weighted mean of marginal contributions, it can also be seen as the expected value of a probability distribution over marginal contributions defined by the weights, thus, establishing mean estimates obtained via sampling as a natural approximation. Generally speaking, the precision of mean estimates, collected by randomly sampling from a distribution, depends on the distribution's variance. The lower the variance, the higher is the precision. Although the population to draw samples from is already fixed, *stratification* can still reduce the variance of the estimate. Instead of sampling from the whole population, one can form a partition, dividing the population into multiple subpopulations, called *strata*. By sampling from each *stratum* separately, mean estimates are obtained for all strata, and these are aggregated to the desired estimate of the whole population. This approach allows a more accurate approximation if the strata are reasonably homogeneous, or in other words, have significantly lower variances than the base population.

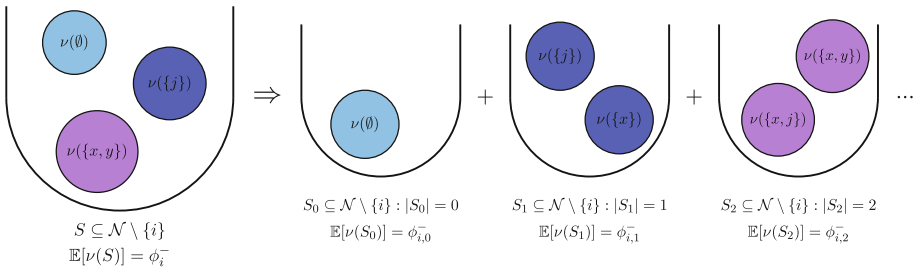

Fig. 1. Stratification at the example of Stratified SVARM: The population of coalitions $S \subseteq \mathcal{N} \setminus \{i\}$, depicted as marbles in an urn, is partitioned into multiple strata by grouping them by size. Aggregating the strata values $\phi_{i,\ell}^-$ yields the desired average ϕ_i^-. The homogeneity of each stratum increases the precision of sample-based estimates.

Stratified approaches have been proposed for two different representations of the Shapley value. First and most obviously, by viewing the individual marginal contributions of a player as elements of the population to be stratified, as first exemplified with *Stratified Sampling* by [21]. And second, [18] took advantage of a representation that dispenses with the ubiquitous notion of marginal contributions [2,7], based on averages of single coalition values. Stratifying this representation, [18] propose *Stratified SVARM*, a mechanism that updates the estimates of all players simultaneously with each sampled coalition.

We recapitulate both approaches in its simplest form proposed: *static stratification*. With this term, we coin the ignorance regarding the strata variances, which is reflected by an a priori allocation of samples over the strata. In other words, the mechanism to select the stratum to be sampled from next is oblivious

of the observations made and determined before the approximation task itself. We assume to sample with replacement as done by most methods for the sake of simplicity. All proofs of our theoretical results are given in Appendix B.

4.1 Stratified Sampling

Stratified Sampling proposed in [21] is the first algorithm to approximate a player's Shapley value based on stratification. The marginal contributions $\Delta_i(S)$ of a fixed player i are grouped by the cardinality of S into n many strata. Fittingly, the definition of the Shapley value uses this partition also from an algebraic view, suggesting a convenient interpretation. For any $\ell \in \mathcal{L}_\Delta := \{0, \ldots, n-1\}$, we define the ℓ-th stratum value of player i as the average of its marginal contributions to all coalitions of size ℓ, i.e.

$$\phi_{i,\ell} := \frac{1}{\binom{n-1}{\ell}} \sum_{\substack{S \subseteq \mathcal{N} \setminus \{i\} \\ |S| = \ell}} \Delta_i(S)\,. \tag{3}$$

Next, by building upon this notion, we retrieve the Shapley value as the uniform average of the n many strata values:

$$\phi_i = \frac{1}{n} \sum_{\ell=0}^{n-1} \phi_{i,\ell}\,. \tag{4}$$

The sampling procedure maintains an estimate $\hat{\phi}_{i,\ell}$ for each stratum, given by the empirical mean of the independently sampled marginal contributions from the according stratum. Note that each sampled marginal contribution $\Delta_i(S)$ consumes two budget tokens as both values $\nu(S)$ and $\nu(S \cup \{i\})$ need to be evaluated. The stratum value estimates are aggregated in the same manner as the true values in order to obtain the final Shapley value estimate $\hat{\phi}_i$.

Since each player has its own population of marginal contributions, we are dealing with n separate approximation problems. In [21], the authors do not specify how to approximate the Shapley values of all players, though one can simply divide the available budget T among them and repeat the procedure for each player. We denote by $m_{i,\ell}$ the number of sampled marginal contributions from player i's ℓ-th stratum after the algorithm's termination and call $(m_{i,\ell})_{i \in \mathcal{N}, \ell \in \mathcal{L}_\Delta}$ the *sample allocation*, capturing the sample numbers of all n^2 many strata. Worth mentioning is the possibility that the sample numbers do not have to be fixed beforehand. Instead, Stratified Sampling employs a *static* sample allocation: each $m_{i,\ell}$ is fixed upon initialization. Given the variance $\sigma^2_{i,\ell} := \mathbb{V}[\Delta_i(S)]$ w.r.t. the uniform probability distribution over all coalitions $S \subseteq \mathcal{N} \setminus \{i\}$ of size ℓ we can express the achieved MSE.

Theorem 1. *The mean squared error of Stratified Sampling using any static sample allocation* $(m_{i,\ell})_{i \in \mathcal{N}, \ell \in \mathcal{L}_\Delta}$ *with* $m_{i,\ell} \geq 1$ *for all* $i \in \mathcal{N}$ *and* $\ell \in \mathcal{L}_\Delta$ *is*

$$\frac{1}{n} \sum_{i=1}^{n} \mathbb{E}\left[\left(\hat{\phi}_i - \phi_i\right)^2\right] = \frac{1}{n^3} \sum_{i=1}^{n} \sum_{\ell=0}^{n-1} \frac{\sigma^2_{i,\ell}}{m_{i,\ell}}\,.$$

The condition $m_{i,\ell} \geq 1$ for all strata implies that each stratum gets assigned at least one sample which is necessary to avoid any bias of the final estimate $\hat{\phi}_i$, i.e. $\mathbb{E}[\hat{\phi}_i] = \phi_i$. In fact, under this condition even the strata estimates themselves are unbiased. For technical reasons, [21] specifies an allocation that increases with coalition size, irrelevant for our objective of MSE minimization. Instead, one could simply distribute the budget uniformly over all players and strata by assigning $m_{i,\ell} = \frac{T}{2n^2}$, leading to an MSE of

$$\frac{1}{n}\sum_{i=1}^{n}\mathbb{E}\left[\left(\hat{\phi}_i - \phi_i\right)^2\right] = \frac{2}{nT}\sum_{i=1}^{n}\sum_{\ell=0}^{n-1}\sigma_{i,\ell}^2\,. \tag{5}$$

For the sake of simplicity, we assume that the assigned fraction to $m_{i,\ell}$ is a natural number and save ourselves the effort of rounding the sample allocation otherwise. Even further, one can save some budget by taking into account that the strata $\ell = 0$ and $\ell = n-1$ contain exactly one marginal contribution, reducing the variances $\sigma_{i,0}^2$ and $\sigma_{i,n-1}^2$ to zero for all i. The shared coalition values $\nu(\emptyset)$ and $\nu(\mathcal{N})$ can be reused saving a budget of $2n-2$. Hence, one could set $m_{i,0} = m_{i,n-1}$ to 1 for all players and split the remaining budget evenly, i.e. $m_{i,\ell} = \frac{\tilde{T}}{2n(n-2)}$ for all $\ell \in \mathcal{L}'_\Delta := \{1, \ldots, n-2\}$ with $\tilde{T} := T - 2n - 2$.

4.2 Stratified SVARM

Departing from marginal contributions, the second representation suitable for stratification separates the Shapley value into two sums ϕ_i^+ and ϕ_i^-:

$$\phi_i = \underbrace{\sum_{S\subseteq\mathcal{N}\setminus\{i\}} \frac{1}{n\cdot\binom{n-1}{|S|}}\nu(S\cup\{i\})}_{=:\phi_i^+} - \underbrace{\sum_{S\subseteq\mathcal{N}\setminus\{i\}} \frac{1}{n\cdot\binom{n-1}{|S|}}\nu(S)}_{=:\phi_i^-}\,. \tag{6}$$

Instead of approximating each sum separately, both weighted averages of coalition values are further stratified by the size of S in [18], illustrated in Fig. 1. For each player i and size $\ell \in \mathcal{L}_\Delta$, the following strata values are thus obtained:

$$\phi_{i,\ell}^+ := \frac{1}{\binom{n-1}{\ell}}\sum_{\substack{S\subseteq\mathcal{N}_i\\|S|=\ell}}\nu(S\cup\{i\}) \quad \text{and} \quad \phi_{i,\ell}^- := \frac{1}{\binom{n-1}{\ell}}\sum_{\substack{S\subseteq\mathcal{N}_i\\|S|=\ell}}\nu(S)\,. \tag{7}$$

The interpretation is simple and appealing: Each stratum value $\phi_{i,\ell}^+$ is the uniform average of all coalitions of size $\ell+1$ that include i. An analogous statement can be made for $\phi_{i,\ell}^-$. The Shapley value can then be written as

$$\phi_i = \frac{1}{n}\sum_{i=0}^{n-1}\phi_{i,\ell}^+ - \frac{1}{n}\sum_{i=0}^{n-1}\phi_{i,\ell}^-\,. \tag{8}$$

Update Mechanism. In contrast to *Stratified Sampling*, which samples from each stratum separately, *Stratified SVARM* reuses each sampled coalition A to update at least one stratum estimate $\hat{\phi}^+_{i,\ell}$ or $\hat{\phi}^-_{i,\ell}$ of each player, thereby getting the most out of the information observed. This is made possible by the observation that for any player i and any player subset A, $\nu(A)$ is either a part of $\phi^+_{i,|A|-1}$ if $i \in A$, or otherwise a part of $\phi^-_{i,|A|}$ if $i \notin A$. As a consequence, this effectively reduces the number of strata to sample from to $n+1$, one for each subset size, captured by $\mathcal{L}_\nu := \{0, \ldots, n\}$. In each time step a subset size to sample from is chosen and the sampled coalition of that size is used for the update mechanism. Note that in comparison to the sampling of marginal contributions, here each sample only consumes one budget token instead of two.

Warmup. Similar to Sect. 4.1 we define $m^+_{i,\ell}$ and $m^-_{i,\ell}$ as the number of sampled coalitions used to update $\phi^+_{i,\ell}$, or $\hat{\phi}^-_{i,\ell}$ respectively, for each $i \in \mathcal{N}$ and $\ell \in \mathcal{L}_\nu$. As we proposed in Sect. 4.1 for Stratified Sampling, [18] made use of the fact that some strata contain only a few coalitions. Stratified SVARM computes the strata values $\phi^-_{i,0}, \phi^+_{i,0}, \phi^-_{i,1}, \phi^+_{i,n-2}, \phi^-_{i,n-1}, \phi^+_{i,n-1}$ exactly for all players by evaluating all coalitions of size 0, 1, $n-1$, and n at the price of $2n+2$ budget tokens. This reduces the number of subset sizes to choose from to $n-3$. To guarantee unbiasedness, [18] introduce a warmup procedure preceding the sampling such that each of the remaining $2n^2 - 2n$ strata is covered by at least one sample, i.e. $m^+_{i,\ell}, m^-_{i,\ell} \geq 1$. It consumes a budget of $W := 2\sum_{s=2}^{n-2} \lceil \frac{n}{s} \rceil$ and we denote the remaining budget left for sampling as $\bar{T} := W + 2n + 2$.

Further, we let m_ℓ be the number of sampled coalitions of size ℓ after the warmup for each $\ell \in \mathcal{L}'_\nu := \{2, \ldots, n-2\}$. Note that although the numbers $m^+_{i,\ell}$ and $m^-_{i,\ell}$ are determining the approximation quality, the algorithm is only fixing an allocation on the level of the subset sizes, i.e. $(m_\ell)_{\ell \in \mathcal{L}'_\nu}$. Here, again, given a static sample allocation and the variances $\sigma^2_{i,\ell,+} := \mathbb{V}[\nu(S \cup \{i\})]$ and $\sigma^2_{i,\ell,-} := \mathbb{V}[\nu(S)]$ w.r.t. the uniform probability distribution over all coalitions $S \subseteq \mathcal{N} \setminus \{i\}$ of size ℓ, we can express a bound on the MSE.

Theorem 2. *The mean squared error of Stratified SVARM using any static sample allocation $(m_\ell)_{\ell \in \mathcal{L}'_\nu}$ is bounded by*

$$\frac{1}{n}\sum_{i=1}^{n} \mathbb{E}\left[\left(\hat{\phi}_i - \phi_i\right)^2\right] \leq \frac{1}{n^2}\sum_{\ell=2}^{n-2}\frac{1}{m_\ell}\sum_{i=1}^{n}\frac{\sigma^2_{i,\ell-1,+}}{\ell} + \frac{\sigma^2_{i,\ell,-}}{n-\ell}\,.$$

A subtle difference is that [18] selects the coalition size randomly according to a fixed probability distribution. The expected sample allocation can be interpreted as the chosen static allocation since it is not influenced by the observations made during sampling. Their proposed distribution is sophisticated, prioritizing subset sizes close to zero and close to n at the cost of those further in the middle of the spectrum in order to achieve a bound that equally weighs all strata variances. If

we instead split the remaining budget equally such that $m_\ell = \frac{\bar{T}}{n-3}$, we obtain

$$\frac{1}{n}\sum_{i=1}^{n}\mathbb{E}\left[\left(\hat{\phi}_i - \phi_i\right)^2\right] \leq \frac{n-3}{n^2\bar{T}}\sum_{i=1}^{n}\sum_{\ell=2}^{n-2}\frac{\sigma_{i,\ell-1,+}^2}{\ell} + \frac{\sigma_{i,\ell,-}^2}{n-\ell}\,. \tag{9}$$

We demonstrate the sharpness of these bounds in Sect. 8.

5 Theoretically Optimal Allocation

The results shown above in Theorems 1 and 2 give rise to the question of how small the MSE can possibly be, and motivate the search for the responsible *optimal* sample allocation. For both sampling approaches *Stratified Sampling* and *Stratified SVARM*, the impact of a stratum's variance on the MSE is directly linked to its sample number. Hence, it comes quite naturally to fine-tune the sample allocation by adjusting it to the underlying variances. Obviously, these variances are unknown such that the resulting MSE can only be achieved in theory. However, this investigation will yield important insights, as we uncover the theoretical limit on the MSE for each approach that no static sample allocation can improve upon, thus showcasing the theoretical potential of stratification.

5.1 Optimal Stratified Sampling

In the pursuit of the optimal allocation, we minimize the MSE given in Theorem 1, while constraining the total number of evaluations according to the given budget. This can be formulated as the following optimization problem:

$$\begin{aligned}(m_{i,\ell}^*)_{i\in\mathcal{N},\ell\in\mathcal{L}'_\Delta} = \underset{(m_{i,\ell})_{i\in\mathcal{N},\ell\in\mathcal{L}'_\Delta}}{\arg\min} \quad & \frac{1}{n^3}\sum_{i=1}^{n}\sum_{\ell=1}^{n-2}\frac{\sigma_{i,\ell}^2}{m_{i,\ell}} \\ \text{s.t.} \quad & 2\sum_{i=1}^{n}\sum_{\ell=1}^{n-2} m_{i,\ell} = \tilde{T} \\ & m_{i,\ell} \in \mathbb{N} \qquad\qquad \forall i \in \mathcal{N}, \ell \in \mathcal{L}'_\Delta\end{aligned}$$

In order to allow for a fair comparison with *Stratified SVARM*, we assume that the strata $\ell = 0$ and $\ell = n-1$ are already computed exactly (see Sect. 4.1). While the constraint on the sample numbers reflects the fact that no fractions of samples can be taken, it also impedes the attempt to derive an analytical solution of the problem. For this reason, we allow non-negative real-valued numbers, i.e. substituting it by $m_{i,\ell} \in \mathbb{R}_{>0}\ \forall i \in \mathcal{N}, \ell \in \mathcal{L}$, and call the resulting optimization problem the *relaxed sample allocation problem for marginal contributions*.

Theorem 3. *The solution to the relaxed sample allocation problem for marginal contributions is given by the allocation*

$$m_{i,\ell}^* = \frac{\sigma_{i,\ell}}{2\sum_{j=1}^{n}\sum_{k=1}^{n-2}\sigma_{j,k}} \cdot \tilde{T} \quad \text{for all } i \in \mathcal{N}, \ell \in \mathcal{L}'_\Delta\,,$$

which yields the following mean squared error for Stratified Sampling:

$$\frac{1}{n}\sum_{i=1}^{n}\mathbb{E}\left[\left(\hat{\phi}_i-\phi_i\right)^2\right]=\frac{2}{n^3\bar{T}}\left(\sum_{i=1}^{n}\sum_{\ell=1}^{n-2}\sigma_{i,\ell}\right)^2 .$$

Theorem 3 reveals that the optimal allocation partitions the budget among the strata in proportion to their fraction of the total sum of stratum standard deviations. This result is also known as the *Neyman allocation* [24]. It was discovered previously by [5] to improve *Stratified Sampling*, though used incorrectly by replacing the standard deviations with the variances. Some sample numbers might fall below 1, although at least one sample is needed per stratum. We fix this by rounding the sample allocation appropriately without violating the constraint.

5.2 Optimal Stratified SVARM

We now transfer the optimization problem in search for the optimal allocation from *Stratified Sampling* to *Stratified SVARM*. The objective function to be minimized is our result on the MSE in Theorem 2. Although this is only an upper bound, it does not significantly harm the meaningfulness of the solution to be derived, since the inequality stems from a minor technical detail introduced for the sake of readability. As a constraint, we impose again that the sum of samples drawn has to equal the budget left after the warmup:

$$\begin{aligned}(m_\ell^*)_{\ell\in\mathcal{L}'_\nu}=\min_{(m_\ell)_{\ell\in\mathcal{L}'_\nu}}\quad & \frac{1}{n^2}\sum_{\ell=2}^{n-2}\frac{1}{m_\ell}\sum_{i=1}^{n}\frac{\sigma_{i,\ell-1,+}^2}{\ell}+\frac{\sigma_{i,\ell,-}^2}{n-\ell}\\ \text{s.t.}\quad & \sum_{\ell=2}^{n-2}m_\ell=\bar{T}\\ & m_\ell\in\mathbb{N}\qquad\qquad\forall\ell\in\mathcal{L}'_\nu\end{aligned}$$

Again, we relax this optimization problem by allowing the sample numbers to be real-valued but non-negative, i.e. $m_\ell\in\mathbb{R}_{\geq 0}\ \forall\ell\in\mathcal{L}'_\nu$. We name this relaxation the *relaxed sample allocation problem for coalitions*.

Theorem 4. *The solution to the relaxed sample allocation problem for coalitions is given by the allocation*

$$m_\ell^*=\frac{\sqrt{\sum_{i=1}^{n}\frac{\sigma_{i,\ell-1,+}^2}{\ell}+\frac{\sigma_{i,\ell,-}^2}{n-\ell}}}{\sum_{k=2}^{n-2}\sqrt{\sum_{i=1}^{n}\frac{\sigma_{i,k-1,+}^2}{k}+\frac{\sigma_{i,k,-}^2}{n-k}}}\cdot\bar{T}\qquad\text{for all } i\in\mathcal{N},\ell\in\mathcal{L}'_\nu\,,$$

which yields the following mean squared error for Stratified SVARM:

$$\frac{1}{n}\sum_{i=1}^{n}\mathbb{E}\left[\left(\hat{\phi}_i-\phi_i\right)^2\right]\leq\frac{1}{n^2\bar{T}}\left(\sum_{\ell=2}^{n-2}\sqrt{\sum_{i=1}^{n}\frac{\sigma_{i,\ell-1,+}^2}{\ell}+\frac{\sigma_{i,\ell,-}^2}{n-\ell}}\right)^2 .$$

Similarly to Theorem 3, the optimal allocation for *Stratified SVARM* assigns each coalition size ℓ a number of samples that depends on the proportion its associated stratum variances contribute to the MSE bound. In contrast to the Neyman allocation, the update mechanism dilutes the relationship between strata and the sample numbers on the level of coalition sizes.

We dispense with an analytical comparison of the MSE given in Theorems 3 and 4 because the different variance notions of marginal contributions $\sigma^2_{i,\ell}$ and coalition values $\sigma^2_{\ell,+/-}$ of a specific game decide which approach offers the better approximation potential. Instead, we compare both optimal allocations empirically in Sect. 8. For the same reason, we consider asymptotic notation as inappropriate since it conceals the effect of small stratum variances, thus doing injustice to the core idea of stratification. Worth mentioning is that if all standard deviations are equal, both optimal allocations lead to the same MSE as the uniform allocations proposed in Sect. 4. The degree to which the former improve upon the latter increases with the variability of the standard deviations.

6 Adaptive Stratification

It comes at no surprise that the MSE achieved by *Stratified Sampling* and *Stratified SVARM* employing their respective optimal allocation (see Sect. 5) is not applicable in practice, since the stratum variances are unknown during approximation. Fortunately, overcoming this lack of information by estimating the stratum variances with the means of the observed samples poses a promising remedy, as it enables the adaptation of the sampling allocation during the approximation process itself. We call this *adaptive stratification*. In contrast to *static stratification*, the mechanism to select the stratum to be sampled from next is now *informed* and utilizes this (possibly inexact) knowledge.

Pursuing this idea, all adaptive methods are confronted with the *exploration-exploitation dilemma*. While adjusting the sample allocation according to the obtained knowledge about the strata variances promises to achieve a more accurate approximation, excessively exploiting it can even be harmful. Since it is based on the variances' estimates, it might lead to a poorly performing allocation if these are not approximated precisely enough. Hence, it is of vital importance to explore, i.e., to collect samples from strata with apparently low variances to ensure a convergence to the optimal allocation. Otherwise, the algorithm might be trapped in sampling too often from falsely assessed strata of high variance, without having the chance to correct its estimated optimal allocation by sampling more often from other strata. Performing exploration and exploitation to the right degree is key to successful adaptive stratification. We shortly recapitulate on current adaptive methods for *Stratified Sampling* in the following.

6.1 The Two-Phase Approach

One way to tackle the exploration-exploitation dilemma is by dividing both into two separate phases. The *Two-Staged-St-ApproShapley-opt* algorithm [5]

follows this motive as it samples in the first phase uniformly from all strata for exploration, and switches to exploitation in the second phase. This is done by estimating the optimal allocation w.r.t. the available budget T on the basis of Theorem 3 in combination with the observed strata variances in the exploration phase. Next, the algorithm calculates for each stratum the number of samples to be drawn in the exploitation phase such that the sum of both phases matches the estimated optimal allocation. Note that this is not guaranteed to be feasible, since sample numbers may already exceed their counterparts within the optimal allocation during exploration. Shortening the exploration phase reduces this risk, but unfortunately, the quality of the strata estimates as well. This concern is not dealt with in [5], where both phases are simply set to consume half of the total budget. We observe that the algorithm wrongly deviates from the Neyman allocation by considering the stratas' variances instead of their standard deviations.

6.2 Bandit-Based Approach

Instead of demanding a strict separation, one can transition from exploration towards exploitation in a more seamless manner. This paradigm is employed for the approximation of the Shapley value of a single player by *Standard Deviation Sampling* [25]. Instead of determining the sample numbers upfront it employs a probability distribution over the strata to select the next draw. The closer this distribution is to the proportions of the Newman allocation, the more exploitation is performed on average. On the other side, full exploration can be modelled by choosing a uniform distribution. The mix of both is achieved via a convex combination with $\epsilon(t) \in [0, 1]$ being the weight for the uniform distribution at timestep $t \in \{1, \ldots, T\}$, and respectively $1 - \epsilon(t)$ for the estimated optimal allocation. The degree of freedom to design the rate of the transition lies with the function ϵ. In [25], a sigmoid function with two parameters is chosen for controlling the percentage of exploration and the transition speed. Obviously, these are hyperparameters to be specified by the practitioner relying on domain-knowledge. In light of our aim to remain model-agnostic, this poses a considerable vulnerability to the robustness of the method.

6.3 Empirical Bernstein Bound

Continuing in the spirit of simultaneously performing exploration and exploitation, the *Stratified Empirical Bernstein Method* [4] combines both in an even more interwoven way. It relies on the Stratified Empirical Bernstein Bound [4], which bounds the probability that an estimate $\hat{\phi}_i$ deviates from ϕ_i by more than some specific degree. The innovation of the algorithm is to greedily select in each timestep the stratum that promises the highest reduction on the deviation probability given by the bound. As the actual sample numbers are also part of the considered degree of the deviation, the compromise between exploration and exploitation is steered automatically by the bound itself without further algorithmic intervention. In contrast to *Standard Deviation Sampling*, it comes with

a different flaw that impedes any effective usage in the absence of rich domain-knowledge: The bound assumes the ranges of all strata to be given, a requirement that is hard to fulfill for cooperative games stemming from explanation tasks.

7 Adaptive Stratification of SVARM

The previously presented techniques are only introduced for the class of methods which sample marginal contributions. Hence, we close this gap in the current literature by applying the two-phased approach to *Stratified SVARM*, as we consider it to be the only adaptive technique without critical deficiencies. This transfer yields *Adaptive SVARM*, and *Continuous Adaptive SVARM* in combination with a conceptual improvement, which we propose as new model-agnostic approximation algorithms for the Shapley value.

7.1 Adaptive SVARM

We present our algorithm (see Algorithm 1) formally building upon *Stratified SVARM* (see Sect. 4.2) and our derived optimal allocation in Sect. 5.2. The pseudocode of the subprocedures is given in Appendix A.

Warmup. To begin with, all coalitions of the sizes $0, 1, n-1, n$ are evaluated to compute all strata values $\phi^-_{i,0}, \phi^-_{i,1}, \phi^-_{i,n-1}, \phi^+_{i,0}, \phi^+_{i,n-2}, \phi^+_{i,n-1}$ exactly, captured by `ExactCalculation` (see Algorithm 4). Hence, *Adaptive SVARM* also maintains stratum estimates $\hat{\phi}^+_{i,\ell}$ for each $i \in \mathcal{N}$ and $\ell \in \mathcal{L}'_\nu$. Next, we keep valid estimates $\hat{\sigma}^2_{i,\ell,+/-}$ of the variances by ensuring to have observed at least two samples from each stratum. This is achieved by calling each procedure `WarmUp`$^+$ and `WarmUp`$^-$ (cf. Algorithm 6 and 7 in [18]) twice. In total this procedure leaves $\bar{T} := T - 2n - 2 - 2W$ budget tokens for the remaining phases.

Exploration Phase. In each timestep the size $s \in \{2, \ldots, n-2\}$ of the next coalition A to sample is determined in Round-robin fashion to mimic the uniform allocation. Afterwards, $A \subseteq \mathcal{N}$ is drawn u.a.r. The `Update` procedure (see Algorithm 3) evaluates the worth of A once and incrementally updates the stratum estimates $\hat{\phi}^{+/-}_{i,\ell}$ using the sample counters $c^{+/-}_{i,\ell}$. The computation of the variance estimates and $\hat{\sigma}^2_{i,\ell,+/-}$ is prepared by maintaining the sum of observed coalition values $\Sigma^{+/-}_{i,\ell}$ and the sum of squared values $\Omega^{+/-}_{i,\ell}$. We specify the length of the exploration phase by a parameter $\lambda \in [0, 1]$, representing the percentage of time steps it consumes of the available budget $\bar{T}$ after the warmup. Subsequently, the phase stretches over $\lambda\bar{T}$ many time steps. Assuming the latter to be a multiple of the number of remaining sizes $n-3$, each size is selected $\frac{\lambda\bar{T}}{n-3}$ many times.

Exploitation Phase. After completing the exploration phase, *Adaptive SVARM* has $(1-\lambda)\bar{T}$ many samples left for the exploitation phase, in which the previously employed uniform allocation is extended to the estimated optimal allocation over the budget $\bar{T}$. The procedure `CalculateAllocation` (see Algorithm 2) computes

Algorithm 1 Adaptive SVARM

Input: $\mathcal{N}$, $T \in \mathbb{N}, \lambda \in [0,1]$

$\hat{\phi}^+_{i,\ell}, \hat{\phi}^-_{i,\ell}, \hat{\sigma}^2_{i,\ell,+}, \hat{\sigma}^2_{i,\ell,-} \leftarrow 0$ for all $i \in \mathcal{N}, \ell \in \mathcal{L}_\nu$
$c^+_{i,\ell}, c^-_{i,\ell}, \Sigma^+_{i,\ell}, \Sigma^-_{i,\ell}, \Omega^+_{i,\ell}, \Omega^-_{i,\ell} \leftarrow 0$ for all $i \in \mathcal{N}, \ell \in \mathcal{L}_\nu$
$c_\ell \leftarrow 0$ for all $\ell \in \mathcal{L}_\nu$
ExactCalculation
2×WarmUp$^+$; 2×WarmUp$^-$
$\bar{T} \leftarrow T - 2n - 2 - 2W$
for $t = 1, \ldots, \lambda\bar{T}$ **do**
  $s \leftarrow (t - 1 \mod n - 3) + 2$
  Draw A from $\{S \subseteq \mathcal{N} \mid |S| = s\}$ uniformly at random
  Update(A)
end for
$(\hat{m}^*_\ell)_{\ell \in \{2,\ldots,n-2\}} \leftarrow$ CalculateAllocation
$t \leftarrow 1$
for $t \leq 1, \ldots, (1-\lambda)\bar{T}$ **do**
  $s \leftarrow \arg\min_{\ell \in \mathcal{L}'_\nu} \frac{c_s}{\hat{m}^*_s}$
  Draw A from $\{S \subseteq \mathcal{N} \mid |S| = s\}$ uniformly at random
  Update(A)
end for
$\hat{\phi}_i \leftarrow \frac{1}{n} \sum_{\ell=0}^{n-1} \hat{\phi}^+_{i,\ell} - \hat{\phi}^-_{i,\ell}$ for all $i \in \mathcal{N}$

Output: $\hat{\phi}_1, \ldots, \hat{\phi}_n$

the variance estimates on the basis of $\Sigma^{+/-}_{i,\ell}$ and $\Omega^{+/-}_{i,\ell}$, and plugs them into Theorem 4 to obtain the estimated optimal allocation:

$$\hat{m}^*_\ell = \frac{\sqrt{\sum_{i=1}^{n} \frac{\hat{\sigma}^2_{i,\ell-1,+}}{\ell} + \frac{\hat{\sigma}^2_{i,\ell,-}}{n-\ell}}}{\sum_{k=2}^{n-2} \sqrt{\sum_{i=1}^{n} \frac{\hat{\sigma}^2_{i,k-1,+}}{k} + \frac{\hat{\sigma}^2_{i,k,-}}{n-k}}} \cdot \bar{T} \qquad \text{for all } i \in \mathcal{N}, \ell \in \mathcal{L}'_\nu\,, \tag{10}$$

where $\hat{m}^*_\ell$ denotes the number of coalitions of size ℓ to be drawn in total in both phases. Let m^{explore}_ℓ and m^{exploit}_ℓ be the number of coalitions of size ℓ drawn during exploration, and exploitation respectively. Then, ideally, the sample number during exploitation is given by $m^{\text{exploit}}_\ell := \hat{m}^*_\ell - m^{\text{explore}}_\ell$. This is not necessarily feasible since one can not exclude $\hat{m}^*_\ell < m^{\text{explore}}_\ell$. In this case, m^{exploit}_ℓ is set to zero and after this altering of the allocation one has to adjust the other samples whose sum exceeds now $\bar{T}$. We perform this by excluding this coalition size from Eq. (10) and compute the allocation again, e.g., the summation in the denominator is taken without the affected size ℓ, while the budget is set to $\bar{T} - m^{\text{explore}}_\ell$. The intuition is to neglect the size ℓ, as more samples have been collected than demanded by the optimal allocation, and solve the optimization problem in Sect. 5.2 again with only the total budget available for the remaining sizes. Since $\hat{m}^*_\ell < m^{\text{explore}}_\ell$ can occur in the newly assigned allocation, this

procedure has to be repeated eventually. However, it is guaranteed to terminate due to n being finite. During each of the remaining $(1-\lambda)\bar{T}$ timesteps, the algorithm proceeds similarly to the exploration phase. With the purpose of sampling evenly, the only exception is that the next size s is chosen as the one with the lowest number of current samples c_s relative to its assignment $\hat{m}_s^*$. Finally, the stratum estimates are aggregated to obtain an estimate $\hat{\phi}_i$ for each player.

7.2 Continuous Adaptive SVARM

We find that the two-phase approach as presented so far does not fully exploit the collected information to estimate the optimal allocation as precisely as possible. While it derives variance estimates during the exploration phase, the observations during the exploitation phase remain untouched. Since these posses further useful information, we propose to update the estimated optimal sample allocation continuously during exploitation. The resulting algorithm *Continuous Adaptive SVARM* poses a straightforward improvement by additionally calling `CalculateAllocation` after each update (line 15) in the exploitation phase.

8 Empirical Results

In order to complement our theoretical work, we empirically compare the approximation qualities depending on the spent budget, assess the algorithms' efficacy, answer whether adaptive stratification improves upon its static counterpart, and quantify the remaining gap of *Stratified SVARM* variants to the theoretical optimum derived in Sect. 5.2. In particular, we are interested in a comparison between *ApproShapley* as a baseline of sampling marginal contributions without stratification, the class of algorithms sampling marginal contributions, including *Stratified Sampling* and its extension *Two-Staged-St-ApproShapley-opt* [5] (Adaptive Sampling), and on the contrary *Stratified SVARM* in combination with our proposed improvements *Adaptive SVARM* and *Continuous Adaptive SVARM*. In addition, we include for each class the optimal algorithm, that a priori knows all stratum variances, given by *Optimal Sampling* and *Optimal SVARM*. We consider cooperative games of different origin and structure: feature explanations (local and global) for real-world data and synthetic games.

Table 1. Used datasets and models for feature explanation tasks.

Type	Dataset	Features	Task	Model
Global	Adult census [17]	14	Classification	Random forest
Global	Bank Marketing [23]	16	Classification	Random forest
Global Local	Bike sharing [11]	12	Regression	Random forest XGBoost
Global	German credit [14]	20	Classification	Random forest
Local	ImageNet [9]	14	Classification	ResNet18

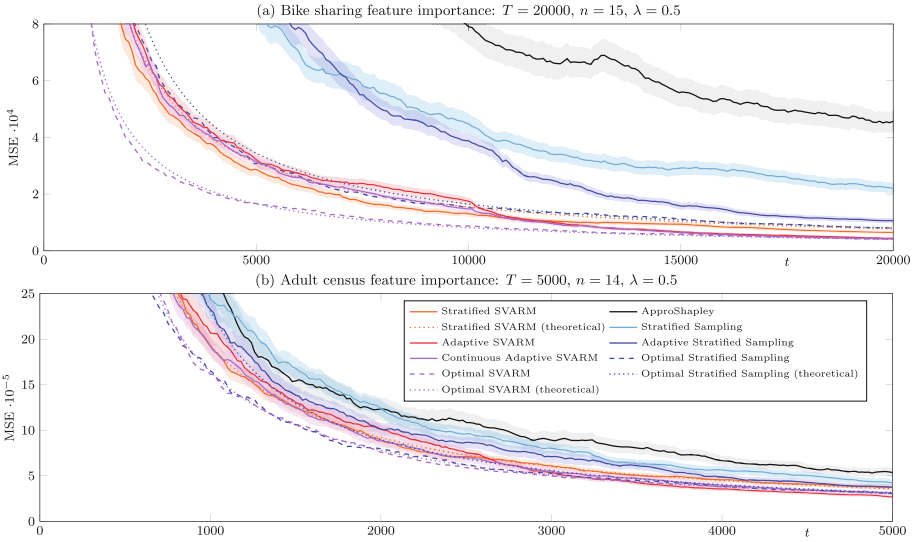

Fig. 2. Averaged MSE and standard error over 50 repetitions during approximation depending on current timestep t for global explanations. All adaptive algorithms use $\lambda = 0.5$. The performance of the hypothetical optimal algorithms (see Sect. 5) and theoretical bounds for Stratified SVARM (Eq. (9)), Optimal SVARM (Theorem 4), and Optimal Stratified Sampling (Theorem 3) are included.

Feature Importance Games. Given a model and dataset, we construct a cooperative game by setting the value function of a feature subset to be the classification accuracy (or R^2 for regression tasks) of the model on a test set (30% of datapoints). For each coalition S the model is fitted on the training set (70% of datapoints) with only the features contained in S. We use a default *sklearn* random forest consisting of 20 trees on all datasets given in Table 1.

Feature Attribution Games. For a model's prediction $h(x)$ on a specified datapoint x, the worth $\nu(S)$ of a feature subset is set to be the difference $h(x_S) - h(x_\emptyset)$, where $h(x_S)$ denotes the prediction with all features $\mathcal{N} \setminus S$ removed from x. For classification we take the class confidence $h_c(x_S) \in [0, 1]$ with $h(x) = c$ instead of the class label. Feature values in the Bike sharing dataset are removed by substituting them with their mean (or mode if categorical) within the dataset. For ImageNet pictures, semantic superpixels to be removed are replaced by their mean color. The used models and datasets are given in Table 1.

Synthetic Games. The experiments based on explanation tasks are limited to relatively low feature numbers because the Shapley values have to be computed in order to track the approximation error. Synthetic games [5,6,18] provide a possibility to investigate approximation algorithms for large player numbers n because their structure elicits a closed-form polynomial expression of the Shapley value.

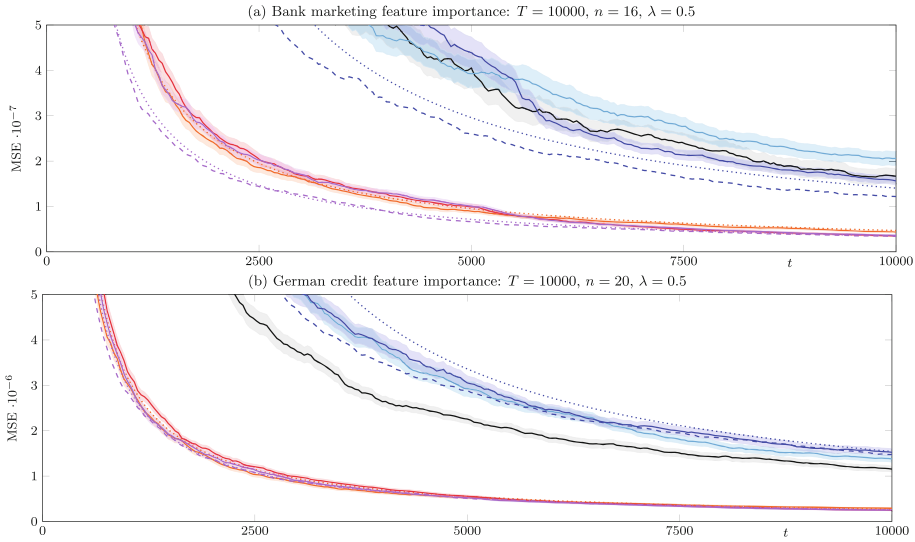

Fig. 3. Averaged MSE and std. error over 50 repetitions during approximation depending on current t for global explanations. For legend and further details see Fig. 2.

Our results show a significant gap between *Stratified Sampling* and its theoretical optimum across all datasets except for Fig. 3(b). Although not closing it completely, Adaptive SVARM reduces it visibly in Fig. 2(a) and Fig. 3(a) after switching to its exploitation phase. However, we also encounter cases in which it fails to have a positive effect on the approximation quality or performs even worse than the baseline *ApproShapley*. Interestingly, the deficit of the optimum to *ApproShapley* in Fig. 3(b) shows that stratifying marginal contributions can even be counterproductive to the goal of precise approximation.

On the other hand, the class of *Stratified SVARM* and our proposed improvements effortlessly outperform its counterpart based on marginal contributions except for feature attribution for regression tasks (see Fig. 4(a)). The optimality gap of *Stratified SVARM* is smaller or even nonexistent in Fig. 3(b). Independent of the gap's extent both of our algorithms close it after sufficient time spent in their exploitation phase. Hence, they not only improve upon state-of-the-art *Stratified SVARM*, but also lead us to conclude that we have reached optimality for this class of stratifying methods. To our surprise, we observe no considerable difference between *Adaptive SVARM* and *Continuous Adaptive SVARM* with $\lambda = 0.5$, implying that the variance estimates are precise enough and that λ can be further decreased. *Adaptive SVARM* reduces the approximation error at termination by (a) 28% and (b) 33% in Fig. 2, 18% in Fig. 4(b), and 70% in Fig. 5(b) compared to *Stratified SVARM*. Finally, the alignment of our theoretical statements with the empirical results, as especially exemplified for *Stratified SVARM* and Optimal SVARM, gives evidence for the precision of our analysis in Sect. 5 and the appropriateness of our approach.

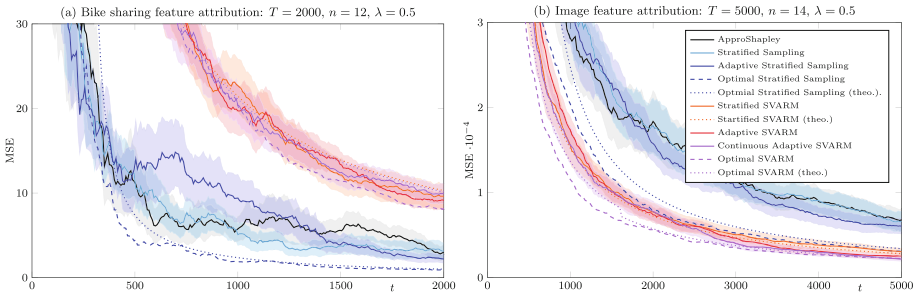

Fig. 4. Averaged MSE and standard error over 50 repetitions during approximation depending on current t for local explanations. All adaptive algorithms use $\lambda = 0.5$.

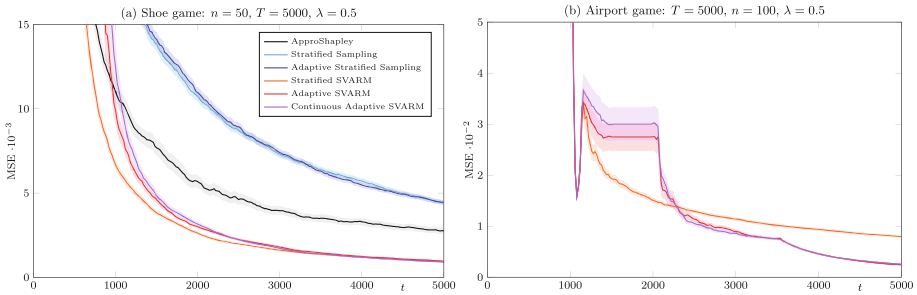

Fig. 5. Averaged MSE and standard error over 50 repetitions during approximation depending on current t for synthetic games. For further game details see [6, 18].

9 Conclusion

We categorized stratified approaches for the approximation of all player's Shapley values in the fixed-budget setting by distinguishing between static and adaptive stratification. Recognizing the lack of more involved techniques for the class of methods that sample coalitions instead of marginals, we derived analytically the optimal allocation for *Stratified SVARM*. Moreover, we transferred the two-phase approach resulting in two novel approximation algorithms. Our empirical evaluation provides a multi-faceted insight. The gap of static methods to the theoretical optimum is of varies depending on the considered game. *Adaptive SVARM* and *Continuous Adaptive SVARM* close that gap and reach class optimality during approximation. The quadratically growing number of strata poses an inherent drawback of stratification for both classes. It prolongs the warmup, increases space complexity, and reduces the number of available samples per stratum. Future work could examine whether coarser strata encompassing multiple coalition sizes offer a reasonable workaround. Lastly, our work is effortlessly transferred to other semi-values proposed in game theory like the Banzhaf value.

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A Pseudocode

CalculateAllocation begins by computing the variance estimates prepared by the observations during the exploration phase. Next, it estimates the optimal allocation in potentially multiple iterations. The sizes ℓ which are still part of the optimization problem are kept in $\mathcal{M}$. Each size whose sample number $\hat{m}_\ell^*$ does not exceed its number c_ℓ from the exploration phase, i.e. $\hat{m}_\ell^* \leq c_\ell$, is assigned after computation to the set of dropouts $\mathcal{F}$. The available budget $\bar{T}'$ considered is reduced by the sum of sample numbers of the current dropouts. We assume Round to return a vector of natural numbers by rounding appropriately.

Update updates the affected stratum estimates identical to [18] by iterating over all players. The variance estimation is prepared by maintaining for each stratum the sum observed coalition values in $\Sigma_{i,\ell}^{+/-}$ and the sum of squared values in $\Omega_{i,\ell}^{+/-}$. Note that the value function is only accessed once.

ExactCalculation iterates over all coalitions of size $0, 1, n-1, n$. Each coalition is used to call the Update procedure. After $2n+2$ spending budget tokens the strata values $\phi_{i,0}^-, \phi_{i,0^+}, \phi_{i,1}^-, \phi_{i,n-2}^+, \phi_{i,n-1}^-, \phi_{i,n-1}^+$ are computed exactly.

Algorithm 2 CalculateAllocation

$\hat{\sigma}_{i,\ell-1,+}^2 \leftarrow \frac{1}{c_{i,\ell-1,}^+ - 1}\left(\Omega_{i,\ell-1}^+ - \frac{1}{c_{i,\ell-1,}^+}\left(\Sigma_{i,\ell-1}^+\right)^2\right)$ for all $i \in \mathcal{N}, \ell \in \mathcal{L}'_\nu$

$\hat{\sigma}_{i,\ell,-}^2 \leftarrow \frac{1}{c_{i,\ell,}^- - 1}\left(\Omega_{i,\ell}^- - \frac{1}{c_{i,\ell,}^-}\left(\Sigma_{i,\ell}^-\right)^2\right)$ for all $i \in \mathcal{N}, \ell \in \mathcal{L}'_\nu$

$\hat{m}_\ell^* \leftarrow 0$ for all $\ell \in \{2, \ldots, n-2\}$

$\mathcal{M}, \mathcal{D} \leftarrow \{2, \ldots, n-2\}$

$\bar{T}' \leftarrow \bar{T}$

while $\mathcal{D} \neq \emptyset$ **do**

  $\hat{m}_\ell^* \leftarrow \frac{\sqrt{\sum_{i=1}^n \frac{\hat{\sigma}_{i,\ell-1,+}^2}{\ell} + \frac{\hat{\sigma}_{i,\ell,-}^2}{n-\ell}}}{\sum_{\ell \in \mathcal{M}} \sqrt{\sum_{i=1}^n \frac{\hat{\sigma}_{i,k-1,+}^2}{k} + \frac{\hat{\sigma}_{i,k,-}^2}{n-k}}} \cdot \bar{T}'$ for all $\ell \in \mathcal{M}$

  $\mathcal{D} \leftarrow \{\ell \in \mathcal{M} \mid \hat{m}_\ell^* \leq c_\ell\}$

  $\bar{T}' \leftarrow \bar{T}' - \sum_{\ell \in \mathcal{D}} c_\ell$

  $M \leftarrow M \setminus \mathcal{D}$

end while

$\hat{m}_\ell^* \leftarrow c_\ell$ for all $\ell \in \{0, \ldots, n-2\} \setminus \mathcal{M}$

Output: Round$(\hat{m}_0^*, \ldots, \hat{m}_{n-2}^*)$

Algorithm 3 UPDATE(A)

$v_A \leftarrow \nu(A)$; $c_{|A|} \leftarrow c_{|A|} + 1$
for $i \in A$ **do**
  $\hat{\phi}^+_{i,|A|-1} \leftarrow \frac{c^+_{i,|A|-1} \cdot \hat{\phi}^+_{i,|A|-1} + v_A}{c^+_{i,|A|-1}+1}$
  $c^+_{i,|A|-1} \leftarrow c^+_{i,|A|-1} + 1$
  $\Sigma^+_{i,|A|-1} \leftarrow \Sigma^+_{i,|A|-1} + v_A$
  $\Omega^+_{i,|A|-1} \leftarrow \Omega^+_{i,|A|-1} + (v_A)^2$
end for
for $i \in \mathcal{N} \setminus A$ **do**
  $\hat{\phi}^-_{i,|A|} \leftarrow \frac{c^-_{i,|A|} \cdot \hat{\phi}^-_{i,|A|} + v_A}{c^-_{i,|A|}+1}$
  $c^-_{i,|A|} \leftarrow c^-_{i,|A|} + 1$
  $\Sigma^-_{i,|A|} \leftarrow \Sigma^-_{i,|A|} + v_A$
  $\Omega^-_{i,|A|} \leftarrow \Omega^-_{i,|A|} + (v_A)^2$
end for

Algorithm 4 EXACTCALCULATION [18]

for $s \in \{0, 1, n-1, n\}$ **do**
  for $A \in \{S \subseteq \mathcal{N} \mid |S| = s\}$ **do**
    UPDATE(A)
  end for
end for

B Proofs

Static Allocation for Stratified Sampling. *Proof of Theorem 1*: Let $x^{(m)}_{i,\ell}$ be the m-th sampled marginal contribution of player i's ℓ-th stratum. Since $m_{i,\ell} \geq 1$ for all $i \in \mathcal{N}$ and $\ell \in \mathcal{L}_\Delta$, and the marginal contributions are drawn uniformly at random from their stratum, each estimate $\hat{\phi}_{i,\ell}$ and $\hat{\phi}_i$ is unbiased:

$$\mathbb{E}\left[\hat{\phi}_i\right] = \frac{1}{n}\sum_{\ell=0}^{n-1} \mathbb{E}\left[\hat{\phi}_{i,\ell}\right] = \frac{1}{n}\sum_{\ell=0}^{n-1} \mathbb{E}\left[\frac{1}{m_{i,\ell}}\sum_{m=1}^{m_{i,\ell}} x^{(m)}_{i,\ell}\right] = \frac{1}{n}\sum_{\ell=0}^{n-1} \phi_{i,\ell} = \phi_i\,.$$

Next, we investigate the variance where make use of the independent samples:

$$\mathbb{V}\left[\hat{\phi}_i\right] = \frac{1}{n^2}\sum_{\ell=0}^{n-1} \mathbb{V}\left[\hat{\phi}_{i,\ell}\right] = \frac{1}{n^2}\sum_{\ell=0}^{n-1} \frac{1}{m^2_{i,\ell}}\sum_{m=1}^{m_{i,\ell}} \mathbb{V}\left[x^{(m)}_{i,\ell}\right] = \frac{1}{n^2}\sum_{\ell=0}^{n-1} \frac{\sigma^2_{i,\ell}}{m_{i,\ell}}\,.$$

The bias-variance decomposition allows to combine both intermediate results and quantify the MSE for a single player. Averaging over all players yields:

$$\frac{1}{n}\sum_{i=1}^{n} \mathbb{E}\left[\left(\hat{\phi}_i - \phi_i\right)^2\right] = \frac{1}{n}\sum_{i=1}^{n} \left(\mathbb{E}\left[\hat{\phi}_i\right] - \phi_i\right)^2 + \mathbb{V}\left[\hat{\phi}_i\right] = \frac{1}{n^3}\sum_{i=1}^{n}\sum_{\ell=0}^{n-1} \frac{\sigma^2_{i,\ell}}{m_{i,\ell}}\,.$$

□

Static Allocation for Stratified SVARM. *Proof of Theorem 2*: The unbiasedness of each estimate $\hat{\phi}^+_{i,\ell}$, $\hat{\phi}^-_{i,\ell}$, and $\hat{\phi}_i$ has been shown in Lemma 8, Lemma 9, and Theorem 5 [18]. Let $\bar{m}_{i,\ell,+}, \bar{m}_{i,\ell,-}$ be the number of times the estimate $\hat{\phi}^+_{i,\ell}$ and respectively $\hat{\phi}^-_{i,\ell}$ got updated after the warmup and $m_{i,\ell,+}, m_{i,\ell,-}$ be the total numbers including the warmup.

Lemma 1. *For any $i \in \mathcal{N}$ and $\ell \in \mathcal{L}'_\nu$ the number of updates are binomially distributed with $\bar{m}_{i,\ell-1,+} \sim Bin\left(m_\ell, \frac{\ell}{n}\right)$ and $\bar{m}_{i,\ell,-} \sim Bin\left(m_\ell, \frac{n-\ell}{n}\right)$.*

The proof is analogous to Lemma 10 in [18]. We utilize Lemma 1 to bound the following expectations for all $i \in \mathcal{N}$ and all $\ell \in \mathcal{L}'_\nu$ (cf. Lemma 13 [18]):

$$\mathbb{E}\left[\frac{1}{m_{i,\ell-1,+}}\right] \leq \frac{n}{m_\ell \cdot \ell} \quad \text{and} \quad \mathbb{E}\left[\frac{1}{m_{i,\ell,-}}\right] \leq \frac{1}{\mathbb{E}\left[\bar{m}_{i,\ell,-}\right]} \leq \frac{n}{m_\ell(n-\ell)}\,.$$

From the proof of Theorem 6 in [18] we extract and reuse:

$$\begin{aligned}
\mathbb{V}\left[\hat{\phi}_i\right] &\leq \mathbb{E}\left[\frac{1}{n^2}\sum_{\ell=2}^{n-2}\frac{\sigma^2_{i,\ell-1,+}}{m_{i,\ell-1,+}} + \frac{\sigma^2_{i,\ell,-}}{m_{i,\ell,-}}\right] \\
&= \frac{1}{n^2}\sum_{\ell=2}^{n-2}\sigma^2_{i,\ell-1,+}\cdot\mathbb{E}\left[\frac{1}{m_{i,\ell-1,+}}\right] + \sigma^2_{i,\ell,-}\cdot\mathbb{E}\left[\frac{1}{m_{i,\ell,-}}\right] \\
&\leq \frac{1}{n}\sum_{\ell=2}^{n-2}\frac{1}{m_\ell}\left(\frac{\sigma^2_{i,\ell-1,+}}{\ell} + \frac{\sigma^2_{i,\ell,-}}{n-\ell}\right).
\end{aligned}$$

The bound on the variance enables us to take advantage of the bias-variance decomposition since the unbiasedness is already given:

$$\mathbb{E}\left[\left(\hat{\phi}_i - \phi_i\right)^2\right] = \left(\mathbb{E}\left[\hat{\phi}_i\right] - \phi_i\right)^2 + \mathbb{V}\left[\hat{\phi}_i\right] \leq \frac{1}{n}\sum_{\ell=2}^{n-2}\frac{1}{m_\ell}\left(\frac{\sigma^2_{i,\ell-1,+}}{\ell} + \frac{\sigma^2_{i,\ell,-}}{n-\ell}\right).$$

Averaging the mean squared error over the players completes the proof:

$$\frac{1}{n}\sum_{i=1}^{n}\mathbb{E}\left[\left(\hat{\phi}_i - \phi_i\right)^2\right] \leq \frac{1}{n^2}\sum_{\ell=2}^{n-2}\frac{1}{m_\ell}\sum_{i=1}^{n}\frac{\sigma^2_{i,\ell-1,+}}{\ell} + \frac{\sigma^2_{i,\ell,-}}{n-\ell}\,.$$

□

Optimal Allocation for Stratified Sampling. *Proof of Theorem 3*: The relaxed sample allocation problem for marginal contributions is given by:

$$\begin{aligned}
(m^*_{i,\ell})_{i\in\mathcal{N},\ell\in\mathcal{L}'_\Delta} = \underset{(m_{i,\ell})_{i\in\mathcal{N},\ell\in\mathcal{L}'_\Delta}}{\arg\min} \quad & \frac{1}{n^3}\sum_{i=1}^{n}\sum_{\ell=1}^{n-2}\frac{\sigma^2_{i,\ell}}{m_{i,\ell}} \\
\text{s.t.} \quad & 2\sum_{i=1}^{n}\sum_{\ell=1}^{n-2} m_{i,\ell} = \tilde{T} \\
& m_{i,\ell} \in \mathbb{R}_{>0} \qquad \forall i \in \mathcal{N}, \ell \in \mathcal{L}'_\Delta\,.
\end{aligned}$$

We derive the solution by employing Lagrangian multipliers, hence we minimize

$$L\left((m_{i,\ell})_{i\in\mathcal{N},\ell\in\mathcal{L}'_\Delta},\lambda\right) = \frac{1}{n^3}\sum_{i=1}^{n}\sum_{\ell=1}^{n-2}\frac{\sigma_{i,\ell}^2}{m_{i,\ell}} + \lambda\left(2\sum_{i=1}^{n}\sum_{\ell=1}^{n-2}m_{i,\ell} - \tilde{T}\right).$$

We solve for $\nabla L = 0$ which requires

$$\frac{\partial}{\partial m_{i,\ell}}L = -\frac{\sigma_{i,\ell}^2}{n^3 m_{i,\ell}^2} + 2\lambda \overset{!}{=} 0 \;\forall i \in \mathcal{N}, \ell \in \mathcal{L}'_\Delta \quad \text{and} \quad \frac{\partial}{\partial\lambda}L = 2\sum_{i=1}^{n}\sum_{\ell=1}^{n-2}m_{i,\ell} - \tilde{T} \overset{!}{=} 0\,.$$

This equation system yields together with the budget constraint

$$m_{i,\ell} = \frac{\sigma_{i,\ell}}{\sqrt{2\lambda n^3}} \quad \forall i \in \mathcal{N}, \ell \in \mathcal{L}'_\Delta \quad \text{and} \quad \lambda = \frac{2}{n^3\tilde{T}^2}\left(\sum_{i=1}^{n}\sum_{\ell=1}^{n-2}\sigma_{i,\ell}\right)^2.$$

from which we derive the optimal allocation to minimize the objective function:

$$m_{i,\ell} = \frac{\sigma_{i,\ell}}{2\sum_{j=1}^{n}\sum_{k=1}^{n-2}\sigma_{j,k}} \cdot \tilde{T}\,.$$

□

Optimal Allocation for Stratified SVARM. *Proof of Theorem 4*: The relaxed sample allocation problem for coalitions is given by:

$$\begin{aligned}(m_\ell^*)_{\ell\in\mathcal{L}'} = \min_{(m_\ell)_{\ell\in\mathcal{L}'}} \quad & \frac{1}{n^2}\sum_{\ell=2}^{n-2}\frac{c_\ell}{m_\ell} \\ \text{s.t.} \quad & \sum_{\ell=2}^{n-2} m_\ell = \bar{T} \\ & m_\ell \in \mathbb{R}_{\geq 0} \qquad\qquad \forall \ell \in \mathcal{L}'_\nu\end{aligned}$$

where we define the coefficients $c_\ell := \sum_{i=1}^{n}\frac{\sigma_{i,\ell-1,+}^2}{\ell} + \frac{\sigma_{i,\ell,-}^2}{n-\ell}$. We derive the solution by employing Lagrangian multipliers, hence we minimize

$$L(m_2,\ldots,m_{n-2},\lambda) = \frac{1}{n^2}\sum_{\ell=2}^{n-2}\frac{c_\ell}{m_\ell} + \lambda\left(\sum_{\ell=2}^{n-2}m_\ell - \bar{T}\right).$$

We solve for $\nabla L = 0$ which requires

$$\frac{\partial}{\partial m_\ell}L = -\frac{c_\ell}{n^2 m_\ell^2} + \lambda \overset{!}{=} 0 \qquad \forall \ell \in \mathcal{L}'_\nu \quad \text{and} \quad \frac{\partial}{\partial\lambda}L = \sum_{\ell=2}^{n-2}m_\ell - \bar{T} \overset{!}{=} 0\,.$$

This equation system yields together with the budget constraint

$$m_\ell = \sqrt{\frac{c_\ell}{\lambda n^2}} \quad \forall \ell \in \mathcal{L}'_\nu \quad \text{and} \quad \lambda = \frac{1}{n^2\bar{T}^2}\left(\sum_{\ell=2}^{n-2}\sqrt{c_\ell}\right)^2.$$

from which we derive the optimal allocation to minimize the objective function:

$$m_\ell = \frac{\sqrt{c_\ell}}{\sum_{k=2}^{n-2} \sqrt{c_k}} \cdot \bar{T} = \frac{\sqrt{\sum_{i=1}^{n} \frac{\sigma^2_{i,\ell-1,+}}{\ell} + \frac{\sigma^2_{i,\ell,-}}{n-\ell}}}{\sum_{k=2}^{n-2} \sqrt{\sum_{i=1}^{n} \frac{\sigma^2_{i,k-1,+}}{k} + \frac{\sigma^2_{i,k,-}}{n-k}}} \cdot \bar{T} \,.$$

□

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